System Design Patterns — Vol 1: Fundamentals and Algorithms

Every basic concept from zero (Module 0) and the algorithms behind real systems, step by step (Module 19)

2 modules · 60+ patterns · every lesson explained, with course diagrams and links to the originals

what it is · problem · how it works · trade-offs · real world · interview · self-check · TL;DR

Personal study notes summarised from the DesignGurus course (Arslan Ahmad). Not for redistribution.

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Module 0 — System Design Fundamentals

Module Introduction

What this module is. Every other module in these notes assumes you already know the basic vocabulary of system design: what *latency* means, what a *load balancer* does, why a *cache* helps, what *sharding* or *eventual consistency* are. This module teaches all of it from zero. If you have only ever written programs that run on one computer, start here.

How it is organised. Six parts, each building on the one before:

Part	What you learn
1 — Foundations and performance	Clients and servers, latency, throughput, percentiles, estimation, availability and the nines, SLAs, scaling up vs out, stateless services, bottlenecks, how a design interview runs
2 — Networking and communication	IP, ports, DNS, TCP/UDP/QUIC, HTTP versions, TLS, REST/gRPC/GraphQL, WebSockets and friends, proxies, API gateways, load balancers (every type and algorithm), CDNs
3 — Data storage	Relational databases, ACID, isolation levels, NoSQL families, indexes, replication, partitioning and sharding, consistent hashing, block/file/object storage, OLTP vs OLAP, durability
4 — Caching, async and consistency	Cache strategies and eviction, queues and streams, delivery guarantees, idempotency, consistency models, CAP/PACELC, quorums, consensus, clocks, 2PC and sagas, Bloom filters, ID generation
5 — Architecture, operations and security	Monolith vs microservices vs serverless, resilience patterns, rate limiting, autoscaling, containers and Kubernetes, CI/CD and deployments, observability, disaster recovery, auth, encryption, web security
6 — Putting it together	A request's journey end to end, a worked design, a decision cheat sheet, 60 interview questions and a master A–Z glossary

How to read it. Each concept follows the same rhythm as the rest of the book: *What it is → Why it matters → How it works → Types → Trade-offs → Common mistakes → Real-world examples → In the interview → TL;DR*. Read Parts 1–5 in order the first time; afterwards use the Part 6 glossary and cheat sheet as a reference. Every part ends with its own glossary table.

> These notes are written in my own words from official documentation, papers and engineering blogs; specific numbers were checked against their sources.

Part 1 — Foundations and performance

This part is the ground floor. It explains what "system design" means, names the basic pieces every web application is built from, and gives you the vocabulary for talking about speed (latency, throughput, percentiles), size (estimation, units), staying up (availability, reliability, durability) and growing (scalability). It ends with a step-by-step method for a system design interview. Nothing here assumes more than this: a website runs on a computer called a **server**, and you reach it from your phone or laptop over the internet.

What system design is

What it is

System design is the work of deciding which parts a piece of software needs, what each part does, how the parts talk to each other, and where data lives, so that the whole thing meets its goals. Those goals

usually include serving a certain number of users, responding within a certain time, staying available when machines break, and costing an acceptable amount.

Writing code answers "how does this function work?". System design answers "how do these twenty programs, running on two hundred machines in three countries, work together without losing anyone's data?".

> **Analogy:** an architect designing a building does not lay bricks. They decide how many floors, where the lifts and fire exits go, how water and electricity flow, and how many people the building can hold. System design is the architect's job for software. The builders (programmers) then fill in each room.

Why it matters

A program that works perfectly for 10 users can collapse at 10 million. The code may be fine; the shape of the system is wrong. Typical shape problems are: one database that every request has to queue for, one machine whose failure takes the whole site down, or data stored so far from users that every page takes seconds. These are fixed by design decisions, not by better loops.

System design also matters for cost. Two designs that both work can differ by ten times in their monthly cloud bill.

How it works

Most system design follows the same loop:

1. **Understand the goal.** What must the system do (features), and how well (speed, size, uptime)?
2. **Estimate the load.** How many users, requests per second, gigabytes per day?
3. **Sketch the parts.** Clients, servers, databases, caches, queues and how requests flow between them.
4. **Find the weak spots.** Which part runs out of capacity first? Which part, if it dies, takes everything with it?
5. **Fix and justify.** Add or change parts, and state what each change costs.

You repeat this loop as the system grows. A design is never "finished"; it is right for a given load and set of goals.

Trade-offs

Almost every design choice gives something up. Faster reads may mean slower writes. More copies of data mean better safety but more cost and more chance that copies disagree. Simpler systems are easier to run but may not scale. A good designer names the trade-off out loud instead of pretending a choice is free.

Common mistakes

- Designing for Google-scale traffic when the product has 1,000 users. Extra parts add cost and failure points.
- Jumping to product names ("use Kafka") before stating the problem the product solves.
- Ignoring failure. Every machine, disk and network link eventually breaks.

In the interview

Q: What is the interviewer actually testing in a system design round?

A: Your ability to turn a vague product idea into a reasoned architecture: asking the right questions, estimating scale, choosing components, and explaining trade-offs. There is rarely one correct answer; the reasoning is graded.

Q: How is system design different from coding?

A: Coding is about correctness inside one program. System design is about how many programs and machines cooperate, how they behave under load and failure, and what that costs.

TL;DR

System design is choosing the parts of a software system and how they fit together so it is fast enough, big enough, reliable enough and cheap enough. Every choice is a trade-off you should be able to explain.

Clients, servers and the request/response cycle

What it is

A **client** is any program that asks for something. Your web browser, a phone app, or even another server can be a client. A **server** is a program (and, loosely, the machine it runs on) that waits for requests and answers them.

The conversation between them follows the **request/response** pattern: the client sends a **request** ("give me the home page", "save this photo"), the server does some work, and sends back a **response** (the page, or "saved, here is its ID"). On the web, these messages usually use **HTTP** (HyperText Transfer Protocol), a standard text format for requests and responses. Part 2 covers HTTP in detail.

> **Analogy:** a restaurant. You (the client) tell the waiter your order (the request). The kitchen (the server) cooks it and the waiter brings the plate (the response). You don't need to know how the kitchen works; you only need to know how to order.

How it works

Here is what happens when you open `https://example.com/profile` in a browser, at the simplest level:

1. The browser finds the server's address on the internet (using **DNS**, the internet's phone book, covered in Part 2).
2. It opens a network connection to that server.
3. It sends an HTTP request such as:

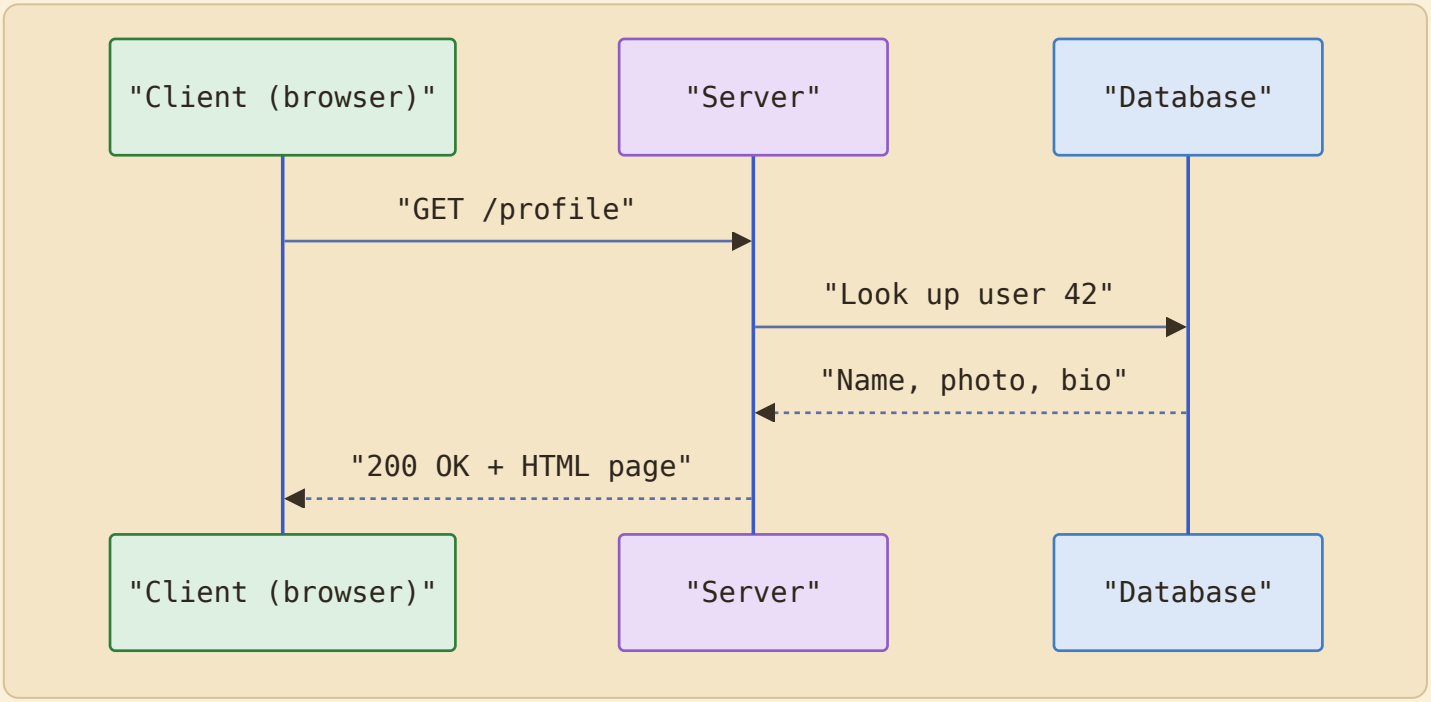
```
GET /profile HTTP/1.1
Host: example.com
```

4. The server reads the request, looks up your profile (perhaps in a database), and builds a response.
5. It sends back something like:

```
HTTP/1.1 200 OK
Content-Type: text/html

<html> ... your profile ... </html>
```

6. The browser draws the page.



Types

- **Thin client vs thick client.** A thin client (a basic web page) does little work itself and relies on the server. A thick client (a mobile game, a desktop app) does a lot locally and talks to the server only for shared data.
- **Server-to-server.** Inside a large system, servers are constantly clients of other servers. The "profile service" may call the "photo service". The same request/response idea applies.
- **Push instead of pull.** Normally the client asks (pulls). Some features need the server to send data unasked (a chat message arriving). Techniques for this (WebSockets, Server-Sent Events) are in Part 2.

Common mistakes

- Thinking "server" always means one physical box. A server is a role. One machine can run many server programs, and one logical "server" (say, `api.example.com`) is often hundreds of machines behind a load balancer.
- Forgetting that the network between client and server is slow and unreliable compared with work inside one machine.

In the interview

Q: Can a server also be a client?

A: Yes. Any program that sends a request is acting as a client for that request. In a microservice system almost every service is both.

TL;DR

A client asks, a server answers. Almost everything on the web is a chain of request/response conversations, often with servers acting as clients of other servers.

The building blocks of a web application

What it is

Nearly every web application, from a blog to a bank, is built from the same handful of parts. Learn these and most architecture diagrams become readable.

Part	What it does	Everyday comparison
Client (browser/app)	Shows the interface, sends requests	The customer
Load balancer	Spreads requests across many servers	The host seating diners at free tables
App server	Runs the business logic	The kitchen
Database	Stores data permanently and lets you query it	The pantry and recipe book
Cache	Keeps a fast copy of popular data	Ingredients kept on the counter
Message queue	Holds jobs to be done later	The order ticket rail
Object storage	Stores large files (images, videos)	The warehouse
CDN	Copies static files close to users	Branch shops in every town

How it works

Client. The browser or mobile app. It renders the user interface and calls the backend over HTTP.

App server (application server). A program written in a language such as Java, Python, Go or JavaScript (Node.js) that receives requests, applies rules ("only friends may see this post"), reads and writes data, and returns a response. You usually run several identical copies so that one crash doesn't stop the site.

Load balancer. A component that sits in front of the app servers and forwards each incoming request to one of them, so no single server is overloaded and dead servers are skipped. (Covered in depth in Part 2.)

Database. Software that stores data durably (it survives power loss) and lets you search it. Examples: PostgreSQL, MySQL, MongoDB, DynamoDB. It is usually the hardest part to scale because it holds **state** (data that must be remembered). (Part 3.)

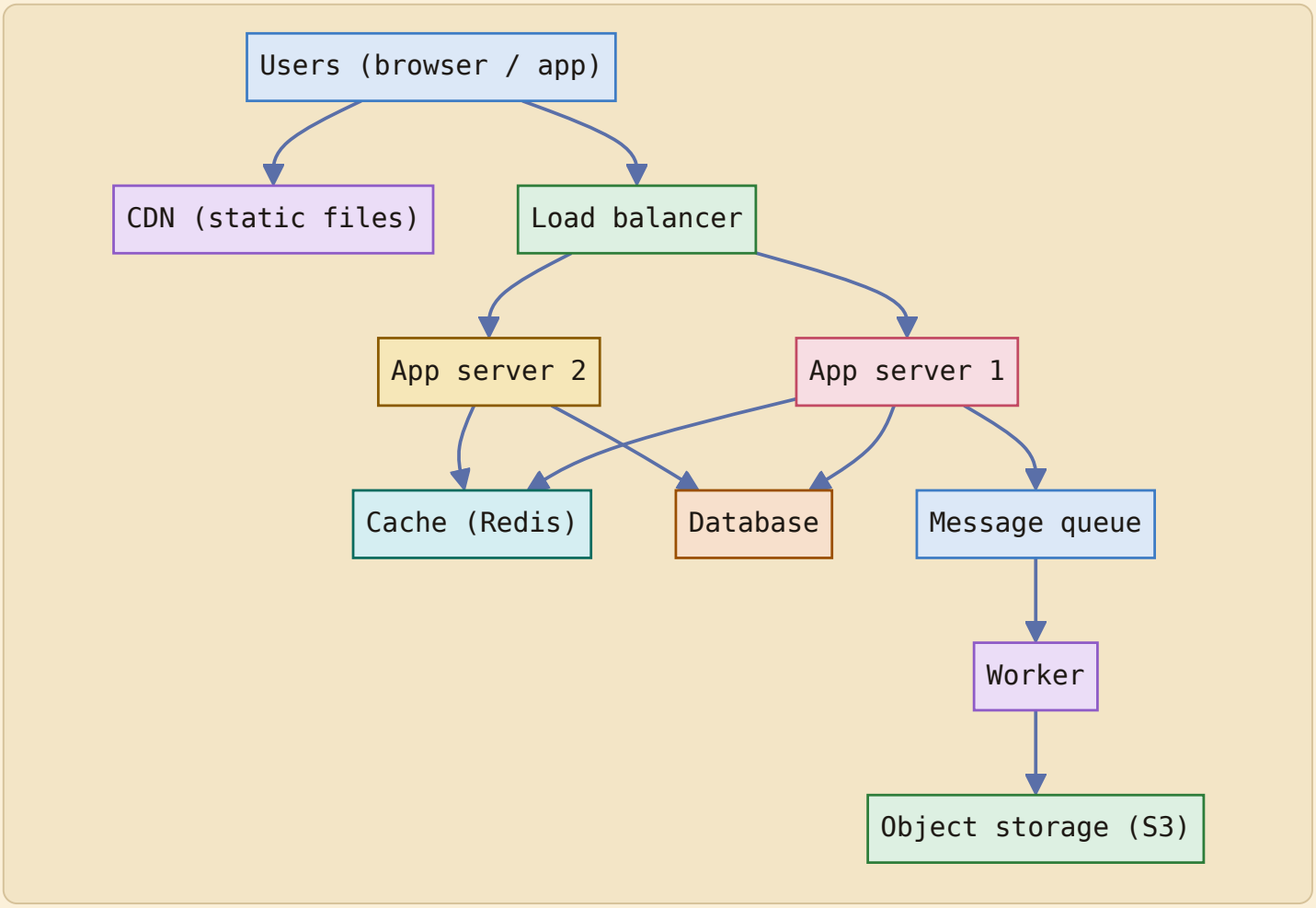
Cache. A small, fast store (usually in memory, RAM) holding copies of data that is read often, so the app doesn't have to ask the slower database every time. Redis and Memcached are common. Reading from RAM takes well under a millisecond; a database query may take several milliseconds or more. (Part 4.)

Message queue. A buffer where one part of the system drops a job ("send a welcome email to user 42") and another part picks it up later. This lets the web request return immediately while slow work happens in the background. Examples: RabbitMQ, Amazon SQS, Apache Kafka. (Part 4.)

Workers. Background programs that take jobs from the queue and do them.

Object storage. A service for storing big files ("objects") cheaply, addressed by a key such as `photos/42/avatar.jpg`. Amazon S3 is the best-known example. Databases are bad at storing large files; object storage is built for it. (Part 3.)

CDN (Content Delivery Network). A network of servers spread around the world that keep copies of static files (images, JavaScript, video) close to users, so a user in Sydney doesn't fetch every image from Virginia. (Part 2.)



Why it matters

In an interview you will draw some version of this diagram within the first fifteen minutes. Knowing what each box is for lets you justify adding it ("we add a cache because reads outnumber writes 100 to 1") rather than adding boxes because "big systems have them".

Common mistakes

- Adding every component by default. A small internal tool may need only one app server and one database.
- Storing images inside the database. Put files in object storage and keep only their key (address) in the database.
- Doing slow work (emails, video encoding) inside the web request. Put it on a queue.

Real-world examples

- Instagram's early stack (2011–2012) was Django app servers, PostgreSQL, Redis and Memcached, with photos in Amazon S3 — almost exactly the diagram above.
- Most cloud tutorials (AWS, Google Cloud, Azure) describe a "three-tier" app: presentation (client), logic (app servers), data (database), with caches and queues added as it grows.

In the interview

Q: Why not keep everything on one server?

A: One server is a single point of failure (if it dies the site is down) and has a hard capacity limit. Splitting into stateless app servers behind a load balancer, plus a separate database, lets you add servers and survive crashes.

Q: When do you add a queue?

A: When some work is slow, can be done later, or comes in bursts. The queue lets the user get a fast response and smooths spikes for the workers behind it.

TL;DR

Client, load balancer, app servers, database, cache, queue with workers, object storage and CDN are the standard building blocks. Each solves one problem; add each one only when you can name that problem.

Latency, throughput and bandwidth

What it is

These three words describe speed, but different kinds of speed.

- **Latency** is how long one thing takes: the delay between asking and getting the answer. Measured in time: milliseconds (ms, thousandths of a second) or microseconds (µs, millionths; written "us" in plain text).
- **Throughput** is how many things get done per unit of time: requests per second, messages per second, megabytes per second.
- **Bandwidth** is the maximum amount of data a network link (or disk, or memory bus) can carry per second. It is a capacity, like the width of a pipe. Measured in bits per second (Mbps, Gbps).

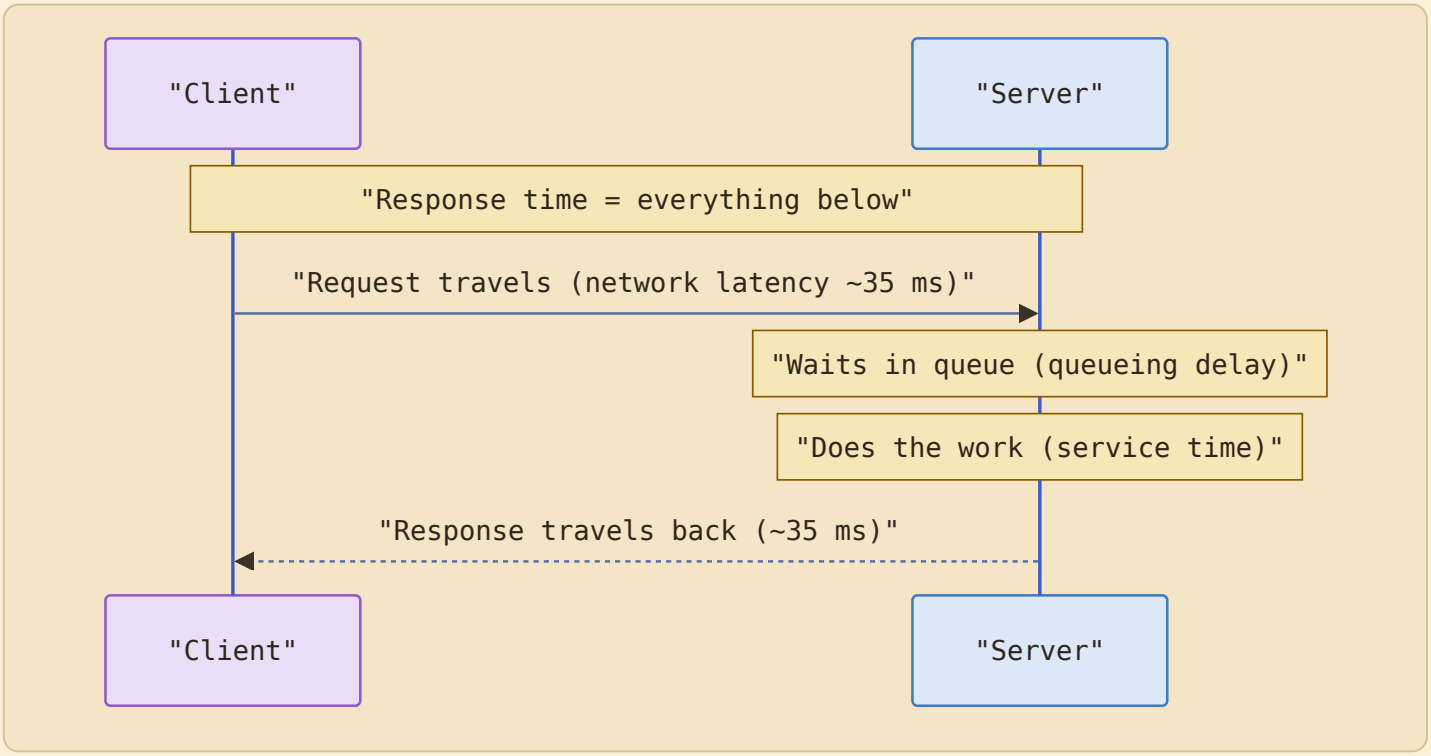
> **Analogy:** a motorway. Latency is how long one car takes to drive from London to Birmingham (about 2 hours). Bandwidth is how many lanes the motorway has. Throughput is how many cars actually arrive per hour, which can be lower than the lanes allow if there is a crash or a toll booth.

Response time vs latency

People often use these as synonyms, but strictly:

- **Response time** is what the client experiences: the full time from sending a request to receiving the complete answer. It includes network travel both ways, time waiting in queues, and the actual processing.
- **Latency**, used strictly, covers only delay: the stretch before a server starts on the request (sitting in a queue, travelling a network hop), not the work itself.
- **Service time** is the time the server spends actually working on it.

So: response time = network delay + queueing delay + service time. In interviews "latency" almost always means response time; just be clear which one you mean when it matters.



Why it matters

- Users feel latency. Interfaces under about 100 ms feel instant; delays over a second break concentration. Google and Amazon have both reported that small added delays measurably reduce searches and sales.
- Throughput decides how many machines you need. If one server handles 1,000 requests per second and you expect 50,000, you need at least 50 (and more for headroom).
- Bandwidth decides whether large transfers (video, backups) are even possible in time.

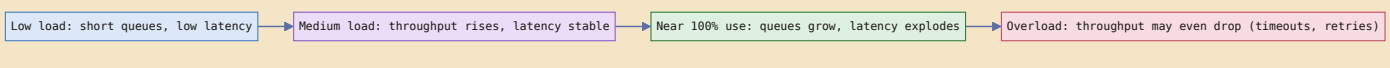
How it works

Where latency comes from. A request picks up delay at every step:

1. **Propagation delay:** signals in fibre travel at roughly two thirds the speed of light, about 200 km per millisecond. London to New York is about 5,600 km, so one way takes at least 28 ms, and a real **round trip** (there and back, RTT) is typically about 70 ms because cables don't run in straight lines and routers add delay.
2. **Transmission delay:** time to push the bits onto the wire. Sending 1 MB (8 megabits) over a 100 Mbps link takes 80 ms; over 1 Gbps, 8 ms. This is where bandwidth matters.
3. **Queueing delay:** waiting behind other requests at routers, load balancers, thread pools, database locks.
4. **Processing delay:** the actual work: parsing, running code, reading disk.

Latency and bandwidth are independent. A satellite link may have huge bandwidth and terrible latency (about 600 ms round trip for geostationary satellites). A small request is limited by latency; a large file download is limited by bandwidth. Adding bandwidth does not make a small API call faster.

Throughput and latency are linked through load. At low load, throughput rises as you send more requests and latency stays flat. As a server approaches full use, requests start waiting for each other and latency shoots up, while throughput levels off. Running a server with its CPU (central processing unit, the processor) 95% busy usually means terrible latency.



Trade-offs

- **Batching** improves throughput but hurts latency: waiting to collect 100 writes and sending them together is efficient, but the first write waits for the other 99.
- **Spare capacity** keeps latency low but costs money. Many teams target 50–70% peak utilisation so that bursts don't create queues.

Common mistakes

- Mixing bits and bytes. Network speeds are in bits (Mbps); file sizes are in bytes (MB). 1 byte = 8 bits, so a 100 Mbps link moves at most 12.5 MB per second.
- Assuming more bandwidth fixes slowness. If the problem is round trips (a page making 40 sequential API calls), only fewer round trips or shorter distances help.
- Measuring throughput without latency (or the reverse). "10,000 requests per second" means nothing if each takes 5 seconds.

Real-world examples

- Video streaming (Netflix, YouTube) is bandwidth-bound: a 4K stream needs roughly 15–25 Mbps.
- High-frequency trading firms pay for microwave links between Chicago and New York because they shave a few milliseconds of propagation delay compared with fibre.
- Online games are latency-bound: players feel the difference between 20 ms and 120 ms "ping" (round-trip time).

In the interview

Q: What is the difference between latency and throughput?

A: Latency is how long one request takes; throughput is how many requests complete per second. You can have high throughput with high latency (a cargo ship) or low latency with low throughput (a motorbike courier).

Q: A user in Australia complains the site is slow; servers are in Virginia. What helps?

A: The round trip is roughly 200 ms, so cut round trips: serve static files from a CDN near them, reduce the number of sequential calls, reuse connections, and consider a regional deployment closer to them. More bandwidth won't help much.

TL;DR

Latency is time per request, throughput is requests per time, bandwidth is the pipe's maximum capacity. They are related through load: push a system near its limit and latency rises sharply.

Percentiles, tail latency and why averages lie

What it is

A **percentile** tells you the value below which a given percentage of measurements fall. If the **p95** (95th percentile) latency is 300 ms, then 95% of requests finished in 300 ms or less, and 5% took longer.

The common ones:

Name	Meaning	1 in N requests slower
p50 (median)	Half are faster, half slower	1 in 2
p90	90% are faster	1 in 10
p95	95% are faster	1 in 20
p99	99% are faster	1 in 100
p999 (p99.9)	99.9% are faster	1 in 1,000

Tail latency means the slow end of the distribution: p99, p999 and the maximum. These are "the tail" because on a graph of latencies the slow requests form a long thin tail to the right.

> **Analogy:** a bus route. "The average trip takes 20 minutes" sounds fine. But if one trip in ten takes 90 minutes because of a broken-down bus, someone who rides twice a day gets stuck several times a month. The p95 (90 minutes) describes their real worry far better than the average.

Why averages lie

The **mean** (average) mixes fast and slow requests into one number that may match no real request. Take ten requests:

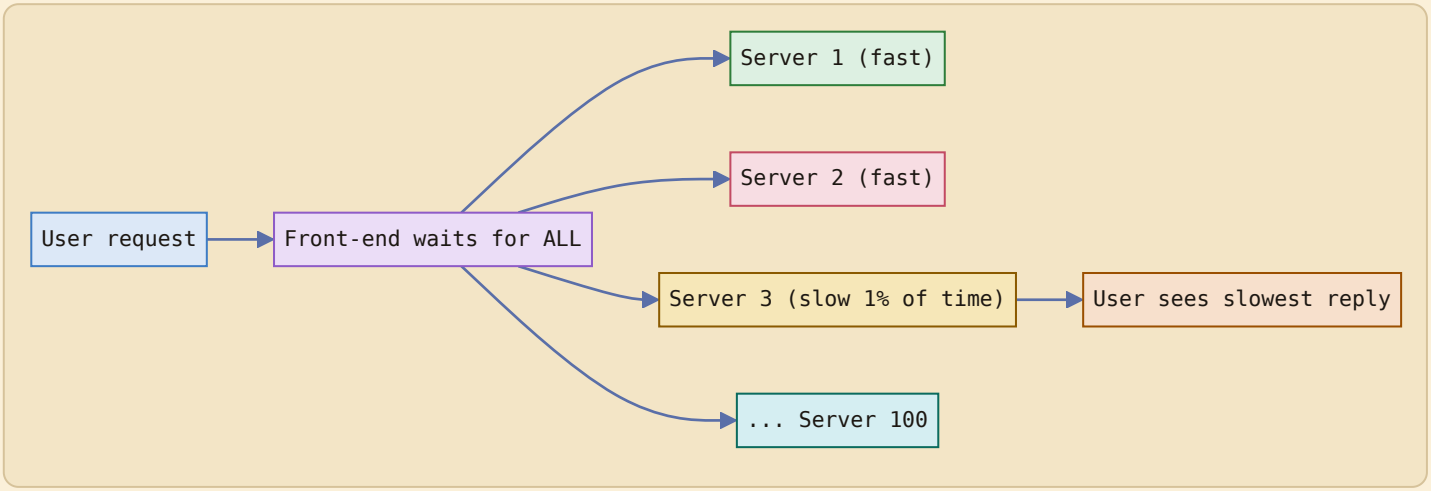
```
10, 10, 10, 10, 10, 10, 10, 10, 10, 1000   (milliseconds)
```

- Mean = 1090 / 10 = 109 ms. No request actually took 109 ms.
- Median (p50) = 10 ms. The typical user is happy.
- p95 and p99 = 1000 ms (they land on the slowest request). One user in ten waits a full second.

The mean hides the 1-second outlier inside a number that looks moderate. In real systems latency is almost never a neat bell curve; it is skewed, with a long tail caused by garbage collection pauses, cache misses, slow disks, retries and queueing. Percentiles show the shape; averages flatten it.

Why the tail matters more than you think

1. **Heavy users hit the tail most.** A user who loads 100 pages a day hits your p99 about once a day. Often these are your most valuable customers (they have the most data, the biggest carts).
2. **Fan-out amplifies the tail.** If one user request calls 100 backend servers in parallel and waits for all of them, the request is as slow as the slowest server. Suppose each server is slow (1 second) just 1% of the time. The chance that *at least one* of 100 is slow is 1 − 0.99^100 ≈ 63%. So a 1-in-100 problem per server becomes a nearly 2-in-3 problem for the user. This effect is described in Google's well-known paper "The Tail at Scale" (Dean and Barroso, 2013).



How it works

Measuring percentiles. Record each request's latency, sort them, and pick the value at the required position (the 990th of 1,000 sorted values is roughly p99). Real monitoring systems can't store every value, so they use compact summaries called **histograms** (counts per latency bucket, e.g. 0–10 ms, 10–25 ms ...) or sketches such as HDRHistogram or t-digest, which estimate percentiles with small error.

Do not average percentiles. The p99 of server A plus the p99 of server B divided by two is not the p99 of the combined traffic. Merge the histograms and compute the percentile from the merged data.

Reducing tail latency. Common techniques:

- **Hedged requests:** if a reply hasn't come back within (say) the p95 time, send the same request to a second replica and use whichever answers first.
- **Timeouts and fewer fan-out calls.**

- Removing sources of pauses: tune garbage collection, avoid shared locks, keep spare capacity so queues stay short.

Common mistakes

- Reporting only the average in dashboards or SLOs.
- Measuring latency only on the server. The client also sees network time, DNS lookups and connection setup.
- **Coordinated omission:** a load-test tool that waits for each slow response before sending the next request quietly sends fewer requests during slow periods, so it under-reports the tail.

Real-world examples

- Amazon has long described setting internal service targets at p99.9 rather than averages, so that even its heaviest customers get good performance.
- Monitoring tools such as Prometheus, Datadog and Grafana show latency as p50/p95/p99 lines by default.

In the interview

Q: Why use p99 instead of the average for an SLO?

A: The average hides slow outliers and doesn't describe any real user. p99 says "99% of requests are at least this fast", which captures the experience of the unlucky but common slow cases.

Q: Your service's p99 is fine but users still complain. Why might that be?

A: Each page may call many services or make many requests, so the chance of hitting at least one slow call per page is much higher than 1%. Look at end-to-end page latency and fan-out, not each service in isolation.

TL;DR

Use percentiles (p50, p95, p99, p999), not averages, to describe latency. The slow tail matters because heavy users and fan-out make rare slowness common.

Little's law

What it is

Little's law is a simple rule that holds for any stable system where items arrive, wait or get served, and leave:

> **$L = \lambda \times W$**

- **L** = the average number of items inside the system at once (requests in flight, people in a shop).
- **λ** (lambda) = the average arrival rate (requests per second). In a stable system this equals the throughput.
- **W** = the average time each item spends inside (latency).

> **Analogy:** a coffee shop. If 2 customers arrive per minute and each stays 10 minutes, then on average $2 \times 10 = 20$ customers are in the shop. You don't need to know anything about how the barista works.

Why it matters

It lets you work out any one of the three numbers from the other two, which is extremely handy for sizing:

- **How many concurrent connections or threads do I need?** If a service gets 1,000 requests per second and each takes 200 ms (0.2 s), then on average $1{,}000 \times 0.2 =$ **200 requests are in flight**. You need at least 200 worker threads or connections (plus headroom), or requests will queue.

- **What throughput can I get?** A database connection pool of 20 connections, with queries taking 5 ms each, can serve at most 20 / 0.005 = **4,000 queries per second**.
- **Why did latency double?** If concurrency limits stay fixed and latency doubles (a slow dependency), maximum throughput halves.

How it works

1. Pick the boundary of your "system": one server, one thread pool, one queue, the whole service.
2. Measure or assume two of the three numbers over a long enough period for the system to be stable (arrivals roughly equal departures).
3. Compute the third.

Rearranged forms:

- W = L / λ (latency from in-flight count and rate)
- λ = L / W (throughput from concurrency limit and latency)

Common mistakes

- Applying it to an unstable system where the queue keeps growing (arrival rate above capacity). Then there is no steady state and L grows without limit.
- Forgetting units. Convert milliseconds to seconds before multiplying by "per second".

In the interview

Q: A service handles 5,000 requests per second with an average latency of 40 ms. How many requests are being processed at any moment?

A: L = 5,000 × 0.04 = 200 requests in flight on average. If the service has 50 instances, that's about 4 per instance.

Q: Your thread pool has 100 threads and each request holds a thread for 250 ms. What is the max throughput?

A: λ = 100 / 0.25 = 400 requests per second. Beyond that, requests queue and latency rises.

TL;DR

Items in the system = arrival rate × time in system. Use it to size thread pools, connection pools and fleets, and to explain why slower dependencies cut throughput.

Latency numbers every engineer should know

What it is

A list of rough costs for common operations, first popularised by Google's Jeff Dean and Peter Norvig. The exact numbers change with hardware (the table below uses rounded figures for current machines; the original list, for example, gave 150 us for an SSD random read); what matters is the **order of magnitude** (whether something takes nanoseconds, microseconds or milliseconds) and the ratios between them.

Units: 1 ns (nanosecond) = one billionth of a second. 1 µs (microsecond) = 1,000 ns. 1 ms (millisecond) = 1,000 µs.

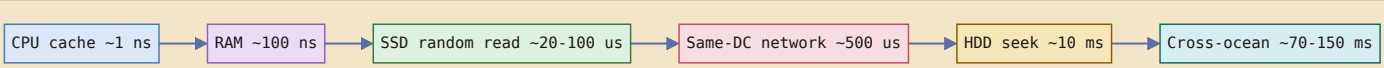
Operation	Approximate time	Scaled: if 1 ns were 1 second
CPU L1 cache read	1 ns	1 second
CPU L2 cache read	4 ns	4 seconds
Main memory (RAM) read	100 ns	about 2 minutes
Compress 1 KB (fast algorithm)	2 us	about half an hour
Send 1 KB over 1 Gbps network	10 us	about 3 hours
Read 4 KB at random from NVMe SSD	20-100 us	6 hours to a day
Read 1 MB sequentially from RAM	50-250 us	14 hours to 3 days
Round trip inside one data centre	0.5 ms (500 us)	about 6 days
Read 1 MB sequentially from SSD	0.2-1 ms	2 to 12 days
Hard disk (HDD) seek	10 ms	about 4 months
Read 1 MB sequentially from HDD	5-20 ms	2 to 8 months
Round trip London to New York	70 ms	about 2 years
Round trip California to Europe	150 ms	about 5 years

(A **CPU cache** is a tiny, very fast memory built into the processor. **RAM** is the computer's main working memory, lost on power-off. An **SSD** is a solid-state drive with no moving parts; an **HDD** is a spinning magnetic disk with a moving arm, which is why "seeking" to a new position is slow.)

Why it matters

These numbers drive design decisions:

- **Memory is about 100 to 1,000 times faster than SSD for random reads**, and SSD is about 100 times faster than spinning disk. That's why caches live in RAM.
- **A network round trip inside a data centre (0.5 ms) costs as much as thousands of memory reads.** A loop that makes 1,000 small database calls spends at least half a second just on round trips. Batch them.
- **Crossing an ocean costs 70–150 ms per round trip**, and the speed of light is not getting faster. Only moving data closer to users (CDNs, regional servers) fixes it.
- **Sequential reads are far faster than random reads**, especially on HDD. Many storage engines (append-only logs, and the log-structured merge trees, or LSM-trees, of Part 3) are designed to write sequentially for this reason.



Common mistakes

- Memorising exact numbers. Interviewers want orders of magnitude: "RAM is nanoseconds, SSD microseconds, network and disk milliseconds".
- Treating a remote cache (Redis over the network) as "as fast as memory". The RAM read is 100 ns, but the network round trip to reach it is ~0.5 ms. It is still far faster than a typical database query, but not free.

In the interview

Q: Why is a cache faster than a database?

A: A cache serves data from RAM (nanoseconds per read plus one network hop), while a database may read from disk, parse a query, take locks and do more work. Typical cache hits take well under 1 ms; database queries often take several ms or more.

Q: Why does a service that calls another service 50 times per request feel slow even though each call is "fast"?

A: Each call costs at least a network round trip (~0.5 ms in the same data centre, more across zones), and sequential calls add up: 50 × 1 ms = 50 ms before any real work. Batch or parallelise the calls.

TL;DR

Memory is nanoseconds, SSD is tens of microseconds, data-centre network is half a millisecond, spinning disks and cross-continent trips are tens to hundreds of milliseconds. Design so the hot path stays at the fast end.

Units, powers of two and data sizes

What it is

Estimation needs a shared vocabulary for size. Data is measured in **bits** (a single 0 or 1) and **bytes** (8 bits, enough for one English letter).

Unit	Bytes (decimal, used in estimates)	Power of two (exact)	Roughly holds
KB (kilobyte)	1,000	2^10 = 1,024	A short text message or tweet with metadata
MB (megabyte)	1,000,000 (10^6)	2^20 ≈ 1.05 million	A high-quality photo, a minute of MP3
GB (gigabyte)	10^9	2^30 ≈ 1.07 billion	An hour of streamed HD video (a few GB)
TB (terabyte)	10^12	2^40 ≈ 1.1 trillion	A typical laptop drive
PB (petabyte)	10^15	2^50	A large company's data warehouse

Above PB come EB (exabyte, 10^18) and ZB (zettabyte, 10^21). Strictly, the power-of-two units have their own names (KiB, MiB, GiB), but in estimates everybody rounds 2^10 to 1,000. The 2–10% error doesn't matter.

How it works

Powers of two to remember:

- 2^10 ≈ 1 thousand (10^3)
- 2^20 ≈ 1 million (10^6)
- 2^30 ≈ 1 billion (10^9)
- 2^40 ≈ 1 trillion (10^12)
- 2^32 ≈ 4.3 billion (the number of IPv4 addresses, and the limit of an unsigned 32-bit integer; a 32-bit ID column runs out at about 2 billion signed or 4 billion unsigned)
- 2^64 ≈ 1.8 × 10^19 (plenty for any ID)

Typical sizes of things:

Item	Rough size
A number (integer)	4 or 8 bytes
A timestamp	8 bytes
A UUID (random unique ID)	16 bytes (36 characters as text)
A short text post with metadata	0.5-1 KB
A web page's HTML	50-100 KB
A compressed photo	200 KB - 2 MB
One minute of HD video	roughly 25-60 MB

Time conversions:

- 1 day = 86,400 seconds ≈ **10^5 seconds** (round it; makes maths easy).
- 1 month ≈ 2.5 million seconds (2.6 × 10^6).
- 1 year ≈ 31.5 million seconds (≈ 3 × 10^7).

Shortcut: 1 million requests per day ≈ 12 requests per second (10^6 / 86,400 ≈ 11.6). So "X million per day" ≈ "12 × X per second".

Common mistakes

- Mixing bits and bytes: a "1 Gbps" network card moves 125 MB per second at best.
- Carrying too many digits. Round to one or two significant figures and say so.

TL;DR

KB, MB, GB, TB, PB each step up by about 1,000 (2^10). A day is about 10^5 seconds, so 1 million per day is about 12 per second.

Back-of-the-envelope estimation

What it is

A **back-of-the-envelope estimate** is a quick, rough calculation (small enough to scribble on an envelope) that tells you the size of a problem: how many requests per second, how much storage, how much bandwidth, how many servers. The goal is the right **order of magnitude** (is it 10, 1,000 or 1,000,000?), not precision.

> **Analogy:** before a road trip you estimate "about 400 km, about 5 hours, need to refuel once". You don't compute it to the minute, but it tells you whether to leave at 6 am or noon.

Why it matters

The estimate decides the architecture. 100 requests per second fits on one server with one database. 100,000 per second needs many servers, caching, and probably sharded data (split across many databases). 10 GB of data fits in memory on one machine; 10 PB needs a distributed storage system. Without numbers you can't tell which design is needed.

How it works

The key terms:

- **DAU (Daily Active Users):** distinct users who use the product on a given day. **MAU** is the same per month. DAU is usually 20–50% of MAU for social apps.
- **QPS (Queries Per Second)** or **RPS (Requests Per Second):** how many requests reach the system each second.
- **Average QPS** = total requests per day / 86,400.

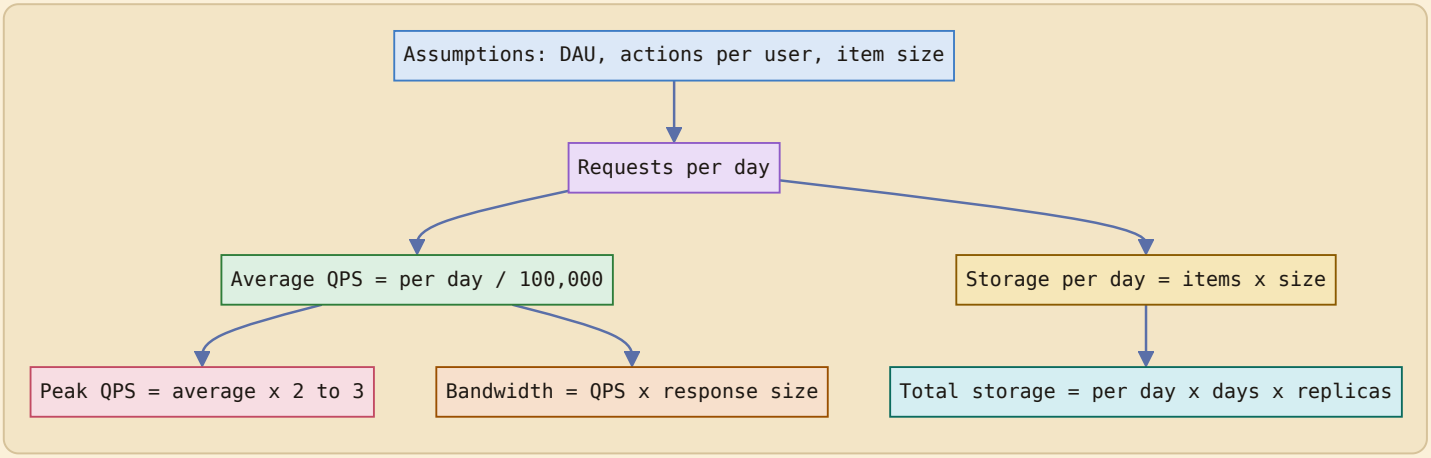
- **Peak factor:** traffic is not flat. Evenings, weekends, or events (a sale, a football final) create peaks. Peak QPS is commonly estimated as **2–3 times** the average, sometimes 5–10× for spiky products.
- **Read/write ratio:** how many reads per write. Social feeds might be 100:1; logging systems are almost all writes.
- **Storage per day** = new items per day × size of each item.
- **Bandwidth** = QPS × size of each response (or request).

Step by step, worked example — a photo-sharing app:

Assumptions (state them out loud):

- 10 million DAU.
- Each user views 50 photos and uploads 0.2 photos per day (one in five users posts once).
- Average photo after compression: 500 KB. Metadata per photo: 1 KB.
- Keep data for 5 years.

1. **Read QPS:** 10M × 50 = 500M views per day. 500M / 10^5 ≈ **5,000 QPS** average. Peak (×3) ≈ **15,000 QPS**.
2. **Write QPS:** 10M × 0.2 = 2M uploads per day ≈ **20 uploads per second** average, ~60 at peak. Read:write ≈ 250:1, so this is a read-heavy system: caching and a CDN will matter a lot.
3. **Storage per day:** 2M × 500 KB = 1,000,000 MB = **1 TB per day** of photos, plus 2M × 1 KB = 2 GB per day of metadata.
4. **Storage for 5 years:** 1 TB × 365 × 5 ≈ **1.8 PB** of photos (more with replicas: typically × 3). Metadata ≈ 3.6 TB. So photos go in object storage; metadata fits in a (sharded) database.
5. **Outgoing bandwidth:** 5,000 views/s × 500 KB = 2,500 MB/s = **2.5 GB/s ≈ 20 Gbps** average. That's a strong argument for a CDN.
6. **Servers:** if one app server handles ~1,000 simple requests per second, peak 15,000 QPS needs ~15 servers, so plan ~20–30 for headroom and failures.



Rules of thumb for capacity (very rough, for simple workloads on modern hardware):

Component	Rough capacity per machine
Web/app server (simple API)	1,000-10,000 requests/s
Relational database (simple indexed queries)	1,000-10,000 queries/s, more for cached reads
Redis/Memcached cache	100,000+ operations/s
Kafka broker	hundreds of MB/s

Trade-offs

More precise estimates take longer and rarely change the design. Spend 3–5 minutes in an interview, write numbers down, and move on.

Common mistakes

- Forgetting the peak. Systems fall over at peak, not at average.
- Forgetting replication: storing 3 copies triples storage.

- Forgetting metadata, indexes and logs, which can add 20–50%.
- False precision ("11,574.07 QPS"). Say "about 12,000".
- Doing maths silently. Say each assumption so the interviewer can correct it.

Real-world examples

- WhatsApp said in late 2020 that it delivers about 100 billion messages per day. 10^11 / 10^5 ≈ 1 million messages per second on average, a useful anchor for "very large" chat scale.
- YouTube said in 2019 that more than 500 hours of video are uploaded every minute. That is about 720,000 hours a day, which makes its storage needs enormous.

In the interview

Q: A service has 100 million DAU, each making 20 requests per day. What QPS should you plan for?

A: 100M × 20 = 2 billion requests per day. 2 × 10^9 / 10^5 = 20,000 QPS average. With a peak factor of 3, plan for about 60,000 QPS.

Q: Why bother estimating if numbers are guesses?

A: The order of magnitude decides the architecture: whether one database is enough, whether data fits in memory, whether you need a CDN. Even rough numbers rule designs in or out.

TL;DR

Start from DAU and per-user actions, convert per day to per second by dividing by about 10^5, multiply by 2–3 for peak, and compute storage and bandwidth from item sizes. Round aggressively and state every assumption.

Availability and the nines

What it is

Availability is the fraction of time a system is up and doing its job (answering requests correctly). It is usually written as a percentage with many 9s, so people talk about "three nines" (99.9%) or "four nines" (99.99%).

Availability = uptime / (uptime + downtime), or, measured by requests, successful requests / total valid requests.

> **Analogy:** a shop that is supposed to be open 24 hours. If it is closed by surprise for 9 hours across a whole year, it was available about 99.9% of the time.

The nines table

(Year = 365 days, month = 30 days.)

Availability	Name	Downtime per year	Downtime per month	Downtime per day
99%	two nines	3.65 days	7.2 hours	14.4 minutes
99.9%	three nines	8.76 hours	43.2 minutes	1.44 minutes
99.95%	three and a half	4.38 hours	21.6 minutes	43 seconds
99.99%	four nines	52.6 minutes	4.32 minutes	8.6 seconds
99.999%	five nines	5.26 minutes	26 seconds	0.86 seconds

Each extra nine cuts allowed downtime by 10×, and usually costs much more than 10× the engineering effort.

Why it matters

- Downtime costs money and trust. An online shop down for an hour on Black Friday loses that hour's sales permanently.
- The target decides the design. 99% can survive manual recovery ("someone restarts the server in the morning"). 99.99% leaves 4 minutes a month, which is less than the time it takes a human to wake up and log in, so recovery must be **automatic**.

How it works

Availability of parts in series (all must work). If a request passes through a load balancer (99.99%), an app server (99.9%) and a database (99.9%), all three must be up:

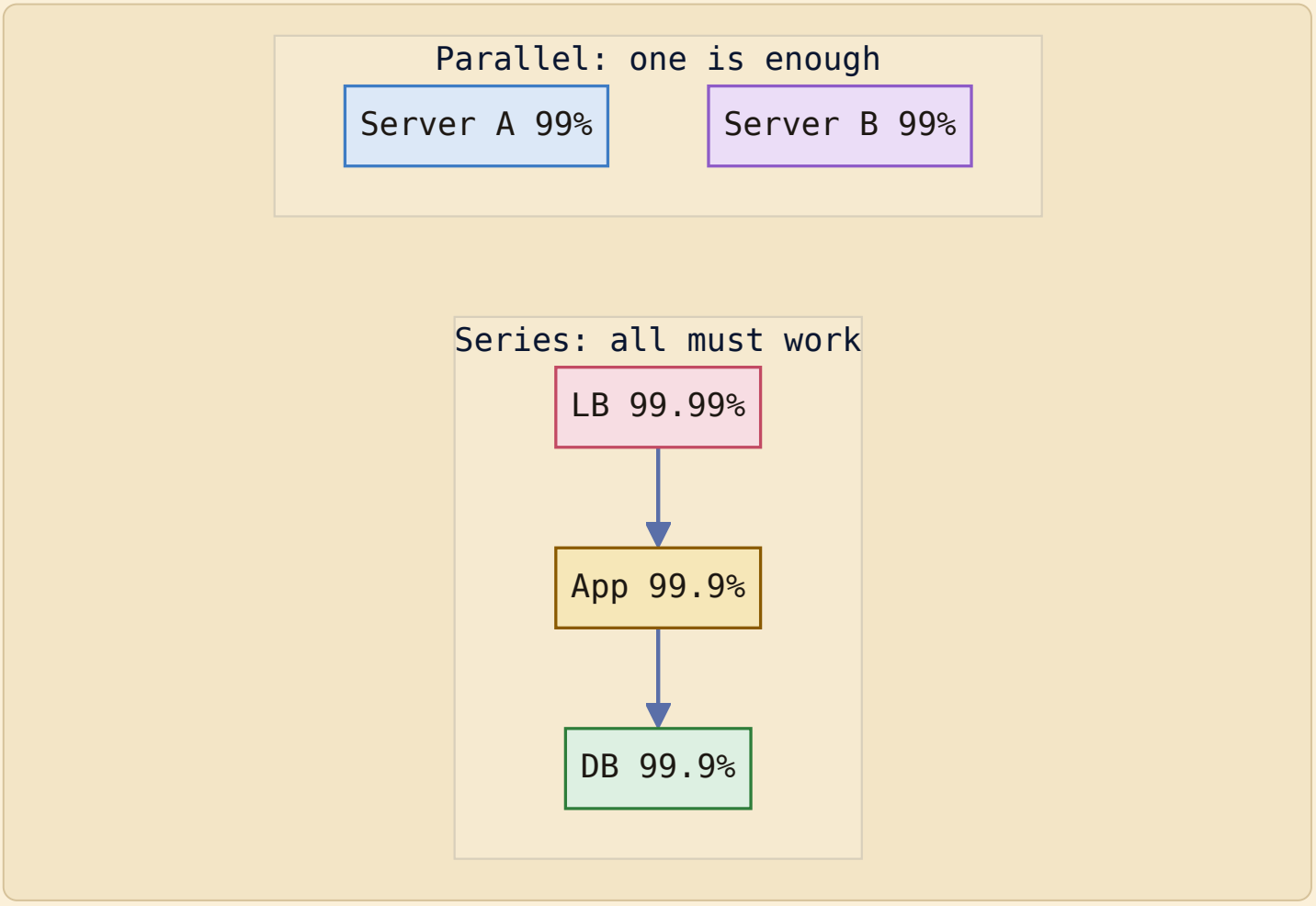
Availability = 0.9999 × 0.999 × 0.999 ≈ 0.9979 → **99.79%**.

The chain is weaker than its weakest link. Every dependency you add in series lowers availability.

Availability of parts in parallel (any one is enough). Two independent servers, each 99%: the system is down only when both are down at the same time.

Availability = 1 − (0.01 × 0.01) = 1 − 0.0001 = **99.99%**.

Redundancy turns two "two nines" machines into a "four nines" pair, assuming failures are independent (they often aren't fully, e.g. both in the same rack lose power together).



Trade-offs

Higher availability needs redundancy (more machines, more data centres), automation, testing and on-call staff. Going from 99.9% to 99.99% might mean running in multiple data centres with automatic failover, which roughly doubles infrastructure and adds complexity. Many internal tools are fine at 99.5%.

Common mistakes

- Promising five nines without the budget. Even the big cloud providers' SLAs for single services are usually 99.9%–99.99%.

- Forgetting dependencies. Your service can't be more available than the things it needs in series (unless you design fallbacks).
- Counting planned maintenance as "not downtime". Users don't care why it's down.

Real-world examples

- Amazon S3 Standard is designed for 99.99% availability, and its SLA promises 99.9% per month (with service credits below that).
- Telephone networks were historically engineered for five nines.

In the interview

Q: How much downtime per month does 99.9% allow?

A: About 43 minutes. 99.99% allows about 4.3 minutes.

Q: How do you get higher availability than any single machine provides?

A: Run redundant copies in parallel with automatic failover, and remove single points of failure, ideally spreading copies across separate failure domains (racks, zones, regions).

TL;DR

Availability is the share of time a system works. Each extra nine is ten times less downtime: 99.9% is about 43 minutes a month, 99.99% about 4. Series dependencies multiply down; parallel redundancy multiplies up.

SLA, SLO, SLI and error budgets

What it is

Three related terms, from most technical to most legal:

- **SLI (Service Level Indicator):** a *measurement* of how the service is doing. Example: "percentage of HTTP requests that returned successfully in under 300 ms over the last 30 days."
- **SLO (Service Level Objective):** an internal *target* for an SLI. Example: "99.9% of requests succeed in under 300 ms, measured over 30 days."
- **SLA (Service Level Agreement):** a *contract* with customers saying what happens if a target is missed, usually refunds or service credits. Example: "If monthly availability falls below 99.9%, customers get 10% of their bill back."

> **Analogy:** a pizza shop. The SLI is "how long deliveries actually take". The SLO is the manager's target: "95% delivered within 30 minutes". The SLA is the printed promise: "over 45 minutes and it's free".

The SLA is normally looser than the SLO, so the team gets warning (missing the SLO) before the company has to pay (missing the SLA).

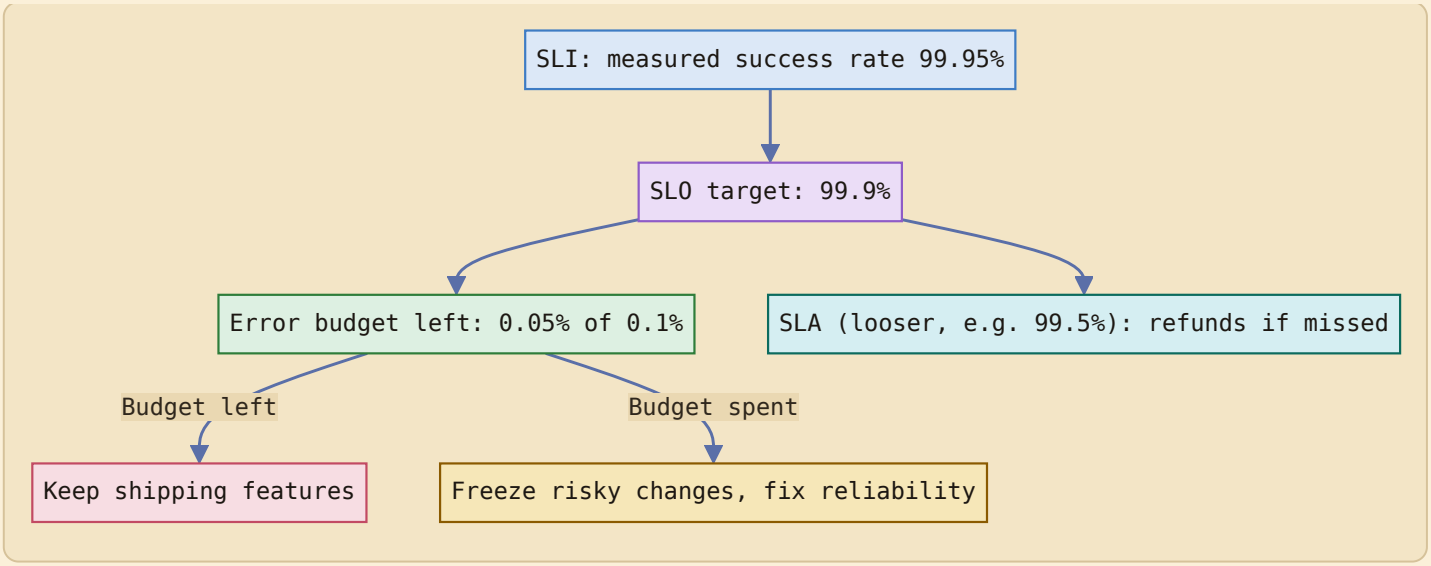
Error budgets

An **error budget** is the amount of unreliability the SLO allows. If the SLO is 99.9% success, the error budget is 0.1%: in a month with 100 million requests, up to 100,000 may fail (or about 43 minutes of full outage).

The budget turns reliability into a shared decision:

- While budget remains, the team can ship features fast and take risks.
- When the budget is spent, the team slows releases and works on reliability until it recovers.

This idea comes from Google's **Site Reliability Engineering (SRE)** practice. SRE is the discipline of running production systems using software engineering methods.



How it works

1. Choose SLIs that reflect what users feel: availability (success rate), latency (p99 under X ms), freshness (data no older than Y), durability.
2. Set an SLO slightly below what users actually need, not "100%". 100% is impossible and leaves no room to change anything.
3. Measure over a window (often 28 or 30 days, rolling).
4. Alert when the budget is being burned too fast (Part 5 covers burn-rate alerts).

Common mistakes

- Setting the SLO to 100%. Any deployment risk then "violates" it.
- Choosing SLIs users don't feel (CPU usage) instead of ones they do (failed checkouts).
- Making the SLA as tight as the SLO, leaving no warning margin.

In the interview

Q: What's the difference between an SLO and an SLA?

A: An SLO is an internal target for a measured indicator; an SLA is an external contract with penalties. SLAs are usually set looser than SLOs.

Q: What is an error budget used for?

A: It quantifies how much failure is acceptable (1 − SLO). Teams spend it on releases and experiments; when it's used up, they prioritise reliability work.

TL;DR

SLI is the measurement, SLO the target, SLA the contract with penalties. The error budget (1 − SLO) is how much failure you can "spend" before you must slow down and fix reliability.

MTBF, MTTR and measuring failure

What it is

Two averages that describe how often things break and how fast they're fixed:

- **MTBF (Mean Time Between Failures):** average time a system runs before it fails. Higher is better.
- **MTTR (Mean Time To Repair / Recover):** average time from a failure until service is restored. Lower is better.

Related: **MTTD (Mean Time To Detect)**, how long before anyone notices, and **MTTF (Mean Time To Failure)**, used for parts that are replaced rather than repaired (like disks).

> **Analogy:** a car. MTBF is how many months it typically goes between breakdowns. MTTR is how long it sits at the garage each time.

How it works

Availability can be estimated as:

```
Availability = MTBF / (MTBF + MTTR)
```

Example: a server fails on average every 1,000 hours (MTBF) and takes 1 hour to recover (MTTR). Availability = 1,000 / 1,001 ≈ 99.9%.

There are two levers:

1. **Increase MTBF:** better hardware, testing, careful deployments.
2. **Decrease MTTR:** fast detection (monitoring and alerts), automatic failover, quick rollback, runbooks.

In large distributed systems, failures are constant, so cutting MTTR is usually the cheaper lever. If recovery is automatic and takes 10 seconds instead of 1 hour, the same failure rate gives about 99.9997% availability.

Common mistakes

- Only trying to prevent failures. You can't prevent them all; plan to recover fast.
- Ignoring detection time. An outage nobody notices for 30 minutes has at least 30 minutes of MTTR.

In the interview

Q: Which is easier to improve at scale, MTBF or MTTR?

A: Usually MTTR. With thousands of machines something always fails, so automatic detection and failover (seconds) give more availability than trying to make each machine fail less often.

TL;DR

MTBF is how long between failures, MTTR how long to recover. Availability ≈ MTBF / (MTBF + MTTR); in practice, shrink MTTR with monitoring and automation.

Reliability, availability and durability

What it is

Three words often mixed up:

- **Reliability:** the system keeps doing the *right thing* over time, even when things go wrong. It returns correct results and doesn't corrupt data. A reliable system is one you can trust.
- **Availability:** the system is *up and responding* when you ask. A system can be available but unreliable (it answers quickly, with wrong data).
- **Durability:** once data is saved, it *stays saved* and isn't lost, even if machines crash or disks die. A system can be unavailable but durable (down for an hour, but no data lost when it returns).

> **Analogy:** a bank. Availability: the cash machine works when you walk up. Reliability: it gives you the right amount and debits the right account. Durability: your balance is still there tomorrow even if the bank's building floods.

Property	Question it answers	Typical measure
Availability	Can I use it right now?	% uptime, % successful requests
Reliability	Does it work correctly over time?	Failure rate, MTBF, correctness
Durability	Will my saved data still exist later?	Probability of losing an object per year

How it works

- Availability is improved by redundancy and failover (more on this below).
- Durability is improved by keeping multiple copies of data on separate disks, machines and locations, writing to disk before confirming (Part 3 covers write-ahead logs), and backups.
- Reliability is the umbrella: testing, validation, handling faults gracefully, and both of the above.

Real-world examples

- Amazon S3 Standard is designed for 99.999999999% ("eleven nines") yearly durability per object: keep ten million objects there and, on average, a single one goes missing roughly once in ten thousand years. It is designed for only 99.99% availability (and its SLA pays credits below 99.9%). Durability and availability are separate promises.

Common mistakes

- Saying "highly available" when you mean "doesn't lose data". Say durable.
- Assuming replication equals backup. If a bug deletes data, replication faithfully deletes every copy. Backups protect against that.

In the interview

Q: Can a system be durable but not available?

A: Yes. If a database cluster is down for maintenance, no one can read the data (unavailable), but all of it is safely on disk (durable).

TL;DR

Availability = can I use it now; reliability = does it work correctly over time; durability = will saved data survive. They are improved by different techniques and promised separately.

Faults, failures and fault tolerance

What it is

- A **fault** is something going wrong with one part: a disk dies, a process crashes, a network cable is cut, a bug returns a wrong value.
- A **failure** is when the *whole system* stops providing its service to users.

Faults are inevitable; failures are what we try to prevent. **Fault tolerance** (or resilience) is the ability of a system to keep working, perhaps in a reduced way, when some of its parts have faults.

> **Analogy:** a plane with two engines. One engine stopping is a fault. The plane crashing would be a failure. Being able to fly and land on one engine is fault tolerance.

Types

Common kinds of faults:

Fault type	Example	Typical defence
Hardware	Disk dies, power supply fails	Redundant machines, replicated data
Software	Bug, memory leak, crash on bad input	Testing, gradual rollouts, restarts
Network	Packets lost, link cut, partition	Timeouts, retries, replicas in other zones
Human	Wrong config pushed, table deleted	Reviews, automation, backups, rollback
Dependency	Third-party API is slow or down	Timeouts, fallbacks, circuit breakers

Faults also differ by behaviour:

- **Crash (fail-stop):** the part stops completely. Easiest to detect.

- **Omission:** some messages are lost.
- **Timing / slow:** the part works but too slowly. Hard to tell from a crash; often worse, because callers keep waiting.
- **Byzantine:** the part sends wrong or even malicious answers. Most business systems assume this doesn't happen inside their own network; blockchains are designed for it.

How it works

Fault tolerance generally follows a pattern: **detect** the fault (health checks, timeouts, heartbeats), **isolate** it so it doesn't spread (timeouts, bulkheads), and **recover** (route around it, restart, fail over to a copy). Parts 4 and 5 cover the specific tools: retries, circuit breakers, replication, consensus.

Common mistakes

- Treating "slow" as fine. A dependency that takes 30 seconds to fail can tie up all your threads; a fast error is often better.
- Assuming faults are independent. A bad deployment or a shared power supply can break many "redundant" copies at once (**correlated failure**).

Real-world examples

- Netflix's Chaos Monkey randomly shuts down production servers so that engineers must build services that tolerate instance faults.
- Large cloud outages are often caused by a configuration change pushed everywhere at once, a reminder that human and software faults matter as much as hardware.

In the interview

Q: What is the difference between a fault and a failure?

A: A fault is a component misbehaving; a failure is the system as a whole not serving users. Fault tolerance stops faults from becoming failures.

TL;DR

Faults happen to parts; failures happen to the system. Design to detect, isolate and recover from faults so they don't become user-visible failures.

Redundancy, single points of failure and failover

What it is

- **Redundancy** means having more than one of something, so a spare can take over if one breaks: extra servers, extra copies of data, extra network links, extra power supplies.
- A **single point of failure (SPOF)** is any part whose failure alone brings down the whole system. One database, one load balancer, one DNS provider, one data centre, even one engineer who knows the password.
- **Failover** is the process of switching work from a failed component to a standby one.

> **Analogy:** a hospital has backup generators. Mains power is a SPOF without them; with them, a blackout triggers failover to the generators within seconds.

How it works

Finding SPOFs: walk through the diagram and ask of every box and arrow, "if this disappears, what happens?". If the answer is "everything stops", it's a SPOF.

Removing them:

1. Run at least two of every critical component, in separate **failure domains**: different racks, different **availability zones** (separate data centres in the same region, with their own power and networking), or different regions.

2. Put something in front that can route around dead copies (a load balancer, DNS, a client library).
3. Add **health checks**: periodic tests ("are you alive?") so dead copies are detected.
4. Automate failover so it happens in seconds, not when a human notices.

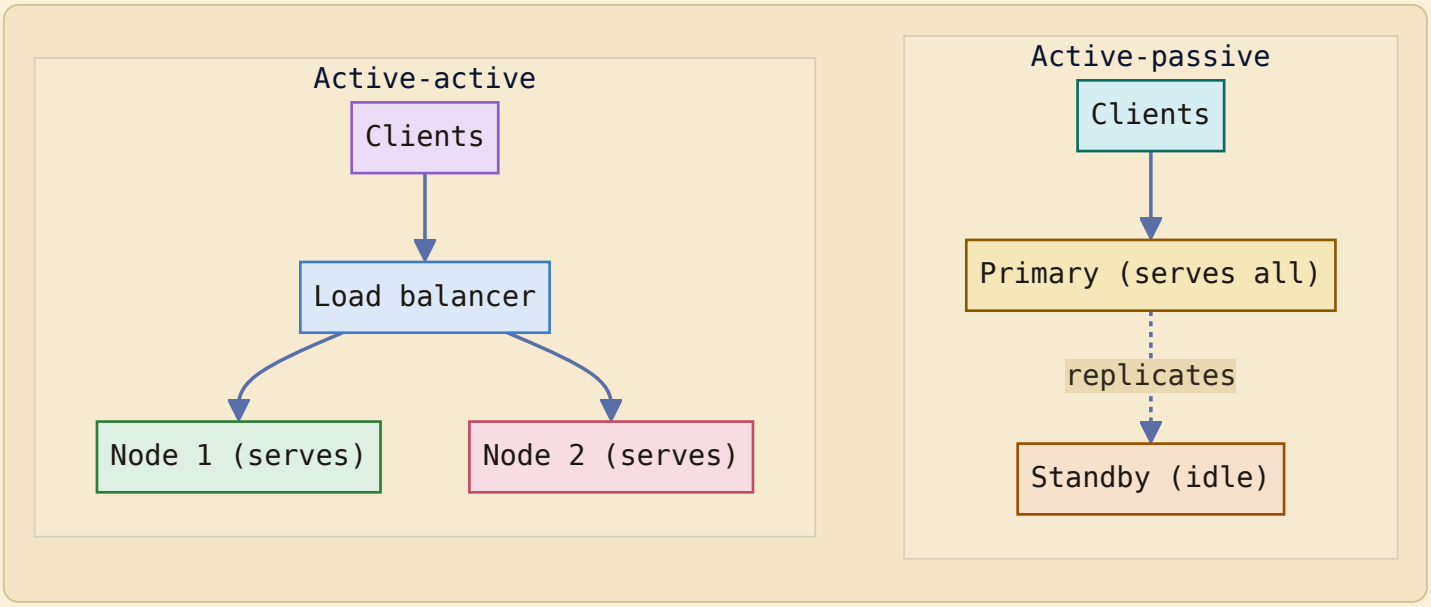
Types

Active-passive (primary-standby): one copy handles all traffic; a standby waits, kept up to date, and takes over when the primary fails. Simple, but the standby is idle capacity, and failover takes some time (seconds to minutes).

Active-active: all copies handle traffic at the same time. If one fails, the others absorb its share. Better use of hardware and faster recovery, but harder for stateful parts like databases (two copies accepting writes can conflict).

Cold, warm and hot standby:

- **Cold:** the spare exists but must be started and loaded with data (minutes to hours).
- **Warm:** running and partially up to date (minutes).
- **Hot:** running and fully in sync, ready immediately (seconds).



N+1 and N+2: if you need N servers to handle peak load, run N+1 (survive one failure) or N+2 (survive one failure during maintenance of another).

Trade-offs

- Redundancy costs money: standby machines, extra copies of data.
- Failover is risky. If the system wrongly decides the primary is dead while it's actually alive (perhaps just slow or cut off), you can end up with two primaries both accepting writes, called **split brain**. Part 3 and Part 4 cover how databases prevent this (quorums, fencing).
- Failover itself must be tested regularly; untested failover often fails when needed.

Common mistakes

- Making app servers redundant but leaving one database or one load balancer.
- Putting all "redundant" copies in the same rack or zone.
- Forgetting that the standby needs enough capacity to take the full load.

Real-world examples

- AWS recommends spreading instances across at least two availability zones; managed databases such as Amazon RDS Multi-AZ keep a synchronous standby in another zone and fail over automatically, typically within a minute or two.
- Load balancers themselves are deployed in pairs (or as a managed, internally redundant service) to avoid being the SPOF.

In the interview

Q: How do you find single points of failure in a design?

A: Go component by component (including DNS, load balancers, databases, config stores, third-party APIs) and ask what happens if it disappears. Anything that stops the whole service needs a redundant copy in a separate failure domain and automatic failover.

Q: Active-passive or active-active?

A: Active-passive is simpler and common for databases with one writer. Active-active uses capacity better and recovers faster, and is natural for stateless services, but needs conflict handling for stateful data.

TL;DR

Redundancy means spares; a SPOF is any part with no spare; failover moves work to the spare. Remove SPOFs by duplicating critical parts across failure domains and automating failover, while guarding against split brain.

Scalability

What it is

Scalability is a system's ability to handle more load (more users, more requests, more data) by adding resources, without a redesign and without performance collapsing. A scalable system's cost and effort grow roughly in proportion to its load; an unscalable one hits a wall.

Load needs to be described with specific numbers, called **load parameters**: requests per second, number of concurrent users, read/write ratio, size of the data, number of followers per user, and so on. "Scalable" only makes sense relative to one of them: "we can scale to 10× the requests per second by adding servers".

> **Analogy:** a restaurant that can serve more diners by adding tables and hiring more cooks is scalable. One whose single tiny kitchen can never cook faster than 50 meals an hour, no matter how many tables you add, is not.

Why it matters

Successful products grow, sometimes overnight. A system that needs a rewrite at each 10× step slows the business down at exactly the wrong moment.

How it works

There are a few dimensions you can scale along:

- **Load scalability:** more requests per second.
- **Data scalability:** more stored data.
- **Geographic scalability:** users spread around the world, needing low latency everywhere.
- **Organisational scalability:** more engineers working on the system without stepping on each other (one reason teams split into microservices, Part 5).

The core techniques, all explained in later parts:

1. Scale out stateless app servers behind a load balancer (next section).
2. Cache hot data (Part 4).
3. Copy data to read replicas for read-heavy loads (Part 3).
4. Split data across shards for write-heavy or huge data (Part 3).
5. Move slow work to queues and workers (Part 4).
6. Put static content on a CDN (Part 2).

Common mistakes

- Calling a system "scalable" without saying which load parameter and by how much.
- Scaling the wrong part. Adding app servers does nothing if the database is the bottleneck.
- Building for 1,000× growth on day one. Design so the next 10× is easy, and re-evaluate.

In the interview

Q: What does it mean for a system to be scalable?

A: That it can handle growing load, described with concrete numbers, by adding resources roughly in proportion, while keeping performance within targets.

TL;DR

Scalability is handling more load by adding resources without a redesign. Always say what load (QPS, data size, users) and by how much.

Vertical and horizontal scaling

What it is

- **Vertical scaling (scale up):** make one machine bigger: more CPU cores, more RAM, faster disks.
- **Horizontal scaling (scale out):** add more machines and spread the work across them.

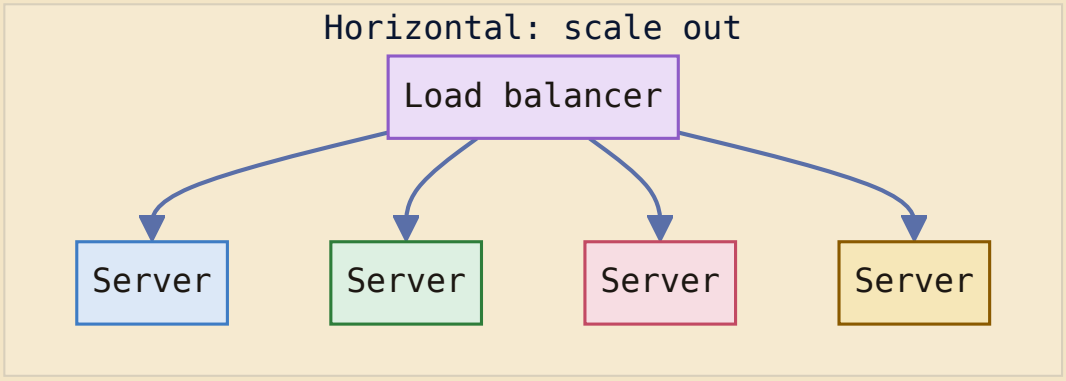
> **Analogy:** to carry more passengers you can buy a bigger bus (vertical) or run more buses (horizontal).

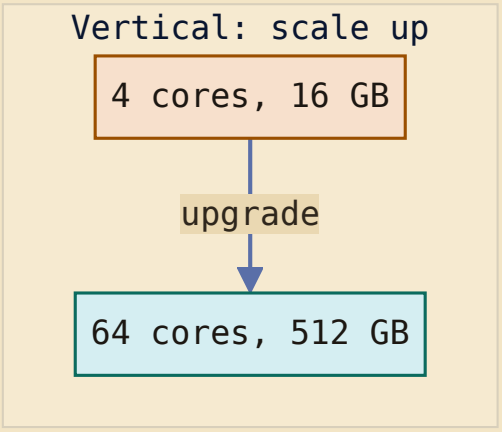
How it works

Vertical scaling is usually just a configuration change on a cloud provider: stop the machine, pick a larger size, start it. No code changes. Modern cloud machines go very large: AWS offers instances with hundreds of CPU cores and multiple terabytes of RAM.

Horizontal scaling requires the work to be divisible:

- Stateless app servers: easy. Put a load balancer in front and add copies.
- Databases: harder. Reads can go to replicas; spreading writes needs sharding (each machine holds part of the data).





Trade-offs

	Vertical (up)	Horizontal (out)
Simplicity	Very simple, no code change	Needs load balancing, stateless design or sharding
Limit	Hard ceiling: the biggest machine	Practically unlimited
Fault tolerance	One machine is still a SPOF	Losing one node loses only a fraction
Cost curve	Big machines cost disproportionately more	Many cheap commodity machines
Downtime to resize	Often needs a restart	Add or remove nodes live
Consistency	Easy (one copy of data)	Harder (data spread across machines)

Common mistakes

- Dismissing vertical scaling. A single large database server handles a lot; scaling up first is often the right, cheap answer.
- Scaling out a stateful service without dealing with its state (see next sections).

Real-world examples

- Stack Overflow famously served a very large site from a small number of powerful servers (in 2016: 11 web servers and 4 SQL Server machines, plus a few Redis, search and tag-engine boxes), a well-known example of vertical scaling done well.
- Google, Meta and Amazon are each estimated to run millions of commodity servers: horizontal scaling at the extreme.

In the interview

Q: When would you choose vertical scaling?

A: Early on, or for components that are hard to distribute (like a relational database's primary). It's simple and buys time. Switch to horizontal when you approach the largest machine size, need fault tolerance, or costs climb steeply.

Q: What makes horizontal scaling hard?

A: State. Stateless servers scale out trivially; anything that stores data must be replicated or partitioned, which brings consistency and coordination problems.

TL;DR

Scale up = bigger machine (simple, limited, still a SPOF). Scale out = more machines (unlimited and fault tolerant, but needs stateless design or data partitioning). Most real systems do both.

Elasticity

What it is

Elasticity is the ability to add and remove resources *automatically and quickly* as load changes, so you pay for roughly what you use. Scalability says "we *can* grow"; elasticity says "we grow and shrink by ourselves, in minutes".

> **Analogy:** a supermarket that opens more checkouts when queues get long and closes them when the shop is quiet.

How it works

1. Watch a metric: CPU use, requests per second, queue length.
2. When it crosses a threshold (say CPU above 70% for 5 minutes), launch more servers.
3. When it drops (CPU below 30%), remove some.
4. The load balancer automatically starts and stops sending them traffic.

This is called **autoscaling** (Part 5 covers kinds: reactive, scheduled, predictive). Cloud platforms (AWS Auto Scaling groups, Kubernetes Horizontal Pod Autoscaler) and **serverless** platforms (AWS Lambda, which runs your code per request and scales to zero) provide it.

Trade-offs

- New servers take time to start (seconds for containers, minutes for virtual machines). A sudden spike can arrive faster than capacity.
- Only stateless parts are easily elastic; databases scale much more slowly.
- Scaling down too eagerly causes "flapping" (adding and removing repeatedly).

In the interview

Q: What's the difference between scalability and elasticity?

A: Scalability is the capability to handle more load by adding resources. Elasticity is doing that automatically, in both directions, to match demand as it changes.

TL;DR

Elasticity is automatic, fast scaling up and down with demand, mostly for stateless parts, so you pay only for what you need.

Stateless and stateful services

What it is

State is data a program remembers between requests: who is logged in, what's in a shopping cart, the contents of a database.

- A **stateless service** keeps no client-specific data in its own memory between requests. Every request carries (or points to) everything needed to handle it, and any copy of the service can handle any request.
- A **stateful service** remembers data between requests locally, so a particular client or a particular piece of data is tied to a particular server.

> **Analogy:** a stateless service is a call centre where every agent can pull up your file from the shared system, so any agent can help you. A stateful service is your family doctor who keeps your history in their head: great until they're on holiday.

Why it matters

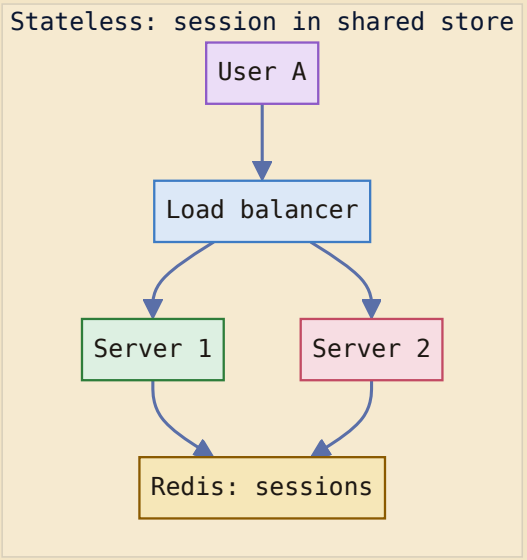
Stateless services are easy to scale out, load balance, restart and replace: kill any copy and nothing is lost. That's why the standard advice is "**make app servers stateless and push state into dedicated stores**" (databases, caches, object storage) that are designed for replication.

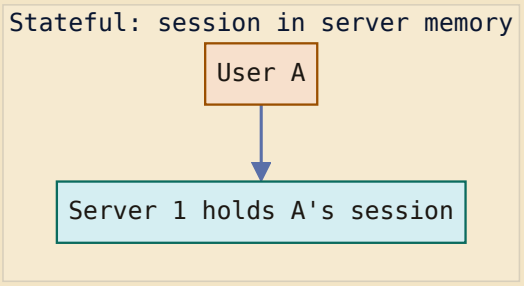
How it works

How to make a web app stateless:

1. Store login sessions in a shared store (Redis) or in a signed token the client sends with each request (like a JWT, JSON Web Token, covered in Part 5), not in server memory.
2. Store uploaded files in object storage, not on the server's local disk.
3. Keep shopping carts and similar data in a database or cache.

Then the load balancer can send each request to any server.





If a service stays stateful (for example a game server holding a live match, or a chat server holding open connections), the load balancer must send each client to the same server, called **sticky sessions** (Part 2), and you must plan for what happens when that server dies.

Trade-offs

- Stateless servers need an extra network hop to fetch state, adding latency (usually under a millisecond to a cache).
- Stateful designs can be faster (data already in memory) and are sometimes necessary (databases, real-time games, WebSocket servers), but are harder to scale and recover.

Common mistakes

- Keeping sessions in local memory, then adding a second server and finding users randomly logged out.
- Writing uploaded files to the local disk of one server.
- Calling a system stateless when it just moved state somewhere else; the state store now needs scaling and redundancy too.

In the interview

Q: Why should application servers be stateless?

A: So any instance can handle any request: you can add or remove instances freely, load balance evenly, and lose an instance without losing user data. State lives in stores built for durability and replication.

TL;DR

Stateless services remember nothing between requests and scale easily; stateful ones hold data and are harder to scale and recover. Keep app servers stateless and put state in databases and caches.

Performance vs scalability

What it is

- **Performance** is how fast the system is for *one* request or user at the current load: low latency, efficient use of resources.
- **Scalability** is how well performance holds up as load *grows*.

A quick diagnostic:

- Send one request to an otherwise idle system. If even that is slow, the trouble is **performance**.
- If that lone request is quick but things drag once traffic piles up, the trouble is **scalability**.

> **Analogy:** a sports car is high-performance (fast for one person) but doesn't scale (two seats). A train is slower to get going but carries a thousand people.

Trade-offs

Some changes help one and hurt the other. Adding a network hop to reach a shared cache slightly hurts single-request performance but lets you scale out. Keeping everything in one process's memory is fast but caps you at one machine.

In the interview

Q: The site is slow even at 3 am with almost no traffic. Performance or scalability?

A: Performance: something is slow per request (an unindexed query, a slow external call, a big page). Adding servers won't help; profile the request.

TL;DR

Performance = fast for one; scalability = stays fast for many. Diagnose which one you have before fixing.

Concurrency vs parallelism

What it is

- **Concurrency** means several tasks are in progress during the same period, with the program switching between them whenever one has to wait. It describes how the work is organised.
- **Parallelism** means several tasks are literally running at the same instant, on multiple CPU cores or machines. It describes how the work is executed.

Go co-creator Rob Pike summed this up as concurrency being about *dealing with* lots of things at once and parallelism about *doing* lots of things at once.

> **Analogy:** one barista taking an order, starting the milk, taking another order while the milk steams, then finishing the first drink, is concurrent. Three baristas each making a drink at the same moment is parallel. You can have concurrency on a single CPU core (by switching) but parallelism needs multiple cores.

Why it matters

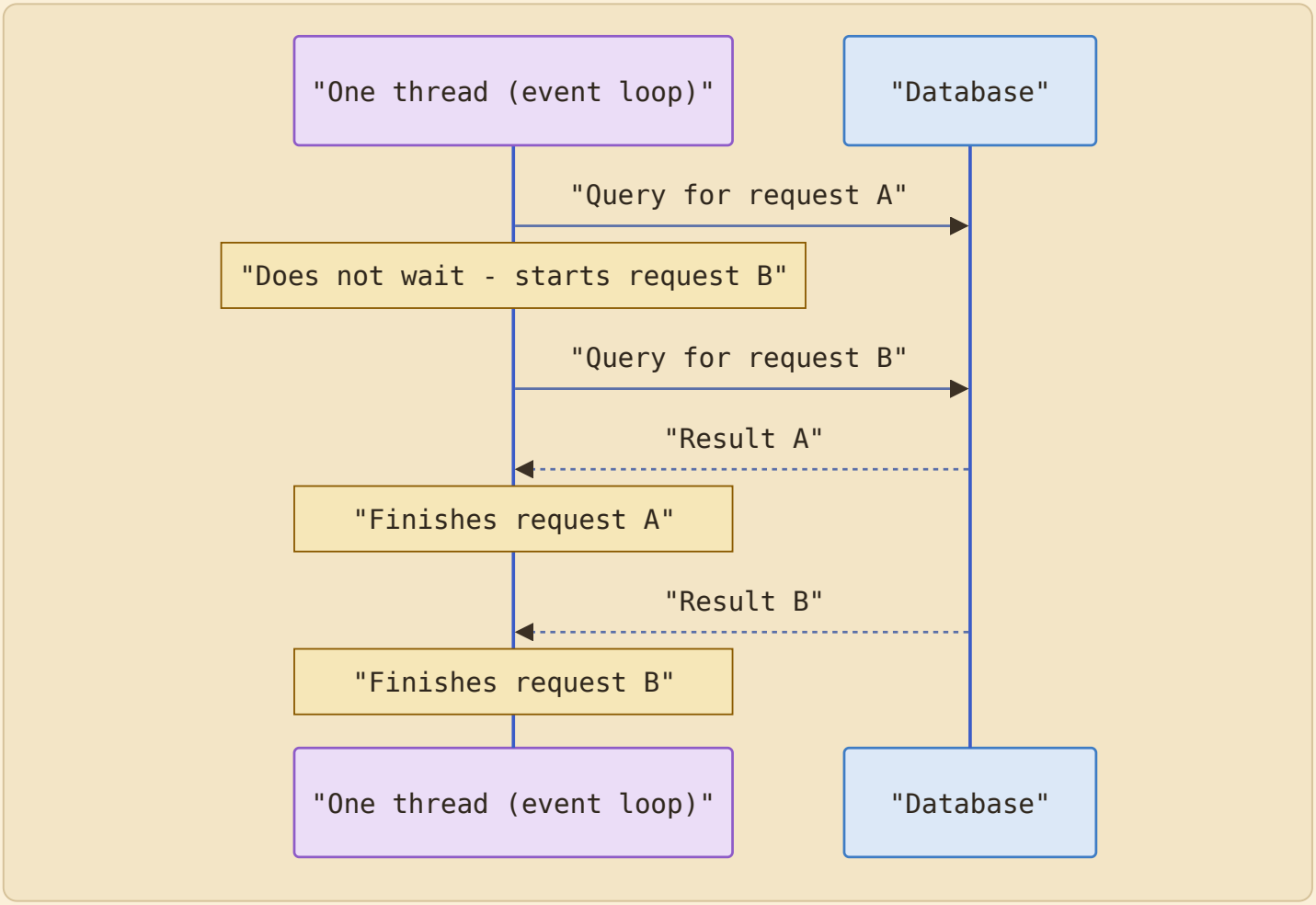
Most web servers spend their time **waiting** (on the network, the database, the disk), not computing. Concurrency lets one server keep thousands of requests in progress while each waits. Parallelism speeds up **CPU-heavy** work (image processing, encryption, analytics) by splitting it across cores.

How it works

Ways to achieve concurrency:

- **Processes:** separate running programs, each with its own memory. Heavy but isolated.

- **Threads:** multiple flows of execution inside one process, sharing memory. Lighter, but shared memory needs locks to avoid errors.
- **Event loop / async I/O:** one thread handles many requests by never blocking: it starts an I/O operation, handles something else, and comes back when the result is ready. Node.js, NGINX and Python's asyncio work this way.
- **Lightweight threads / coroutines:** cheap tasks scheduled by the language runtime (Go's goroutines, Java's virtual threads), letting you write simple blocking-style code that runs with event-loop efficiency.



Concurrency hazards: when tasks share data you can get a **race condition** (the result depends on timing, like two requests both reading a balance of 100 and both writing 90 after a withdrawal of 10, losing one withdrawal), **deadlock** (two tasks each wait for a lock the other holds, forever) and contention (tasks queue for the same lock). Locks, atomic operations and database transactions (Part 3) address them.

Amdahl's law (a related rule) says the speed-up from parallelism is limited by the part of the work that must run one step at a time. If 10% of a job is serial, even infinite cores give at most a 10× speed-up.

Common mistakes

- Adding threads to a CPU-bound task beyond the number of cores. It just adds switching overhead.
- Assuming parallel code is automatically faster; coordination and locking can make it slower.

In the interview

Q: Is a single-threaded Node.js server concurrent?

A: Yes, it handles many requests concurrently using an event loop and non-blocking I/O. It isn't parallel for JavaScript code on one core; you run several processes to use multiple cores.

TL;DR

Concurrency is juggling many tasks (great for waiting on I/O); parallelism is doing several at the same instant on multiple cores (great for CPU-heavy work). Shared data under concurrency needs care to avoid

races and deadlocks.

Bottlenecks: CPU, memory, disk and network

What it is

A **bottleneck** is the part of a system that limits its overall capacity, like the narrow neck of a bottle that limits how fast you can pour. Improving anything else won't help until the bottleneck is fixed. Every machine has four basic resources that can become one.

Resource	What it limits	Signs it's the bottleneck	Usual fixes
CPU	Computation (encryption, parsing, compression)	CPU near 100%, latency rises with load	Optimise code, cache results, add cores or servers
Memory (RAM)	How much data fits in fast storage	Swapping to disk, out-of-memory crashes, cache misses	More RAM, smaller data, evict wisely
Disk	Reading and writing stored data	High disk wait time, IOPS or MB/s at limit	SSDs, caching, sequential writes, sharding
Network	Moving data between machines	Link saturated, packet loss, high RTT	Compression, CDNs, fewer calls, closer servers

How it works

Finding the bottleneck:

1. Measure each resource under load (monitoring dashboards, `top`, `iostat`, cloud metrics).
2. Look for the one at or near its limit, or for queues building up in front of something (thread pools full, database connections exhausted, lock waits).
3. Fix it, then measure again. The bottleneck moves: fix the database and the network may be next.

Non-hardware bottlenecks are just as common: a single lock everybody waits for, a database connection pool that's too small, one "hot" row that every request updates, or an external API with a rate limit.

Bound terms: a task is **CPU-bound** when it's limited by computation, and **I/O-bound** when limited by waiting for disk or network. Most web services are I/O-bound.

IOPS and disk performance

IOPS (Input/Output Operations Per Second) is how many separate read or write operations a storage device can perform per second. It matters most for **random** access (many small reads at different locations), such as database lookups.

Disks have two separate limits:

- **IOPS:** number of operations per second (matters for small random reads/writes).
- **Throughput:** megabytes per second (matters for large sequential reads/writes, like streaming a video file or a backup).

Device	Rough random IOPS	Rough sequential throughput
Spinning HDD (7,200 rpm)	about 75-150	100-250 MB/s
SATA SSD	tens of thousands	about 500 MB/s
NVMe SSD	hundreds of thousands to about 1 million	several GB/s
Cloud block disk (e.g. AWS gp3 baseline)	3,000 (can buy more)	125 MB/s (can buy more)

An HDD's IOPS is low because each random read must physically move the arm and wait for the platter to spin (roughly 5–10 ms). That's why databases moved to SSDs, and why systems that must use HDDs favour sequential writes.

Example: a database needs 20,000 random reads per second that miss the cache. One HDD (~150 IOPS) would need over 100 disks; one NVMe SSD handles it comfortably.

Common mistakes

- Guessing the bottleneck instead of measuring.
- Looking only at averages. A resource at 50% average can be at 100% for the busy minute that matters.
- Forgetting cloud disks have provisioned IOPS limits: a database can be throttled by its disk while CPU looks idle.

In the interview

Q: Latency rises under load but CPU is only at 30%. Where would you look?

A: Something else is the bottleneck: disk IOPS, network, a full database connection pool, a lock, or a slow dependency. Look for where requests are queueing.

Q: What's the difference between IOPS and disk throughput?

A: IOPS counts operations per second (key for many small random reads, like database lookups); throughput counts bytes per second (key for large sequential transfers). A disk can hit one limit without hitting the other.

TL;DR

The bottleneck is whichever resource (CPU, memory, disk, network, or a lock or pool) runs out first. Measure to find it; disks are limited separately by IOPS and by MB/s.

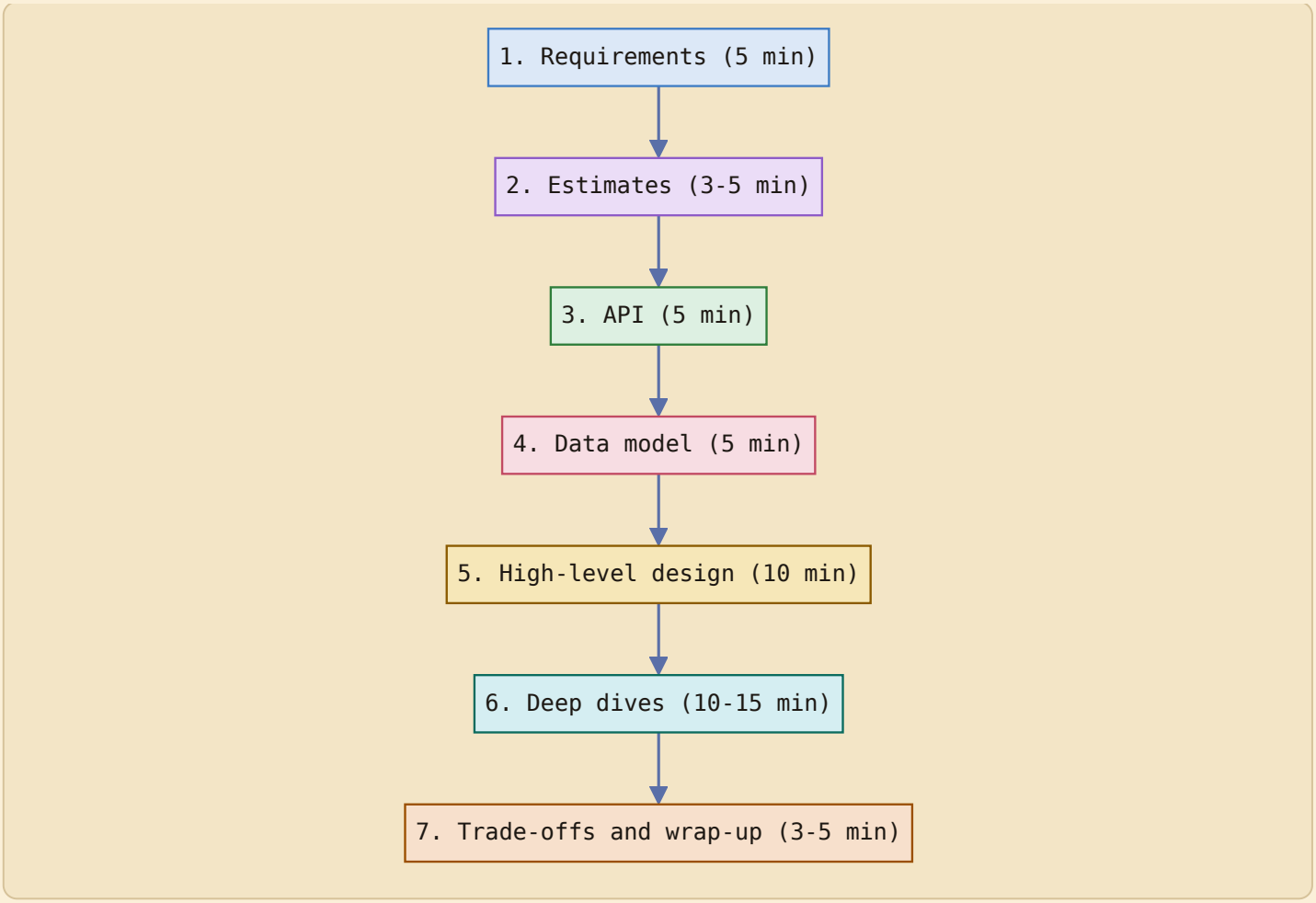
How to approach a system design interview

What it is

A system design interview is usually a 45–60 minute conversation where you're asked to design something like "a URL shortener", "a chat app" or "a news feed". It is open-ended on purpose. A clear, repeatable structure helps you cover everything and shows the interviewer how you think.

How it works

A widely used sequence, with rough timings for a 45-minute interview:



1. Clarify requirements. Never start drawing straight away. Ask questions and write down two lists:

- **Functional requirements:** *what* the system does, the features. Example for a URL shortener: "create a short link from a long URL", "redirect a short link to the original", "optional custom aliases", "links expire after a year".
- **Non-functional requirements:** *how well* it does it, the qualities. Scale (100M new links per month), latency (redirect in under 50 ms at p99), availability (99.99% for redirects), durability (never lose a link), consistency (is a newly created link visible everywhere immediately?), security, cost.
- Also agree on what's **out of scope** (analytics dashboards, user accounts), so you don't waste time.

2. Estimate scale. Use the back-of-the-envelope method: QPS (average and peak), read/write ratio, storage over time, bandwidth. Keep it quick and say how the numbers affect the design ("100:1 reads, so caching matters").

3. Define the API. List the main endpoints the client calls, with inputs and outputs. For example:

```
POST /links        body: {long_url, custom_alias?}  -> {short_code}
GET  /{short_code}                                  -> 301 redirect to long_url
```

This pins down exactly what the system must do.

4. Sketch the data model. What entities are stored, their key fields, and how they're accessed. For example a `links` table: `short_code` (primary key), `long_url`, `created_at`, `expires_at`. Mention which kind of database fits the access pattern (Part 3).

5. Draw the high-level design. The boxes and arrows: clients, load balancer, app servers, cache, database, queue, etc. Walk through one write and one read end to end. Keep it simple at first; it should meet the functional requirements.

6. Deep dives. Pick the hardest or most interesting parts, often guided by the interviewer, and go deeper: how to generate unique short codes without collisions, how to shard the database, how to cache

hot links, what happens when a data centre fails, how to handle a celebrity's viral link. Use your estimates to justify choices.

7. Discuss trade-offs and wrap up. Name the weaknesses of your design and what you'd do next: bottlenecks, single points of failure, monitoring, what changes at 10× scale.

Trade-offs

Breadth vs depth: covering every box shallowly looks junior; going deep on one box and never finishing the design is worse. Aim for a complete simple design first, then depth.

Common mistakes

- Jumping into a detailed design before clarifying requirements.
- Staying silent while thinking. Think out loud.
- Naming technologies without reasons ("use Cassandra"). Always say why ("write-heavy, simple key lookups, needs multi-region").
- Ignoring the non-functional requirements, which usually decide the design.
- Over-engineering: microservices, Kafka and five databases for a system with 100 QPS.
- Not checking the design against the original requirements and numbers at the end.

In the interview

Q: What is the difference between functional and non-functional requirements?

A: Functional requirements are features: what the system does (post a photo, follow a user). Non-functional requirements are qualities: how well it does them (latency, availability, scale, consistency, durability, cost). Non-functional requirements usually drive architecture choices.

Q: The interviewer says "design Twitter". What do you do first?

A: Ask clarifying questions to narrow scope: which features (post, follow, home timeline, search?), how many users, read/write mix, latency and consistency expectations. Write the agreed requirements down before estimating or drawing.

Q: How do you choose what to deep-dive into?

A: The parts that are hardest given the requirements and estimates (usually the biggest scale or consistency challenge), or whatever the interviewer steers toward. Ask if unsure.

TL;DR

Requirements (functional and non-functional), estimates, API, data model, high-level design, deep dives, trade-offs. Think out loud, justify every component with a requirement or a number, and finish with weaknesses and next steps.

Part 1 glossary

Term	Meaning in one line
Active-active	All redundant copies serve traffic at once and absorb a failed copy's share
Active-passive	One copy serves traffic while a standby waits to take over
Amdahl's law	Parallel speed-up is limited by the part of the work that must run serially
App server (application server)	Program that runs business logic and answers client requests
Availability	Fraction of time (or requests) the system is up and working
Availability zone	A separate data centre within a cloud region with its own power and network
Back-of-the-envelope estimate	Quick rough calculation to find the order of magnitude of load or storage
Bandwidth	Maximum data a link or device can carry per second
Batching	Grouping many small operations into one to raise throughput at some latency cost
Bit / byte	A bit is a single 0 or 1; a byte is 8 bits
Bottleneck	The resource or component that limits overall capacity
Cache	Fast, usually in-memory store holding copies of frequently used data
CDN	Content Delivery Network; servers worldwide that serve static files close to users
Client	Program that sends requests (browser, app, or another server)
Cold / warm / hot standby	Spares that need starting, are partly ready, or are ready instantly
Concurrency	Making progress on many tasks by interleaving them
Coordinated omission	Load-test flaw that under-reports slow responses
Correlated failure	Several "independent" parts failing together from a shared cause
CPU-bound	Limited by computation speed
CPU cache	Tiny, very fast memory inside the processor
DAU / MAU	Daily / Monthly Active Users
Deadlock	Tasks waiting forever for locks held by each other
Durability	Saved data is not lost, even after crashes or disk failures
Elasticity	Automatically adding and removing resources to match demand
Error budget	Amount of failure an SLO allows (1 minus the SLO)
Event loop	Single thread handling many requests by never blocking on I/O
Failover	Switching work from a failed component to a standby
Failure	The whole system stops providing its service
Failure domain	A group of parts that can fail together (rack, zone, region)
Fan-out	One request triggering many parallel downstream requests
Fault	One component misbehaving (crash, slowness, wrong output)
Fault tolerance	Continuing to work despite faults in some components
Functional requirement	A feature the system must provide
HDD	Hard disk drive; spinning magnetic disk with slow seeks

Term	Meaning in one line
Health check	Periodic test of whether a component is alive and working
Hedged request	Sending a duplicate request to another replica if the first is slow
Histogram	Counts of measurements per value bucket, used to compute percentiles
Horizontal scaling (scale out)	Adding more machines to share the load
HTTP	Standard protocol for web requests and responses
I/O-bound	Limited by waiting for disk or network
IOPS	Input/output operations per second a storage device can perform
KB / MB / GB / TB / PB	Data sizes, each about 1,000 times the previous
Latency	Time delay for one request or one hop
Little's law	Items in system = arrival rate x time in system (L = lambda x W)
Load balancer	Component that spreads requests across servers and skips dead ones
Load parameters	The numbers that describe load (QPS, data size, users)
Mean (average)	Sum divided by count; hides outliers in skewed data
Median (p50)	Middle value: half are faster, half slower
Message queue	Buffer where jobs wait to be processed later by workers
MTBF	Mean time between failures
MTTD	Mean time to detect a failure
MTTF	Mean time to failure for parts that are replaced, not repaired
MTTR	Mean time to repair or recover
N+1 redundancy	Running one more unit than peak load needs
Nines	Shorthand for availability levels (three nines = 99.9%)
Non-functional requirement	A quality the system must have (latency, availability, scale)
NVMe	Fast interface for SSDs with very high IOPS
Object storage	Service storing large files as objects under keys (e.g. Amazon S3)
Order of magnitude	Rough size as a power of ten
Parallelism	Executing several tasks at the same instant on multiple cores or machines
Peak factor	Ratio of peak traffic to average traffic
Percentile	Value below which a given percentage of measurements fall
Performance	How fast the system is for a single request at current load
Propagation delay	Time for a signal to travel the physical distance
QPS / RPS	Queries / requests per second
Queueing delay	Time spent waiting behind other work
RAM	Main working memory of a computer; fast but volatile
Race condition	Bug where the result depends on the timing of concurrent tasks
Read/write ratio	How many reads happen per write
Redundancy	Having spare copies so one can replace another
Reliability	Working correctly over time despite faults

Term	Meaning in one line
Request / response	A client's message asking for something and the server's reply
Response time	Total time the client waits, including network, queueing and service
Round trip time (RTT)	Time for a message to go to a destination and back
Scalability	Handling more load by adding resources without redesign
Server	Program (or machine) that waits for and answers requests
Service time	Time the server spends actually working on a request
SLA	Service Level Agreement; contract with penalties for missed targets
SLI	Service Level Indicator; a measurement of service health
SLO	Service Level Objective; internal target for an SLI
Single point of failure (SPOF)	A part whose failure alone brings the whole system down
Split brain	Two nodes both believing they are the primary
SRE	Site Reliability Engineering; running production with engineering methods
SSD	Solid-state drive; flash storage with no moving parts
State	Data a program remembers between requests
Stateful service	Service that keeps client or data state locally
Stateless service	Service that keeps no per-client state between requests
Sticky sessions	Load balancer sending a client to the same server every time
System design	Choosing a system's parts and how they interact to meet its goals
Tail latency	Latency of the slowest requests (p99, p999, max)
Thread	Independent flow of execution inside a process
Throughput	Amount of work completed per unit of time
Transmission delay	Time to push the bits of a message onto a link
Vertical scaling (scale up)	Making one machine bigger
Worker	Background program that processes jobs from a queue

Part 2 — Networking and communication

Every system design diagram is boxes joined by arrows. Part 1 was about the boxes. This part is about the arrows: how one computer finds another, how bytes travel between them, which languages (protocols) they speak, and which helpers sit in the middle, such as proxies, load balancers and CDNs. You do not need to be a network engineer for a system design interview. You do need to know what happens when a browser types `www.example.com`, why HTTP/2 is faster than HTTP/1.1, and what an L7 load balancer can do that an L4 one cannot. This part teaches exactly that, from zero.

A **protocol** is an agreed set of rules for how two programs talk, the way "say hello, wait for a reply, then speak" is a protocol for a phone call. Networking is a stack of protocols, each one solving one problem and handing the rest to the layer above or below.

IP addresses, NAT, ports and sockets

What it is

An **IP address** (Internet Protocol address) is the number that identifies a machine's network connection, like a street address on an envelope. Routers — the boxes that forward traffic across the internet —

read only this address to decide where to send each **packet** (a small chunk of data, usually up to about 1,500 bytes on Ethernet).

- **IPv4** addresses are 32 bits, written as four numbers from 0 to 255: `93.184.215.14`. That gives about 4.3 billion addresses (2^32), which the world ran out of in the 2010s.
- **IPv6** addresses are 128 bits, written as eight groups of hex digits: `2606:2800:21f:cb07:6820:80da:af6b:8b2c` (runs of zeros can be shortened with `::`). That is about 3.4 × 10^38 addresses — effectively unlimited.

A **port** is a number from 0 to 65535 that picks which program on a machine gets the data. If the IP address is the building, the port is the apartment number. Well-known ports: 22 (SSH, remote login), 53 (DNS), 80 (HTTP), 443 (HTTPS), 5432 (PostgreSQL), 6379 (Redis).

A **socket** is one end of a conversation, identified by IP address plus port. A single TCP connection is uniquely identified by a **5-tuple**: protocol, source IP, source port, destination IP, destination port. That is why one server on port 443 can hold hundreds of thousands of connections at once — each client brings a different source IP or source port.

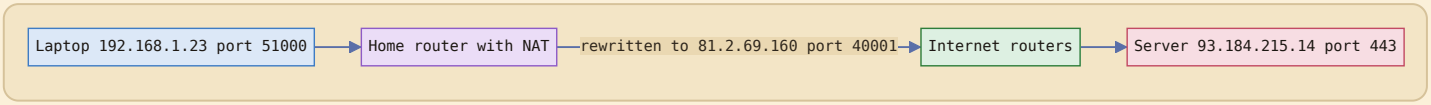
Public vs private IP. A **public IP** is reachable from anywhere on the internet and is unique worldwide. A **private IP** is only meaningful inside one local network and is reused by millions of networks. The reserved private IPv4 ranges are `10.0.0.0/8`, `172.16.0.0/12` and `192.168.0.0/16`. (The `/8` is **CIDR notation**: the number of leading bits that are fixed. `10.0.0.0/8` means "every address starting with 10", about 16.7 million addresses.) Your laptop at home is almost certainly something like `192.168.1.23`.

NAT (Network Address Translation) lets many private machines share one public IP. Your home router rewrites each outgoing packet's source from `192.168.1.23:51000` to its own public IP and a free port, say `81.2.69.160:40001`, remembers the mapping in a table, and reverses the rewrite on replies. In the cloud, the same job is done by a **NAT gateway** so that servers in a private subnet can download updates without being reachable from outside.

Why it matters

- Servers you design live mostly on private IPs inside a **VPC** (Virtual Private Cloud — your own isolated network inside a cloud provider). Only the load balancer or gateway gets a public address. This is the first layer of security.
- NAT explains why a server cannot simply "call" a phone or a browser: the client is behind NAT and has no stable public address. That is why push systems keep a long-lived connection open *from* the client (see WebSockets below).
- The 5-tuple explains connection limits. A proxy that opens connections from one IP to one backend IP and port can use at most about 64,000 source ports (only about 28,000 with Linux's default range), which is a real limit on busy proxies (called **ephemeral port exhaustion**).

How it works



1. Your browser asks the operating system for a socket to `93.184.215.14:443`. The OS picks a random **ephemeral port** (a temporary source port, typically from 32768–60999 on Linux) such as 51000.
2. The router's NAT swaps the private source address for its public one and records the mapping.
3. Internet routers forward the packet hop by hop based only on the destination IP. A typical path crosses 10–20 routers.
4. The server replies to `81.2.69.160:40001`. The router looks up its table and forwards the reply to `192.168.1.23:51000`.

Common mistakes

- Thinking an IP identifies a person or even a device. Behind **carrier-grade NAT** (NAT run by a mobile operator) thousands of users share one public IP, so rate limiting by IP alone can block an entire office or mobile network.
- Forgetting that IPs change. Cloud instances get new private IPs when they restart, which is why systems use DNS names and service discovery instead of hard-coded addresses.

TL;DR

An IP address finds the machine, a port finds the program, and the 5-tuple identifies one connection. Private IPs are hidden behind NAT, and only a few edge components in your design should have public IPs.

DNS — the internet's phone book

What it is

DNS (Domain Name System) turns human-friendly names like `www.example.com` into IP addresses. It is like your phone's contacts list: you tap "Mum", and the phone dials the number. Without DNS you would type `93.184.215.14` into the browser, and a company could never move its servers without telling every user the new number.

A **domain name** is read right to left as a hierarchy: `.` (the root, usually invisible) → `com` (the **top-level domain**, TLD) → `example` (the registered domain) → `www` (a **subdomain**). Each level is managed by a different organisation, which is what lets DNS scale to hundreds of millions of domains.

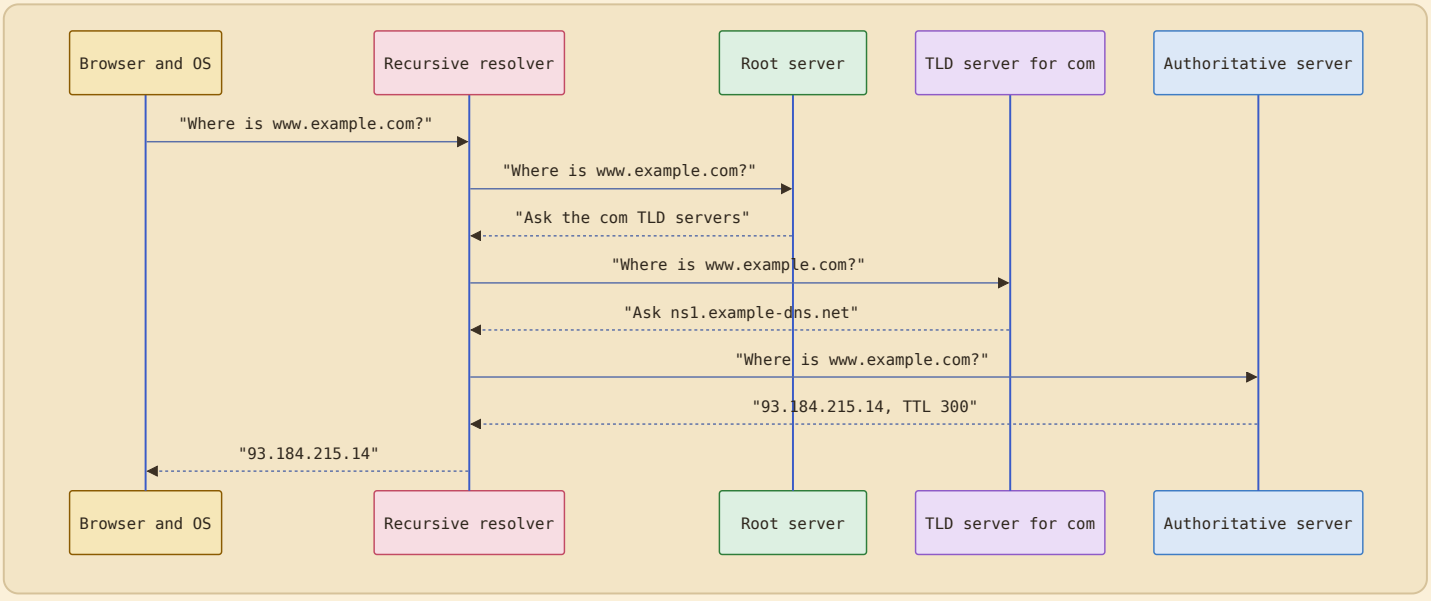
Why it matters

- DNS is the very first step of almost every request, so its speed adds directly to page load time (a cold lookup can cost 20–120 ms).
- DNS is a cheap, global traffic-steering tool: by answering with different IPs, you can spread users across data centres or send them to the closest one.
- When DNS breaks, everything looks down even though servers are healthy. Large outages (for example the 2016 Dyn attack, and Facebook's 2021 outage, where its DNS servers became unreachable after a routing change) show it is a critical dependency.

How it works

The players:

- **Stub resolver** — the tiny DNS client inside your OS. It does not do the hard work; it just asks someone else.
- **Recursive resolver** — the server that does the legwork on your behalf and caches answers. It is usually run by your ISP (internet service provider), your company, or a public service like Google `8.8.8.8` or Cloudflare `1.1.1.1`.
- **Root name servers** — the top of the tree. There are 13 named root server identities (`a.root-servers.net` to `m.root-servers.net`), but each is copied to many locations worldwide using anycast (explained in the CDN section), so there are close to 2,000 physical root server instances.
- **TLD name servers** — know who is in charge of each domain under `.com`, `.org`, `.uk` and so on.
- **Authoritative name server** — holds the real records for a domain (for example Route 53, Cloudflare DNS, or your own servers). Its answer is the final truth.



Step by step:

1. The browser checks its own cache, then the OS cache (and the `hosts` file).
2. On a miss, the stub resolver sends a **recursive query** to the recursive resolver: "give me the final answer".
3. If the resolver has nothing cached, it asks a root server, which replies with a **referral**: "I don't know, but here are the `.com` servers". These follow-up questions are **iterative queries** — each server just points to the next.
4. The `.com` TLD server refers it to the domain's authoritative servers (the domain's **NS records**).
5. The authoritative server returns the answer.
6. The resolver caches every step (root and TLD referrals are cached for days), so the next lookup for any `.com` name skips steps 3–4.

In practice, a busy resolver almost always has the root and TLD answers cached, so most lookups are one hop to the authoritative server or a pure cache hit (under 1 ms on the same machine, ~10–30 ms to a public resolver).

Types

DNS record types you should know:

Record	Maps	Example
A	name → IPv4 address	`www → 93.184.215.14`
AAAA	name → IPv6 address	`www → 2606:2800::1`
CNAME	name → another name (alias)	`shop → shops.myplatform.com`
MX	domain → mail servers, with priority	`10 mail1.example.com`
NS	domain → its authoritative name servers	`ns1.example-dns.net`
TXT	name → free text	domain-ownership proofs, SPF rules listing which servers may send a domain's email

Others you may meet: **SOA** (start of authority — admin data for a zone), **SRV** (service location with port), **PTR** (reverse lookup, IP → name), **CAA** (which certificate authorities may issue certificates for the domain). A **zone** is the set of records one authoritative server is responsible for.

A CNAME cannot sit at the bare domain (`example.com`, called the **apex**) because the apex must also hold NS and SOA records. Providers work around this with non-standard "ALIAS" or "ANAME" records, or Cloudflare's "CNAME flattening".

Here is a real-looking lookup with the `dig` tool:

```
$ dig www.example.com A +noall +answer
www.example.com.   300   IN   A   93.184.215.14
```

`300` is the **TTL**, `IN` means internet class, and `A` is the record type.

TTL and DNS caching. Every record carries a **TTL (time to live)** in seconds: how long any resolver or browser may cache it. A TTL of 300 means "reuse this for 5 minutes". Long TTLs (hours or a day) mean fewer lookups and faster pages, but changes take that long to reach everyone (what people call "propagation" is nothing being pushed out; resolvers simply keep the old answer until their copy times out). Short TTLs (30–60 s) let you fail over quickly but cause more queries. A common trick: lower the TTL a day before a planned migration, switch the record, then raise it again. Some clients and resolvers ignore TTLs and cache longer, so never rely on DNS alone for instant failover.

DNS-based load balancing. The authoritative server can return several A records (for example three IPs) and rotate their order on each query — **round-robin DNS**. Clients usually try the first one, so load spreads roughly evenly. It is free and simple but blind: DNS does not know whether a server is busy or dead unless the DNS provider runs health checks and removes bad IPs.

GeoDNS answers differently depending on where the query comes from: a user in Paris gets the IP of the Frankfurt data centre, a user in Tokyo gets the Tokyo one. The DNS server guesses location from the resolver's IP, or from the **EDNS Client Subnet** extension, which passes part of the user's own IP so the guess is better. **Latency-based routing** (offered by AWS Route 53) picks the region with the lowest measured latency rather than the closest on a map.

Trade-offs

Choice	Benefit	Cost
Long TTL	Fewer lookups, resilient if DNS provider has a blip	Slow to change or fail over
Short TTL	Fast failover and traffic shifts	More DNS traffic and lookup latency
DNS load balancing	Global, cheap, no extra hop	Coarse, cached, no knowledge of server load

Common mistakes

- Expecting a DNS change to take effect instantly. Cached answers live until their TTL expires, sometimes longer.
- Using one DNS provider with no redundancy. Many large sites now use two providers.
- Treating DNS round robin as a real load balancer. It cannot react to a server being slow.

Real-world examples

Amazon Route 53, Cloudflare DNS, Google Cloud DNS and NS1 are authoritative DNS services with health checks, GeoDNS and latency routing. Netflix and many others use DNS to steer users between AWS regions during regional failures.

In the interview

Q: What happens when you type a URL in the browser, up to the IP address?

A: Browser cache, OS cache, then a recursive resolver. If the resolver has nothing cached, it walks root → TLD → authoritative server, caches each answer for its TTL, and returns the IP. Then the browser opens a TCP (or QUIC) connection to that IP.

Q: How would you move traffic away from a failed region using DNS?

A: Use health-checked records with a short TTL (say 60 s) so the provider stops returning the failed region's IP. Accept that some clients will still hit the old IP for a while, so the failed region's load balancer should also redirect or fail fast.

TL;DR

DNS maps names to IPs through a cached hierarchy of root, TLD and authoritative servers. TTL controls the trade-off between speed and how quickly changes spread, and DNS can do coarse global load balancing (round robin, GeoDNS, latency-based).

Network layers — OSI and TCP/IP models

What it is

Networking is split into **layers**, each solving one job and relying on the layer below. Think of posting a parcel: you write a letter (application), put it in an envelope with an apartment number (transport), the post office adds a city and street address (network), and a van physically drives it (link and physical). Each layer only cares about its own label.

There are two models. The **OSI model** has seven layers and is mainly a teaching vocabulary. The **TCP/IP model** has four and describes what the internet actually runs. Engineers mix them: when someone says "L4" or "L7", they mean OSI layer numbers.

OSI layer	Name	What it handles	Examples
7	Application	Meaning of messages	HTTP, DNS, gRPC, SMTP
6	Presentation	Encoding, encryption	TLS (roughly), JSON (a text data format)
5	Session	Conversations	(rarely discussed)
4	Transport	Process-to-process delivery, ports	TCP, UDP, QUIC
3	Network	Machine-to-machine routing	IP, ICMP
2	Data link	One hop on a local network	Ethernet, Wi-Fi, MAC addresses
1	Physical	Bits on wire, light or radio	Copper, fibre, radio

The TCP/IP model merges 5–7 into "application" and 1–2 into "link": Application, Transport, Internet, Link.

Why it matters

The layer a component works at decides what it can see and therefore what it can do:

- An **L3** device (router, firewall rule on IPs) sees only source and destination IPs.
- An **L4** device (L4 load balancer) also sees ports and TCP/UDP state, but not the content. It can say "send port 443 traffic to these servers" but not "send `/api/` to these".
- An **L7** device (L7 load balancer, API gateway) reads the HTTP request itself: URL path, headers, cookies. It can route by path, rewrite headers, and block bad requests — at the cost of more CPU and needing to decrypt TLS (the encryption behind HTTPS, explained below).

How it works

When your app sends an HTTP request, each layer wraps the data with its own header — **encapsulation**. HTTP request → TCP segment (adds ports, sequence numbers) → IP packet (adds IPs) → Ethernet frame (adds MAC addresses, the hardware address of a network card). The receiver unwraps in reverse order.

TL;DR

Layers separate concerns: L3 is IP addresses, L4 is ports and connections (TCP/UDP), L7 is application content (HTTP). "L4 vs L7" in system design always means "can it read the request or only the connection?"

TCP, UDP and QUIC

What it is

These are the **transport-layer** protocols: they carry data between two programs.

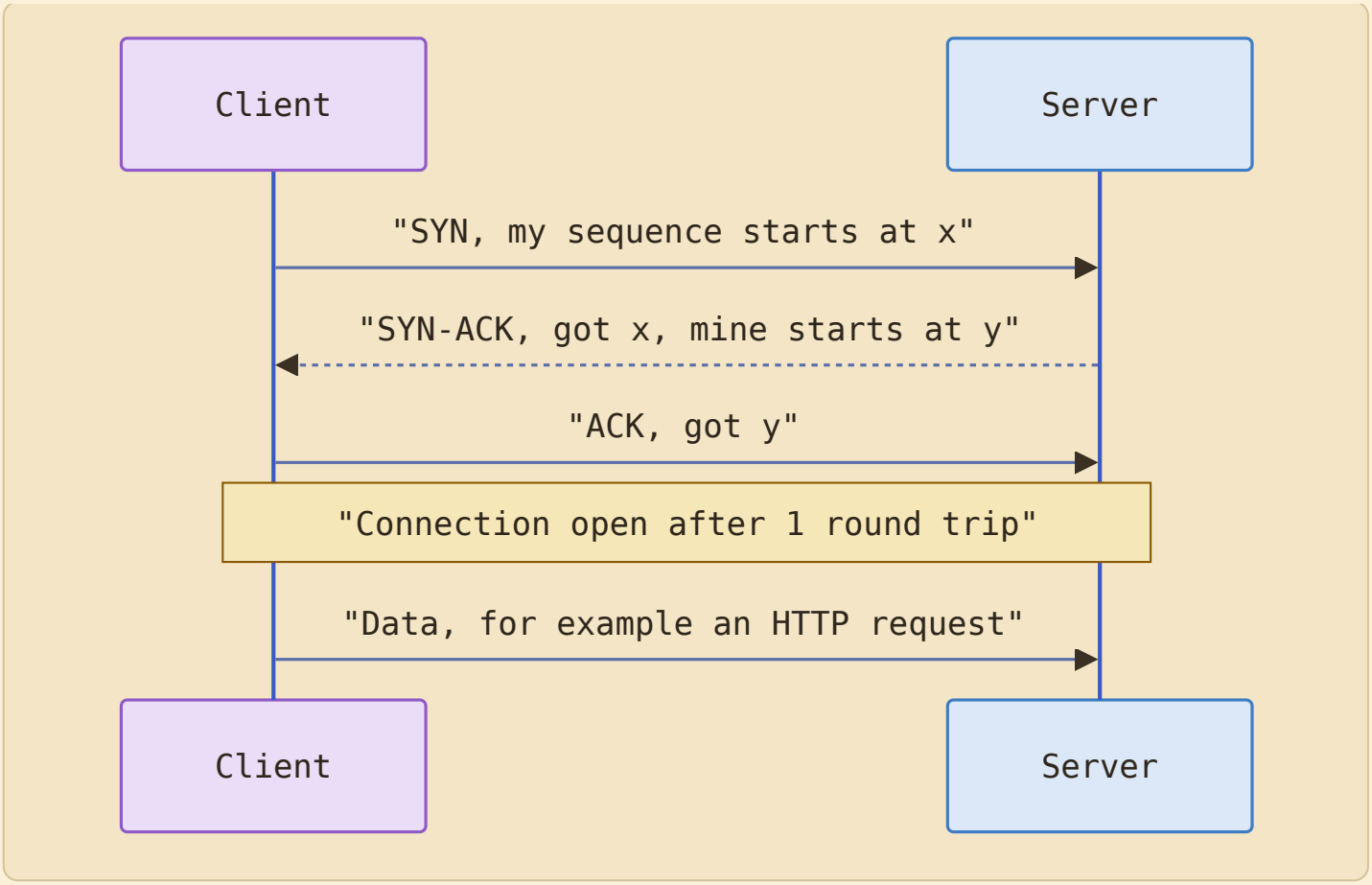
- **TCP (Transmission Control Protocol)** is like a phone call with a careful note-taker: you connect first, everything arrives, in order, and lost words are repeated.
- **UDP (User Datagram Protocol)** is like shouting postcards across a room: no setup, each message is independent, some may be lost or arrive out of order, nobody repeats anything.
- **QUIC** is a newer protocol built on top of UDP that gives TCP-like reliability plus built-in encryption, without some of TCP's weaknesses. HTTP/3 runs on it.

Why it matters

The choice shapes latency and behaviour under packet loss. Web pages, APIs and databases want every byte correct, so they use TCP (or QUIC). Live voice, video calls, online games and DNS lookups prefer UDP because a late packet is useless — better to skip it than to wait.

How it works

TCP three-way handshake. Before any data, the two sides agree on starting sequence numbers:



This costs one **round trip (RTT)** before the first byte of the request. London to New York is about 70 ms RTT, so the handshake alone is 70 ms — and TLS adds more on top (see HTTPS). This is why reusing connections matters so much.

Reliability and ordering. TCP numbers every byte (**sequence numbers**). The receiver sends **acknowledgements (ACKs)**. If an ACK does not arrive in time, the sender **retransmits**. The receiver reorders out-of-order segments and hands the application a clean, ordered byte stream.

Flow control. The receiver advertises a **receive window** — how much buffer space it has — so a fast sender cannot overwhelm a slow receiver.

Congestion control. The network in between can also be overwhelmed. TCP starts slowly (**slow start**: send a few packets, roughly double each round trip while ACKs come back), and when it sees loss it backs off. Algorithms include Reno, CUBIC (default on Linux) and Google's BBR, which models bandwidth and RTT instead of waiting for loss. A practical result: a brand-new connection cannot use the full bandwidth immediately, another reason to keep connections warm.

Closing. Either side sends **FIN**; the other acknowledges. The side that closes first keeps the socket in **TIME_WAIT** for a short time (on Linux, 60 s) to catch stray packets — a source of port exhaustion on busy clients that open and close many connections.

Head-of-line (HOL) blocking. Because TCP delivers bytes strictly in order, one lost packet holds up everything behind it until it is retransmitted, even data that already arrived. Picture a single-lane road where one stalled car stops all traffic behind. If several independent requests share one TCP connection (as in HTTP/2), a single loss stalls all of them.

UDP has no handshake, no ACKs, no ordering and no congestion control. Each **datagram** is sent and forgotten. The header is 8 bytes versus TCP's 20+. Applications that need some reliability add it themselves.

QUIC (standardised by the IETF, the Internet Engineering Task Force, in 2021, originally developed at Google) runs over UDP and puts into one protocol what used to take TCP + TLS:

1. **Faster setup** — the transport and TLS 1.3 handshakes are combined, so a new connection takes 1 RTT, and a returning client can send data in the first packet (**0-RTT**).

2. **Independent streams** — many streams inside one connection, and a lost packet only stalls the stream it belongs to. No transport-level HOL blocking between streams.
3. **Connection migration** — a connection is identified by a **connection ID**, not the 5-tuple, so your phone switching from Wi-Fi to 4G keeps the same connection.
4. **Always encrypted** — even most of the transport headers.
5. **Lives in user space** — it can evolve without operating-system upgrades. The downside is more CPU per byte than kernel TCP, and some corporate firewalls block UDP, so clients fall back to TCP.

Types

	TCP	UDP	QUIC
Connection setup	1 RTT (plus TLS)	none	1 RTT incl. TLS, 0-RTT on resume
Reliable, ordered	Yes, one stream	No	Yes, per stream
HOL blocking	Yes	No	Only within a stream
Encryption	Separate (TLS)	None built in	Built in
Typical use	Web, APIs, databases	DNS, video calls, games	HTTP/3

Trade-offs

TCP gives correctness at the cost of setup latency and stalls under loss. UDP gives speed and control at the cost of building whatever reliability you need yourself. QUIC fixes TCP's main pains for the web but costs more CPU and is sometimes blocked.

Common mistakes

- Saying "UDP is faster" without context. On a clean network, a long-lived TCP connection is just as fast for bulk data. UDP wins on setup time and on not waiting for retransmissions.
- Forgetting that every new TCP + TLS connection costs 2–3 RTTs before the first byte. Use connection pooling and keep-alive.

Real-world examples

Zoom, WebRTC (browser video calls), and most online games use UDP. DNS uses UDP for normal queries and switches to TCP for large responses. Google, YouTube, Meta and Cloudflare serve a large share of their traffic over QUIC / HTTP/3.

In the interview

Q: Why would a video-calling app use UDP instead of TCP?

A: Because a frame that arrives 300 ms late is worthless. TCP would pause all later frames while retransmitting the lost one (HOL blocking). With UDP the app just skips the lost frame and stays real-time.

Q: What problem does QUIC solve?

A: TCP's head-of-line blocking across multiplexed streams and its slow combined TCP + TLS handshake. QUIC gives independent streams, a 1-RTT (or 0-RTT) encrypted handshake, and survives network changes.

TL;DR

TCP = connected, reliable, ordered, congestion-controlled, but has handshake cost and head-of-line blocking. UDP = fire-and-forget datagrams. QUIC = reliable, encrypted, multi-stream transport over UDP that powers HTTP/3.

HTTP — the language of the web

What it is

HTTP (HyperText Transfer Protocol) is the request–response protocol browsers and APIs use. The client sends a **request** (a method, a path, headers, optional body); the server sends a **response** (a status code, headers, optional body). It is like ordering at a counter: "GET me the menu" → "200 OK, here it is".

```
GET /users/42 HTTP/1.1
Host: api.example.com
Accept: application/json
Authorization: Bearer eyJhbGciOi...

HTTP/1.1 200 OK
Content-Type: application/json
Cache-Control: max-age=60

{"id": 42, "name": "Asha"}
```

Why it matters

Almost every API you design in an interview is HTTP. Choosing the right method and status code makes APIs predictable, cache-friendly and safe to retry. The HTTP version decides how many connections you need and how much latency you pay.

How it works

Methods (verbs).

Method	Meaning	Safe?	Idempotent?
GET	Read a resource	Yes	Yes
HEAD	Like GET, headers only	Yes	Yes
POST	Create, or run an action	No	No
PUT	Replace a resource fully	No	Yes
PATCH	Change part of a resource	No	Not guaranteed
DELETE	Remove a resource	No	Yes
OPTIONS	Ask what is allowed (used by CORS, the browser's cross-origin permission check)	Yes	Yes

Safe means it does not change anything on the server. **Idempotent** means doing it once or ten times leaves the server in the same state — so a client can safely retry after a timeout. `DELETE /orders/7` twice still leaves order 7 deleted (the second call may return 404, but the state is the same). `POST /orders` twice creates two orders, which is why payment APIs add an **idempotency key** header so the server can recognise a retry.

Status codes — the first digit is the category:

Range	Meaning	Common codes
1xx	Informational	101 Switching Protocols (WebSocket upgrade)
2xx	Success	200 OK, 201 Created, 202 Accepted, 204 No Content
3xx	Redirect	301 Moved Permanently, 302 Found, 304 Not Modified
4xx	Client's fault	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict, 429 Too Many Requests
5xx	Server's fault	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable, 504 Gateway Timeout

401 means "we don't know who you are" (not logged in); 403 means "we know who you are and you may not do this". 502 and 504 typically come from a proxy or load balancer: the backend gave a bad answer (502) or no answer in time (504). 202 Accepted is for "queued, will be processed later".

Headers are key–value metadata. Important ones: `Host` (which site — many sites share one IP), `Content-Type` (format of the body), `Authorization` (credentials), `Cache-Control` (who may cache and for how long), `ETag` and `If-None-Match` (let the server reply 304 Not Modified if the client's copy is current), `User-Agent`, `Accept-Encoding: gzip, br` (compression), `X-Forwarded-For` (original client IP, added by proxies).

Statelessness and cookies. HTTP is **stateless**: each request carries everything the server needs, and the server does not remember the previous request by itself. This is what makes it easy to put many identical servers behind a load balancer. To build logins and shopping carts on top, servers use **cookies**: the server sends `Set-Cookie: session=abc123; HttpOnly; Secure; SameSite=Lax`, and the browser automatically sends `Cookie: session=abc123` on every later request to that site. The session data itself lives in a shared store (like Redis) or inside a signed token, so any server can handle the next request.

Keep-alive (persistent connections). HTTP/1.0 opened a new TCP connection per request. HTTP/1.1 keeps the connection open by default for reuse, saving the handshake on every request.

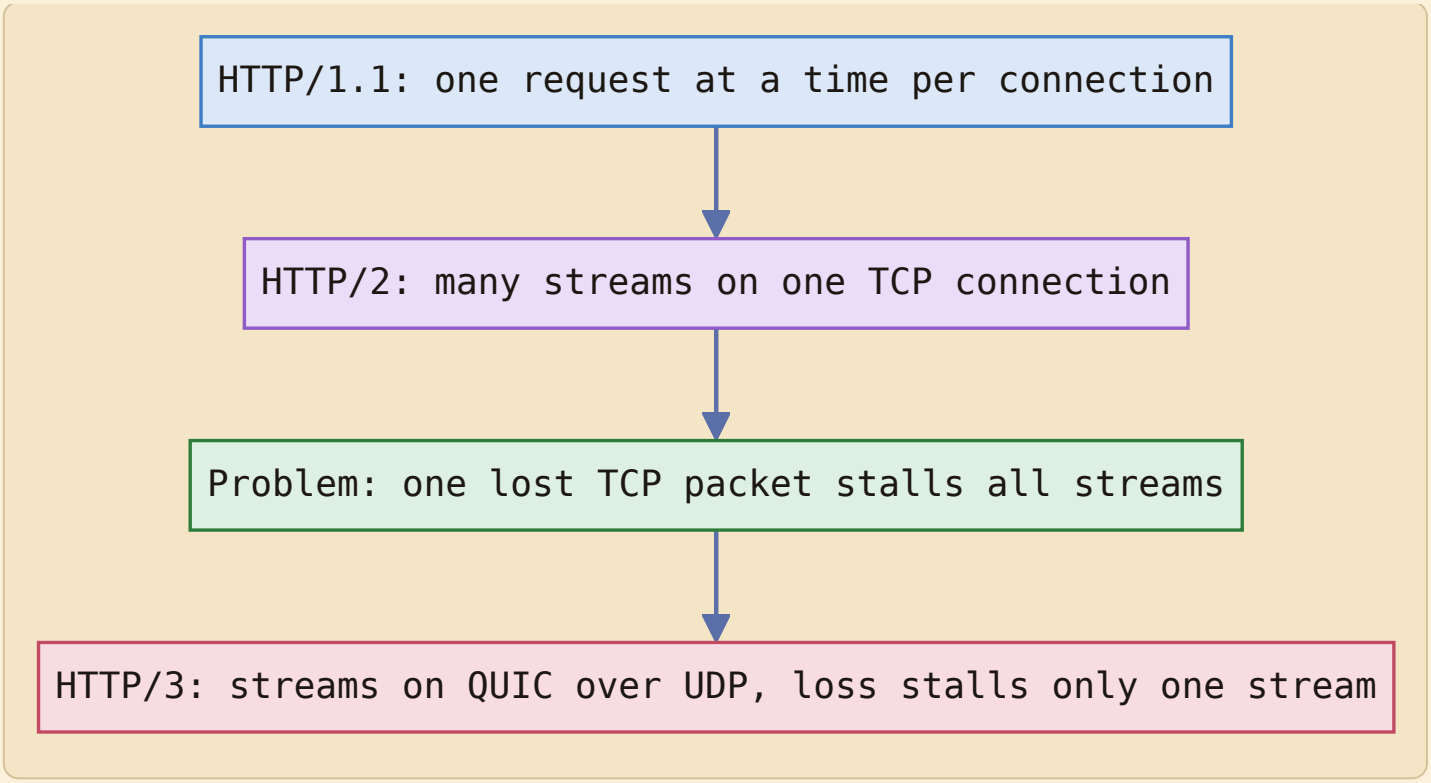
Types

HTTP/1.1 (1997). Text-based. One request at a time per connection; the next request waits for the previous response (**HTTP-level HOL blocking**). "Pipelining" was specified but never worked reliably. Browsers work around it by opening about 6 connections per host, and sites used tricks like domain sharding and sprite images.

HTTP/2 (2015). Binary framing. **Multiplexing**: many requests and responses interleave as **streams** over one TCP connection, so one slow response no longer blocks others at the HTTP level. **Header compression** (HPACK) shrinks repetitive headers. Server push existed but was little used and has been removed from Chrome. Weakness: it still runs on one TCP connection, so one lost packet stalls every stream (TCP-level HOL blocking). On lossy mobile networks this can be worse than six HTTP/1.1 connections.

HTTP/3 (2022). Same meaning as HTTP/2 but runs on QUIC over UDP. Independent streams remove TCP HOL blocking, setup is faster, and connections survive network switches. Header compression is QPACK. Browsers discover HTTP/3 through an `Alt-Svc` header or a DNS **HTTPS record**, and fall back to HTTP/2 if UDP is blocked.

	HTTP/1.1	HTTP/2	HTTP/3
Transport	TCP	TCP	QUIC (UDP)
Format	Text	Binary frames	Binary frames
Parallel requests	~6 connections	Many streams, 1 connection	Many streams, 1 connection
HOL blocking	HTTP and TCP level	TCP level	Per stream only
Encryption	Optional	Required by browsers	Always



Common mistakes

- Using GET for actions that change data (a crawler or prefetcher may trigger them).
- Returning 200 with `{"error": ...}` in the body. Proxies, caches, monitoring and retries all rely on real status codes.
- Retrying non-idempotent POSTs blindly. Use idempotency keys.
- Storing session state in one server's memory, which breaks as soon as there are two servers.

In the interview

Q: Which HTTP methods are idempotent and why does it matter?

A: GET, HEAD, PUT, DELETE, OPTIONS. It matters because networks time out: if the client does not know whether the request succeeded, it can retry an idempotent request safely. For POST you need an idempotency key.

Q: What did HTTP/2 improve and what did HTTP/3 fix on top?

A: HTTP/2 multiplexed many requests over one connection with header compression, removing HTTP-level head-of-line blocking. HTTP/3 moved to QUIC to remove the remaining TCP-level head-of-line blocking and speed up connection setup.

TL;DR

HTTP is a stateless request–response protocol of methods, status codes and headers. Cookies add sessions on top, keep-alive reuses connections, HTTP/2 multiplexes streams over TCP, and HTTP/3 does the same over QUIC.

HTTPS and TLS

What it is

HTTPS is HTTP sent inside a **TLS (Transport Layer Security)** encrypted tunnel. TLS gives three guarantees: **confidentiality** (eavesdroppers see gibberish), **integrity** (nobody can change the bytes undetected) and **authentication** (you really are talking to `bank.com`, not an impostor). Its older name is **SSL**, which is why people still say "SSL certificate" and "SSL termination". The current version is TLS 1.3 (2018); TLS 1.2 is still common; anything older is deprecated.

Analogy: a sealed, tamper-evident envelope, plus a passport check of the person you are handing it to.

Why it matters

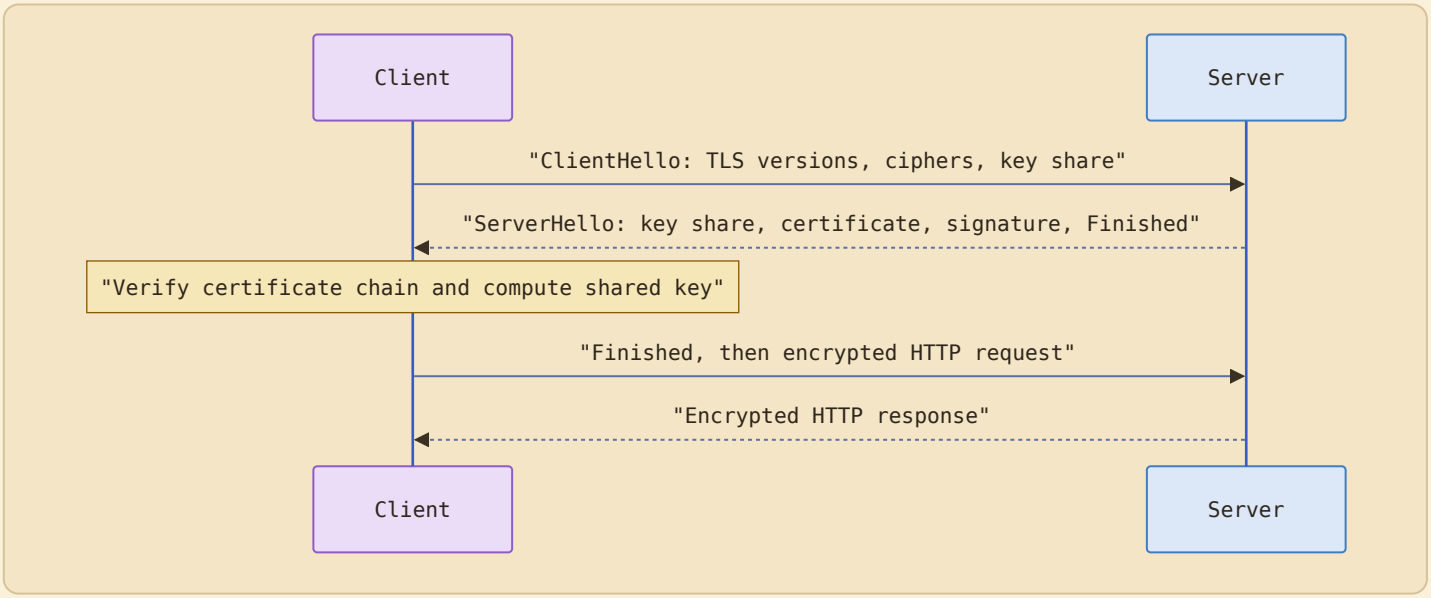
Browsers mark plain HTTP as "Not secure", HTTP/2 in browsers and HTTP/3 always require it, and any password, token or personal data on the wire must be encrypted. This is **encryption in transit** (as opposed to **encryption at rest**, which protects stored data on disk). TLS also costs latency and CPU, which affects where you decrypt (see TLS termination in the load balancer section).

How it works

Certificates. A server proves its identity with a **certificate**: a file containing the domain name, the server's **public key**, an expiry date, and a digital signature from a **Certificate Authority (CA)** — an organisation trusted by browsers and operating systems, such as Let's Encrypt or DigiCert. The browser follows the **chain of trust**: server certificate → intermediate CA → **root CA** that is pre-installed in the browser. If the chain is valid, the name matches and it is not expired, the server is trusted. Let's Encrypt issues free certificates, renewed automatically with the ACME protocol. They have been valid for 90 days, and lifetimes across the industry are shrinking: Let's Encrypt plans 64 days in 2027 and 45 days in 2028, so automated renewal is mandatory.

Public-key vs symmetric encryption. Asymmetric (public-key) cryptography uses a key pair: anything signed by the private key can be verified with the public key. It is slow, so TLS uses it only during the handshake to authenticate the server and agree on a shared secret. After that, all data is encrypted with fast **symmetric** encryption (one shared key, e.g. AES-GCM or ChaCha20).

TLS 1.3 handshake (simplified):



1. The client sends supported versions and cipher suites and a **key share** (its half of a Diffie–Hellman key exchange — a method where two sides each mix in a secret and arrive at the same shared key without ever sending it).
2. The server answers with its key share, its certificate, and a signature proving it owns the certificate's private key.
3. Both sides derive the same symmetric keys. The client sends Finished and can immediately send the encrypted request.

TLS 1.3 needs **1 RTT** (TLS 1.2 needed 2). Combined with TCP's 1 RTT, a fresh HTTPS connection over TCP costs about 2 RTTs before the request goes out — 140 ms from London to New York. **Session resumption** and **0-RTT** let returning clients skip work (0-RTT data can be replayed by an attacker, so it should only be used for idempotent requests). Using fresh Diffie–Hellman keys per session gives **forward secrecy**: stealing the server's private key later does not decrypt old recorded traffic.

SNI (Server Name Indication). The ClientHello includes the hostname so one IP can serve certificates for many domains — essential for load balancers and CDNs.

mTLS (mutual TLS) — both sides present certificates. Used between internal services (service meshes) so each service proves its identity.

Common mistakes

- Letting certificates expire. It has caused outages at large companies; automate renewal and alert on expiry.
- Terminating TLS at the edge and sending plaintext across untrusted networks. Inside one VPC it may be acceptable; across data centres, re-encrypt.

In the interview

Q: Why does HTTPS use both asymmetric and symmetric encryption?

A: Asymmetric crypto solves authentication and key agreement without a shared secret, but it is slow. Once both sides share a key, symmetric encryption is fast enough for bulk data.

Q: How many round trips does a new HTTPS request cost?

A: About 1 RTT for TCP, 1 for TLS 1.3, then 1 for the request and response — 3 in total. With HTTP/3 over QUIC the first two merge into one, and resumed connections can send data immediately.

TL;DR

TLS encrypts and authenticates connections using certificates signed by trusted CAs. The handshake uses public-key crypto to agree on a symmetric key, TLS 1.3 takes 1 RTT, and HTTPS is simply HTTP inside TLS.

API styles — REST, RPC and gRPC, GraphQL

What it is

An **API (Application Programming Interface)** is the menu a service offers to other programs: which operations exist, what inputs they take and what they return. Three styles dominate.

- **REST (Representational State Transfer)** models everything as **resources** (nouns) with URLs, manipulated with standard HTTP methods (verbs). `GET /users/42`, `POST /users/42/orders`. Like a library: every book has a shelf address, and there are a few standard things you can do with any book (look, add, replace, remove).
- **RPC (Remote Procedure Call)** models the API as functions you call on a remote machine as if they were local: `getUser(42)`, `chargeCard(order, amount)`. Like phoning a colleague and asking them to do a specific task.
- **GraphQL** gives clients one endpoint and a query language to ask for exactly the fields they want, possibly across several related objects, in one round trip. Like a form where you tick exactly which details you need.

Why it matters

The API is the contract between teams and between your system and its users. A good style choice makes clients simple and fast; a poor one causes chatty clients, over-fetching (downloading data you don't need) or under-fetching (needing many calls to build one screen).

How it works

REST in practice.

1. Name resources with plural nouns: `/users`, `/users/42`, `/users/42/orders`.
2. Use methods for actions: GET reads, POST creates, PUT replaces, PATCH updates part, DELETE removes.
3. Return proper status codes (201 Created with a `Location` header, 404, 409 Conflict).
4. Keep requests stateless, so any server can answer and GET responses can be cached by browsers and CDNs.
5. Usually JSON over HTTP/1.1 or HTTP/2.

Idempotent methods (GET, PUT, DELETE) can be retried safely; this is a big reason to prefer `PUT /users/42/avatar` over `POST /changeAvatar`.

gRPC. An open-source RPC framework from Google that runs on HTTP/2. You define services and messages in a `.proto` file using **Protocol Buffers (Protobuf)** — a compact binary format with a schema:

```
service UserService {
  rpc GetUser (GetUserRequest) returns (User) {}
}
message GetUserRequest { int64 id = 1; }
message User { int64 id = 1; string name = 2; }
```

The numbers (`= 1`, `= 2`) are field tags that identify each field on the wire. A code generator produces client and server code in many languages, so calling another service looks like calling a local function, with types checked at compile time. Protobuf messages are typically several times smaller and faster to parse than the same JSON, and the numbered fields allow **backward-compatible** schema changes (add new field numbers, never reuse old ones). gRPC supports four call shapes: unary (one request, one response), server streaming, client streaming and bidirectional streaming. Browsers cannot speak native gRPC directly, so it is mostly used **service-to-service**, with gRPC-Web or a REST gateway for browsers.

GraphQL. Created at Facebook (open-sourced in 2015). The server publishes a typed **schema**; the client sends a query such as:

```
query {
  user(id: 42) {
    name
    orders(last: 3) { id total }
  }
}
```

and gets back exactly that shape. Writes are **mutations**, and live updates are **subscriptions**. Each field is filled by a **resolver** function on the server. It is ideal when many different clients (web, iOS, Android) need different slices of the same data. Downsides: usually one POST endpoint, so ordinary HTTP caching does not work; a careless query can be very expensive, so servers limit query depth and cost; and naive resolvers cause the **N+1 query problem** (one query for the list, then one per item), usually fixed with batching (the DataLoader pattern).

Types

	REST	gRPC	GraphQL
Model	Resources + HTTP verbs	Functions	Typed graph, client picks fields
Format	Usually JSON	Protobuf (binary)	JSON
Transport	HTTP/1.1 or 2	HTTP/2	HTTP, one endpoint
Best for	Public APIs, CRUD (create/read/update/delete), caching	Internal service calls, streaming	Many client types, complex screens
Weak spot	Over/under-fetching	Browser support, readability	Caching, query cost control

Trade-offs

REST is universally understood and cache-friendly, but may need many calls per screen. gRPC is fast, strongly typed and streams well, but is binary (harder to debug with `curl`) and needs a proxy for browsers. GraphQL gives clients flexibility but moves complexity and performance risk to the server.

Common mistakes

- Verbs in REST URLs (`/getUser?id=42`) instead of methods on resources.
- Using GraphQL "because it is modern" for a simple public API where REST with caching would be cheaper.
- Changing Protobuf field numbers, which silently breaks old clients.

Real-world examples

Stripe and GitHub's v3 API are classic REST. GitHub also offers a GraphQL API (v4), and Shopify's storefront API is GraphQL. Google, Netflix and many companies use gRPC between internal microservices;

Kubernetes uses gRPC for its plug-in interfaces, such as the one between the kubelet and the container runtime.

In the interview

Q: You are designing a mobile app's backend and an internal ranking service. Which API styles?

A: REST (or GraphQL if screens need many related objects) for the public mobile API, because it is easy to consume, cache and debug. gRPC between the backend and the ranking service, for compact binary messages, typed contracts and streaming.

Q: What makes a REST call safe to retry?

A: Using an idempotent method (GET, PUT, DELETE), or attaching an idempotency key to a POST so the server returns the original result instead of repeating the action.

TL;DR

REST = resources and HTTP verbs, simple and cacheable. gRPC = typed remote functions over HTTP/2 with Protobuf, great inside the data centre. GraphQL = clients query exactly the fields they need from a typed schema.

Real-time communication — polling, long polling, WebSockets, SSE, webhooks

What it is

Plain HTTP only lets the client ask. Many features need the server to *tell* the client something as soon as it happens: a chat message, a new notification, a stock price, a driver's location. Because the client is usually behind NAT and the server cannot open a connection to it, every solution is some way of keeping a channel open from the client side — or, between servers, of calling back a public URL.

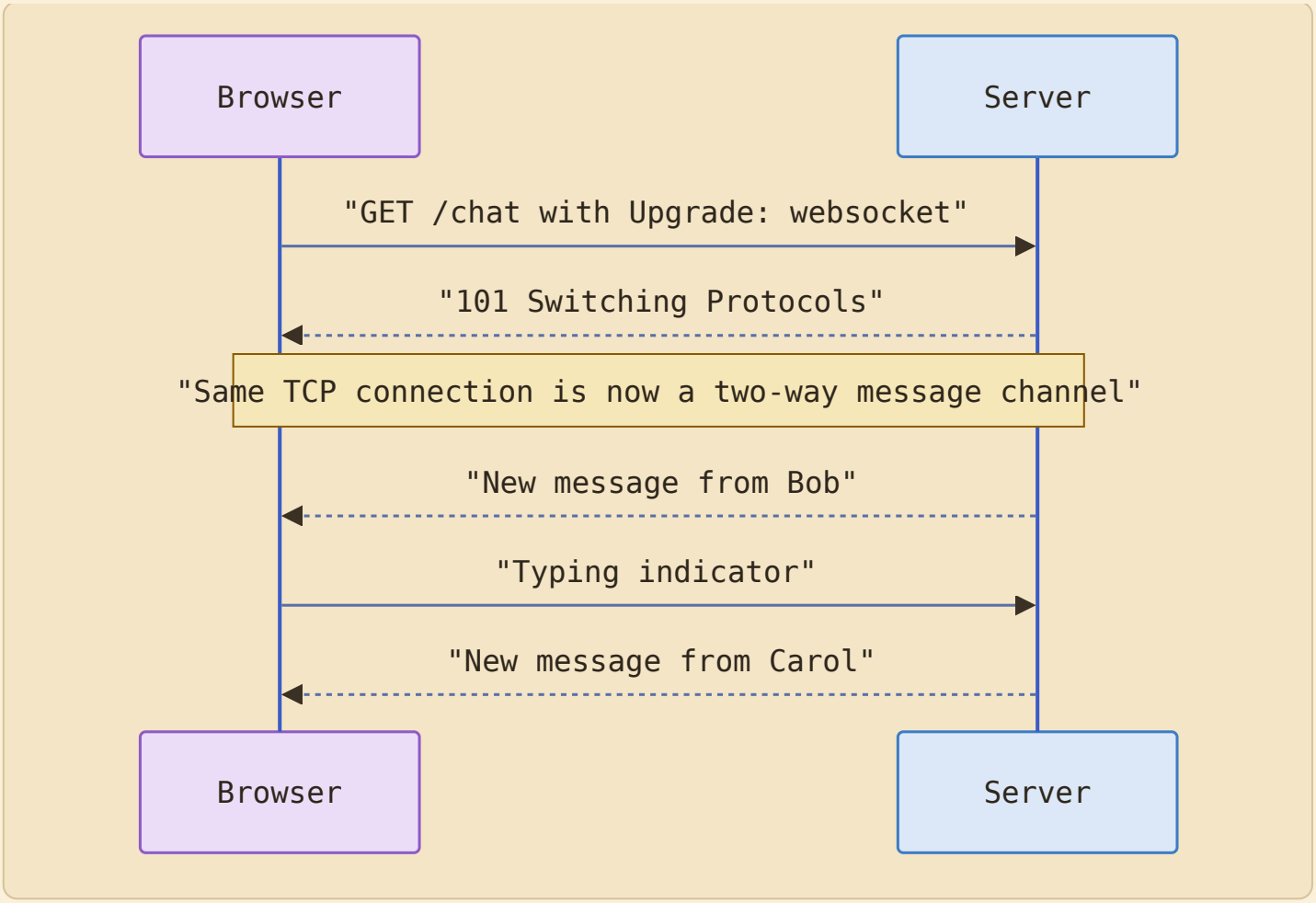
- **Short polling** — the client asks "anything new?" every few seconds. Like a child asking "are we there yet?" every minute.
- **Long polling** — the client asks, and the server holds the request open until there is news or a timeout (say 30 s), then the client immediately asks again. Like asking a shopkeeper to call you over when your order is ready.
- **WebSockets** — one long-lived, two-way connection over which both sides send messages any time. Like an open phone line.
- **Server-Sent Events (SSE)** — one long-lived HTTP response that the server keeps writing events into; one-way, server to client. Like a radio broadcast you tune into.
- **Webhooks** — server-to-server: when an event happens, service A sends an HTTP POST to a URL that service B registered. Like leaving your phone number so the shop calls you.

Why it matters

Choosing wrongly wastes resources (polling a million clients every second is a million requests per second, mostly answering "nothing new") or adds needless complexity. Long-lived connections also change capacity planning: a chat server's limit is often the number of open connections (memory per connection, file descriptors) rather than requests per second.

How it works

WebSocket upgrade. It starts as an ordinary HTTP/1.1 request, so it passes through normal ports and proxies:



1. The client sends `GET` with `Upgrade: websocket` and `Connection: Upgrade` headers (URL scheme `ws://` or encrypted `wss://`).
2. The server answers `101 Switching Protocols`.
3. From then on, both sides exchange small **frames** with only a few bytes of overhead each, until either side closes.
4. Applications send **heartbeats** (ping/pong) every 20–60 s so that idle connections are not silently cut by NAT routers or load balancers, and clients reconnect with backoff if the link drops.

SSE. The client opens `GET /events` with `Accept: text/event-stream`. The server responds and never finishes, writing lines like `data: {"price": 101.5}` followed by a blank line for each event. The browser's built-in `EventSource` reconnects automatically and sends `Last-Event-ID` so the server can resume. It is plain HTTP, so it works well with HTTP/2 and existing infrastructure. Large language model chat interfaces commonly stream tokens this way.

Webhooks. The receiver registers `https://myapp.com/hooks/payments`. When a payment succeeds, the provider POSTs a JSON event there. Good practice: sign each request (an HMAC header: a hash of the body computed with a shared secret key) so the receiver can verify the sender; retry with backoff on failure; and make the receiver idempotent because the same event may arrive twice.

Types

Technique	Direction	Latency	Cost / complexity	Use when
Short polling	Client pulls	Up to the interval	Many wasted requests	Rare updates, simplicity first
Long polling	Server push (emulated)	Low	Held requests, reconnect loops	Fallback where WebSockets are blocked
WebSocket	Two-way	Lowest	Stateful connections, sticky routing	Chat, games, collaboration
SSE	Server to client	Low	Simple, auto-reconnect	Feeds, notifications, live scores, token streaming
Webhook	Server to server	Low	Receiver needs public URL	Payment and Git events

Trade-offs

Persistent connections give instant delivery but make servers **stateful**: each user is attached to one server, so you need a way to route messages to the right server (often a pub/sub layer such as Redis Pub/Sub or Kafka between servers), and deployments must drain connections gracefully. Polling is stateless and trivially scalable but wasteful and slower.

Common mistakes

- Choosing WebSockets for a one-way feed where SSE would be simpler.
- Forgetting that a load balancer's idle timeout (for example 60 s by default on AWS ALB) will close quiet WebSockets unless you send heartbeats.
- Assuming webhooks arrive exactly once and in order. They do not.

Real-world examples

Slack and Discord use WebSockets for their real-time clients. GitHub, Stripe and Shopify deliver events by webhooks. Many dashboards and AI chat apps stream with SSE.

In the interview

Q: How would you push new chat messages to 10 million online users?

A: WebSocket connections terminated on a fleet of connection servers (each holding perhaps 50,000–100,000 connections). A registry maps user → connection server, and a pub/sub layer delivers each message to the right server, which writes it to the socket. Clients reconnect with backoff and fetch missed messages using their last seen ID.

Q: When would you choose SSE over WebSockets?

A: When only the server needs to push (notifications, live scores, streaming text). SSE is plain HTTP, reconnects automatically and needs no protocol upgrade.

TL;DR

Polling asks repeatedly, long polling waits for news, WebSockets keep a two-way channel, SSE streams one-way from server to client, and webhooks are server-to-server callbacks. Persistent connections are fast but make servers stateful.

API design basics

What it is

The small, practical rules that make an API pleasant and safe to evolve: how you version it, how clients page through big lists, how you tell clients about limits and errors.

Why it matters

Once a public API is used by thousands of clients, you cannot change it freely. Pagination and rate limits protect your servers from clients asking for too much at once.

How it works

Versioning. When you must make a breaking change (removing or renaming a field, changing meaning), publish a new version and keep the old one running for a deprecation period.

- **URL versioning**: `/v1/orders`, `/v2/orders`. Most common, very visible.
- **Header versioning**: `Accept: application/vnd.example.v2+json`, or a custom header. Cleaner URLs, harder to test in a browser.
- **Date-based versions**: Stripe pins each account to a dated version and lets clients opt in to newer ones.

Prefer **additive** changes (new optional fields) that need no new version at all, and tell clients to ignore fields they do not recognise.

Pagination. Never return an unbounded list.

- **Offset pagination**: `GET /posts?limit=20&offset=40`. Simple, allows "jump to page 5". But the database still reads and skips the first 40 rows, which gets slow at offset 1,000,000, and if a new post is inserted while you page, items shift and you see duplicates or miss items.
- **Cursor (keyset) pagination**: `GET /posts?limit=20&after=eyJpZCI6MTIzNH0`. The cursor is an opaque token encoding the last item's sort key (e.g. `id=1234`). The server runs `WHERE id < 1234 ORDER BY id DESC LIMIT 20`, which uses an index and is equally fast on page 1 and page 50,000, and stays stable when new items arrive. You cannot jump to an arbitrary page. Feeds and timelines (Twitter/X, Facebook, Slack) use cursors.

	Offset	Cursor
Jump to page N	Yes	No
Speed deep into list	Gets slower	Constant
Stable under inserts	No	Yes
Best for	Admin tables, small data	Feeds, large or changing data

Rate-limit headers. When you limit clients (say 5,000 requests per hour), tell them where they stand. Common headers: `X-RateLimit-Limit` (quota), `X-RateLimit-Remaining`, `X-RateLimit-Reset` (when the window resets). GitHub uses exactly these. When a client is over the limit, return **429 Too Many Requests** with a `Retry-After: 30` header (seconds to wait). The IETF is standardising unprefixed `RateLimit` headers for the same purpose. Rate-limiting algorithms themselves are covered in Part 5.

Other basics. Consistent error bodies (a machine-readable code plus a human message), idempotency keys on POST, filtering and sorting via query parameters (`?status=paid&sort=-created_at`), and documenting the API with an **OpenAPI** (formerly Swagger) specification.

Common mistakes

- Breaking changes without a version bump.
- Offset pagination on an infinite-scroll feed with millions of rows.
- Returning 500 for bad input (that is a 400) or hiding rate limiting behind 503.

TL;DR

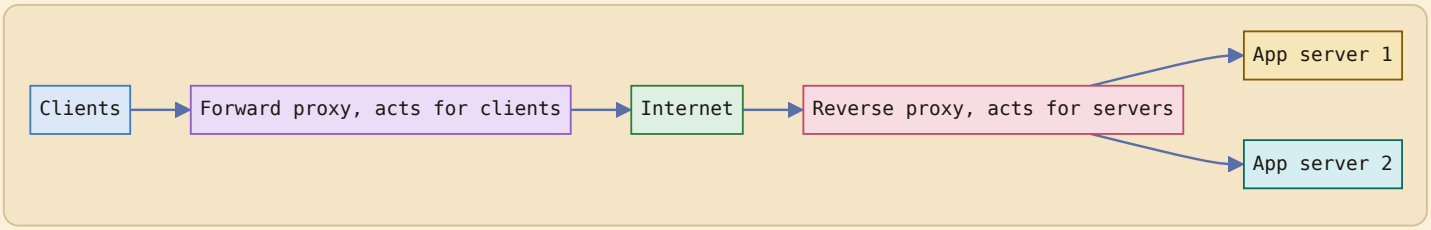
Version only for breaking changes, prefer cursor pagination for big or changing lists, and communicate limits with rate-limit headers and 429 + Retry-After.

Proxies and API gateways

What it is

A **proxy** is a middleman that receives a request and forwards it on someone else's behalf.

- A **forward proxy** acts for **clients**. The client is configured to send all traffic through it, and the internet sees the proxy instead of the client. Like an assistant who makes calls for you. Used by companies to filter and log employee browsing, by VPN-like services (virtual private networks that tunnel your traffic through another machine) to hide the user's IP, and for caching outbound downloads.
- A **reverse proxy** acts for **servers**. Clients think they are talking to the website; in reality the reverse proxy receives the request and forwards it to one of the hidden servers behind it. Like a hotel receptionist who passes your request to the right staff member. Used for load balancing, TLS termination, caching, compression and hiding internal structure.



An **API gateway** is a reverse proxy specialised for APIs. It is the single front door to many backend services (often microservices) and handles cross-cutting concerns once instead of in every service.

Why it matters

Reverse proxies and gateways are where you put shared concerns: authentication, rate limiting, TLS, logging, routing by path. They also let you change the backend (split a monolith, move a service) without clients noticing.

How it works

An API gateway receiving `GET /orders/7` typically:

1. Terminates TLS.
2. Authenticates the caller (validates an API key, a JWT (JSON Web Token, a signed token carrying the user's identity) or an OAuth access token issued by a login service).
3. Applies rate limits and quotas per client.
4. Routes by path, host or header: `/orders/*` → order service, `/users/*` → user service.
5. Optionally transforms the request (REST to gRPC, adds headers like a request ID), or **aggregates** several backend calls into one response.
6. Records logs, metrics and traces, then returns the response.

A variant is **BFF (Backend for Frontend)**: a separate gateway per client type (web, mobile) that shapes responses for that client.

Types

Component	Main job	Examples
Forward proxy	Act for clients, filter, hide identity	Squid, corporate proxies
Reverse proxy	Act for servers, TLS, caching, routing	NGINX, HAProxy, Envoy
Load balancer	Spread traffic over healthy servers	AWS ALB/NLB, HAProxy
API gateway	Auth, rate limits, routing for APIs	AWS API Gateway, Kong, Apigee

The lines blur: NGINX is a reverse proxy, load balancer and basic gateway at once. Think of a load balancer as a reverse proxy whose main job is distribution, and an API gateway as a reverse proxy whose main job is API policy.

Trade-offs

Every proxy adds a network hop (typically well under 1 ms inside a data centre, but more with heavy logic) and is a potential bottleneck and single point of failure, so it must itself be redundant and scaled. A gateway that grows too much business logic becomes a hard-to-change "god service".

Common mistakes

- Confusing forward and reverse proxies. Ask: whom does it represent — the client or the server?
- Putting business rules in the gateway. Keep it to cross-cutting policy.
- Losing the client's real IP: proxies must pass it on in `X-Forwarded-For` (or the PROXY protocol for L4), and backends must trust that header only from known proxies.

In the interview

Q: What is the difference between a reverse proxy, a load balancer and an API gateway?

A: All three sit in front of servers. A reverse proxy is the general idea (forward on behalf of servers). A load balancer focuses on spreading traffic across healthy instances. An API gateway focuses on API concerns — auth, rate limits, routing to many services, transformation. One product often plays all three roles.

TL;DR

A forward proxy represents clients, a reverse proxy represents servers, and an API gateway is a reverse proxy that enforces API policy (auth, rate limits, routing) at the front door.

Load balancers

What it is

A **load balancer (LB)** is a component that receives incoming traffic and spreads it across a group of servers (often called a **pool**, **backend set** or **target group**), sending each request or connection only to servers that are healthy. Clients see one address; behind it there may be 3 or 3,000 servers.

Analogy: the host at a busy restaurant. Guests do not pick a table themselves; the host looks at which waiters are free, skips the one who has gone home sick, and seats each party so no single waiter is swamped.

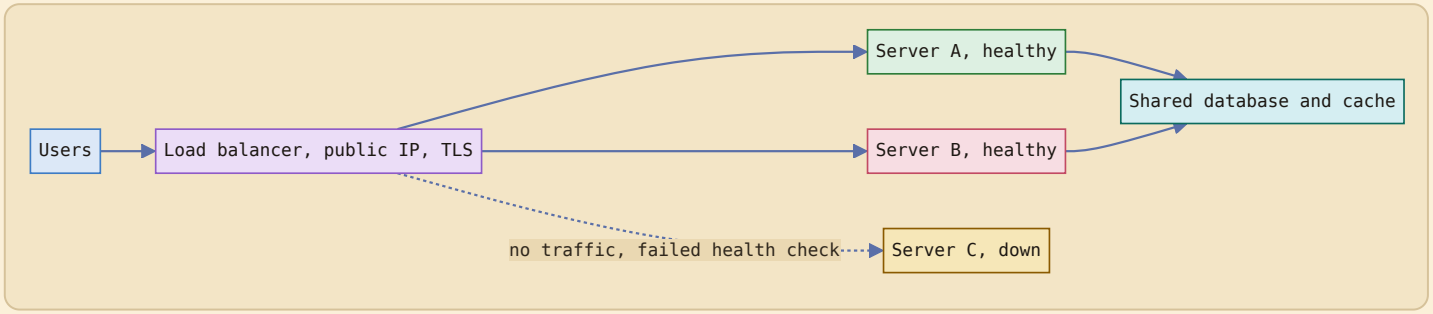
Why it matters

The load balancer is what makes **horizontal scaling** (Part 1) actually work:

- **Scalability** — add servers to the pool and capacity grows, with no change for clients.
- **Availability** — if a server dies, health checks notice and traffic flows to the rest. A single server is a single point of failure; a pool behind an LB is not.
- **Zero-downtime deployments** — take servers out one by one, update them, put them back.
- **Central place for shared work** — TLS termination, compression, request logging, basic security.
- **Hiding the backend** — servers keep private IPs; only the LB is public.

Almost every system design answer has a load balancer in front of each tier of stateless servers, and often internal ones between services too.

How it works



1. DNS resolves `api.example.com` to the load balancer's address (a **virtual IP**, VIP — an address that belongs to the service, not to any one machine).
2. The client connects to the LB. An L7 LB terminates TLS and reads the HTTP request; an L4 LB just sees the TCP or UDP connection.
3. The LB picks a backend using its **algorithm** (below), considering only servers that currently pass **health checks**.
4. It forwards the request (L7 opens or reuses its own connection to the backend) or the connection's packets (L4).
5. The response flows back through the LB to the client (in most designs; some L4 modes let servers reply directly — see "direct server return").
6. The LB records metrics: requests per second, errors, latency per backend.

Because the servers behind an LB may each receive any request, they should be **stateless** — sessions and data live in a shared cache or database, not in one server's memory.

Types

By network layer: L4 vs L7

Layer 4 (transport) load balancer. Works on TCP/UDP connections using IPs and ports. It picks a backend when a connection opens and then shuttles that connection's packets back and forth (either by rewriting addresses, NAT-style, or by proxying the TCP stream), never looking inside. It cannot see URLs, headers or cookies, and with TLS it sees only encrypted bytes (unless it terminates TLS itself, which some L4 products can do).

- Pros: very fast, low latency (microseconds of overhead), handles millions of connections, works for any protocol (databases, MQTT messaging for IoT devices, game servers, raw TCP), cheap.
- Cons: no content-based routing; balancing is per connection, so one long-lived connection carrying lots of requests (HTTP/2, gRPC) is stuck on one backend.

Layer 7 (application) load balancer. Terminates the client connection, parses the HTTP request, then makes a separate request to the chosen backend. Because it understands the content, it can:

- Route by path (`/api` → API servers, `/images` → image servers), host name (`shop.example.com` vs `blog.example.com`), headers, cookies or query string.
- Balance **per request**, so many HTTP/2 or gRPC requests on one client connection are spread across many backends.
- Do TLS termination, compression, caching, retries, redirects (HTTP → HTTPS), add headers like `X-Forwarded-For` and request IDs, enforce auth and rate limits, block malicious patterns (web application firewall, or WAF, features).
- Pros: smart routing and rich features. Cons: more CPU per request, more latency (typically still around a millisecond), must hold TLS keys, HTTP-family protocols only.

	L4 load balancer	L7 load balancer
Sees	IPs, ports, TCP/UDP	Full HTTP request
Decides per	Connection	Request
Routing by URL, header, cookie	No	Yes
Speed and cost	Faster, cheaper	More CPU per request
Protocols	Any TCP/UDP	HTTP, HTTP/2, gRPC, WebSocket
AWS example	NLB	ALB

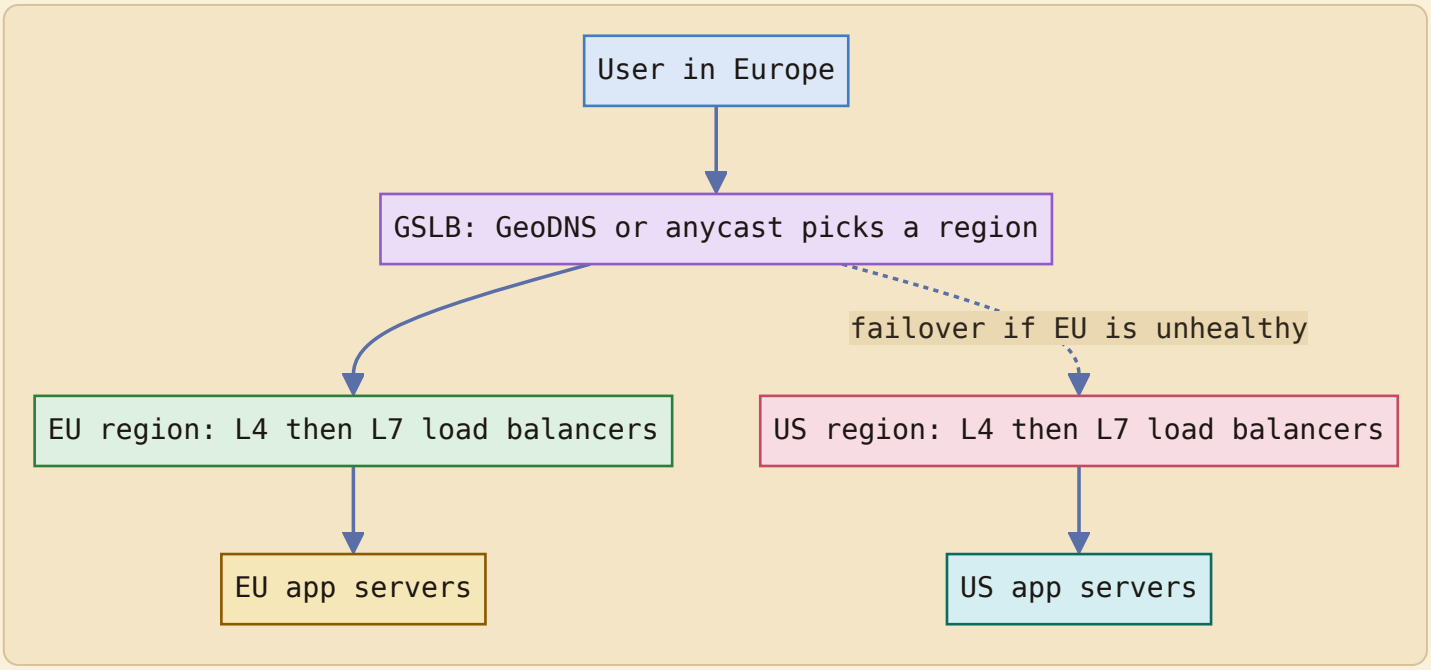
A common layered setup: an L4 balancer in front (cheap, huge scale, static IPs) feeding a fleet of L7 proxies (smart routing), which feed the app servers.

By form: hardware, software, cloud, DNS, global

- **Hardware load balancers** — dedicated appliances with specialised chips (for example F5 BIG-IP, Citrix NetScaler, now NetScaler ADC). Very high throughput and vendor support, but expensive (tens of

thousands of dollars each), scaled by buying more boxes, and slow to change. Common in traditional enterprises and banks.

- **Software load balancers** — programs on ordinary servers: **NGINX**, **HAProxy**, **Envoy**, Traefik. Cheap, flexible, configurable as code, scale by adding instances. Hyperscalers built their own software L4 balancers too: Google's **Maglev** and Meta's **Katran** run on commodity Linux servers.
- **Cloud (managed) load balancers** — a service you rent, where the provider runs and scales the fleet: AWS **ALB** (L7), **NLB** (L4), and **Gateway Load Balancer** (for routing through firewalls and appliances); Google Cloud Load Balancing; Azure Load Balancer (L4) and Application Gateway (L7). You pay per hour plus per traffic unit, and get automatic scaling and high availability across zones.
- **DNS load balancing** — the DNS server spreads users by returning different IPs (round robin, weighted, GeoDNS, latency-based), as described in the DNS section. Coarse and cached, but global and with no extra hop.
- **Global server load balancing (GSLB)** — distributing users across **multiple data centres or regions** rather than across servers in one place. It decides which region a user goes to based on geography, latency, region health and capacity, and fails over whole regions. It is built from DNS-based steering (Route 53, Cloudflare Load Balancing, NS1, F5 DNS) and/or **anycast** (one IP announced from many locations, explained in the CDN section), as in Google Cloud's global load balancer and Cloudflare. Inside each region, regular L4/L7 balancers take over.



Where it runs: client-side vs server-side

Most LBs are **server-side** (a separate box or service in the middle). In **client-side load balancing**, each client gets the list of healthy instances from a service registry and picks one itself (gRPC clients, Netflix's Ribbon library, service-mesh sidecars like Envoy next to each service). This removes a hop and a bottleneck but puts logic in every client. Service discovery is covered in Part 5.

Load-balancing algorithms

The **algorithm** decides which healthy backend gets the next request or connection. They fall into two families: **static** (ignore current load, decide by a fixed rule) and **dynamic** (look at live load signals).

Round robin. Send requests to servers in turn: A, B, C, A, B, C... Like dealing cards. Simple and fair when all servers are identical and requests cost about the same. Weakness: blind to load — if some requests are heavy (a 10-second report) a server can pile up work while still getting its "turn".

Weighted round robin. Each server gets a weight proportional to its capacity. With weights A=3, B=1, A receives 3 of every 4 requests. Used when servers differ in size (a 16-core next to an 8-core machine), and to send a small share of traffic to a new version (**canary release**: weight 1 for new, 99 for old).

Least connections. Send the new request or connection to the server with the fewest active connections right now. Good when requests vary a lot in duration or connections are long-lived (WebSockets, database

proxies), since busy servers naturally get fewer new ones. **Weighted least connections** divides the count by each server's weight.

Least response time (least latency). Prefer the server with the lowest recent response time, often combined with fewest active connections. Routes around a server that is slow because of a noisy neighbour, garbage collection or a bad disk. Needs the LB to measure latency, and can over-react (everyone rushes to the fastest server, which then becomes slow — **herding**), so implementations smooth the measurements. Envoy's and NGINX Plus's "least time" style policies and the **EWMA** (exponentially weighted moving average) balancer used by some proxies are examples.

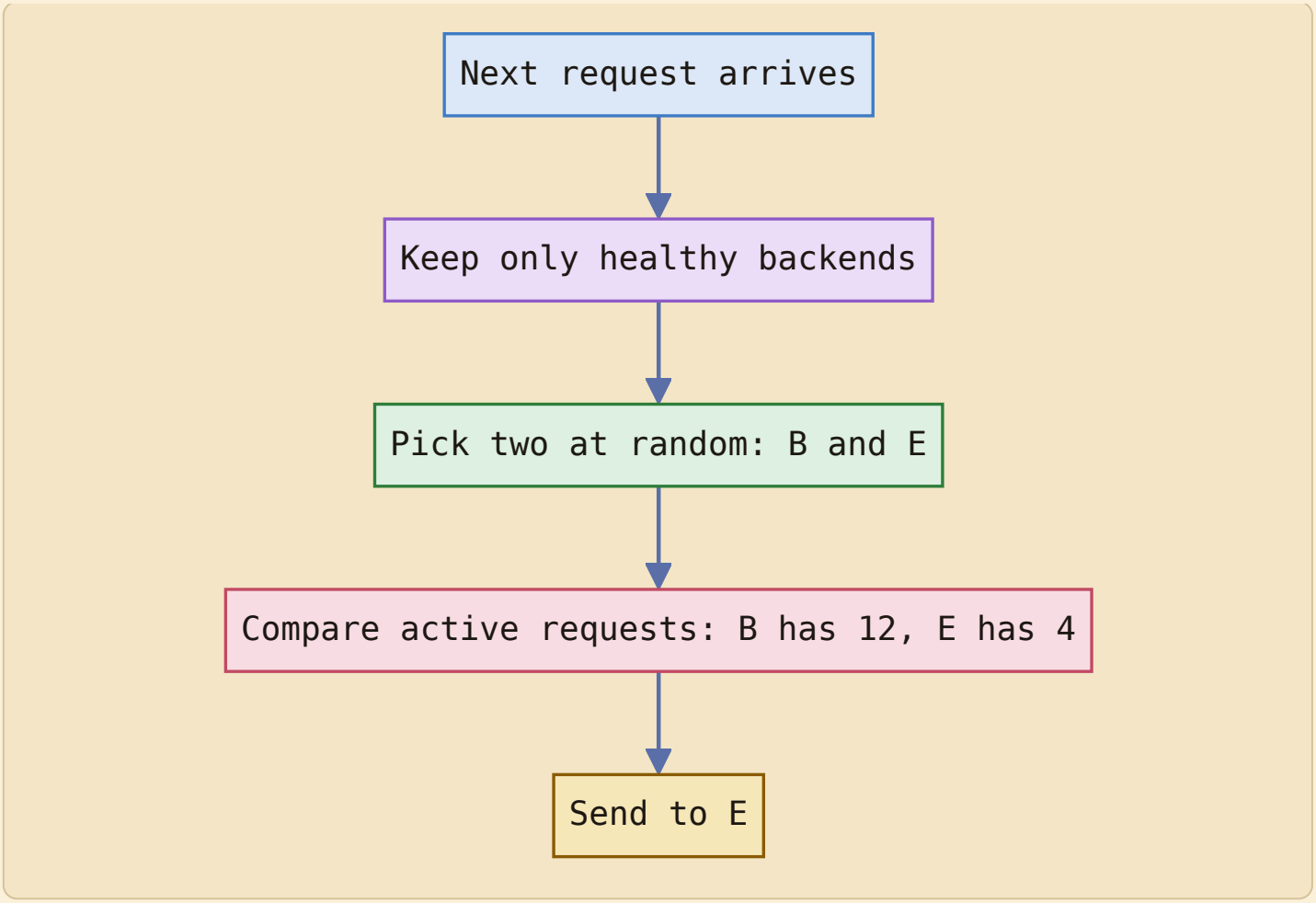
IP hash (source hash). Compute `hash(client IP) mod N` to pick a server, so the same client always lands on the same server — stickiness with no cookies. Problems: many users behind one NAT all hit one server, and when N changes (a server is added or removed) almost every client is remapped to a different server.

Consistent hashing. Fixes the remapping problem. Imagine a ring of hash values from 0 to 2^32. Each server is placed at several points on the ring (**virtual nodes**). A key — client IP, user ID, URL — is hashed onto the ring and goes to the next server clockwise. When a server is added or removed, only the keys in its slice of the ring move, about 1/N of them, instead of almost all. Used when the *same key should reach the same server* so that server's local cache stays useful: caching proxies, sharded caches, chat servers. Google's Maglev uses its own consistent-hashing variant, and HAProxy and Envoy offer **ring hash** and **Maglev** policies. (Consistent hashing is explained more fully with sharding in Part 3.)

Random. Pick a server at random. With many requests it evens out statistically, needs no shared state between multiple LB instances, and is surprisingly good. Its weakness is short-term imbalance.

Power of two choices (P2C). Pick **two** servers at random, then send the request to the less loaded of the two (fewest active requests or lowest latency). A famous result from probability (Mitzenmacher's analysis of "two choices") shows this tiny bit of information sharply reduces the worst-case load compared with pure random, while avoiding the herding problem of always choosing the global least-loaded server — because many independent LB instances with slightly stale data will not all pick the same "best" server. Envoy's least-request policy uses P2C by default (Envoy's overall default is round robin), and NGINX supports `random two least_conn`.

Others you may hear. Least bandwidth (fewest bytes per second), **resource-based / adaptive** (backends report CPU or load via an agent), **URL hash** (same URL to the same cache server), and **locality-aware** (prefer backends in the same zone to save cross-zone latency and cost).



Algorithm	Uses live load?	Sticky by key?	Best for
Round robin	No	No	Identical servers, similar requests
Weighted round robin	No	No	Mixed server sizes, canaries
Least connections	Yes	No	Long or uneven requests, WebSockets
Least response time	Yes	No	Avoiding slow servers
IP hash	No	Yes, by client IP	Simple stickiness, fixed pool
Consistent hashing	No	Yes, by any key	Cache affinity, changing pool
Random	No	No	Many LB instances, simplicity
Power of two choices	Yes, sampled	No	Large fleets, distributed LBs

Health checks

A load balancer must know which backends are alive. Two kinds:

- **Active health checks** — the LB itself probes each backend on a schedule: a TCP connect (L4), or an HTTP `GET /health` expecting a 200 (L7). Typical settings: every 10 s, timeout 5 s, mark **unhealthy after 2–3 consecutive failures**, mark **healthy again after 2–5 consecutive successes**. These thresholds stop one blip from flapping a server in and out.
- **Passive health checks (outlier detection)** — the LB watches real traffic. If a backend returns several 5xx errors in a row or times out, it is ejected for a while (say 30 s) and then retried. Envoy calls this outlier detection; NGINX uses `max_fails` and `fail_timeout`.

Good practice: combine both. Make the health endpoint meaningful but cheap. A **liveness** check says "the process is running"; a **readiness** check says "I can serve traffic now" (warm caches loaded, database reachable). Be careful not to make health checks depend deeply on shared dependencies: if the database blips and every server's health check fails, the LB removes all servers at once and turns a partial

problem into a total outage. Many LBs therefore **fail open** — if every backend looks unhealthy, they send traffic to all of them anyway (AWS target groups and Envoy's "panic threshold" both do this).

Sticky sessions (session affinity)

Sticky sessions make the LB send all requests from one user to the same backend. Implemented by:

- **Cookie-based** — the LB sets a cookie naming the backend (AWS ALB's `AWSALB` cookie) or uses the app's own session cookie.
- **IP-based** — hashing the client IP (L4 or L7).

Why use it: the server holds in-memory state (an old app storing sessions locally, a WebSocket connection, a shopping cart in memory) or a warm per-user cache. Why avoid it: load becomes uneven (a few heavy users stuck on one server), scaling in or failing a server loses those users' sessions, and deployments are harder. The better design is to make servers stateless and keep session data in a shared store like Redis, or inside a signed token on the client. Use stickiness only where state is unavoidable, such as WebSockets — and even then design for reconnection to a different server.

SSL/TLS termination

TLS termination means the load balancer decrypts HTTPS traffic and forwards plain HTTP to the backends inside the private network.

- **Benefits**: certificates live in one place and renew in one place; backends save the CPU cost of handshakes; and the LB can read requests for L7 routing, caching and WAF rules.
- **Risk**: traffic between LB and servers is unencrypted. Acceptable inside a trusted private network for many companies, not for regulated data or zero-trust designs.

Variants:

- **TLS re-encryption (bridging)** — decrypt at the LB, inspect, then open a new TLS connection to the backend. Secure end to end, costs more CPU.
- **TLS passthrough** — the LB (typically L4) does not decrypt at all; the backend terminates TLS. The LB cannot route by content, but keys stay only on the servers. It can still route by the SNI host name, which is visible in the ClientHello.

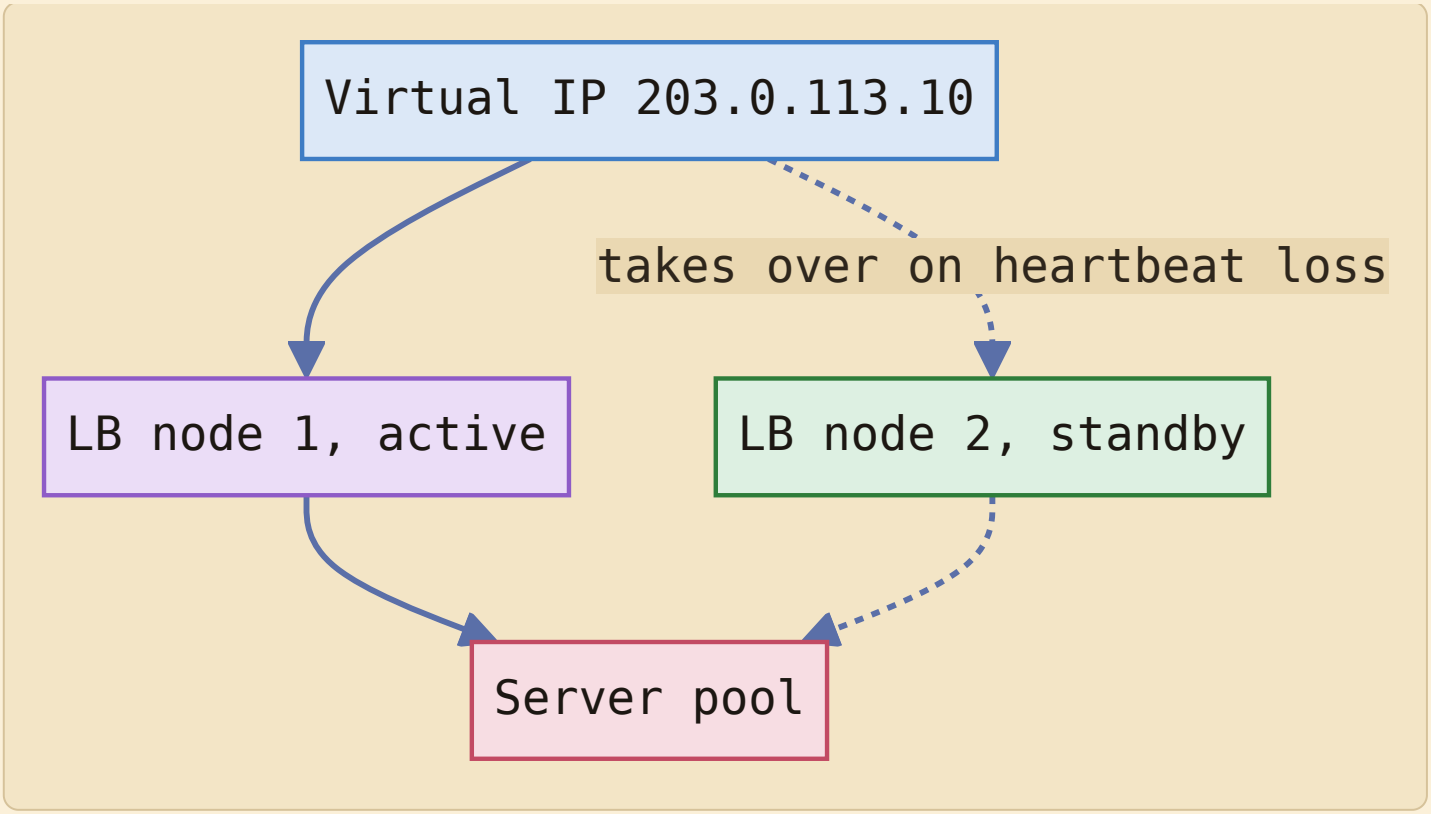
Connection draining (deregistration delay)

When a server is removed — for a deployment, scale-in or maintenance — cutting it off instantly would break in-flight requests. **Connection draining** tells the LB to stop sending **new** requests to that server but let **existing** requests and connections finish, up to a timeout. AWS calls this the **deregistration delay** (default 300 s, often lowered to 30–60 s for short API requests). For long-lived WebSockets, the server should also tell clients to reconnect elsewhere, because they may never finish on their own. Draining plus health checks is what makes **rolling deployments** invisible to users.

Making the load balancer itself highly available

A single load balancer is itself a single point of failure. Options:

- **Active–passive** — two LB nodes share one virtual IP. The active node serves all traffic; the passive one watches it via heartbeats and takes over the VIP if the active one dies (commonly using **VRRP**, the Virtual Router Redundancy Protocol, via the `keepalived` tool). Simple, but half the hardware sits idle and failover takes a second or more.
- **Active–active** — all LB nodes serve traffic at once. Traffic is spread across them by DNS (multiple IPs), by anycast, or by **ECMP** (equal-cost multi-path routing, where the network's routers spread flows across several LB machines that all announce the same IP). Better use of capacity and no idle standby; each node must be able to absorb the others' load if they fail. This is how Maglev, Katran and cloud LBs operate.
- **Managed cloud LBs** handle this for you: AWS ALB and NLB run nodes in every Availability Zone you enable, and NLB offers one static IP per zone.



Direct server return (bonus detail)

Responses are usually much larger than requests (a small GET returns a big page or video). In **direct server return (DSR)**, an L4 LB forwards incoming packets to the server, and the server sends the response **straight to the client**, bypassing the LB. The LB handles only the small inbound half, so one LB can front far more traffic. Used by large-scale L4 systems such as Maglev, Katran and some LVS (Linux Virtual Server) setups.

Trade-offs

Decision	Option A	Option B
Layer	L4: fast, any protocol, blind to content	L7: smart routing, per-request, more CPU
Form	Managed cloud LB: no operations, pay per use	Self-run NGINX/HAProxy/Envoy: full control, you operate it
TLS	Terminate at LB: simpler, inspectable	Passthrough or re-encrypt: end-to-end encryption
Algorithm	Static (round robin): predictable, simple	Dynamic (least conn, P2C): adapts to load
Sessions	Sticky: supports local state	Stateless: even load, easy failover

Common mistakes

- Drawing a single LB box and forgetting it needs redundancy.
- Using round robin for long-lived WebSocket connections; least connections is usually better.
- Sticky sessions as a substitute for a shared session store.
- Health checks that only test "process is up" — the server may be up but unable to reach its database — or, the opposite, deep checks that make the whole fleet fail at once.
- Using an L4 LB in front of gRPC and wondering why one backend is overloaded: gRPC multiplexes many requests on one long connection, so you need per-request (L7 or client-side) balancing.
- Forgetting that backends see the LB's IP, not the client's, unless `X-Forwarded-For` or the PROXY protocol is used.

Real-world examples

- **NGINX** — popular web server and L7 reverse proxy/LB; round robin by default, also least connections, IP hash, generic hash with consistent option, and random with two choices.
- **HAProxy** — very fast L4/L7 load balancer; round robin, static round robin, least connections, source hash, URI hash, consistent hashing; used at GitHub, Stack Overflow and many others.
- **Envoy** — modern L7 proxy created at Lyft, used as the data plane in service meshes like Istio. Policies include round robin, least request (P2C), ring hash, Maglev and random, plus outlier detection and automatic retries.
- **AWS Application Load Balancer (ALB)** — L7; routes by path, host, headers and query; supports HTTP/2, gRPC and WebSockets; round robin, least outstanding requests or weighted random; cookie stickiness.
- **AWS Network Load Balancer (NLB)** — L4 TCP/UDP/TLS; millions of requests per second; static IP per zone; preserves the client source IP.
- **Google Maglev** and **Meta Katran** — software L4 balancers on commodity servers, using consistent hashing and ECMP to scale out.

In the interview

Q: What is the difference between an L4 and an L7 load balancer, and when would you pick each?

A: L4 balances TCP/UDP connections by IP and port without reading content: fast, cheap, any protocol. L7 parses HTTP, so it can route by path or header, balance per request, terminate TLS and apply policies. I would put an L7 balancer in front of HTTP services and gRPC, and use L4 for non-HTTP protocols, extreme throughput, or as the first tier in front of L7 proxies.

Q: Which algorithm would you choose for a fleet of identical stateless API servers? For a cache tier?

A: For stateless API servers, least outstanding requests or power of two choices — they adapt to uneven request cost and slow servers. For a cache tier, consistent hashing on the cache key, so each key goes to the same node and adding a node moves only about 1/N of the keys.

Q: How do you deploy a new version without dropping requests?

A: Rolling deployment behind the LB: for each server, deregister it with connection draining (stop new requests, let in-flight ones finish), update it, wait for readiness checks to pass, then re-register it. For extra safety, use weighted routing to canary the new version on a small share of traffic first.

Q: Isn't the load balancer a single point of failure?

A: Only if there is one. Run LB nodes as active–passive with a floating virtual IP, or active–active behind DNS, anycast or ECMP, across multiple availability zones. Managed cloud LBs already do this.

TL;DR

A load balancer spreads traffic across healthy servers so you can scale out and survive failures. Know L4 vs L7, the main forms, the algorithms (from round robin to power of two choices), health checks, stickiness, TLS termination, draining, and how to make the LB itself redundant.

CDNs and anycast

What it is

A **CDN (Content Delivery Network)** is a worldwide network of caching servers that keep copies of your content close to users. Instead of every user in Sydney fetching an image from your server in Virginia (about 200 ms round trip), they fetch it from a CDN server in Sydney (a few milliseconds).

Analogy: rather than everyone driving to one central warehouse, a retailer stocks local shops with the most popular items.

Key terms:

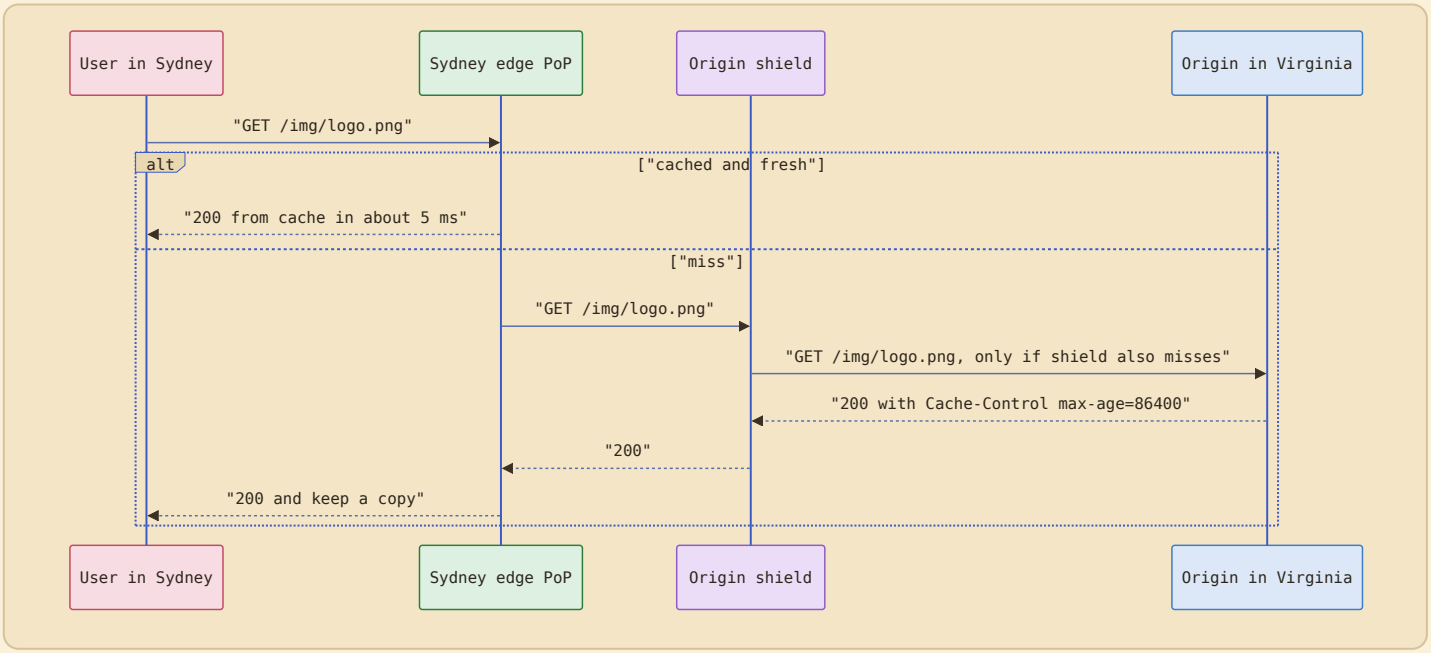
- **Origin** — your real server (or storage like Amazon S3) that holds the master copy.
- **Edge server** — a CDN cache server near users.
- **PoP (Point of Presence)** — a CDN location (a data centre site in a city) containing many edge servers. Large CDNs have hundreds of PoPs.

- **Cache hit** — the edge already has a fresh copy and answers directly. **Cache miss** — it must fetch from the origin (or an upper tier) first.
- **Cache hit ratio** — hits ÷ total requests. 95% means only 1 in 20 requests reaches the origin.

Why it matters

- **Lower latency** — static files (images, JavaScript, CSS, video) are served from nearby, and TLS handshakes finish against a nearby edge, which shortens every round trip.
- **Origin offload** — at 95% hit ratio the origin handles 20 times less traffic, so you need far fewer servers and less egress bandwidth.
- **Absorbing spikes and attacks** — the CDN's huge capacity soaks up traffic surges and DDoS (distributed denial-of-service) attacks, where many machines flood a site with junk traffic, before they reach you.
- **Video and large downloads** at global scale would be impossible from one origin.

How it works



1. The user's DNS lookup for `cdn.example.com` returns a nearby edge (via GeoDNS or anycast).
2. The edge checks its cache for the URL (the **cache key**, usually host + path + some query parameters and headers).
3. On a hit with a fresh copy, it answers immediately.
4. On a miss, it fetches from the origin, stores the copy for as long as the **TTL** allows, and returns it.
5. The origin controls caching through HTTP headers: `Cache-Control: public, max-age=86400` (cache for one day), `s-maxage` (TTL for shared caches like CDNs only), `no-store` (never cache), and validators (`ETag`, `Last-Modified`) so an expired copy can be re-checked cheaply with a 304 Not Modified.

Types

Pull CDN vs push CDN.

- **Pull (origin pull)** — the CDN fetches content from the origin on the first request, lazily. Zero upload effort, and only requested content is cached. Whoever asks for a file first at a given PoP waits for the trip to the origin; later visitors there get the cached copy. Most websites use pull.
- **Push** — you upload content to the CDN's storage ahead of time. Good for large files you know will be requested (game patches, video libraries) and for content that rarely changes; you manage what is stored and pay for storage.

	Pull CDN	Push CDN
How content arrives	Fetched on first miss	Uploaded in advance
Effort	Low	You manage uploads
First request	Slower (miss)	Fast
Best for	Websites, mixed traffic	Big known files, video

Invalidation (purging). When content changes before its TTL expires you can:

1. **Purge** — ask the CDN to delete a URL, a prefix or a tag from all edges (seconds to minutes depending on CDN).
2. **Versioned file names (cache busting)** — publish `app.3f9a1c.js` instead of overwriting `app.js`; the HTML references the new name, so there is nothing to invalidate. This is the standard for static assets and lets you use very long TTLs (a year) safely.
3. **Short TTLs** for content that changes often, accepting more origin traffic.

Origin shield (tiered caching). An extra cache layer between edges and the origin. Without it, a new file that is popular worldwide causes a miss at every one of hundreds of PoPs, each hitting the origin. With a shield, edges miss to the shield, so the origin sees roughly one request per object. It raises the overall hit ratio and protects the origin.

Static vs dynamic content. CDNs mostly cache static, shared content. For personalised or dynamic responses they still help by terminating TLS near the user and using optimised, kept-warm connections back to the origin, and some run code at the edge (Cloudflare Workers, Lambda@Edge, Fastly Compute).

Anycast. Normally one IP address belongs to one location (**unicast**). With **anycast**, the same IP address is announced from many locations at once using **BGP** (Border Gateway Protocol — the protocol internet networks use to tell each other which IP ranges they can reach). Routers send each user to the "nearest" location in network terms. Benefits: automatic proximity routing with no DNS tricks, natural spreading of DDoS traffic across all sites, and fast failover — if a site stops announcing, routes shift to the next site. Cloudflare's whole network, Google's public DNS `8.8.8.8`, Cloudflare's `1.1.1.1` and the DNS root servers all use anycast. "Nearest" means the best path according to BGP routing policy (often the fewest networks crossed), which is usually but not always the lowest-latency site.

Trade-offs

Long TTLs give a high hit ratio but stale content; short TTLs give freshness but more origin load. CDNs cost money per GB served, and caching anything personalised risks leaking one user's data to another if the cache key is wrong.

Common mistakes

- Caching responses that contain user-specific data under a shared key (use `Cache-Control: private` or vary the key).
- Overwriting files in place with long TTLs, then being unable to update them. Use versioned file names.
- Forgetting the origin must survive a full cache flush (a "cold" CDN) — the origin shield and gradual warm-up help.

Real-world examples

Akamai, Cloudflare, Fastly, Amazon CloudFront, Google Cloud CDN. Netflix runs its own CDN, **Open Connect**, placing caching appliances inside internet providers' networks so most video bytes never cross the wider internet. YouTube uses Google's global cache network in a similar way.

In the interview

Q: How does a CDN reduce latency and origin load?

A: It serves cached copies from PoPs near users, cutting round-trip time, and at a high hit ratio most requests never reach the origin. An origin shield collapses many edge misses into one origin request.

Q: You deploy a new version of your JavaScript and some users still get the old one. Why and how do you fix it?

A: Edges and browsers are still serving the cached file until its TTL expires. Use content-hashed file names so each version has a new URL, and keep long TTLs; purge only as an emergency tool.

TL;DR

A CDN caches content at edge PoPs near users to cut latency and offload the origin. Pull fetches on demand, push uploads ahead; manage freshness with TTLs, versioned names and purges; an origin shield protects the origin, and anycast routes users to the nearest site using one IP.

Bandwidth vs latency, and network partitions

What it is

Part 1 defined both terms: **bandwidth** is how much data a link can carry per second (1 Gbps is about 125 MB/s), and **latency** is how long one piece of data takes to get from A to B. A wider road does not make the trip shorter. This section looks at what that means for networked systems.

A **network partition** is when the network splits so that some machines cannot reach others, even though all of them are still running. Picture an office building where a corridor door jams: people on both sides keep working, but can no longer talk to each other.

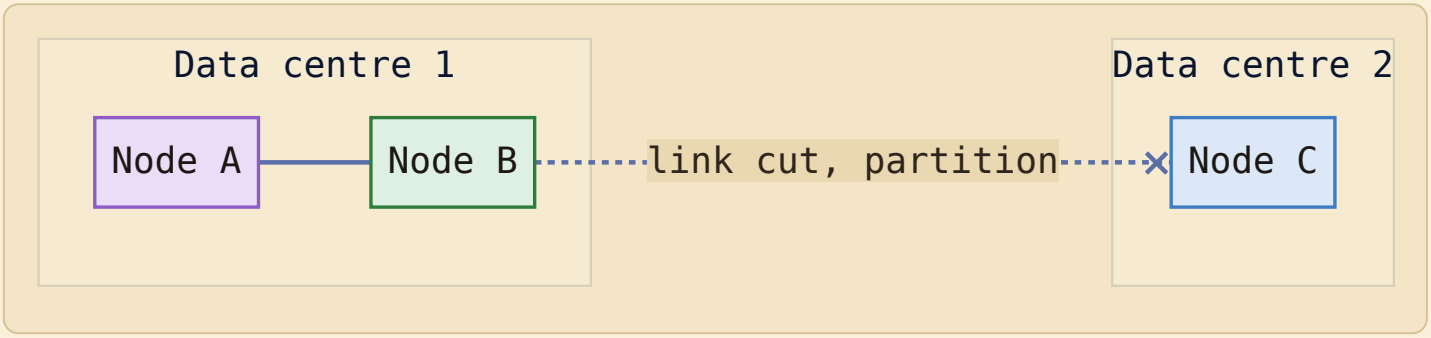
Why it matters

- Many system designs are latency-bound, not bandwidth-bound. A page that needs 10 sequential round trips across the Atlantic takes at least 700 ms no matter how fast the connection is. Fixes: fewer round trips (batching, HTTP/2, connection reuse), moving data closer (CDN, regional replicas).
- Large transfers are bandwidth-bound: copying 10 TB over a 1 Gbps link takes about 22 hours (10 × 8 × 10^12 bits ÷ 10^9 bits/s ≈ 80,000 s). That is why cloud providers offer physical data-transfer devices (AWS Snowball).
- Latency has a physical floor set by the speed of light in fibre (about 200 km per millisecond, see Part 1), so no protocol trick removes it.
- **Bandwidth-delay product** (bandwidth × RTT) is how much data must be "in flight" to fill a link. A 1 Gbps link with 100 ms RTT needs 12.5 MB in flight, which is why TCP window sizes and congestion control matter on long links.
- Partitions are not rare: a failed switch, a misconfigured firewall, a cut cable between data centres. Every distributed system must decide what to do during one, which leads to the CAP theorem (Part 4).

How it works

During a partition, a node cannot tell the difference between "the other node crashed" and "the other node is fine but unreachable" — both look like silence. Systems detect trouble with **timeouts** and **heartbeats** and then choose:

1. **Stay consistent** — refuse writes (or reads) on the side that cannot reach a majority, so data never diverges. The minority side becomes unavailable.
2. **Stay available** — keep accepting requests on both sides and reconcile the differences after the network heals, accepting temporarily inconsistent data.



A partition between nodes that both think they are the leader can cause **split brain** (two leaders accepting conflicting writes); quorum-based systems prevent it by requiring a majority (Part 3 and Part 4).

Common mistakes

- Assuming "the network is reliable" — the first of the classic **fallacies of distributed computing** (others: latency is zero, bandwidth is infinite, the network is secure, topology does not change).
- Treating a timeout as proof that the remote operation failed. It may have succeeded, and only the reply was lost — hence idempotent retries.
- Buying more bandwidth to fix a latency problem.

In the interview

Q: Our API is slow for users in Asia but fine in the US. More bandwidth?

A: Probably not — it is likely latency from distance multiplied by round trips. Reduce round trips (keep-alive, HTTP/2 or HTTP/3, batching) and move things closer: a CDN for static content and TLS termination at the edge, then regional deployments or read replicas.

Q: What is a network partition and why can't you just avoid it?

A: It is a break in communication between groups of healthy nodes. Hardware and configuration failures make it unavoidable at scale, so the design must choose between consistency and availability for the duration of the partition.

TL;DR

Bandwidth is capacity, latency is delay — round trips usually dominate. Partitions split healthy nodes from each other, cannot be avoided, and force a choice between consistency and availability.

Part 2 glossary

Term	Meaning in one line
Protocol	Agreed rules for how two programs exchange messages
Packet	A small chunk of data (about 1,500 bytes max on Ethernet) routed independently
Router	Device that forwards packets toward their destination IP
IP address	Number identifying a machine's network interface
IPv4 / IPv6	32-bit addresses (~4.3 billion) / 128-bit addresses (practically unlimited)
CIDR	Notation like `10.0.0.0/8` giving an address range by its fixed leading bits
Public IP	Globally unique address reachable from the internet
Private IP	Address valid only inside a local network (10/8, 172.16/12, 192.168/16)
NAT	Rewriting addresses so many private machines share one public IP
NAT gateway	Cloud service giving private servers outbound internet access
Carrier-grade NAT	NAT run by an ISP, so many customers share one public IP
VPC	Your own isolated private network inside a cloud provider
Port	Number (0–65535) selecting which program on a machine receives data
Ephemeral port	Temporary source port the OS assigns to an outgoing connection
Socket	One endpoint of a connection: IP address plus port
5-tuple	Protocol, source IP/port, destination IP/port — identifies a connection
Port exhaustion	Running out of source ports for new outgoing connections
DNS	System that translates domain names into IP addresses
Domain name	Human-readable hierarchical name such as `www.example.com`
TLD	Top-level domain such as `.com` or `.uk`
Subdomain	A name under a domain, such as `api.example.com`
Apex domain	The bare registered domain without a subdomain, such as `example.com`
Stub resolver	Minimal DNS client in the OS that asks a recursive resolver
Recursive resolver	DNS server that finds the full answer for clients and caches it
Root name server	Top of the DNS tree, refers resolvers to TLD servers
TLD name server	Refers resolvers to a domain's authoritative servers
Authoritative name server	Server holding the official records for a domain
Zone	Set of DNS records one authoritative server is responsible for
Recursive vs iterative query	"Give me the final answer" vs "point me to who knows"
A / AAAA record	Name to IPv4 / IPv6 address
CNAME record	Alias from one name to another name
MX record	Mail servers for a domain, with priorities
NS record	Authoritative name servers for a domain
TXT record	Free-text record used for verification and email policy

Term	Meaning in one line
SOA, SRV, PTR, CAA	Zone admin data, service location, reverse lookup, allowed certificate issuers
TTL (DNS)	Seconds a DNS answer may be cached
Round-robin DNS	Returning several IPs in rotating order to spread load
GeoDNS	Answering DNS queries differently by the user's location
Latency-based routing	DNS steering to the region with the lowest measured latency
EDNS Client Subnet	DNS extension passing part of the user's IP for better geo answers
OSI model	Seven-layer reference model for networking
TCP/IP model	Four-layer model the internet actually uses
L3 / L4 / L7	Network (IP), transport (TCP/UDP, ports), application (HTTP) layers
Encapsulation	Each layer wrapping data with its own header
MAC address	Hardware address of a network card used on the local link
TCP	Connection-based, reliable, ordered byte-stream transport
UDP	Connectionless, unreliable, unordered datagram transport
Datagram	Self-contained UDP message
Three-way handshake	SYN, SYN-ACK, ACK exchange that opens a TCP connection
RTT	Round-trip time for a message to go and its reply to return
Sequence number	Counter TCP uses to order bytes and detect loss
ACK	Acknowledgement that data was received
Retransmission	Resending data whose acknowledgement did not arrive
Flow control	Receiver limiting how fast the sender sends (receive window)
Congestion control	Sender slowing down to avoid overloading the network
Slow start	TCP ramp-up that grows the sending rate each round trip
CUBIC / BBR	Common TCP congestion-control algorithms
TIME_WAIT	Short wait state after closing a TCP connection
Head-of-line blocking	One delayed item holding up everything queued behind it
QUIC	Encrypted, multi-stream, reliable transport over UDP
0-RTT	Sending data in the very first packet when resuming a connection
Connection migration	QUIC keeping a connection alive across network changes
Connection ID	QUIC identifier for a connection independent of IPs and ports
HTTP	Stateless request–response protocol of the web
HTTP method	Verb such as GET, POST, PUT, PATCH, DELETE
Safe method	Method that does not change server state
Idempotent	Repeating the operation leaves the same result as doing it once
Idempotency key	Client-supplied ID letting a server detect and ignore retries
Status code	Three-digit result code (2xx success, 4xx client error, 5xx server error)
Header	Key–value metadata on a request or response

Term	Meaning in one line
Cache-Control	Header telling caches whether and how long to store a response
ETag	Version tag of a resource, used to answer 304 Not Modified
X-Forwarded-For	Header carrying the original client IP through proxies
Stateless	Each request carries all needed context, server keeps no memory between them
Cookie	Small value the browser stores and sends back on each request to a site
Session	Server-side record of a logged-in user, referenced by a cookie or token
Keep-alive	Reusing one TCP connection for many HTTP requests
HTTP/1.1	Text protocol, one request at a time per connection
HTTP/2	Binary protocol multiplexing many streams over one TCP connection
HTTP/3	HTTP over QUIC, removing TCP head-of-line blocking
Multiplexing	Interleaving many independent streams over one connection
HPACK / QPACK	Header compression for HTTP/2 / HTTP/3
Alt-Svc	Header advertising that a server also speaks HTTP/3
HTTPS	HTTP inside a TLS-encrypted connection
TLS / SSL	Protocol for encrypted, authenticated connections (SSL is the old name)
Encryption in transit	Protecting data while it moves over the network
Certificate	Signed file binding a domain name to a public key
Certificate Authority	Trusted organisation that signs certificates
Chain of trust	Certificate signed by intermediate, signed by a root the browser trusts
Public-key cryptography	Key pair where the public key verifies or encrypts for the private key
Symmetric encryption	One shared key both encrypts and decrypts, fast
Diffie–Hellman	Method for two sides to agree on a secret key without sending it
Forward secrecy	Stolen long-term keys cannot decrypt past sessions
Session resumption	Reusing earlier TLS secrets to skip handshake work
SNI	Host name sent in the TLS ClientHello so one IP can serve many sites
mTLS	Mutual TLS where both client and server present certificates
API	Defined set of operations a service offers to other programs
REST	API style of resources (URLs) manipulated with HTTP verbs
Resource	A noun the API exposes, addressed by a URL
RPC	Calling a function on another machine as if it were local
gRPC	Google's RPC framework over HTTP/2 using Protobuf
Protocol Buffers	Compact binary serialisation format with a schema
Streaming RPC	gRPC call where client, server or both send a stream of messages
GraphQL	Query language letting clients request exactly the fields they need
Schema (GraphQL)	Typed description of all data and operations an API offers
Resolver	Server function that fetches one GraphQL field
Mutation / subscription	GraphQL write / live-update operation

Term	Meaning in one line
Over-fetching / under-fetching	Getting more data than needed / needing several calls for one screen
N+1 query problem	One query for a list plus one extra query per item
Short polling	Client repeatedly asking the server for updates on a timer
Long polling	Server holding a request open until data is available
WebSocket	Persistent two-way message channel upgraded from HTTP
Server-Sent Events	One-way server-to-client event stream over HTTP
Webhook	HTTP callback a service sends to your URL when an event occurs
Heartbeat (ping/pong)	Periodic small message keeping a connection alive and detecting failure
API versioning	Running old and new API contracts side by side for breaking changes
Offset pagination	Paging with limit and offset, simple but slow and unstable deep in a list
Cursor pagination	Paging from an opaque marker of the last item seen, fast and stable
Rate-limit headers	Headers telling clients their quota, remaining calls and reset time
429 Too Many Requests	Status code for a client that exceeded its rate limit
Retry-After	Header telling the client how long to wait before retrying
OpenAPI	Standard format for describing HTTP APIs
Proxy	Middleman that forwards requests on someone's behalf
Forward proxy	Proxy acting for clients toward the internet
Reverse proxy	Proxy acting for servers, receiving client traffic in front of them
API gateway	Reverse proxy handling auth, rate limits and routing for APIs
BFF	Backend for Frontend, a gateway tailored to one client type
Load balancer	Component spreading traffic across healthy servers
Pool / target group	The set of backend servers behind a load balancer
Virtual IP (VIP)	Address owned by a service rather than one machine
L4 load balancer	Balances TCP/UDP connections without reading content
L7 load balancer	Parses HTTP and routes or balances per request
Hardware load balancer	Dedicated appliance such as F5 BIG-IP
Software load balancer	LB program on ordinary servers such as NGINX, HAProxy, Envoy
Cloud load balancer	Provider-managed LB service such as AWS ALB or NLB
GSLB	Global server load balancing across regions or data centres
Client-side load balancing	Each client picks a backend from a list of healthy instances
Round robin	Send requests to servers in turn
Weighted round robin	Round robin where bigger servers get proportionally more
Least connections	Send to the server with the fewest active connections
Least response time	Send to the server answering fastest recently
EWMA	Exponentially weighted moving average, a smoothed recent value
IP hash	Choose a server by hashing the client IP
Consistent hashing	Hash ring where adding or removing a node moves only about 1/N of keys

Term	Meaning in one line
Virtual nodes	Several ring positions per server for even spread
Random (LB)	Pick a backend at random
Power of two choices	Pick two random backends and use the less loaded one
Herding	Many balancers sending traffic to the same seemingly best server at once
Canary release	Sending a small share of traffic to a new version first
Health check	Test that decides whether a backend should receive traffic
Active health check	LB probes backends on a schedule
Passive health check	LB ejects backends based on errors in real traffic (outlier detection)
Liveness vs readiness	"Process is running" vs "ready to serve traffic"
Fail open	Sending traffic to all backends when all appear unhealthy
Sticky sessions	Routing one user's requests to the same backend
TLS termination	Decrypting HTTPS at the load balancer or proxy
TLS passthrough	LB forwards encrypted traffic, backend decrypts
TLS re-encryption	LB decrypts, inspects, then encrypts again to the backend
Connection draining	Stop new traffic to a server but let in-flight work finish
Deregistration delay	AWS name for the connection-draining timeout
Active–passive	One node serves, a standby takes over on failure
Active–active	All nodes serve traffic simultaneously
VRRP / keepalived	Protocol and tool that move a virtual IP to a standby node
ECMP	Routers spreading flows across several equal-cost paths or machines
Direct server return	Backend replies straight to the client, bypassing the LB
PROXY protocol	Header an L4 proxy prepends to pass the original client address
CDN	Global network of edge caches serving content near users
Origin	Your server or storage that holds the master copy of content
Edge server	CDN cache server close to users
PoP	CDN site in a city containing many edge servers
Cache hit / miss	Content served from cache / fetched from upstream
Cache hit ratio	Hits divided by total requests
Cache key	Values (host, path, some headers) identifying a cached object
Pull CDN	CDN fetching content from origin on first request
Push CDN	You upload content to the CDN in advance
Purge / invalidation	Removing content from CDN caches before its TTL ends
Cache busting	Changing a file's name per version so caches fetch the new one
Origin shield	Middle cache tier that collapses edge misses before the origin
Edge computing	Running code on CDN edge servers
Unicast	One IP address belongs to one location
Anycast	Same IP announced from many locations, routed to the nearest

Term	Meaning in one line
BGP	Protocol networks use to announce which IP ranges they can reach
Bandwidth	Data a link can carry per second
Bandwidth-delay product	Bandwidth times RTT, the data in flight needed to fill a link
Network partition	Network failure splitting healthy nodes into groups that cannot talk
Split brain	Two sides of a partition each acting as leader
Fallacies of distributed computing	False assumptions such as "the network is reliable" and "latency is zero"

Part 3 — Data storage

Almost every system you will design exists to remember things: who the users are, what they bought, which messages they sent, which photos they posted. The program code can be thrown away and redeployed in minutes. The data cannot. If you lose it, the business may be over. This part explains, from zero, where data lives, how it is organised, how it is kept safe when machines crash, and how it is spread over many machines once one is no longer enough.

Read it in order. Each section uses words defined in the sections before it. Every term is also collected in the glossary at the end of this part.

What a database is

What it is

A **database** is a program whose job is to store data, keep it safe, and let other programs find it again quickly. The program you talk to is properly called a **database management system (DBMS)**. People say "database" for both the program and the data it holds, and so will we.

> **Analogy.** A database is like a well-run library. Books (data) arrive and are shelved in an agreed place. A catalogue (an index) lets you find a book without walking every aisle. A librarian (the DBMS) makes sure two people cannot check out the same copy at the same moment, and that a book is not lost if the lights go out halfway through shelving it.

Why not just write data to a plain file? You can, for a tiny program. But as soon as you have many users at once, plain files give you none of these:

- **Fast lookup.** Finding one customer among 50 million by reading the whole file takes seconds. A database index finds it in well under a millisecond.
- **Safe concurrent access.** Two web servers writing to the same file at the same moment can corrupt it. A database coordinates them.
- **Crash safety.** If the power fails halfway through a write, a database guarantees that the write either fully happened or did not happen at all.
- **A query language.** Instead of writing a loop for every question, you ask "give me all orders over $100 from last week" and the database works out how.

Why it matters

In a typical web application, the app servers are **stateless**: they keep nothing important in their own memory between requests, so you can add or remove them freely. All the important **state** (stored facts that must survive) lives in the database. That makes the database the hardest part to scale, the slowest part to change, and the most expensive part to get wrong. In interviews, the data model and the choice of database are usually where most of the deep questions land.

How it works

At a high level every database does the same few things when your program sends it a request:

1. **Parse** the request (for example a SQL statement) and check it is valid.
2. **Plan** the cheapest way to answer it, such as "use the index on `email`" instead of "read every row". The part that does this is the **query planner** (or optimizer).
3. **Execute** the plan, reading pages of data from memory if they are already there (the **buffer cache**) or from disk if not.
4. For writes, **log** the change to a durable file first, then apply it (explained in the Durability section).
5. **Return** the result to the caller over the network.

Two words you will see everywhere: **read** means fetching data without changing it, and **write** means inserting, updating or deleting data. Most consumer apps read far more than they write; a ratio of 10 reads to 1 write, or even 100 to 1, is common for social feeds and product catalogues.

TL;DR

A database is a program that stores data safely, finds it quickly, and coordinates many users at once. It holds the state that the rest of the system can afford to forget.

Relational databases and SQL

What it is

A **relational database** stores data in **tables**. A table looks like a spreadsheet:

- Each **row** (also called a record or tuple) is one thing, such as one user.
- Each **column** (also called a field or attribute) is one property of that thing, such as `email`.
- The **schema** is the formal description of the tables: their names, their columns, and each column's **data type** (integer, text, date and so on). The database refuses data that does not fit the schema.

The word "relational" comes from mathematics (a table is a "relation"), not from "relationships", although tables do relate to each other through keys.

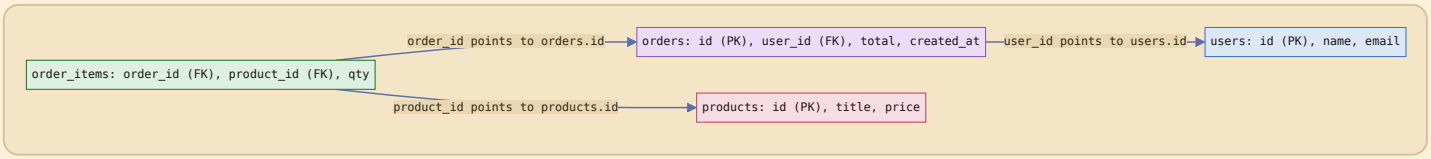
You talk to a relational database with **SQL (Structured Query Language)**, a language for describing *what* data you want rather than *how* to fetch it. Well-known relational databases include PostgreSQL, MySQL, MariaDB, Oracle Database, Microsoft SQL Server and SQLite. Cloud versions include Amazon RDS and Aurora, Google Cloud SQL and Azure SQL Database.

> **Analogy.** Think of a school office with filing cabinets. One cabinet holds a card per student. Another holds a card per class enrolment. Each enrolment card does not copy the student's address; it just writes the student's ID number, and you look the rest up in the student cabinet.

How it works

Keys tie the tables together:

- A **primary key** is a column (or set of columns) whose value is unique for every row and never empty. It is the row's identity. Example: `users.id`.
- A **foreign key** is a column in one table that holds the primary key of a row in another table. Example: `orders.user_id` points to `users.id`. The database can enforce that the referenced row exists (this is called **referential integrity**), so you cannot create an order for a user who does not exist.
- A **unique constraint** says that no two rows may share a value, for example no two users with the same `email`.



A **join** combines rows from two tables that match on a key. Here is a short query that lists each order with the buyer's name:

```sql
SELECT o.id, u.name, o.total
FROM orders o
JOIN users u ON u.id = o.user_id
WHERE o.created_at > '2026-01-01'
ORDER BY o.total DESC
LIMIT 10;
```

The main kinds of join are: **inner join** (only rows that match on both sides), **left join** (every row from the left table, with empty values where the right side has no match), **right join** (the mirror image) and **full outer join** (every row from both sides).

SQL statements fall into a few groups. `SELECT` reads. `INSERT`, `UPDATE` and `DELETE` change rows. `CREATE TABLE`, `ALTER TABLE` and `DROP TABLE` change the schema; this kind of statement is called **DDL (data definition language)**.

Why it matters

Relational databases are the default choice for most applications, and have been for about 50 years, because they give you:

- A **flexible query language**: you can ask questions you did not plan for when you designed the tables.
- **Strong guarantees** (transactions and ACID, next section) that protect money, inventory and anything else that must add up.
- **Enforced structure**: bad data is rejected at the door.
- **Mature tools**: backups, monitoring, replication and decades of operational knowledge.

Trade-offs

- **Schema changes take care.** Adding a column to a 2-billion-row table can lock it or take hours unless you use online migration tools (for example `gh-ost` or `pt-online-schema-change` for MySQL).
- **Scaling writes past one machine is hard.** A classic relational database is built around one main server. Spreading it across many machines (sharding, covered later) is possible but you lose cross-machine joins and transactions unless you use a distributed SQL database such as Google Spanner, CockroachDB, YugabyteDB or TiDB.
- **Joins get expensive** when the tables are huge and the join cannot use an index.

Common mistakes

- Choosing a NoSQL database "for scale" when the data is a few hundred gigabytes and fits comfortably in one PostgreSQL server. A single modern server with 64 cores, 512 GB of memory and fast SSDs can handle tens of thousands of simple transactions per second.
- Storing a list as a comma-separated string inside one column. It cannot be indexed or joined properly; use a separate table.
- Using a natural value such as an email address as a primary key. Emails change; IDs should not.

Real-world examples

Shopify, GitHub and Basecamp run large parts of their products on MySQL. Instagram began on PostgreSQL and sharded it as it grew. Stack Overflow is well known for running one of the web's busiest sites on a handful of SQL Server machines.

TL;DR

A relational database stores data in tables with a fixed schema, links them with primary and foreign keys, and lets you combine them with joins in SQL. It is the safe default for most systems.

Transactions and ACID

What it is

A **transaction** is a group of reads and writes that the database treats as one unit: either all of them take effect, or none do. You start one with `BEGIN`, and finish with `COMMIT` (make it permanent) or

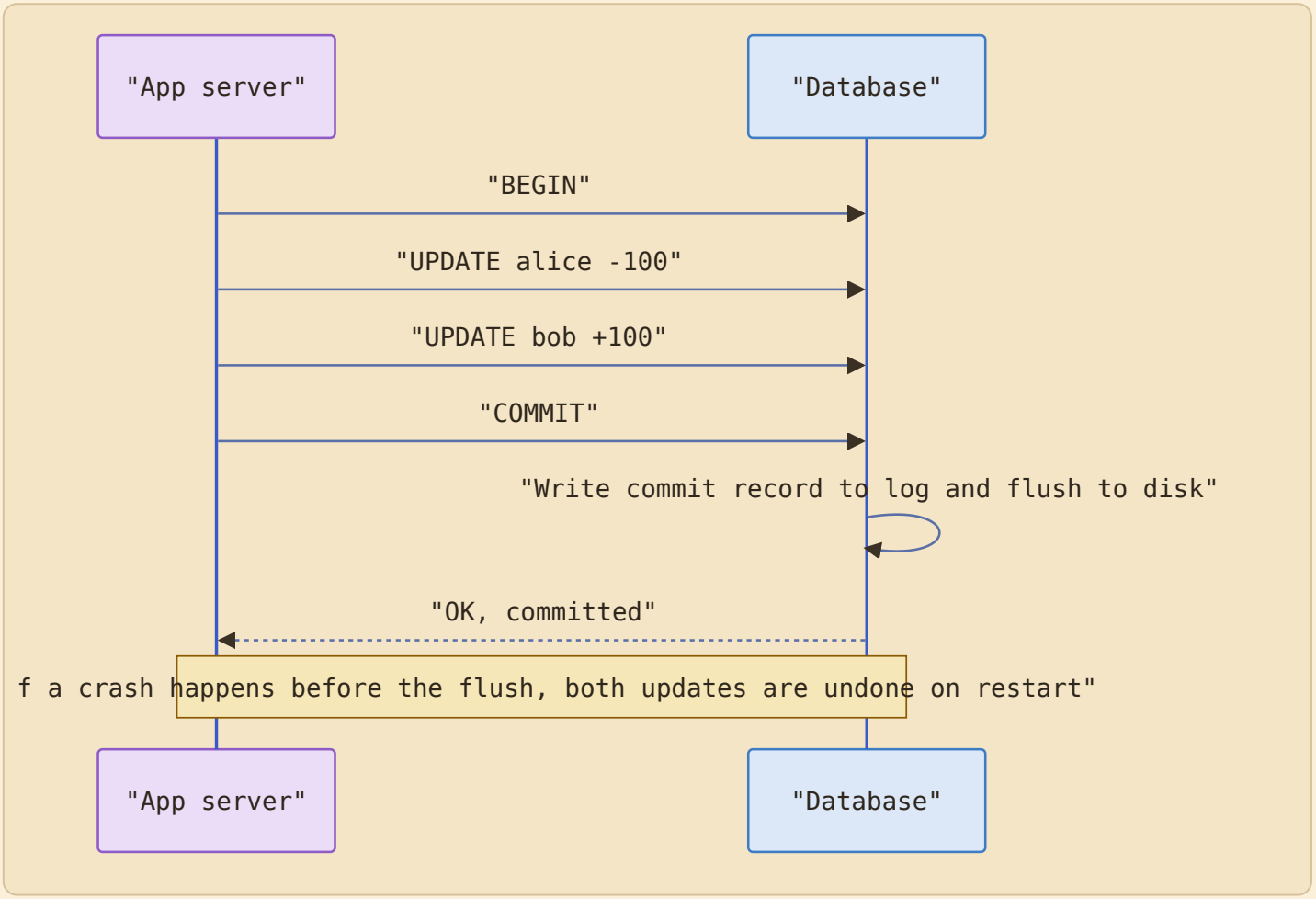
`ROLLBACK` (throw it all away).

> **Analogy.** Moving money between two bank accounts is two steps: take $100 from Alice, add $100 to Bob. If the system crashes between the steps, $100 vanishes. A transaction is like a sealed envelope: the bank only opens and acts on it when both instructions are inside, and if anything goes wrong the whole envelope is discarded.

```
BEGIN;
UPDATE accounts SET balance = balance - 100 WHERE id = 'alice';
UPDATE accounts SET balance = balance + 100 WHERE id = 'bob';
COMMIT;
```

ACID is the name for the four promises a transactional database makes:

- **A — Atomicity.** All or nothing. If any step fails, or the server crashes before `COMMIT`, every change in the transaction is undone. "Atomic" here means "cannot be split", not anything to do with speed or concurrency.
- **C — Consistency.** The transaction moves the data from one valid state to another valid state, where "valid" means it obeys the rules you declared: constraints, foreign keys, "balance must not go below zero" checks. Note that this C is mostly the application's job (you define the rules); the database only enforces the ones you tell it about. It is a different meaning from the "C" in the CAP theorem (Part 4).
- **I — Isolation.** Transactions running at the same time do not see each other's half-finished work. Ideally the result is the same as if they had run one after another. In practice databases offer several weaker **isolation levels** for speed (next section).
- **D — Durability.** Once the database says "committed", the data survives a crash or power cut. This is done by writing the change to a log on disk and forcing it to the physical device before replying (see the Durability section).



Why it matters

Without transactions, every piece of application code would have to handle "what if we crash halfway?" and "what if someone else changed this row while I was reading it?". Those bugs are rare, hard to reproduce and often cost real money: double-charged customers, oversold concert tickets, negative stock.

Trade-offs

Stronger guarantees cost speed. Isolation needs locks or version tracking, and durability needs a disk flush on every commit (often 0.1–2 ms on an SSD, more on a spinning disk or a network disk). Spreading a transaction across several machines (a **distributed transaction**, Part 4) is far slower again, because the machines must agree over the network.

Common mistakes

- Holding a transaction open while calling a slow external service (for example a payment API). The transaction keeps its locks the whole time and blocks other users. Do the external call outside the transaction.
- Assuming that a database labelled "ACID" gives you full isolation by default. Most do not (see next section).

In the interview

Q: What does ACID stand for and which letter is the database's responsibility versus the application's?

A: Atomicity, Consistency, Isolation, Durability. Atomicity, isolation and durability are mechanisms the database provides. Consistency is mostly a property of the application's rules; the database helps by enforcing the constraints you declare.

Q: Why not use transactions for everything across services?

A: A transaction that spans several databases needs a coordination protocol such as two-phase commit, which is slow and blocks if the coordinator fails. Across services people usually use sagas and idempotent operations instead (Part 4).

TL;DR

A transaction groups several operations so they all succeed or all fail. ACID means all-or-nothing, rules respected, no interference, and nothing lost after commit.

Isolation levels and anomalies

What it is

Concurrency means several transactions running at overlapping times. When they touch the same data, strange results called **anomalies** (or race conditions) can appear. An **isolation level** is a setting that chooses which anomalies the database prevents. Stronger levels prevent more but run slower or abort more transactions.

> **Analogy.** Imagine two people editing the same shared shopping list on paper. If one reads the list while the other is halfway through crossing items out, they see a messy in-between state. Isolation levels are house rules about when you are allowed to read or write the list while someone else is using it.

Types

The anomalies, from simplest to subtlest:

- **Dirty read.** Transaction A reads a value that transaction B has written but not yet committed. If B then rolls back, A has acted on data that never officially existed.
- **Dirty write.** A overwrites B's uncommitted write. Almost every database prevents this at every level.
- **Non-repeatable read (read skew).** A reads a row, B updates and commits that row, A reads it again and gets a different value within the same transaction.

- **Phantom read.** A runs a query such as "all bookings for room 7 on Friday" and gets 2 rows. B inserts a new matching booking and commits. A runs the same query again and now sees 3 rows. The new row is the "phantom".
- **Lost update.** A and B both read a counter as 10, both add 1, both write 11. One increment is lost; the answer should be 12.
- **Write skew.** A and B each read the same set of rows, each decide something is safe based on what they read, and each write to a *different* row. Classic example: a hospital rule says at least one doctor must be on call. Two doctors on call both click "go off call" at the same moment. Each transaction sees "two doctors on call, so it is fine for me to leave" and each updates only its own row. Result: zero doctors on call. No single row was written twice, so simple row locks do not catch it.

The standard isolation levels (defined in the SQL standard), weakest to strongest:

Level	Dirty read	Non-repeatable read	Phantom read
Read uncommitted	Possible	Possible	Possible
Read committed	Prevented	Possible	Possible
Repeatable read	Prevented	Prevented	Possible (per standard)
Serializable	Prevented	Prevented	Prevented

Snapshot isolation is a widely used level that is not in the original standard. Each transaction reads from a consistent **snapshot**: a frozen picture of the database as of the moment the transaction started. It prevents dirty reads, non-repeatable reads and most phantoms, but it still allows write skew. It is usually built with **MVCC (multi-version concurrency control)**: instead of overwriting a row, the database keeps several versions of it, each stamped with the transaction that created it, so readers can see the old version while a writer prepares a new one. The practical pay-off: a long report and a busy stream of updates can run side by side without either waiting on the other's locks.

Serializable means the end result is the same as if the transactions had run one at a time in some order. It prevents every anomaly above, including write skew. Databases achieve it by literally running one at a time (Redis, VoltDB), by **two-phase locking** (readers and writers take locks and hold them until commit), or by **serializable snapshot isolation (SSI)**, which runs optimistically and aborts a transaction if a dangerous pattern is detected (PostgreSQL does this).

How it works

Defaults in popular databases (worth knowing because they surprise people):

- **PostgreSQL**: default is *read committed*. Its *repeatable read* is actually snapshot isolation. *Serializable* uses SSI.
- **MySQL (InnoDB engine)**: default is *repeatable read*, implemented with MVCC snapshots plus some locking.
- **Oracle**: default is *read committed*; its level called "serializable" is really snapshot isolation.

Tools for preventing lost updates and write skew without going fully serializable:

1. **Atomic operations**: `UPDATE counters SET n = n + 1 WHERE id = 7` lets the database do the read-modify-write in one step.
2. **Explicit locks**: `SELECT ... FOR UPDATE` locks the rows you read so no one else can change them until you commit.
3. **Optimistic concurrency control**: store a `version` number in the row and write with `UPDATE ... SET ..., version = 6 WHERE id = 7 AND version = 5`. If zero rows were updated, someone else got there first, so retry. Also called **compare-and-set**.
4. **Constraints**: a unique index on `(room_id, date)` makes the database reject the second double-booking.

Trade-offs

Higher isolation means more aborted transactions (which the application must retry) or more waiting on locks, so lower throughput. Most web applications run at read committed or snapshot isolation and add explicit locks or constraints only where money or inventory is involved.

Common mistakes

- Doing "read the balance in the app, check it, then write the new balance" with no lock or version check. That is a lost-update bug waiting to happen.
- Believing "repeatable read" means the same thing in every database. It does not.

In the interview

Q: Two users try to buy the last ticket at the same time. How do you stop overselling?

A: Make the check and the decrement one atomic step, for example `UPDATE tickets SET left = left - 1 WHERE id = 9 AND left > 0` and check that one row was affected. Alternatives are `SELECT ... FOR UPDATE` or an optimistic version column with retry.

Q: What is write skew?

A: Two transactions read overlapping data, each makes a decision based on it, and each writes a different row, so together they break a rule that each alone respects. Snapshot isolation allows it; serializable isolation prevents it.

TL;DR

Isolation levels decide which concurrency bugs the database blocks. Know the anomalies (dirty read, non-repeatable read, phantom, lost update, write skew) and the fact that most databases default to something weaker than serializable.

NoSQL database families

What it is

NoSQL ("not only SQL") is an umbrella name for databases that do not use the relational table model as their main way of storing data. The term became popular around 2009, when companies such as Google (Bigtable), Amazon (Dynamo) and Facebook (Cassandra) published or open-sourced systems built to spread huge amounts of data across hundreds of cheap machines.

There is no single "NoSQL design". Each family is shaped around one access pattern and gives up something (often joins, flexible queries or strong transactions) to be very good at it.

> **Analogy.** A relational database is a general-purpose kitchen. NoSQL databases are specialist tools: a pizza oven, a deep fryer, an espresso machine. Each is far better than the general kitchen at one thing and poor at everything else.

Types

Key-value stores. The simplest model: a giant dictionary. You `put(key, value)` and `get(key)`. The database does not look inside the value. Extremely fast (often well under 1 ms) and easy to spread across machines, because each key can live anywhere. Weak at "find all items where X". Examples: Redis (in memory, also has lists, sets, sorted sets), Memcached (cache only), Amazon DynamoDB (key-value plus document features), etcd (small, strongly consistent configuration data), Riak. Typical use: sessions, caches, shopping carts, feature flags, rate-limit counters.

Document stores. Each record is a **document**, usually JSON or a binary form of it, which can hold nested fields and arrays. Documents in the same **collection** do not have to share a schema. You can query and index fields inside documents. Examples: MongoDB (maximum document size 16 MB), Couchbase, Amazon DocumentDB, Google Firestore. Good when each item is naturally a self-contained tree, such as a product with its variants or a blog post with its tags, and you usually read the whole thing at once.

```
{ "_id": "p42", "title": "Trail shoe", "price": 89.0,
  "sizes": [40, 41, 42], "seller": { "id": "s7", "name": "RunCo" } }
```

Wide-column stores (also called column-family stores). Data is grouped by a **partition key**; inside one partition, rows are sorted by a **clustering key**, and each row can have many columns, with different rows having different columns. You design one table per query. Built to absorb huge write volumes spread over many machines. Examples: Apache Cassandra, ScyllaDB, Apache HBase, Google Bigtable. Typical use: message

histories, activity logs, IoT readings, anything that is "write a lot, read by key and time range". Note: "wide-column" is not the same as "columnar" (column-oriented analytics storage, covered later).

Graph databases. Data is stored as **nodes** (things, such as people) and **edges** (relationships, such as "follows"), both of which can carry properties. Following edges is cheap no matter how big the graph is, so questions like "friends of friends who live in Paris" are natural. Examples: Neo4j (query language Cypher), Amazon Neptune, JanusGraph. Typical use: social graphs, fraud rings, recommendations, knowledge graphs.

Time-series databases. Optimised for data points stamped with a time, arriving in time order, such as "CPU 73% at 10:00:05 on host 12". They compress such data heavily, make "average per minute over the last 6 hours" fast, and expire old data automatically. Examples: InfluxDB, TimescaleDB (a PostgreSQL extension), Prometheus (monitoring), QuestDB. Typical use: metrics, sensor data, stock ticks.

Search engines. Built around an **inverted index**: for each word, a list of the documents that contain it, like the index at the back of a book. To find documents containing "red" and "shoe", the engine intersects two lists instead of scanning everything. They also handle **tokenising** (splitting text into words), stemming ("running" matches "run"), typo tolerance and **relevance ranking** (which results are best, for example with the BM25 scoring formula). Examples: Elasticsearch and OpenSearch (both built on the Apache Lucene library), Apache Solr. Usually a *secondary* store: the source of truth is another database and data is copied into the search engine.

Vector databases. Store **embeddings**: lists of numbers (vectors), often hundreds or thousands long, produced by a machine-learning model so that similar items (images, sentences) get nearby vectors. The key query is **nearest-neighbour search**: "find the 10 vectors closest to this one". Doing that exactly over millions of vectors is slow, so they use **approximate nearest neighbour (ANN)** indexes such as HNSW (a layered graph you walk towards the target). Examples: Pinecone, Milvus, Weaviate, Qdrant, and the pgvector extension for PostgreSQL. Typical use: semantic search, recommendations, and retrieval-augmented generation (feeding relevant documents to a large language model).



Trade-offs

NoSQL databases usually give up some of: joins, ad hoc queries, multi-record transactions and enforced schemas. In return they gain easy horizontal scaling, high write throughput, flexible data shapes, or a very fast specialised query. The lines have blurred: MongoDB added multi-document transactions in version 4.0, DynamoDB has transactions, and PostgreSQL stores and indexes JSON with its `jsonb` type.

Real-world examples

Discord stores trillions of chat messages in a wide-column store (it moved from Cassandra to ScyllaDB around 2022–2023). Amazon's shopping cart was the motivating example in the 2007 Dynamo paper. LinkedIn and Uber have published about their graph and time-series systems. Most large sites run Elasticsearch or OpenSearch next to their main database for search.

TL;DR

NoSQL is a family of specialist databases: key-value, document, wide-column, graph, time-series, search and vector. Each trades general querying for speed or scale on one access pattern.

SQL vs NoSQL and BASE

What it is

Choosing between a relational (SQL) database and a NoSQL one is one of the most common interview decisions. There is no universal winner; the question is "which access patterns and guarantees does this data need?".

BASE is a tongue-in-cheek opposite of ACID used to describe many early NoSQL systems:

- **Basically Available**: the system keeps answering requests even when some machines fail, even if some answers are a bit old.
- **Soft state**: the data on different copies may be changing even without new writes, because copies are still catching up with each other.
- **Eventually consistent**: if writes stop, all copies will eventually agree. Until then, two readers may see different values.

> **Analogy.** ACID is a bank teller who will not hand over cash until the ledger is updated and checked. BASE is a busy coffee shop that takes your order immediately and sorts out the odd mix-up afterwards.

Trade-offs

Need	Lean towards
Money, inventory, bookings, strong correctness	SQL (or distributed SQL)
Complex queries, reporting, joins across many entities	SQL
Data shape changes often, nested records read whole	Document store
Massive write volume, simple key lookups, many regions	Wide-column or key-value
Relationship-heavy queries many hops deep	Graph
Team already knows PostgreSQL and data fits on one machine	SQL, almost always

A good interview answer names the access pattern first: "Users are read by ID and email, orders need transactions with payments, so I will use PostgreSQL. The activity feed is append-heavy and read by user and time, so I will use Cassandra for it." Using more than one database type in one system is normal and is called **polyglot persistence**.

Common mistakes

- "NoSQL is faster." It is faster for the pattern it was designed for, and often slower or impossible for others.
- "NoSQL has no schema." The schema still exists; it just lives in the application code instead of the database (often called **schema-on-read** versus SQL's **schema-on-write**).
- Picking eventual consistency for data where users would notice, such as an account balance.

In the interview

Q: When would you choose NoSQL over SQL?

A: When the access pattern is simple (mostly by key), the volume of data or writes is beyond one machine, the schema varies a lot, or you need multi-region writes with high availability, and you can live without joins and multi-row transactions.

TL;DR

Pick from the access pattern and the correctness needs, not from fashion. SQL is the default; reach for NoSQL when scale, shape or availability needs justify the lost features.

Indexes

What it is

An **index** is an extra data structure the database keeps next to a table so it can find rows by a column's value without reading the whole table. Without an index, finding `email = 'a@b.com'` means a **full table scan**: reading every row. With 100 million rows, that could take many seconds. With an index, it takes a few page reads and well under a millisecond if the pages are cached.

> **Analogy.** The index at the back of a textbook. To find "photosynthesis" you look it up alphabetically and jump to page 212, instead of reading all 600 pages.

How it works

The index stores the indexed column's values in a structure that is fast to search, each paired with a pointer to where the full row lives (or with the row's primary key). The database keeps it up to date on every insert, update and delete.

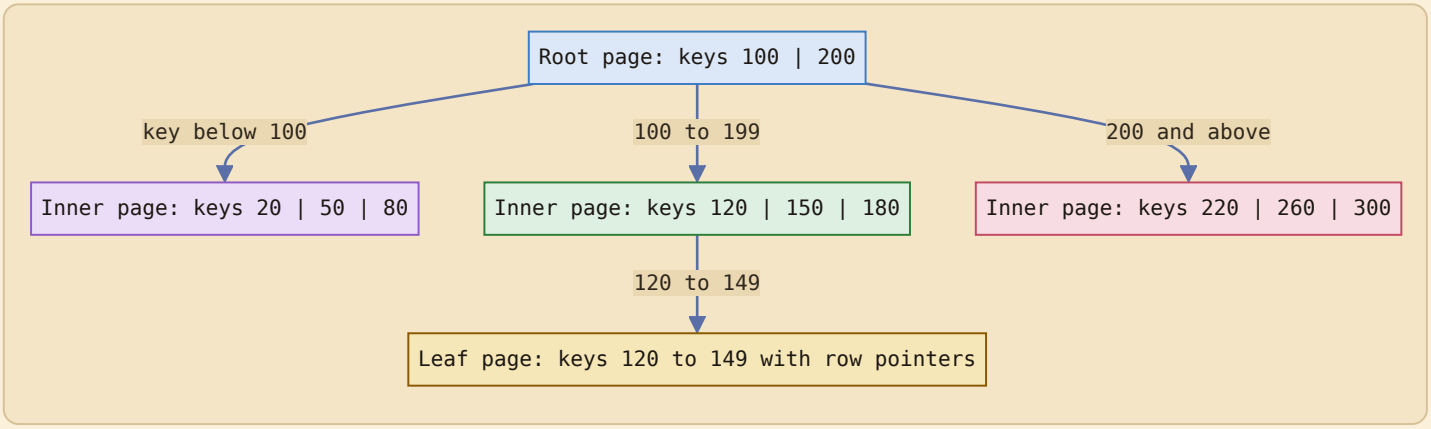
You create one like this:

```
CREATE INDEX idx_orders_user_created ON orders (user_id, created_at);
```

You can check whether a query uses an index with `EXPLAIN` (PostgreSQL, MySQL), which shows the plan the query planner chose.

Types

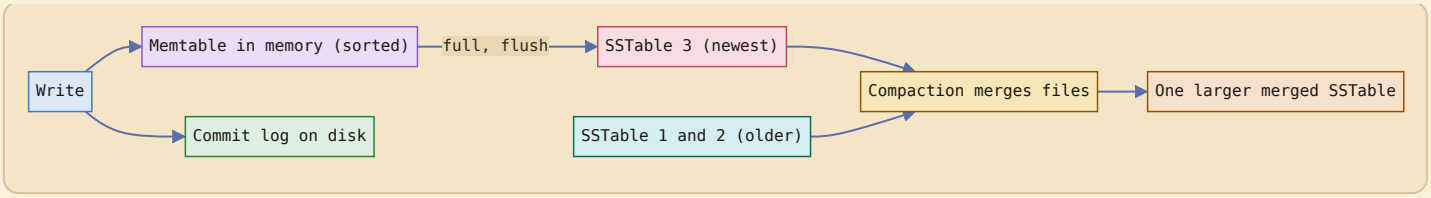
B-tree index. The default in almost every relational database. A **B-tree** is a balanced tree of fixed-size **pages** (blocks of storage, typically 4–16 KB: PostgreSQL uses 8 KB, MySQL InnoDB 16 KB). Each page holds hundreds of sorted keys and pointers to child pages. Because each page has so many children (the **branching factor**), the tree is very shallow: about 4 levels are enough for billions of rows. A lookup walks from the root to a **leaf** page, reading one page per level. Because keys are sorted, B-trees support both exact matches (`=`) and **range queries** (`BETWEEN`, `>`, `ORDER BY`). Updates are done in place, which means random writes on disk.



LSM-tree (log-structured merge tree). Used by write-heavy stores such as Cassandra, ScyllaDB, HBase, RocksDB and LevelDB. Writes never modify data in place:

1. A write is appended to a log on disk (for crash safety) and inserted into an in-memory sorted structure called the **memtable**.
2. When the memtable reaches a size limit (say 64 MB), it is written out as an immutable sorted file called an **SSTable (sorted string table)**. Writing a whole file in order is a fast **sequential write**.
3. Over time there are many SSTables. A background process called **compaction** merges them, keeping only the newest version of each key and dropping deleted ones (a deletion is first recorded as a marker called a **tombstone**).
4. A read checks the memtable, then the SSTables from newest to oldest. To avoid opening every file, each SSTable has a **Bloom filter** (a small structure that can say "this key is definitely not here", Part 4) and a sparse index.

LSM-trees give very high write throughput but reads can be slower, and compaction uses background disk and CPU. B-trees give predictable reads but slower writes. This is the classic **read amplification versus write amplification** trade-off: how many extra disk reads a lookup needs versus how many times each byte is rewritten.



Hash index. Keeps a hash map from each key to the location of its value. Exact lookups are O(1) (constant time), but it cannot do range queries or sorting, because hashing scatters neighbouring keys. Example: Bitcask, the storage engine used by Riak, keeps every key in an in-memory hash map pointing into append-only data files. PostgreSQL also offers a `HASH` index type.

Composite (multi-column) index. One index on several columns, such as `(user_id, created_at)`. It is sorted by the first column, then by the second within each first value, like a phone book sorted by last name then first name. It helps queries that filter on the **leftmost prefix**: `user_id`, or `user_id` plus `created_at`. It does *not* help a query that filters only on `created_at`. Put the column you filter by equality first and the range or sort column after it.

Covering index. An index that contains every column a query needs, so the database can answer from the index alone without visiting the table (an **index-only scan**). PostgreSQL and SQL Server let you add extra non-key columns with `INCLUDE`.

Clustered vs non-clustered (secondary) index. In a **clustered index** the table rows themselves are stored in index order; a table can have only one. MySQL InnoDB always clusters rows by the primary key. A **secondary index** is any other index; it points back to the row (in InnoDB, by storing the primary key).

Other kinds you may meet: **unique index** (also enforces uniqueness), **partial index** (only rows matching a condition, such as `WHERE deleted = false`), **full-text index** (an inverted index inside a relational database), **GIN/GiST** indexes in PostgreSQL for JSON, arrays and geometry, and **geospatial indexes** (R-trees, geohashes).

Trade-offs

Indexes are not free:

- **Every write gets slower.** An insert into a table with 6 indexes means 7 structures to update.
- **They use space**, sometimes as much as the table itself.
- **They use memory.** Indexes are fast only when their hot pages fit in the buffer cache.
- **Low-selectivity columns** (few distinct values, such as a boolean `is_active`) rarely benefit, because the database would read most of the table anyway.

The rule of thumb: index the columns you filter, join and sort on in your frequent queries, and nothing else. Read-heavy tables can afford more indexes than write-heavy ones.

Common mistakes

- Wrapping an indexed column in a function, such as `WHERE LOWER(email) = ...`, which stops the plain index from being used (create an index on the expression instead).
- A leading wildcard `LIKE '%shoe'` cannot use a B-tree index; use full-text search.
- Creating an index on a big production table in a way that locks writes (PostgreSQL offers `CREATE INDEX CONCURRENTLY`).

In the interview

Q: Why do LSM-tree databases handle writes better than B-tree databases?

A: LSM-trees buffer writes in memory and flush them as large sequential files, never updating in place, while B-trees do random in-place page updates. The cost is slower reads and background compaction.

Q: You have an index on `(country, city)`. Does `WHERE city = 'Lyon'` use it?

A: Generally no, because the index is sorted by country first. It helps queries on `country` or on `country` and `city` together.

TL;DR

An index is a sorted (or hashed) lookup structure that turns a full scan into a few page reads. B-trees suit reads and ranges, LSM-trees suit heavy writes, and every index makes writes slower.

Normalization and denormalization

What it is

Normalization means organising tables so that each fact is stored in exactly one place. **Denormalization** means deliberately storing the same fact in more than one place so that reads are faster.

> **Analogy.** A normalized office keeps one master address book; everyone looks addresses up there. A denormalized office writes the customer's address on every invoice. Reading an invoice is faster (no lookup), but when the customer moves, every invoice copy must be updated or it goes stale.

How it works

Normalization is described by **normal forms**. A beginner needs three:

1. **First normal form (1NF)**: each cell holds one value, not a list. No `tags = "red,blue,green"`.
2. **Second normal form (2NF)**: every non-key column depends on the *whole* primary key, not part of it. In an `order_items(order_id, product_id, qty, product_title)` table, `product_title` depends only on `product_id`, so it belongs in `products`.
3. **Third normal form (3NF)**: non-key columns depend only on the key, not on other non-key columns. If `users` has `zip_code` and `city`, and the city follows from the zip code, the city belongs in a zip-code table.

A classic mnemonic, usually credited to William Kent (1983), sums up all three: every column should describe "the key, the whole key, and nothing but the key".

Normalization prevents **update anomalies**: if the product title is stored in 10,000 order rows and you rename the product, you must update 10,000 rows, and if one fails they disagree.

Denormalization techniques include: copying a column (store `author_name` on each post), storing a precomputed total (`comment_count` on a post instead of counting each time), storing a nested document with its children (a document database's natural style), and **materialized views** (a stored, periodically refreshed query result).

Trade-offs

	Normalized	Denormalized
Storage	Smaller	Larger (duplicates)
Reads	May need joins	Fewer or no joins, faster
Writes	One place to update	Many places to update
Risk	Slow joins at scale	Copies drift out of sync

Start normalized. Denormalize specific hot read paths once you measure a problem, and decide how the copies will be kept in sync (same transaction, a background job, or events). At large scale and in NoSQL stores without joins, denormalization is the norm: in Cassandra you often keep one table per query, each holding a copy of the data arranged for that query.

Common mistakes

- Denormalizing before there is a performance problem, and then spending months fixing drift bugs.
- Forgetting that a counter like `like_count` updated by many users at once needs an atomic increment.

TL;DR

Normalize so each fact lives once; denormalize selected data to make hot reads fast, accepting extra write work and the risk of stale copies.

Replication

What it is

Replication means keeping copies of the same data on several machines, each called a **replica** (or node). The copies are kept in sync by sending every change from one machine to the others.

> **Analogy.** A company keeps its master price list at head office and sends updated copies to every branch. Customers at any branch can read prices locally. If head office burns down, a branch copy can become the new master.

Why it matters

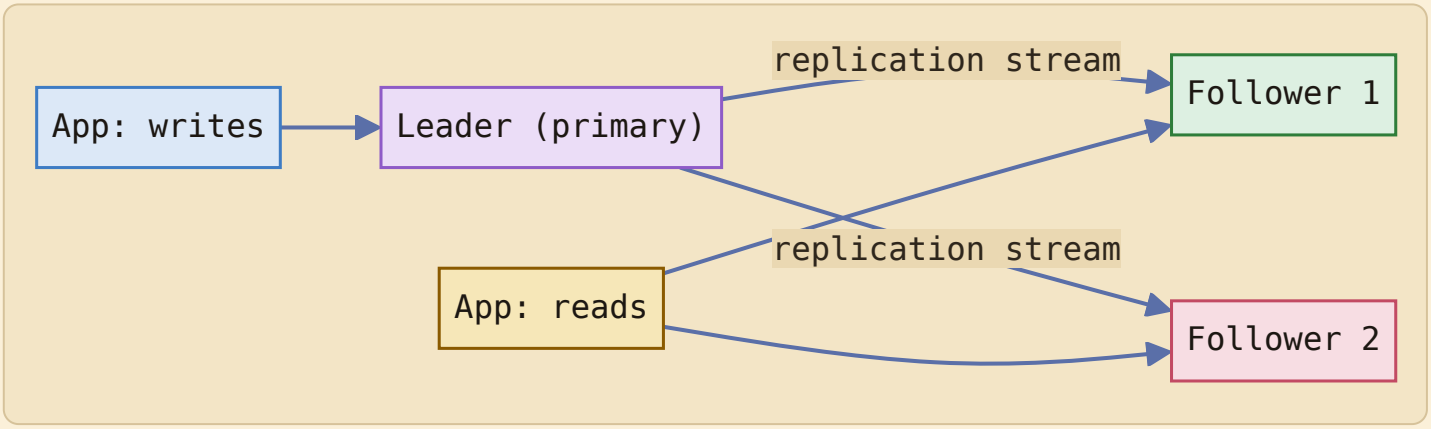
Replication gives you three things:

1. **Availability and fault tolerance.** If one machine dies, another copy can serve requests.
2. **Read scaling.** Spread read traffic over many copies. Ten replicas can serve roughly ten times the reads.
3. **Lower latency.** Put a copy near users: a replica in Frankfurt answers European readers in about 10–20 ms instead of the roughly 80–100 ms round trip to Virginia.

Remember what replication does *not* do: it does not help write throughput, because every copy must still apply every write. To scale writes you split the data (partitioning, next section). A useful rule: **copy for reads, split for writes**.

Types

Leader-follower (also called primary-replica, master-slave, or active-passive replication). One node, the **leader** (primary), accepts all writes. It records each change in its replication log and streams it to the **followers** (replicas, secondaries), which apply the changes in the same order. Clients may read from the leader or the followers. This is the default in PostgreSQL, MySQL, MongoDB replica sets, SQL Server and Redis.



Read replicas is the common name for followers used mainly to serve reads, for example Amazon RDS read replicas. They are ideal for read-heavy workloads such as product pages, reports and search indexing.

Multi-leader (multi-primary, active-active). Several nodes accept writes, usually one per data centre or region, and replicate to each other. Users in each region get fast local writes, and a region can keep working if it loses contact with the others. The hard part is **write conflicts**: two leaders change the same record at the same time. Ways to resolve them:

- **Last write wins (LWW)**: keep the write with the latest timestamp, silently dropping the other. Simple but loses data, and clocks on different machines disagree (Part 4).
- **Merge** the values with application logic, or ask the user.

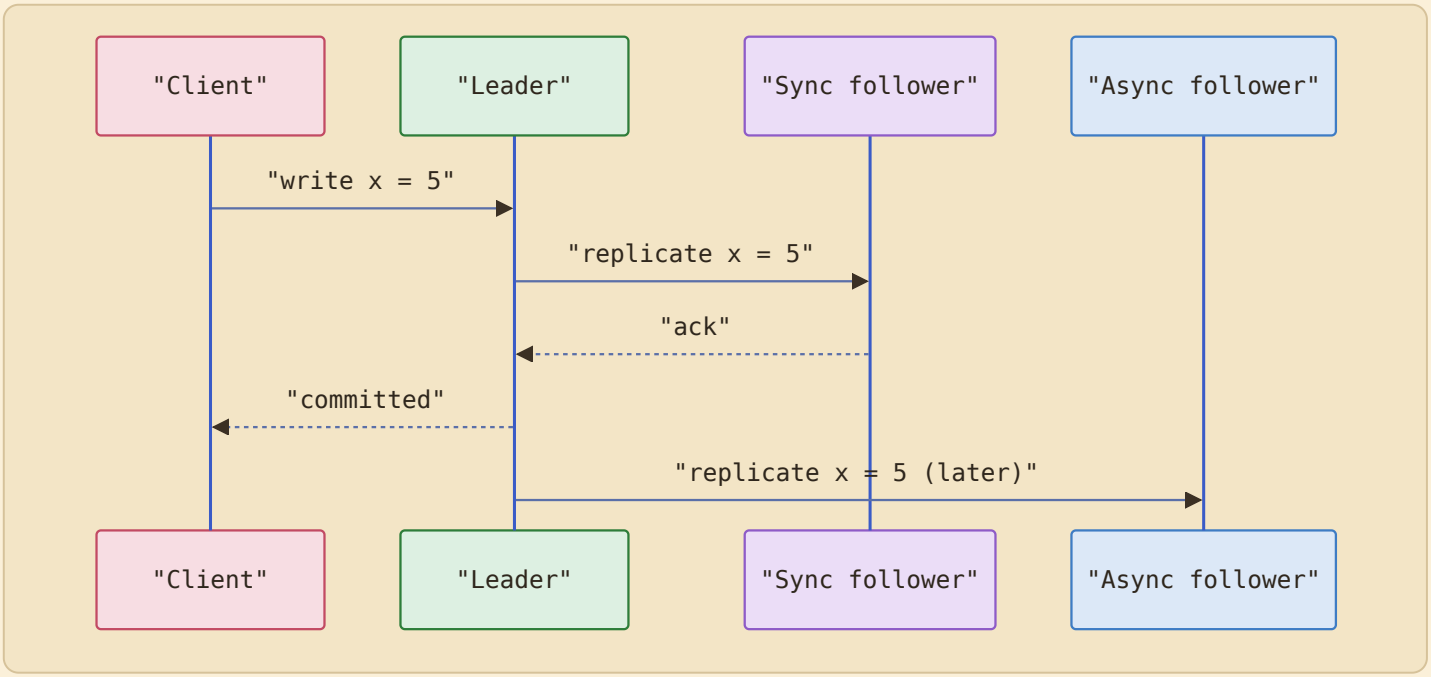
- **CRDTs (conflict-free replicated data types)**: special data types, such as counters and sets, designed so that concurrent changes always merge to the same result.

Examples: MySQL and PostgreSQL with multi-primary tools, CouchDB, collaborative editors and offline-first mobile apps (each device is effectively a leader).

Leaderless. Any replica accepts reads and writes. A client (or a coordinator node) sends each write to several replicas at once and considers it successful when enough of them confirm. Reads also query several replicas and take the newest value. With N copies, writing to W of them and reading from R of them, choosing **R + W > N** means every read overlaps at least one up-to-date copy (this is a **quorum**, covered in Part 4). A common setting is N=3, W=2, R=2. Stale copies are fixed by **read repair** (the reader writes back the newest value to replicas that returned an old one), **anti-entropy** (a background process that compares replicas, often using Merkle trees) and **hinted handoff** (a healthy node temporarily holds writes meant for a down node and passes them on later). Examples: Amazon's original Dynamo, Apache Cassandra, ScyllaDB, Riak.

Synchronous vs asynchronous replication. This is about *when* the leader tells the client "done":

- **Synchronous**: the leader waits until a follower confirms it has the change. If the leader then dies, no committed data is lost. Cost: every write waits for a network round trip, and if the follower is down or slow, writes stall.
- **Asynchronous**: the leader replies as soon as its own write is durable and ships the change to followers in the background. Fast and unaffected by slow followers, but if the leader dies, the last few changes that had not reached any follower are lost.
- **Semi-synchronous**: a common middle ground. Wait for *one* follower (or a quorum) to confirm, and replicate to the rest asynchronously. MySQL has a semi-synchronous mode; PostgreSQL lets you list synchronous standbys.



How it works

Replication lag is how far behind the leader a follower is, usually milliseconds but sometimes seconds or minutes when a follower is overloaded or a huge transaction is being applied. Lag causes visible bugs:

- **Read-your-own-writes violation**: a user edits their profile, the page reloads from a lagging replica, and their change seems to have vanished. Fix: read the user's own data from the leader for a short time after they write, or track the write's position in the log and only read from a replica that has caught up to it.
- **Monotonic reads violation**: two refreshes hit different replicas and the second shows *older* data than the first, as if time went backwards. Fix: pin each user to one replica.

Failover is the process of promoting a follower to leader when the leader fails:

1. **Detect** the failure, usually because the leader misses **heartbeats** (regular "I am alive" messages) for a timeout such as 10–30 seconds.
2. **Choose** a new leader, ideally the follower with the most up-to-date data. This may be done by an election among replicas or by an external coordinator.
3. **Reconfigure**: point clients and the other followers at the new leader (via DNS, a proxy, or a service-discovery system).

Failover can be **automatic** (tools such as Patroni for PostgreSQL, Orchestrator for MySQL, or managed cloud databases that fail over in under a minute or two) or **manual** (a human decides).

Split brain is the nightmare case: two nodes both believe they are the leader and both accept writes, so the data diverges. It happens when the old leader was not dead, just cut off by a network problem, and comes back still thinking it is in charge. Defences: require a **majority (quorum)** of nodes to agree on who the leader is, so an isolated minority cannot elect one; **fencing**, which forcibly stops or blocks the old leader (sometimes called STONITH, "shoot the other node in the head"); and **fencing tokens**, increasing numbers that let storage reject writes from an out-of-date leader (Part 4).

Trade-offs

Style	Writes go to	Strength	Weakness
Leader-follower	One leader	Simple, no write conflicts	Leader limits writes, failover needed
Multi-leader	Several leaders	Local writes per region, survives region loss	Conflicts must be resolved
Leaderless	Any W replicas	No failover, high availability	Weaker consistency, more complex reads

Common mistakes

- Treating a replica as a backup. If someone runs `DELETE FROM users` by mistake, the delete replicates to every copy within milliseconds. You also need real backups (see Durability).
- Sending all reads to replicas without thinking about read-your-own-writes.
- Setting the failover timeout too short, so a brief network hiccup triggers a needless and risky failover.

Real-world examples

GitHub's October 2018 incident is a famous lesson: a network interruption of about 43 seconds between its US East Coast hub and data centre triggered an automatic MySQL failover to its West Coast data centre; some writes had reached only the old primary, and untangling the two sets of data left the site degraded for just over 24 hours. Amazon Aurora keeps six copies of each piece of data across three **availability zones** (separate data centres within one cloud region, two copies in each) and treats a write as durable after four of the six confirm; reads and repairs need three of the six.

In the interview

Q: Synchronous or asynchronous replication for a payments database?

A: At least one synchronous (or quorum) copy so a committed payment survives the loss of the leader, with further asynchronous replicas for reads and other regions. Fully synchronous to all copies would make every write as slow as the slowest replica.

Q: A user updates their bio and immediately sees the old one. Why, and how do you fix it?

A: The read went to a lagging replica. Route that user's reads of their own data to the leader for a short window after a write, or read from a replica only once it has applied the write's log position.

TL;DR

Replication keeps copies for availability, read scaling and locality. Know leader-follower, multi-leader and leaderless; sync versus async; replication lag; failover; and split brain.

Partitioning and sharding

What it is

Partitioning means splitting one large dataset into smaller pieces, called **partitions**. **Sharding** is partitioning where the pieces, called **shards**, live on *different machines*. In everyday speech the two words are often used interchangeably, but strictly, you can partition a table inside one database server (PostgreSQL's declarative partitioning splits a table by month on the same machine), whereas sharding always spreads the data across servers.

> **Analogy.** A library that outgrows its building opens branches: A–F books at the north branch, G–M at the east branch, and so on. Each branch handles only its share of visitors. A catalogue tells you which branch to go to.

Why it matters

Replication copies everything onto every node, so it cannot help when:

- The data is too big for one machine's disk or memory (say 50 TB).
- The *writes* per second are more than one machine can handle (say 200,000 writes per second).

Sharding divides both data and write load. With 10 shards, each holds roughly a tenth of the data and takes a tenth of the writes. In practice each shard is also replicated, so you get both: split for writes, copy for reads and safety.

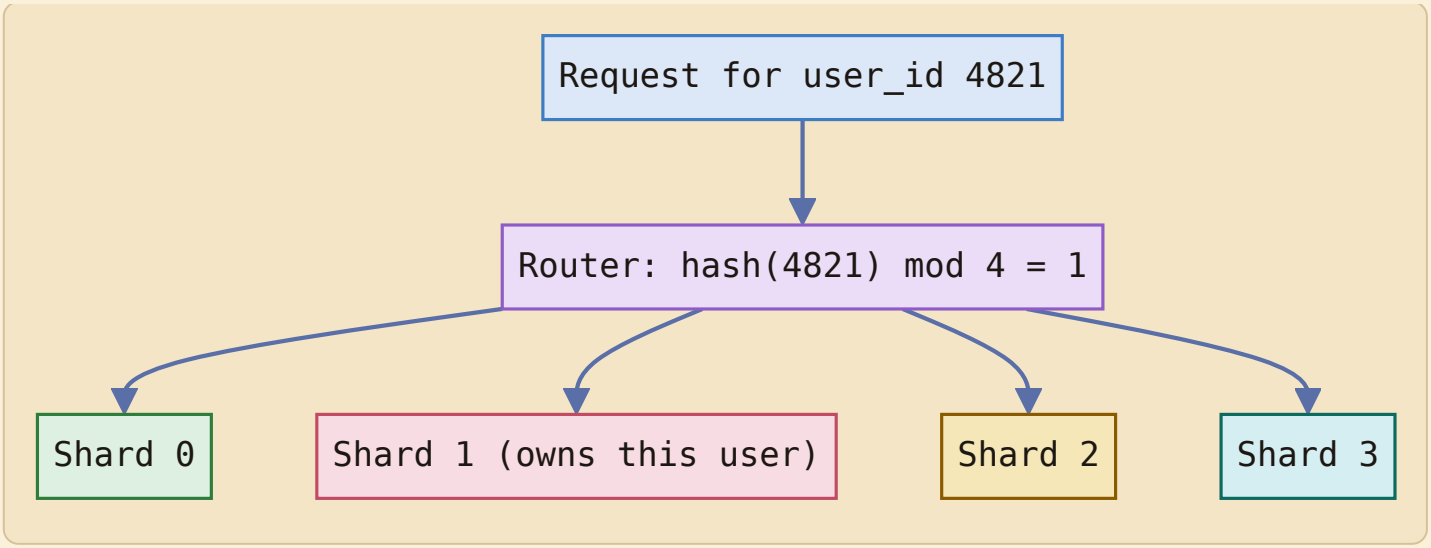
Types

Horizontal vs vertical partitioning.

- **Horizontal partitioning** splits *rows*: users 1–1,000,000 in one partition, 1,000,001–2,000,000 in another. Every partition has the same columns. Sharding is horizontal partitioning across machines.
- **Vertical partitioning** splits *columns*: frequently read small columns (`id`, `name`, `avatar_url`) in one table, rarely read large ones (`bio`, `settings_json`) in another, joined by the same ID. It keeps hot data compact. Splitting whole tables or features onto different databases is related and is called federation (below).

Strategies for deciding which shard a row belongs to. The column used to decide is the **shard key** (also called partition key).

- **Range-based.** Each shard owns a continuous range of key values: A–F, G–M; or January, February. Range queries ("all orders from March") hit one or a few shards. Danger: uneven load. If the key is a timestamp, *all* new writes go to the newest shard. Used by HBase, Bigtable, MongoDB (ranged sharding) and CockroachDB, which split ranges automatically as they grow.
- **Hash-based.** Compute a hash of the key (a function that turns any input into a number that looks random) and use it to pick the shard, for example `shard = hash(user_id) mod 8`. Keys spread evenly, so load is balanced. Cost: neighbouring keys land on different shards, so range queries must ask every shard. Also, with plain `mod N`, changing N from 8 to 9 moves almost every key, which is why consistent hashing exists (next section).
- **Directory-based (lookup table).** A separate service stores a map from key (or key range) to shard. Fully flexible: you can move a single big customer to its own shard. Cost: the directory is an extra lookup on every request and must itself be highly available and cached.
- **Geo-based (geographic).** Shard by the user's region: European users on EU shards, US users on US shards. Lowers latency and helps with data-residency laws such as the EU's GDPR. Load is uneven if regions differ in size.



Where the routing happens. The application can compute the shard itself; a proxy or **router** can do it (Vitess for MySQL, `mongos` for MongoDB, Citus for PostgreSQL); or any node can accept the request and forward it (Cassandra).

Federation (functional partitioning). Instead of splitting one table, split the database *by feature*: a users database, a products database, an orders database, each on its own server. Each is smaller and gets its own share of traffic. You lose joins and transactions between them, and it only goes as far as the number of features. It is often a first step before true sharding, and it maps naturally onto microservices, each owning its database.

How it works

Choosing a shard key is the most important decision in sharding, because changing it later is very expensive. A good shard key:

1. **Has high cardinality** (many distinct values), so data can be split finely. `country` is poor (about 200 values, very uneven); `user_id` is good.
2. **Spreads load evenly**, both data size and traffic.
3. **Matches the main query**, so most requests touch a single shard. If you almost always look up "orders for a user", shard orders by `user_id`, so all of one user's orders sit together.
4. **Does not grow in one direction** if you use range sharding (avoid auto-increment IDs and timestamps as the range key).

Queries that include the shard key are **single-shard** and fast. Queries that do not must be sent to every shard and the results combined (**scatter-gather**), which is slower and gets worse as shards are added.

Hot spots and hot keys. A **hot spot** is a shard that gets far more load than the others. A **hot key** is a single key that causes it, such as a celebrity's account with 300 million followers, or one viral product during a flash sale. Even perfect hashing cannot split a single key. Fixes:

- **Key salting**: append a small random number to the hot key (`celebrity_42#0` to `celebrity_42#9`) to spread its writes over 10 shards, then read all 10 and combine.
- **Caching** the hot item's reads in front of the database.
- **Dedicated shards** for known whales (big customers), using a directory.
- **Write buffering**: collect increments in memory and apply them in batches.

Rebalancing (resharding). Over time you must add shards because data grows, or move data because some shards are hotter. Approaches:

- **Fixed number of logical partitions**: create many more partitions than machines from day one (say 1,024 partitions on 8 machines, 128 each). To add a machine, move whole partitions to it; keys never change partition. Used by Elasticsearch, Couchbase and Riak, and in spirit by many hand-rolled systems.
- **Dynamic splitting**: when a range partition grows past a size limit (by default 512 MB per range in CockroachDB, 128 MB per chunk in recent MongoDB versions, 10 GB per region in HBase), split it in two and move one half (HBase, MongoDB, CockroachDB).
- **Consistent hashing** (next section).

Rebalancing must move data while the system keeps serving traffic, usually by copying in the background, catching up on new writes, then switching the routing. Doing this automatically is convenient but risky; many teams keep a human in the loop.

Trade-offs

What you give up when you shard:

- **Cross-shard joins** become application code or scatter-gather.
- **Cross-shard transactions** need distributed transactions or sagas (Part 4).
- **Globally unique IDs** cannot come from one auto-increment counter; you need UUIDs or a scheme like Snowflake IDs (Part 4).
- **Secondary indexes** become harder: either each shard indexes its own rows (**local index**, queries must hit all shards) or a separate **global index** is itself sharded by the indexed value (reads are fast, writes must update another shard).
- **Operations** (backups, schema changes, monitoring) multiply by the number of shards.

So shard late. First optimize queries, add indexes, add caching, add read replicas, and scale up the machine. Shard when one primary truly cannot handle the writes or the data size.

Common mistakes

- Sharding by a timestamp with range partitioning, so today's shard takes every write.
- Choosing a shard key that does not appear in the main queries, making most reads scatter-gather.
- Using `hash mod N`, then discovering that adding one server moves nearly all data.
- Forgetting that a single hot key cannot be fixed by adding shards.

Real-world examples

Instagram sharded PostgreSQL into thousands of logical shards mapped onto fewer physical servers, so shards could be moved without changing keys. Notion split its single PostgreSQL database into 480 logical shards on 32 physical databases in 2021, then in 2023 spread the same 480 shards over 96 machines without changing any keys. YouTube created Vitess to shard MySQL; it is now used by Slack, GitHub and others. Pinterest shards MySQL by putting a shard ID inside every object ID.

In the interview

Q: How would you shard a chat application's messages table?

A: Shard by `conversation_id` (hashed), because messages are almost always read per conversation, so each read hits one shard; inside the shard, order by message time. Watch for hot conversations, such as huge group channels.

Q: What makes a good shard key?

A: High cardinality, even spread of data and traffic, present in most queries so they hit one shard, and not monotonically increasing when range-sharded.

TL;DR

Sharding splits rows across machines so data and writes scale. The shard key decides everything: pick one with many values, even load and alignment with your main queries, and plan for hot keys and rebalancing.

Consistent hashing and virtual nodes

What it is

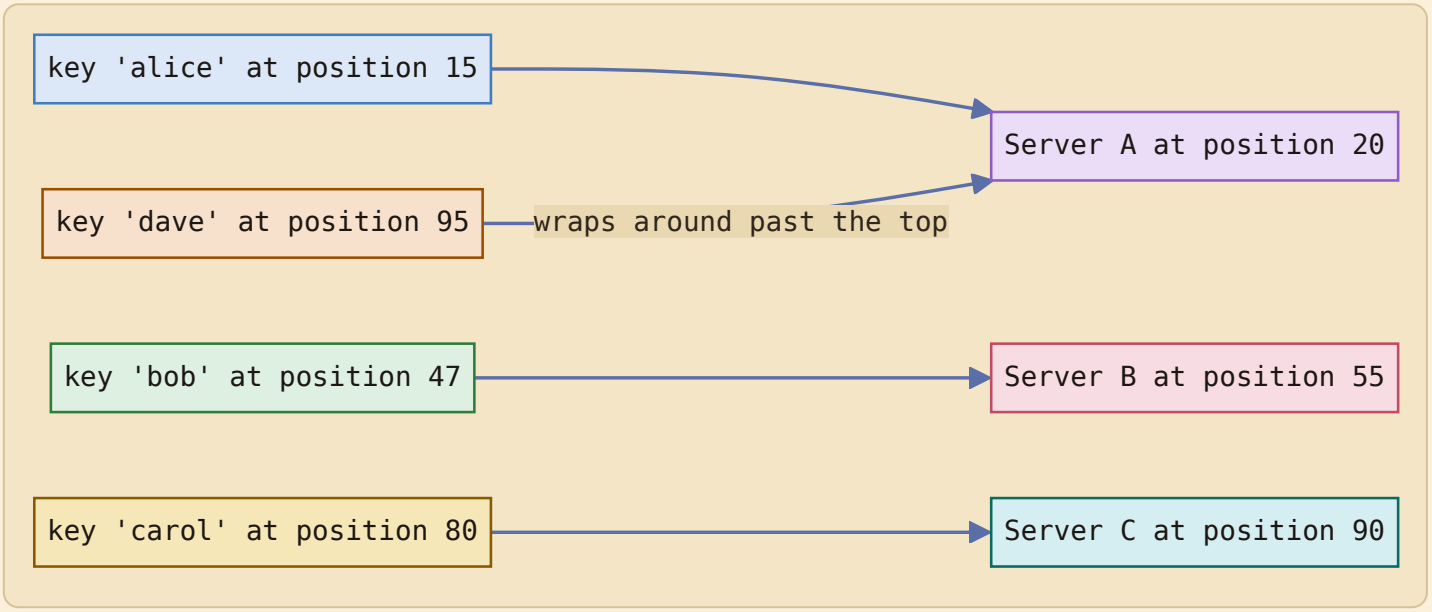
Consistent hashing is a way of assigning keys to servers so that adding or removing a server moves only a small share of the keys, about 1/N of them with N servers, instead of almost all of them.

Why plain modulo fails. With `server = hash(key) mod 4`, a key whose hash is 10 goes to server 2. Add a fifth server and it becomes `10 mod 5 = 0`. In general about 80% of keys change servers when going from 4 to 5. For a cache, that is a sudden flood of misses; for a database, a huge data migration.

Analogy. Picture a round clock face. Servers stand at a few positions around the rim. Each key is also placed on the rim, then walks clockwise until it meets a server: that server owns it. If a new server steps onto the rim, it only takes over the keys between itself and the previous server. Everyone else keeps their keys.

How it works

1. Imagine the output range of a hash function (for example 0 to 2^32 − 1) bent into a circle, the **hash ring**.
2. Hash each server's name to get its position on the ring.
3. Hash each key to get its position, then move clockwise to the first server. That server stores the key.
4. **Adding a server**: it lands at one point and takes over only the keys between its predecessor and itself. On average 1/N of all keys move.
5. **Removing a server**: its keys move to the next server clockwise; nothing else changes.
6. **Replication**: store each key on the first server it meets plus the next 2 distinct servers clockwise to keep 3 copies (Cassandra and Dynamo do this).



Virtual nodes (vnodes). With only a few servers, their random positions leave some with big arcs of the ring and some with small arcs, so load is uneven. Also, when a server leaves, *all* its load lands on one neighbour. The fix: give each physical server many positions on the ring, for example 100–256 **virtual nodes** each, by hashing "A-1", "A-2", ... "A-256". Now each server owns many small arcs scattered around the ring, so:

- Load evens out statistically.
- When a server leaves, its many small arcs are picked up by many different servers, spreading the extra load.
- A more powerful machine can be given more virtual nodes to take a bigger share.

Cassandra used 256 vnodes per node by default for years and lowered the default to 16 in version 4.0 with a smarter token-allocation algorithm.

Types

Other algorithms solve the same problem: **rendezvous hashing** (highest random weight: compute a score for each key-server pair and pick the highest), **jump consistent hash** (Google, very fast and memory-free, but servers can only be added or removed at the end), and **Maglev hashing** (Google's load balancer, which uses a lookup table for even spread).

Trade-offs

Consistent hashing minimises data movement and needs no central directory, but plain consistent hashing balances poorly without virtual nodes, and like any hashing it destroys key ordering, so range queries become scatter-gather.

Real-world examples

Amazon Dynamo (2007 paper), Apache Cassandra, ScyllaDB, Riak, Memcached client libraries (ketama), Discord's routing of users and guilds to servers, and many load balancers (for sticky routing, Part 2) use consistent hashing.

In the interview

Q: Why use consistent hashing instead of `hash mod N`?

A: With `mod N`, changing N remaps almost every key. With consistent hashing, adding or removing one of N servers moves only about 1/N of the keys.

Q: What problem do virtual nodes solve?

A: Uneven arcs and uneven load. Many positions per server spread each server's ownership in small pieces, smooth the load, and let a departing server's keys spread across many others.

TL;DR

Consistent hashing places servers and keys on a ring so a membership change moves only about 1/N of keys; virtual nodes make the spread even.

Storage types: block, file and object

What it is

Beneath every database and every uploaded photo is raw storage. Cloud providers sell it in three shapes, and picking the right one is a common design decision.

> **Analogy.** **Block storage** is an empty plot of land: you decide how to build on it. **File storage** is an office building with floors, rooms and labelled cabinets that several people can share. **Object storage** is a giant valet coat check: you hand over an item, get a ticket (the key), and later hand back the ticket to get the item. You cannot rearrange things inside the cloakroom; you can only check in, retrieve or throw away whole items.

Types

Block storage. The storage looks like a raw disk divided into fixed-size **blocks**, each with an address. The operating system puts a **file system** (such as ext4 or NTFS) on top, or a database manages the blocks directly. It is attached to one server at a time (usually), offers low latency (well under a millisecond for local NVMe SSDs, roughly one to a few milliseconds for network block storage) and lets you change a few bytes in the middle of a file. Measured in **IOPS** (input/output operations per second) and throughput. Examples: a laptop's SSD, Amazon EBS, Google Persistent Disk, Azure Managed Disks. Use for: database data files, virtual machine boot disks.

File storage. A shared file system with folders and files, accessed over the network by many servers at once using protocols such as **NFS** (Network File System, common on Linux) or **SMB** (common on Windows). Familiar to programs that expect files and paths. Examples: Amazon EFS, Azure Files, Google Filestore, a NAS (network-attached storage) box. Use for: shared content between servers, home directories, legacy applications.

Object storage. Stores **objects**: a blob of bytes plus metadata, addressed by a unique key inside a **bucket** (a top-level container). There are no real folders (a key like `photos/2026/cat.jpg` only looks like a path), and you cannot edit part of an object; you replace the whole thing. Accessed over HTTP with simple calls such as `PUT`, `GET` and `DELETE`. It scales to practically unlimited size, is the cheapest per gigabyte, and is extremely durable. Latency is higher, often tens of milliseconds for the first byte.

Examples: Amazon S3, Google Cloud Storage, Azure Blob Storage, Cloudflare R2, and the open-source MinIO and Ceph.

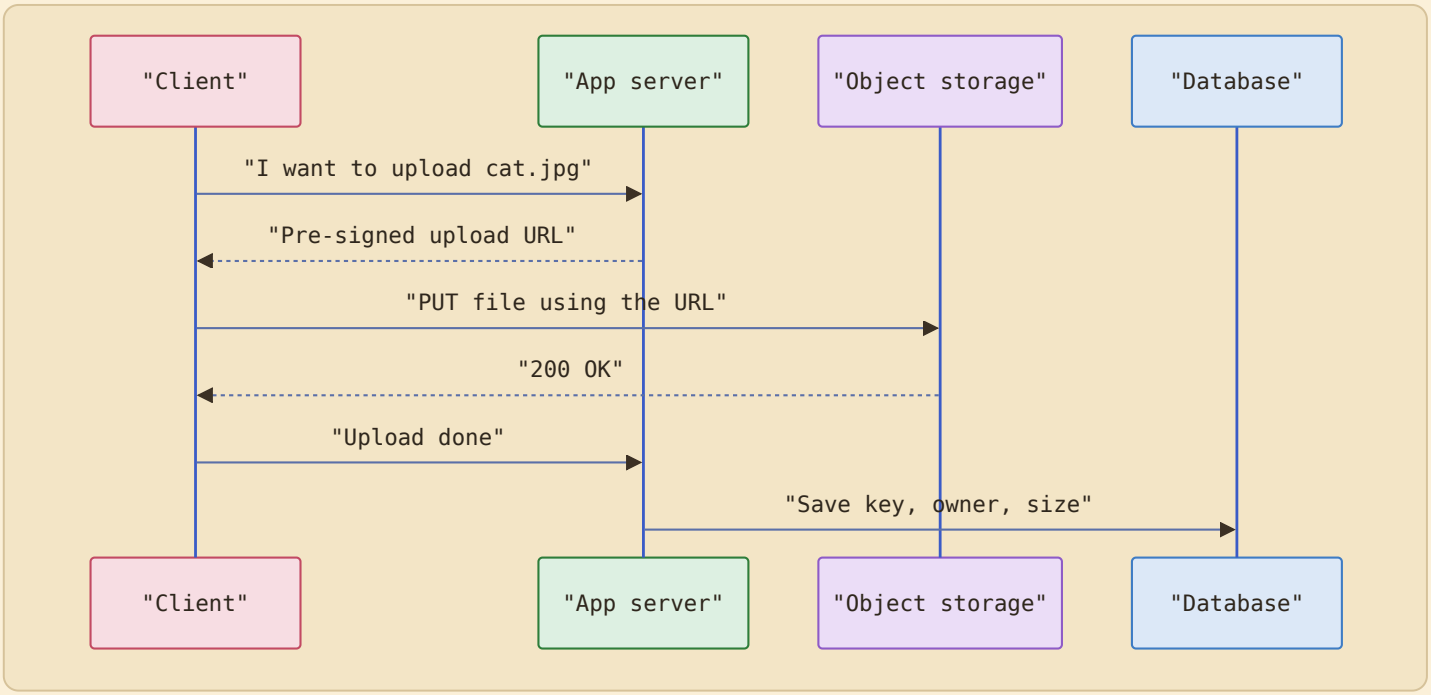
A **blob** (binary large object) is any large chunk of unstructured binary data: an image, video, PDF or backup file. "Blob storage" is another name for object storage (Azure literally calls its service Blob Storage).

	Block	File	Object
Unit	Fixed-size block	File in folders	Whole object with key
Access	Attached disk	Network file protocol	HTTP API
Shared by many servers	Usually no	Yes	Yes
Edit part in place	Yes	Yes	No, replace whole
Typical use	Databases, VM disks	Shared files	Media, backups, data lakes

How it works

The standard pattern for user uploads (photos, videos, documents):

1. The client asks the app server for permission to upload.
2. The app server returns a **pre-signed URL**: a time-limited link that allows one upload to one key in the bucket.
3. The client uploads the file straight to object storage, so gigabytes never pass through your app servers.
4. The app server stores only the object's key and metadata (owner, size, type) in the database.
5. Downloads are served through a CDN (Part 2) in front of the bucket.



Trade-offs

Never store large files inside the relational database: it bloats backups, wastes the buffer cache and makes replication slow. Store the file in object storage and the pointer in the database.

Real-world examples

Amazon S3 is designed for 99.999999999% ("eleven nines") durability of objects, stores each object across at least three availability zones in its standard storage classes, allows single objects up to 50 TB (raised from 5 TB in December 2025), and has offered strong read-after-write consistency since December 2020. Netflix, Dropbox (which later moved much of its data to its own Magic Pocket system), and almost every photo-sharing app store media in object storage.

TL;DR

Block storage is a raw fast disk for one server, file storage is a shared folder tree, and object storage is cheap, huge, durable key-to-blob storage over HTTP. Put media in object storage and keep only its key in the database.

OLTP vs OLAP, row vs column storage, lakes and warehouses

What it is

Databases serve two very different kinds of work:

- **OLTP (online transaction processing)**: the everyday work of an application. Many small, fast reads and writes that each touch a few rows: "load user 42", "insert this order". Latency matters (milliseconds), and thousands run per second. PostgreSQL, MySQL and DynamoDB are OLTP systems.
- **OLAP (online analytical processing)**: analysis and reporting. Few, big queries that scan millions or billions of rows but only a few columns: "total revenue per country per month for the last three years". Each query may take seconds. Analysts and dashboards run them.

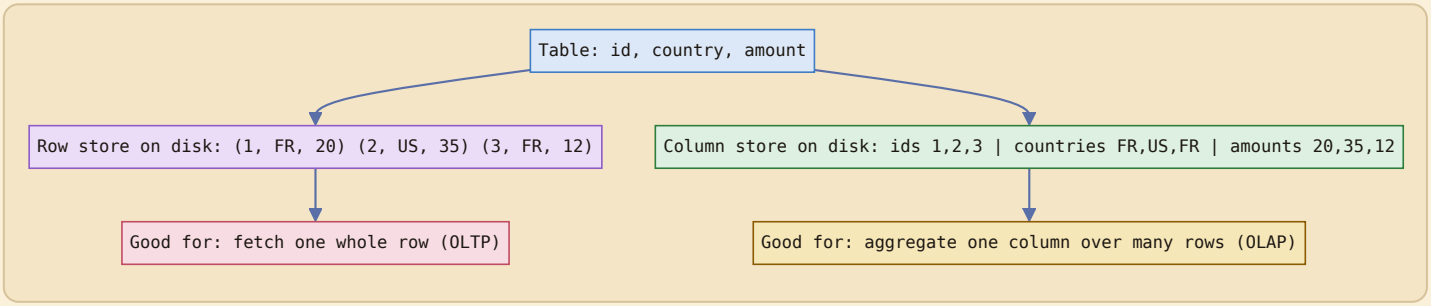
> **Analogy.** OLTP is the checkout counter at a supermarket: fast, one customer at a time. OLAP is the head office accountant going through a year of receipts to find trends.

Running heavy OLAP queries on the OLTP database is a classic mistake: one big report can slow down every customer's checkout. So companies copy the data into separate analytical systems.

Types

Row-oriented vs column-oriented storage.

- A **row store** keeps all the columns of one row together on disk. Great for OLTP: "fetch user 42" reads one place and gets everything.
- A **column store** (columnar storage) keeps all values of one column together: every `country` value in one file, every `amount` value in another. Great for OLAP: "sum of `amount` by `country`" reads just those two columns and skips the other 50. Values in one column are similar, so they compress very well (often several times smaller), and the CPU can process them in batches.



Columnar databases and formats: ClickHouse, Apache Druid, Apache Pinot, Amazon Redshift, Google BigQuery, Snowflake, DuckDB; and the file formats **Apache Parquet** and **ORC**.

Data warehouse. A central analytical database that holds cleaned, structured data from many sources (orders, payments, marketing), organised for reporting. Data is usually loaded on a schedule, and the schema is defined in advance (schema-on-write). Often modelled as a **star schema**: a big **fact table** of events (each sale) surrounded by **dimension tables** (product, customer, date). Examples: Snowflake, BigQuery, Amazon Redshift, Azure Synapse.

Data lake. A huge, cheap store of raw data in any format (logs, JSON, CSV, images, Parquet files), usually in object storage such as S3. You store first and decide structure when you read (schema-on-read). Flexible and cheap, but without discipline it becomes a "data swamp" nobody can trust. Query engines such as Apache Spark, Trino/Presto and Amazon Athena read it directly.

Lakehouse. A newer middle ground: data lake files plus a table format that adds transactions, schemas and versioning on top, such as Delta Lake, Apache Iceberg and Apache Hudi.

ETL vs ELT. Data moves from OLTP systems into analytics through a **pipeline**. **ETL (extract, transform, load)** cleans and reshapes the data before loading it into the warehouse. **ELT (extract, load, transform)** loads raw data first and transforms it inside the warehouse, which became popular as cloud warehouses got cheap and powerful. **Change data capture (CDC)** streams every change from the OLTP database's log (for example with Debezium into Kafka) so analytics stay minutes or seconds behind instead of a day.

Trade-offs

	OLTP	OLAP
Query shape	Few rows, by key	Many rows, few columns
Users	Customers via the app	Analysts, dashboards
Storage layout	Row-oriented	Column-oriented
Latency target	Milliseconds	Seconds to minutes
Examples	PostgreSQL, MySQL	Snowflake, BigQuery, ClickHouse

In the interview

Q: Why are column stores faster for analytics?

A: An analytic query reads a few columns across many rows. A column store reads only those columns from disk, compresses them well, and processes them in tight batches, so it moves far less data than a row store.

Q: Data lake or data warehouse?

A: A warehouse holds curated, structured data with a fixed schema for reliable reporting; a lake holds raw data of any type cheaply for exploration and machine learning. Many companies have both, or a lakehouse.

TL;DR

OLTP is many small transactional queries on row stores; OLAP is big analytical scans on column stores. Warehouses hold curated structured data, lakes hold raw data cheaply, and pipelines (ETL, ELT, CDC) move data between them.

Durability: WAL, fsync, backups and recovery

What it is

Durability means that once data is saved (committed), it stays saved, even through crashes, power cuts, disk failures and human mistakes. Different tools protect against different threats:

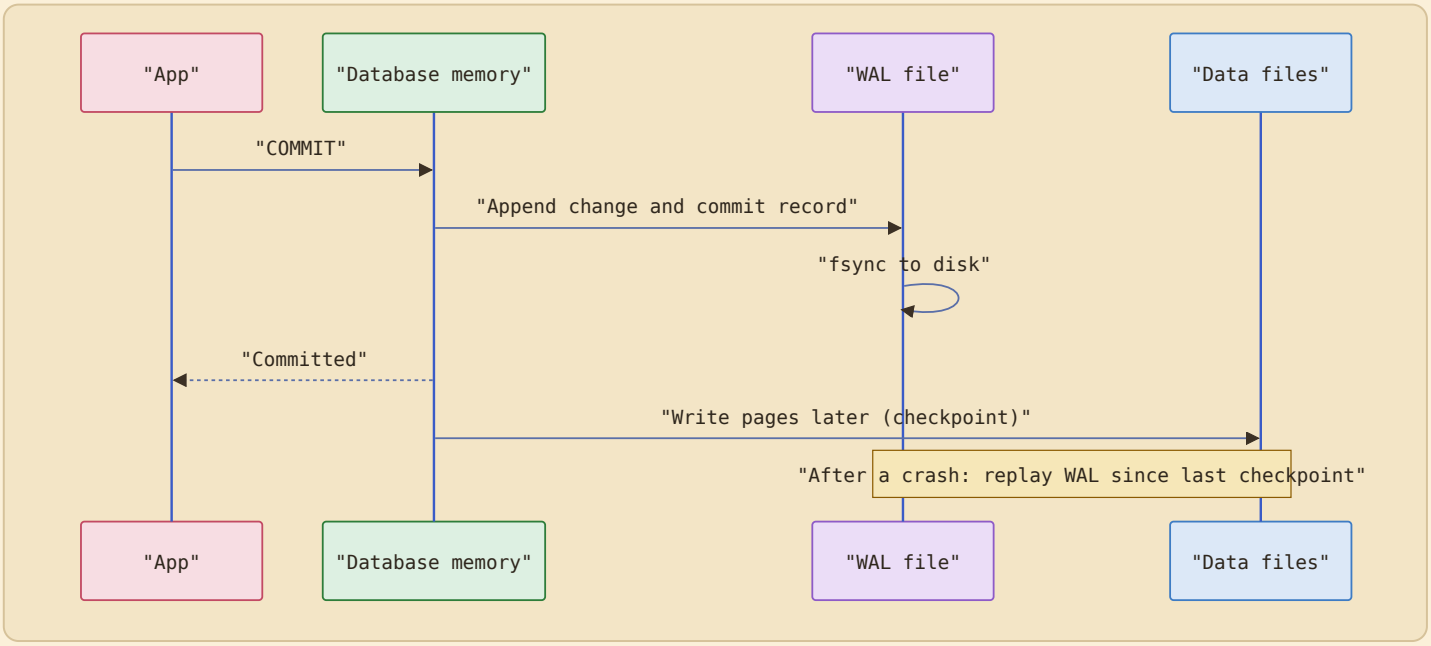
Threat	Protection
Process crash or power cut	Write-ahead log plus fsync
One disk or machine dies	Replication, RAID
Whole data centre lost	Copies in another zone or region
Bug or human deletes data	Backups and point-in-time recovery
Ransomware or account takeover	Immutable, separate-account backups

Analogy. A careful accountant writes each transaction in a pen-and-ink journal *before* updating the neat ledger. If the ledger is ruined halfway through an update, the journal shows exactly what to redo. And every night she photocopies the books and stores the copy in another building.

How it works

Write-ahead log (WAL). Updating the real data files in place is slow (random writes) and dangerous (a crash midway leaves a half-written page). So the database first appends a description of each change to a sequential log file, the **write-ahead log** (also called a redo log in MySQL InnoDB, or a commit log in Cassandra). The rule is: *log first, then data*.

1. A transaction changes some rows in memory.
2. On `COMMIT`, the database appends the changes and a commit record to the WAL.
3. It forces the WAL to physical storage with **fsync**, a system call that tells the operating system "really write this to the device now, do not just keep it in your memory buffer".
4. Only then does it tell the client "committed".
5. Later, a background process writes the changed pages to the data files. A **checkpoint** records how far the data files are up to date, so old WAL can be recycled.
6. After a crash, the database replays the WAL from the last checkpoint, redoing committed changes and discarding uncommitted ones.



The same log is reused for replication (followers replay the leader's WAL stream) and for change data capture.

Why fsync matters. Operating systems keep written data in a memory **page cache** and flush it to disk later, for speed. Without fsync, a power cut can lose writes that the program believed were saved. fsync is slow-ish (from tens of microseconds on enterprise SSDs with power-loss protection to several milliseconds on cheaper or network disks), so databases batch many commits into one fsync (**group commit**). Some systems let you trade safety for speed: PostgreSQL's `synchronous_commit = off` and MySQL's `innodb_flush_log_at_trx_commit = 2` can lose the last moment of commits on a crash. Redis's append-only file defaults to fsync once per second, so up to about one second of writes can be lost.

RAID (redundant array of independent disks) combines several disks so the failure of one does not lose data: RAID 1 mirrors two disks, RAID 5/6 store parity, RAID 10 stripes across mirrored pairs. It protects against disk failure only.

Backups. A **backup** is a copy of the data stored separately, kept for days or months, so you can recover from things replication would happily copy, such as an accidental `DROP TABLE`.

- **Full backup**: a complete copy. Simple, slow and large.
- **Incremental backup**: only what changed since the last backup. Small and fast, but restoring needs the chain of increments.
- **Logical backup**: the data exported as SQL statements or rows (`pg_dump`, `mysqldump`). Portable but slow for big databases.
- **Physical backup**: a copy of the raw data files (`pg_basebackup`, Percona XtraBackup).

- **Snapshot**: a point-in-time image of a disk or volume (for example an EBS snapshot), usually taken in seconds using copy-on-write, where only blocks changed after the snapshot need extra space.

Point-in-time recovery (PITR) lets you restore to any moment, say 14:31:07, just before a bad deploy deleted data. It works by restoring the last full backup and then replaying archived WAL up to the chosen moment. Managed databases such as Amazon RDS offer it with a retention window of up to 35 days.

A widely used rule is **3-2-1**: keep 3 copies of the data, on 2 different kinds of media, with 1 copy off-site (today often: another region or another cloud account).

Test your restores. Until you have actually rebuilt a database from a backup, you do not know that the backup works. Measure how long a restore takes, because that number feeds your recovery time objective (RTO, Part 5).

Common mistakes

- Keeping backups in the same account or region as production, where one compromised credential or one regional outage takes both.
- Never testing restores. In January 2017 GitLab accidentally deleted a production database directory and found that several of its backup and replication mechanisms had not been working; it restored from a copy about six hours old and lost roughly six hours of changes.
- Disabling fsync for benchmarks and forgetting to turn it back on.

In the interview

Q: How does a database guarantee durability without writing every change to the data files immediately?

A: It appends each change to a sequential write-ahead log and fsyncs the log before acknowledging the commit. The data files are updated later; after a crash, the log is replayed.

Q: You have three replicas. Do you still need backups?

A: Yes. Replicas protect against hardware failure, but a bad deploy, a bug or a human error is copied to every replica in milliseconds. Backups with point-in-time recovery let you go back to before the mistake.

TL;DR

Durability comes in layers: WAL and fsync survive crashes, replication survives machine loss, cross-region copies survive site loss, and tested backups with point-in-time recovery survive mistakes.

Connection pooling, ORMs and N+1 queries

What it is

These three topics are about how application code talks to the database, and they cause a surprising share of real-world performance problems.

Connection pooling. Opening a database connection is expensive: a TCP handshake, often a TLS handshake, authentication, and the database setting up memory for the session. That can cost several milliseconds, sometimes tens. PostgreSQL even starts a whole operating-system process per connection, using several megabytes of memory each, and its default limit (`max_connections`) is 100. A **connection pool** keeps a set of open connections ready; a request borrows one, uses it and returns it, instead of opening and closing one each time.

> **Analogy.** A taxi rank. Instead of building a new car for every passenger, a fixed set of taxis waits at the rank; passengers take one and it comes back.

How it works

There are two places a pool can live:

1. **In the application** (client-side pool): a library inside each app process, such as HikariCP for Java, or the pools built into most database drivers. Typical settings are a minimum and maximum size, and a timeout for waiting when all connections are busy.

2. **As a separate proxy** (server-side pool): a small service between all app servers and the database, such as **PgBouncer** or Pgpool-II for PostgreSQL, ProxySQL for MySQL, or Amazon RDS Proxy. Useful when you have hundreds of app instances or serverless functions, which would otherwise open thousands of connections. PgBouncer's **transaction pooling** mode hands a server connection to a client only for the duration of one transaction, so a few dozen real connections can serve thousands of clients.

Sizing tip: bigger is not better. A database with 16 CPU cores cannot run 500 queries at once; extra connections just wait and compete for memory. Pools of a few times the core count often perform best. Remember the multiplication: 50 app servers × 20 connections each = 1,000 connections at the database.

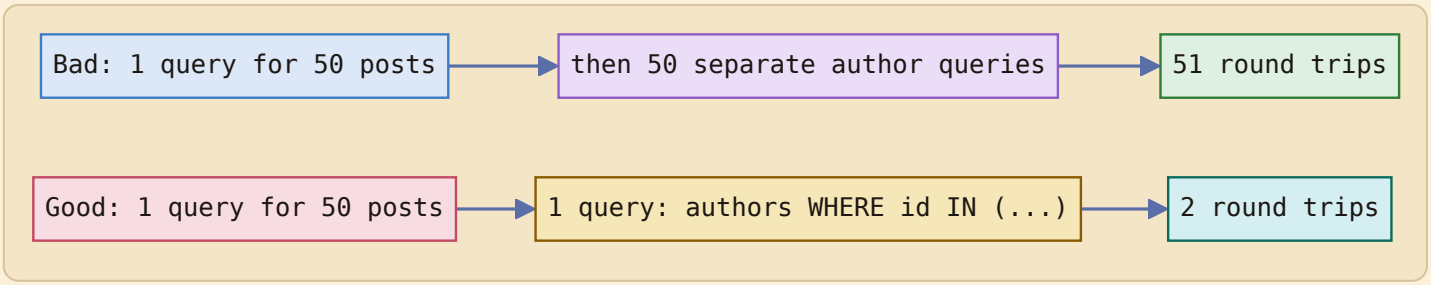
ORMs. An **ORM (object-relational mapper)** is a library that maps database rows to objects in your programming language, so you can write `user.orders` instead of SQL. Examples: Hibernate (Java), Django ORM and SQLAlchemy (Python), ActiveRecord (Ruby on Rails), Entity Framework (.NET), Prisma and Sequelize (JavaScript/TypeScript). ORMs speed up development, protect against SQL injection by using parameters, and handle schema **migrations** (versioned scripts that change the schema). The cost is that they hide the queries actually being run, which makes it easy to write very slow code without noticing. Good teams use the ORM for everyday work, read the generated SQL for hot paths, and drop to hand-written SQL where needed.

The N+1 query problem. The most famous ORM trap. You load a list of N items with 1 query, then loop over them and the ORM quietly runs 1 more query per item to load a related object: N+1 queries in total.

```
posts = Post.objects.all()[:50]       # 1 query
for p in posts:
    print(p.author.name)               # 50 more queries, one per post
```

At 1 ms per round trip that is about 51 ms instead of 2 ms, and with 1,000 items it is a full second. The fixes:

- **Eager loading**: tell the ORM to fetch the related data up front, in a join or one extra query (`select_related` / `prefetch_related` in Django, `includes` in Rails, `JOIN FETCH` in Hibernate).
- **Batching**: collect all the IDs and fetch them in one `WHERE id IN (...)` query. The **DataLoader** pattern does this automatically in GraphQL servers.
- **Detect it**: log query counts per request in development; tools such as Bullet (Rails) flag N+1 patterns.



Common mistakes

- Opening a new connection per request with no pool, then hitting "too many connections" at peak traffic.
- Pool size × number of instances exceeding the database limit after autoscaling adds servers.
- Serverless functions each holding their own connections, without a proxy such as RDS Proxy or PgBouncer.
- Trusting the ORM blindly and shipping N+1 queries on a list page.

In the interview

Q: A list page gets slower as the list grows, and the database shows thousands of tiny identical queries. What is happening?

A: An N+1 query problem: one query for the list and one per item for related data. Use eager loading or batch the lookups into a single `IN` query.

Q: Why put PgBouncer in front of PostgreSQL?

A: PostgreSQL connections are heavy (one process each) and limited. PgBouncer lets thousands of client connections share a small number of real database connections.

TL;DR

Reuse connections through a pool, sized to what the database can actually run in parallel. Use ORMs for productivity but watch the SQL they generate, especially for N+1 queries.

Choosing a database

What it is

This section pulls the part together into a quick decision aid. In an interview, always justify the choice from the **access pattern** (how data is read and written), the **consistency** needed, and the **scale** (data size and requests per second), not from brand names.

How it works

Ask these questions, in order:

1. **What are the main queries?** By key, by range, full-text, joins across entities, aggregations, graph traversal, similarity?
2. **How much data, and how fast does it grow?** Gigabytes fit anywhere. Tens of terabytes push towards sharding or a distributed store.
3. **What is the read-to-write ratio, and the peak writes per second?** Read-heavy suggests replicas and caches; write-heavy suggests LSM-based or sharded stores.
4. **How strong must consistency be?** Money and inventory need transactions. Likes and view counts can be eventually consistent.
5. **Does it need to work across regions?** Multi-region writes push towards multi-leader, leaderless or globally distributed databases.
6. **What does the team know how to run?** Operating a database well is a major cost; a managed service often wins.

Real-world examples

Need	Good choice	Examples
General app data, transactions	Relational	PostgreSQL, MySQL, Aurora
Relational at global scale	Distributed SQL	Spanner, CockroachDB, YugabyteDB
Cache, sessions, counters	In-memory key-value	Redis, Memcached
Simple key access at huge scale	Managed key-value	DynamoDB, Bigtable
Flexible nested records	Document	MongoDB, Firestore, Couchbase
Heavy writes by key and time	Wide-column	Cassandra, ScyllaDB, HBase
Relationships many hops deep	Graph	Neo4j, Neptune
Metrics and sensor data	Time-series	Prometheus, InfluxDB, TimescaleDB
Full-text search, logs	Search engine	Elasticsearch, OpenSearch
Semantic or image similarity	Vector	pgvector, Pinecone, Milvus
Analytics and reporting	Columnar warehouse	Snowflake, BigQuery, ClickHouse
Photos, videos, backups	Object storage	S3, GCS, Azure Blob
Small strongly consistent config	Coordination store	etcd, ZooKeeper

Common mistakes

- Naming a database without saying why.

- Using one database type for everything, or a different exotic one for every table.
- Forgetting the operational cost: backups, upgrades, monitoring and on-call.

In the interview

Q: Which database would you use for a Twitter-like timeline?

A: Users, follows and tweets as the source of truth in a relational database or a wide-column store sharded by user ID; precomputed home timelines in an in-memory store like Redis for fast reads; media in object storage behind a CDN; and a search engine for tweet search. Each choice follows the access pattern.

TL;DR

Start with PostgreSQL or MySQL unless the access pattern or scale clearly needs something else, and add specialist stores (cache, search, object storage, analytics) beside it as needs appear.

Part 3 glossary

Term	Meaning in one line
Database / DBMS	A program that stores data safely, finds it quickly and coordinates concurrent users
State	Stored facts that must survive between requests
Query planner (optimizer)	The part of a database that picks the cheapest way to run a query
Buffer cache	Database memory holding recently used data pages
Read / write	Fetching data without changing it / inserting, updating or deleting data
Relational database	A database storing data in tables of rows and columns with a fixed schema
Table, row, column	A set of records, one record, one property of a record
Schema	The declared structure of tables, columns and data types
Primary key	A column whose unique value identifies each row
Foreign key	A column holding another table's primary key, linking the two
Referential integrity	The guarantee that foreign keys point to rows that exist
Unique constraint	A rule that no two rows share a value in a column
Join	Combining rows from two tables that match on a key
SQL	The standard language for querying relational databases
DDL	SQL statements that change the schema, such as CREATE TABLE
Distributed SQL	Relational databases spread over many machines, such as Spanner or CockroachDB
Transaction	A group of operations that all succeed or all fail
Commit / rollback	Make a transaction permanent / discard all its changes
ACID	Atomicity, Consistency, Isolation, Durability
Atomicity	All-or-nothing execution of a transaction
Consistency (ACID)	A transaction leaves data obeying all declared rules
Isolation	Concurrent transactions do not see each other's unfinished work
Durability	Committed data survives crashes
Concurrency	Several operations running at overlapping times
Anomaly	An incorrect result caused by concurrent transactions
Dirty read	Reading another transaction's uncommitted data
Dirty write	Overwriting another transaction's uncommitted data
Non-repeatable read	Re-reading a row in one transaction and getting a different value
Phantom read	A repeated query returning new rows inserted by another transaction
Lost update	Two read-modify-write cycles overwrite each other, losing one change
Write skew	Two transactions each write different rows based on a shared read and together break a rule
Isolation level	A setting choosing which anomalies the database prevents
Read committed	Isolation level preventing dirty reads only
Repeatable read	Isolation level also preventing non-repeatable reads

Term	Meaning in one line
Snapshot isolation	Each transaction reads a frozen consistent picture of the data
Serializable	Result equals some one-at-a-time order of the transactions
MVCC	Keeping multiple row versions so readers and writers do not block each other
Two-phase locking	Taking locks and holding them until commit to get serializability
SSI	Serializable snapshot isolation, which aborts transactions on dangerous patterns
SELECT FOR UPDATE	Locking rows that were read until the transaction ends
Optimistic concurrency control	Detecting conflicting writes with a version check and retrying
Compare-and-set	Writing only if the current value still equals what was read
NoSQL	Non-relational databases built around specific access patterns
Key-value store	A database mapping keys to opaque values
Document store	A database storing self-contained JSON-like documents
Collection	A group of documents in a document store
Wide-column store	A database grouping rows by partition key and sorting them by clustering key
Clustering key	The key that sorts rows inside one partition in a wide-column store
Graph database	A database storing nodes and edges for relationship queries
Time-series database	A database optimised for timestamped measurements
Search engine	A store built on an inverted index for full-text queries
Inverted index	A map from each word to the documents containing it
Tokenising	Splitting text into searchable words
Relevance ranking	Ordering search results by how well they match
Vector database	A store for embeddings supporting nearest-neighbour search
Embedding	A list of numbers representing an item so similar items are close
ANN / HNSW	Approximate nearest-neighbour search / a popular graph-based ANN index
BASE	Basically available, soft state, eventually consistent
Eventual consistency	Copies agree eventually once writes stop
Polyglot persistence	Using several database types in one system
Schema-on-read / on-write	Structure applied when reading / enforced when writing
Index	An extra structure that finds rows by value without a full scan
Full table scan	Reading every row to answer a query
B-tree	A shallow balanced tree of sorted pages, the default index
Page	A fixed-size block of storage, typically 4 to 16 KB
Branching factor	Number of children per B-tree page
Range query	A query for values between two bounds
LSM-tree	A write-optimised structure of memtable plus merged sorted files
Memtable	The in-memory sorted buffer of an LSM-tree
SSTable	An immutable sorted file written by an LSM-tree

Term	Meaning in one line
Compaction	Merging SSTables and dropping old versions
Tombstone	A marker recording that a key was deleted
Read / write amplification	Extra disk reads per lookup / extra rewrites per byte written
Hash index	An index mapping keys to locations via a hash map
Composite index	An index over several columns in order
Leftmost prefix	The rule that a composite index helps queries on its leading columns
Covering index	An index containing every column a query needs
Index-only scan	Answering a query from the index without touching the table
Clustered index	An index whose order is the physical order of the rows
Secondary index	Any index other than the clustered or primary one
Partial index	An index over only rows matching a condition
Selectivity	How few rows a filter value matches
EXPLAIN	A command showing the plan the database will use
Normalization	Storing each fact in exactly one place
Normal forms (1NF, 2NF, 3NF)	Progressive rules removing repeated and misplaced data
Update anomaly	Inconsistency caused by a fact stored in many places
Denormalization	Duplicating data deliberately to speed up reads
Materialized view	A stored, refreshed result of a query
Replication	Keeping copies of data on several machines
Replica	One copy of the data
Leader / follower	The node accepting writes / nodes copying from it
Read replica	A follower used mainly to serve reads
Multi-leader replication	Several nodes accept writes and replicate to each other
Write conflict	Two concurrent writes to the same record on different leaders
Last write wins	Conflict rule keeping the write with the latest timestamp
CRDT	A data type whose concurrent updates always merge to the same result
Leaderless replication	Any replica accepts writes, with quorums for reads and writes
Quorum	The minimum number of replicas that must respond for success
Read repair	Fixing stale replicas during a read
Anti-entropy	Background comparison and repair of replicas
Hinted handoff	Holding writes for a down node and delivering them later
Synchronous replication	Leader waits for a follower before confirming a write
Asynchronous replication	Leader confirms before followers have the write
Semi-synchronous replication	Leader waits for one follower, the rest are asynchronous
Replication lag	How far a follower is behind the leader
Read-your-own-writes	A user always sees their own latest changes
Monotonic reads	A user never sees data go backwards in time

Term	Meaning in one line
Failover	Promoting a follower to leader after the leader fails
Heartbeat	A periodic "I am alive" message
Split brain	Two nodes both acting as leader and accepting writes
Fencing	Forcibly stopping an old leader from acting
Partitioning	Splitting a dataset into smaller pieces
Sharding	Partitioning across different machines
Shard / partition	One piece of the split data
Horizontal partitioning	Splitting by rows
Vertical partitioning	Splitting by columns
Shard key	The column that decides which shard a row goes to
Range sharding	Each shard owns a continuous range of keys
Hash sharding	The shard is chosen from a hash of the key
Directory-based sharding	A lookup service maps keys to shards
Geo sharding	Sharding by user location
Router	A component that sends each query to the right shard
Cardinality	Number of distinct values in a column
Scatter-gather	Sending a query to all shards and merging results
Hot spot	A shard receiving far more load than others
Hot key	A single key receiving disproportionate traffic
Key salting	Adding a suffix to spread one hot key over several shards
Rebalancing / resharding	Moving data between shards as the cluster changes
Local vs global index	Index per shard / index sharded by the indexed value
Federation	Splitting databases by feature or function
Hash function	A function turning any input into a fixed-size, random-looking number
Consistent hashing	Key placement on a ring so membership changes move about 1/N of keys
Hash ring	The hash output range arranged as a circle
Virtual node	One of many ring positions owned by a single physical server
Rendezvous hashing	Picking the server with the highest score for each key
Block storage	Raw fixed-size blocks presented as a disk
IOPS	Input/output operations per second
File storage	A shared network file system with folders
NFS / SMB	Network file-sharing protocols for Linux / Windows
Object storage	Keyed blobs with metadata accessed over HTTP
Bucket	A top-level container in object storage
Blob	A large chunk of unstructured binary data
Pre-signed URL	A time-limited link allowing one direct upload or download
OLTP	Many small transactional queries from an application

Term	Meaning in one line
OLAP	Large analytical queries over many rows
Row store	Storage keeping each row's columns together
Column store	Storage keeping each column's values together
Parquet / ORC	Columnar file formats for analytics
Data warehouse	Curated structured store for reporting
Star schema	A fact table surrounded by dimension tables
Data lake	Cheap store of raw data in any format
Lakehouse	A data lake with transactional table formats on top
ETL / ELT	Transform before loading / load then transform
Change data capture (CDC)	Streaming every change from a database's log
Write-ahead log (WAL)	A sequential log of changes written before the data files
fsync	A system call forcing written data onto the physical device
Page cache	Operating-system memory buffering file writes
Group commit	Flushing many commits with one fsync
Checkpoint	A point up to which data files reflect the log
RAID	Combining disks so one failure loses no data
Backup	A separate copy kept to recover from loss or mistakes
Full / incremental backup	Complete copy / only changes since the last backup
Logical / physical backup	Exported rows or SQL / copied raw data files
Snapshot	A point-in-time image of a disk or volume
Point-in-time recovery	Restoring to any moment by replaying logs over a backup
3-2-1 rule	Three copies, two media types, one off-site
Connection pool	A set of reusable open database connections
PgBouncer	A popular connection-pooling proxy for PostgreSQL
Transaction pooling	Lending a database connection only for one transaction
ORM	A library mapping rows to programming-language objects
Migration	A versioned script that changes the schema
N+1 query problem	One query for a list plus one per item for related data
Eager loading	Fetching related data up front in few queries
DataLoader	A pattern that batches per-item lookups into one query
Access pattern	How an application reads and writes its data

Part 4 — Caching, async processing and consistency

Parts 1 to 3 built a system that works: clients send requests, servers answer them, and databases remember things. This part covers the three ideas that let that system stay fast and correct once it grows past one machine:

1. **Caching**: keep a copy of popular data somewhere quicker to reach, so you do less slow work.

2. **Asynchronous processing**: hand slow work to a queue and answer the user straight away, instead of making them wait.
3. **Consistency in distributed systems**: once data lives on many machines, decide what "the current value" means and how the machines agree on it.

Most of the vocabulary that makes system design sound hard (CAP, quorum, Raft, idempotency, saga, vector clock, Bloom filter, Snowflake ID) is here. Every term is explained from zero.

Caching basics

What it is

A **cache** is a small, fast store that keeps copies of data that is slow or expensive to get from its real home. The real home is called the **source of truth** or **origin**, usually a database or another service.

Analogy: a chef keeps a small bowl of chopped onions next to the stove instead of going to the cold room for one onion at a time. The bowl is the cache, the cold room is the source of truth. The bowl is within arm's reach but holds little, and what is in it slowly stops being fresh.

Some words you will hear constantly:

- **Cache hit**: the data you asked for was in the cache. Fast path.
- **Cache miss**: it was not, so you go to the source of truth (slow path) and usually put the answer into the cache for next time.
- **Hit ratio** (or hit rate): hits divided by total lookups. A hit ratio of 0.95 means 95 out of every 100 requests never touch the database.
- **TTL (time to live)**: how long an entry may stay in the cache before it is treated as expired and dropped or refreshed.
- **Stale data**: a cached copy that no longer matches the source of truth because the source changed after the copy was taken.
- **Warm cache / cold cache**: a warm cache already holds the popular data. A cold cache (just started, or just emptied) misses on almost everything until it fills up.

Why it matters

Reading from memory is many times faster than asking a database across the network: a local in-memory lookup takes well under a microsecond, while even a quick indexed query takes about a millisecond. Typical numbers:

Where the data is	Typical time to read a small item
CPU cache / process memory	well under 1 microsecond
Redis or Memcached on the same data-centre network	about 0.2–1 ms
Database query hitting an index on SSD	about 1–10 ms
Database query doing a big join or scan	50 ms to seconds
Call to a service in another continent	70–200 ms

Caching gives you three wins:

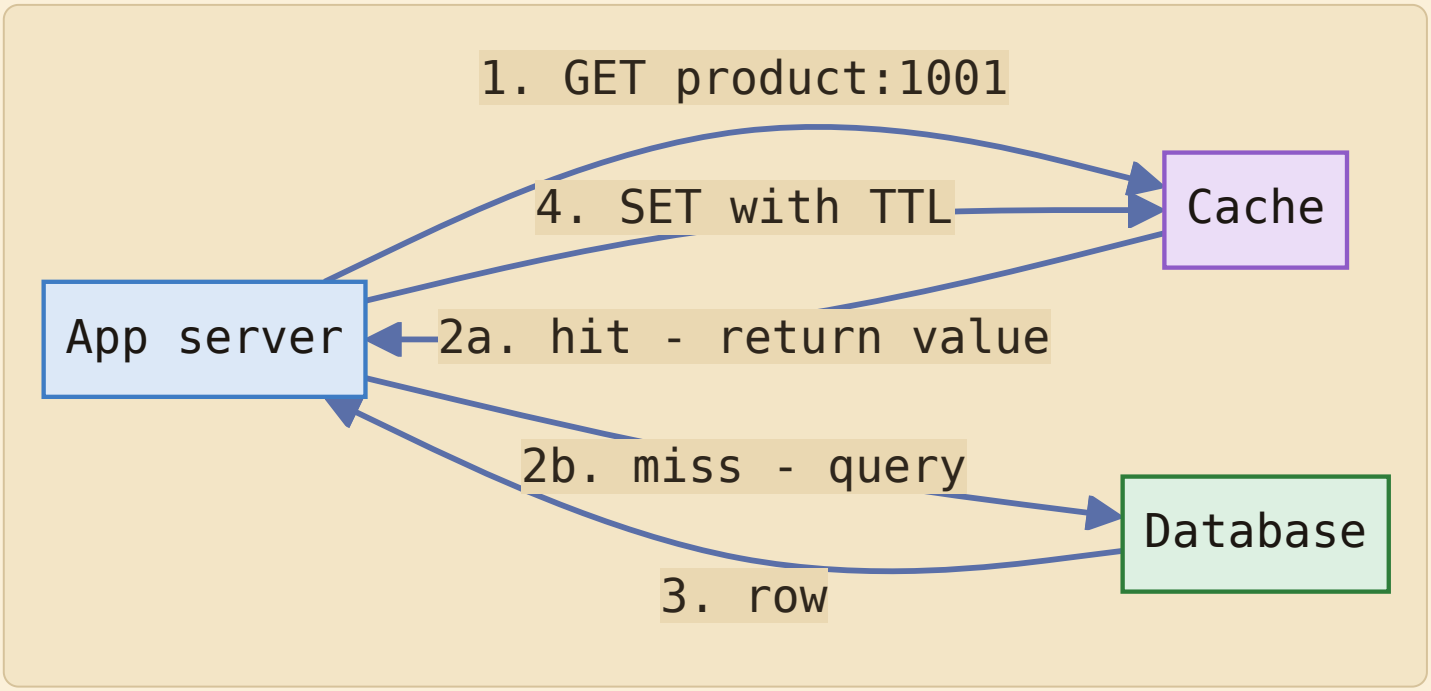
1. **Lower latency** for the user: a 1 ms cache read instead of a 30 ms query.
2. **Less load on the database**: with a 95% hit ratio the database sees only 5% of the read traffic, which can be the difference between needing one database and needing twenty.
3. **Lower cost**: memory reads are cheaper than recomputing a page or paying for database capacity.

The maths of hit ratio is worth remembering. Average read time = hit ratio x cache time + miss ratio x (cache time + database time). With a 1 ms cache, a 20 ms database and a 90% hit ratio the average is about 0.9 x 1 + 0.1 x 21 = 3 ms. Raise the hit ratio to 99% and it falls to about 1.2 ms. Watch the miss ratio, because that is what the database feels: moving from 90% to 99% hits takes misses from 10 in every 100 reads to 1, a tenfold drop in database reads.

How it works

A cache is usually a **key-value store**: you look things up by a key such as `user:42` and get back a value such as the JSON of user 42's profile. The basic read path:

1. The application builds a key from the request, for example `product:1001`.
2. It asks the cache for that key.
3. On a hit, it returns the cached value.
4. On a miss, it reads from the database, stores the result in the cache with a TTL (say 5 minutes), and returns it.



Caches work because real traffic is **skewed**: a small fraction of items gets most of the requests (the most popular 20% of products might get 80% of the views). You only need to cache that small hot fraction to get a high hit ratio.

Trade-offs

- **Speed vs freshness**: the longer you cache, the faster and cheaper, but the more likely users see stale data.
- **Memory cost**: RAM costs much more per gigabyte than disk, so you cannot cache everything.
- **Extra moving part**: the cache can fail, fill up, or disagree with the database. You now have two copies of the truth to keep in line.
- **Cold-start risk**: if the cache is wiped, every request falls through to a database that was sized for 5% of the traffic.

Common mistakes

- Caching data that is rarely read twice (low hit ratio, wasted memory).
- Forgetting a TTL, so a wrong value lives forever.
- Treating the cache as the source of truth. A cache may lose data at any time; the database must still have it.
- Sizing the database only for the cache-miss traffic and then crashing when the cache restarts.

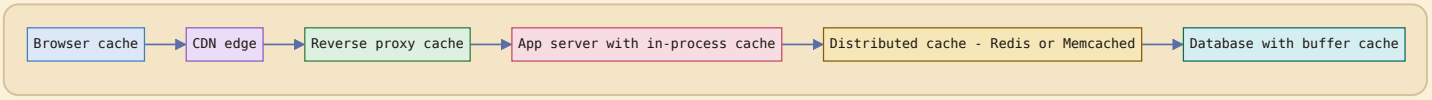
TL;DR

A cache is a fast copy of slow data. It wins on latency and database load, and the price is memory and the risk of serving stale data.

Where caches live

What it is

Caching is not one box in the diagram. Copies of data can be kept at every hop between the user and the database. The closer to the user, the faster the hit, but the harder it is to update or delete.



Types

Client or browser cache. The browser keeps copies of files (images, CSS, JavaScript, sometimes API responses) on the user's own device. The server controls this with HTTP headers: `Cache-Control: max-age=86400` means "you may reuse this for 86,400 seconds (one day) without asking me". An `ETag` header gives the file a version fingerprint so the browser can ask "has it changed?" and get a tiny `304 Not Modified` reply instead of the whole file. Mobile apps do the same thing with their own local storage. Fastest possible hit (no network at all), but you cannot reach into a user's device to delete a stale copy, so you rely on TTLs and on changing the file name when content changes (for example `app.3f9a1c.js`, called **cache busting**).

CDN cache. A **CDN (content delivery network)** is a network of servers placed in many cities (Part 2). Each edge server caches responses near users so a viewer in Sydney gets the image from Sydney, not from a server in Virginia. Great for static files and for public pages that are the same for everyone.

Reverse proxy cache. A **reverse proxy** is a server that sits in front of your app servers and forwards requests to them (Part 2). NGINX, Varnish and similar proxies can store whole HTTP responses and serve repeated requests without touching the app at all.

Application or in-process cache. A plain in-memory map inside the application process itself (for example a Java Caffeine cache or a Python `functools.lru_cache`). Reads take nanoseconds because there is no network hop. Downsides: each app server has its own copy (ten servers means ten copies that can disagree), the memory is limited to that process, and the cache is lost every time the process restarts or is redeployed.

Distributed cache. A separate cluster of cache servers shared by all app servers, reached over the network. The two famous ones:

- **Memcached**: a simple, very fast, multi-threaded key-value cache holding plain strings or bytes. No persistence, no replication built in. Facebook ran one of the largest Memcached deployments in the world.
- **Redis**: a key-value server with rich data types (strings, lists, hashes, sets, sorted sets, streams, HyperLogLog), optional saving to disk, replication and a cluster mode that splits keys across nodes. Used for caching and also for leaderboards, rate limiters, session stores and simple queues. (Valkey is a community fork of Redis created in 2024 when Redis changed its licence.)

All app servers see the same copy, it survives app restarts, and it can hold hundreds of gigabytes across nodes. The cost is a network hop (well under a millisecond inside a data centre) and another system to run.

Database buffer cache. Databases keep recently used disk pages in RAM themselves: PostgreSQL's **shared buffers**, MySQL InnoDB's **buffer pool**. You get this for free. It is why the second run of the same query is often much faster than the first.

Other caches you may meet: **DNS caches** (your laptop and resolver cache name lookups for their TTL, Part 2), CPU caches (L1/L2/L3, inside the processor), and the operating system **page cache** that keeps recently read file data in spare RAM.

Trade-offs

Layer	Speed of hit	Shared by all users?	Easy to invalidate?
Browser	fastest (no network)	no, one user	no
CDN	very fast (nearby city)	yes, per region	yes, with purge API, takes seconds
In-process	nanoseconds	no, one server	hard across many servers
Distributed (Redis)	under 1 ms	yes	yes, delete the key
DB buffer cache	fast inside DB	yes	automatic

Real-world examples

- Facebook described in its 2013 paper "Scaling Memcache at Facebook" a Memcached layer handling billions of requests per second in front of MySQL.
- Twitter (X) and GitHub use Redis and Memcached heavily for timelines, sessions and counters.
- Netflix runs EVCache, its own distributed cache built on Memcached, replicated across AWS regions.

In the interview

Q: Where would you put a cache in this design?

A: Name the layers from outside in: browser and CDN for static assets and public pages, a distributed cache like Redis in front of the database for hot reads such as user profiles, and rely on the database's own buffer cache. Explain what you cache, the key format and the TTL.

Q: In-process cache or Redis?

A: In-process is fastest but each server has its own copy, so it suits small, rarely changing data such as config or feature flags. Redis is shared, larger and survives deploys, so it suits user data and anything that must look the same from every server.

TL;DR

Caches exist at every hop: browser, CDN, reverse proxy, app memory, a shared cache like Redis, and inside the database. Closer to the user is faster but harder to invalidate.

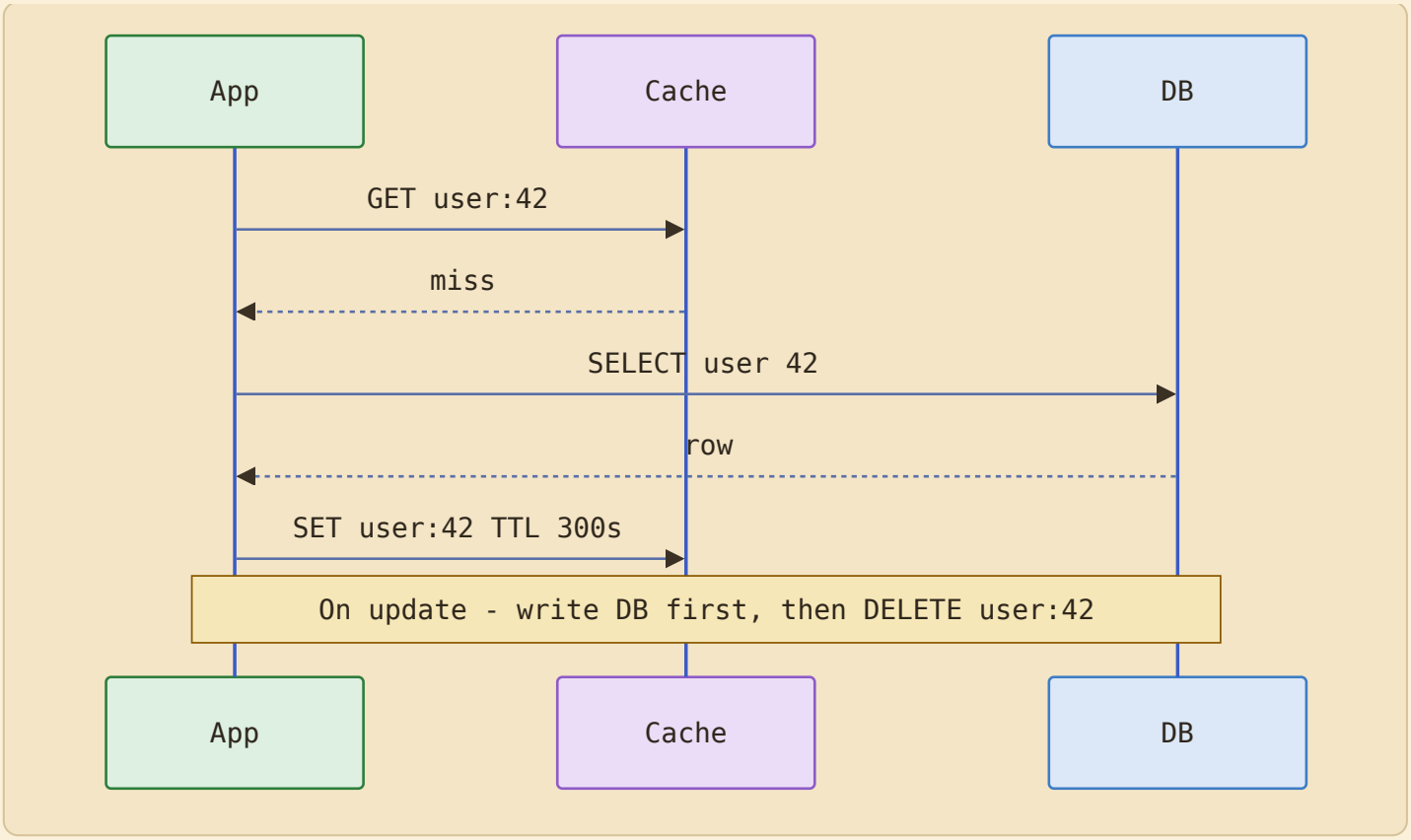
Caching strategies

What it is

A **caching strategy** (or caching pattern) is the rule for who reads and writes the cache and in what order relative to the database. The choice decides how stale data can get and what happens when something fails half-way.

Types

Cache-aside (lazy loading). The application talks to both the cache and the database. On read: check the cache, on a miss read the database and fill the cache. On write: update the database, then **delete** (invalidate) the cache key so the next read reloads it. This is the most common pattern. Only data that is actually requested gets cached, and if the cache dies the app still works (slower).



Why delete instead of updating the cache on writes? If two writers race, "update cache" can leave the older value in the cache last. Deleting means the next reader fetches the fresh row. A small race still exists (a slow reader can put an old value back just after the delete), which is why you keep a TTL as a safety net.

Read-through. Like cache-aside, but the cache itself loads from the database on a miss. The application only ever talks to the cache. Simpler application code, but you need a cache library or product that knows how to call your database (for example a loader function in Caffeine, or Amazon DynamoDB Accelerator, DAX, in front of DynamoDB).

Write-through. Every write goes to the cache and the cache synchronously writes it to the database before confirming. Cache and database stay in step, and reads right after a write are hits. Cost: every write pays for two writes, and you fill the cache with data that may never be read.

Write-behind (write-back). The write goes to the cache only, the user gets "done" immediately, and the cache flushes changes to the database later, often in batches (for example every second). Very fast writes and fewer database writes (ten updates to the same counter can become one). Danger: if the cache node crashes before flushing, those writes are lost. Used for things like view counters where losing a few is acceptable. CPUs and disk controllers use write-back caches internally too.

Write-around. Writes go straight to the database and skip the cache entirely. The cache is filled only by reads (usually combined with cache-aside). Good when written data is rarely read soon after, such as logs or bulk imports, so the cache isn't flooded with cold data.

Refresh-ahead. The cache reloads popular entries **before** they expire. If an item has a 60-second TTL and is read at second 50, the cache returns the current value and refreshes it in the background. Users never wait on a miss for hot keys. Cost: you may refresh items nobody will ask for again.

Strategy	Who writes the DB	Read after write	Main risk
Cache-aside	App	miss, then reload	brief stale window
Read-through	App (cache loads reads)	miss, then reload	same as cache-aside
Write-through	Cache, synchronously	hit, fresh	slower writes
Write-behind	Cache, later	hit, fresh	data loss on crash
Write-around	App, cache skipped	miss	first reads slow
Refresh-ahead	n/a (read side)	depends	wasted refreshes

Trade-offs

Read-heavy, tolerant of a few seconds of staleness (product pages, profiles): cache-aside with a TTL. Must read your own write immediately: write-through. Extremely write-heavy and loss-tolerant (likes, view counts): write-behind. Data written once and rarely read: write-around.

Common mistakes

- Updating the cache and the database in two steps with no TTL, so a failure between the steps leaves them different forever.
- Using write-behind for money or orders, where a lost write is unacceptable.
- Caching `null` results never, so lookups for things that don't exist always hit the database (an attacker can exploit this by requesting random IDs; caching a short-lived "not found" marker, called **negative caching**, fixes it).

In the interview

Q: Which caching pattern would you use for user profiles?

A: Cache-aside: read from Redis, on miss load from the database and set a TTL of a few minutes. On profile update, write to the database then delete the key. It's simple, survives cache failure, and only caches what's read.

Q: When is write-behind acceptable?

A: When speed matters more than losing a few recent writes, such as counters or analytics events. Never for payments or anything a user was told is saved.

TL;DR

Cache-aside is the default: app reads cache, falls back to the database, and deletes the key on writes. Write-through trades write speed for freshness, write-behind trades safety for speed.

Eviction policies

What it is

A cache has limited memory. When it is full and a new item arrives, it must throw something out. The rule that picks the victim is the **eviction policy**. Eviction (removing because the cache is full) is different from **expiration** (removing because the TTL ran out), though many caches do both.

Analogy: a small bookshelf next to your bed. When it's full and you buy a new book, you move one to the attic. Which one? The one you haven't touched in longest (LRU), the one you read least often (LFU), or the one that's been there longest (FIFO).

Types

- **LRU (least recently used)**: evict the item that has gone longest without being read. Implemented with a hash map plus a doubly linked list so every read moves the item to the front in O(1) time. The default choice almost everywhere because recent use is a good predictor of near-future use.
- **LFU (least frequently used)**: evict the item with the fewest reads. Better when some items are steadily popular for a long time. Weakness: an item that was hugely popular yesterday keeps a high

count and squats in the cache, so real LFU implementations decay counts over time.

- **FIFO (first in, first out)**: evict the oldest inserted item regardless of use. Simple, but can throw out a hot item just because it arrived early.
- **TTL-based**: items leave when their time to live expires. Not strictly a "who to evict when full" policy, but it bounds staleness and frees memory.
- **Random**: evict a random item. Surprisingly decent and very cheap.
- Others you may meet: **MRU** (most recently used, useful for some scan patterns), and modern designs like **W-TinyLFU** (used by the Java Caffeine library), which combine recency and frequency.

Redis lets you choose with the `maxmemory-policy` setting, for example `allkeys-lru`, `allkeys-lfu`, `volatile-ttl` or `noeviction` (refuse new writes when full). Redis uses an approximate LRU: it samples a handful of keys and evicts the best candidate among them, which saves memory compared with a full linked list.

Common mistakes

- Using `noeviction` for a cache, so it starts rejecting writes when full.
- One huge scan (like a nightly report reading every row) flushing all the hot items out of an LRU cache. LFU or scan-resistant policies avoid this.

TL;DR

When a cache is full it evicts by a rule: LRU (least recently used) is the safe default, LFU for steady favourites, FIFO and random for simplicity, TTL to bound staleness.

Cache invalidation, stale data and stampedes

What it is

Cache invalidation means removing or updating a cached copy when the source of truth changes. There is a well-known joke, credited to Phil Karlton, that "there are only two hard things in computer science: cache invalidation and naming things". It is hard because the cache and the database are two separate systems and there is no single moment where both change together.

How it works

Ways to keep caches fresh, from simplest to most precise:

1. **TTL only.** Accept that data may be up to TTL seconds old. A 60-second TTL on a product price means a price change can take up to a minute to show.
2. **Delete on write.** After writing to the database, the app deletes the cache key (cache-aside).
3. **Event-driven invalidation.** The database publishes a change event (for example through **change data capture, CDC**, which reads the database's own change log) and a consumer deletes or updates the affected cache keys. Works even when many services write to the database.
4. **Versioned keys.** Put a version in the key, such as `user:42:v7`. A write bumps the version to `v8`, so readers look for a new key and the old one simply ages out. No delete needed.
5. **CDN purge.** CDNs offer an API to purge a URL or a tag; it usually propagates worldwide in seconds.

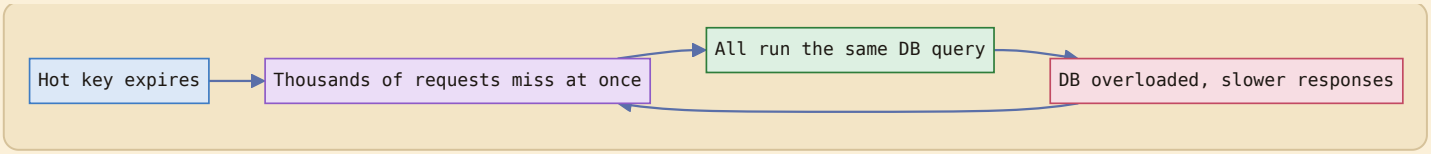
Stale data

Stale reads happen because of TTLs, lost delete messages, races between readers and writers, and replication lag (the cache was filled from a replica that was behind). Decide per feature how stale is acceptable: a follower count can be minutes old, a bank balance on the transfer screen cannot.

The thundering herd / cache stampede

A **cache stampede** (also called **thundering herd** or **dog-piling**) happens when a very popular key expires or is deleted and thousands of requests miss at the same instant. They all query the database for the same value at once, the database slows down or falls over, and since nothing has refilled the cache yet, more requests pile in.

Example: the home page key gets 10,000 reads per second. It expires. In the 200 ms it takes to rebuild, 2,000 requests all hit the database with the same expensive query.



Fixes for stampedes

- **Request coalescing / single flight**: only one request per key is allowed to rebuild. Others wait for its result (inside one process) or briefly retry the cache. CDNs such as Fastly and Cloudflare do the same at the edge and call it **request collapsing**.
- **Lock or lease**: the first miss takes a short lock in the cache (for example Redis `SET key:lock 1 NX PX 5000`, meaning "set only if not present, expire in 5 s"). Whoever gets the lock rebuilds, the rest serve the old value or wait. Facebook's Memcache paper calls this a **lease**.
- **Serve stale while revalidating**: keep the expired value and return it while one background job refreshes it. HTTP has a header for exactly this: `Cache-Control: stale-while-revalidate=30`.
- **Early or probabilistic refresh**: each reader, as expiry approaches, has a small and growing chance to refresh early, so one request refreshes before the deadline instead of everyone at the deadline.
- **TTL jitter**: add randomness to TTLs (300 s plus or minus 30 s) so keys written together don't all expire together. This also prevents a **cache avalanche**, where a mass expiry or a cache restart sends everything to the database at once.
- **Warm-up**: pre-load hot keys after a restart or deploy before sending real traffic.

Two related problems: **cache penetration** (requests for keys that don't exist anywhere, so every one hits the database; fixed with negative caching or a Bloom filter, see later) and **hot key** (one key so popular that the single cache node holding it is overloaded; fixed by replicating that key to several nodes or caching it in-process too).

Common mistakes

- Setting the same fixed TTL on millions of keys loaded at the same moment.
- Letting every miss rebuild independently with no coalescing.
- Assuming "delete after write" is atomic. If the delete fails, the stale value lives until TTL, so always have a TTL.

In the interview

Q: A celebrity's profile key expires and the database spikes. What happened and how do you fix it?

A: A cache stampede: thousands of concurrent misses on one hot key all query the database. Fix with single-flight or a short lock so only one request rebuilds, serve the stale value meanwhile, refresh hot keys early, and add jitter to TTLs.

Q: How do you keep a cache consistent with the database?

A: You can't make it perfectly consistent cheaply. Use cache-aside with delete-on-write plus a TTL as a safety net. If many services write, invalidate from a change-data-capture stream. Pick the TTL per data type based on how much staleness users can tolerate.

TL;DR

Invalidation is the hard part of caching: use delete-on-write plus TTLs, and protect hot keys from stampedes with locks, stale-while-revalidate, early refresh and TTL jitter.

Message queues

What it is

A **message queue** is a middleman that holds messages (small pieces of data describing work or an event) until another program is ready to handle them. Three roles:

- **Producer** (or publisher): the program that sends a message, for example the web server saying "send welcome email to user 42".

- **Broker**: the queue server that stores messages safely until they are taken. Examples: RabbitMQ, Amazon SQS, Apache Kafka.
- **Consumer** (or worker, subscriber): the program that takes messages and does the work, for example an email worker.

Analogy: a restaurant's order rail. The waiter (producer) clips an order ticket to the rail (broker) and goes back to the customers. The cooks (consumers) take tickets when they have a free hand. The waiter never waits for the food to be cooked, and if the kitchen is busy the tickets just queue up.

Why it matters

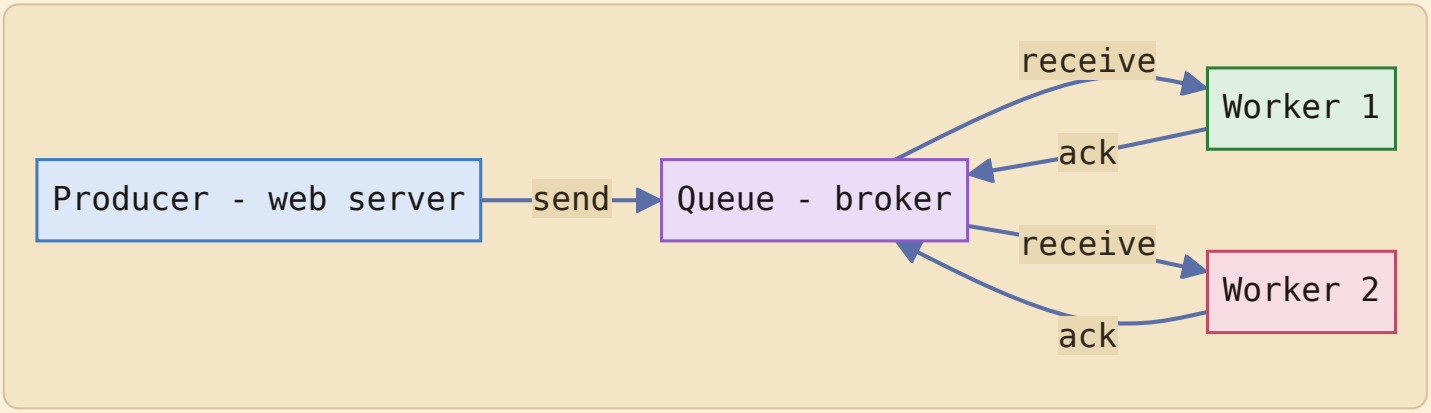
Without a queue, the web server would call the email service directly and the user would wait for the email to send. Queues give you:

1. **Faster responses**: the user gets "signed up!" in 50 ms while the slow work (sending email, resizing a photo, charging a card, generating a PDF) happens in the background. This is **asynchronous** processing: the caller doesn't wait for the result.
2. **Decoupling**: the producer doesn't need to know who consumes, how many consumers there are, or whether they are up right now. You can add a new consumer (say, analytics) without touching the producer.
3. **Absorbing spikes (load levelling)**: if 10,000 orders arrive in one minute but workers can handle 2,000 a minute, the queue holds the backlog and workers catch up over five minutes instead of the system falling over.
4. **Reliability**: if a worker crashes mid-task, the message is still in the broker and another worker picks it up.
5. **Independent scaling**: add more workers when the queue grows, remove them when it shrinks.

How it works

A typical lifecycle in a work queue (like SQS or RabbitMQ):

1. The producer sends the message. The broker writes it to disk (and often to replicas) and replies "accepted".
2. A consumer asks for messages and receives one. The broker hides it from other consumers for a while (SQS calls this the **visibility timeout**, 30 seconds by default).
3. The consumer does the work.
4. The consumer sends an **acknowledgement (ack)**: "done, delete it".
5. If no ack arrives in time (the consumer crashed or was too slow), the broker makes the message visible again and another consumer gets it. This is why messages can be delivered more than once, discussed below.

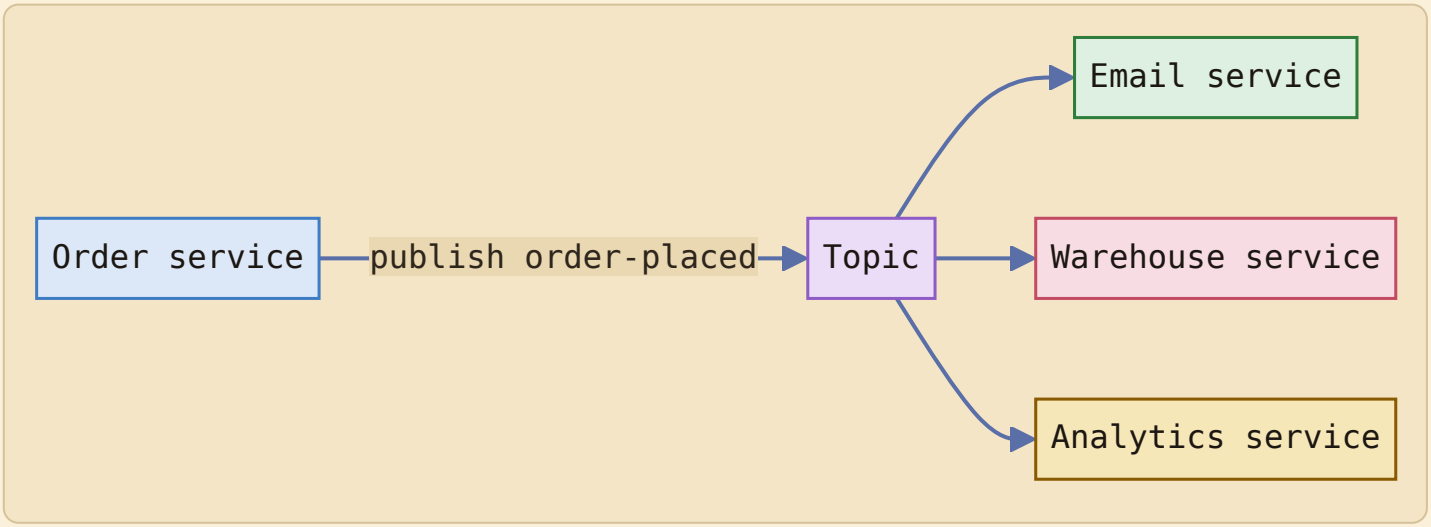


Types

Point-to-point (work queue). Each message is processed by exactly one consumer out of many. Workers compete for messages (the **competing consumers** pattern). Used for jobs: "resize this image" should happen once, not once per worker.

Publish/subscribe (pub/sub). A message is published to a **topic** (a named channel, like "order-placed") and **every** subscriber gets its own copy. The order service publishes "order 9 placed" once, and the email

service, the warehouse service and the analytics service each receive it. Used for events that many parts of the system care about.



Many systems combine both: a topic fans out to several queues, and each queue has competing workers (for example Amazon SNS topic feeding several SQS queues, or a RabbitMQ fanout exchange feeding several queues).

Trade-offs

- You gain resilience and speed but lose the immediate answer: the user can't be told "email sent", only "email will be sent".
- Debugging is harder because work happens later, somewhere else. You need tracing and monitoring of **queue depth** (how many messages are waiting) and **consumer lag** (how far behind consumers are).
- Messages may arrive twice or out of order (see delivery guarantees), so consumers must be written carefully.
- The broker is another system to run and size.

Common mistakes

- Using a queue for something the user must see immediately (like "is my password correct?").
- Not monitoring queue depth, so a dead worker goes unnoticed while millions of messages pile up.
- Putting huge payloads (videos) in messages. Store the file in object storage and send its URL in the message instead (the **claim check** pattern).

TL;DR

A message queue lets producers hand work to consumers through a broker, so slow tasks run in the background, spikes are absorbed and services are decoupled. Point-to-point gives each message to one worker, pub/sub gives every subscriber a copy.

Event streaming: topics, partitions, consumer groups and offsets

What it is

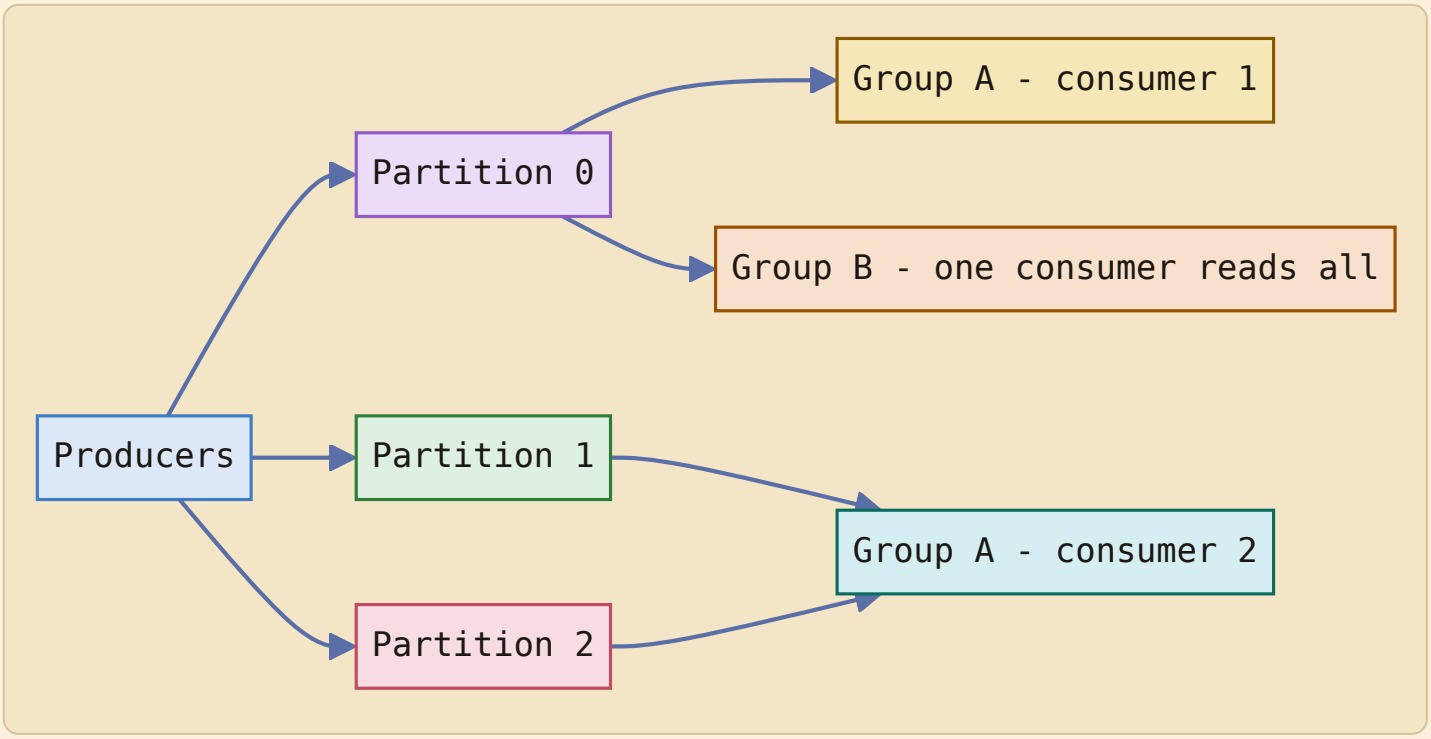
Classic queues delete a message once it is acknowledged. An **event streaming platform** (also called a **log-based broker**) such as **Apache Kafka** instead keeps every message in an append-only **log** (a file you only add to the end of) for a set retention period (Kafka's default is 7 days), whether or not anyone has read it. Consumers just remember how far they have read. Many independent consumers can read the same log, and a consumer can rewind and re-read history.

Analogy: a classic queue is a to-do pile where you throw away each note once it's done. A log is a newspaper archive: every reader keeps a bookmark, the papers stay on the shelf, and a new reader can start from last week's issue.

How it works

Kafka vocabulary, one term at a time:

- **Topic**: a named stream of events, such as `payments`.
- **Partition**: a topic is split into partitions, each an ordered log stored on one broker (with copies on others for safety). Partitions are how Kafka scales: a topic with 12 partitions can be written and read by 12 machines in parallel.
- **Key**: each message can carry a key, such as a user ID. Kafka hashes the key to choose the partition, so all events for user 42 always land in the same partition and stay in order.
- **Offset**: the position number of a message within a partition: 0, 1, 2, ... It never changes. Ordering is guaranteed only **within** one partition, not across the whole topic.
- **Consumer group**: a set of consumers sharing the work for one application. Kafka assigns each partition to exactly one consumer in the group. With 12 partitions and 4 consumers, each consumer reads 3 partitions. A second group (say, analytics) gets its own full copy of the stream independently.
- **Committed offset**: each group periodically records "for partition 3 I have processed up to offset 1,500". After a crash, the replacement consumer resumes from the committed offset.
- **Rebalance**: when a consumer joins or leaves the group, partitions are reassigned.
- **Consumer lag**: newest offset minus committed offset, the main health metric.



Note that adding more consumers than partitions doesn't help: extra consumers sit idle. Choose the partition count with future parallelism in mind.

Types

	Classic queue (RabbitMQ, SQS)	Event stream (Kafka, Kinesis, Pulsar)
After consumption	message deleted	message kept until retention ends
Re-reading history	no	yes, reset the offset
Many independent readers	needs one queue per reader	natural, one group per reader
Ordering	weak (FIFO queues exist)	strict per partition
Best for	background jobs, task distribution	event pipelines, analytics, CDC, replay

RabbitMQ is a traditional broker with flexible routing: producers send to an **exchange**, which routes to queues by rules (direct, fanout, topic patterns). **Amazon SQS** is a fully managed queue: standard queues give at-least-once delivery and best-effort ordering, FIFO queues add strict ordering and de-

duplication. **Kafka** was created at LinkedIn and is now an Apache project; it now runs its own Raft-based metadata quorum (KRaft) instead of ZooKeeper, and Kafka 4.0 (2025) removed ZooKeeper support entirely. **Amazon Kinesis**, **Apache Pulsar** and **Redpanda** are similar streaming systems.

In the interview

Q: Kafka or RabbitMQ/SQS?

A: For background jobs where each task runs once and is then forgotten, a queue like SQS or RabbitMQ is simpler. For a high-volume stream of events that several services must read independently, need to replay, or need ordered per key, choose Kafka.

Q: How do you keep all events for one order in order?

A: Use the order ID as the message key, so they all go to the same partition, which is strictly ordered, and have one consumer per partition.

TL;DR

Kafka-style streams keep an ordered, replayable log split into partitions. Consumer groups share partitions and track their position with offsets. Ordering holds only within a partition.

Delivery guarantees, idempotency and retries

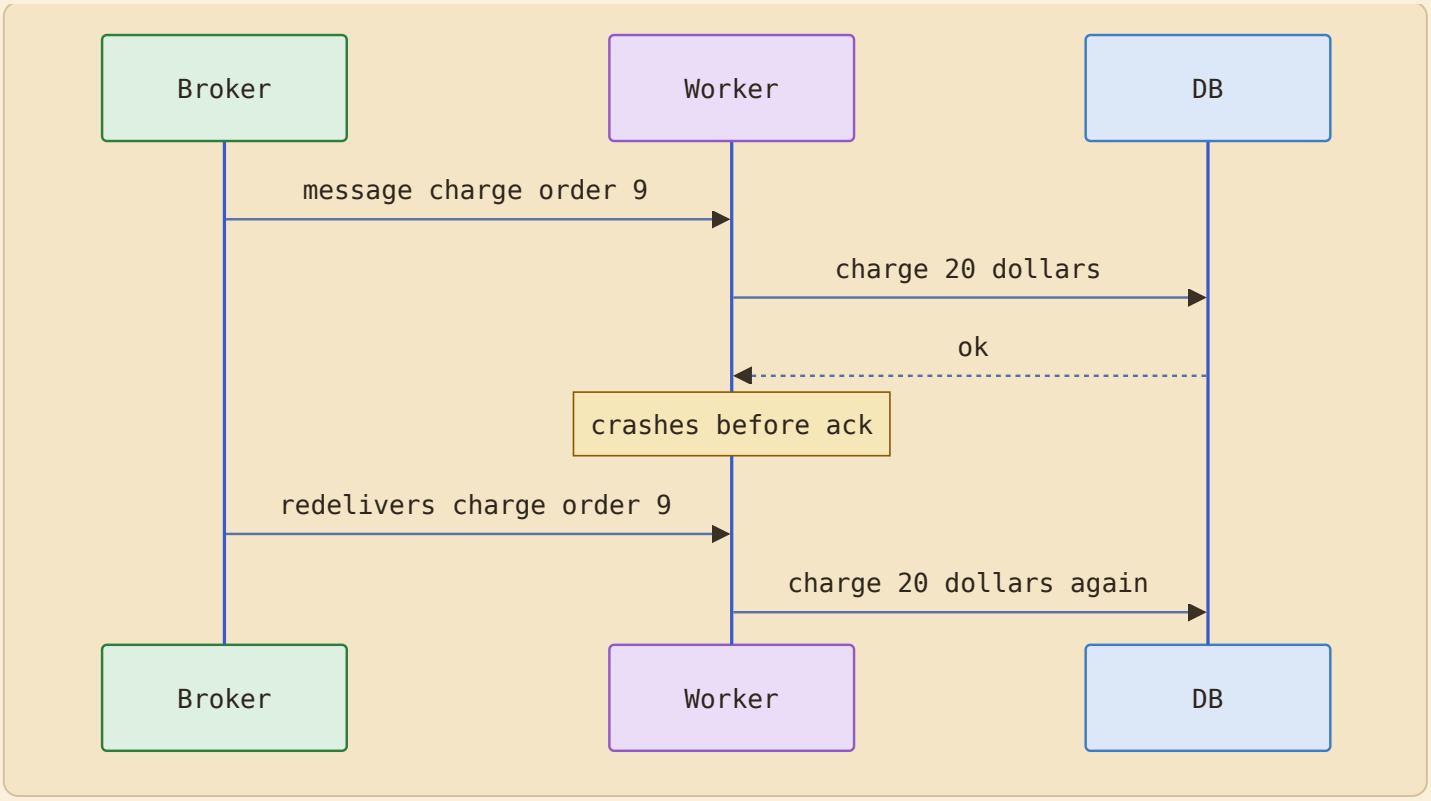
What it is

Networks drop packets and machines crash, so a broker can never be sure a message was processed exactly once. A **delivery guarantee** states what the system promises:

- **At-most-once**: each message is delivered zero or one times. If anything fails, it is lost, never repeated. Done by acknowledging (or committing the offset) **before** processing. Fine for metrics where a lost data point doesn't matter.
- **At-least-once**: each message is delivered one or more times. Never lost, but may be repeated. Done by acknowledging **after** processing: if the consumer crashes after doing the work but before the ack, the message comes back and is processed again. This is the default for most systems.
- **Exactly-once**: each message affects the result exactly one time. True exactly-once **delivery** over an unreliable network is impossible in general. What systems offer is **exactly-once processing** or **effectively-once**: messages may be delivered more than once, but duplicates are detected and ignored, so the outcome is as if each was processed once.

How it works

Why duplicates are unavoidable with at-least-once:



The fix is to make the consumer **idempotent**.

Idempotency means doing an operation twice has the same effect as doing it once. "Set balance to 100" is idempotent; "add 20 to balance" is not. "Delete order 9" is idempotent; "create a new order" is not.

How to make non-idempotent work idempotent:

1. Give each operation a unique **idempotency key**, such as a UUID the client generates once for "pay for order 9" and reuses on every retry.
2. Before doing the work, the consumer (or API) checks a table of processed keys.
3. If the key is new, do the work **and** record the key in the same database transaction.
4. If the key was seen, skip the work and return the saved result.

Payment APIs such as Stripe accept an `Idempotency-Key` HTTP header for exactly this reason: a phone that loses signal and retries "pay $20" must not charge twice. Other techniques: **upserts** (insert or overwrite by a unique ID), unique constraints in the database, and conditional updates ("set status to SHIPPED only if it is PAID").

Kafka provides exactly-once semantics inside Kafka (since version 0.11, 2017) by combining an **idempotent producer** (the broker drops duplicate sends using a producer ID and sequence number) with **transactions** that write output messages and the consumer's offset atomically. Once a side effect leaves Kafka (sending an email, calling a bank), you are back to needing idempotency in that system. SQS FIFO queues de-duplicate messages with the same deduplication ID inside a 5-minute window.

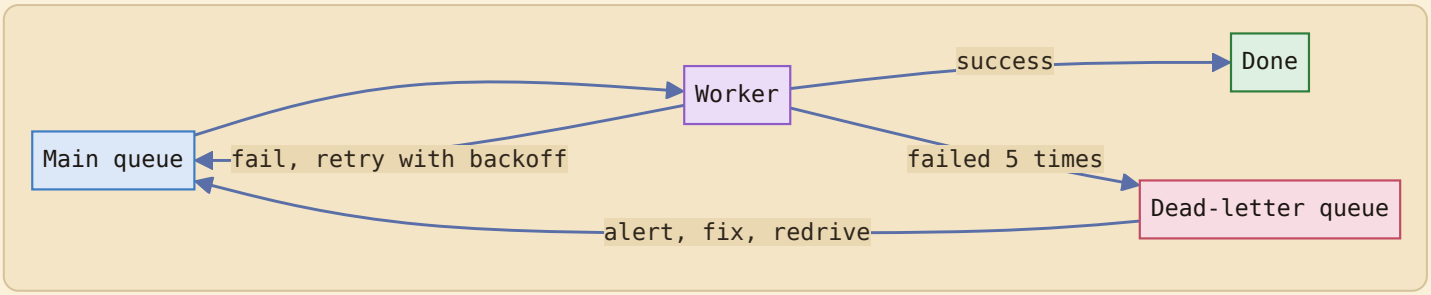
Ordering

Ordering costs throughput, because strict global order means one line of processing. Practical systems guarantee order **per key** (per user, per order) by routing a key to one partition or FIFO group. Retries can also break order: if message 5 fails and is retried after message 6 succeeds, they are now out of order. Consumers that care can carry a version number and ignore an older version arriving after a newer one.

Retries, dead-letter queues and backpressure

Retries. When processing fails because of something temporary (a timeout, a database briefly unavailable), try again. Use **exponential backoff**: wait 1 s, then 2 s, 4 s, 8 s, so a struggling dependency isn't hammered, and add **jitter** (random variation) so thousands of clients don't retry at the same instant. Cap the number of attempts. Part 5 covers this in depth.

Dead-letter queue (DLQ). A message that keeps failing (a **poison message**, for example malformed JSON, or an order referring to a deleted product) would otherwise be retried forever and block or waste workers. After N attempts (say 5), the broker moves it to a separate dead-letter queue. Engineers get an alert, inspect it, fix the bug, and **redrive** (replay) the messages back to the main queue. SQS, RabbitMQ and most Kafka frameworks support DLQs.



Backpressure. When consumers can't keep up, something must give. Backpressure is any way for a slow downstream part to signal "slow down" to the upstream part, instead of silently drowning. Options:

- **Buffer**: let the queue grow (fine for spikes, dangerous if it never drains).
- **Block or throttle the producer**: the broker or a bounded queue refuses or delays new messages, for example HTTP `429 Too Many Requests` or `503 Service Unavailable`.
- **Drop**: discard low-priority messages (load shedding).
- **Scale out consumers**: autoscale workers based on queue depth.

TCP flow control (Part 2) and reactive streams libraries are low-level forms of backpressure.

Common mistakes

- Assuming "exactly-once" in a vendor's docs covers your email sends and database writes. It rarely does; design idempotent consumers anyway.
- Retrying instantly in a tight loop, making an outage worse.
- No DLQ, so one poison message blocks a partition (Kafka consumers process a partition in order, so a stuck message stalls everything behind it).
- Unbounded in-memory queues that grow until the process runs out of memory.

Real-world examples

- Stripe's API idempotency keys, described in its public docs.
- Uber and LinkedIn run Kafka at trillions of messages per day, relying on per-key partitioning for ordering.
- AWS SQS redrive policies send messages to a DLQ after a configured `maxReceiveCount`.

In the interview

Q: How do you guarantee a payment is not charged twice when the queue is at-least-once?

A: Make the consumer idempotent. Each payment carries an idempotency key; the worker records processed keys in the same transaction as the charge and skips any key it has seen. Duplicates are then harmless.

Q: What is a dead-letter queue?

A: A side queue where messages go after failing a set number of times, so a poison message doesn't block or waste workers. It's monitored, and messages are replayed after the bug is fixed.

TL;DR

Assume at-least-once delivery, make consumers idempotent with idempotency keys, retry with backoff and jitter, park repeated failures in a dead-letter queue, and apply backpressure when consumers fall behind.

Batch vs stream processing

What it is

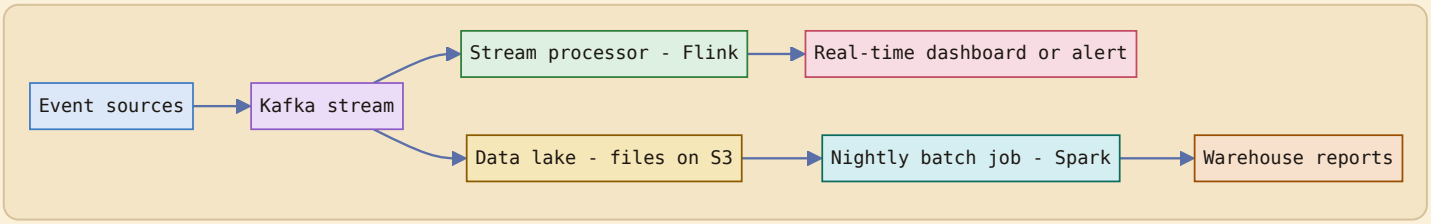
Once data is flowing through a system, you often need to compute things from it: daily revenue, recommendations, fraud scores, search indexes.

- **Batch processing** takes a large, bounded pile of data that has already been collected (say, yesterday's 2 TB of logs) and processes it all in one job, usually on a schedule (nightly, hourly). Think of a bakery that bakes the whole day's bread before the shop opens.
- **Stream processing** handles each event (or small window of events) continuously as it arrives, within milliseconds or seconds. Think of a barista who makes each coffee the moment it is ordered.

How it works

MapReduce in one paragraph. MapReduce, published by Google in 2004 and copied by Apache Hadoop, is the classic batch model. The input files are split across many machines. In the **map** step, each machine reads its chunk and emits key-value pairs (for counting words: `("cat", 1)` for every "cat"). In the **shuffle** step, the framework sends all pairs with the same key to the same machine. In the **reduce** step, each machine combines the values for its keys (`("cat", 1+1+1) = ("cat", 3)`). If a machine dies, its piece is simply re-run. Modern engines such as **Apache Spark** keep the same idea but hold intermediate data in memory and are much faster; SQL warehouses like BigQuery, Snowflake and Redshift run batch-style queries over huge tables.

Stream processing engines such as **Apache Flink**, **Kafka Streams** and **Spark Structured Streaming** read from a stream (usually Kafka), keep state (such as running counts per user), and group events into **windows**, for example "orders per minute" (a tumbling one-minute window) or "logins in the last 5 minutes, updated every 10 seconds" (a sliding window). A subtle point: events can arrive late or out of order, so engines distinguish **event time** (when it happened) from **processing time** (when it arrived) and use **watermarks** to decide when a window is complete.



Trade-offs

	Batch	Stream
Latency of results	minutes to hours	milliseconds to seconds
Input	bounded, complete	unbounded, arriving
Complexity	simpler, easy to re-run	harder: state, late events, failures
Typical uses	reports, billing, ML training, backfills	fraud detection, live metrics, alerts

Two named architectures: **Lambda architecture** runs both a batch layer (accurate, slow) and a speed layer (fast, approximate) and merges them. **Kappa architecture** uses only a stream and handles reprocessing by replaying the log from the start. Many companies now prefer Kappa-style designs to avoid writing logic twice.

In the interview

Q: Should a "trending hashtags" feature use batch or streaming?

A: Streaming, because users expect it to reflect the last few minutes. Count hashtags in sliding windows with a stream processor reading from Kafka, possibly with an approximate structure like count-min sketch, and recompute exact daily numbers in batch.

TL;DR

Batch processes a complete pile of data on a schedule (MapReduce, Spark); streaming processes events continuously as they arrive (Flink, Kafka Streams). Choose by how fresh the result must be.

Event-driven architecture, event sourcing and CQRS

What it is

Event-driven architecture (EDA) is a style where services communicate by publishing **events**, facts about something that already happened ("OrderPlaced", "PaymentFailed"), rather than calling each other with commands ("ship this now"). The publisher doesn't know or care who listens.

Analogy: a town crier announces "the market opens at noon" to the square. Anyone who cares acts on it; the crier doesn't keep a list of who should be told.

An **event** reports something that is already done, so a listener can ignore it but cannot say no to it. A **command** is a request that the receiver may reject. Name them to match: events in the past tense (`OrderPlaced`), commands as orders (`ShipOrder`).

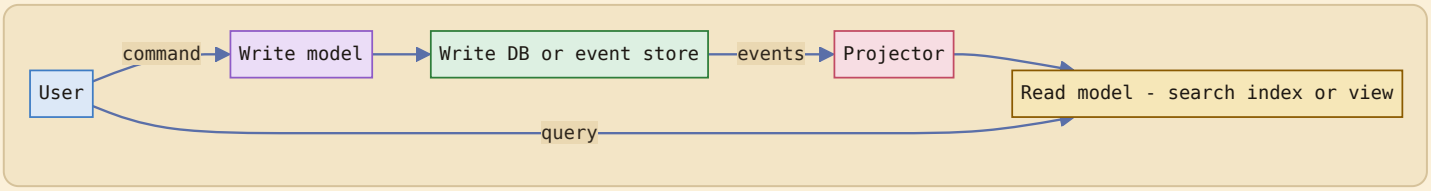
How it works

When an order is placed, the order service saves it and publishes `OrderPlaced`. The payment, inventory, email and analytics services each subscribe and react. Adding a loyalty-points service later needs no change to the order service.

A practical problem: the service must both update its database and publish the event. If it does one and crashes before the other, they disagree (the **dual write** problem). The standard fix is the **transactional outbox**: in the same database transaction as the business change, insert the event into an `outbox` table. A separate relay process (or CDC tool like Debezium) reads the outbox and publishes to the broker, retrying until it succeeds. Combined with idempotent consumers, nothing is lost.

Event sourcing. Instead of storing only the current state (balance = 70), store the full sequence of events that led there (Deposited 100, Withdrew 50, Deposited 20) as the source of truth. Current state is computed by replaying events, usually with periodic **snapshots** so you don't replay millions every time. Benefits: a complete audit trail, the ability to rebuild state or answer "what was the balance last Tuesday?", and easy new projections. Costs: more complex, events can't be changed once written so schema evolution is harder, and queries need separate read models. Bank ledgers and accounting books have always worked this way.

CQRS (Command Query Responsibility Segregation). Use one model (and often one database) for writes (commands) and a different model for reads (queries). Writes go to a normalized database; events or CDC update one or more read-optimised views, such as a denormalized table, a search index in Elasticsearch, or a cache. Reads scale independently and each screen gets data in exactly the shape it needs. Cost: the read side lags the write side slightly (eventual consistency), and there are more moving parts. CQRS pairs naturally with event sourcing but doesn't require it.



Trade-offs

EDA gives loose coupling and easy extension, but flows become harder to follow (no single place shows the whole order process), and you live with eventual consistency, duplicates and ordering issues. Use it where many services react to the same facts; keep simple request-response for things needing an immediate answer.

In the interview

Q: How do you reliably update the database and publish an event?

A: Transactional outbox: write the event to an outbox table in the same transaction as the change, then a relay publishes it with retries. Consumers are idempotent because the relay may publish twice.

Q: What is CQRS and when would you use it?

A: Separate write and read models. Use it when reads and writes have very different shapes or scale, for example writes to a relational DB and searches served from Elasticsearch kept up to date by events.

TL;DR

Event-driven systems react to published facts instead of direct calls. Event sourcing stores the events themselves as the truth; CQRS splits the write model from read-optimised views. Both add flexibility at the cost of eventual consistency and complexity.

Consistency models

What it is

When data is copied to several machines (replication, Part 3), a read might go to a copy that hasn't received the latest write yet. A **consistency model** is the promise a system makes about what a read can return after a write. It is a contract between the database and the programmer: "if you do X, you will never see Y".

Note: this "consistency" is different from the C in ACID (Part 3), which means "transactions keep the data valid according to your rules". Same word, different idea.

Analogy: a family shares a shopping list copied onto three fridges in three houses. Strong consistency means that the moment anyone adds "milk", every fridge shows it before anyone can read any list. Eventual consistency means the copies are updated by phone calls over the next few minutes; for a while, different fridges show different lists, but if people stop editing, all three end up the same.

Types

From strongest (easiest to reason about, most expensive) to weakest:

Linearizability (strong consistency). The system behaves as if there is only one copy of the data and every operation happens at a single instant between its start and its end. Once any read has returned the new value, no later read anywhere can return the old one. Plain words: "it looks like one machine". Needed for things like a username uniqueness check, a lock, or a bank balance check before withdrawal. Cost: every write must reach a majority or a leader before it's confirmed, which adds latency, especially across regions, and the system may refuse requests during a network partition.

Sequential consistency. All clients see all operations in the same order, and each client's own operations appear in the order it issued them, but that order needn't match real time. Mostly a term you'll meet in theory and CPU memory models.

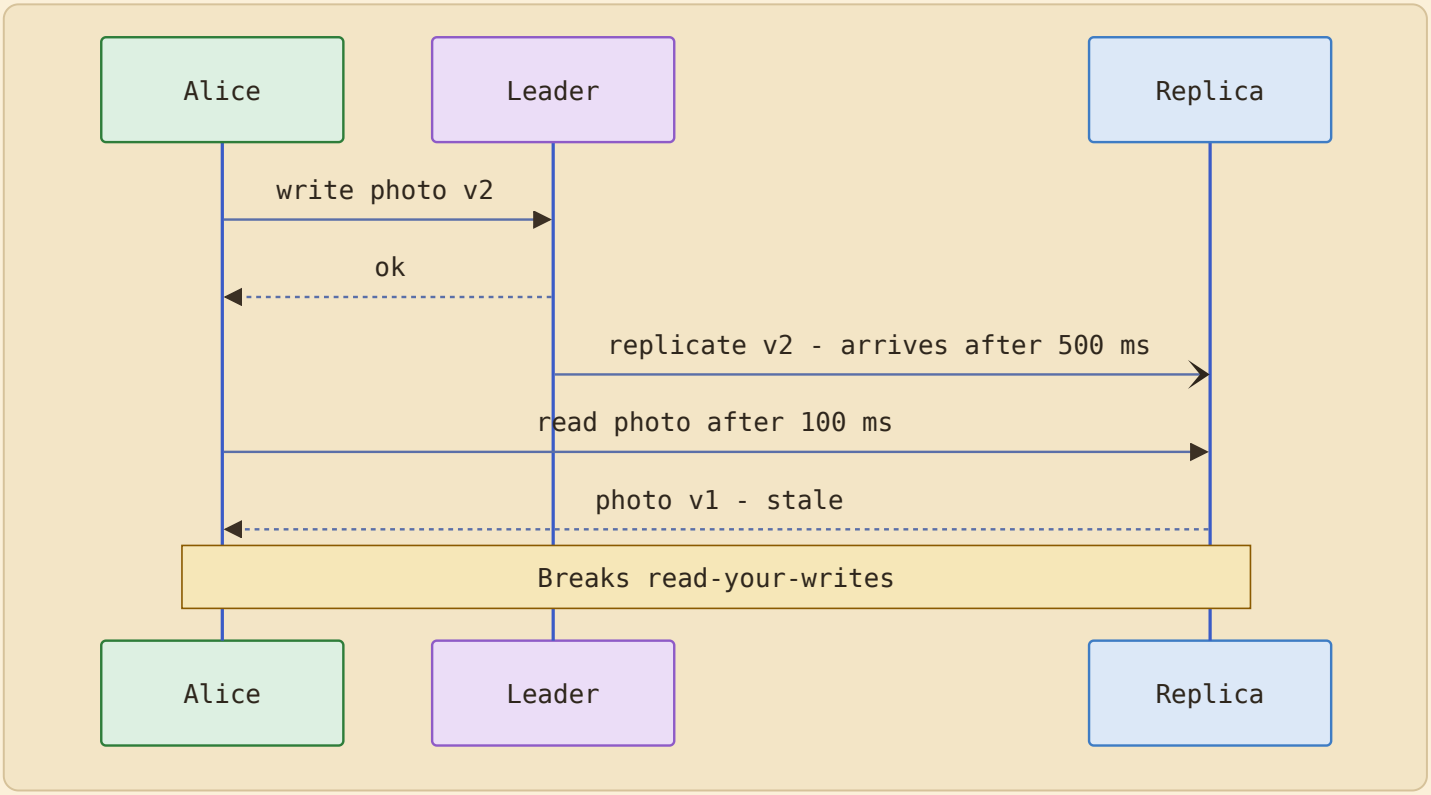
Causal consistency. If one operation could have caused another, everyone sees them in that order. If Alice posts "I'm engaged!" and Bob replies "Congratulations!", nobody should see Bob's reply without Alice's post. Unrelated operations (Carol's post in another thread) may appear in different orders to different people. Research has shown that causal consistency (in a slightly strengthened form) is about the strongest model a system can offer while staying available during a network partition, so it's attractive for global systems.

Session guarantees (also called client-centric consistency) are promises to a single user, cheap to provide and often enough:

- **Read-your-writes** (read-your-own-writes): after you change your profile photo, you always see the new photo, even if other users see the old one for a few seconds. Implemented by reading your own data from the leader for a short time after writing, or by tracking the write's position and only reading from replicas that have caught up.
- **Monotonic reads**: once you've seen a value, you never see an older one later. Without it, refreshing a page could show a comment, then hide it, then show it, because each refresh hit a different replica. Implemented by sticking each user to one replica.
- **Monotonic writes**: your writes are applied in the order you made them.

- **Writes-follow-reads**: if you read something and then write based on it, your write is ordered after what you read.

Eventual consistency. If no new writes happen, all replicas will eventually return the same value. No promise about how long "eventually" is (usually milliseconds to seconds) or what you see in the meantime. DNS is a classic example: change a record and different resolvers return old or new answers until TTLs expire. Cheap, fast, and highly available.



Strong vs eventual in practice. Many databases let you choose per request. DynamoDB reads are eventually consistent by default, and you can ask for a strongly consistent read at double the read cost. Cassandra lets you set a consistency level per query (ONE, QUORUM, ALL). Google Spanner offers externally consistent (linearizable) transactions globally.

Trade-offs

Model	Promise	Cost	Good for
Linearizable	looks like one copy	high latency, less available	locks, balances, unique names
Causal	cause before effect	moderate	comments, chat, social feeds
Read-your-writes	you see your own changes	low	profiles, settings
Eventual	converges eventually	lowest	likes, view counts, DNS

Common mistakes

- Saying "eventually consistent" as if it means "consistent after a short fixed time". It promises no bound.
- Assuming reading from replicas is safe for a flow like "save, then redirect to the page that shows it". Users will see their own change missing. Use read-your-writes.
- Confusing consistency in CAP (linearizability) with consistency in ACID (valid data).

In the interview

Q: Where is eventual consistency acceptable?

A: Where brief staleness is harmless: like counts, follower counts, view counts, recommendations, search indexes. Not for balances, inventory at checkout, or uniqueness checks.

TL;DR

Consistency models describe what reads may return after writes, from linearizable (acts like one copy) through causal and session guarantees to eventual (converges sometime). Stronger means simpler reasoning but slower and less available.

CAP theorem and PACELC

What it is

The **CAP theorem**, proposed by Eric Brewer in 2000 and proved by Gilbert and Lynch in 2002, says that a distributed data store facing a **network partition** must choose between consistency and availability.

- **C, consistency**: here it means linearizability: every read sees the latest write, as if there were one copy.
- **A, availability**: every request to a working (non-crashed) node gets a non-error answer, without waiting forever.
- **P, partition tolerance**: the system keeps operating even when the network between nodes drops or delays messages.

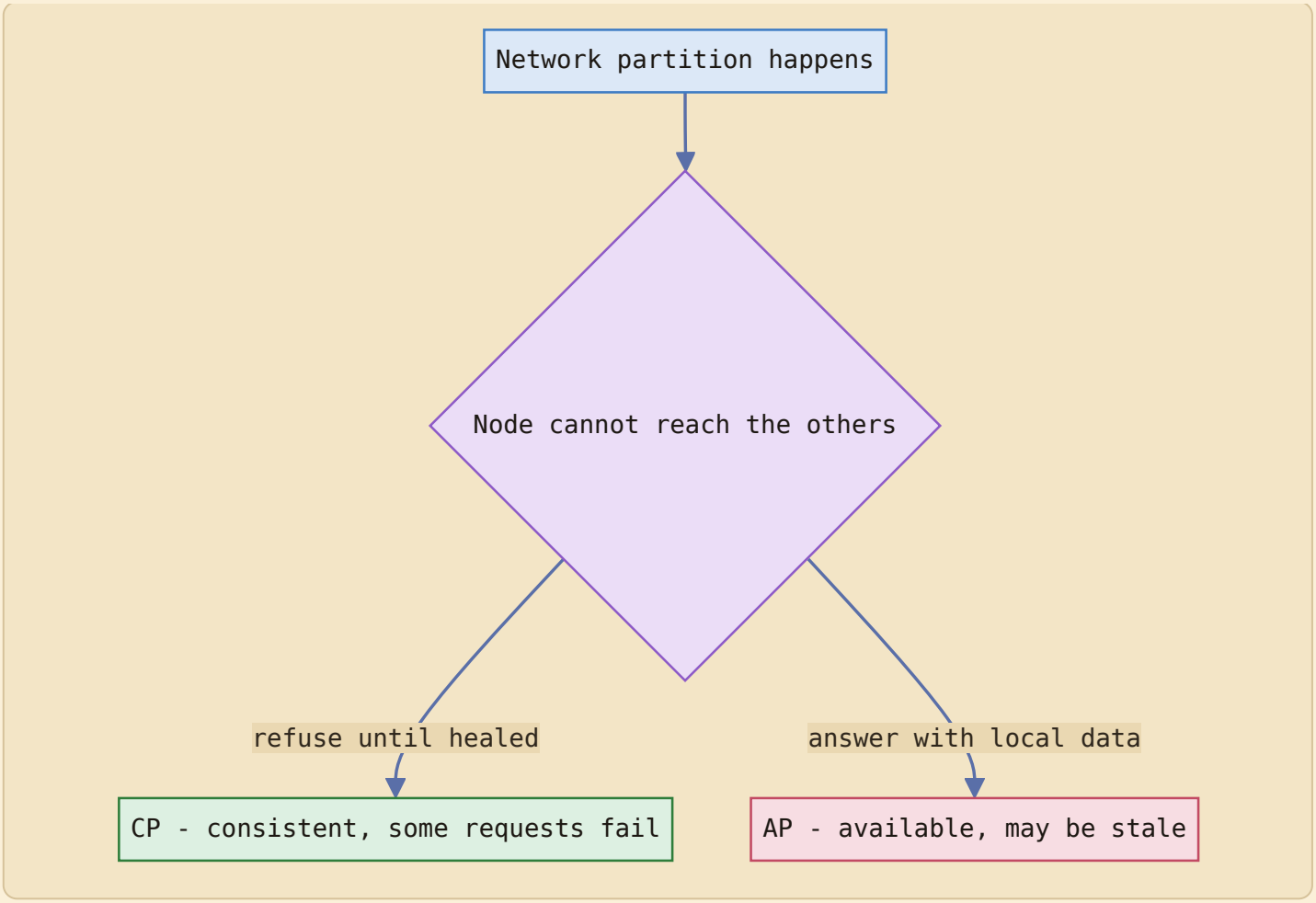
A **network partition** is when the network splits the nodes into groups that can't talk to each other, even though each node is still running. Example: the link between your US and EU data centres fails. Each side is healthy and still receiving user requests, but neither can tell what the other is doing.

How it works

Suppose two replicas, one in the US and one in the EU, and the link between them is cut. A user in the EU writes `x = 2`. Now a user in the US reads `x`. The US replica has two options:

1. **Refuse or wait** (return an error or block) because it can't confirm it has the latest value. It stays **consistent** but is not **available**. This is **CP**.
2. **Answer with what it has** (`x = 1`, old). It stays **available** but is not **consistent**. This is **AP**.

There is no third option: it can't know about a write it can't hear about. Because partitions in real networks are not optional (cables get cut, switches fail, cloud zones lose connectivity), the real choice is between C and A **during a partition**. When the network is healthy, a system can provide both.



Types

CP examples: systems built on consensus or a single leader, such as ZooKeeper, etcd, Consul, Google Spanner, HBase, and MongoDB with majority reads and writes. On the minority side of a partition they stop accepting writes.

AP examples: Cassandra, Riak, DynamoDB-style leaderless stores at low consistency levels, CouchDB, DNS. Both sides keep serving and reconcile afterwards (using timestamps, version vectors or merge rules).

Many systems are tunable, so "Cassandra is AP" really means "Cassandra with consistency level ONE behaves AP".

PACELC

CAP only talks about the rare partition case. **PACELC**, proposed by Daniel Abadi in 2010, adds the everyday case: **if** there is a **P**artition, choose **A** or **C**; **E**lse (normal operation), choose **L**atency or **C**onsistency. Even with a perfect network, keeping replicas strongly consistent means waiting for other machines, which costs latency.

- **PA/EL**: Cassandra, DynamoDB (default settings): available in partitions, low latency normally, consistency relaxed both times.
- **PC/EC**: Spanner, traditional single-leader setups with synchronous replication: consistent always, pay in availability and latency.
- **PA/EC** and **PC/EL** exist too (for example, MongoDB is often described as PA/EC under some settings; descriptions vary with configuration).

Common mistakes (misreadings of CAP)

- **"Pick any two of three."** You can't pick CA in a distributed system, because you can't opt out of partitions. A single-node database is "CA" only because it isn't distributed.
- **"AP means no consistency ever."** It means the system may return stale data during a partition. Normally it's usually consistent within milliseconds.

- **"CP means the system goes down."** Only requests that can't be safely served fail, typically on the minority side.
- **Mixing CAP's C with ACID's C.** CAP's C is linearizability.
- **Treating CAP as a whole-system label.** Different features of one product can make different choices (a shopping cart can be AP, checkout CP).

In the interview

Q: Explain CAP in one breath.

A: When the network splits, a replicated system must either refuse some requests to stay consistent (CP) or keep answering with possibly stale data (AP). Partitions are a fact of life, so the real choice is C or A during them.

Q: Is your design CP or AP?

A: Say per component: payment and inventory reservation are CP (we'd rather fail than double-sell), while the product catalogue, likes and feeds are AP (stale is fine, downtime is not). Then mention PACELC: even without partitions, we accept slightly stale reads for lower latency on the AP parts.

TL;DR

CAP: during a network partition, choose consistency (refuse) or availability (answer, maybe stale). PACELC adds: otherwise, choose latency or consistency.

Quorums

What it is

A **quorum** is the minimum number of replicas that must take part in an operation for it to count. In **leaderless replication** (Part 3: Dynamo, Cassandra, Riak), clients write to and read from several replicas at once, and quorums decide how many must answer.

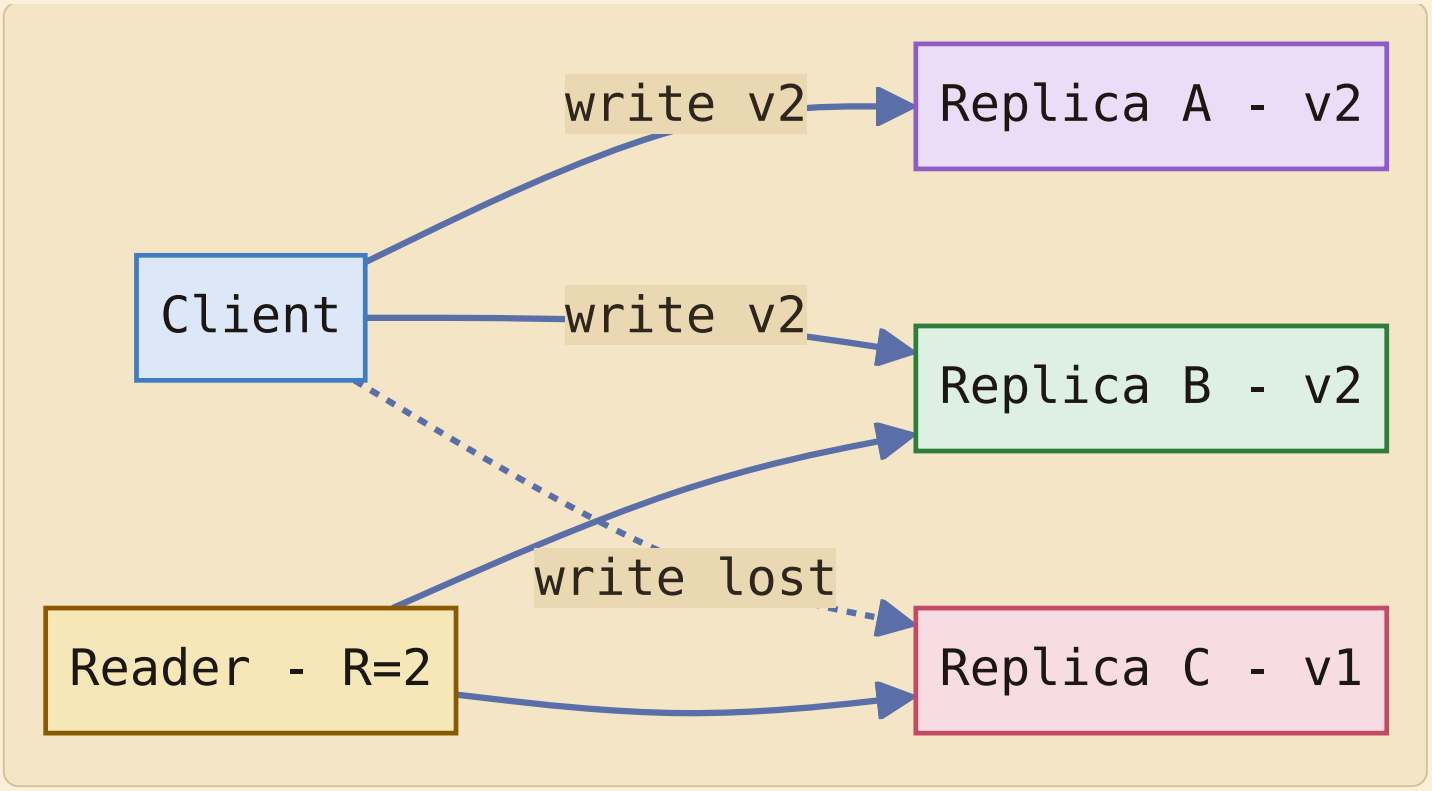
Three numbers:

- **N**: number of replicas that store each piece of data (often 3).
- **W**: number of replicas that must confirm a write before it's considered successful.
- **R**: number of replicas that must answer a read.

How it works

The key rule: if **R + W > N**, every read set overlaps every write set in at least one replica, so a read is guaranteed to reach at least one replica holding the latest successful write. The reader picks the newest version among the answers (by version number or timestamp).

Example with N = 3, W = 2, R = 2: a write reaches replicas A and B. A later read asks B and C. B has the new value, so the reader sees it. 2 + 2 = 4 > 3.



The reader gets v2 from B and v1 from C and keeps v2. It may also fix C on the spot (**read repair**). Background **anti-entropy** processes compare replicas (often with Merkle trees, below) to fix the rest.

Common settings for N = 3:

W	R	Effect
2	2	balanced, tolerates 1 replica down for both reads and writes
3	1	fast reads, any replica down blocks writes
1	3	fast writes, any replica down blocks reads
1	1	fastest, R + W = 2 is not > 3, so reads may be stale

A **majority quorum** is more than half: 2 of 3, 3 of 5. Two majorities always overlap, which is why consensus algorithms use them. With N nodes, a majority quorum tolerates `floor((N - 1) / 2)` failures: 1 failure with 3 nodes, 2 with 5. That's why clusters use odd sizes: 4 nodes tolerate only 1 failure, the same as 3.

Sloppy quorum and hinted handoff. If some home replicas are unreachable, Dynamo-style systems may temporarily write to other nodes, which later hand the data back. This boosts availability but breaks the R + W > N guarantee for a while.

Trade-offs and caveats

Quorums give tunable consistency, but R + W > N is not full linearizability on its own: concurrent writes, sloppy quorums and failed partial writes can still produce odd results. Larger W makes writes slower and less available; larger R does the same for reads.

In the interview

Q: With 3 replicas, how do you make sure reads see the latest write?

A: Use quorums with R + W > N, for example W = 2 and R = 2, so read and write sets always overlap. That also tolerates one replica being down.

TL;DR

Quorums: write to W replicas and read from R out of N. When R + W > N, reads overlap writes and see the latest value; tune R and W to trade read speed, write speed and availability.

Consensus, leader election and distributed locks

What it is

Consensus is getting several machines to agree on one value (or one sequence of values) even when some of them crash or messages are lost. "Who is the leader?", "Was this transaction committed?", "Who holds the lock?", "What is the next entry in the log?" are all consensus problems.

Analogy: five friends choosing a restaurant over unreliable text messages, where some phones die mid-conversation. They need a method so that, once a choice is announced, nobody ends up at a different restaurant, and the group can still decide if one or two phones die.

Why it matters

Without consensus you get **split brain** (Part 3): two nodes both think they're the leader, both accept writes, and the data diverges. Consensus systems are the backbone of coordination services like **etcd** (used by Kubernetes to store all cluster state), **ZooKeeper** (used by Hadoop, older Kafka), **Consul**, and of databases like CockroachDB, TiDB, YugabyteDB and Spanner.

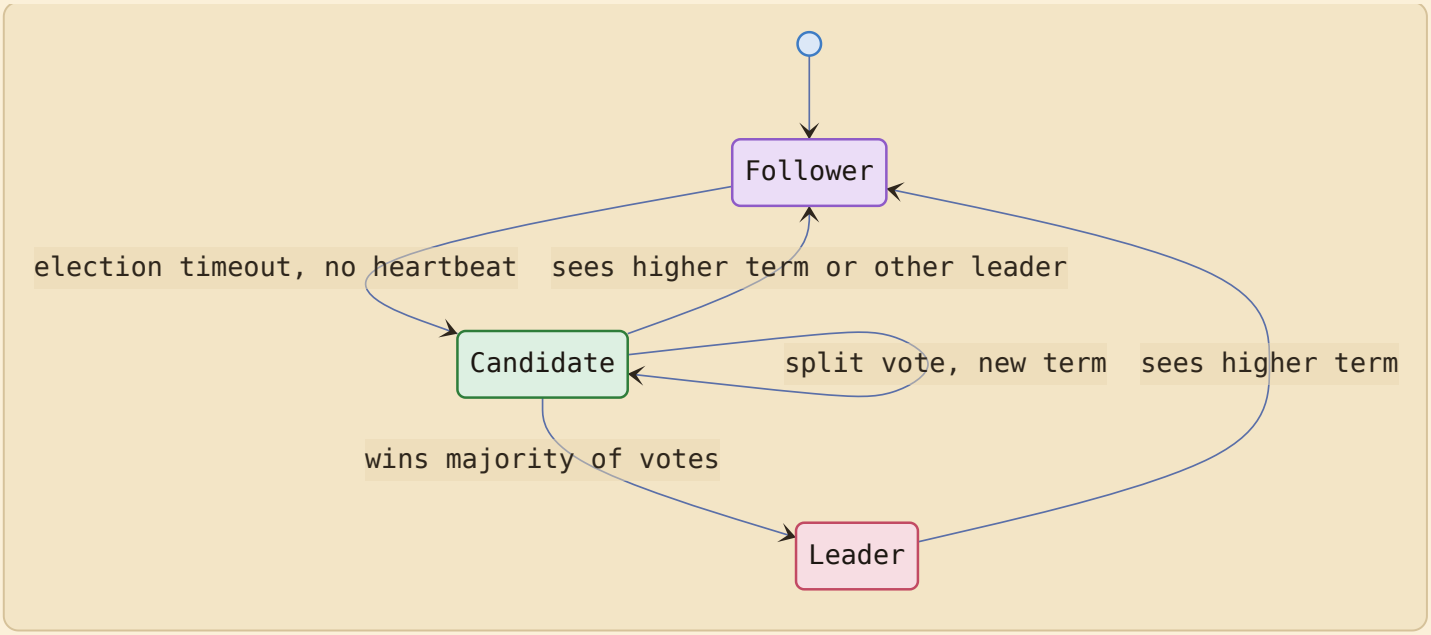
How it works

Two famous algorithms:

- **Paxos**, by Leslie Lamport (first written up around 1989–1990, formally published in 1998), the first widely known proven solution. Correct but famously hard to understand and implement.
- **Raft**, by Diego Ongaro and John Ousterhout (2014), designed to be understandable. Most new systems use Raft (etcd, Consul, CockroachDB, Kafka's KRaft).

Raft intuition, step by step:

1. Each node is a **follower**, a **candidate** or the **leader**. Time is divided into numbered **terms** (also called **epochs** or generations in other systems). Each term has at most one leader.
2. The leader sends regular **heartbeats** to followers. If a follower hears nothing for its **election timeout** (randomised, for example 150 to 300 ms, so nodes don't all time out together), it assumes the leader is dead.
3. That follower becomes a candidate, increments the term (say from 4 to 5), votes for itself, and asks others for votes.
4. Each node votes for at most one candidate per term, and only for a candidate whose log is at least as up-to-date as its own. A candidate with votes from a **majority** becomes leader for term 5.
5. Clients send writes to the leader. The leader appends the write to its log and sends it to followers. Once a majority have stored it, the entry is **committed**, applied, and the client gets "ok".
6. If an old leader from term 4 comes back, any message it sends carries term 4; nodes reject it because they've seen term 5, and it steps down. Terms are how stale leaders are detected.



Because a majority is needed, a 5-node Raft cluster keeps working with 2 nodes down; with 3 down it stops accepting writes (it chooses consistency, so it's CP). Consensus clusters are usually 3 or 5 nodes: more nodes means more fault tolerance but slower writes.

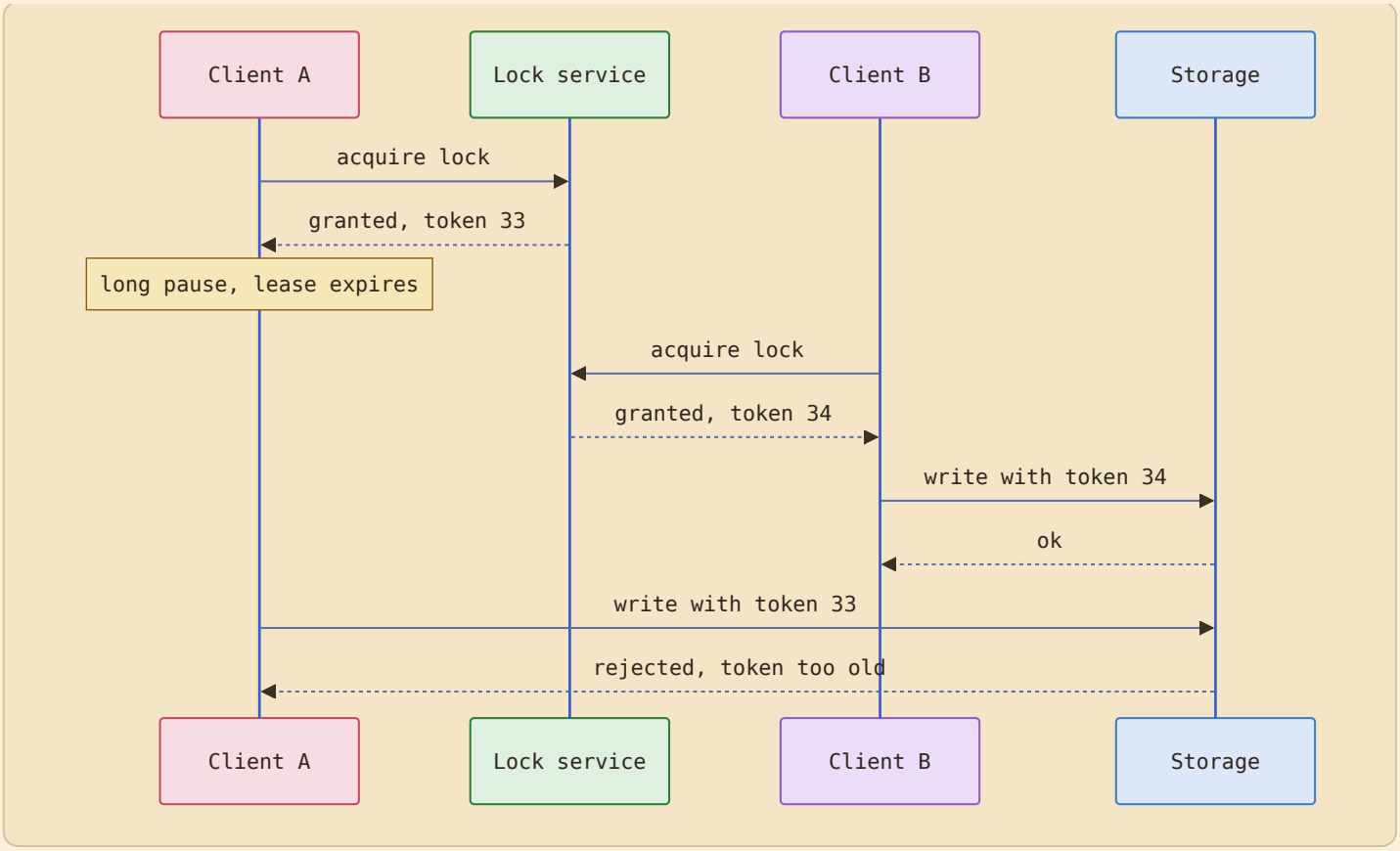
Leader election is the most common use: many replicas of a service, but only one should run the nightly billing job or act as database primary. You use a coordination service (etcd, ZooKeeper, Consul) rather than writing your own consensus.

Distributed locks and fencing tokens

A **distributed lock** ensures that only one process across many machines does something at a time (for example, only one worker processes refund 77). It's usually a key in a coordination service or Redis with a **lease** (an expiry time) so that a crashed holder doesn't block everyone forever.

The danger: a process takes the lock with a 10-second lease, then pauses (a long garbage-collection pause, a slow disk, its virtual machine being moved to another host) for 15 seconds. The lease expires, another process takes the lock and starts writing. The first process wakes up, still believes it holds the lock, and writes too. Now two "lock holders" are writing.

The fix is a **fencing token**: every time the lock is granted, the lock service hands out a number that always increases (33, then 34). The holder sends its token with every write to the storage system, and storage rejects any write with a token lower than one it has already seen. The paused process with token 33 is rejected because 34 has already written.



ZooKeeper's zxid and etcd's revision numbers can serve as fencing tokens. A simple Redis `SET lock NX PX` lock is fine for efficiency (avoiding duplicate work) but should not be trusted alone for correctness, because it provides no fencing token.

Common mistakes

- Writing your own consensus or leader election with timestamps and hope. Use etcd, ZooKeeper or Consul.
- Running consensus with 2 or 4 nodes (no extra fault tolerance over 1 or 3).
- Trusting a lease-based lock without fencing for data correctness.
- Spreading a consensus cluster across continents without realising every write now waits for an intercontinental round trip (70+ ms).

In the interview

Q: How do you ensure only one instance runs a scheduled job?

A: Leader election through a coordination service such as etcd or ZooKeeper, or a lock with a lease. Make the job idempotent and, if it writes shared data, pass a fencing token so a stale leader's writes are rejected.

Q: Why are consensus clusters 3 or 5 nodes?

A: They need a majority to make progress. 3 nodes tolerate 1 failure, 5 tolerate 2. Even numbers add cost without extra tolerance, and more nodes slow every write.

TL;DR

Consensus (Raft, Paxos) lets machines agree on a leader or a log despite failures by requiring a majority and numbering leadership terms. Use a coordination service for leader election and locks, and protect locks with fencing tokens.

Clocks and ordering

What it is

To decide "which write happened last?" it's tempting to compare timestamps. But each machine has its own clock, and clocks are not in perfect agreement.

- **Clock skew**: the difference between two machines' clocks at the same moment, for example server A says 12:00:00.000 and server B says 12:00:00.150.
- **Clock drift**: a clock running slightly fast or slow. A quartz clock can drift by tens of microseconds per second, which adds up to seconds per day if never corrected.
- **NTP (Network Time Protocol)**: machines periodically ask time servers for the correct time and adjust. Over the internet it typically keeps clocks within a few to tens of milliseconds; inside a good data centre, better than a millisecond. Clocks can still jump backwards after a correction.

Two kinds of clock in every machine: the **wall clock** (time of day, can jump) and the **monotonic clock** (only moves forward, good for measuring durations like timeouts, meaningless across machines).

Why it matters

Last-write-wins (LWW) conflict resolution keeps the write with the highest timestamp. If server B's clock is 150 ms fast, a write on B can "win" over a later write on A, and A's write silently disappears. Ordering events by wall-clock timestamps from different machines is unsafe.

How it works

Logical clocks order events without real time.

Lamport clock (Leslie Lamport, 1978). Each process keeps a counter.

1. Before each local event, increment the counter.
2. When sending a message, attach the counter.
3. When receiving a message with counter `c`, set your counter to `max(yours, c) + 1`.

If event X caused event Y (directly or through a chain of messages), then X's Lamport time is smaller than Y's. Ties are broken by process ID to get a total order. Limitation: a smaller number does not prove X caused Y, so Lamport clocks can't detect concurrent events.

Vector clock. Each process keeps a list of counters, one per process, for example `[A:2, B:0, C:1]`. You increment your own entry on each event and, on receiving, take the element-wise maximum and then increment your own. Comparing two vectors tells you one of three things: X happened before Y (every entry of X is less than or equal to Y's, and at least one is smaller), Y before X, or they are **concurrent** (each has some entry larger). Concurrency detection is exactly what you need to spot conflicting writes, for example two people editing the same shopping cart offline. Dynamo-style stores use similar **version vectors** to keep both conflicting versions ("siblings") and let the application merge them. Cost: the vector grows with the number of participants.

Clock	Detects causality?	Detects concurrency?	Size
Wall clock	no (skew)	no	one number
Lamport	if X caused Y, then L(X) < L(Y)	no	one number
Vector	yes	yes	one number per node

Hybrid logical clocks (HLC) combine wall time with a logical counter, giving timestamps close to real time that still respect causality; CockroachDB and YugabyteDB use them. **Google TrueTime** (in Spanner) uses GPS receivers and atomic clocks in every data centre and returns an interval "now is between T1 and T2" with a typical uncertainty of a few milliseconds; Spanner waits out that uncertainty before committing so timestamps are globally ordered.

Common mistakes

- Using `System.currentTimeMillis()` from different servers to decide which update wins.

- Measuring timeouts with the wall clock (an NTP jump can make a duration negative). Use a monotonic clock.

In the interview

Q: Two replicas accepted conflicting writes. How do you tell which came first?

A: Wall clocks can't be trusted across machines. Vector clocks (version vectors) tell you whether one write happened before the other or whether they were concurrent; concurrent writes need a merge rule or the user to choose. Lamport clocks give a consistent total order but can't detect concurrency.

TL;DR

Machine clocks disagree (skew) and drift, even with NTP, so don't order cross-machine events by timestamps. Lamport clocks give an order consistent with causality; vector clocks also detect concurrent, conflicting updates.

Distributed transactions: two-phase commit and sagas

What it is

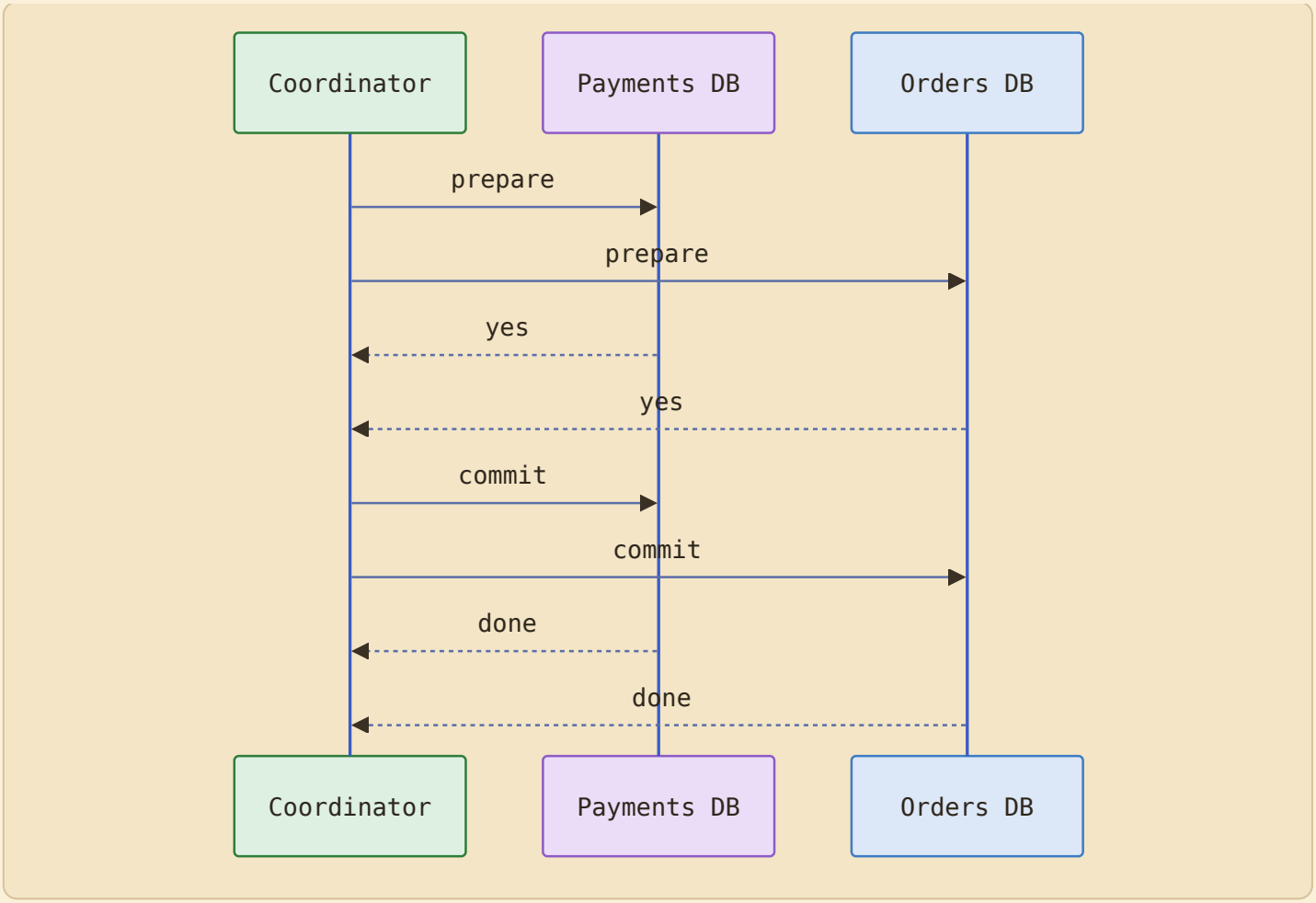
A **distributed transaction** is a change that must happen across several databases or services as one unit: either all parts happen or none do. Example: booking a trip charges the card (payment service), reserves a seat (flight service) and books a room (hotel service). Charging the card without the seat is a bug that costs real money.

Inside one database, ACID transactions (Part 3) handle this. Across several, there are two main approaches.

Types

Two-phase commit (2PC). A **coordinator** (also called transaction manager) runs the transaction across **participants** (the databases).

1. **Prepare phase**: the coordinator asks every participant "can you commit?". Each one does the work, writes it durably, locks the rows involved, and votes **yes** (promising it can commit even after a crash) or **no**.
2. **Commit phase**: if all voted yes, the coordinator writes "commit" to its own log and tells everyone to commit. If any voted no or didn't answer, it tells everyone to abort.



Strength: real atomicity and isolation across databases. Weaknesses: it's **blocking**. If the coordinator crashes after participants voted yes, they must keep their rows locked and wait, possibly for a long time, because they promised to obey whatever the coordinator decides. Every transaction needs several round trips and holds locks meanwhile, so throughput and availability suffer. It's used inside some distributed databases (Spanner runs 2PC over Paxos-replicated groups, so the coordinator itself is replicated and cannot simply vanish) and via the XA standard (a common interface that lets one transaction manager drive several databases) between traditional databases, but rarely across microservices. (Three-phase commit was proposed to reduce blocking but is little used in practice.)

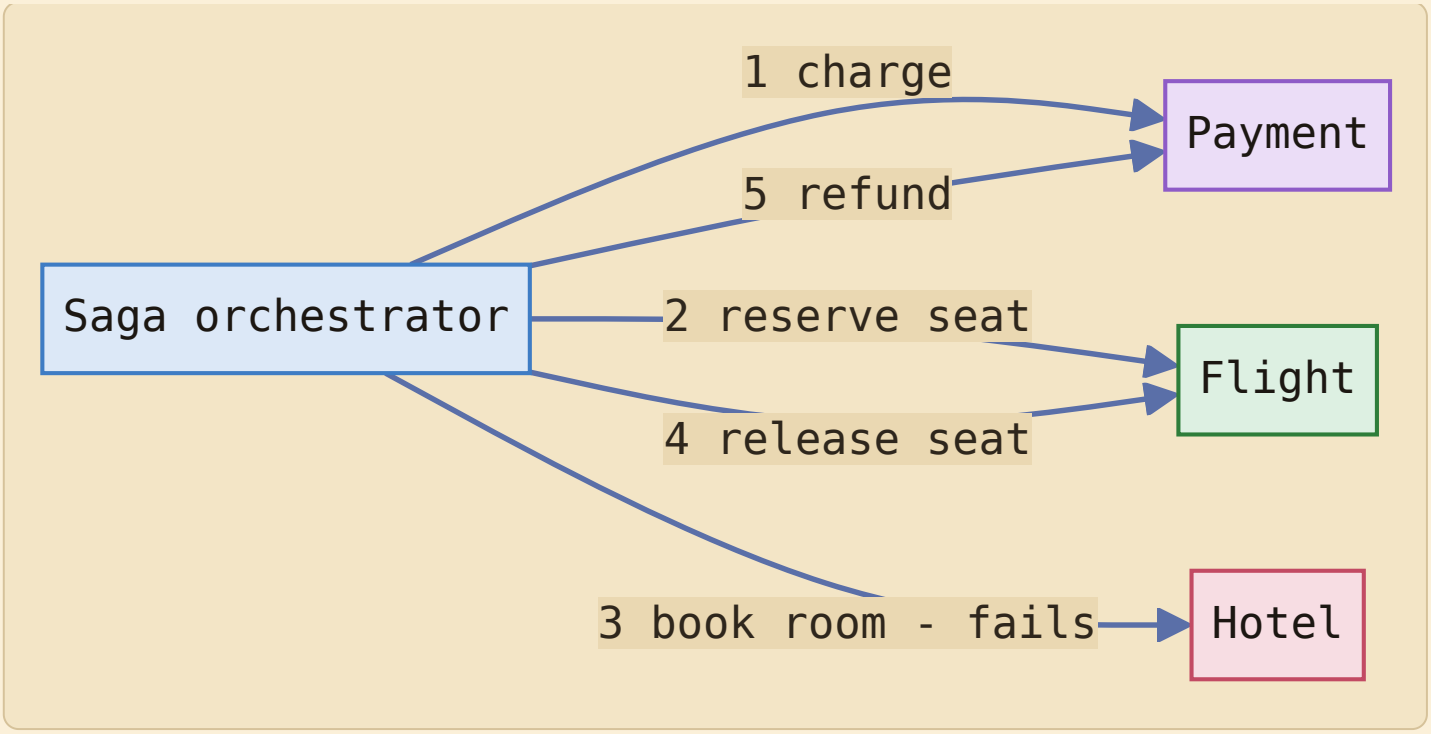
Saga. Break the big transaction into a sequence of **local transactions**, one per service, each committed on its own. If a later step fails, run **compensating transactions** that semantically undo the earlier steps, in reverse order.

Trip example:

1. Payment service charges the card. (Compensation: refund.)
2. Flight service reserves a seat. (Compensation: release the seat.)
3. Hotel service books a room. Fails, no rooms.
4. Run compensations: release the seat, refund the card.

Two ways to coordinate a saga:

- **Choreography**: no central controller. Each service listens for events and reacts ("PaymentCompleted" makes the flight service reserve, "HotelFailed" makes the flight service release). Simple for 2–3 steps, hard to follow with many.
- **Orchestration**: a central **orchestrator** (a workflow engine such as Temporal, AWS Step Functions or Camunda, or your own service) tells each service what to do next and triggers compensations on failure. Easier to see and change the whole flow.



Trade-offs

	2PC	Saga
Atomicity	true all-or-nothing	eventual, via compensations
Isolation	yes, locks held	no, others can see partial results
Availability	blocks if coordinator fails	each step independent
Fits	few databases, short transactions	microservices, long business flows

Sagas give up isolation: between steps, someone could see "card charged, no seat yet". Designs cope with it using states like `PENDING`, and compensations must be idempotent and retryable because they may run more than once. Some actions can't be undone (an email sent), so you put them last or compensate with another action (a "sorry" email).

In the interview

Q: How do you keep data consistent across order, payment and inventory microservices?

A: A saga: each service does a local transaction and publishes an event (via outbox), with compensations on failure, coordinated by an orchestrator for visibility. Avoid 2PC across services because it blocks and couples availability.

Q: What's the main weakness of 2PC?

A: It's blocking: if the coordinator fails after participants vote yes, they hold locks and wait. It also needs every participant up for every transaction.

TL;DR

2PC gives true atomic commits across databases but blocks on coordinator failure; sagas chain local transactions with compensating undo steps, trading isolation for availability, and are the norm in microservices.

Heartbeats, gossip and failure detection

What it is

In a cluster, nodes need to know which other nodes are alive. But over a network you can never tell the difference between a node that's dead, one that's slow, and a network that's dropping messages. **Failure**

detection is the art of making a good guess.

- **Heartbeat**: a small "I'm alive" message a node sends regularly, say every second. If the monitor misses several in a row (say, nothing for 5 seconds), it marks the node as **suspected dead**. Load balancer health checks (Part 2) and Raft leader heartbeats are the same idea.
- **Timeout choice**: short timeouts detect failures quickly but cause **false positives** (declaring a slow-but-alive node dead, triggering needless failovers). Long timeouts are safe but slow to react.

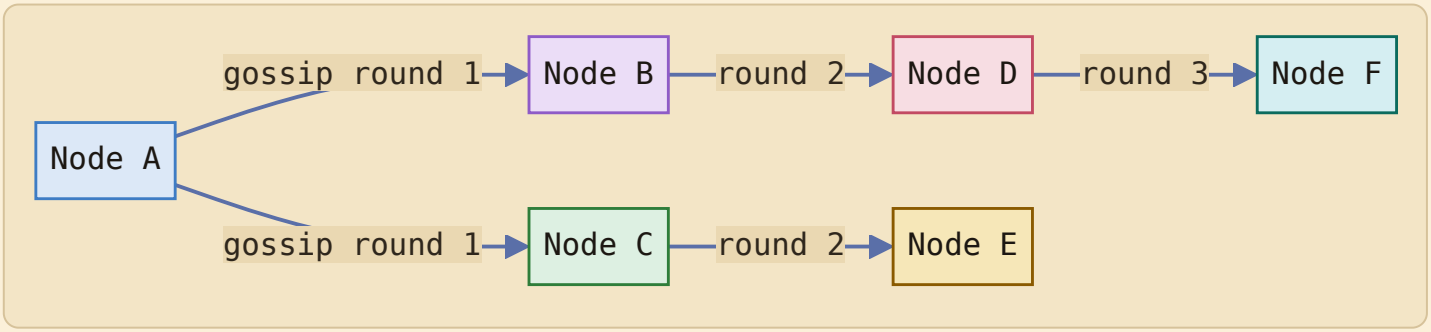
How it works

Centralised monitoring (every node heartbeats to one coordinator, such as ZooKeeper sessions) is simple, but the monitor must itself be highly available.

Gossip protocol. In large clusters with no central monitor, nodes spread information the way rumours spread in an office:

1. Every second, each node picks one or a few random other nodes.
2. It sends them what it knows: its membership list, with a heartbeat counter for every node it has heard of.
3. The receiver merges the lists, keeping the higher counter for each node.
4. If a node's counter hasn't gone up for a while, it is suspected dead, and that suspicion spreads too.

Information reaches all N nodes in about log(N) rounds: with 1,000 nodes, around 10 rounds. No single point of failure, and load per node stays constant. Cassandra, DynamoDB's original design, Consul (via memberlist, a library based on the SWIM protocol described below) and Redis Cluster use gossip for membership and failure detection.



Smarter detectors. The **phi accrual failure detector** (used by Cassandra and Akka) outputs a suspicion level based on the history of heartbeat arrival times, instead of a fixed yes/no timeout, so it adapts to networks that are normally slow. **SWIM** asks a few other nodes to ping a suspected node indirectly before declaring it dead, cutting false positives caused by one bad link.

Common mistakes

- Very short failure timeouts that cause **flapping**: nodes marked dead and alive over and over, triggering repeated failovers.
- Assuming a node declared dead has stopped. It may still be running and writing (the paused lock holder from earlier). Use fencing.

TL;DR

Nodes send heartbeats; missing heartbeats mean "suspected dead". Gossip spreads membership and liveness through random peer-to-peer exchanges, reaching everyone in about log N rounds with no central monitor.

Probabilistic data structures and Merkle trees

These are clever structures that answer questions using far less memory than an exact answer would need, in exchange for a small, controlled error. Each appears often in interviews.

Bloom filter

What it is. A compact structure that answers "is X in the set?" with either **"definitely not"** or **"probably yes"**. It never gives a false negative, but it can give a **false positive**.

How it works.

1. Start with an array of m bits, all 0 (say 10 million bits, about 1.2 MB).
2. Pick k hash functions (say 7). A hash function turns any input into a number; here each one maps an item to a bit position.
3. To add "alice", compute her 7 positions and set those bits to 1.
4. To check "bob", compute his 7 positions. If any bit is 0, bob was **definitely never added**. If all are 1, bob is **probably** in the set (those bits could have been set by other items).

With about 10 bits per item and 7 hash functions, the false-positive rate is about 1%. You can't delete items from a basic Bloom filter (a **counting Bloom filter** variant allows it).

Uses. Databases with LSM-trees (Cassandra, RocksDB, HBase) keep a Bloom filter per data file so a read for a missing key skips files without touching disk. Caches use one to block lookups for keys that don't exist (cache penetration). Web crawlers use them to skip URLs already visited. Chrome once used one to check URLs against a list of malicious sites before a slower exact check.

HyperLogLog

What it is. Estimates the **number of distinct items** (cardinality), such as unique visitors today, using a tiny, fixed amount of memory.

How it works (intuition). Hash each item to a random-looking binary number. Rare patterns such as a long run of leading zeros only show up when you've seen many distinct items: seeing a hash that starts with 20 zeros suggests you've seen roughly 2^20 (about a million) distinct items. HyperLogLog splits items into many buckets, records the longest run in each, and averages them to reduce luck. Duplicates hash the same, so they don't change the estimate.

Numbers. Redis's HyperLogLog uses at most 12 KB per counter, has a standard error of 0.81%, and can count billions of distinct items. An exact set of a billion user IDs would need gigabytes. Sketches from different servers can be merged (`PFMERGE`) to get a combined count. Used for unique visitors, distinct search queries, and approximate `COUNT(DISTINCT ...)` in BigQuery and other warehouses.

Count-min sketch

What it is. Estimates **how many times** each item has appeared in a stream (frequency), in fixed memory, for example "how many times was #worldcup posted in the last hour?".

How it works. Keep a small table of counters with d rows and w columns, and one hash function per row. To add an item, hash it in each row and increment that counter. To query, hash it in each row and take the **minimum** of the d counters. Collisions only ever add counts, so the estimate can be too high but never too low; taking the minimum picks the least-polluted counter.

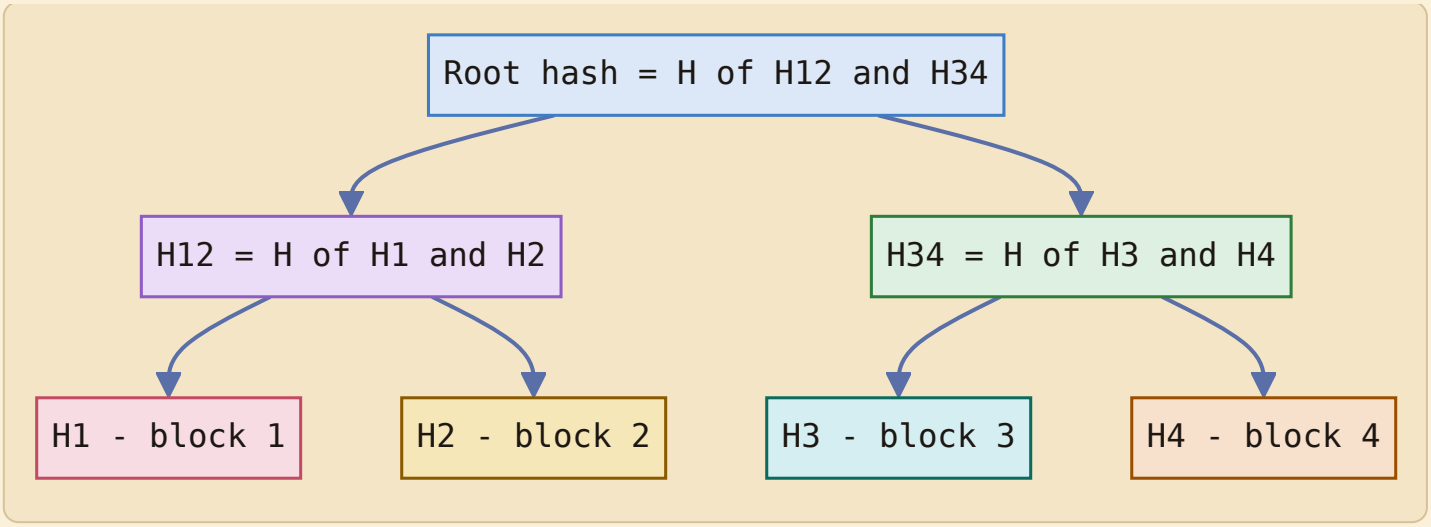
Uses. Finding heavy hitters and trending topics, detecting hot keys in a cache, network traffic monitoring, approximate top-K lists.

Merkle tree

What it is. A tree of hashes. Each leaf is the hash of a block of data; each parent is the hash of its children's hashes; the single hash at the top is the **root hash**. If any byte of any block changes, its leaf hash changes, and so does every hash on the path up to the root.

How it works for comparing replicas. Two replicas each build a Merkle tree over their data, split into ranges.

1. Compare root hashes. Equal means all the data is identical: done, with one tiny message.
2. If different, compare the two children, and only descend into the halves whose hashes differ.
3. At the bottom you find exactly which ranges differ and copy only those.



Finding a few differences among billions of records costs about log(number of blocks) comparisons instead of shipping everything. **Uses**: anti-entropy repair in Cassandra and Dynamo-style stores, Git (commit and tree objects are built from hashes), Bitcoin and other blockchains (proving a transaction is in a block), certificate transparency logs, and file sync tools.

Summary table

Structure	Answers	Error	Memory
Bloom filter	is X in the set?	false positives only	~10 bits per item for 1%
HyperLogLog	how many distinct items?	~0.81% in Redis	12 KB per counter
Count-min sketch	how often has X appeared?	overestimates only	fixed table
Merkle tree	do two datasets differ, and where?	none (exact)	one hash per block

In the interview

Q: How would you count unique daily visitors for a site with 500 million users?

A: A HyperLogLog per day (for example Redis `PFADD` and `PFCOUNT`): about 12 KB per day with under 1% error, and daily sketches can be merged into weekly counts. An exact set would take gigabytes.

Q: How can a database avoid disk reads for keys that don't exist?

A: A Bloom filter per data file: if it says "definitely not here", skip the file. Only "probably yes" answers cost a disk read.

TL;DR

Bloom filter: set membership with false positives only. HyperLogLog: distinct counts in kilobytes. Count-min sketch: approximate frequencies. Merkle tree: find differences between large datasets by comparing hashes top-down.

Unique ID generation

What it is

Every order, user, message and photo needs an **ID** that is unique. On one database this is easy; across many servers and databases, generating unique IDs without them checking with each other is a design problem of its own. Nice extra properties: IDs that sort roughly by creation time (good for database indexes and "newest first" queries), short IDs (64 bits fits a standard integer column), and IDs that don't leak business information.

Types

Auto-increment (database sequence). The database hands out 1, 2, 3, ... Simple, short and ordered. Problems: one database becomes a bottleneck and single point of failure; after sharding, two shards would both issue ID 1000 (workarounds: each shard steps by the number of shards, so shard 1 issues 1, 3, 5 and shard 2 issues 2, 4, 6, or a central ticket server like Flickr used); and sequential IDs reveal volume (a competitor can see you got 500 orders today) and let attackers guess other users' URLs.

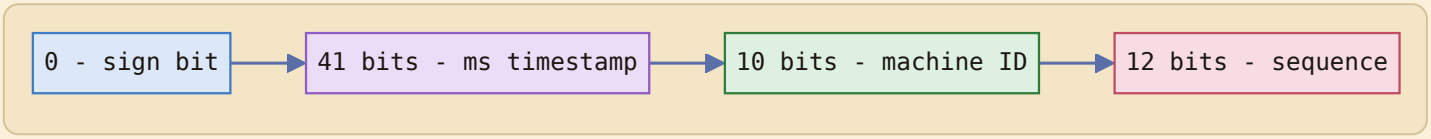
UUID (universally unique identifier). A 128-bit value, usually written as 36 characters like `550e8400-e29b-41d4-a716-446655440000`. Any machine can generate one with no coordination.

- **UUIDv4** is 122 random bits. The chance of a collision is negligible (you'd need billions of IDs per second for decades). Downside: random order scatters inserts across a B-tree index, causing more page splits and cache misses, and 128 bits is twice the size of a 64-bit integer.
- **UUIDv7** (standardised in RFC 9562, 2024) puts a millisecond timestamp in the first 48 bits and randomness after, so IDs are unique **and** sort by time, fixing the index problem. A good modern default. (**ULID** is a similar earlier format.)

Snowflake ID. Twitter's 2010 design for 64-bit, time-sortable IDs generated independently by many machines. The 64 bits are split:

Bits	Field	Meaning
1	sign	always 0
41	timestamp	milliseconds since a custom epoch, lasts about 69 years
10	machine ID	up to 1,024 generators
12	sequence	up to 4,096 IDs per machine per millisecond

So each machine can create about 4 million IDs per second with no coordination, and IDs sort by time. Requirements: each machine needs a unique machine ID (assigned through config or a coordination service like ZooKeeper), and clocks must not go backwards (if they do, the generator waits or refuses). Discord, Instagram (a variant generated inside PostgreSQL shards) and many others use Snowflake-style IDs. Sonyflake and others change the bit split.



Other approaches. Segment / range allocation: a central service hands each server a block of 1,000 IDs at a time, so servers rarely talk to it. **Hashing** (for example for URL shorteners, Part 6) derives a short ID from content, with collision checks.

In the interview

Q: How would you generate unique IDs for a sharded system with high write volume?

A: Snowflake-style 64-bit IDs: timestamp, machine ID and per-millisecond sequence. No coordination per ID, roughly time-ordered, fits a BIGINT (64-bit integer) column. If 128 bits is fine, UUIDv7 gives the same ordering with no machine IDs to manage.

Q: Why not just use UUIDv4?

A: It works and needs no coordination, but random IDs fragment B-tree indexes and are larger. Time-ordered IDs (UUIDv7, Snowflake) insert at the end of the index.

TL;DR

Auto-increment is simple but centralised and guessable; UUIDs need no coordination (v7 also sorts by time); Snowflake packs time, machine and sequence into 64 sortable bits.

Geospatial indexing: geohash and quadtree

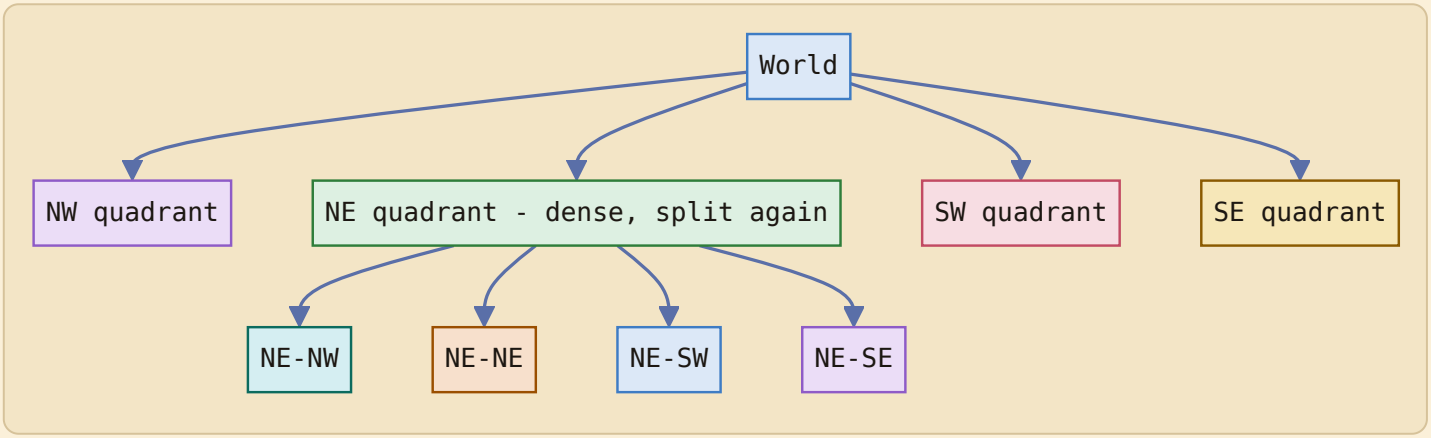
What it is

"Find restaurants within 2 km" or "find drivers near me" can't be answered efficiently with a normal index on latitude and longitude, because a B-tree index sorts by one value at a time. **Geospatial indexes** turn 2D locations into something you can search quickly.

Types

Geohash. Divide the world map in half by longitude, then by latitude, and keep halving, writing down a bit for which half you're in at each step. The bits are encoded in base-32 letters and digits. Longer strings mean smaller boxes: a 5-character geohash is a cell roughly 5 km x 5 km, 6 characters roughly 1.2 km x 0.6 km. Places that share a prefix are near each other (`gcpvj` and `gcpvm` are side-by-side cells in central London), so "nearby" becomes a **prefix search** on a normal string index. To avoid missing places just over a cell edge, you search the target cell plus its 8 neighbours. Redis's `GEOADD`/`GEOSEARCH` and Elasticsearch support geohash-style indexing.

Quadtree. A tree where each node covers a rectangle of the map. When a node holds too many points (say more than 100 businesses), it splits into four children (north-west, north-east, south-west, south-east). Dense cities get small, deep cells; empty oceans stay as one big cell. To find nearby places, walk down to the leaf containing the user and look at it and its neighbours. Quadtrees adapt to uneven density better than fixed-size geohash cells and are typically kept in memory.



Others you'll hear: **R-tree** (bounding-box tree used by PostGIS), Google's **S2** (cells on a sphere, used in Google Maps and by MongoDB) and Uber's **H3** (a grid of hexagons, good for ride-demand heatmaps because every neighbour is the same distance away).

In the interview

Q: How do you find drivers near a rider?

A: Drivers send locations every few seconds; store them in an in-memory geo index (geohash cells in Redis, or a quadtree) keyed by cell. Query the rider's cell and its neighbours, then compute exact distances for the handful of candidates.

TL;DR

Geohash turns locations into strings where a shared prefix means nearby; a quadtree splits the map into four recursively where it's dense. Both turn "what's near me?" into a fast lookup.

Part 4 glossary

Term	Meaning in one line
Cache	Small, fast store holding copies of data from a slower source
Source of truth / origin	The authoritative home of the data, usually a database
Cache hit / miss	Requested item was / was not found in the cache
Hit ratio	Fraction of lookups served by the cache
TTL (time to live)	How long an entry may live before it expires
Stale data	A cached or replicated copy older than the latest write
Warm / cold cache	Cache already full of hot data / empty and missing a lot
Cache-Control, ETag	HTTP headers controlling browser/CDN caching and validating versions
Cache busting	Changing a file name or URL when content changes so old copies aren't used
In-process cache	Cache inside the application's own memory
Distributed cache	Shared cache cluster reached over the network (Redis, Memcached)
Redis	In-memory key-value server with rich data types, often used as a cache
Memcached	Simple, fast, multi-threaded in-memory key-value cache
Buffer cache / buffer pool	Database's own in-memory copy of recently used disk pages
Page cache	Operating system's RAM cache of recently read file data
Cache-aside (lazy loading)	App checks cache, loads from DB on miss, deletes key on write
Read-through	Cache loads from the DB itself on a miss
Write-through	Writes go to cache and DB synchronously
Write-behind / write-back	Writes go to cache, flushed to DB later
Write-around	Writes skip the cache and go only to the DB
Refresh-ahead	Cache reloads hot entries before they expire
Negative caching	Caching "not found" results for a short time
Eviction policy	Rule for what to remove when the cache is full
LRU / LFU / FIFO	Evict least recently used / least frequently used / oldest inserted
Cache invalidation	Removing or updating cached copies when the source changes
Versioned keys	Putting a version in the cache key so new writes use new keys
Change data capture (CDC)	Streaming a database's changes from its log to other systems
Cache stampede / thundering herd	Many simultaneous misses on one key overloading the DB
Request coalescing / single flight	Letting only one request rebuild a missing key
Lease (cache)	Token granting one client the right to refill a key
Stale-while-revalidate	Serve the expired value while refreshing in the background
TTL jitter	Randomising TTLs so keys don't expire together
Cache avalanche	Mass expiry or cache loss sending all traffic to the DB
Cache penetration	Requests for non-existent keys always hitting the DB
Hot key	One key so popular it overloads the node that holds it

Term	Meaning in one line
Message queue	Broker that holds messages until consumers process them
Producer / consumer / broker	Sender / processor / storage middleman of messages
Asynchronous processing	Doing work later without the caller waiting for it
Decoupling	Letting components work without knowing about each other directly
Load levelling	Using a queue to smooth spikes into a steady processing rate
Acknowledgement (ack)	Consumer's signal that a message is done and can be removed
Visibility timeout	Time a received message is hidden before being redelivered (SQS)
Point-to-point / competing consumers	Each message processed by one of many workers
Publish/subscribe (pub/sub)	Each message delivered to every subscriber of a topic
Topic	Named channel or stream of messages
Exchange	RabbitMQ component that routes messages to queues
Claim check	Store a big payload elsewhere and send only its reference
Queue depth	Number of messages waiting in a queue
Event streaming / log-based broker	Broker keeping an ordered, replayable log of events (Kafka)
Partition (stream)	Ordered sub-log of a topic, the unit of parallelism
Message key	Value hashed to pick a partition, keeping related events ordered
Offset	Position number of a message in a partition
Consumer group	Consumers sharing a topic's partitions for one application
Committed offset	Saved position a consumer group has processed up to
Rebalance	Reassigning partitions when group membership changes
Consumer lag	How far a consumer is behind the newest message
Retention	How long a log keeps messages
At-most-once	Delivered zero or one times, may be lost
At-least-once	Delivered one or more times, may be duplicated
Exactly-once / effectively-once	Each message affects the result once, via de-duplication
Idempotency	Doing an operation twice has the same effect as once
Idempotency key	Unique ID per operation used to detect and skip repeats
Upsert	Insert a row or update it if it already exists
Idempotent producer	Producer whose duplicate sends the broker discards
Exponential backoff	Waiting longer after each failed retry
Jitter	Random variation added to waits or TTLs
Poison message	A message that fails every time it's processed
Dead-letter queue (DLQ)	Side queue for messages that failed too many times
Redrive	Moving DLQ messages back to the main queue
Backpressure	Downstream signalling upstream to slow down
Load shedding	Dropping some work on purpose to stay alive under overload
Batch processing	Processing a bounded pile of data in one scheduled job

Term	Meaning in one line
Stream processing	Processing events continuously as they arrive
MapReduce	Batch model of map, shuffle and reduce across many machines
Window (streaming)	Time range grouping events, e.g. tumbling or sliding
Event time / processing time	When an event happened / when it was processed
Watermark	Stream engine's estimate that all events up to a time have arrived
Lambda / Kappa architecture	Batch plus speed layer / stream-only with replay
Event-driven architecture	Services reacting to published events instead of direct calls
Event vs command	Fact that happened vs request that something happen
Dual write problem	Updating DB and broker separately, risking disagreement
Transactional outbox	Writing events to an outbox table in the same DB transaction
Event sourcing	Storing the sequence of events as the source of truth
Snapshot (event sourcing)	Saved state to avoid replaying every event
CQRS	Separate models for writes (commands) and reads (queries)
Projection / read model	Read-optimised view built from events
Consistency model	Promise about what reads can return after writes
Linearizability / strong consistency	System behaves as if there were one copy, in real-time order
Sequential consistency	Everyone sees the same order, not necessarily real time
Causal consistency	Causes are always seen before their effects
Read-your-writes	A user always sees their own changes
Monotonic reads	A user never sees data go back in time
Monotonic writes	A user's writes apply in the order made
Eventual consistency	Replicas converge if writes stop, no time bound
CAP theorem	Under a partition, choose consistency or availability
Network partition	Network failure splitting nodes into groups that can't talk
CP / AP	Refuses requests to stay consistent / answers with maybe-stale data
PACELC	If partition: A or C, else: latency or consistency
Quorum	Minimum number of replicas that must take part in an operation
N, R, W	Replica count, read quorum, write quorum
R + W > N	Condition for read and write sets to overlap
Majority quorum	More than half the nodes
Read repair	Fixing stale replicas noticed during a read
Anti-entropy	Background process comparing and syncing replicas
Sloppy quorum / hinted handoff	Temporarily writing to stand-in nodes that hand data back later
Consensus	Getting nodes to agree on a value despite failures
Paxos	First widely known proven consensus algorithm
Raft	Consensus algorithm designed to be understandable
Leader / follower / candidate	Raft node roles

Term	Meaning in one line
Term / epoch	Number that increases with each new leader
Election timeout	Time without heartbeats before a follower starts an election
Leader election	Choosing one node to act as leader
Split brain	Two nodes both acting as leader
Coordination service	System like etcd, ZooKeeper or Consul for consensus-backed coordination
Distributed lock	Lock that works across machines
Lease	Lock or role that automatically expires after a time
Fencing token	Increasing number that lets storage reject stale lock holders
Clock skew	Difference between two machines' clocks
Clock drift	A clock gradually running fast or slow
NTP	Protocol for synchronising clocks with time servers
Wall clock / monotonic clock	Time of day (can jump) / always-forward timer for durations
Last-write-wins (LWW)	Conflict rule keeping the write with the highest timestamp
Logical clock	Counter-based ordering of events without real time
Lamport clock	Single counter consistent with causality
Vector clock / version vector	Per-node counters that also detect concurrent events
Hybrid logical clock	Timestamp mixing wall time with a logical counter
TrueTime	Spanner's clock API returning a bounded time interval
Distributed transaction	Change spanning several databases or services, all or nothing
Two-phase commit (2PC)	Prepare then commit across participants via a coordinator
Coordinator / participant	Runner of 2PC / database taking part in it
Saga	Sequence of local transactions with compensations on failure
Compensating transaction	Action that semantically undoes an earlier step
Choreography / orchestration	Saga steps triggered by events / by a central controller
Heartbeat	Periodic "I'm alive" message
Failure detection	Deciding whether a node is probably dead
False positive (failure)	Declaring a slow but alive node dead
Flapping	Node repeatedly marked dead and alive
Gossip protocol	Nodes spreading state to random peers each round
Phi accrual detector	Failure detector giving a suspicion level, not yes/no
SWIM	Gossip membership protocol with indirect pings
Bloom filter	Set membership test with false positives but no false negatives
Hash function	Function mapping any input to a fixed-size number
HyperLogLog	Distinct-count estimator using kilobytes of memory
Cardinality	Number of distinct items
Count-min sketch	Fixed-size table estimating item frequencies
Heavy hitters	The most frequent items in a stream

Term	Meaning in one line
Merkle tree	Tree of hashes for detecting where datasets differ
Root hash	Top hash of a Merkle tree, summarising all data
Auto-increment / sequence	Database-generated increasing integer IDs
Ticket server	Central service issuing unique IDs
UUID	128-bit identifier generated without coordination
UUIDv4 / UUIDv7	Random UUID / time-ordered UUID
ULID	Time-sortable 128-bit ID format
Snowflake ID	64-bit ID of timestamp, machine ID and sequence
Geospatial index	Index for searching by location
Geohash	String encoding of a map cell, shared prefix means nearby
Quadtree	Tree splitting map regions into four as density grows
R-tree / S2 / H3	Bounding-box tree / sphere cells / hexagon grid geo indexes

Part 5 — Architecture, reliability, operations and security

Parts 1 to 4 covered the building blocks: servers, networks, databases, caches and queues. This part is about how you arrange those blocks into a whole system, how you keep that system running when pieces break, how you ship changes to it safely, how you see what it is doing, and how you stop the wrong people from getting in. These are the topics that separate "it works on my laptop" from "it has served a billion requests this month and nobody noticed when a data centre lost power".

Architectural styles: monolith, modular monolith, SOA, microservices and serverless

What it is

An **architectural style** is the big-picture answer to "how is our code split into running programs, and how do those programs talk?". Think of a restaurant. One chef can cook every dish in one kitchen (a monolith). A big kitchen can be divided into stations, grill, pastry, salad, each with its own cook but still one kitchen (a modular monolith). A food court has independent stalls, each with its own staff, menu and till, and customers walk between them (microservices). A catering agency that sends a cook only when an order comes in, and bills you per dish, is serverless.

- **Monolith**: the whole application (user accounts, product catalogue, checkout, emails) is one codebase, built into one deployable program, usually sharing one database. You run several identical copies behind a load balancer to scale.
- **Modular monolith**: still one deployable program, but the code is split into modules with strict boundaries (the checkout module may only call the catalogue module through a defined interface, never reach into its tables). It gives most of the tidiness of microservices without the network.
- **Service-oriented architecture (SOA)**: a 2000s-era style where large enterprise systems were split into services that talked through a central **enterprise service bus (ESB)**, a piece of middleware that routed, transformed and orchestrated messages. Services were often coarse (a whole "billing system") and shared data models.
- **Microservices**: the application is split into many small services, each owning one business capability (for example "payments" or "search"), each with its **own database**, deployed independently, and talking over the network using HTTP/REST, gRPC or messages. "Smart endpoints, dumb pipes": the logic lives in the services, not in a central bus.
- **Serverless / Function as a Service (FaaS)**: you upload individual functions; the cloud provider runs them only when an event arrives (an HTTP request, a file upload, a queue message), scales them

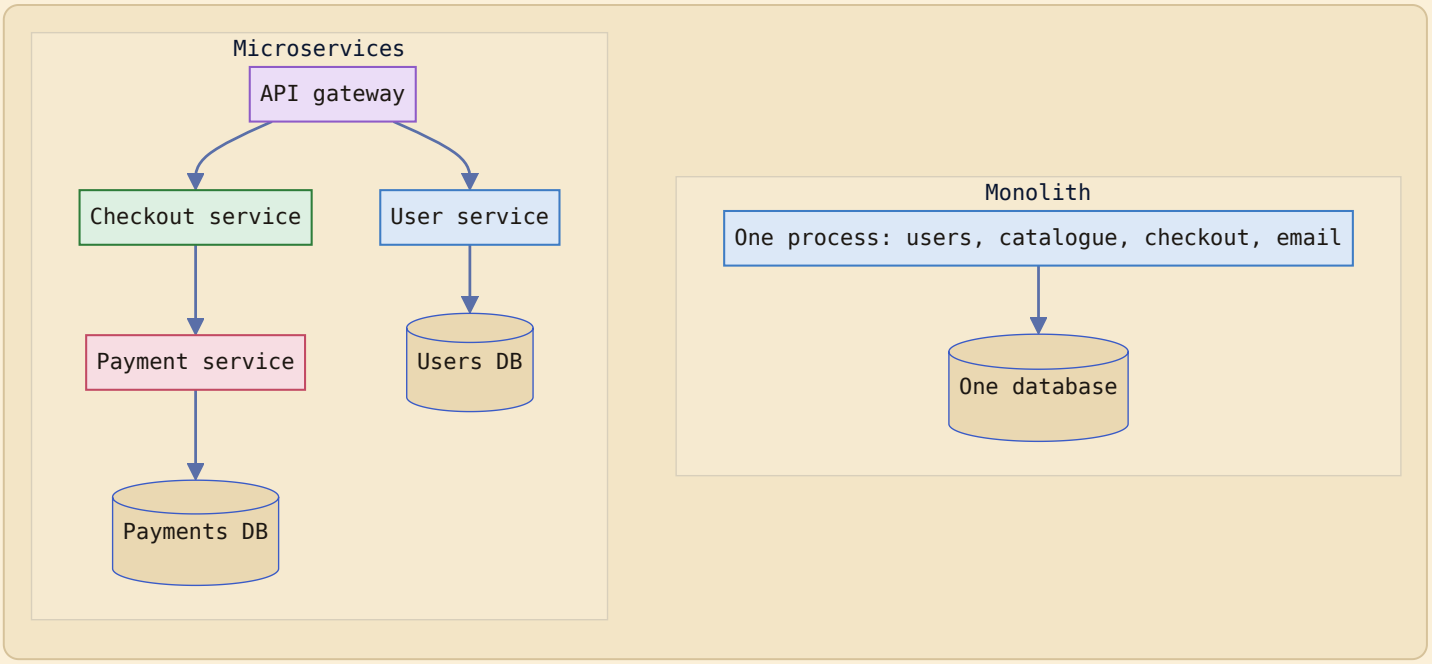
automatically from zero to thousands of copies, and bills per invocation and per millisecond of run time. You never manage a server. Examples: AWS Lambda, Google Cloud Functions, Azure Functions, Cloudflare Workers. "Serverless" also loosely covers managed services you never size yourself, such as DynamoDB or S3.

Why it matters

The style decides how fast teams can move, how failures spread, how much the system costs and how hard it is to debug. A five-person startup and a 2,000-engineer company need different answers. **Conway's law** says a system's structure ends up mirroring the communication structure of the organisation that builds it, so architecture is partly a decision about teams.

How it works

A request in a monolith is a chain of in-process function calls: `checkout()` calls `reserveStock()` calls `chargeCard()`, each taking nanoseconds to microseconds and all inside one database transaction. The same request in microservices becomes network calls: the checkout service sends an HTTP request to the inventory service (about 0.5 to 2 ms inside one data centre) and another to the payment service, and there is no single transaction covering both, so you need patterns like sagas (Part 4) to undo work if a later step fails.



Types

Style	Deploy unit	Data	Best for
Monolith	One app	One shared DB	Small teams, new products
Modular monolith	One app, strict modules	One DB, owned schemas	Growing teams wanting order
SOA	Coarse services + ESB	Often shared	Legacy enterprise integration
Microservices	Many small services	DB per service	Many teams, independent scaling
Serverless (FaaS)	Single functions	Managed stores	Spiky or event-driven work

Trade-offs

- **Monolith**: simple to build, test, deploy and debug; one stack trace shows everything; transactions are easy. But as it grows past a few hundred thousand lines and dozens of developers, builds slow down, one bad change can take everything down, and you must scale the whole thing even if only search is busy.
- **Microservices**: teams deploy independently many times a day, each service scales and fails on its own, and each can pick its own language or database. The price is the **distributed systems tax**: network

calls fail and add latency, data is spread across databases so consistency is harder, you need service discovery, tracing, per-service CI/CD and on-call, and a single user action may touch 20 services.

- **Serverless**: zero servers to manage, scales to zero so idle costs nothing, scales up in seconds. Downsides: **cold starts** (the first call to a fresh copy waits for the runtime to load, from about 100 ms to a few seconds), execution time limits (AWS Lambda caps a single run at 15 minutes), harder local testing, vendor lock-in, and at steady high load it is often more expensive than always-on servers.

Common mistakes

- Starting a new product with 30 microservices. You do not yet know where the boundaries are, and you pay the tax from day one. The common advice is "monolith first", split later.
- A **distributed monolith**: services that must all be deployed together, share one database or call each other synchronously in long chains. You pay the network and operations bill of microservices while still moving in lockstep like a monolith.
- Splitting by technical layer ("database service", "UI service") instead of business capability.
- Thinking microservices are faster. A network hop is thousands of times slower than a function call.

Real-world examples

- **Shopify** runs one of the largest Ruby on Rails monoliths and deliberately chose a modular monolith, enforcing component boundaries with tooling, instead of breaking it into microservices.
- **Netflix** and **Amazon** are the classic microservices stories; Uber has written about running around 2,200 critical microservices and then grouping them into "domains" to tame the complexity.
- In 2023 the **Amazon Prime Video** team publicly described moving an audio/video quality-monitoring tool from a distributed serverless design (AWS Step Functions plus Lambda) into a single process running as one container task, cutting that tool's infrastructure cost by more than 90%. The lesson: pick the style per workload.
- The **strangler fig pattern** is how most companies migrate: put a proxy in front of the monolith and move one feature at a time into a new service until the old code can be deleted.

In the interview

Q: Would you start this design with microservices?

A: Usually no. I would start with a well-modularised monolith, then extract a service when there is a clear reason: a part that needs to scale independently, a team that needs to deploy on its own schedule, or a component with very different reliability needs.

Q: What does "database per service" buy you and what does it cost?

A: It lets each team change its schema and scale its store without coordinating with others, and stops one service's bad query from hurting another. It costs you cross-service joins and transactions: you need APIs, events and sagas, and you accept eventual consistency between services.

Q: When is serverless a good fit?

A: Spiky or unpredictable, event-driven work such as image thumbnails on upload, webhooks or nightly jobs, where scaling to zero saves money. It is a poor fit for long-running jobs, latency-critical paths that cannot tolerate cold starts, or steady high traffic.

TL;DR

Monoliths are simple and fast to start; microservices trade simplicity for team independence and independent scaling; serverless trades control for zero operations. Choose by team size and workload, not fashion.

Synchronous vs asynchronous communication

What it is

When one service needs another, it can either **wait for the answer** or **leave a message and carry on**. A phone call is synchronous: you speak, you wait, you get a reply. A text message is asynchronous: you send it and continue your day; the reply comes whenever it comes.

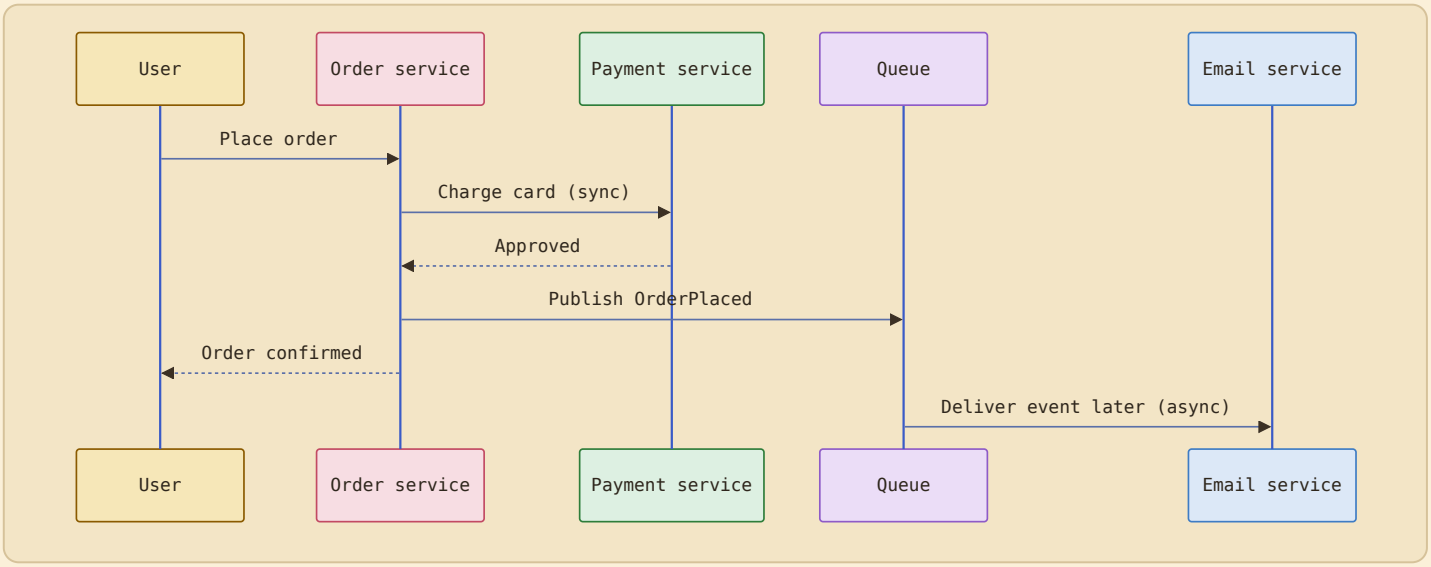
- **Synchronous**: service A calls service B (REST, gRPC) and blocks until B responds or a timeout fires. The caller needs the answer to continue.
- **Asynchronous**: A puts a message on a queue or event stream (Part 4) and returns immediately. B processes it later. A may get a result through a callback, a webhook or by checking a status endpoint.

Why it matters

A synchronous chain ties the callers' fate to the callee: if B is slow or down, A is slow or down. Availability multiplies, too. If a request must pass through five services that are each 99.9% available, the chain is at best 0.999^5, about 99.5%, which is roughly 44 hours of downtime a year instead of 9. Asynchronous messaging breaks that chain: if the email service is down, orders still succeed and the emails go out when it recovers.

How it works

1. User clicks "Place order". The order service must check the card synchronously because the user is waiting for "approved" or "declined".
2. Once the order is saved, the order service publishes an `OrderPlaced` event to a queue and replies to the user in, say, 200 ms.
3. Email, analytics and warehouse services each consume the event at their own pace. If the warehouse system is down for ten minutes, events wait in the queue.



Trade-offs

	Synchronous	Asynchronous
Simplicity	Easy to reason about	Harder: eventual results
Coupling	Tight in time	Loose
Latency for user	Sum of all calls	Only the critical path
Failure spread	Cascades	Absorbed by the queue
Debugging	One request, one trace	Needs correlation IDs

Common mistakes

- Making every call synchronous "because it is simpler", building deep call chains.
- Using async for something the user must see immediately (like "is my payment approved?") and then building awkward polling.
- Forgetting that async consumers must be idempotent, because queues usually deliver at least once.

In the interview

Q: How do you decide sync vs async between two services?

A: Ask whether the caller needs the result to answer the user right now. If yes, call synchronously with a timeout. If the work can finish later (notifications, analytics, search indexing), publish an event so failures and slowness do not reach the user.

Q: How do you tell the user about an async job's result?

A: Return a job ID immediately (HTTP 202 Accepted), then let the client poll a status endpoint, or push the result with WebSockets, server-sent events or a webhook.

TL;DR

Synchronous calls are simple but chain failures and latency together; asynchronous messages decouple services in time at the cost of eventual results and harder debugging.

Service discovery, sidecars and service mesh

What it is

In a system with many services, each running many copies that start and stop all the time, a caller needs to know "which IP addresses currently run the payment service?". **Service discovery** answers that. It is like a hospital switchboard: you ask for "cardiology" rather than memorising which doctor is on shift today.

- **Service registry**: a database of live service instances and their addresses. Instances **register** when they start, send **heartbeats** to say they are alive, and are removed when they stop or miss heartbeats. Examples: Consul, etcd, ZooKeeper, Netflix Eureka, and the built-in registry inside Kubernetes.
- **Sidecar**: a helper process deployed right next to each service instance (in the same pod or host) that handles cross-cutting jobs such as networking, TLS, retries, logging. Like a motorbike sidecar, it rides alongside and goes everywhere the main process goes.
- **Service mesh**: a fleet of sidecar proxies, one per service instance, that intercept all service-to-service traffic, plus a **control plane** that configures them all. The sidecars form the **data plane**. Examples: Istio and Linkerd; Envoy is the most common sidecar proxy.

Why it matters

Hard-coding IP addresses breaks the moment autoscaling adds a machine or a container is rescheduled. And every service needs the same plumbing (timeouts, retries, encryption between services, metrics); writing it into every codebase in every language is slow and inconsistent. A mesh moves that plumbing out of application code.

How it works

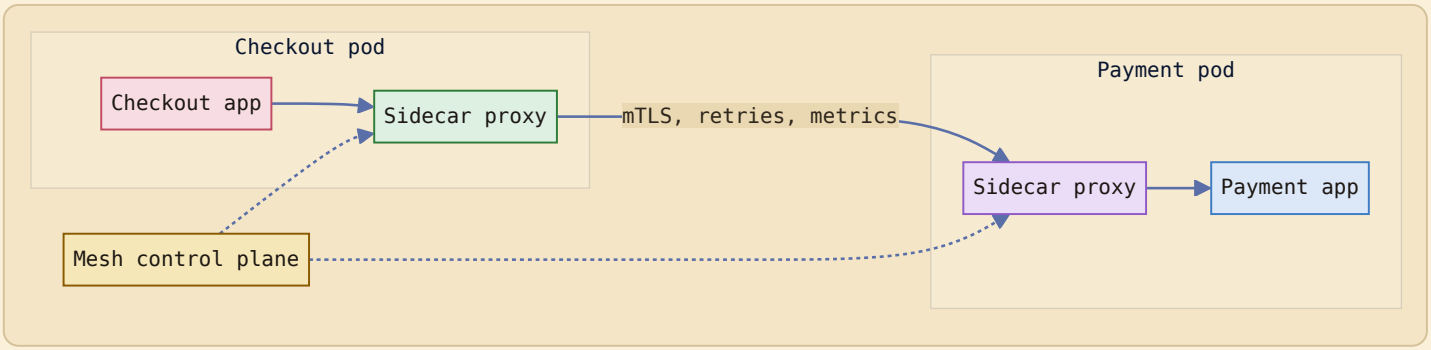
1. A new payment-service instance starts on `10.0.3.17:8080` and registers itself in the registry.
2. The registry health-checks it every few seconds.
3. The checkout service wants to call payments. It either asks the registry itself or goes through a router that asks for it.
4. The call goes to one healthy instance. If that instance dies, it misses heartbeats and is removed, usually within seconds.

Types

- **Client-side discovery**: the caller queries the registry, gets the list of instances and picks one with its own load-balancing logic. Fewer hops, but every client needs a discovery library. Netflix Eureka

with the Ribbon client library worked this way.

- **Server-side discovery**: the caller sends the request to a fixed address (a load balancer or router), which looks up the registry and forwards it. Clients stay simple. AWS load balancers with target groups and Kubernetes Services work this way.
- **DNS-based discovery**: the registry publishes instances as DNS records (Consul and Kubernetes both do), so any program can find a service by a name such as `payments.default.svc.cluster.local`. Simple, but DNS caching can leave stale addresses.



Trade-offs

A service mesh gives you, without code changes, mutual TLS (both sides prove their identity with certificates) between all services, uniform retries and timeouts, traffic splitting for canary releases and per-call metrics and traces. It costs an extra proxy hop on each side of every call (usually well under a millisecond, but not zero), extra memory and CPU per pod, and a complex control plane your team must understand when things go wrong. For ten services it is often overkill.

Common mistakes

- Running a registry with no health checks, so dead instances keep receiving traffic.
- Adopting a full service mesh before you have the operational skills to debug it.
- Configuring retries both in application code and in the mesh, multiplying retries (see the next section).

In the interview

Q: Client-side or server-side discovery?

A: Server-side keeps clients thin and language-neutral, so it is the usual default, especially on Kubernetes. Client-side saves a hop and gives the caller control over load balancing, which matters at very high call rates.

Q: What is a service mesh in one sentence?

A: A sidecar proxy beside every service instance that handles service-to-service networking (discovery, load balancing, mTLS, retries, metrics), all configured centrally by a control plane.

TL;DR

Service discovery is a live phone book of service instances; a sidecar is a helper process beside each instance; a service mesh is a centrally managed fleet of sidecars that handles networking for you.

Timeouts, retries, exponential backoff and jitter

What it is

- A **timeout** is the maximum time a caller will wait for a response before giving up. Like hanging up after the phone rings 10 times.
- A **retry** is trying the same request again after a failure, in case the failure was temporary (a **transient fault**, such as a dropped packet or a server that was restarting).
- **Exponential backoff** means waiting longer after each failed attempt, doubling the wait: 100 ms, 200 ms, 400 ms, 800 ms, up to a cap.

- **Jitter** means adding randomness to that wait so that thousands of clients do not all retry at exactly the same moment.

Why it matters

Without a timeout, a caller waiting on a dead service holds a thread, a connection and memory forever; soon it runs out and dies too. That is how one slow service takes down everything upstream, a **cascading failure**. Retries turn many transient failures into successes, but careless retries are dangerous: if a struggling service gets every request three times, its load triples exactly when it is weakest. This is a **retry storm**.

How it works

Setting a timeout. Base it on the callee's measured latency, not a guess. If payment calls have a p99 of 300 ms, a timeout around 500 ms to 1 s is sensible. Use a **deadline** across the whole request: if the user-facing budget is 2 s and 1.5 s has already gone, the next call gets at most 0.5 s. gRPC propagates deadlines between services for exactly this reason.

Retrying safely.

1. Retry only errors that may be temporary: timeouts, connection resets, HTTP 502/503/504, HTTP 429 (after the `Retry-After` delay). Never retry 400 Bad Request or 403 Forbidden; they will fail again.
2. Retry only **idempotent** operations (doing them twice has the same effect as once), or make them idempotent with an idempotency key (Part 4). Retrying "charge card $50" without a key can charge twice.
3. Limit attempts, typically 2 to 3 retries in total.
4. Wait with exponential backoff and jitter between attempts.
5. Retry at one layer only. If the browser, gateway, service A and service B each make up to 3 attempts, one user click can become 3 x 3 x 3 x 3 = 81 calls to the database.

Backoff with jitter. With "full jitter", the wait before attempt n is a random number between 0 and `min(cap, base * 2^n)`.

```
base = 100 ms, cap = 10 s
attempt 1: wait random(0, 200 ms)
attempt 2: wait random(0, 400 ms)
attempt 3: wait random(0, 800 ms)
```

Without jitter, 10,000 clients that failed at the same second all retry at exactly +100 ms, then +200 ms, in synchronised waves. With jitter, their retries spread across the window and the server sees a smooth trickle.

A **retry budget** is a further guard: allow retries to add at most, say, 10% extra load. If more than 10% of calls are already retries, stop retrying and fail fast.

Trade-offs

Short timeouts free resources quickly but may cut off requests that would have succeeded; long ones risk resource exhaustion. More retries improve success rates for rare blips but amplify load during real outages. **Hedged requests** (sending a second copy to another replica if the first has not answered by the p95 time, and taking whichever answers first) cut tail latency at the cost of a few percent extra load.

Common mistakes

- No timeout at all. Many HTTP client libraries default to waiting forever or for minutes.
- A caller timeout shorter than the callee's, so the callee keeps working on an answer nobody will read.
- Retrying non-idempotent writes.
- Retrying immediately in a tight loop.

Real-world examples

AWS SDKs retry throttled and transient errors with exponential backoff and jitter by default, and the AWS Architecture Blog popularised the "full jitter" formula. Google's **Site Reliability Engineering (SRE)**

book describes per-client retry budgets to prevent retry storms.

In the interview

Q: Why add jitter to backoff?

A: To break synchronisation. When many clients fail together, plain backoff makes them retry together in waves that knock the service over again. Random jitter spreads them out.

Q: Where should retries live in a deep call chain?

A: At one layer, ideally close to the user or at the edge of each service boundary, with a small attempt limit, so retries do not multiply across layers.

TL;DR

Always set timeouts; retry only transient failures on idempotent operations, a few times, with exponential backoff plus random jitter, at a single layer.

Circuit breakers, bulkheads and fallbacks

What it is

- A **circuit breaker** is a wrapper around calls to a dependency that "trips" after too many failures and then fails calls instantly for a while instead of waiting on a broken service. It is named after the electrical breaker in your house that cuts power when a wire overloads, protecting the rest of the house.
- A **bulkhead** isolates resources so one failing part cannot consume everything. Ships are divided into watertight compartments (bulkheads) so a hole floods one compartment, not the whole hull.
- A **fallback** is what you do instead when a call fails: return cached data, a default, or a reduced answer.

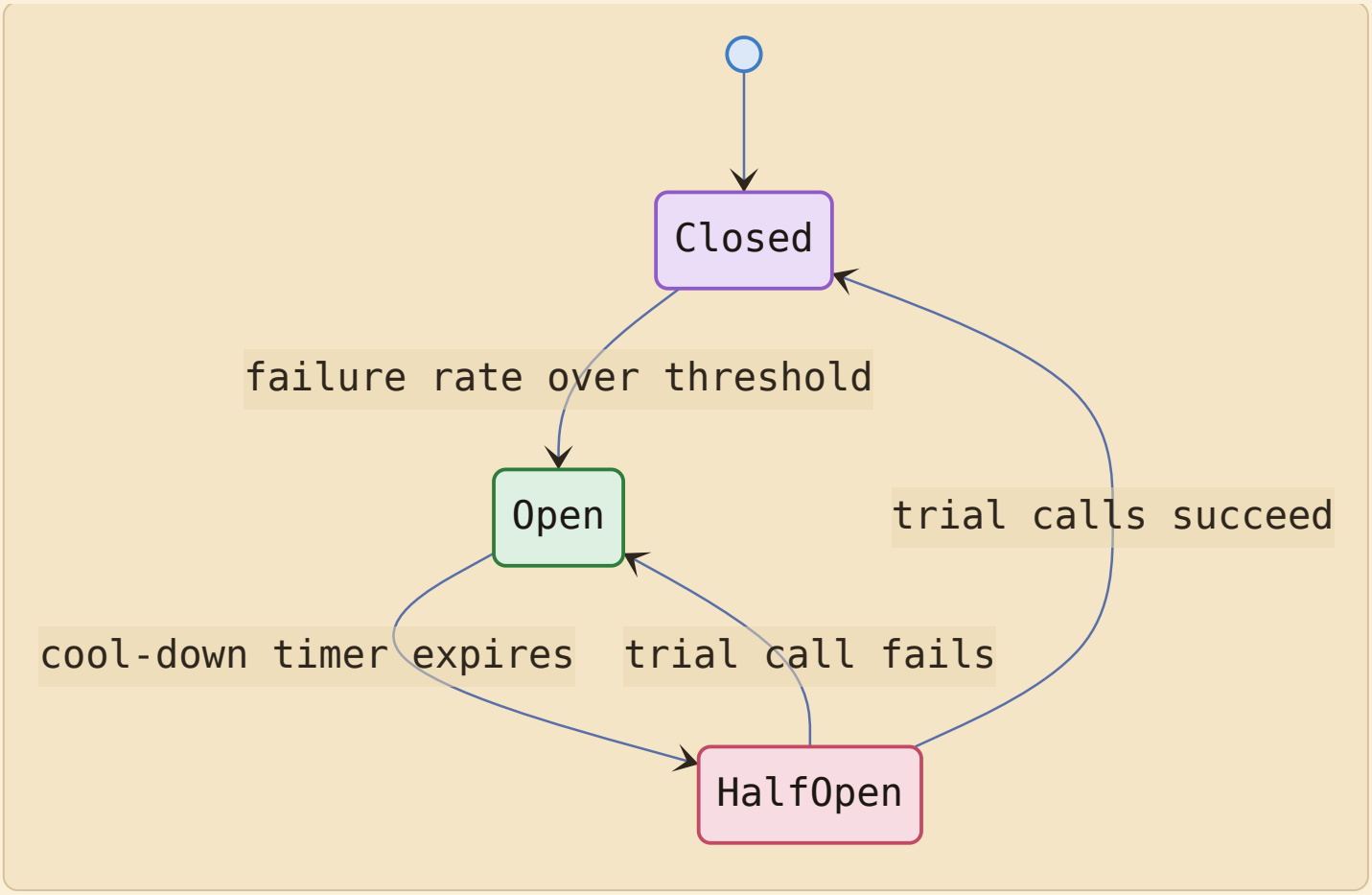
Why it matters

Suppose a recommendation service hangs. Every product page waits 2 seconds for its timeout. Web server threads pile up waiting, the thread pool fills, and now product pages, which could have rendered without recommendations, fail too. A circuit breaker notices recommendations are broken and stops calling them; a bulkhead makes sure recommendation calls can only use, say, 20 of the 200 threads; a fallback shows "popular items" instead.

How it works

A circuit breaker has three states.

1. **Closed** (normal): calls pass through. The breaker counts failures in a rolling window, for example the last 100 calls or last 10 seconds.
2. If failures exceed a threshold (say 50% of at least 20 calls), it moves to **Open**: every call fails immediately with an error or goes straight to the fallback, without touching the dependency. This gives the dependency time to recover and frees the caller's threads.
3. After a cool-down (say 30 seconds) it moves to **Half-open**: it lets a few trial calls through. If they succeed, it returns to Closed. If they fail, back to Open for another cool-down.



Bulkheads in practice: separate thread pools or connection pools per dependency, separate queues per customer tier, separate clusters for internal batch jobs and user-facing traffic, or **cell-based architecture**, where users are split into independent copies of the whole stack ("cells") so a bad deploy or poison request only affects one cell's users.

Fallback options: a cached last-known value, a static default ("free shipping info unavailable"), a simpler algorithm, an empty section of the page, or queueing the work to finish later.

Trade-offs

Breakers add a little state and configuration per dependency, and thresholds tuned badly either trip on normal noise or never trip. Bulkheads waste some capacity, because a reserved pool sits idle even when another is overloaded. Fallbacks give users stale or partial data, which is fine for recommendations but not for a bank balance.

Common mistakes

- Using a fallback that calls another fragile service, which then fails too.
- One global breaker for a dependency with many endpoints, so one bad endpoint blocks healthy ones.
- Never testing the open state, so the fallback code path has a bug nobody has seen.

Real-world examples

Netflix popularised circuit breakers with its Hystrix library (now in maintenance mode); Resilience4j is the common Java successor, and service meshes such as Istio offer breakers and outlier ejection at the proxy. Amazon describes cell-based architectures to limit the "blast radius" of failures.

In the interview

Q: Explain the half-open state.

A: After the breaker has been open for a cool-down period, it lets a small number of test requests through. Success closes the breaker and resumes normal traffic; failure re-opens it. It is how the breaker discovers that the dependency has recovered without flooding it.

Q: How would you stop a slow non-critical service from taking down your homepage?

A: Tight timeout on the call, a circuit breaker around it, a separate small thread pool (bulkhead) for it, and a fallback such as cached or default content, so the page renders without that section.

TL;DR

A circuit breaker stops calling a failing dependency (Closed, Open, Half-open); a bulkhead caps how many resources any one dependency can consume; a fallback gives a degraded answer instead of an error.

Rate limiting and throttling

What it is

Rate limiting caps how many requests a client may make in a time window, for example 100 requests per minute per user. **Throttling** is often used as a synonym; strictly, throttling slows requests down (delays or queues them) while rate limiting rejects the excess. A nightclub bouncer who lets in only 50 people per hour is rate limiting; one who makes people wait in line until there is room is throttling.

Why it matters

- **Protection**: stops one buggy script or abusive client from overwhelming the service for everyone.
- **Fairness**: each customer gets a fair share.
- **Cost control**: expensive operations (sending SMS, calling a paid AI model) cannot run away.
- **Security**: slows brute-force password guessing and scraping.
- **Business tiers**: free plan 1,000 calls per day, paid plan 100,000.

How it works

When a client exceeds its limit, the server returns **HTTP 429 Too Many Requests**, ideally with a `Retry-After` header saying how many seconds to wait. Many APIs also send headers so clients can pace themselves:

```
HTTP/1.1 429 Too Many Requests
Retry-After: 30
X-RateLimit-Limit: 5000
X-RateLimit-Remaining: 0
X-RateLimit-Reset: 1767225600
```

The limit is applied per **key**: user ID, API key, IP address, or a combination such as (user, endpoint).

Types

Algorithms.

- **Token bucket**: a bucket holds up to B tokens and is refilled at R tokens per second. Each request takes one token; if the bucket is empty, the request is rejected. It allows short **bursts** up to B, then a steady rate R. Example: B = 10, R = 1 per second lets a user fire 10 requests at once, then one per second. Only two numbers per key (token count, last refill time), so it is cheap. The most widely used algorithm.
- **Leaky bucket**: requests enter a queue (the bucket) and leave at a fixed rate, like water dripping from a hole. If the bucket is full, new requests are dropped. It smooths bursts into a perfectly steady outflow, good for protecting a downstream that cannot handle bursts, but adds queueing delay.
- **Fixed window counter**: count requests in each calendar window (12:00:00 to 12:00:59). Reset at the boundary. Very simple, one counter per key, but a client can send the full limit at 12:00:59 and again at 12:01:00, doubling the rate across the boundary.
- **Sliding window log**: store the timestamp of every request; on each new request, drop timestamps older than the window and count the rest. Exact, but stores one entry per request, which is memory-hungry at high limits.
- **Sliding window counter**: keep counters for the current and previous fixed windows and estimate the sliding count as `current + previous x (fraction of previous window still inside the sliding window)`. If the limit is 100 per minute, we are 15 s into the current minute with 30 requests, and the previous

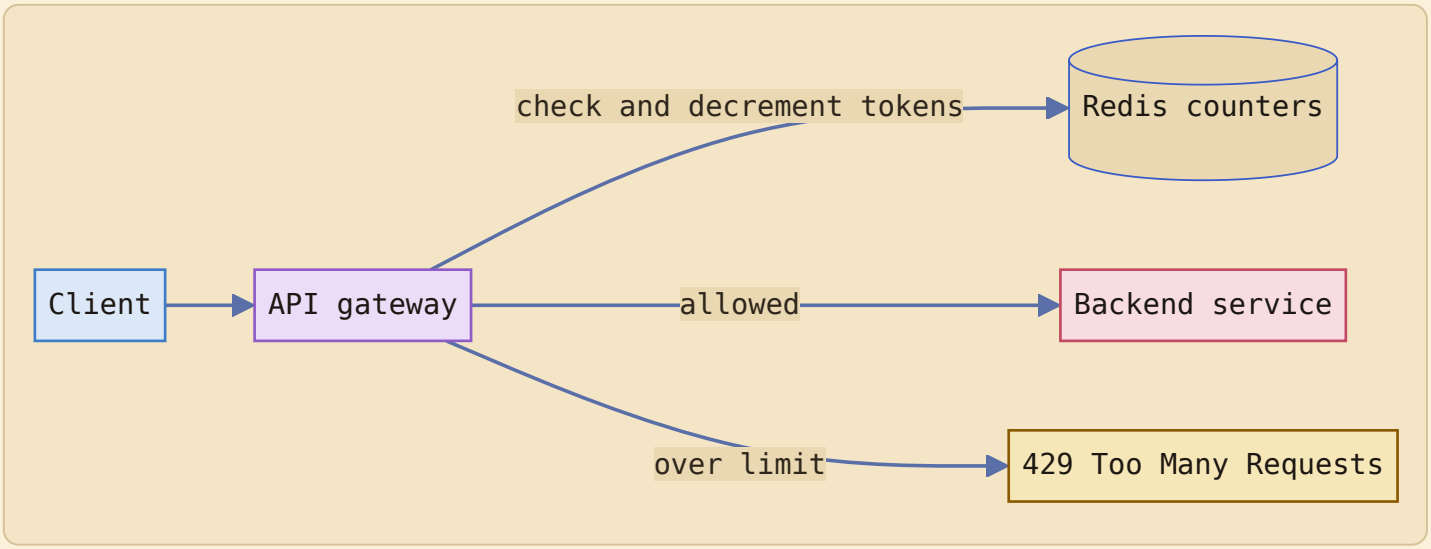
minute had 80, the estimate is 30 + 80 x 0.75 = 90, so the request is allowed. Nearly as accurate as the log with the memory of fixed windows.

Algorithm	Allows bursts	Memory per key	Accuracy
Token bucket	Yes, up to B	Tiny	Good
Leaky bucket	No, smooths	Queue size	Good
Fixed window	Yes, 2x at edges	Tiny	Rough
Sliding log	No	One entry per request	Exact
Sliding counter	Little	Two counters	Very good

Where to rate limit.

- **Client side**: polite clients self-limit, but you cannot trust them.
- **Edge / CDN / WAF**: block floods by IP before they reach you (Cloudflare, AWS WAF rate rules).
- **API gateway or load balancer**: the common place for per-API-key limits.
- **Inside each service**: protects the service itself, for example limiting calls to an expensive endpoint.
- **In front of a fragile dependency**: an outbound limiter so you never exceed a partner API's quota.

Distributed rate limiting. With 20 gateway servers, each must see the same count. The usual answer is a shared fast store such as Redis, with an atomic increment-and-check (often a small Lua script so the read and write happen in one step). To avoid a Redis call on every request, some systems give each node a local share of the limit and sync periodically, accepting slight inaccuracy.



Trade-offs

Strict, globally exact limits need a central store on the hot path (added latency, a dependency that can fail). Local per-node limits are fast but approximate. If the limiter's store is down, you must choose **fail open** (allow traffic, risk overload) or **fail closed** (reject traffic, guaranteed outage); most choose fail open for user traffic.

Common mistakes

- Limiting only by IP address, which punishes many users behind one corporate NAT and is dodged by attackers with many IPs.
- Returning 500 or 503 instead of 429, so clients do not know to back off.
- Forgetting to tell clients their remaining quota.

Real-world examples

GitHub's REST API allows 5,000 requests per hour for an authenticated user and 60 per hour unauthenticated, and reports usage in `x-ratelimit-*` headers. Stripe has written about using token buckets

for its API rate limiters. NGINX's `limit_req` module implements a leaky bucket. Cloudflare has described using the sliding window counter approximation at its edge.

In the interview

Q: Design a rate limiter for an API used by millions of users.

A: Token bucket per API key, stored in Redis with an atomic script, enforced at the API gateway. Return 429 with `Retry-After` and rate-limit headers. Shard Redis by key, keep limits in config so they change without deploys, and fail open if Redis is unavailable.

Q: Token bucket vs leaky bucket?

A: Token bucket permits bursts up to the bucket size while enforcing an average rate, which suits APIs. Leaky bucket outputs a constant rate, which suits protecting a downstream that cannot absorb bursts.

TL;DR

Rate limiting caps requests per client per time window and answers excess with 429; token bucket is the default algorithm; enforce it at the edge or gateway with a shared counter store.

Load shedding and graceful degradation

What it is

- **Load shedding** means deliberately rejecting some requests when the system is overloaded so that the rest still succeed. Like an electricity company switching off some neighbourhoods for an hour so the whole grid does not collapse.
- **Graceful degradation** means turning off non-essential features under stress so the core keeps working. An online shop at peak may hide personalised recommendations and reviews but keep search and checkout.

Why it matters

When a server receives more work than it can do, things do not just get a bit slower. Queues grow, every request waits longer, many time out after the server has already done the work, clients retry, and the server ends up spending all its effort on requests nobody is waiting for. Throughput can collapse to near zero. This is **congestion collapse**. Shedding the excess early keeps **goodput** (useful completed work) high.

How it works

1. Measure a sign of overload: CPU above 90%, concurrent in-flight requests above a limit, queue wait time above, say, 100 ms.
2. When overloaded, reject new low-priority requests immediately and cheaply with 503 Service Unavailable (with `Retry-After`) rather than queueing them.
3. Prioritise: shed background jobs and bots first, then logged-out traffic, and keep checkout and login last.
4. Drop requests that have already waited longer than the client's timeout; working on them is wasted.
5. Combine with degradation flags: a "kill switch" turns off expensive features during an incident.

Backpressure (Part 4) is the cooperative cousin: a consumer signals producers to slow down rather than silently dropping.

Trade-offs

Some users see errors on purpose. But controlled, partial failure beats total failure. The hard parts are picking the right overload signal and making the rejection path cheap: if rejecting a request costs as much as serving it, shedding does not help.

Common mistakes

- Unbounded queues: they hide overload until latency is minutes long.
- Shedding at random, dropping a checkout while serving a crawler.

- No degradation switches, so the only lever during an incident is "add servers and hope".

Real-world examples

Google's SRE book describes per-request criticality levels so overloaded servers shed the least important traffic first. Amazon has written about using in-flight request limits and dropping requests whose callers have already timed out. Netflix serves a generic, non-personalised home page row when its personalisation services are degraded.

In the interview

Q: What happens if traffic is 3x your capacity and autoscaling takes 5 minutes?

A: Without protection, queues grow and the service collapses. I would shed load at the edge by priority with fast 503s, rate limit heavy clients, turn off non-essential features, and serve cached content where possible until new capacity arrives.

TL;DR

When overloaded, reject some work early and cheaply (load shedding) and switch off extras (graceful degradation) so the core stays up.

Autoscaling

What it is

Autoscaling automatically adds servers (or containers, or function copies) when load rises and removes them when it falls. Adding capacity is **scaling out**; removing it is **scaling in**. A supermarket that opens more tills when queues get long and closes them in the quiet afternoon is autoscaling its cashiers. It is the practical form of the **elasticity** introduced in Part 1.

Why it matters

Traffic is rarely flat. A food-delivery app may see 5x more orders at 7 pm than at 3 pm. Without autoscaling you either buy for the peak (and pay for idle machines most of the day) or buy for the average (and fall over at dinner time). In the cloud you pay by the second or hour, so matching capacity to demand saves real money.

How it works

1. Choose a **metric** that tracks load: average CPU utilisation, requests per second per instance, queue length (messages waiting), p95 latency, or a custom business metric.
2. Choose a **target**: for example "keep average CPU at 60%". The headroom (40%) absorbs spikes while new instances boot.
3. A controller checks the metric periodically (Kubernetes' Horizontal Pod Autoscaler checks every 15 seconds by default) and computes the needed count: `desired = current x (current metric / target)`. With 10 instances at 90% CPU and a 60% target, desired = 10 x 90/60 = 15.
4. It launches 5 more instances. They boot, pass health checks and join the load balancer. This takes seconds for containers and roughly 1 to 5 minutes for virtual machines.
5. When load drops, it scales in slowly, after a **cool-down** period (often 5 to 10 minutes), so it does not flap up and down, and it **drains** connections from instances before terminating them.
6. **Minimum and maximum** bounds stop it from scaling to zero by mistake or to 10,000 machines during an attack.

Types

- **Reactive (dynamic) scaling**: responds to current metrics as above. Simple, but always slightly late, because capacity arrives minutes after the spike begins.
- **Scheduled scaling**: scale by the clock. "Every weekday at 08:30 set minimum to 40 instances." Perfect for known patterns such as business hours, a TV ad slot or a ticket sale opening at 10:00.
- **Predictive scaling**: a model learns daily and weekly patterns from history and adds capacity before the forecasted rise. AWS Auto Scaling offers this.

- **Horizontal vs vertical autoscaling**: horizontal adds instances (the common case); vertical gives an instance more CPU or memory (Kubernetes' Vertical Pod Autoscaler), which usually needs a restart.
- **Cluster autoscaling**: in Kubernetes, adds whole machines (nodes) when pods cannot fit on the existing ones.

Trade-offs

Reactive scaling is cheap but lags; keep headroom or combine it with scheduled or predictive scaling. Scaling on CPU is easy but misleading for services that wait on I/O; requests per instance or queue depth is often better. Autoscaling the stateless web tier is easy; databases and other stateful systems scale far more slowly because data must be copied.

Common mistakes

- Scaling the web tier to 200 instances that then exhaust the database's connection limit, moving the bottleneck rather than fixing it.
- Targets set too high (90% CPU), leaving no buffer for the minutes it takes to boot.
- Slow startup (large images, long cache warm-up) making autoscaling useless for sudden spikes.
- No maximum, so a bug or a DDoS attack produces a huge bill.

Real-world examples

AWS EC2 Auto Scaling groups, Google Cloud managed instance groups, Kubernetes Horizontal Pod Autoscaler and KEDA (Kubernetes Event-driven Autoscaling, which scales on event sources such as Kafka consumer lag). Netflix built a predictive autoscaler called Scryer to pre-provision capacity before its daily evening peak.

In the interview

Q: Your service gets a 10x spike every evening at 8 pm. How do you scale?

A: Scheduled or predictive scaling to raise capacity before 8 pm, reactive scaling on requests per instance to handle surprises, and make sure the database, caches and connection pools can handle the higher instance count.

Q: Which metric would you autoscale a queue worker on?

A: Queue depth or consumer lag divided by the rate one worker can process, because CPU does not tell you how far behind you are.

TL;DR

Autoscaling adds and removes capacity based on a metric and a target; reactive is late, scheduled and predictive are early; set min and max bounds and keep headroom.

Virtual machines, containers and Docker

What it is

- A **virtual machine (VM)** is a software-emulated computer. A program called a **hypervisor** (such as KVM, Xen, VMware ESXi, or AWS's Nitro) splits one physical server into several VMs, each with its own complete operating system (its own Linux or Windows kernel). It is like dividing a house into flats, each with its own kitchen, plumbing and front door.
- A **container** is an isolated process (or group of processes) that shares the host machine's operating-system kernel but has its own view of files, network and process list, and its own limits on CPU and memory. It is like rooms in a shared house: private rooms, shared kitchen and plumbing. Linux implements it with **namespaces** (what a process can see) and **cgroups** (how much it can use).
- A **container image** is a packaged, read-only snapshot of an application and everything it needs (runtime, libraries, config defaults). Run the image and you get a container.
- **Docker** is the tool that made containers popular from 2013: a simple file format (the `Dockerfile`) to build images, a way to run them and **registries** (Docker Hub, Amazon ECR) to share them.

Why it matters

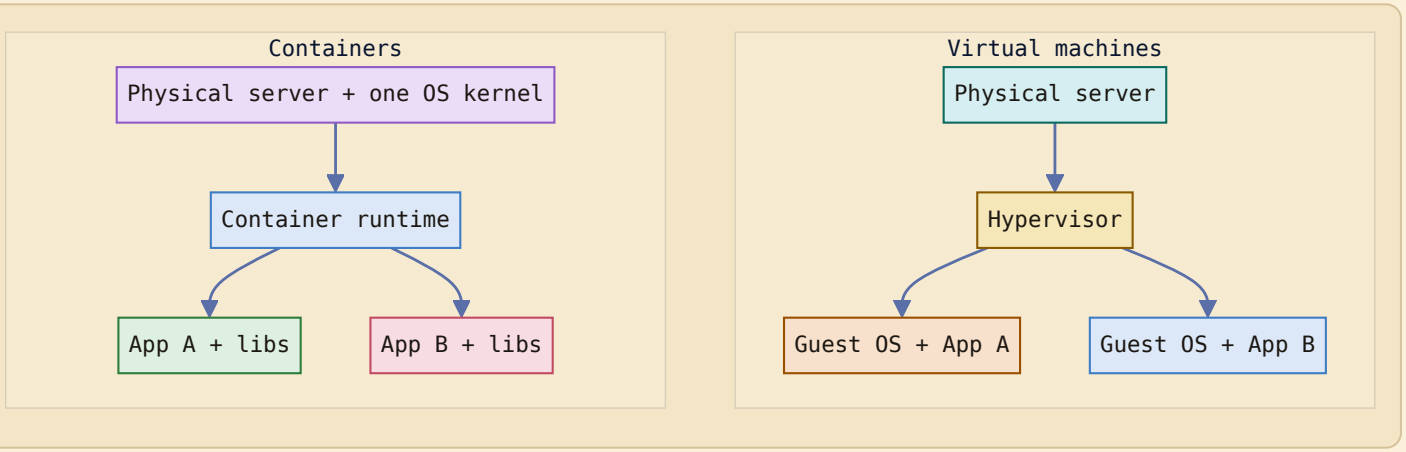
"It works on my machine" disappears: the same image runs on a laptop, in testing and in production. Containers start in under a second to a few seconds and use megabytes rather than gigabytes of overhead, so you can pack dozens onto a server and scale them quickly. They are the unit that modern platforms like Kubernetes schedule.

How it works

A tiny Dockerfile for a web service:

```
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
CMD ["gunicorn", "-b", "0.0.0.0:8080", "app:server"]
```

1. `docker build` turns this into an image made of stacked **layers** (base OS files, then the installed libraries, then your code). Unchanged layers are cached, so rebuilding after a code change takes seconds.
2. The image is pushed to a registry with a tag such as `shop-api:1.4.2`.
3. Any server pulls it and runs it as a container, with its own isolated filesystem and network interface.



Trade-offs

	Virtual machine	Container
Isolation	Strong: separate kernel	Weaker: shared kernel
Start time	Tens of seconds to minutes	Milliseconds to seconds
Size	Gigabytes	Megabytes
Density per host	Few to tens	Tens to hundreds
Typical use	Multi-tenant clouds, different OSes	Packaging and running apps

In practice they are combined: cloud providers give you VMs, and you run containers inside them. Lightweight "micro-VMs" such as AWS Firecracker (used by Lambda and Fargate) blend the two, booting a minimal VM in around 125 ms.

Common mistakes

- Storing important data inside the container's filesystem; it vanishes when the container is replaced. Use volumes or external storage.
- Huge images (several GB) that slow down every deploy and scale-out.
- Running containers as root and baking secrets into images.

In the interview

Q: Container vs VM in one line?

A: A VM virtualises hardware and runs its own OS kernel; a container virtualises the OS, sharing the host kernel, so it is lighter and faster to start but less isolated.

TL;DR

VMs are full emulated computers with their own OS; containers are isolated processes sharing one kernel; Docker packages apps as portable container images.

Kubernetes basics

What it is

Kubernetes (often written **K8s**) is a **container orchestrator**: software that runs and manages containers across a fleet of machines. You tell it *what* you want ("run 5 copies of `shop-api:1.4.2`, each with 1 CPU and 512 MB of memory, reachable at the name `shop-api`") and it continuously works to make reality match. Picture a harbour master for a busy port: it assigns each shipping container to a free berth, notices when one falls into the water and orders a replacement, and moves cargo elsewhere when a crane breaks. Google open-sourced it in 2014, based on its internal system Borg; it is now run by the Cloud Native Computing Foundation (CNCF).

Why it matters

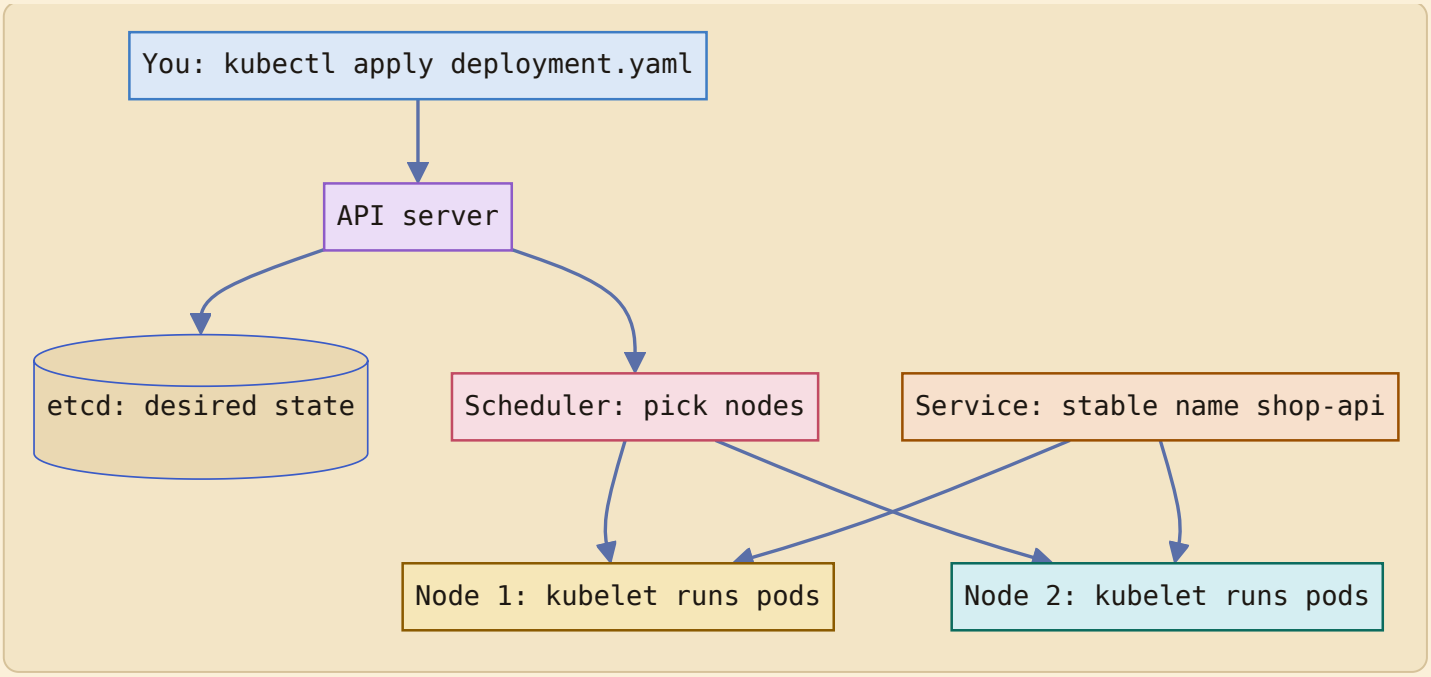
With 300 containers across 40 machines, nobody can place, restart and wire them up by hand. Kubernetes automates scheduling, self-healing, scaling, rolling updates, service discovery and configuration, and it works the same on every major cloud (Amazon EKS, Google GKE, Azure AKS) and on your own servers.

How it works

Key words, beginner level:

- **Cluster**: the whole Kubernetes installation, a set of machines plus the brain that controls them.
- **Node**: one machine (physical or VM) in the cluster that runs containers. Each node runs an agent called the **kubelet**.
- **Control plane**: the brain. The **API server** receives your instructions, **etcd** stores the cluster's desired and current state, the **scheduler** decides which node each pod goes on, and **controllers** keep fixing differences.
- **Pod**: the smallest unit Kubernetes runs. One or more tightly coupled containers that share a network address and storage, for example your app plus its sidecar. Pods are disposable: they are replaced, not repaired.
- **Deployment**: a description of a stateless app ("5 replicas of this pod template"). It creates and replaces pods and performs rolling updates when you change the image.
- **Service**: a stable name and virtual IP address that load-balances traffic across the current set of matching pods, even as pods come and go. This is Kubernetes' built-in server-side service discovery.
- **Ingress** (or the newer **Gateway API**): rules that route outside HTTP traffic (`shop.com/api`) to Services.
- **ConfigMap** and **Secret**: configuration values and sensitive values injected into pods.
- **Namespace**: a folder-like grouping to separate teams or environments.
- **StatefulSet**: like a Deployment but for stateful apps (databases), giving each pod a stable name and its own persistent storage.
- **Probes**: a **liveness probe** tells Kubernetes to restart a stuck container; a **readiness probe** tells it whether to send the pod traffic yet.

The heart of Kubernetes is the **reconciliation loop**: controllers repeatedly compare desired state with actual state and act on any difference. If a node dies and 2 of your 5 pods vanish, the Deployment's controller sees "3, want 5" and creates 2 new pods on healthy nodes.



Trade-offs

Kubernetes is powerful but complex: networking, storage, security policies and upgrades all need expertise, and a small team can spend more time running the platform than building the product. Managed offerings (EKS, GKE, AKS) and simpler platforms (AWS ECS, Google Cloud Run, Fly.io, or a Heroku-style **platform as a service (PaaS)**, where you hand over code and the provider runs it) are often the better first choice.

Common mistakes

- No resource requests and limits, so one pod hogs a node's memory and others get killed.
- Missing readiness probes, so traffic arrives before the app has finished starting.
- Running a production database on Kubernetes without understanding StatefulSets, persistent volumes and backups.

In the interview

Q: What happens if a node crashes?

A: The control plane notices the node has stopped reporting, marks its pods as lost, and the Deployment controller creates replacement pods on healthy nodes. The Service automatically stops routing to the dead pods.

Q: Pod vs container?

A: A container is one isolated process image; a pod is Kubernetes' wrapper around one or more containers that share an IP address and volumes and are scheduled together.

TL;DR

Kubernetes runs containers across many machines, keeping the actual state equal to the desired state you declare: pods run on nodes, Deployments manage replicas and rollouts, Services give them a stable address.

Infrastructure as code

What it is

Infrastructure as code (IaC) means describing your servers, networks, load balancers, databases and permissions in text files that are kept in version control and applied by a tool, instead of clicking around a cloud console. It is like a recipe instead of cooking from memory: anyone can reproduce the same dish, and you can see exactly what changed between versions.

Why it matters

Hand-built environments drift apart ("staging has a setting production does not"), cannot be rebuilt quickly after a disaster and have no review trail. With IaC, a new region or a test environment is one command away, every change is code-reviewed, and the files themselves document the system.

How it works

1. Write a **declarative** description of the desired resources, for example in Terraform:

```
resource "aws_s3_bucket" "images" {
  bucket = "shop-product-images"
}
```

2. Run a **plan**: the tool compares the files with what actually exists and shows the diff ("will create 1 bucket").
3. A teammate reviews the plan in a pull request.
4. **Apply** it; the tool calls the cloud APIs to create or change resources and records what it built in a **state file**.

Declarative tools describe the end state (Terraform, AWS CloudFormation, Kubernetes YAML, Pulumi); **imperative** scripts describe the steps. **Configuration management** tools (Ansible, Chef, Puppet) configure software on existing servers. **Immutable infrastructure** is the related idea of never patching a running server: build a new image and replace the old server instead.

Trade-offs and common mistakes

IaC has a learning curve and its state files must be stored and locked carefully (two people applying at once can corrupt them). The classic mistake is mixing IaC with manual console edits, which creates **drift** the next apply may silently undo.

TL;DR

Infrastructure as code keeps your cloud setup in reviewed, versioned files that a tool applies, so environments are reproducible and changes are auditable.

CI/CD

What it is

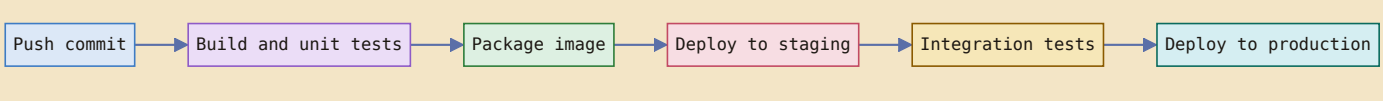
- **Continuous integration (CI)**: every developer merges small changes into the main branch frequently (at least daily), and each change automatically triggers a build and test run. It catches conflicts and bugs within minutes.
- **Continuous delivery (CD)**: every change that passes the pipeline is automatically packaged and ready to release to production at the push of a button.
- **Continuous deployment**: one step further, every passing change goes to production automatically, with no human button.

A **pipeline** is the automated sequence of steps. Think of a car factory line: each station checks or adds something, and a faulty car is pulled off before it reaches the showroom.

Why it matters

Small, frequent releases are less risky than big, rare ones: if a release contains one change and breaks something, you know exactly what to roll back. The research programme behind the **DORA metrics** (deployment frequency, lead time for changes, change failure rate, time to restore service) found that high-performing teams deploy more often *and* have fewer failures.

How it works



1. A developer pushes a commit or opens a pull request.
2. The CI server (GitHub Actions, GitLab CI, Jenkins, CircleCI) compiles the code, runs linters and unit tests and security scans, typically in under 10 minutes.
3. It builds one versioned artifact, such as a container image, and stores it. The **same artifact** is promoted through every environment, never rebuilt.
4. It deploys to a staging environment and runs integration or end-to-end tests.
5. It deploys to production using a safe strategy (next section), watches health metrics and rolls back automatically if they worsen.

Common mistakes

- Flaky tests that fail randomly, so people learn to ignore red builds.
- Rebuilding separately for production, so what you tested is not what you ship.
- Long-lived feature branches merged once a month, which defeats the point of integration.
- Database schema changes that are not backward compatible, so the new code and old code cannot run side by side during a rollout. Use **expand and contract**: add the new column first, deploy code that writes both, migrate, then remove the old column in a later release.

TL;DR

CI builds and tests every change automatically; CD keeps every passing build ready to release (or releases it automatically); small frequent releases are safer releases.

Deployment strategies and rollback

What it is

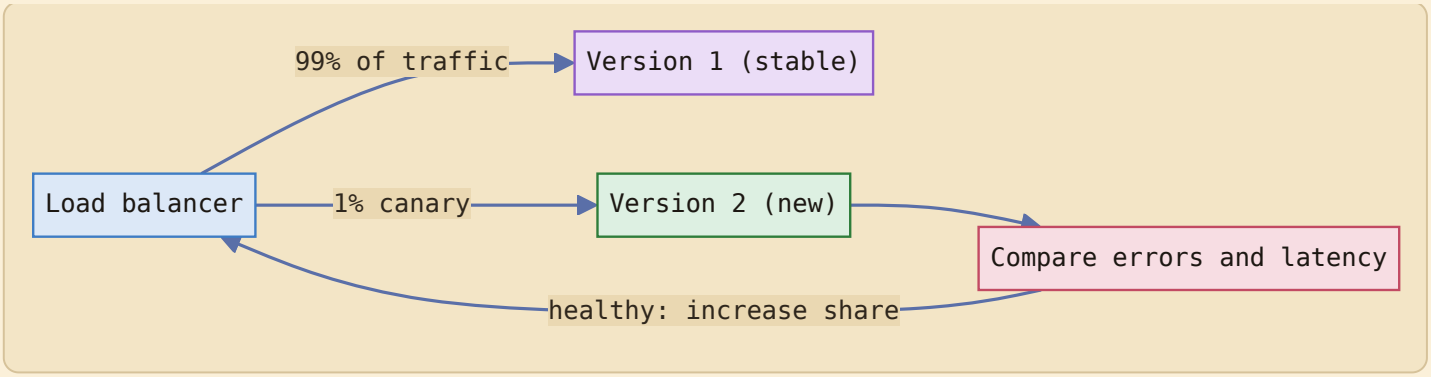
A **deployment strategy** is how you replace the old version of a running service with a new one without taking the site down, and how you limit damage if the new version is bad. The **blast radius** is how many users a bad change can hurt; good strategies keep it small.

Why it matters

Most outages are caused by changes: a new code release, a config edit, a schema migration. How you release decides whether a bug hits 1% of users for two minutes or 100% of users for an hour.

Types

- **Recreate (big bang)**: stop all old instances, start new ones. Simple, but there is downtime. Only acceptable for internal tools or maintenance windows.
- **Rolling deployment**: replace instances a few at a time (say 2 of 20). Each new instance must pass health checks before the next batch goes. No extra servers needed and no downtime, but for a while both versions serve traffic side by side, and rollback means rolling back through the batches. Kubernetes Deployments do this by default.
- **Blue-green deployment**: run two full environments. **Blue** serves all live traffic on the old version; **green** gets the new version and is tested privately. Then the load balancer (or DNS) switches all traffic to green in one step. Rollback is instant: switch back to blue. The cost is double capacity during the release, and databases are shared, so schema changes must work for both.
- **Canary release**: send a small slice of real traffic, say 1%, to the new version, compare its error rate and latency with the old version, then step up (1%, 5%, 25%, 50%, 100%) if healthy, or pull it back automatically if not. Named after canaries that miners carried to detect gas before it hurt people. Smallest blast radius, but needs good metrics and automation.
- **Shadow (dark) launch**: copy real traffic to the new version but throw away its responses. Tests performance under real load with zero user impact; you must be careful that it does not write data or send emails.
- **Feature flags (feature toggles)**: ship new code turned off, then turn it on at runtime through configuration, for internal users first, then 5% of customers, then everyone. This separates **deploy** (code is on the servers) from **release** (users see the feature). A flag is also a kill switch: turn it off in seconds without a deploy. Tools: LaunchDarkly, Unleash, Flagsmith, or in-house systems.



Rollback means returning to the last known-good version. During an incident, going back beats going forward: restore the last good version, then investigate calmly. Make it boring: keep the previous artifact, automate the button, practise it. Rollback is easy for stateless code, hard when the release changed data or the schema in a non-backward-compatible way; then you need **roll forward** (ship a quick fix) instead.

Trade-offs

Strategy	Downtime	Extra capacity	Rollback speed	Blast radius
Recreate	Yes	None	Slow	Everyone
Rolling	No	Small	Medium	Grows per batch
Blue-green	No	2x briefly	Instant	Everyone at switch
Canary	No	Small	Fast	Tiny at first
Feature flag	No	None	Seconds	Chosen users

Common mistakes

- Canary without automated metric comparison, so nobody notices the 1% is failing.
- Old feature flags never cleaned up, leaving confusing dead code paths.
- Treating config changes as safe and pushing them globally at once.

Real-world examples

In 2012 **Knight Capital** deployed new trading code to only 7 of its 8 servers; a reused flag activated old dormant code on the eighth, which sent more than 4 million orders into the market and lost the firm about $440 million in roughly 45 minutes (the SEC later put the loss above $460 million). In July 2024 a faulty **CrowdStrike** content update was pushed to all customers at once and crashed about 8.5 million Windows machines (Microsoft's estimate); staged, canary-style rollout was the main lesson. Large companies such as Google and Meta release most changes through progressive rollouts with automatic rollback.

In the interview

Q: Blue-green or canary?

A: Blue-green gives an instant switch and instant rollback but exposes everyone at once and needs double capacity. Canary exposes a small percentage first and grows only if metrics stay healthy, so a bad release hurts far fewer users. For high-traffic user-facing services I prefer canary, ideally combined with feature flags.

Q: How do you deploy a database schema change safely?

A: Make it backward compatible with expand and contract: add new columns or tables first, deploy code that works with both shapes, backfill data, switch reads, and only drop the old structure in a later release once nothing uses it.

TL;DR

Release gradually (rolling, canary, feature flags) or switch whole environments (blue-green), watch metrics, and keep rollback one fast, practised step.

Observability: logs, metrics and traces

What it is

Monitoring is watching known numbers to check the system is healthy. **Observability** is the broader ability to understand *why* a system is behaving as it is, including problems you never predicted, purely from the data it emits. A car dashboard (speed, fuel, warning lights) is monitoring; a mechanic's diagnostic computer that lets you ask any question about the engine is closer to observability. The data comes in three main kinds, often called the **three pillars**.

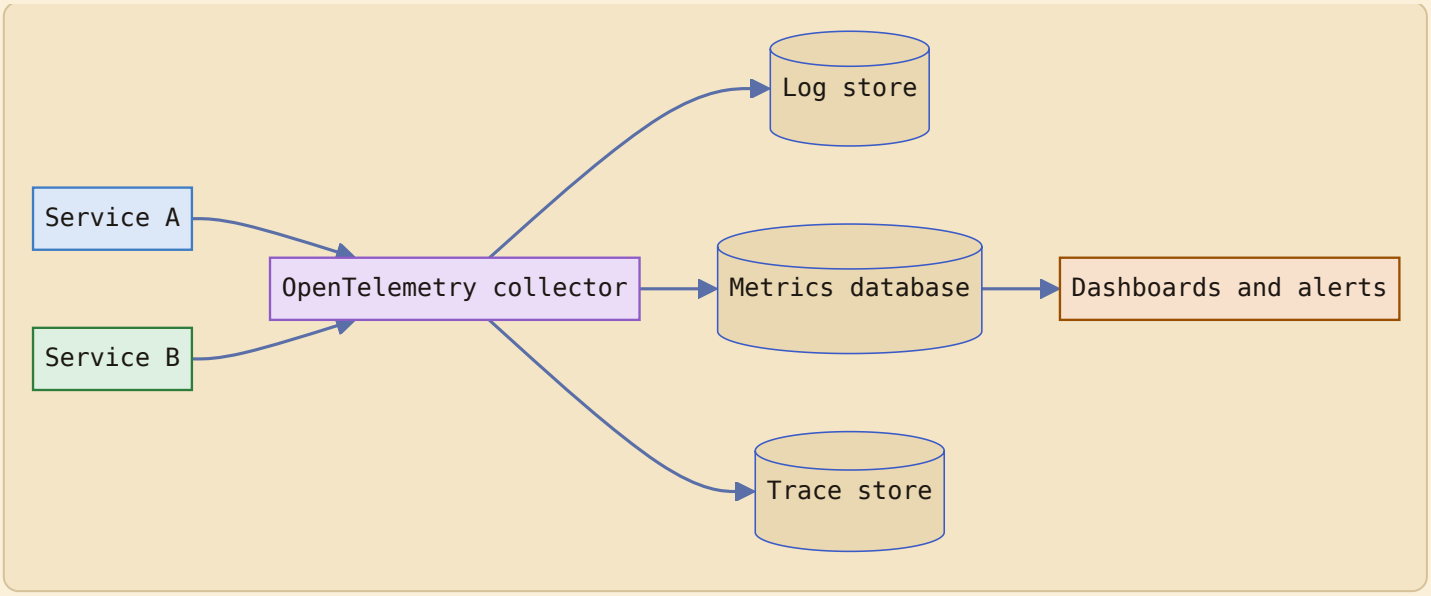
- **Logs**: timestamped text records of individual events. `2026-10-07T09:14:03Z ERROR order=8812 payment declined: card expired`. Best written as **structured logs** (JSON with named fields) so tools can search and filter them.
- **Metrics**: numbers measured over time, cheap to store and fast to graph. Examples: requests per second, error count, p99 latency, CPU %, queue depth. Common shapes are **counters** (only go up, such as total requests), **gauges** (go up and down, such as memory in use) and **histograms** (distribution of values, such as latency buckets, from which percentiles are computed).
- **Traces**: the path of one request through many services. A trace is a tree of **spans**; each span is one operation (a service handling the request, a database query) with a start time and duration. All spans share a **trace ID** passed along in request headers, so you can see that a 2.1 s checkout spent 1.8 s waiting on the tax service.

Why it matters

In a monolith you can read one log file. In microservices, one click touches 15 services on 200 machines. Without metrics you do not know something is wrong; without traces you do not know where; without logs you do not know why. Observability also drives decisions: capacity planning, SLOs and whether a canary is healthy.

How it works

1. **Instrument** code with a library, today usually **OpenTelemetry**, the vendor-neutral standard for emitting logs, metrics and traces.
2. Every incoming request gets a **correlation ID** or trace ID, which is logged and forwarded to every downstream call (the W3C `traceparent` header is the standard).
3. **Collect and ship**: agents or collectors send logs to a store (Elasticsearch/OpenSearch, Loki, Splunk), metrics to a time-series database (Prometheus, Datadog, CloudWatch) and traces to a tracing backend (Jaeger, Zipkin, Tempo, Honeycomb).
4. **Visualise**: dashboards (Grafana, Datadog, Kibana) graph key metrics.
5. **Alert**: rules on metrics notify a human when something needs attention.



Sampling. Storing every trace at 100,000 requests per second is too expensive, so systems keep a sample, for example 1% of normal requests but 100% of errors and slow requests (**tail-based sampling**, which decides after seeing the whole trace).

Cardinality. Each unique combination of metric labels is a separate time series. A label like `user_id` with 10 million values creates 10 million series and can crash a metrics system. Put high-cardinality details in logs and traces, not metric labels.

Types: what to measure

- **Four golden signals** (from Google's SRE book), for any user-facing service:
  - **Latency**: how long requests take, tracking successful and failed requests separately (a fast error is not a success).
  - **Traffic**: demand, such as requests per second.
  - **Errors**: rate of failed requests (explicit 5xx, wrong answers, too slow).
  - **Saturation**: how full the most constrained resource is (CPU, memory, connection pool, queue).
- **RED method** (Tom Wilkie), for request-driven services: **Rate** (requests per second), **Errors** (failed requests per second), **Duration** (latency distribution).
- **USE method** (Brendan Gregg), for resources like CPU, disks and network: **Utilisation** (% of time busy), **Saturation** (extra work queued, waiting), **Errors** (error events).

RED looks at a service from the outside (what users feel); USE looks at the machines underneath (why it might be happening).

Trade-offs

Observability data can cost a lot: large companies spend a noticeable share of their infrastructure bill on logging and monitoring. Logs are rich but expensive per event; metrics are cheap but aggregated (they say *that* p99 rose, not *which* request); traces are precise but sampled. Use all three, linked by trace ID.

Common mistakes

- Logging everything at debug level in production, then paying to store terabytes nobody reads.
- Logging secrets, passwords or personal data.
- Averages instead of percentiles on latency dashboards (Part 1).
- No correlation ID, so logs from different services cannot be stitched together.

Real-world examples

Google's **Dapper** paper (2010) introduced large-scale distributed tracing and inspired Zipkin (Twitter) and Jaeger (Uber). Prometheus, created at SoundCloud, is the standard open-source metrics system in the Kubernetes world, usually paired with Grafana dashboards.

In the interview

Q: A user says checkout is slow. How do you find out why?

A: Check the RED metrics for checkout to confirm and size the problem (which percentile, since when, which regions), correlate with recent deploys, then open slow traces to see which span is slow, and read that service's logs for those trace IDs. Then use USE metrics on the slow component's machines or database.

Q: Logs vs metrics vs traces?

A: Metrics are the cheap early warning: a number moved, and by how much. A trace pinpoints which hop in the call chain was slow or failed. Logs give the detailed story of what one component did at that moment.

TL;DR

Emit metrics (cheap numbers), logs (detailed events) and traces (one request across services), tied together by a trace ID; watch the golden signals, RED for services and USE for resources.

Dashboards, alerting, on-call and SLO-based alerts

What it is

- A **dashboard** is a page of graphs showing a system's key metrics at a glance.
- An **alert** is an automated notification fired when a rule on a metric is broken, for example "error rate above 2% for 5 minutes".
- **On-call** is the rota of engineers responsible for responding to alerts, including at night. A **page** is an urgent alert that wakes someone up (via PagerDuty, Opsgenie and similar tools); a **ticket** is a non-urgent alert handled in working hours.
- A **runbook** (or playbook) is a written guide for what to do when a specific alert fires.
- A **postmortem** (incident review) is a written analysis after an incident: timeline, impact, root causes and follow-up actions, written **blameless**, focusing on how the system allowed the mistake rather than who made it.

Why it matters

Alerts are how humans learn something is broken before customers flood support. But too many alerts cause **alert fatigue**: if on-call is paged 30 times a night for things that fix themselves, they start ignoring pages and miss the real one.

How it works

Good alerting rules:

1. Page on **symptoms users feel** (high error rate, high latency, checkout failures), not on causes (one server at 90% CPU, which may be fine).
2. Every page must be urgent, actionable and needing human judgement. Otherwise make it a ticket or automate the fix.
3. Attach a runbook link and a dashboard to every alert.

SLO-based alerting. Recall from Part 1: an **SLI** is a measured indicator (fraction of successful requests), an **SLO** is the target (99.9% over 30 days), and the **error budget** is the allowed failure (0.1%, about 43 minutes of full downtime per 30 days). Instead of alerting on any blip, alert on the **burn rate**: how fast you are spending the budget. A burn rate of 1 uses exactly the whole budget in 30 days; a burn rate of 14.4 sustained for one hour uses 2% of the monthly budget in that hour, and would exhaust the budget in about two days.

A common scheme from Google's SRE workbook:

Window	Burn rate	Budget used	Action
1 hour	14.4	2%	Page now
6 hours	6	5%	Page
3 days	1	10%	Ticket

Pairing each long window with a short one (for example 1 hour and 5 minutes) makes the alert fire fast and also stop quickly once the problem is fixed.

Incident response roles. In a serious incident, one **incident commander** coordinates, others investigate, one person communicates with stakeholders and customers. The first goal is **mitigation** (stop the bleeding: roll back, fail over, shed load), not finding the root cause.

Common mistakes

- Alerting on every metric someone thought of, and never deleting noisy alerts.
- Dashboards with 80 graphs and no clear "is it healthy?" panel at the top.
- Debugging the root cause during an outage instead of rolling back first.
- Postmortems that blame a person, which teaches people to hide mistakes.

In the interview

Q: What would you alert on for this service?

A: SLO burn rate for availability and latency of the main user journeys, plus a few critical saturation signals that predict imminent failure, such as disk nearly full or queue lag growing. Everything else goes on dashboards or tickets, not pages.

TL;DR

Page on user-facing symptoms and fast error-budget burn, keep every page actionable with a runbook, mitigate first, and write blameless postmortems.

Disaster recovery

What it is

Disaster recovery (DR) is your plan for getting the service and its data back after a large failure: a data-centre fire, a whole cloud region going down, a ransomware attack, or an engineer deleting the production database. **Business continuity** is the wider plan for keeping the business running. Two numbers define a DR target:

- **RPO (recovery point objective)**: how much data, measured in time, you can afford to lose. RPO of 1 hour means after recovery you may be missing at most the last hour of writes. Think "how far back is my last save point?".
- **RTO (recovery time objective)**: how long the service may be down before it is restored. RTO of 4 hours means you must be running again within 4 hours. Think "how long until I am playing again?".

Why it matters

Redundancy within one site (Part 1) protects against a disk or server failing. It does not help if the whole site goes, or if a bug or human corrupts data that is then faithfully replicated to every copy. Only backups and DR plans protect you there. Lower RPO and RTO cost more, so they must be agreed with the business: a bank's payment ledger may need RPO near zero and RTO of minutes; an internal wiki can accept a day.

How it works

Backups. Copies of data stored separately from the original, ideally following the **3-2-1 rule**: 3 copies, on 2 different media or services, with 1 off-site (another region or account). Kinds:

- **Full backup**: everything, every time. Simple, large.
- **Incremental backup**: only what changed since the last backup. Small, but restoring needs the chain.

- **Snapshot**: a point-in-time image of a disk or database, usually fast and incremental under the hood.
- **Continuous archiving** of the write-ahead log (Part 3), which allows **point-in-time recovery** to any second.
- **Immutable / air-gapped backups** that cannot be deleted or altered for a set period, protecting against ransomware and rogue deletes.

Until you have actually restored from a backup, you do not know it works. Rehearse restores on a schedule and time them, because restore time is part of your RTO.

Availability zones and regions. A cloud **region** is a geographic area (for example AWS `eu-west-1` in Ireland). Each region contains several **availability zones (AZs)**: one or more separate data centres with independent power, cooling and networking, a few to tens of kilometres apart, linked by fast connections with round trips of roughly 1 to 2 ms. AZs fail independently; regions are hundreds or thousands of kilometres apart.

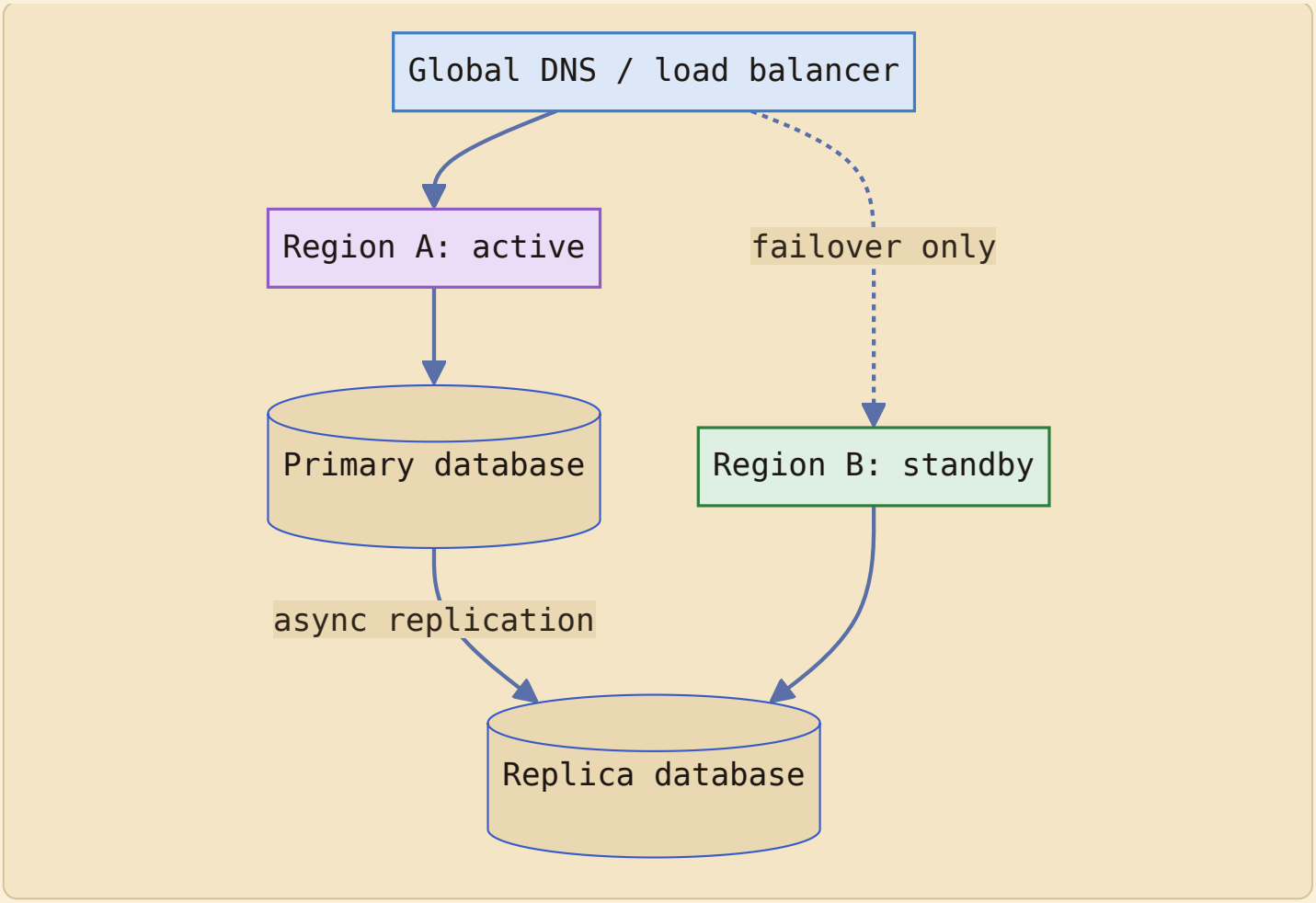
- **Multi-AZ**: run copies in 2 or 3 AZs of one region. Protects against a data-centre failure. Synchronous database replication is practical because AZs are close. This is the normal baseline for production.
- **Multi-region**: run in two or more regions. Protects against a whole-region outage and serves users nearer home, but cross-region round trips are 50 to 150 ms or more, so replication is usually asynchronous (RPO greater than zero), and it costs much more and is much more complex.

Types: DR strategies

From cheapest and slowest to most expensive and fastest:

Strategy	What runs in the DR site	Typical RTO	Typical RPO
Backup and restore	Nothing, only backups	Hours to a day	Hours
Pilot light	Data replicated, servers off	Tens of minutes	Minutes
Warm standby	Scaled-down full copy	Minutes	Seconds to minutes
Active-active	Full copy serving traffic	Near zero	Near zero to seconds

- **Active-passive**: one site serves all traffic; the other is on standby (pilot light or warm) and takes over on **failover**. Simpler, but standby capacity sits idle and failover is a risky moment that is rarely exercised.
- **Active-active**: all sites serve live traffic at the same time; if one fails, the others absorb its share. Fast recovery and no idle capacity, but each site must have spare headroom, and writes in several places raise hard consistency problems (Part 3 multi-leader replication, Part 4 conflict resolution).



Chaos engineering is the practice of deliberately injecting failures into a system (killing servers, adding network latency, blocking a dependency, even taking out a whole AZ) in a controlled way to prove that redundancy, failover and alerting actually work, and to find weaknesses before real outages do. Start small: a hypothesis ("if one cache node dies, p99 stays under 300 ms"), a limited blast radius, an abort button. **Game days** are scheduled team exercises of this kind, including practising a full regional failover.

Common mistakes

- Backups stored in the same account and region as the data, so the same disaster or attacker removes both.
- Never testing restores; discovering during the disaster that backups were silently failing.
- Assuming replication is a backup. A `DELETE` replicates too.
- A DR region that depends on a service in the failed region (shared DNS, auth, or deployment tooling).

Real-world examples

In 2017 **GitLab** accidentally deleted a production database directory; none of its five backup and replication methods turned out to be working reliably, and it restored from a six-hour-old copy, losing about six hours of data. Netflix created **Chaos Monkey** (open-sourced in 2012) to randomly terminate production instances, and later tools that simulate whole-region outages; it runs active-active across several AWS regions. The October 2021 **Facebook** outage, where a mistaken maintenance command cut its backbone network and its DNS servers then withdrew their **BGP** routes (the announcements that tell the rest of the internet how to reach them), lasted about six hours and also locked engineers out of internal tools they needed to fix it, a reminder that recovery tooling must not depend on the thing that failed.

In the interview

Q: Explain RPO and RTO with an example.

A: If the database is backed up every 6 hours and a restore takes 2 hours, the worst-case RPO is 6 hours of lost data and RTO is at least 2 hours of downtime. Adding continuous log shipping to another region could bring RPO to seconds, and a warm standby could bring RTO to minutes.

Q: Multi-AZ or multi-region?

A: Multi-AZ is the standard baseline: it survives a data-centre failure with low latency and synchronous replication. Multi-region adds protection against a regional outage and lower latency for global users, but costs much more and usually means asynchronous replication, so I would choose it only when the SLA or user geography demands it.

TL;DR

RPO is how much data you can lose, RTO is how long you can be down; backups (tested!) and multi-AZ are the baseline, multi-region active-passive or active-active buys more at a higher price, and chaos engineering proves it works.

Authentication and authorization

What it is

- **Authentication (AuthN)** answers "who are you?". It proves identity, usually with something you know (a password), something you have (a phone or security key) or something you are (a fingerprint). Using two or more of these is **multi-factor authentication (MFA)**.
- **Authorization (AuthZ)** answers "what are you allowed to do?". It checks permissions after identity is known.

At an airport, showing your passport is authentication; your boarding pass deciding which plane and seat you may board is authorization. Failing authentication returns HTTP **401 Unauthorized** (confusingly named; it really means "unauthenticated"); being known but not permitted returns **403 Forbidden**.

Why it matters

Nearly every system has users, and nearly every security breach involves either stolen identities or missing permission checks. Getting the vocabulary right is also one of the most common interview probes.

How it works

Authorization models:

- **Access control list (ACL)**: each resource lists who may do what ("doc 42: Alice can edit, Bob can read"). Fine-grained but hard to manage at scale.
- **Role-based access control (RBAC)**: permissions are attached to roles (admin, editor, viewer) and users are given roles. Simple and auditable; most business software uses it.
- **Attribute-based access control (ABAC)**: rules evaluate attributes of the user, resource and context: "a doctor may read a patient record if they are in the same hospital department and it is during their shift". Very flexible, harder to reason about.
- **Relationship-based access control (ReBAC)**: permissions follow relationships in a graph ("can view if member of a group that owns the folder containing the file"). Google's Zanzibar system, which powers sharing in Drive and other products, is the famous example.

Authorization must be checked on the server for every request, never only by hiding buttons in the user interface.

In the interview

Q: Authentication vs authorization?

A: Authentication proves who the caller is (login, token validation, returns 401 on failure); authorization decides whether that identity may perform this action on this resource (roles, policies, returns 403 on failure).

TL;DR

AuthN is who you are; AuthZ is what you may do; RBAC grants permissions through roles, ABAC through attribute rules.

Sessions, cookies, tokens and JWT

What it is

HTTP is stateless (Part 2): each request stands alone, so after you log in the server needs a way to recognise you on the next request.

- **Session-based authentication**: after login, the server creates a **session** record (user ID, expiry) in its memory, a database or Redis, and gives the browser a random **session ID** inside a **cookie** (a small piece of data the browser stores and automatically sends back to the same site on every request). Like a cloakroom ticket: the ticket itself means nothing, the server looks it up.
- **Token-based authentication**: after login, the server gives the client a **token**, a string the client sends with each request, usually in the header `Authorization: Bearer <token>`. An **opaque token** is a random string the server must look up; a **self-contained token** carries the user's details inside it and is signed so it cannot be forged.
- **JWT (JSON Web Token)**, pronounced "jot", is the most common self-contained token format: three Base64URL-encoded parts separated by dots, `header.payload.signature`.

```
header:    {"alg":"RS256","typ":"JWT"}
payload:   {"sub":"user_8812","role":"editor","exp":1767225600}
signature: RSA signature over header + payload, made with the server's private key
```

How it works

JWT verification: any service with the issuer's key (a shared secret for HS256, an HMAC-based signature, or the public key for RS256, an RSA signature) recomputes or checks the signature and the `exp` (expiry) claim. If valid, it trusts the claims without calling a database. That is why JWTs suit microservices: every service can verify a request locally.

A JWT is **signed, not encrypted**. Anyone can Base64-decode the payload and read it, so never put secrets in it. And because no database lookup happens, a stolen JWT works until it expires and cannot easily be revoked. The usual answer is short-lived **access tokens** (5 to 15 minutes) plus a long-lived **refresh token**, stored more carefully and checked against a database, used to obtain new access tokens; revoking the refresh token logs the user out within minutes.

Cookie safety flags: `HttpOnly` (JavaScript cannot read the cookie, protecting it from XSS theft), `Secure` (sent only over HTTPS), `SameSite=Lax` or `Strict` (not sent on most cross-site requests, protecting against CSRF), plus an expiry.

Trade-offs

	Server sessions	JWT access tokens
Server state	Session store needed	None for verification
Revocation	Instant: delete session	Hard until expiry
Per-request cost	Store lookup	Signature check only
Size	Small ID	Hundreds of bytes or more
Best fit	Classic web apps	APIs, microservices, mobile

Common mistakes

- Accepting JWTs with `alg: none` or not checking the signature and expiry.
- Storing tokens in browser `localStorage`, where any XSS script can steal them; an `HttpOnly` cookie is safer for browser apps.
- Very long-lived access tokens.

TL;DR

Sessions keep login state on the server and give the browser an ID in a cookie; tokens like JWT carry signed claims the client presents on each request, easy to verify anywhere but hard to revoke, so keep them short-lived.

OAuth 2.0, OpenID Connect, SSO and API keys

What it is

- **OAuth 2.0** is a standard for **delegated authorization**: letting one app access your data in another app *without giving it your password*. "Allow PhotoPrinter to read your Google Photos?" is OAuth. Analogy: a hotel key card that opens only your room and expires at checkout, instead of handing over the master key.
- **OpenID Connect (OIDC)** is a thin identity layer on top of OAuth 2.0 that adds **authentication**: it returns an **ID token** (a JWT) saying who the user is. "Sign in with Google" or "Sign in with Apple" is OIDC.
- **Single sign-on (SSO)**: log in once with one identity provider and get access to many applications. At work you sign in to Okta or Microsoft Entra ID once and reach email, chat and dashboards without logging in again. Enterprise SSO uses OIDC or the older XML-based **SAML**.
- **API key**: a long random secret string that identifies a calling program (not a person), sent in a header such as `X-API-Key`. Simple for server-to-server use, metering and rate limiting, but it is a bearer secret with no built-in expiry or user identity.

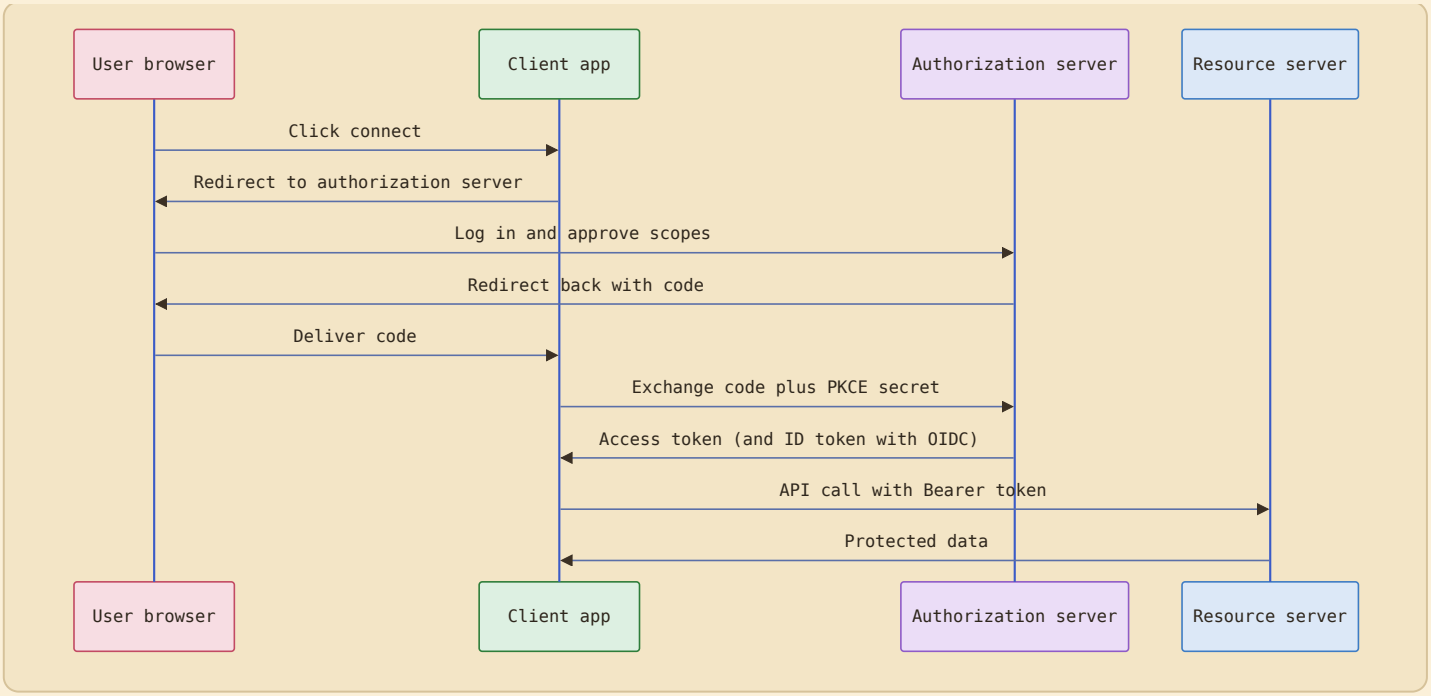
How it works

The four OAuth roles:

- **Resource owner**: the user who owns the data.
- **Client**: the app that wants access (PhotoPrinter).
- **Authorization server**: issues tokens after the user consents (Google's login and consent service).
- **Resource server**: the API holding the data (Google Photos API), which accepts the access token.

Authorization code flow (with PKCE), the recommended flow for web and mobile apps:

1. The user clicks "Connect Google Photos". The client redirects the browser to the authorization server with its client ID, the **scopes** it wants (for example `photos.read`) and a redirect URL. With **PKCE** (Proof Key for Code Exchange) it also sends a hash of a one-time random secret.
2. The user logs in at Google (the client never sees the password) and approves the scopes.
3. The authorization server redirects back to the client with a short-lived, single-use **authorization code**.
4. The client exchanges the code (plus the PKCE secret, and for server apps its client secret) at the authorization server's token endpoint, server to server.
5. It receives an **access token** (and often a refresh token; with OIDC also an ID token).
6. The client calls the Photos API with `Authorization: Bearer <access token>`; the resource server validates it and checks the scopes.



Other flows: **client credentials** for machine-to-machine calls with no user, and **device code** for TVs and devices without a keyboard. The old "implicit" and "password" flows are discouraged.

Common mistakes

- Using OAuth access tokens as proof of identity. OAuth says what a client may access; OIDC's ID token says who logged in.
- Requesting far broader scopes than needed.
- Committing API keys to Git repositories. Automated scanners find leaked keys within minutes.

In the interview

Q: OAuth vs OpenID Connect?

A: OAuth 2.0 is for authorization: it gives an app a scoped access token to call an API on the user's behalf. OIDC builds on it for authentication, adding a signed ID token that tells the app who the user is, which is what "Sign in with Google" uses.

Q: When would you use an API key instead of OAuth?

A: For simple server-to-server access by a known customer's backend, where you mainly need identification, metering and rate limiting. For acting on behalf of users, or anything needing scopes and expiry, use OAuth.

TL;DR

OAuth 2.0 delegates scoped access without sharing passwords (owner, client, authorization server, resource server; authorization code flow with PKCE); OIDC adds an ID token for login; SSO reuses one login across apps; API keys identify programs.

Encryption, hashing, passwords and secrets

What it is

- **Encryption** scrambles data with a **key** so only someone with the right key can unscramble (**decrypt**) it. It is two-way.
  - **Symmetric encryption**: the same key locks and unlocks (AES). Fast; used for bulk data.
  - **Asymmetric (public-key) encryption**: a **public key** anyone can use to encrypt or verify, and a **private key** only the owner holds to decrypt or sign (RSA, elliptic-curve). Slower; used for key exchange and digital signatures, as in TLS.

- **Encryption in transit**: protecting data while it travels over a network, almost always with **TLS** (Part 2), the "S" in HTTPS. Inside a data centre, service-to-service **mutual TLS (mTLS)** has both sides present certificates.
- **Encryption at rest**: protecting stored data on disks, databases, backups and object storage, so a stolen disk or backup file is unreadable. Cloud services typically do this with AES-256 and keys from a **key management service (KMS)**.
- **Hashing** turns any input into a fixed-size **fingerprint** (a **hash** or **digest**) that cannot be reversed. SHA-256 of any file is 256 bits. The same input always gives the same hash; a tiny change gives a completely different one. It is one-way: you can grind coffee beans into powder, but you cannot rebuild the beans from the powder.

How it works

Hashing vs encryption. Use encryption when you need the original back (a stored credit-card number you will charge later). Use hashing when you only need to check a match (passwords, file integrity, deduplication).

Password hashing. Never store passwords in plain text or encrypted (if the key leaks, all passwords leak). Store a **slow, salted hash**:

1. Generate a random **salt** (for example 16 bytes) per user, so two users with the same password get different hashes and precomputed "rainbow tables" are useless.
2. Hash password + salt with a deliberately slow algorithm: **bcrypt** (with a cost factor), **scrypt**, or **Argon2id** (a variant of Argon2, which won the 2015 Password Hashing Competition; it is the current first choice of OWASP, the Open Worldwide Application Security Project), tuned so one hash takes tens to hundreds of milliseconds and, for scrypt and Argon2, lots of memory.
3. Store the algorithm, parameters, salt and hash together.
4. At login, recompute with the stored salt and compare.

Fast hashes like SHA-256 or MD5 are wrong for passwords: a single modern GPU can try billions of SHA-256 guesses per second, but only thousands to tens of thousands of bcrypt guesses at typical cost settings.

Envelope encryption. A KMS holds a master key that never leaves it. Each piece of data is encrypted with its own **data key**, and the data key is itself encrypted by the master key and stored beside the data. Rotating or revoking the master key controls access to everything.

Secrets management. Secrets are passwords, API keys, database credentials, private keys and tokens. Keep them out of source code, container images and logs. Store them in a secrets manager (HashiCorp Vault, AWS Secrets Manager, Google Secret Manager, Kubernetes Secrets backed by a KMS), give each service access only to its own secrets, **rotate** them regularly and automatically, and prefer short-lived dynamically issued credentials where possible.

Common mistakes

- Inventing your own cryptography; always use well-known libraries.
- Confusing encoding (Base64, reversible by anyone) with encryption.
- Encrypting data at rest but keeping the key on the same disk.
- Passwords hashed with plain SHA-256 and no salt.

In the interview

Q: How should you store user passwords?

A: Hash each one with a unique random salt using a deliberately slow algorithm such as Argon2id (which is also memory-hard) or bcrypt with a tuned cost, store only the hash and parameters, and compare by re-hashing at login. Never store them plain or reversibly encrypted.

Q: Encryption in transit vs at rest?

A: In transit protects data moving over networks, using TLS (and mTLS between services). At rest protects stored data on disks, databases and backups, usually AES with keys managed by a KMS. You need both.

TL;DR

Encryption is reversible with a key (in transit via TLS, at rest via AES and a KMS); hashing is one-way; passwords get a unique salt and a slow hash like Argon2id or bcrypt; secrets live in a secrets manager and rotate.

Network security: firewalls, WAF and DDoS

What it is

- A **firewall** filters network traffic by rules on IP addresses, ports and protocols (layers 3 and 4): "allow TCP 443 from anywhere, allow TCP 5432 only from the app servers' subnet, deny everything else". Cloud **security groups** and network ACLs are firewalls. It is the building's security desk checking which doors visitors may use.
- A **web application firewall (WAF)** inspects HTTP requests (layer 7) and blocks those that look like attacks: SQL injection patterns, cross-site scripting payloads, known bad bots, too many requests from one source. Examples: AWS WAF, Cloudflare WAF, ModSecurity, often using the OWASP Core Rule Set.
- A **DDoS (distributed denial-of-service) attack** floods a service with traffic from many machines at once (often a **botnet** of hijacked devices) to make it unavailable to real users.
- **Network segmentation**: placing databases and internal services in **private subnets** with no route from the internet, reachable only through load balancers or a **bastion host** (a hardened jump server).

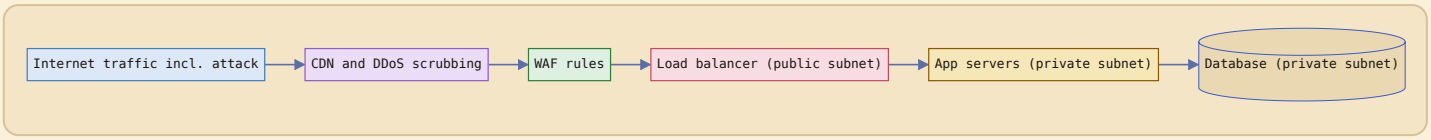
How it works

Kinds of DDoS:

- **Volumetric** (layer 3/4): saturate the network pipe, for example UDP floods or **amplification** attacks that send small spoofed requests to open DNS or memcached servers, which reply with much larger responses to the victim.
- **Protocol**: exhaust connection state, for example a **SYN flood** sending millions of half-open TCP handshakes.
- **Application layer** (layer 7): many legitimate-looking HTTP requests to expensive pages such as search.

Mitigation:

1. Absorb with scale: a CDN or DDoS-protection network with huge capacity, often using anycast (Part 2) to spread the attack across hundreds of data centres (Cloudflare, AWS Shield, Google Cloud Armor, Akamai).
2. Filter at the edge: drop malformed packets, use SYN cookies, block known bad sources.
3. Rate limit and challenge suspicious clients (CAPTCHAs, JavaScript challenges).
4. Cache aggressively so most requests never reach the origin, and hide origin IP addresses so attackers cannot bypass the CDN.
5. Autoscale, with a cost ceiling, and keep graceful degradation switches ready.



Real-world examples

In October 2023 Google disclosed that, that August, it had mitigated an attack peaking at about 398 million HTTP requests per second, exploiting an HTTP/2 weakness called "Rapid Reset"; Cloudflare and AWS reported record attacks from the same technique. The 2016 **Mirai** botnet of hacked cameras and routers took down the DNS provider Dyn and, with it, many major websites for hours.

TL;DR

Firewalls filter by IP and port, WAFs filter malicious HTTP, private subnets hide internal systems, and DDoS is absorbed by large edge networks, filtering, rate limiting and caching.

Common web vulnerabilities: CORS, CSRF, XSS and SQL injection

These four come up constantly. One-line ideas plus the fix.

Same-origin policy and CORS. Browsers stop a page from one **origin** (scheme + domain + port, such as `https://shop.com`) reading responses from another origin. **CORS (Cross-Origin Resource Sharing)** is how a server opts in: it replies with headers such as `Access-Control-Allow-Origin: https://shop.com`, and for non-simple requests the browser first sends an `OPTIONS` **preflight** request to ask permission. CORS is a browser safety relaxation, not server protection: `curl` ignores it. Mistake: `Access-Control-Allow-Origin: *` on APIs that use cookies or private data.

CSRF (cross-site request forgery). An evil site tricks your logged-in browser into sending a request to another site, and the browser attaches your cookies, for example a hidden form that posts "transfer $1,000" to your bank. Fixes: `SameSite` cookies, anti-CSRF tokens (a random value the real page includes and the server checks), and never changing state on GET.

XSS (cross-site scripting). An attacker gets their JavaScript to run inside your site in other users' browsers, for example by posting a comment containing `<script>`, which can then steal data or act as the user. Fixes: escape all user content when rendering (modern frameworks like React do by default), a **Content Security Policy (CSP)** header that restricts which scripts may run, and `HttpOnly` cookies so scripts cannot read session cookies.

SQL injection. User input is glued into a SQL string, so input like `' OR '1'='1` changes the query's meaning and can read or delete data. Fix: **parameterised queries** (prepared statements), where the input is always treated as data, never as SQL.

```
Unsafe: "SELECT * FROM users WHERE email = '" + input + "'"
Safe:   "SELECT * FROM users WHERE email = ?"   with parameter [input]
```

The **OWASP Top 10** (from the Open Worldwide Application Security Project) is the widely used list of the most critical web application risks; broken access control has been at the top of its most recent editions.

In the interview

Q: What is the difference between XSS and CSRF?

A: XSS runs the attacker's script inside your site, so it can do anything the user can. CSRF does not run code in your site; it makes the victim's browser send a forged request to your site with their cookies. XSS is fixed by output escaping and CSP; CSRF by SameSite cookies and CSRF tokens.

TL;DR

CORS lets servers allow cross-origin browser reads; CSRF forges requests with the victim's cookies; XSS injects scripts; SQL injection injects query text; use SameSite cookies, CSRF tokens, escaping, CSP and parameterised queries.

Least privilege and zero trust

What it is

- **Principle of least privilege**: every user, service and process gets only the permissions it needs for its job, for only as long as it needs them. A cleaner gets keys to the offices, not the safe. If the image-resizing service is hacked, it should only be able to read one bucket, not delete the customer database.
- **Defence in depth**: several independent layers of protection (firewall, WAF, authentication, authorization, encryption, monitoring), so one failed layer is not fatal.
- **Zero trust**: never trust a request just because it comes from "inside" the network. The old **perimeter model** was a castle with a moat: once inside the office network or the **VPN** (virtual private network, an encrypted tunnel into that network), everything trusted you. Zero trust verifies every request: authenticate the user and device, check authorization, encrypt the connection (often mTLS between services), and log it, wherever it comes from.

Why it matters

Attackers who get one foothold (a phished laptop, a vulnerable server) then move sideways ("lateral movement") through a flat, trusting network. Least privilege and zero trust shrink what that foothold can reach.

How it works

Practical steps: separate **IAM** (identity and access management) roles per service with narrowly scoped policies; no shared admin accounts; just-in-time access for engineers that expires; private subnets; mTLS through a service mesh; device health checks; audit logs reviewed for unusual access. Google's **BeyondCorp** (described publicly from 2014) moved its employees off a privileged corporate network so that access to internal apps depends on user and device identity, not network location.

Common mistakes

- Giving every service an all-powerful cloud role "to get it working", then never narrowing it.
- Long-lived engineer credentials on laptops.
- Treating the VPN as the only security boundary.

TL;DR

Grant the minimum access needed, layer defences, and verify every request regardless of network location.

Cost awareness

Interviewers increasingly ask "and what would this cost?". A few rules of thumb:

- **Big cost drivers**: compute (servers, containers, functions), storage (per GB-month, with cheaper "cold" tiers for rarely read data), managed databases, and **data transfer**. Data entering a cloud is usually free; data leaving to the internet (**egress**) is charged, roughly $0.05 to $0.09 per GB on AWS depending on volume, and even traffic between availability zones costs around $0.01 per GB in each direction. Heavy video or download traffic is mostly a bandwidth bill, which is one reason CDNs matter.
- **Pricing models**: **on-demand** (pay per second or hour, flexible), **reserved instances or savings plans** (commit for 1 to 3 years for large discounts), **spot or preemptible instances** (spare capacity at up to around 90% off, but can be taken back at short notice, fine for batch jobs).
- **Levers**: autoscale and turn off idle environments at night; right-size oversized instances; cache to cut database and egress costs; move old data to cheaper tiers with lifecycle rules; sample logs and traces; compare serverless vs always-on at your real traffic level.
- **Trade-off to mention**: every extra nine of availability, extra region and lower RPO costs real money. Engineering time is a cost too, so a managed service that costs more per hour may still be cheaper overall.
- **FinOps** is the practice of making teams see and own their cloud spend, with tags per team and per service and regular reviews.

In the interview: state the expensive component of your design (for a video site, egress bandwidth;

for a logging system, storage), and one way you would reduce it.

Part 5 glossary

Term	Meaning in one line
Architectural style	The overall way code is split into running programs and how they communicate
Monolith	Whole application built and deployed as one program
Modular monolith	One deployable program with strictly separated internal modules
SOA	Older style of coarse services connected through a central service bus
Enterprise service bus (ESB)	Middleware that routes and transforms messages between SOA services
Microservices	Many small, independently deployed services, each owning one capability and its data
Database per service	Each microservice owns its private data store
Distributed monolith	Microservices so coupled they must change and deploy together
Serverless	Running code or using services without managing servers, billed per use
FaaS	Function as a Service: upload functions, provider runs them per event
Cold start	Extra delay when a serverless function runs on a freshly created instance
Conway's law	Systems mirror the communication structure of the teams that build them
Strangler fig pattern	Migrating a monolith piece by piece behind a routing layer
Synchronous / asynchronous call	Caller waits for the response / sends a message and carries on
Service discovery	Finding the current network addresses of a service's instances
Service registry	Live database of service instances and their addresses
Client-side / server-side discovery	Caller queries the registry itself / a router or load balancer does it for the caller
Sidecar	Helper process deployed beside each service instance
Service mesh	Centrally configured sidecar proxies handling all service-to-service traffic
Data plane / control plane	The proxies that carry traffic / the component that configures them
mTLS	Mutual TLS: both client and server prove identity with certificates
Timeout	Maximum time a caller waits before giving up
Deadline propagation	Passing the remaining time budget along to downstream calls
Retry	Repeating a failed request in case the fault was temporary
Transient fault	A short-lived failure likely to succeed on retry
Exponential backoff	Doubling the wait between successive retries
Jitter	Randomness added to retry waits to avoid synchronised retries

Term	Meaning in one line
Retry storm	Retries multiplying load on an already struggling service
Retry budget	Cap on the share of traffic that may be retries
Hedged request	Sending a duplicate request to another replica if the first is slow
Cascading failure	One component's failure spreading to the components that depend on it
Circuit breaker	Wrapper that stops calling a failing dependency for a while
Closed / Open / Half-open	Circuit breaker states: normal, failing fast, testing recovery
Bulkhead	Isolating resource pools so one failure cannot consume everything
Cell-based architecture	Splitting users across independent copies of the stack to limit blast radius
Fallback	Alternative response used when a call fails
Rate limiting	Capping requests per client per time window and rejecting the excess
Throttling	Slowing or queuing excess requests instead of rejecting them
HTTP 429	Status code meaning too many requests
Retry-After	Header telling a client how long to wait before retrying
Token bucket	Rate limiter with refilling tokens that allows bursts up to bucket size
Leaky bucket	Rate limiter that outputs requests at a fixed steady rate
Fixed window counter	Rate limiter counting requests per calendar time window
Sliding window log	Rate limiter storing each request timestamp for an exact count
Sliding window counter	Rate limiter estimating a sliding count from two fixed windows
Fail open / fail closed	Allowing or denying traffic when a safety component itself fails
Load shedding	Deliberately rejecting some work when overloaded so the rest succeeds
Graceful degradation	Turning off non-essential features under stress to keep the core working
Goodput	Useful work completed, as opposed to raw throughput
Congestion collapse	Overload where throughput falls towards zero due to wasted work
Kill switch	Flag that instantly disables a feature
Autoscaling	Automatically adding or removing capacity based on load
Scale out / scale in	Adding or removing instances
Reactive / scheduled / predictive scaling	Scaling on current metrics / at preset times / ahead of forecast demand
Cool-down period	Wait after a scaling action before another, to avoid flapping
Cluster autoscaler	Adds or removes Kubernetes nodes as pods need room
Virtual machine (VM)	Emulated computer with its own operating system
Hypervisor	Software that runs and isolates virtual machines on a physical host

Term	Meaning in one line
Container	Isolated process sharing the host's OS kernel
Namespaces and cgroups	Linux features that isolate what a process sees and limit what it uses
Container image	Read-only package of an app and its dependencies
Docker	Popular tool for building, sharing and running container images
Dockerfile	Text recipe for building a container image
Container registry	Storage service for container images
Kubernetes (K8s)	Container orchestrator that keeps declared state running across machines
Cluster	The set of machines and control plane managed by Kubernetes
Node	One machine in a Kubernetes cluster
Kubelet	Agent on each node that runs its pods
etcd	Consistent key-value store holding Kubernetes cluster state
Pod	Smallest Kubernetes unit: one or more containers sharing network and storage
Deployment (Kubernetes)	Object managing replicas and rolling updates of stateless pods
Service (Kubernetes)	Stable name and virtual IP load-balancing to matching pods
Ingress	Kubernetes rules routing external HTTP traffic to services
ConfigMap / Secret	Kubernetes objects for configuration and sensitive values
StatefulSet	Kubernetes object for stateful pods with stable identity and storage
Liveness / readiness probe	Checks for whether to restart a container or send it traffic
Reconciliation loop	Controllers continually driving actual state towards desired state
Infrastructure as code (IaC)	Defining infrastructure in versioned files applied by tools
Terraform	Popular declarative infrastructure-as-code tool
Configuration drift	Real infrastructure diverging from its definition
Immutable infrastructure	Replacing servers with new images instead of modifying them
Continuous integration (CI)	Automatically building and testing every merged change
Continuous delivery	Keeping every passing build releasable on demand
Continuous deployment	Automatically releasing every passing build to production
Pipeline	Automated sequence of build, test and deploy steps
DORA metrics	Deployment frequency, lead time, change failure rate, time to restore
Expand and contract	Backward-compatible schema change done in several releases
Deployment strategy	Method for replacing a running version with a new one
Blast radius	How much of the system or user base a failure can affect
Recreate deployment	Stop old version, start new, with downtime
Rolling deployment	Replacing instances in small batches

Term	Meaning in one line
Blue-green deployment	Switching traffic between two full environments
Canary release	Sending a small share of traffic to a new version first
Shadow launch	Copying real traffic to a new version and discarding its responses
Feature flag	Runtime switch that turns code paths on or off for chosen users
Rollback	Returning to the previous known-good version
Roll forward	Fixing a bad release by shipping a new fix instead of reverting
Monitoring	Watching known metrics to check health
Observability	Ability to understand system behaviour from its emitted data
Log	Timestamped record of a single event
Structured logging	Logs written as fields (such as JSON) for easy querying
Metric	Numeric measurement tracked over time
Counter / gauge / histogram	Metric that only rises / goes up and down / records a distribution
Trace	Record of one request's path across services
Span	One timed operation inside a trace
Trace ID / correlation ID	Identifier passed along to link all data for one request
OpenTelemetry	Vendor-neutral standard and libraries for telemetry
Sampling	Keeping only a fraction of traces or logs to control cost
Cardinality	Number of distinct label combinations in metrics
Four golden signals	Latency, traffic, errors, saturation
RED method	Rate, Errors, Duration for services
USE method	Utilisation, Saturation, Errors for resources
Saturation	How full the most constrained resource is
Dashboard / alert	Page of graphs showing key metrics / automated notification when a metric rule breaks
Page / ticket	Urgent alert waking someone / non-urgent work item
Alert fatigue	Ignoring alerts because there are too many noisy ones
On-call	Rota of engineers responding to alerts
Runbook	Step-by-step guide for handling a specific alert
Burn rate	Speed at which the error budget is being consumed
Incident commander	Person coordinating the response to an incident
Mitigation	Action that stops user impact before the root cause is fixed
Postmortem	Blameless written review of an incident
Disaster recovery (DR)	Plan for restoring service and data after a major failure
RPO	Maximum acceptable data loss, measured in time
RTO	Maximum acceptable downtime before restoration
Backup	Separate copy of data for restoration

Term	Meaning in one line
Full / incremental backup	Copy of everything / only changes since the last backup
Snapshot	Point-in-time image of a disk or database
Point-in-time recovery	Restoring data to any chosen moment using logs
3-2-1 rule	Three copies, two media, one off-site
Immutable backup	Backup that cannot be altered or deleted for a set period
Region	Geographic cloud area containing several availability zones
Availability zone (AZ)	Independent data centre group within a region
Multi-AZ	Running across several zones in one region
Multi-region	Running across several geographic regions
Backup and restore / pilot light / warm standby	DR strategies from cheapest and slowest to faster and costlier
Active-passive	One site serves traffic while another waits to take over
Active-active	All sites serve traffic at the same time
Chaos engineering	Deliberately injecting failures to prove resilience
Game day	Planned exercise practising failures and recovery
Authentication (AuthN)	Proving who someone is
Authorization (AuthZ)	Deciding what an identity may do
MFA	Multi-factor authentication using two or more kinds of proof
401 / 403	Not authenticated / authenticated but not permitted
ACL	List of who may access a specific resource
RBAC	Permissions granted through roles
ABAC	Permissions decided by rules over attributes
ReBAC	Permissions derived from relationships between users and objects
Session	Server-side record of a logged-in user
Cookie	Small data stored by the browser and sent back to the site automatically
HttpOnly / Secure / SameSite	Cookie flags: no script access / HTTPS only / limit cross-site sending
Token	String a client presents to prove identity or access
Bearer token	Token that grants access to whoever holds it
JWT	Signed, self-contained JSON token: header.payload.signature
Access token / refresh token	Short-lived API credential / long-lived credential to get new access tokens
OAuth 2.0	Standard for delegated, scoped access without sharing passwords
Resource owner / client / authorization server / resource server	The four OAuth roles
Scope	Named permission an OAuth token grants
Authorization code flow	OAuth flow exchanging a one-time code for tokens
PKCE	One-time secret proving the code exchanger started the flow

Term	Meaning in one line
Client credentials flow	OAuth flow for machine-to-machine access
OpenID Connect (OIDC)	Identity layer on OAuth that adds an ID token for login
ID token	JWT stating who the user is
SSO	Single sign-on: one login for many applications
SAML	Older XML-based standard for enterprise SSO
API key	Secret string identifying a calling program
Encryption	Reversible scrambling of data with a key
Symmetric / asymmetric encryption	Same key both ways / public and private key pair
Encryption in transit / at rest	Protecting data moving over a network (TLS) / stored on disks and backups
KMS	Key management service that stores and uses encryption keys
Envelope encryption	Encrypting data keys with a master key held in a KMS
Hashing	One-way conversion of data into a fixed-size fingerprint
Salt	Random per-user value added before hashing a password
bcrypt / scrypt / Argon2id	Deliberately slow password-hashing algorithms
Secrets management	Storing, distributing and regularly rotating (replacing) credentials securely
Firewall	Filters traffic by IP, port and protocol
Security group	Cloud firewall attached to instances
WAF	Web application firewall filtering malicious HTTP requests
DDoS	Flooding a service from many machines to make it unavailable
Botnet	Network of hijacked devices controlled by an attacker
Amplification attack	Spoofed small requests causing large replies to hit a victim
SYN flood	Attack exhausting servers with half-open TCP connections
Private subnet	Network segment with no direct route from the internet
Bastion host	Hardened server used as the only entry point to private machines
Same-origin policy	Browser rule blocking pages from reading other origins' responses
CORS	Headers letting a server allow cross-origin browser requests
Preflight request	Browser OPTIONS request checking CORS permission first
CSRF	Forged request sent by the victim's browser with their cookies
XSS	Attacker's script running inside a trusted site
Content Security Policy (CSP)	Header restricting which scripts and resources a page may load
SQL injection	Input altering the meaning of a SQL query
Parameterised query	Query where inputs are passed separately as data
OWASP Top 10	Widely used list of top web application security risks
Least privilege	Granting only the minimum access needed
Defence in depth	Multiple independent layers of security

Term	Meaning in one line
Lateral movement	Attacker spreading from one compromised system to others
Zero trust	Verifying every request regardless of network location
Perimeter security	Trusting everything inside the network boundary
Egress	Data leaving a cloud network, usually billed
On-demand / reserved / spot instances	Pay as you go / committed discount / cheap but reclaimable
FinOps	Practice of making teams see and manage their cloud costs

Part 6 — Putting it together

Parts 1 to 5 took the system apart and looked at each piece by itself: latency in Part 1, DNS and load balancers in Part 2, databases in Part 3, caches and queues in Part 4, and reliability and security in Part 5. This part puts the pieces back together. It has five sections, and each one is useful on its own:

1. **A request's journey** follows one click from the browser all the way to the database and back, and names every concept along the way. If you can tell this story without notes, you understand the fundamentals.
2. **Worked example: a URL shortener** designs a small but real system from scratch, step by step, with real numbers. It shows the interview method from Part 1 in action.
3. **Decision cheat sheet** is a lookup table: "if you need X, reach for Y".
4. **Top 60 interview questions** are the beginner questions interviewers ask most, with short model answers.
5. **Master A–Z glossary** lists close to 300 of the most important terms, one line each, so you can quickly look up a word you meet in this module or in an interview.

When this part says "see *Load balancers* in Part 2", it means the section with that name in Part 2 of this module. Go back to it whenever a one-line reminder here is not enough.

A request's journey

What it is

A **request's journey** is the full path one user action takes through a system. You click "Like" on a photo, and about a tenth of a second later the heart turns red. In that tenth of a second your request may touch fifteen different machines and use almost every concept in this module. Following that path is the best way to see how the pieces fit.

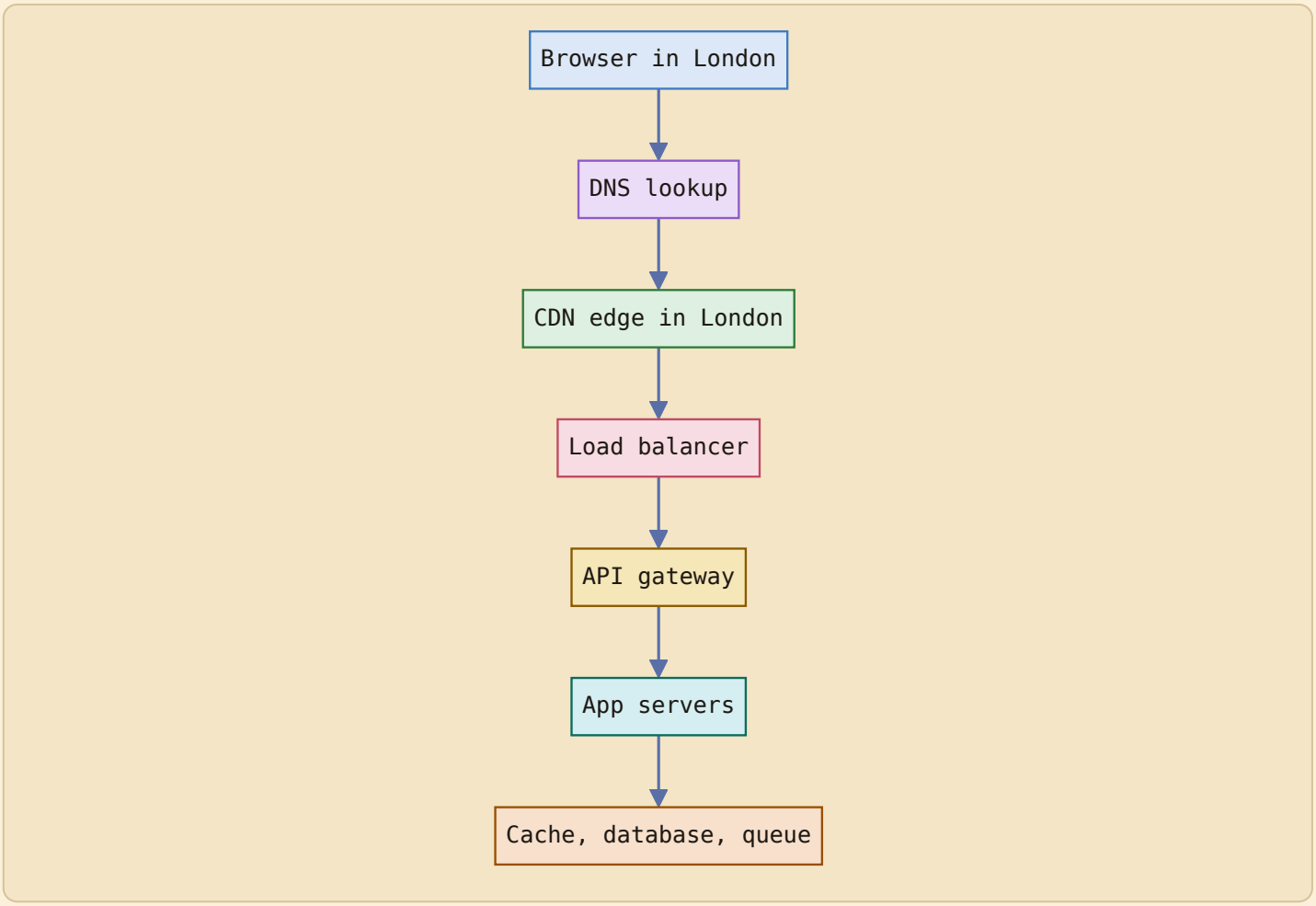
> **Analogy:** sending a parcel. You write an address (a domain name), the post office looks up the route (DNS), the parcel goes to a local sorting office near you (a CDN edge), then to the main depot (the load balancer), where a clerk checks the label (the API gateway) and hands it to a worker (an app server). The worker looks in a quick-access drawer first (the cache) before going down to the archive (the database). Anything that can wait, like sending you a receipt, goes into an out-tray (a queue) for someone else to handle later (a worker).

Why it matters

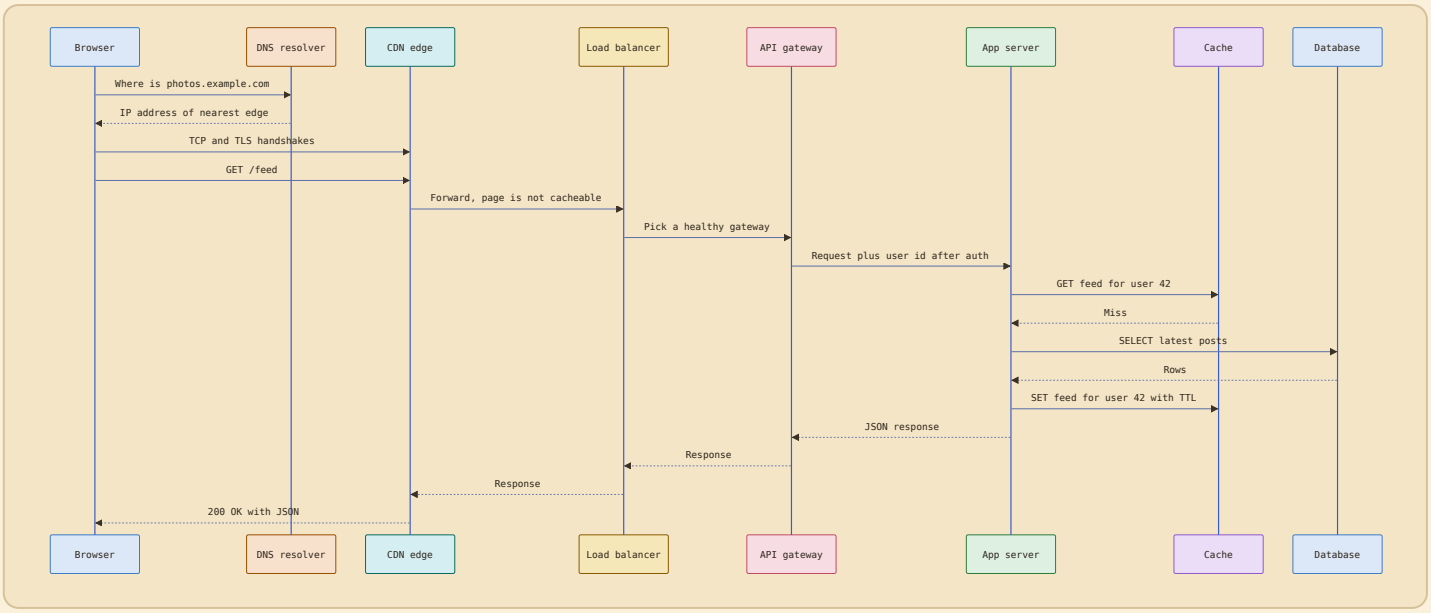
- **It is the most common warm-up question.** "What happens when you type a URL into the browser and press Enter?" is asked at every level, from internships to senior roles.
- **It shows where time goes.** Every hop adds latency. Knowing the hops tells you which ones to remove (with caches, CDNs and connection reuse) when a page is slow.
- **It shows where things break.** Every hop is something that can fail. Knowing the path lets you ask "what if this hop is down?" one hop at a time, which is how you find single points of failure.

The map

Here is the whole path for our example. The user is in London and opens `https://photos.example.com/feed`, a page that shows the latest photos from people they follow.



And here is the same journey as a sequence of messages over time. Read it top to bottom.



How it works

We walk through it one step at a time. The times are typical figures for a user in London talking to a well-run site; your numbers will differ, but the order of magnitude is what matters.

Step 1 — The browser checks what it already knows (0 ms). You type the address and press Enter. The browser splits the URL into a **scheme** (`https`, meaning "HTTP over TLS"), a **host** (`photos.example.com`) and a

path (`/feed`). Before using the network it checks its own **browser cache**: maybe it already has a fresh copy of this page, or at least a remembered IP address for the host. This is caching at the client, the cheapest cache of all because it involves no network at all. See *Where caches live* in Part 4 and *Clients, servers and the request/response cycle* in Part 1.

Step 2 — DNS turns the name into an IP address (0–50 ms). Computers talk using **IP addresses** such as `203.0.113.10`, not names. The browser asks the operating system, which asks a **recursive resolver** (usually run by your internet provider, or a public one like 1.1.1.1 or 8.8.8.8). If the resolver has the answer cached and its **TTL** (time to live) has not run out, it replies in a few milliseconds. If not, it asks a **root server**, then the **TLD server** for `.com`, then the **authoritative server** for `example.com`. Because `photos.example.com` uses a CDN, the authoritative answer is often a **CNAME** (an alias) pointing at the CDN, and the CDN's DNS returns the IP of an edge server close to the user. This is **GeoDNS** or **anycast** at work. See *DNS — the internet's phone book* and *CDNs and anycast* in Part 2.

Step 3 — TCP opens a connection (about 1 round trip). The browser opens a **TCP** connection to the edge server on **port** 443. TCP needs a **three-way handshake** (SYN, SYN-ACK, ACK) before any data flows. The edge is in London, so a round trip takes maybe 5–10 ms. If the server were in New York it would be around 70 ms, which is the whole reason CDNs exist. See *TCP, UDP and QUIC* in Part 2.

Step 4 — TLS sets up encryption (about 1 more round trip). Because the scheme is `https`, the browser and edge now run a **TLS handshake**. The server shows a **certificate** that proves it really is `photos.example.com`, and the two sides agree on encryption keys. With TLS 1.3 this costs one extra round trip (zero for a repeat visit that uses 0-RTT resumption). If the browser and site support **HTTP/3**, it runs over **QUIC**, which combines the transport and TLS handshakes into one round trip. See *HTTPS and TLS* in Part 2.

Step 5 — The browser sends the HTTP request. Now the real request goes out:

```
GET /feed HTTP/2
Host: photos.example.com
Cookie: session=7f3a9c...
Accept: application/json
```

`GET` is the **HTTP method** (read, do not change anything), the **headers** carry extra information, and the **cookie** carries the session token that says who you are. HTTP itself is **stateless**: the server remembers nothing between requests, so every request must carry its own identity. See *HTTP — the language of the web* in Part 2.

Step 6 — The CDN edge decides: serve or forward (1–5 ms at the edge). A **CDN** (content delivery network) is a set of servers spread around the world, each location called a **point of presence (PoP)**. The edge checks its cache. Static files such as images, JavaScript and CSS are almost always a **cache hit**, served straight from London in a few milliseconds. But `/feed` is personal to you, so it is marked as not cacheable and the edge forwards the request to the **origin** (the company's own servers), often over a fast, already-open connection. Many CDNs also absorb **DDoS** attacks and run a **WAF** (web application firewall) here. See *CDNs and anycast* in Part 2 and *Network security: firewalls, WAF and DDoS* in Part 5.

Step 7 — The load balancer picks a server (under 1 ms). The request reaches the origin's **load balancer**, which spreads traffic across many identical machines. A **layer 7** load balancer can read the HTTP path and headers. It often does **TLS termination** (decrypting here so the servers behind it do not have to). It only sends traffic to targets that pass **health checks**, and it chooses among them with an algorithm such as **round robin** or **least connections**. The load balancer is itself run as a redundant pair or a managed cloud service, so it is not a **single point of failure**. See *Load balancers* in Part 2 and *Redundancy, single points of failure and failover* in Part 1.

Step 8 — The API gateway checks the request (1–5 ms). The **API gateway** is the front door for all the company's APIs. It **authenticates** the request (checks the session or **JWT** to learn that you are user 42), applies **rate limiting** (are you sending too many requests? If so, reply **429 Too Many Requests**), and **routes** `/feed` to the feed service. In smaller systems the load balancer and gateway are the same product. See *Proxies and API gateways* in Part 2, and *Authentication and authorization* and *Rate limiting and throttling* in Part 5.

Step 9 — An app server runs the business logic. The feed service runs on many **stateless** app servers, usually as **containers** managed by **Kubernetes**. Stateless means any server can handle any request, because nothing about you is stored in the server's memory between requests. That is what lets the system **scale horizontally**: add more servers when traffic grows, and let **autoscaling** remove them when it drops. The

service may find other services it needs through **service discovery**, and every call it makes has a **timeout**. See *Stateless and stateful services* and *Vertical and horizontal scaling* in Part 1, and *Service discovery, sidecars and service mesh* and *Autoscaling* in Part 5.

Step 10 — Check the cache first (about 0.5 ms). The feed service asks a **distributed cache** such as **Redis**: "do you have the feed for user 42?" This is the **cache-aside** pattern. If it is a **cache hit**, the answer comes back in well under a millisecond and steps 11 and 12 are skipped. A well-tuned cache serves 90–99% of reads this way, which is the main reason big sites are fast and their databases survive. See *Caching strategies* in Part 4.

Step 11 — On a miss, query the database (2–20 ms). On a **cache miss** the service asks the **database**. Reads usually go to a **read replica** (a copy of the data kept in sync from the **primary**), so the primary is free for writes. The query uses an **index** so the database does not scan every row. If the data is too big for one machine, it is **sharded**, and the service (or a routing layer) uses the **shard key** — here, the user ID — to pick the right shard. The service talks to the database through a **connection pool** so it does not pay for a new connection every time. See *Indexes*, *Replication* and Partitioning and sharding in Part 3.

Step 12 — Fill the cache. The service stores the result in the cache with a **TTL** (say 60 seconds), so the next request for the same feed is a hit. Because replicas can be slightly behind (**replication lag**) and the cache can be up to 60 seconds old, the user sees **eventual consistency**: new posts appear within a minute, not instantly. For a feed that is fine. For a bank balance it would not be. See *Consistency models* in Part 4.

Step 13 — The response travels back. The service builds a **JSON** response, and it travels back through the gateway, load balancer and CDN edge. The edge may **compress** it. The browser receives `200 OK`, renders the page, and then fetches the photo files themselves, which are almost all cache hits at the CDN edge.

Step 14 — Asynchronous work happens later. Suppose you now click "Like". That is a **write** (`POST /photos/991/likes`). The service writes the like to the **primary** database and replies at once. But several other things should happen too: notify the photo's owner, update the like counter, feed the recommendation system. None of those need to finish before you see the red heart, so the service puts an event on a **message queue** or **event stream** (such as **Kafka**) and moves on. **Workers** (consumers) pick up the event in the background. Because queues usually give **at-least-once delivery**, the workers must be **idempotent** so a duplicate event does not send two notifications. See *Message queues* and *Delivery guarantees, idempotency and retries* in Part 4.

Step 15 — Everything is watched. Every hop writes **logs**, emits **metrics** (request rate, errors, duration) and passes along a **trace ID**, so an engineer can follow this exact request through all the services later. If the error rate rises above the **SLO**, an **alert** pages the **on-call** engineer. See *Observability: logs, metrics and traces* in Part 5.

Where the time goes

A rough budget for the feed request on a first visit, user in London, origin in the same region:

Step	Typical time	How to shrink it
DNS lookup	0–50 ms	DNS caching, long TTL
TCP + TLS handshakes	10–20 ms to a nearby edge	CDN edge nearby, HTTP/3, keep-alive
Edge to origin	5–80 ms	Origin in the user's region
Gateway + app logic	5–20 ms	Fewer service hops, parallel calls
Cache hit / DB query	0.5 ms / 5–20 ms	Higher cache hit ratio, indexes
Browser rendering	50–200 ms	Smaller pages, fewer scripts

The lesson: on a first visit, the **network round trips** often cost more than the server work. That is why repeat visits feel faster: the DNS answer is cached, the connection is reused (**keep-alive**), and static files come from the browser cache.

What can go wrong at each hop

Hop	Failure	Usual protection
DNS	Provider outage, wrong record	Two DNS providers, sensible TTLs
CDN	Edge outage, stale content	Multi-CDN, cache invalidation
Load balancer	LB node dies	Redundant pair or managed LB
App server	Crash, bad deploy	Health checks, canary deploys, rollback
Cache	Cache node dies, stampede	Replicas, request coalescing
Database	Primary dies	Replica failover, backups
Queue/worker	Worker crashes mid-job	At-least-once delivery, retries, DLQ
Any call	Slow dependency	Timeouts, circuit breaker, fallback

Common mistakes

- **Forgetting DNS and TLS.** Beginners jump straight from "browser" to "server". The handshakes are often the slowest part of a first visit.
- **Making everything synchronous.** If the like request waits for the notification email to send, a slow email provider makes liking slow. Push side effects to a queue.
- **Treating the cache as the source of truth.** The database is the truth. The cache is a copy that can be lost or stale at any moment.
- **Forgetting that every box is redundant.** In a design drawing, one "load balancer" box really means at least two machines.

In the interview

Q: What happens when you type a URL and press Enter?

A: The browser checks its cache, then resolves the name with DNS (resolver, root, TLD, authoritative) to get an IP, usually a nearby CDN edge. It opens a TCP connection and does a TLS handshake, then sends an HTTP GET. The edge serves cached static content or forwards to the origin, where a load balancer picks a healthy server, a gateway authenticates and rate-limits, and the app checks a cache before the database. The response returns the same way, and the browser renders it and fetches images and scripts, mostly from the CDN.

Q: Where would you put a cache in this path?

A: At every layer where it pays: the browser cache for static files, the CDN for public content, a distributed cache like Redis in front of the database for hot data, and the database's own buffer cache. The closer the cache is to the user, the more latency it saves, but the harder it is to invalidate.

Q: Which of these steps would you make asynchronous?

A: Anything the user does not need to see before getting a response: notifications, emails, counters, analytics, search indexing, thumbnails. The request writes the essential data, publishes an event, and returns. Workers do the rest with retries.

TL;DR

One click goes browser cache → DNS → TCP/TLS → CDN → load balancer → API gateway → stateless app server → cache → database, and back; side effects go through a queue to workers. Each hop costs time and can break, so good designs cache it, duplicate it or take it off the user's critical path.

Worked example: designing a URL shortener

What it is

A **URL shortener** turns a long web address into a short one. You give it `https://www.example.com/blog/2026/10/a-very-long-article-title?ref=newsletter` and it gives back

`https://sho.rt/aZ3kQ9b`. When anyone opens the short link, the service **redirects** their browser to the long one. Bitly and TinyURL are well-known examples.

Interviewers love it as a first problem: you can finish it in 45 minutes, yet it still needs estimates, an API, a data model, ID generation, caching, sharding and a read-heavy design. Below we design it with the six-step method from *How to approach a system design interview* in Part 1: requirements, estimates, API, data model, high-level design, deep dives.

Step 1 — Requirements

Always start by asking questions. Write down the answers as two lists.

Functional requirements (what the system does):

- Given a long URL, create a short URL.
- Opening the short URL redirects to the long URL.
- Optional: a user may choose a **custom alias** (`sho.rt/my-party`).
- Optional: links can **expire** after a date.
- Optional: show **click counts** to the link's owner.

Non-functional requirements (how well it does it):

- **Very high availability** for redirects. If redirects fail, every link anyone ever shared is broken. Target 99.99% (about 52 minutes of downtime a year — see *Availability and the nines* in Part 1).
- **Low latency** for redirects: p99 under 50 ms at the server.
- **Short links must not be guessable in order**, so people cannot walk through everyone's links by counting.
- **Durability**: once created, a link must never be lost.
- **Read-heavy**: far more redirects than creations. We assume 100 reads per write.

Also say what is **out of scope**: user accounts, a full analytics dashboard, link previews.

Step 2 — Estimates (back of the envelope)

Assume **100 million new short links per month**. A month has about 2.6 million seconds (30 × 86,400 = 2,592,000).

Quantity	Calculation	Result
Writes per second (average)	100 M ÷ 2.6 M s	about 40 /s
Reads per second (average)	40 × 100	about 4,000 /s
Peak reads (3× average)	4,000 × 3	about 12,000 /s
Links over 5 years	100 M × 12 × 5	6 billion
Storage per link	URL + code + metadata	about 500 bytes
Total storage over 5 years	6 B × 500 B	about 3 TB
Read bandwidth	4,000 × 500 B	about 2 MB/s

What these numbers tell us:

- **40 writes per second is tiny.** One ordinary database handles that easily.
- **12,000 reads per second at peak is moderate.** One database could perhaps cope, but with no headroom. A cache will take most of it.
- **3 TB is too much for one cheap machine's memory, but fine for a few database machines.** Sharding is a "later" concern, not day one.
- **How many characters do we need?** If each character can be one of 62 symbols (a–z, A–Z, 0–9, called **base62**), then 6 characters give $62^6 \approx 56.8$ billion codes and 7 characters give $62^7 \approx 3.5$ trillion. We need 6 billion, so 7 characters gives over 500 times headroom. We choose 7.

Cache sizing: clicks are very uneven: a small minority of links collects most of the traffic (people often quote this as the 80/20 rule). A day brings about 350 million redirects (4,000 × 86,400), but far fewer distinct links; suppose 50 million. Keeping the busiest fifth of those, 10 million entries at

roughly 500 bytes each, needs about 5 GB. That fits in the memory of a single cache node, and we run a few for redundancy.

Step 3 — API

Two endpoints are enough. See *API styles — REST, RPC and gRPC, GraphQL* and *API design basics* in Part 2.

```
POST /api/v1/links
Content-Type: application/json
Idempotency-Key: 5d1c...

{"long_url": "https://www.example.com/blog/...", "custom_alias": null, "expires_at": null}

-> 201 Created
{"short_url": "https://sho.rt/aZ3kQ9b", "code": "aZ3kQ9b"}
```

```
GET /aZ3kQ9b

-> 302 Found
Location: https://www.example.com/blog/...
```

Design notes:

- The `v1` in the path is **API versioning**, so we can change the API later without breaking old clients.
- The **idempotency key** means that if the client times out and retries, it gets the same short link back instead of creating two. See *Delivery guarantees, idempotency and retries* in Part 4.
- The create endpoint is **rate limited** per API key or IP (for example 100 per hour) to stop spammers. Over the limit, it replies `429`. See *Rate limiting and throttling* in Part 5.
- A missing or expired code returns `404 Not Found` (or `410 Gone` for an expired link).

301 or 302? Both are **HTTP redirect** status codes. A **301 Moved Permanently** tells the browser it may remember the redirect and skip our server next time. That saves load, but we stop seeing repeat clicks, so click counts are wrong and we cannot change or expire the link for browsers that cached it. A **302 Found** (temporary) makes the browser ask us every time. Choose 302 if analytics or expiry matter, 301 if pure speed and low cost matter. Saying this trade-off out loud is worth points in the interview.

Step 4 — Data model

One main table. A relational database such as PostgreSQL works well; a key-value store such as DynamoDB or Cassandra works too, because every read is "look up one row by its code". See *SQL vs NoSQL and BASE* in Part 3.

Column	Type	Notes
`code`	varchar(16)	Primary key, e.g. `aZ3kQ9b`
`long_url`	text	Up to about 2 KB
`owner_id`	bigint	Nullable for anonymous links
`created_at`	timestamp	
`expires_at`	timestamp	Nullable

The primary key on `code` gives us an **index** for free, so a redirect is a single index lookup. Clicks are **not** stored by updating a counter in this row on every redirect — that would turn every read into a write. Instead they go to a separate analytics pipeline (see the deep dives).

Step 5 — Generating the short code

This is the heart of the problem. Four common options:

Approach	How	Pros	Cons
Hash the URL	MD5/SHA-256 of the long URL, take 7 base62 chars	Same URL gets same code	Collisions, must check and retry
Random code	7 random base62 chars	Simple, not guessable	Must check for collisions
Counter + base62	Global counter 1, 2, 3 … encoded in base62	No collisions	Counter is a bottleneck, guessable
Key generation service	Pre-make unused codes, hand out in batches	Fast, no collisions	One more service to run

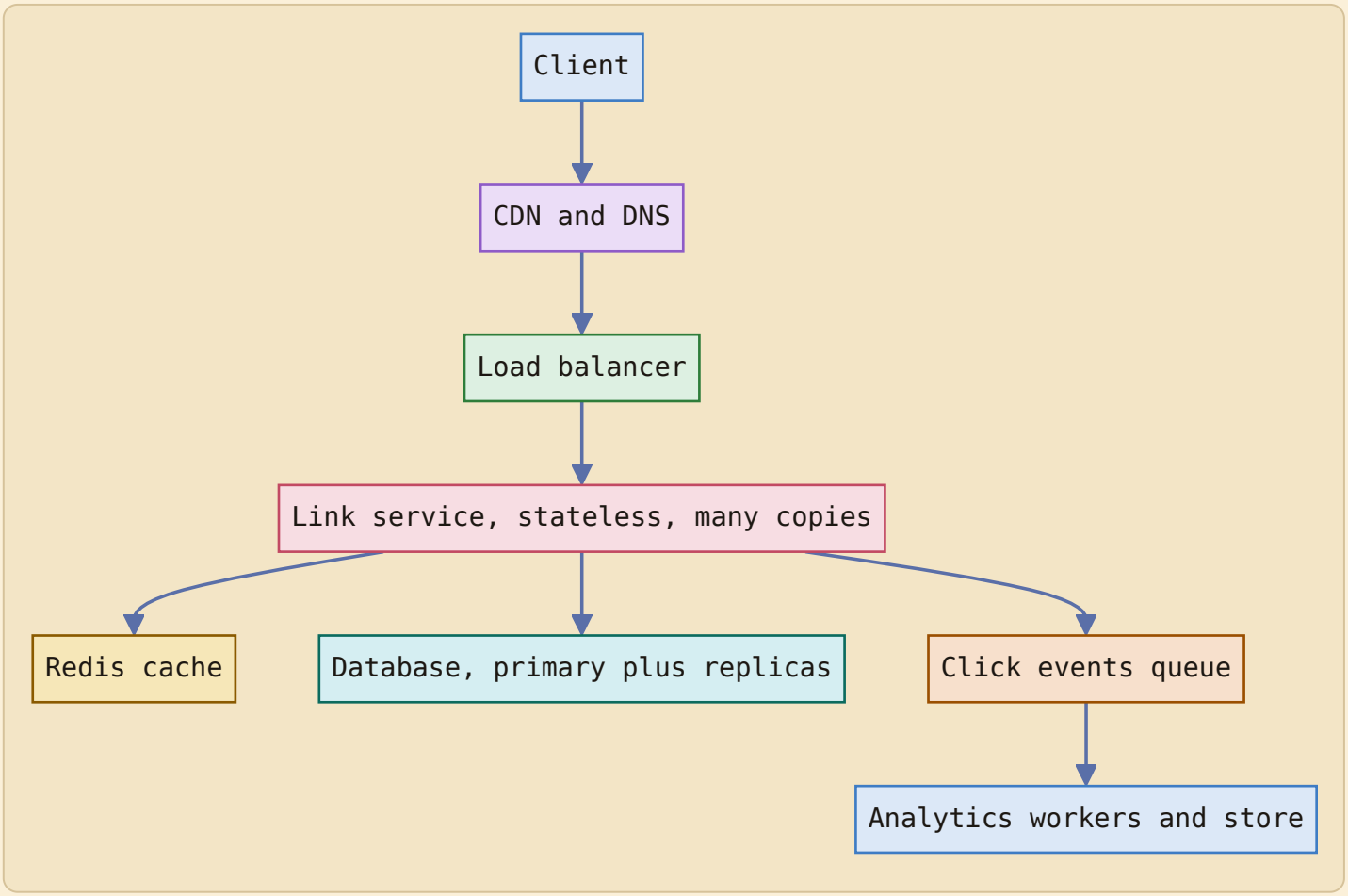
A good beginner answer is **counter ranges plus base62, with shuffling**:

1. A small coordination store (a database row, or ZooKeeper/etcd) hands each app server a **range** of numbers, for example server A gets 1,000,000–1,999,999 and server B gets 2,000,000–2,999,999.
2. Each server counts through its own range in memory, with no network call per link. When it runs out, it asks for a new range. If a server crashes, the rest of its range is simply skipped — we have trillions to spare.
3. The number is **scrambled** with a reversible mixing function (or a block cipher) before encoding, so consecutive links do not get consecutive codes. That meets the "not guessable in order" requirement.
4. The scrambled number is written in **base62**, giving a 7-character code.

Alternatively a **Snowflake-style ID** (a 64-bit number built from timestamp + machine ID + sequence) needs no coordination at all, but the resulting code is 11 characters in base62, longer than we want. See *Unique ID generation* in Part 4.

Custom aliases skip the generator. We simply try to insert the alias as the primary key; if the insert fails because the key exists, we reply `409 Conflict`. The database's uniqueness check is what makes this safe when two people race for the same alias.

Step 6 — High-level design



Redirect path (read, 99% of traffic):

1. `GET /aZ3kQ9b` arrives through DNS, the CDN and the load balancer at any link-service instance.
2. The service looks up `aZ3kQ9b` in **Redis** (cache-aside). On a hit (most of the time) it has the long URL in under a millisecond.
3. On a miss it reads from a **database replica**, then writes the result into Redis with a TTL of, say, 24 hours.
4. It checks `expires_at`, publishes a small "click" event to the queue without waiting, and returns `302` with the `Location` header.

Create path (write, 1% of traffic):

1. `POST /api/v1/links` passes the gateway's authentication and rate limiter.
2. The service validates the URL (well-formed, not on a malware blocklist), takes the next number from its range, scrambles and encodes it.
3. It inserts the row into the **primary** database and returns `201` with the short URL. No cache write is needed; the first redirect fills the cache.

Step 7 — Deep dives

Caching. With a 90%+ hit ratio, the 12,000 peak reads per second become about 1,200 database reads per second, which two read replicas handle easily. Links almost never change, so the cache rarely needs invalidating. When a link is deleted or expires, we delete its cache key. To stop a viral link causing a **cache stampede** when its entry expires, we let only one request refill it (request coalescing). See *Cache invalidation, stale data and stampedes* in Part 4.

Even faster: cache at the edge. For links without analytics we can let the CDN cache the redirect response itself for a few minutes, so most clicks never reach our servers at all.

Scaling the database. 3 TB over five years fits in a primary with a few replicas at first. When it outgrows that, we **shard by hash of `code`**. Every redirect knows its code, so every query goes to exactly one shard — the ideal shard key. Using **consistent hashing** makes adding shards move only a small share of the keys. See Partitioning and sharding in Part 3.

Availability. Every tier runs at least two copies across **availability zones**. The link service is stateless, so a dead instance is simply removed by the load balancer's health checks. The database fails over to a replica if the primary dies. Because the redirect path only needs reads, it can keep working from replicas and cache even while the primary is being replaced — a form of **graceful degradation**: creating links pauses for a minute, but existing links keep working.

Analytics. Writing a counter on every click would put 12,000 writes per second on the database. Instead each click becomes an event on **Kafka**. Workers batch them and update counts once a minute in an analytics store (for example a column store such as ClickHouse). Counts are a little delayed — **eventual consistency** — which owners accept. If a worker crashes, the events are re-read from the queue (**at-least-once**); de-duplicating by event ID keeps counts honest.

Abuse. Short links are popular with spammers. Rate-limit creation, check long URLs against malware lists, and allow links to be disabled.

Trade-offs we made

- **302 over 301:** more load on us, but accurate analytics and the ability to expire links.
- **Counter ranges over hashing:** no collision checks, at the cost of a small coordination store.
- **Eventual consistency for click counts:** much cheaper writes, slightly stale numbers.
- **Cache with long TTL:** fast and cheap, safe because links are effectively immutable.

Common mistakes

- **Skipping the estimates.** Without them you cannot justify 7 characters, the cache size or whether you need sharding at all.
- **Using a single auto-increment column in one database as the generator** and calling it scalable. It works at 40 writes per second, but say out loud that it is a bottleneck and a single point of failure, and how you would fix it.

- **Taking the first 7 characters of an MD5 hash and ignoring collisions.** With billions of links, collisions are certain. You must detect and retry.
- **Updating a click counter on the redirect path.** It makes the cheapest, most common operation expensive.
- **Over-engineering.** Do not open with 50 shards and three regions for 40 writes per second. Start simple, then show how it grows.

In the interview

Q: How long should the short code be?

A: It depends on how many links we must hold. With base62, 7 characters give about 3.5 trillion codes; we expect about 6 billion in five years, so 7 leaves plenty of headroom. 6 characters (about 57 billion) would also work but with less margin.

Q: How would you handle one link going viral with millions of clicks a minute?

A: It is a hot key, but a read-only one, so caching solves it: the entry sits in Redis, we can add a small in-process cache on each app server for the very hottest keys, and we can let the CDN cache the redirect for a short time. Request coalescing prevents a stampede when the entry expires.

Q: SQL or NoSQL?

A: Either works because access is a simple key lookup and the write rate is low. I would start with PostgreSQL for its familiarity and strong uniqueness guarantees, and note that a key-value store such as DynamoDB scales out more simply if we reach many terabytes.

TL;DR

At heart the service maps a code to a URL and is read about a hundred times per write: generate a unique 7-character base62 code (counter ranges plus scrambling), store code → URL in a database, put Redis and optionally the CDN in front for the 100:1 reads, return a 302, and push click analytics through a queue.

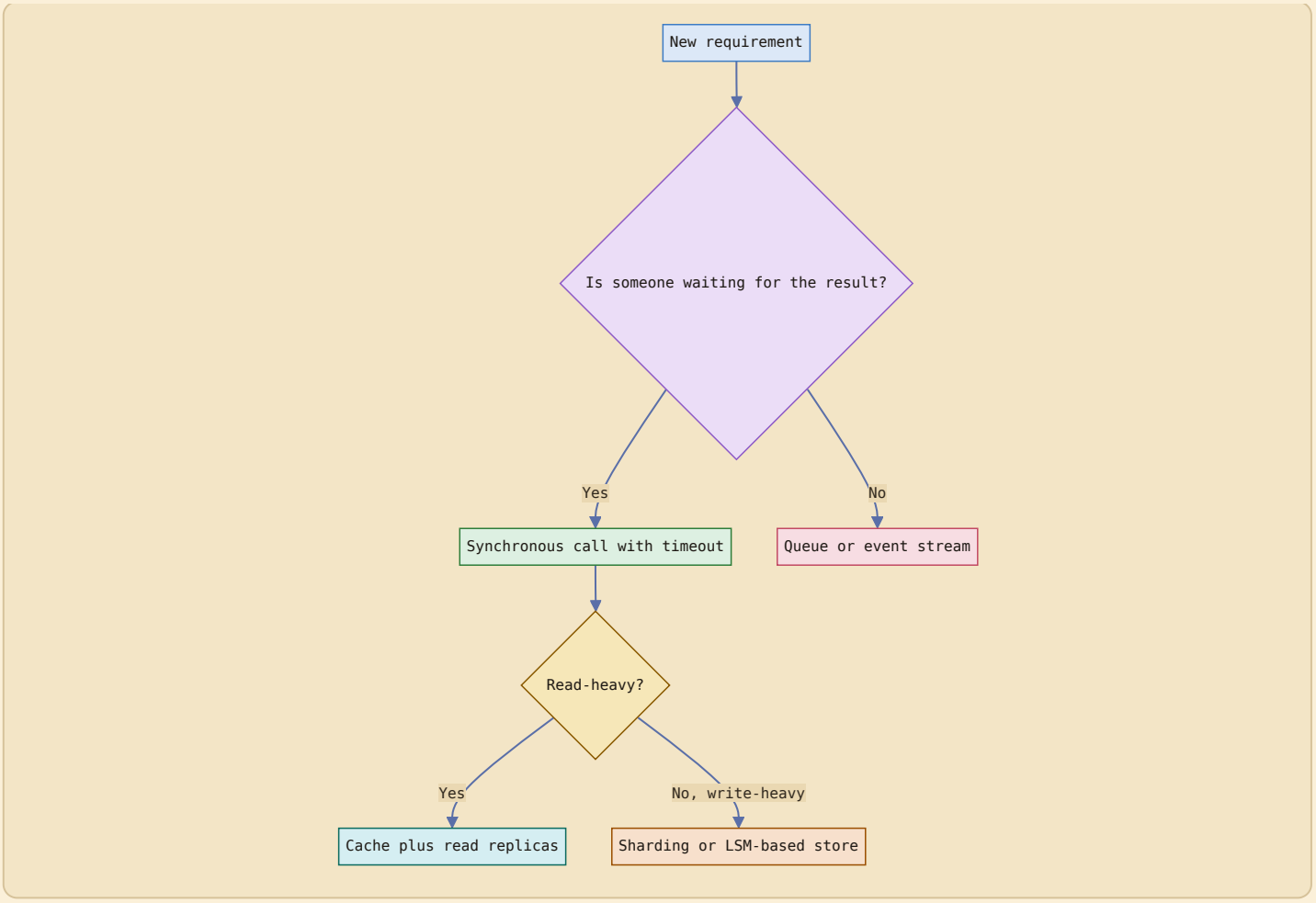
Decision cheat sheet

What it is

A lookup table of the most common design needs and the tool or pattern people usually reach for. It is a starting point, not a rule: in an interview, name the tool, then say **why** and **what it costs**. The "Watch out for" column is the part that earns the marks.

How to use it

Read the left column as "the interviewer just told me…". For example, "the system is read-heavy" → add a cache and read replicas. Then check the right column before committing.


New requirement
Is someone waiting for the result?
Yes
No
Synchronous call with timeout
Queue or event stream
Read-heavy?
Yes
No, write-heavy
Cache plus read replicas
Sharding or LSM-based store

The table

If you need…	Reach for…	Watch out for…
Faster reads of the same data	Cache (Redis, Memcached), cache-aside	Stale data, invalidation, stampedes
Fast static files worldwide	CDN	Invalidation delay, cost per GB
Lower latency for far-away users	CDN, edge, multi-region deployment	Data consistency across regions
More capacity for stateless servers	Horizontal scaling behind a load balancer	Must move session state out of servers
A quick capacity bump, small system	Vertical scaling (bigger machine)	Hard ceiling, still one machine
To spread traffic across servers	Load balancer (L7 for HTTP, L4 for raw TCP)	LB must be redundant itself
Traffic that grows and shrinks daily	Autoscaling	Slow start-up, scale on the right metric
One public entry point for many services	API gateway	Can become a bottleneck or a "god" layer
Strong consistency and transactions	Relational database (PostgreSQL, MySQL)	Harder to scale writes beyond one primary
Flexible, nested records	Document store (MongoDB)	Joins and multi-document transactions are weaker
Simple lookups by key at huge scale	Key-value store (DynamoDB, Redis)	Few query patterns besides by key
Very high write rate, time-ordered data	Wide-column or LSM-based store (Cassandra)	Design tables around queries, eventual consistency
Metrics and sensor readings over time	Time-series database (Prometheus, InfluxDB)	Retention and downsampling planning
Relationship queries (friends of friends)	Graph database (Neo4j)	Harder to shard
Full-text search, fuzzy matching	Search engine (Elasticsearch, OpenSearch)	It is a copy: keep it in sync, not the source of truth
Similarity search on embeddings	Vector database or vector index (pgvector)	Approximate results, memory hungry
Store large files (images, video, backups)	Object storage (Amazon S3)	Not for frequent small updates
More read capacity on the database	Read replicas	Replication lag, read-your-writes issues
More write capacity or data than one machine	Sharding with a good shard key	Hot spots, cross-shard queries, resharding
Add or remove nodes without moving most data	Consistent hashing with virtual nodes	Still some data movement, needs balancing
Fast queries on a column	Index (B-tree)	Slower writes, extra storage
Do slow work without making the user wait	Message queue + workers (SQS, RabbitMQ)	Duplicates, need idempotency and a DLQ
Many services react to one event	Pub/sub or event stream (Kafka, SNS)	Ordering, schema changes
Replay past events, high-throughput logs	Event streaming (Kafka)	Partitions limit parallelism, ops effort

If you need…	Reach for…	Watch out for…
Safe retries of a payment or order	Idempotency key	Storing keys, choosing an expiry
Absorb traffic spikes	Queue as a buffer, backpressure	Queue growth hides overload
Protect a service from abusive clients	Rate limiting (token bucket) returning 429	Fair limits per user, distributed counters
Stop a failing dependency dragging you down	Timeouts + circuit breaker + fallback	Picking thresholds, testing the open state
Retry transient failures	Retries with exponential backoff and jitter	Retry storms, only retry idempotent calls
Keep one failure from sinking everything	Bulkheads (separate pools)	Unused reserved capacity
Survive a whole data-centre outage	Multi-AZ, then multi-region, failover	Cost, replication lag, split brain
A consistent operation across services	Saga with compensating actions	Complex failure handling, interim states
Atomic commit across two databases	Two-phase commit (2PC)	Blocking if the coordinator fails, slow
Only one worker doing a job at a time	Distributed lock (with fencing tokens) or leader election	Lock expiry while still working
Nodes agreeing on a value or leader	Consensus (Raft via etcd/ZooKeeper)	Needs a majority, latency of writes
Unique IDs without coordination	UUID or Snowflake-style IDs	UUIDv4 is random (poor index locality), clock issues
"Have we seen this before?" with little memory	Bloom filter	False positives possible
Count distinct users cheaply	HyperLogLog	Approximate (about 1% error)
Find nearby places	Geohash or quadtree, geo index	Edge cases at cell borders
Real-time updates to the browser, server to client	Server-Sent Events	One direction only
Two-way real-time (chat, games)	WebSockets	Stateful connections, harder load balancing
Notify another company's system	Webhooks	Retries, signatures, duplicates
Fast internal service-to-service calls	gRPC with Protobuf	Browser support needs a proxy
Clients choosing exactly which fields they get	GraphQL	Caching, expensive queries
Large data analysis overnight	Batch processing (Spark), data warehouse	Results are hours old
Results within seconds of events	Stream processing (Flink, Kafka Streams)	Harder to get right, state management
Know what is happening in production	Metrics, logs, traces, alerts on SLOs	Alert fatigue, cost of logs
Release safely	Canary or blue-green deploys, feature flags	Database changes still need care

If you need…	Reach for…	Watch out for…
Users log in once for many apps	SSO with OpenID Connect	Central identity provider is critical
Let an app act for a user on another service	OAuth 2.0	Token storage and scopes
Stateless auth across services	Signed tokens (JWT)	Hard to revoke before expiry
Store passwords	bcrypt, scrypt or Argon2 (slow, salted hashing)	Never encrypt or plain-hash passwords
Protect secrets like API keys	Secrets manager (Vault, AWS Secrets Manager)	Rotation, access audit

TL;DR

Every "reach for" brings a "watch out for". Naming both in one sentence — "I'd add read replicas, accepting some replication lag" — is the habit that interviewers look for.

Top 60 interview questions

What it is

These are the beginner-level questions that come up most in system design interviews and phone screens. Each answer is 2–4 sentences: about what you would say out loud before the interviewer asks a follow-up. The section in brackets tells you where to read more.

Foundations and performance

Q1: What is system design?

A: System design is deciding how the parts of a software system — servers, databases, caches, queues, networks — fit together so it meets its requirements for features, scale, speed and reliability. It is about trade-offs rather than one right answer. (*What system design is*, Part 1.)

Q2: What is the difference between latency and throughput?

A: Latency is how long one request takes, for example 120 ms. Throughput is how many requests the system completes per unit of time, for example 5,000 requests per second. The two can move apart: a cargo ship carries thousands of containers per trip (huge throughput) yet each one takes weeks to arrive (high latency). (*Latency, throughput and bandwidth*, Part 1.)

Q3: Why do we measure p99 latency instead of the average?

A: The average hides the slow requests: a few 5-second requests disappear among thousands of fast ones. p99 means 99% of requests are faster than this value, so it shows what the unluckiest 1% of users feel. A page that makes 100 backend calls will almost always hit at least one p99-slow call, so tail latency matters a lot at scale. (*Percentiles, tail latency and why averages lie*, Part 1.)

Q4: What is Little's law and why is it useful?

A: It says the average number of requests in a system equals the arrival rate times the average time each spends inside: $L = \lambda \times W$. If 1,000 requests per second each take 0.2 seconds, about 200 are in flight at any moment, which tells you how many threads or connections you need. (*Little's law*, Part 1.)

Q5: How do you estimate QPS for a system?

A: Start from daily active users and actions per user: 10 million DAU × 10 requests each = 100 million requests per day. Divide by about 86,400 seconds (round to 100,000) to get about 1,000 QPS on average, then multiply by a peak factor of 2–3 for busy hours. (*Back-of-the-envelope estimation*, Part 1.)

Q6: What does "four nines" availability mean?

A: 99.99% availability: the service may be down for about 52.6 minutes a year, or about 4.4 minutes a month. Each extra nine cuts allowed downtime by ten times and usually costs much more to achieve. (*Availability and the nines*, Part 1.)

Q7: What is the difference between an SLA, an SLO and an SLI?

A: An SLI (indicator) is a measurement, such as the percentage of requests that succeed. An SLO (objective) is the internal target for it, such as 99.9% over 30 days. An SLA (agreement) is a contract with customers, usually looser than the SLO, with refunds or penalties if it is missed. (*SLA, SLO, SLI and error budgets*, Part 1.)

Q8: What is an error budget?

A: It is the amount of failure the SLO allows: a 99.9% SLO leaves 0.1% of requests (or about 43 minutes a month) that may fail. While budget remains, teams can ship fast; when it is spent, they slow down and focus on reliability. (*SLA, SLO, SLI and error budgets*, Part 1.)

Q9: What is the difference between vertical and horizontal scaling?

A: Vertical scaling (scale up) means a bigger machine: more CPU, memory or disk. Horizontal scaling (scale out) means more machines sharing the work. Vertical is simpler but has a ceiling and stays a single point of failure; horizontal grows further and adds redundancy but needs load balancing and stateless services. (*Vertical and horizontal scaling*, Part 1.)

Q10: What does "stateless service" mean and why does it help scaling?

A: A stateless service keeps no per-user data in its own memory between requests; session data lives in a shared store such as Redis or a database, or in a token the client sends. Any instance can then serve any request, so you can add, remove or replace instances freely behind a load balancer. (*Stateless and stateful services*, Part 1.)

Q11: What is a single point of failure and how do you remove one?

A: It is any component whose failure takes the whole system down, such as one database server or one load balancer. You remove it with redundancy — two or more copies — plus automatic failover and health checks so traffic moves to a healthy copy. (*Redundancy, single points of failure and failover*, Part 1.)

Q12: What is the difference between reliability, availability and durability?

A: Availability is whether the system answers right now. Reliability is whether it keeps doing the right thing correctly over time. Durability is whether saved data survives — Amazon S3, for example, is designed for 99.999999999% (eleven nines) annual durability of stored objects. (*Reliability, availability and durability*, Part 1.)

Q13: How do you approach a system design interview question?

A: Clarify functional and non-functional requirements, do quick estimates, define the API and data model, draw a simple high-level design, then deep-dive into the hardest parts and discuss trade-offs and failures. Keep talking through your reasoning, and start simple before adding complexity. (*How to approach a system design interview*, Part 1.)

Networking and communication

Q14: How does DNS work?

A: Your device asks a recursive resolver for a name. If not cached, the resolver asks a root server, which points to the TLD server (for `.com`), which points to the domain's authoritative server, which returns the IP address. Every answer is cached for its TTL so most lookups take only milliseconds. (*DNS — the internet's phone book*, Part 2.)

Q15: What is the difference between TCP and UDP?

A: TCP sets up a connection with a handshake and guarantees that bytes arrive complete, in order, with congestion control; it is used for web pages, APIs and databases. UDP just sends individual packets with no guarantees, so it has less overhead and no waiting for lost packets; it is used for video calls, games and DNS queries, and as the base of QUIC. (*TCP, UDP and QUIC*, Part 2.)

Q16: What changed between HTTP/1.1, HTTP/2 and HTTP/3?

A: HTTP/1.1 handles one request at a time per connection, so browsers open several connections. HTTP/2 multiplexes many requests over one TCP connection with compressed headers, but a lost TCP packet still stalls all of them (head-of-line blocking). HTTP/3 runs over QUIC on UDP, so streams are independent and connection set-up is faster. (*HTTP — the language of the web*, Part 2.)

Q17: What does HTTPS add to HTTP?

A: HTTPS is HTTP inside a TLS connection. TLS encrypts the traffic so others on the network cannot read or change it, and the server's certificate, signed by a trusted certificate authority, proves you are talking to the real site. (*HTTPS and TLS*, Part 2.)

Q18: Which HTTP methods are idempotent and why does it matter?

A: GET, PUT, DELETE, HEAD and OPTIONS are idempotent: repeating them has the same effect as doing them once. POST (and usually PATCH) is not. It matters because networks fail and clients retry; idempotent requests are safe to retry, while non-idempotent ones need an idempotency key. (*HTTP — the language of the web*, Part 2.)

Q19: REST vs gRPC vs GraphQL — when would you use each?

A: REST over JSON is the simple, widely understood default for public APIs. gRPC uses Protobuf over HTTP/2 and is fast and strongly typed, so it suits internal service-to-service calls. GraphQL lets clients ask for exactly the fields they need in one request, which helps mobile apps and complex UIs with many data sources. (*API styles — REST, RPC and gRPC, GraphQL*, Part 2.)

Q20: How do you push real-time updates to a browser?

A: Short polling asks repeatedly on a timer; long polling holds a request open until there is news; Server-Sent Events keep one HTTP response open for a server-to-client stream; WebSockets give a full two-way connection. Use SSE for one-way feeds such as live scores and WebSockets for chat or games. (*Real-time communication — polling, long polling, WebSockets, SSE, webhooks*, Part 2.)

Q21: What is a load balancer and why do you need one?

A: It sits in front of a group of servers and spreads incoming requests across them. It lets you scale horizontally, sends traffic only to healthy servers (via health checks), and lets you deploy or remove servers without users noticing. (*Load balancers*, Part 2.)

Q22: What is the difference between an L4 and an L7 load balancer?

A: An L4 load balancer routes by IP address and port without looking inside the traffic, so it is very fast and works for any TCP or UDP protocol (for example AWS NLB). An L7 load balancer understands HTTP, so it can route by path, header or cookie, terminate TLS and retry requests, at somewhat higher cost (for example AWS ALB, NGINX, Envoy). (*Load balancers*, Part 2.)

Q23: Name some load-balancing algorithms.

A: Round robin sends requests to each server in turn; weighted round robin gives bigger servers more; least connections picks the server with the fewest open connections; IP hash or consistent hashing sends the same client or key to the same server; power of two choices picks two random servers and uses the less busy one. (*Load balancers*, Part 2.)

Q24: What are sticky sessions and why are they often avoided?

A: Sticky sessions make the load balancer send a user to the same server every time, usually via a cookie, so in-memory session data keeps working. They cause uneven load and lost sessions when a server dies, so the better fix is stateless servers with sessions in a shared store. (*Load balancers*, Part 2.)

Q25: What is a CDN and how does it help?

A: A content delivery network is a set of servers in many cities that cache your content close to users. It cuts latency (a few milliseconds instead of a cross-ocean round trip), offloads traffic from your origin servers, and helps absorb DDoS attacks. (*CDNs and anycast*, Part 2.)

Q26: What is the difference between a forward proxy and a reverse proxy?

A: A forward proxy sits in front of clients and makes requests on their behalf, for example a company proxy that filters web access. A reverse proxy sits in front of servers and receives requests on their behalf, providing load balancing, TLS termination, caching and protection; NGINX is a common one. (*Proxies and API gateways*, Part 2.)

Q27: What is an API gateway?

A: It is a reverse proxy that acts as the single entry point for many backend services. It handles shared concerns such as authentication, rate limiting, routing, request transformation and logging so each service does not have to. (*Proxies and API gateways*, Part 2.)

Q28: Offset or cursor pagination?

A: Offset pagination (`?page=5` or `OFFSET 100`) is simple but gets slow on deep pages and can skip or repeat items when data changes. Cursor pagination passes a pointer to the last item seen (`?after=abc`), stays fast and stable, and suits feeds and large lists. (*API design basics*, Part 2.)

Data storage

Q29: SQL or NoSQL — how do you choose?

A: Choose SQL when you need transactions, joins and a well-defined schema, such as orders and payments. Choose a NoSQL family when you need a specific shape at large scale: key-value for simple lookups, document for flexible records, wide-column for heavy writes, graph for relationships. Many systems use both. (*SQL vs NoSQL and BASE*, Part 3.)

Q30: What does ACID mean?

A: Atomicity: a transaction happens completely or not at all. Consistency: it moves the database from one valid state to another, respecting rules. Isolation: concurrent transactions do not see each other's half-finished work. Durability: once committed, data survives crashes. (*Transactions and ACID*, Part 3.)

Q31: What is an index and what does it cost?

A: An index is an extra data structure, usually a B-tree, that lets the database find rows by a column without scanning the whole table — like the index at the back of a book. It makes reads much faster but slows writes, because every insert or update must also update the index, and it uses storage. (*Indexes*, Part 3.)

Q32: What is the difference between replication and sharding?

A: Replication keeps full copies of the same data on several machines, which improves availability and read capacity. Sharding splits the data so each machine holds a different part, which increases write capacity and total storage. Large systems usually do both: each shard is replicated. (*Replication* and Partitioning and sharding, Part 3.)

Q33: Synchronous vs asynchronous replication?

A: With synchronous replication, the primary waits for the replica to confirm before telling the client the write succeeded: no data loss on failover, but slower writes. With asynchronous replication, the primary confirms immediately: faster, but recent writes can be lost if the primary dies and replicas may lag behind. (*Replication*, Part 3.)

Q34: What is replication lag and how does it affect users?

A: It is the delay before a write on the primary shows up on a replica, usually milliseconds but sometimes seconds. A user who posts a comment and then reads from a lagging replica may not see it. Fixes include reading your own recent writes from the primary or tracking the write position. (*Replication*, Part 3, and *Consistency models*, Part 4.)

Q35: How do you choose a shard key?

A: Pick a key that spreads data and traffic evenly and that most queries include, so each query hits one shard. User ID is a common choice; a timestamp is usually bad because all new writes go to the same shard, creating a hot spot. (*Partitioning and sharding*, Part 3.)

Q36: What is consistent hashing?

A: Servers and keys are placed on the same hash ring, and each key belongs to the next server clockwise. When a server is added or removed, only the keys next to it move, about 1/N of them, instead of almost all keys as with `hash mod N`. Virtual nodes give each server many points on the ring for an even spread. (*Consistent hashing and virtual nodes*, Part 3.)

Q37: Normalization or denormalization?

A: Normalization stores each fact once and links tables with keys, which avoids inconsistent copies but needs joins. Denormalization copies data into the shape you read it, making reads fast at the cost of more storage and keeping copies in sync. Read-heavy systems often denormalize their hottest paths. (*Normalization and denormalization*, Part 3.)

Q38: When would you use object storage?

A: For large, unstructured files that are written once and read many times: images, videos, backups, logs, data-lake files. Services like Amazon S3 are cheap, extremely durable and scale without limit; you

store the file there and keep only its URL and metadata in the database. (*Storage types: block, file and object*, Part 3.)

Q39: What is the difference between OLTP and OLAP?

A: OLTP (online transaction processing) handles many small, fast reads and writes for the live application, such as placing an order. OLAP (online analytical processing) runs large queries over lots of historical data for reports, usually in a column-oriented data warehouse. (*OLTP vs OLAP, row vs column storage, lakes and warehouses*, Part 3.)

Q40: What is the N+1 query problem?

A: Code fetches a list with one query, then runs one more query per item to fetch related data: 1 + N queries instead of 1 or 2. It often hides behind an ORM and makes pages slow. Fix it with a join, an `IN (...)` batch query or eager loading. (*Connection pooling, ORMs and N+1 queries*, Part 3.)

Caching, async and consistency

Q41: Explain cache-aside.

A: The application checks the cache first. On a hit it returns the cached value; on a miss it reads the database, stores the result in the cache with a TTL, and returns it. On writes, it updates the database and deletes the cache entry. It is the most common caching pattern. (*Caching strategies*, Part 4.)

Q42: Write-through vs write-back caching?

A: Write-through writes to the cache and the database together, so the cache is always fresh but each write pays both costs. Write-back (write-behind) writes to the cache and saves to the database later in batches, giving fast writes but risking data loss if the cache fails before flushing. (*Caching strategies*, Part 4.)

Q43: What is LRU eviction?

A: Least Recently Used: when the cache is full, it removes the item that has gone longest without being read. It works well because recently used data tends to be used again soon. Alternatives include LFU (least frequently used) and simple TTL expiry. (*Eviction policies*, Part 4.)

Q44: What is a cache stampede and how do you prevent it?

A: When a popular cache entry expires, thousands of requests miss at once and all hit the database together, which can knock it over. Prevent it with request coalescing (only one request refills, others wait), a lock around the refill, random jitter on TTLs, or refreshing popular entries before they expire. (*Cache invalidation, stale data and stampedes*, Part 4.)

Q45: Why use a message queue?

A: It decouples the sender from the worker: the sender drops a message and returns immediately, and workers process it at their own pace. This smooths traffic spikes, lets the worker be down for a while without losing work, and lets you scale producers and consumers separately. (*Message queues*, Part 4.)

Q46: Queue vs publish/subscribe?

A: In a queue (point-to-point), each message is handled by exactly one consumer — good for jobs. In publish/subscribe, every subscriber gets its own copy of each message — good for events that several services care about. Kafka can do both using consumer groups. (*Message queues*, Part 4.)

Q47: At-most-once, at-least-once, exactly-once — what do they mean?

A: At-most-once may lose messages but never duplicates them. At-least-once never loses them but may deliver some twice, which is the common default. Exactly-once is very hard end to end; in practice it means at-least-once delivery plus idempotent processing, sometimes called effectively-once. (*Delivery guarantees, idempotency and retries*, Part 4.)

Q48: What is idempotency and how do you implement it?

A: An operation is idempotent if doing it twice has the same effect as doing it once. Implement it by giving each request or message a unique ID (an idempotency key), recording processed IDs, and skipping or returning the stored result when an ID repeats. (*Delivery guarantees, idempotency and retries*, Part 4.)

Q49: What is a dead-letter queue?

A: A separate queue where messages go after failing processing a set number of times. It stops one bad message from blocking or endlessly retrying, and lets engineers inspect and replay failures later. (*Delivery guarantees, idempotency and retries*, Part 4.)

Q50: Strong vs eventual consistency?

A: With strong consistency, every read sees the latest successful write, as if there were one copy of the data. With eventual consistency, replicas may briefly disagree, but if writes stop they all converge. Strong is simpler to reason about but costs latency and availability; eventual is faster and more available. (*Consistency models*, Part 4.)

Q51: Explain the CAP theorem.

A: When a network partition splits a distributed system, it must choose between consistency (refuse or delay some requests so no one sees stale data) and availability (answer everything, possibly with stale data). Partitions cannot be ruled out, so the real choice is CP vs AP during a partition; when there is no partition you can have both. (*CAP theorem and PACELC*, Part 4.)

Q52: What is a quorum?

A: With N replicas, a write must be confirmed by W of them and a read must ask R of them. If R + W > N, every read overlaps at least one replica holding the latest write. A common setting is N = 3, W = 2, R = 2. (*Quorums*, Part 4.)

Q53: Why do distributed systems need consensus algorithms like Raft?

A: Several nodes must agree on one value — such as who the leader is or the order of log entries — even when some nodes crash or messages are delayed. Raft and Paxos do this safely as long as a majority is reachable. Tools like etcd and ZooKeeper provide it so you do not have to build it. (*Consensus, leader election and distributed locks*, Part 4.)

Q54: 2PC or saga for a transaction across services?

A: Two-phase commit has a coordinator ask everyone to prepare, then commit; it gives atomicity but blocks if the coordinator fails and couples services tightly. A saga runs a chain of local transactions and, on failure, runs compensating actions to undo earlier steps; it is the usual choice for microservices, accepting temporary inconsistency. (*Distributed transactions: two-phase commit and sagas*, Part 4.)

Architecture, reliability and security

Q55: Monolith or microservices?

A: A monolith is one deployable application: simple to build, test and deploy, and the right default for small teams. Microservices split the system into independently deployable services owned by separate teams, helping large organizations scale but adding network calls, distributed failures and operational work. Many teams start with a modular monolith. (*Architectural styles: monolith, modular monolith, SOA, microservices and serverless*, Part 5.)

Q56: What is a circuit breaker?

A: It wraps calls to a dependency and counts failures. After too many, it "opens" and fails fast without calling, giving the dependency time to recover; after a pause it goes "half-open" and lets a few test calls through, closing again if they succeed. It stops one slow service from tying up all your threads. (*Circuit breakers, bulkheads and fallbacks*, Part 5.)

Q57: Why add jitter to exponential backoff?

A: Exponential backoff waits 1, 2, 4, 8 seconds between retries so a struggling service gets breathing room. Without jitter, thousands of clients that failed together retry together, in synchronized waves. Adding randomness spreads retries out. (*Timeouts, retries, exponential backoff and jitter*, Part 5.)

Q58: How does the token bucket rate limiter work?

A: A bucket holds up to B tokens and is refilled at R tokens per second. Each request takes one token; if the bucket is empty the request is rejected with 429. It allows short bursts up to B while enforcing an average rate of R. (*Rate limiting and throttling*, Part 5.)

Q59: What is the difference between authentication and authorization?

A: Authentication answers "who are you?" — for example by checking a password or a token. Authorization answers "what are you allowed to do?" — for example whether user 42 may delete this post. Authentication

always comes first. (*Authentication and authorization*, Part 5.)

Q60: What are RPO and RTO?

A: RPO (recovery point objective) is how much data you can afford to lose, measured in time: an RPO of 5 minutes means backups or replication must be at most 5 minutes behind. RTO (recovery time objective) is how long you can be down before service is restored. Lower values of either cost more. (*Disaster recovery*, Part 5.)

TL;DR

If you can answer these 60 in your own words, with a number or an example in each answer, you have the fundamentals an interviewer expects from a beginner.

Master A–Z glossary

What it is

The most important terms from Parts 1 to 5, plus a few related words you will meet in system design reading and interviews, in alphabetical order with a one-line meaning. Closely related terms share a row (for example "LRU / LFU"). Use it as a quick reminder; each part's own glossary has the complete list, and the parts themselves have the full explanations.

A

Term	Meaning in one line
A / AAAA record	DNS records that map a name to an IPv4 / IPv6 address
ABAC	Attribute-based access control: permissions decided by attributes of user, resource and context
ACID	Transaction guarantees: atomicity (all or nothing), consistency (rules kept), isolation (no interference), durability (survives crashes)
Active-active	Two or more sites or nodes all serving traffic at once
Active-passive	One node serves traffic while a standby waits to take over
Active / passive health check	Load balancer probes servers on a schedule / ejects them after real requests fail
Anycast	Many servers announce the same IP address and the network routes each user to the nearest one
API	Application programming interface: the defined way one program asks another to do something
API gateway	Single entry point for APIs that handles auth, rate limits and routing
API key	Secret string that identifies a calling application
At-least-once / at-most-once delivery	Messages never lost but maybe duplicated / never duplicated but maybe lost
Authentication	Proving who you are
Authoritative name server	DNS server that holds the real records for a domain
Authorization	Deciding what an identified user may do
Autoscaling	Automatically adding or removing servers based on load
Availability	Share of time a system is up and answering correctly
Availability zone (AZ)	One or more isolated data centres with independent power and networking inside a cloud region

B

Term	Meaning in one line
Back-of-the-envelope estimate	Quick rough calculation of load, storage and bandwidth
Backpressure	A slow consumer signalling producers to slow down
Bandwidth	Maximum data a link can carry per second
BASE	Basically available, soft state, eventually consistent: the NoSQL alternative to ACID
Base62	Writing numbers using 62 symbols (a–z, A–Z, 0–9) to make short codes
Batch processing	Processing a large set of stored data in one job, often on a schedule
bcrypt	Deliberately slow, salted password-hashing function
Block storage	Raw fixed-size disk blocks attached to one machine, like a virtual hard drive
Bloom filter	Compact structure that says "definitely not present" or "probably present"
Blue-green deployment	Run old (blue) and new (green) versions side by side and switch traffic at once
B-tree	Balanced tree structure used by most database indexes
Bulkhead	Separate resource pools so one failing part cannot exhaust everything
Burn rate	How fast the error budget is being spent; 1 means exactly on budget for the SLO window

C

Term	Meaning in one line
Cache	Fast storage holding copies of data to serve repeat reads quickly
Cache-aside	App checks cache, on miss loads from database and fills the cache
Cache hit / miss	Requested data was / was not found in the cache; hits ÷ all requests is the hit ratio
Cache invalidation	Removing or updating cached data when the source changes
Cache stampede	Many requests miss the same expired key and hit the database at once
Canary deployment	Release a new version to a small share of traffic first
CAP theorem	During a network partition a system must choose consistency or availability
Causal consistency	Effects are never seen before their causes
CDN	Content delivery network: servers worldwide that cache content near users
Certificate (TLS)	Document signed by a trusted certificate authority (CA) proving a server owns a domain name
Chaos engineering	Deliberately injecting failures to test resilience
CI/CD	Continuous integration and continuous delivery/deployment: automated build, test and release
Circuit breaker	Stops calling a failing dependency for a while so it can recover
CNAME record	DNS record that makes one name an alias of another
Cold start	Extra delay when a new instance or serverless function starts
Column-oriented storage	Stores each column together, great for analytics scans
Compensating action	Step that undoes an earlier step in a saga
Concurrency / parallelism	Many tasks in progress at once by taking turns / tasks literally running at the same moment
Connection draining	Letting a server finish in-flight requests before removing it
Connection pool	Set of reusable open database connections
Consensus	Nodes agreeing on a single value despite failures
Consistency (ACID)	Transactions keep the data valid under all rules
Consistency (distributed)	Whether all readers see the same, latest data
Consistent hashing	Hash ring placement that moves few keys when nodes change
Consumer	Program that reads messages from a queue or stream
Consumer group	Set of consumers that share a topic's partitions
Container	Lightweight packaged app with its dependencies, sharing the host OS kernel
Cookie	Small piece of data a site stores in the browser and receives with each request
CORS	Cross-origin resource sharing: headers by which a server lets pages from other origins read its responses
Count-min sketch	Compact structure that estimates how often items appear
Covering index	Index that contains every column a query needs
CQRS	Command query responsibility segregation: separate models for writes and reads

Term	Meaning in one line
CSRF	Cross-site request forgery: tricking a logged-in browser into sending a request

D

Term	Meaning in one line
Data lake	Cheap storage of raw data in many formats for later analysis
Data warehouse	Structured store optimized for analytical queries
DAU	Daily active users
DDoS	Distributed denial of service: flooding a system from many machines
Dead-letter queue (DLQ)	Queue for messages that keep failing
Denormalization	Duplicating data to make reads faster
Dirty / non-repeatable / phantom read	Isolation anomalies: seeing uncommitted data / a row changing / new rows appearing on re-read
Disaster recovery (DR)	Plans and systems to restore service after a major outage
Distributed cache	Cache spread over several servers and shared by many app servers
Distributed lock	Lock that coordinates processes across machines
Distributed system	Many computers working together over a network as one system
DNS	Domain name system: turns names into IP addresses
Document database	NoSQL store of JSON-like documents (e.g. MongoDB)
Durability	Committed data survives crashes and failures

E

Term	Meaning in one line
Elasticity	Ability to grow and shrink resources quickly with demand
Encryption in transit / at rest	Protecting data moving over a network (TLS) / stored on disks and backups
Error budget	Amount of failure allowed by an SLO
Event-driven architecture	Services communicate by publishing and reacting to events
Event sourcing	Storing every change as an event and rebuilding state from them
Event streaming	Durable, ordered, replayable log of events (e.g. Kafka)
Eventual consistency	Replicas may differ briefly but converge if writes stop
Exactly-once	Each message's effect happens once; in practice at-least-once plus idempotency
Exponential backoff	Doubling the wait after each failed retry (100, 200, 400 ms …), up to a cap

F

Term	Meaning in one line
FaaS	Function as a service: run code per event without managing servers
Failover	Switching to a standby when the active component fails
Fallback	Alternative response used when the normal path fails
Fault / failure	One component going wrong / the whole system no longer delivering its service
Fault tolerance	Continuing to work correctly despite faults
Feature flag	Switch to turn a feature on or off without redeploying
Federation	Splitting databases by function (users DB, orders DB)
Fencing token	Increasing number that lets storage reject requests from a stale lock holder
File storage	Shared files and folders over the network (e.g. NFS)
Firewall	Filter that allows or blocks network traffic by rules
Foreign key	Column that refers to a primary key in another table
Forward proxy	Proxy acting on behalf of clients
fsync	System call that forces written data onto disk
Functional requirement	What the system must do

G

Term	Meaning in one line
GeoDNS	DNS that answers differently depending on the user's location
Geohash	Encoding of a location as a short string where shared prefixes mean nearby
Global server load balancing (GSLB)	Spreading traffic across data centres or regions
Golden signals	Latency, traffic, errors and saturation
Gossip protocol	Nodes share state by chatting with random peers
Graceful degradation	Keeping core features working when parts fail
Graph database	Store of nodes and edges for relationship queries (e.g. Neo4j)
GraphQL	Query language where clients ask for exactly the fields they need
gRPC	High-performance RPC framework using Protobuf over HTTP/2

H

Term	Meaning in one line
Hashing	One-way function that turns data into a fixed-size fingerprint
Head-of-line blocking	One stuck item delays everything queued behind it
Health check	Test that tells whether a server can take traffic
Heartbeat	Periodic "I'm alive" message
Horizontal scaling	Adding more machines (scale out)
Horizontal / vertical partitioning	Splitting a table by rows / by columns
Hot key / hot spot	One key or shard receiving far more traffic than others
HTTP	Hypertext transfer protocol: request/response protocol of the web
HTTP/2 / HTTP/3	HTTP multiplexing many streams over one TCP connection / HTTP over QUIC (UDP), avoiding TCP head-of-line blocking
HTTPS	HTTP over TLS encryption
HyperLogLog	Compact structure for estimating distinct counts

I

Term	Meaning in one line
Idempotency	Doing an operation twice has the same effect as once
Idempotency key	Unique request ID used to detect and ignore repeats
Index	Data structure that speeds up lookups on a column
Infrastructure as code (IaC)	Defining servers and networks in versioned files (e.g. Terraform)
In-process cache	Cache inside the application's own memory
Inverted index	Map from each word to the documents containing it, used by search engines
IOPS	Input/output operations per second a disk can do
IP address	Numeric address of a device on a network: 32-bit IPv4 or 128-bit IPv6
IP hash	Load-balancing by hashing the client IP to pick a server
Isolation level	How strictly a database isolates transactions (read committed, serializable …)

J

Term	Meaning in one line
Jitter	Random variation, added to retry delays or TTLs to avoid synchronized spikes
JWT	JSON Web Token: signed token carrying user claims

K

Term	Meaning in one line
Kafka	Distributed event streaming platform built on a partitioned log
Keep-alive	Reusing one TCP connection for many HTTP requests
Key-value store	Database that stores values looked up by a key
Kubernetes	System for running and managing containers across many machines

L

Term	Meaning in one line
L4 / L7	Transport layer (TCP/UDP) / application layer (HTTP) of the network model
Lamport clock	Logical counter that orders events without real time
Latency	Time for one operation or message to complete or arrive
Leader election	Choosing one node to coordinate the others
Leader-follower replication	One node (leader, or primary) takes writes and copies them to followers (replicas)
Leaderless replication	Any replica accepts writes, using quorums (e.g. Cassandra)
Leaky bucket	Rate limiter that processes requests at a fixed outflow rate
Least connections	Send each request to the server with the fewest active connections
Least privilege	Give each user or service only the access it needs
Linearizability	Strongest single-object consistency: behaves like one copy with instant operations
Little's law	Items in system = arrival rate × time in system
Load balancer	Spreads requests across a group of servers
Load shedding	Rejecting some requests on purpose to protect the system under overload
Long polling	Client request held open until the server has news
Lost update	Two concurrent writes where one silently overwrites the other
LRU / LFU	Eviction of the least recently / least frequently used item
LSM-tree	Log-structured merge tree: write-optimized storage structure

M

Term	Meaning in one line
MapReduce	Batch model: map data into key-value pairs, then reduce each key's values
Merkle tree	Tree of hashes used to find differences between data sets quickly
Message queue	Buffer where producers leave messages for consumers
Microservices	System built from small independently deployable services
Modular monolith	One deployable app with strong internal module boundaries
Monolith	Whole application built and deployed as one unit
Monotonic reads	A user never sees data go backwards in time
MTBF	Mean time between failures
MTTR	Mean time to recovery (or repair)
Multi-leader replication	Several nodes accept writes and sync with each other
Multi-region	Running in several geographic cloud regions

N

Term	Meaning in one line
N+1 query	One query for a list plus one per item
NAT	Network address translation: many private IPs share one public IP
Network partition	Network failure that splits nodes into groups that cannot talk
Nines	Shorthand for availability (99.9% = three nines)
Node	One machine (or process) in a cluster or distributed system
Non-functional requirement	How well the system must work: speed, scale, availability
Normalization	Organizing tables so each fact is stored once
NoSQL	Non-relational databases: key-value, document, wide-column, graph …
NS record	DNS record naming a domain's authoritative servers
NTP	Network time protocol: keeps machine clocks in sync

O

Term	Meaning in one line
OAuth 2.0	Standard for granting an app limited access to a user's resources
Object storage	Flat store of files with metadata, accessed over HTTP (e.g. S3)
Observability	Ability to understand a system's internal state from its outputs
Offset (messaging)	A consumer's position in a partition
OLAP	Online analytical processing: large queries for analysis
OLTP	Online transaction processing: many small live reads and writes
On-call	Engineer responsible for responding to alerts in a given period
OpenID Connect (OIDC)	Identity layer on OAuth 2.0 for login
Origin	Your own servers or storage holding the master copy of content behind a CDN
Origin shield	Extra CDN cache layer that protects the origin
ORM	Object-relational mapper: library mapping objects to database rows
OSI model	Seven-layer reference model of networking

P

Term	Meaning in one line
p50 / p95 / p99 / p999	Latency below which 50 / 95 / 99 / 99.9% of requests fall
PACELC	Extends CAP: else (no partition), choose latency or consistency
Pagination	Returning large results in pages, by offset (skip N rows) or by cursor (after the last item seen)
Partition	A subset of data stored separately; in Kafka, an ordered sub-log of a topic and the unit of parallelism
Paxos	Classic consensus algorithm
Peak factor	Ratio of peak load to average load
Percentile	Value below which a given share of measurements fall
PKCE	Proof Key for Code Exchange: one-time secret proving the app that started an OAuth flow is the one finishing it
Point of presence (PoP)	A CDN or network location in a city
Port	Number identifying a service on a machine (443 for HTTPS)
Power of two choices	Pick two random servers and use the less loaded
Primary key	Column(s) uniquely identifying each row
Private / public IP	Address valid only inside a private network / reachable from the internet
Protobuf	Protocol Buffers: compact binary format for structured data
Pub/sub	Publish/subscribe: every subscriber gets a copy of each message
Push / pull CDN	You upload content to the CDN in advance / the CDN fetches it from origin on first request

Q

Term	Meaning in one line
QPS	Queries (requests) per second
Quadtree	Tree that splits a map into four squares recursively
QUIC	UDP-based transport with built-in TLS, used by HTTP/3
Quorum	Minimum replicas that must agree for a read or write

R

Term	Meaning in one line
Raft	Understandable consensus algorithm with leaders and terms
Rate limiting	Capping how many requests a client may make per time window, rejecting the excess (HTTP 429)
RBAC	Role-based access control: permissions attached to roles
Reactive / scheduled / predictive scaling	Scaling on current metrics / at set times / ahead of forecast load
Read replica	Copy of a database used to serve reads
Read-through / write-through / write-around	Cache loads from the DB on a miss / writes go to cache and DB together / writes skip the cache
Read-your-writes	A user always sees their own recent writes
Recursive resolver	DNS server that does the full lookup for clients
Redis	In-memory data store used as a cache, queue and more
RED method	Rate, errors, duration: metrics for each service
Redundancy	Extra copies of components so one failure is survivable
Region	Geographic area of a cloud provider with several AZs
Relational database	Stores data in tables with schemas and supports SQL
Reliability	Working correctly over time, even when things go wrong
Replication	Keeping copies of data on multiple nodes
Replication lag	Delay before a write reaches replicas
Resharding	Changing the number of shards and moving data
Response time	What the client sees: latency plus queueing and processing
REST	API style using resources, URLs and HTTP methods
Reverse proxy	Proxy acting on behalf of servers
Rollback	Returning to the previous version after a bad deploy
Rolling deployment	Updating servers a few at a time
Root server	Top of the DNS hierarchy, points to TLD servers
Round robin	Send requests to each server in turn
Round trip time (RTT)	Time for a message to go there and back
RPC	Remote procedure call: calling a function on another machine
RPO	Recovery point objective: maximum acceptable data loss
RTO	Recovery time objective: maximum acceptable downtime

S

Term	Meaning in one line
Saga	Sequence of local transactions with compensations on failure
Salt	Random per-user value mixed into a password before hashing
Saturation	How full a resource is (CPU, queue length)
Scalability	Ability to handle more load by adding resources
Secrets management	Securely storing and rotating passwords and keys
Serverless	Running code without managing servers, billed per use
Server-Sent Events (SSE)	One-way stream of events from server to browser over HTTP
Service discovery	How services find each other's current addresses
Service mesh	Network of sidecar proxies handling service-to-service traffic
Session	Server-side record of a logged-in user
Sharding	Splitting data across machines by a shard key
Shard key	Field that decides which shard holds a record
Sidecar	Helper process deployed next to each service instance
SLA	Service level agreement: promise to customers with penalties
SLI	Service level indicator: a measured value
Sliding window	Rate limiter that counts requests in a moving time window
SLO	Service level objective: internal target for an SLI
Snowflake ID	64-bit, roughly time-ordered ID made from timestamp, machine ID and sequence
SOA	Service-oriented architecture: earlier, coarser form of services
Split brain	Two nodes both believe they are the leader
SQL injection	Attack that smuggles SQL commands into input
SSO	Single sign-on: log in once for many apps
Stateless / stateful service	Keeps no client data between requests / keeps client or data state locally
Sticky session	Load balancer sends a user to the same server each time
Stream processing	Processing events continuously as they arrive
Strong consistency	Every read sees the latest write
Synchronous / asynchronous	Caller waits for the result / caller hands off the work and carries on
Synchronous replication	Primary waits for replicas before confirming a write

T

Term	Meaning in one line
Tail latency	The slowest requests (p99 and above)
TCP	Reliable, ordered, connection-based transport protocol
Three-way handshake	SYN, SYN-ACK, ACK exchange that opens a TCP connection
Throttling	Slowing or queuing requests that exceed a rate instead of rejecting them
Throughput	Work completed per unit of time
Thundering herd	Many clients waking or retrying at the same moment
Timeout	Maximum time a caller waits before giving up on a call
Time-series database	Store optimized for timestamped measurements
TLD	Top-level domain such as .com or .org
TLS	Transport layer security: encryption and identity for connections
TLS termination	Decrypting TLS at a proxy or load balancer
Token bucket	Rate limiter allowing bursts up to a bucket size
Topic	Named stream of messages in pub/sub or Kafka
Trace	Record of one request's path across services
Transaction	Group of operations treated as one unit
TTL	Time to live: how long cached data or a DNS answer stays valid
Two-phase commit (2PC)	Prepare then commit protocol for atomic distributed transactions

U

Term	Meaning in one line
UDP	Connectionless transport without delivery or ordering guarantees
USE method	Utilization, saturation, errors: metrics for each resource
UUID	128-bit universally unique identifier

V

Term	Meaning in one line
Vector clock	Per-node counters that detect concurrent updates
Vector database	Store that finds items with similar embedding vectors
Vertical scaling	Making one machine bigger (scale up)
Virtual machine (VM)	Software-emulated computer with its own OS
Virtual node	Many ring positions per server in consistent hashing

W

Term	Meaning in one line
WAF	Web application firewall: filters malicious HTTP traffic
WAL	Write-ahead log: changes written to a log before applying, for crash recovery
Webhook	HTTP callback a system sends to another when an event happens
WebSocket	Persistent two-way connection between browser and server
Wide-column store	NoSQL store that groups rows by a partition key and sorts them within it (e.g. Cassandra)
Worker	Background process that consumes and handles jobs
Write-back / write-behind	Writes go to the cache and reach the database later
Write skew	Two transactions read the same data, each update different rows, and together break a rule

X, Y, Z

Term	Meaning in one line
XSS	Cross-site scripting: injecting malicious script into pages others view
Zero trust	Security model that verifies every request, inside the network or not
ZooKeeper	Coordination service for config, locks and leader election

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Part 6 glossary

Term	Meaning in one line
Request's journey	The full path one user action takes through every component and back
Scheme	First part of a URL, such as `https`, saying which protocol to use
Host	The domain-name part of a URL, such as photos.example.com
Path	The part of a URL after the host, such as /feed
Browser cache	Copies of pages, files and DNS answers kept by the browser
0-RTT resumption	TLS 1.3 feature letting a returning client send data in its first message
URL shortener	Service that maps a short code to a long URL and redirects
Short code	The few characters (e.g. aZ3kQ9b) that identify a short link
Custom alias	User-chosen short code
Redirect	HTTP response telling the browser to go to another URL
301 Moved Permanently	Redirect the browser may cache forever
302 Found	Temporary redirect the browser re-checks each time
409 Conflict	Status code for a request that clashes with existing data
410 Gone	Status code for a resource that existed but was removed
Counter range	Block of numbers handed to one server for ID generation
Key generation service (KGS)	Service that pre-creates unused unique keys
Hash collision	Two different inputs producing the same short hash
Request coalescing	Letting one request refill a cache entry while others wait for it
80/20 rule	Rough rule that 20% of items get 80% of the traffic
Decision cheat sheet	Table mapping common needs to standard tools and their costs
Master glossary	A–Z list of all system design terms in this module

Module 19 — Algorithms Deep Dive

Module Introduction

What this module is. The earlier modules name many algorithms in passing: "use consistent hashing", "a HyperLogLog counts distinct users", "the database uses MVCC". This module opens each one up and shows how it works step by step, with small worked examples and numbers, so you can explain it at a whiteboard and know its limits. It assumes the basics from Module 0 — System Design Fundamentals (Raft, the hash ring with virtual nodes, Bloom filters, Merkle trees, gossip, token buckets and Lamport/vector clocks are explained there).

The four groups

Group	Algorithms
Consensus, coordination and time	Paxos and Multi-Paxos vs Raft and ZAB, 2PC vs 3PC, leases and leader election, failure detection (heartbeats, phi-accrual, SWIM), chain replication and CRAQ, hybrid logical clocks, TrueTime and commit-wait, version vectors
Partitioning and streaming sketches	Jump, rendezvous and Maglev hashing, consistent hashing with bounded loads, HyperLogLog internals, count-min sketch, heavy hitters (Misra-Gries, Space-Saving), reservoir sampling, t-digest percentiles, trending detection
Storage engines, caching and scheduling	LSM trees and compaction, B+-trees, MVCC, two-phase locking and SSI, skip lists, CLOCK / ARC / TinyLFU eviction, hierarchical timing wheels, Chandy-Lamport snapshots and Flink barriers
Search, similarity and routing	Inverted indexes, TF-IDF and BM25, SimHash / MinHash / LSH, vector search (HNSW, IVF, PQ), Dijkstra / A* / contraction hierarchies, feed ranking

Every topic follows the same rhythm: *What it is → The problem it solves → How it works (step by step) → Trade-offs → Real-world examples → In the interview → TL;DR*, with its sources at the end.

Paxos and Multi-Paxos (compared with Raft and ZAB)

What it is

Paxos is a **consensus algorithm**: a set of rules that lets a group of machines agree on one value even when some machines crash, restart or lose messages. Leslie Lamport first described it in "The Part-Time Parliament" (submitted 1990, published 1998) and then rewrote it plainly in "Paxos Made Simple" (2001). The basic version, **single-decree Paxos**, decides exactly one value. **Multi-Paxos** runs many Paxos instances in a row, one per position (called a **slot**) in a replicated log, and adds a stable leader so most slots need only one round trip.

Module 00 (*Consensus, leader election and distributed locks*) already walks through **Raft** step by step: terms, elections, log replication and the majority rule. This section explains the older algorithm Raft was designed to simplify, and then compares the two with **ZAB** (ZooKeeper Atomic Broadcast), the protocol inside Apache ZooKeeper.

The problem it solves

Imagine three database replicas that must agree which transaction is number 42 in the log. If two replicas pick different transactions, their data diverges forever. Messages can be delayed, duplicated or lost, and any machine can crash between two steps. Paxos guarantees **safety** (no two machines ever decide different values for the same slot) under all of those conditions, and **liveness** (it eventually decides) once a majority can talk to each other for long enough.

How it works

Paxos has three roles. One process often plays all three.

- **Proposer**: suggests a value.

- **Acceptor**: votes. A **majority** (also called a **quorum**: more than half, so 2 of 3 or 3 of 5) of acceptors must agree.
- **Learner**: finds out which value was chosen.

Each proposal carries a **proposal number** (also called a **ballot**) that is unique and can be ordered, for example `round.serverId` such as 1.1, 2.3. Each acceptor stores two things on disk: the highest number it has **promised** to respect, and the last proposal it **accepted** (number and value).

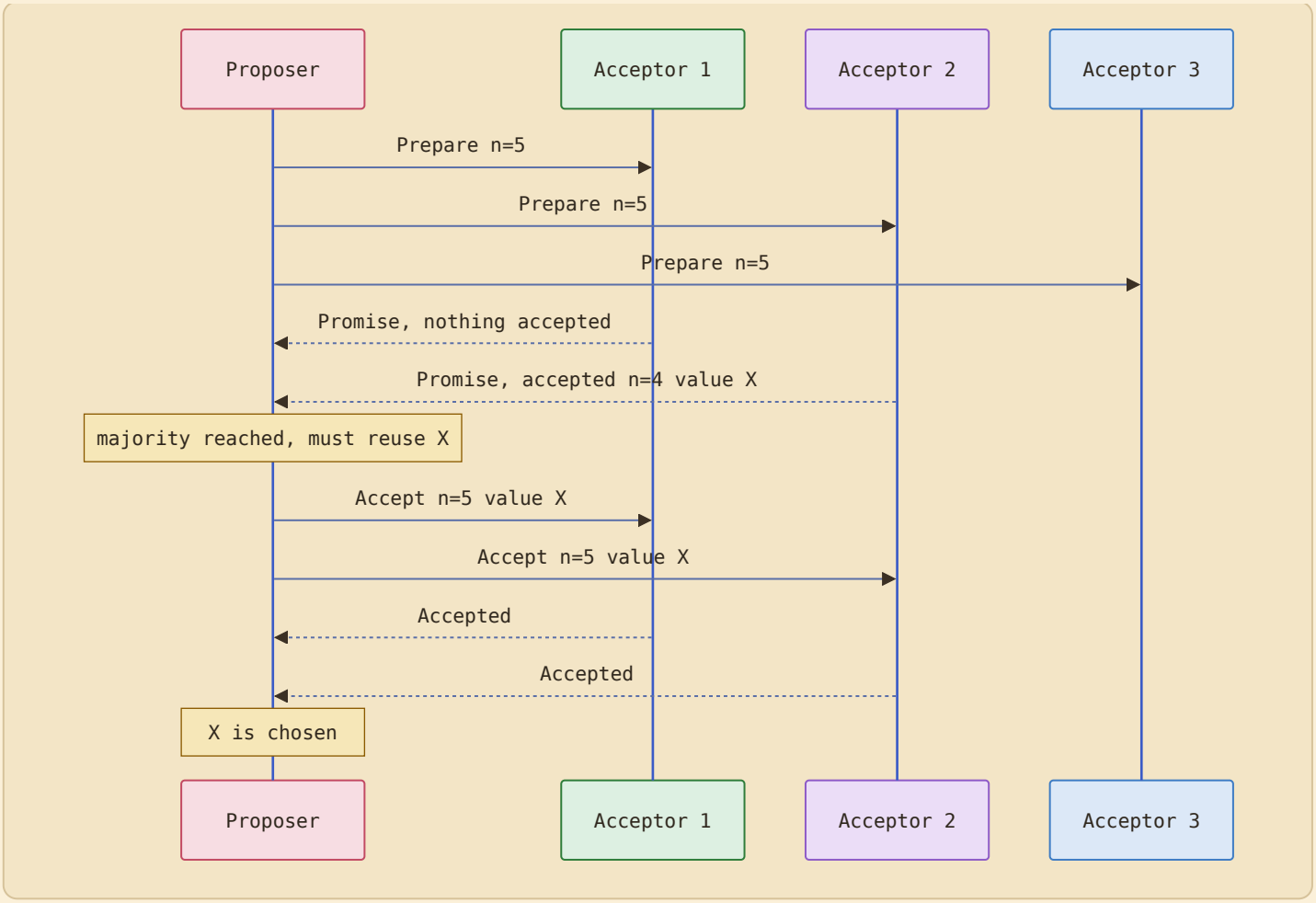
Single-decree Paxos, step by step

1. **Phase 1a, Prepare.** A proposer picks a number `n` higher than any it has used and sends `Prepare(n)` to the acceptors.
2. **Phase 1b, Promise.** An acceptor that has not promised anything higher than `n` replies "I promise to ignore proposals numbered below `n`" and includes the highest-numbered proposal it has already accepted, if any.
3. **Choose the value.** When the proposer has promises from a majority, it looks at the accepted proposals inside those replies. If any exist, it **must** use the value of the highest-numbered one. Only if none exist may it use its own value. This rule is the heart of Paxos safety.
4. **Phase 2a, Accept.** The proposer sends `Accept(n, value)`.
5. **Phase 2b, Accepted.** An acceptor accepts unless it has since promised a higher number. It records `(n, value)` and tells the learners.
6. **Decision.** Once a majority has accepted the same `(n, value)`, that value is **chosen**. It can never change.

Worked example (3 acceptors A1, A2, A3).

- Proposer P1 sends `Prepare(1)`. All three promise; none has accepted anything. P1 sends `Accept(1, "X")`. A1 accepts. The messages to A2 and A3 are delayed, then P1 crashes. Nothing is chosen yet (1 of 3).
- Proposer P2 sends `Prepare(2)` and hears back from A1 and A2 (a majority). A1 reports "I accepted (1, X)". By step 3, P2 must propose X, not its own value "Y". It sends `Accept(2, "X")`, A1 and A2 accept, and X is chosen.
- The delayed `Accept(1, "X")` now reaches A3. A3 never received `Prepare(2)`, so it has promised nothing higher than 1 and accepts (1, X). That is harmless, because it is the same value. Had A3 promised 2 first, it would simply have rejected the old message.

The key insight: any two majorities of 3 overlap in at least one acceptor. That overlap carries forward any value that might already have been chosen.



Pseudo-code for an acceptor:

```
on Prepare(n):
    if n > promised:
        promised = n                      # write to disk before replying
        reply Promise(n, accepted_n, accepted_value)
on Accept(n, v):
    if n >= promised:
        promised = n
        accepted_n, accepted_value = n, v # write to disk before replying
        reply Accepted(n, v)
```

Multi-Paxos. Running both phases for every log slot costs two round trips per decision. Multi-Paxos notices that Phase 1 does not depend on the value, so a proposer that wins Phase 1 for "all slots from here on" becomes the **distinguished proposer**, in practice a leader. For every later slot it only sends Phase 2 (`Accept(n, slot, value)`), one round trip. If the leader dies, another node runs Phase 1 with a higher number, learns any values already accepted in unfinished slots, re-proposes them, and continues.

Cost. One decision in steady-state Multi-Paxos is one round trip to a majority: O(N) messages for N acceptors. Each acceptor stores O(1) state per slot. Classic single-decree Paxos is two round trips. With 2F+1 acceptors, it tolerates F crashed acceptors.

Duelling proposers. Two proposers can keep pre-empting each other with ever-higher numbers (P1 prepares 3, P2 prepares 4, P1 prepares 5, and so on), and nothing is ever chosen. Paxos is safe but this **livelock** stops progress. The fix is the same as in Raft: one leader at a time plus randomised back-off.

Paxos vs Raft vs ZAB

	Multi-Paxos	Raft	ZAB
Leadership	Optional optimisation, not part of the core proof	Built in: strong leader, one per term	Built in: one primary per epoch
Era counter	Proposal number	Term	Epoch (high 32 bits of the 64-bit zxid)
Log holes	Allowed: slot 7 can be decided before slot 6	Not allowed: log is contiguous	Not allowed: strict order
Who can become leader	Anyone; new leader recovers values via Phase 1	Only a node whose log is at least as up to date (election restriction)	Node with latest epoch and highest zxid in a quorum
Leader catch-up	New leader learns from acceptors	Leader overwrites followers' conflicting entries	Synchronisation phase before new broadcasts
Spec completeness	Many details (membership, snapshots) left to implementers	Full spec, including membership change	Tailored to ZooKeeper's primary-backup model

In practice, all three do the same thing in the normal case: a leader sends each entry to followers and commits it when a majority acknowledges. They differ mostly in how a new leader cleans up after the old one.

Trade-offs

- **Paxos is flexible but underspecified.** Allowing holes and out-of-order decisions helps parallelism, but every real system must invent its own leader election, log compaction and membership change. Google's "Paxos Made Live" (2007) describes how hard that was for Chubby.
- **Raft is easier to get right** because it fixes those choices. The price is less flexibility (for example, no holes in the log).
- **Every write costs a majority round trip.** A 5-node group across three regions pays a cross-region round trip on each commit.
- **More nodes means more fault tolerance but slower commits.** 3 nodes tolerate 1 failure, 5 tolerate 2.

Real-world examples

- **Google Chubby** (lock service) and **Spanner** (each data shard is replicated by its own Paxos group).
- **Apache ZooKeeper** uses ZAB, and was the coordination layer for older Kafka, HBase and Hadoop.
- **etcd**, **Consul**, **CockroachDB**, **TiKV** and Kafka's **KRaft** controller use Raft.
- **Apache Cassandra** uses single-decree Paxos rounds per key for its lightweight transactions (`IF NOT EXISTS`).

In the interview

Q: Why must a Paxos proposer reuse a previously accepted value?

A: That value might already have been chosen by a majority. Any majority it hears from overlaps the earlier majority, so at least one promise will report it. Reusing it guarantees that a later proposal can never overwrite a decision.

Q: What does Multi-Paxos add to basic Paxos?

A: A stable leader that runs Phase 1 once and then only Phase 2 for every log slot. That cuts each decision from two round trips to one, which is why it is the version used in practice.

Q: Raft or Paxos for a new system?

A: Use an existing, tested library or service (etcd, ZooKeeper, a Raft library). If choosing, Raft has a complete specification and many production implementations. Paxos variants are worth it only for special needs, such as decisions that may complete out of order.

TL;DR

Paxos reaches agreement with two phases (promise, then accept) and one golden rule: reuse the highest value already accepted by the majority you heard from. Multi-Paxos adds a stable leader so each log entry costs one round trip. Raft and ZAB are leader-centred cousins with a fuller specification. Refer to Module 00 for Raft's step-by-step flow.

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Two-phase commit vs three-phase commit

What it is

An **atomic commit protocol** makes a transaction that touches several machines either commit everywhere or abort everywhere. The machines holding data are **participants** (also called resource managers or cohorts); one machine, the **coordinator** (or transaction manager), runs the protocol.

- **Two-phase commit (2PC)**: a vote phase then a decision phase.
- **Three-phase commit (3PC)**: adds a middle "pre-commit" phase so participants can finish without the coordinator in some failure cases. Dale Skeen described non-blocking commit protocols in 1981.

Module 00 introduces 2PC briefly. Here we focus on exactly where it blocks, how 3PC tries to fix that, and why industry instead puts 2PC on top of consensus.

The problem it solves

A money transfer debits account 17 on shard A and credits account 99 on shard B. If A commits and B aborts, money disappears. Each shard can make its own change durable, but neither can decide alone, because the other might fail its checks (a constraint violation, a full disk, a deadlock).

How it works

2PC step by step (coordinator C, participants A and B):

1. **Prepare.** C sends `PREPARE` to A and B.
2. **Vote.** Each participant checks it can commit, writes all changes plus a "prepared" record to its durable log, keeps its locks, and answers `YES`. Or it answers `NO` and aborts. After voting YES a participant has given up the right to decide alone: it must wait for C.
3. **Decide.** If every vote is YES, C writes "commit" to its own log (this write is the actual **commit point**) and sends `COMMIT`. If any vote is NO or missing after a timeout, it sends `ABORT`.
4. **Finish.** Participants apply the decision, release locks and acknowledge. C can forget the transaction when all acknowledgements arrive.

Messages: 4 per participant (prepare, vote, decision, ack), so 4N for N participants, and 2 round trips plus 2 forced disk writes on the critical path.

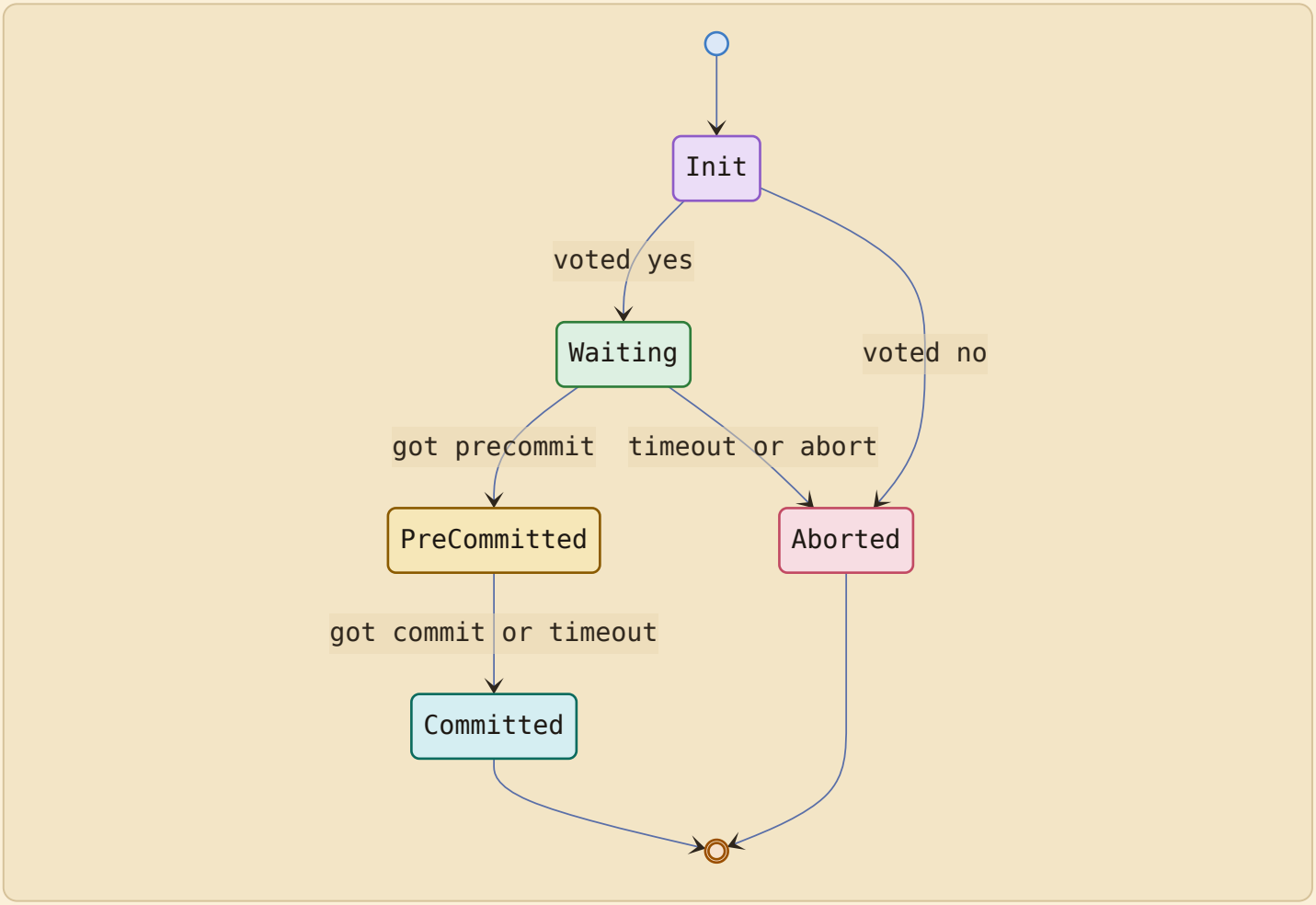
Where it blocks. Suppose A and B both voted YES and then C crashes before sending the decision. A and B are **in doubt**: C may already have told someone COMMIT, or may have decided ABORT. They cannot guess. They hold their locks until C recovers, which could be minutes. Rows 17 and 99 are frozen for every other transaction meanwhile. This is the **blocking problem**.

3PC step by step:

1. **CanCommit.** C asks "can you commit?" Participants vote YES or NO, but do not lock themselves in yet.

2. **PreCommit.** If all said YES, C sends `PRECOMMIT`. Participants acknowledge. Now everyone knows that everyone voted YES.
3. **DoCommit.** After all PreCommit acks, C sends `COMMIT`.

The recovery rule: if C disappears, the surviving participants talk among themselves. If any of them has received PRECOMMIT, nobody can have aborted, so they commit. If none has, nobody can have committed, so they abort. Participants use timeouts to move forward without waiting for C.



Worked example of the timeout logic. Five participants, C crashes after sending PRECOMMIT to P1 only. P2 to P5 time out and elect a recovery coordinator, say P2. P2 asks everyone their state, sees that P1 is pre-committed, and commits all. If P1 had also crashed, the survivors see nobody pre-committed and abort. With a reliable network, both cases end consistently.

Why 3PC is rarely used

- **It is not safe under network partitions.** 3PC's timeouts assume a **synchronous network**: messages arrive within a known bound, so "no reply" really means "crashed". Real networks are asynchronous. If a partition separates P1 (pre-committed) from P2 to P5 (not), P1's side times out and commits while the other side times out and aborts. The two halves disagree, which is worse than blocking.
- **It costs an extra round trip** on every transaction, even in the common case where nothing fails.
- **A better fix exists.** Jim Gray and Leslie Lamport's "Consensus on Transaction Commit" (2006) showed that if the coordinator's decision is itself replicated with Paxos (**Paxos Commit**), commit no longer blocks while a majority of coordinators is alive. Plain 2PC turns out to be the special case with zero tolerated coordinator failures. Modern systems do this: they run ordinary 2PC, but the coordinator and every participant are consensus groups, not single machines, so no single crash leaves anyone in doubt.

Trade-offs

	2PC	3PC	2PC over consensus groups
Round trips (normal case)	2	3	2, each also replicated
Blocks if coordinator dies	Yes	Not with reliable network	No, while a majority survives
Safe under partition	Yes (it blocks instead)	No	Yes
Typical use	XA, database-internal	Textbooks	Spanner, CockroachDB-style systems

2PC also holds locks for the whole protocol, so cross-shard transactions are slower and lower-throughput than single-shard ones. That is why designers try to keep related data on one shard, or use **sagas** (a chain of local transactions with compensations, taught in the microservices modules) when they can tolerate temporary inconsistency.

Real-world examples

- **XA transactions** (the X/Open standard used by Java EE and many databases) and PostgreSQL's `PREPARE TRANSACTION` are plain 2PC.
- **Google Spanner** runs 2PC across Paxos groups: the coordinator leader and the participant leaders each replicate their prepare and commit records through Paxos.
- **Apache Kafka transactions** use a transaction coordinator that writes a prepare marker to its replicated log and then commit markers to each partition, a 2PC-style flow on replicated logs.

In the interview

Q: What is the blocking problem in 2PC?

A: After voting YES, a participant cannot decide alone. If the coordinator crashes before telling it the result, it must hold its locks until the coordinator returns.

Q: Why don't we use 3PC to fix it?

A: 3PC only works if a missing reply reliably means a crash. With network partitions, two groups can time out in different directions, one committing and one aborting. It also adds a round trip. Replicating the coordinator with Paxos or Raft solves blocking safely.

Q: How would you design a cross-shard transaction today?

A: 2PC where the coordinator and participants are consensus-replicated shards, a durable transaction record as the commit point, and idempotent retries. Better still, design the data so most transactions hit one shard.

TL;DR

2PC is correct but blocks when the coordinator dies after participants vote YES. 3PC adds a pre-commit phase to avoid blocking, but becomes unsafe under partitions and costs a round trip, so almost nobody ships it. The industry answer is 2PC whose coordinator and participants are each replicated by consensus.

Sources

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- Gray, Lamport, "Consensus on Transaction Commit", ACM TODS 31(1), 2006: https://arxiv.org/abs/cs/0408036
- Corbett et al., "Spanner: Google's Globally-Distributed Database", OSDI 2012: https://www.usenix.org/conference/osdi12/technical-sessions/presentation/corbett
- Apache Kafka documentation, message delivery semantics and transactions: https://kafka.apache.org/documentation/

Leases and lease-based leader election

What it is

A **lease** is a lock with an expiry time. The holder gets a right (to cache a file, to be the leader, to serve reads) for a fixed period, say 10 seconds. To keep it, the holder must **renew** before it expires. If the holder crashes or is cut off, it stops renewing, and after the period ends the right can be given to someone else without asking the old holder. Cary Gray and David Cheriton introduced leases for file-cache consistency at SOSP 1989.

Lease-based leader election uses a lease as "the leadership token": whoever holds the lease is the leader.

The problem it solves

A plain lock has a fatal flaw in a distributed system: if the holder crashes, the lock is never released, and everyone waits forever. You cannot tell a crashed holder from a slow one. A lease converts "wait until the holder says it's done" into "wait until a known time has passed", which you can always do on your own. Leases also make reads cheap: a leader that holds a valid lease knows no other leader exists, so it can answer reads from local state without a majority round trip.

How it works

Basic lease leader election, step by step (with a coordination store such as etcd):

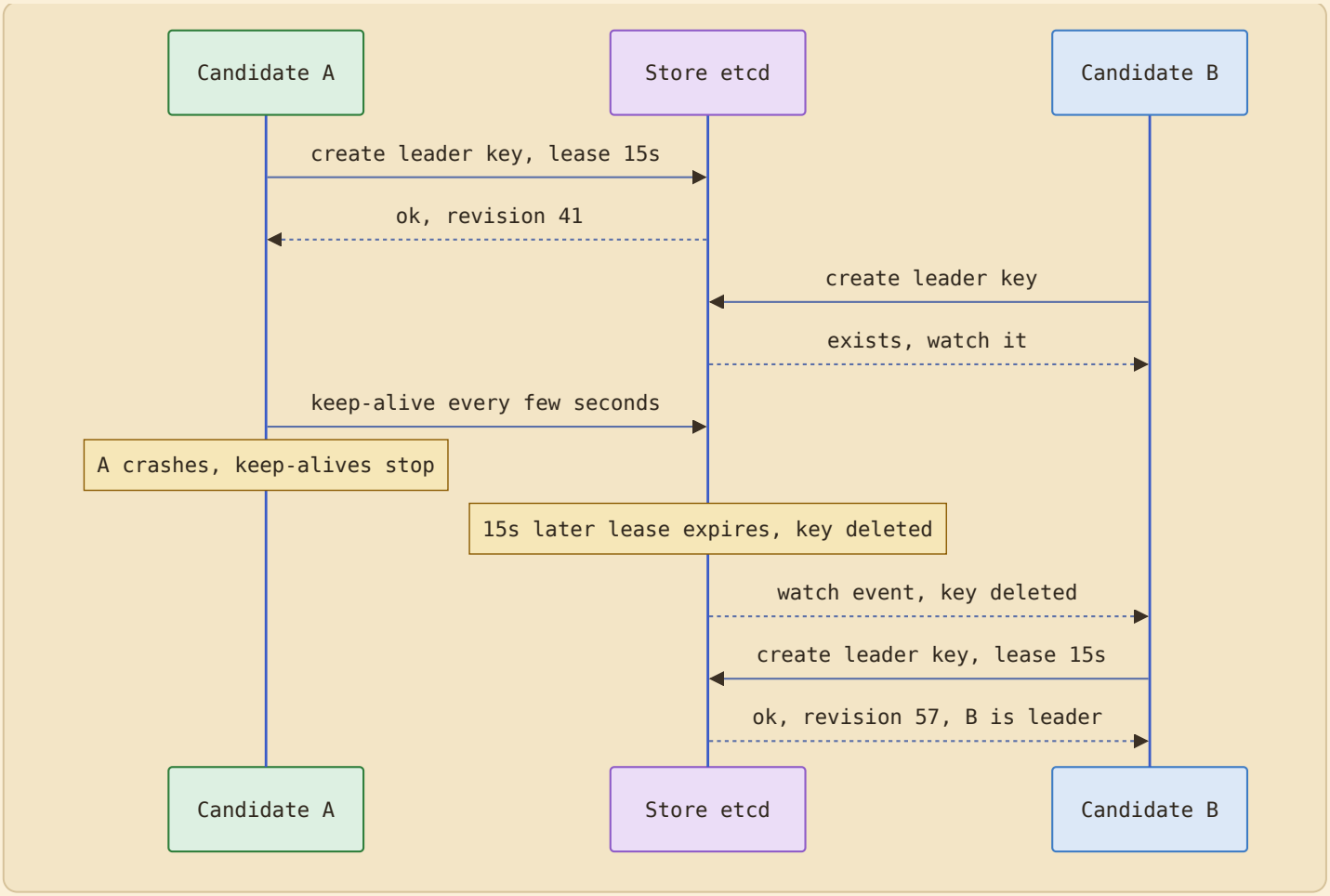
1. Every candidate tries to create the key `/service/leader` with its own ID, attached to a lease with a **TTL** (time to live) of, say, 15 seconds. The store allows only one create to succeed (a compare-and-swap: "create only if absent").
2. The winner is the leader. It renews (sends **keep-alives**) well before expiry, for example every 2 to 5 seconds.
3. The losers **watch** the key. If the leader dies, its renewals stop, the store deletes the key when the TTL runs out, and the watchers race to create it again.
4. Each successful create gets a new, larger **revision number** from the store. The leader attaches it as a **fencing token** to every write, so storage can reject writes from an old leader (Module 00 explains fencing tokens).
5. A leader that cannot renew before its own **renew deadline** must stop acting as leader immediately, even if it is not sure it has lost.

Clocks and the safety margin. A lease is measured with each machine's own clock. Clocks **drift** (run fast or slow). The rule is that the holder must believe the lease ends *earlier* than the granter believes it does.

Worked example: lease length L = 10 s, maximum drift rate 0.1% (1 ms per second), and the holder measures from when it *sent* the request.

- Holder sends the request at its local time 0 and gets the grant at local time 0.05 s. It treats the lease as running from 0, not 0.05, so network delay only makes it more cautious.
- Over 10 s, its clock could be off by 10 ms. It stops acting at local 10 − 0.01 = 9.99 s, minus a safety margin, say at 9.5 s.
- The granter will not hand the lease to anyone else until its own clock shows 10 s plus margin, say 10.5 s.
- The two windows can never overlap, so at most one leader exists at any instant, provided no process pauses longer than the margin without checking time. That last condition is why a fencing token is still needed: a 20-second garbage-collection pause makes the old leader act on stale beliefs.

Kubernetes defaults as a concrete example. Components like `kube-controller-manager` use a `Lease` object with lease duration 15 s, renew deadline 10 s and retry period 2 s. The leader tries to renew every 2 s, gives up leadership if it has not renewed within 10 s, and others may take over 15 s after the last renewal. Worst-case failover is therefore about 15 s.



Leader leases inside consensus. Raft and Multi-Paxos leaders can also hold a lease granted by a majority of followers: followers promise not to vote for anyone else for, say, 10 s after acknowledging the leader. During that window the leader can serve **linearizable reads** (reads guaranteed to see the latest committed write) locally. Spanner's Paxos leaders use leases of 10 seconds by default.

Complexity. Each holder sends one renewal per interval: O(1) messages per period. Failover time is bounded by the lease length plus detection and election time.

Trade-offs

- **Short leases** (a few seconds) mean fast failover but more renewal traffic and more false failovers when the network hiccups.
- **Long leases** mean fewer renewals but a longer outage after a real crash: nobody can lead until the lease expires.
- **Leases depend on bounded clock drift**, not synchronised clocks. Gray and Cheriton found little extra benefit beyond about 10 seconds for their workload.
- **Process pauses break the assumption** that the holder checks time often. Always pair leases with fencing tokens if correctness matters.

Real-world examples

- **Google Chubby**: sessions are leases (default extension 12 s); after a lease lapses the client enters "jeopardy" and waits a 45-second grace period for a new master. Chubby's master itself holds a master lease of a few seconds from its replicas.
- **etcd** leases and **ZooKeeper** ephemeral nodes (deleted when the session times out) are the standard building blocks for leader election.
- **Kubernetes** control-plane components and many operators use `coordination.k8s.io` Lease objects.
- **HDFS** uses leases to give one client at a time the right to write a file.

In the interview

Q: How does a lease differ from a lock?

A: A lease expires on its own. If the holder disappears, the system waits a bounded time and moves on, instead of waiting forever.

Q: Can two leaders exist at once with leases?

A: Not if clocks drift within the assumed bound and the old leader stops before its lease ends. But a long process pause can make an old leader act after expiry, so writes should carry a fencing token that storage checks.

Q: How do you choose the lease length?

A: It is the worst-case failover time. Pick it from the recovery target (for example 10 to 15 s), then set renewals at a fraction of it (every 2 to 5 s) so a couple of lost renewals don't cost leadership.

TL;DR

A lease is a lock that expires. Leader election with leases means "whoever holds the unexpired lease leads"; others take over after expiry. Make the holder stop early, the granter wait late, and use fencing tokens for pauses.

Sources

- Gray, Cheriton, "Leases: An Efficient Fault-Tolerant Mechanism for Distributed File Cache Consistency", SOSP 1989: https://courses.cs.duke.edu/spring15/cps210/readings/leases-sosp89.pdf
- Burrows, "The Chubby lock service for loosely-coupled distributed systems", OSDI 2006: https://static.usenix.org/events/osdi06/tech/full_papers/burrows/burrows_html/
- Kubernetes, `kube-controller-manager` reference (leader-elect flags): https://kubernetes.io/docs/reference/command-line-tools-reference/kube-controller-manager/
- etcd documentation, lease API: https://etcd.io/docs/latest/learning/api/
- Corbett et al., "Spanner", OSDI 2012 (Paxos leader leases, 10 s default): https://www.usenix.org/conference/osdi12/technical-sessions/presentation/corbett

Failure detection: heartbeats, phi-accrual and SWIM

What it is

A **failure detector** is the part of a distributed system that decides "that node is probably dead". It cannot be certain: from the outside, a crashed node, a slow node and a lost network link look identical. So every failure detector is a guess, tuned between two properties:

- **Completeness**: every crashed node is eventually suspected.
- **Accuracy**: healthy nodes are rarely suspected by mistake (few **false positives**).

Three common designs: fixed-timeout **heartbeats**, the adaptive **phi accrual** detector (Hayashibara and colleagues, 2004), and **SWIM** (Das, Gupta and Motivala, DSN 2002), which combines probing with gossip for large clusters.

The problem it solves

Leader election, replica placement, load balancing and membership lists all need to know who is alive. A detector that is too eager causes needless failovers and data movement (a "flapping" node is declared dead and alive repeatedly). A detector that is too slow leaves clients waiting on a dead node. In a cluster of 1,000 nodes, the detection traffic itself must also stay small.

How it works

1. Fixed-timeout heartbeats.

1. Each node sends "I'm alive" to a monitor every interval, for example every 1 s.
2. The monitor records the arrival time of the last heartbeat.
3. If no heartbeat arrives for the timeout, for example 5 s, the node is declared dead.

Simple, but a single fixed number cannot fit every network. Five seconds is far too slow inside one rack and too twitchy across a congested WAN link. If every node heartbeats every other node, traffic grows as

O(N²): 1,000 nodes means about a million messages per interval.

2. Phi accrual detector. Instead of a yes/no answer, it outputs a suspicion level **phi (φ)** that rises the longer a heartbeat is overdue, scaled by how regular the heartbeats have been.

1. Keep a sliding window of recent **inter-arrival times** (gaps between heartbeats), say the last 1,000.
2. Estimate their mean μ and standard deviation σ.
3. At any moment, let t be the time since the last heartbeat. Compute the probability that a healthy node's next heartbeat would be *this* late, P = 1 − F(t), where F is the cumulative distribution of the observed gaps.
4. φ = −log10(P). So φ = 1 means a 10% chance it is just late; φ = 3 means 0.1%; φ = 8 means 1 in 100 million.
5. The application chooses a threshold. Cassandra's `phi_convict_threshold` defaults to 8.

Worked example with gaps averaging μ = 1,000 ms and σ = 100 ms (normal distribution):

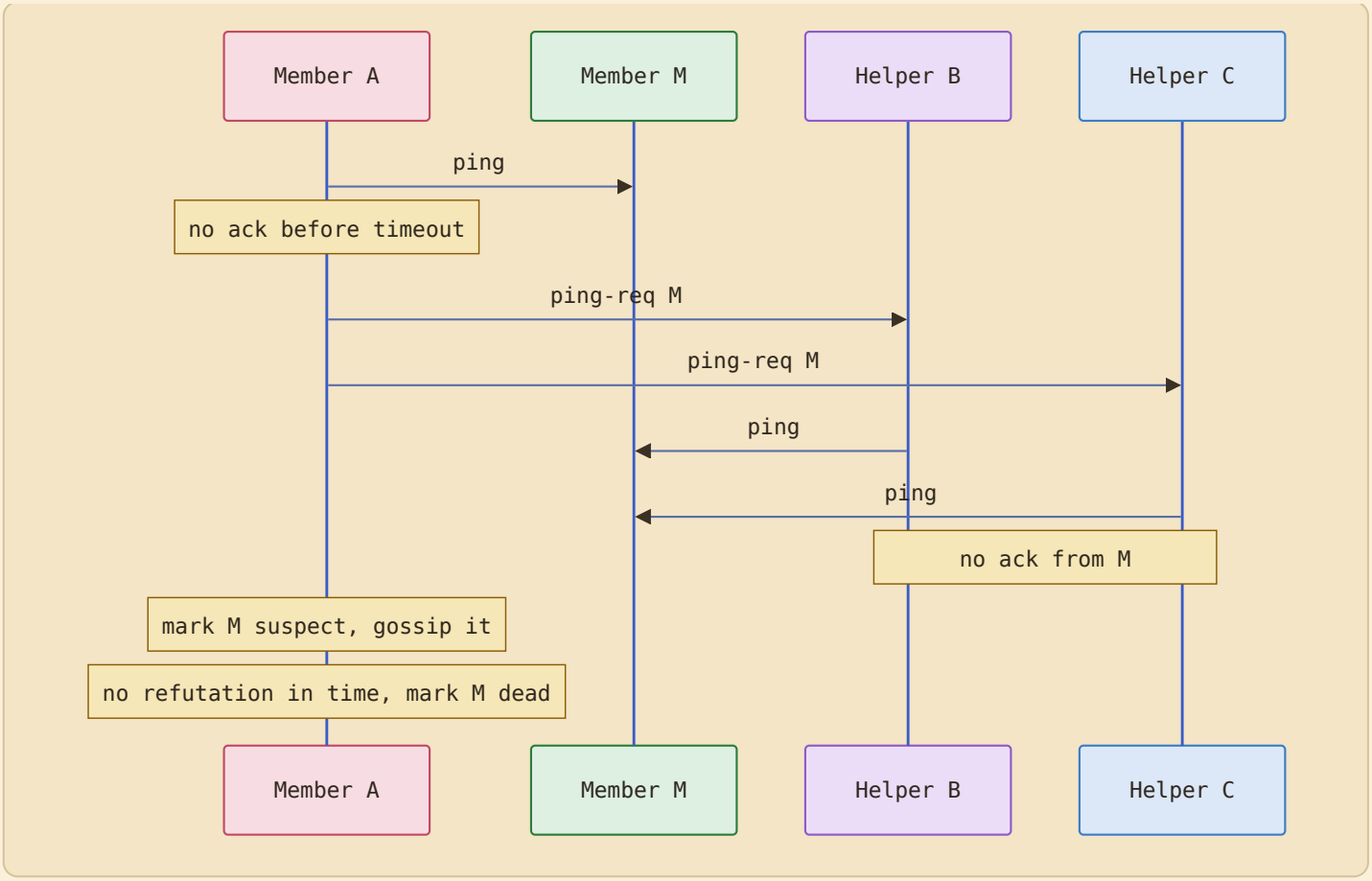
Time since last heartbeat	z = (t − μ)/σ	P(just late)	φ
1,100 ms	1	0.159	0.8
1,200 ms	2	0.023	1.6
1,300 ms	3	0.0013	2.9
1,500 ms	5	2.9e-7	6.5
1,560 ms	5.6	1.1e-8	8.0

So with these statistics a node is convicted about 1.56 s after its last heartbeat. If the network becomes jittery and σ grows to 300 ms, the same φ = 8 is reached only after about 2.7 s. The detector adapts by itself. Implementations differ in which distribution they assume; the paper used a normal distribution, and some systems use a simpler exponential model.

3. SWIM (Scalable Weakly-consistent Infection-style process group Membership). SWIM separates *detecting* failures from *spreading* the news.

1. Every **protocol period** T (say 1 s), each member picks one random member M and sends it `ping`.
2. If M replies `ack` within a short timeout, done.
3. If not, the member asks k other random members (say k = 3) to `ping-req` M on its behalf. If any of them gets an ack, it relays it back. This **indirect probe** avoids blaming M for a bad link between just two nodes.
4. If nobody hears from M, it is marked **suspect**, not dead. The suspicion is gossiped.
5. If M hears it is suspected, it **refutes** by gossiping "alive" with a higher **incarnation number** (a counter only M itself may increase).
6. If the suspicion is not refuted within a suspicion timeout, M is declared **dead**.
7. Membership updates (join, suspect, alive, dead) are **piggybacked** on the ping and ack messages, so news spreads like an infection, reaching all N members in about O(log N) periods without extra messages.

Cost per member per period is constant: one ping plus at most k indirect probes. Total traffic is O(N), not O(N²), and the expected time to first detect a failure does not grow with cluster size.



Trade-offs

- **Fixed timeouts** are simple and predictable but must be tuned per environment and cause either slow detection or false alarms.
- **Phi accrual** adapts to network conditions and gives a tunable confidence level, but needs history (warm-up) and can be fooled when the heartbeat pattern suddenly changes.
- **SWIM** scales to thousands of nodes with constant per-node load, at the cost of eventual (not instant) agreement on membership. A node that is itself overloaded (CPU starved) can wrongly accuse healthy peers. HashiCorp's **Lifeguard** extensions address this, and report about 50 times fewer false positives.
- **No detector is perfect.** Pair it with fencing and idempotency so a wrong guess costs a failover, not corrupt data.

Real-world examples

- **Apache Cassandra** and **Akka Cluster** use phi accrual detectors.
- **HashiCorp memberlist** (inside Consul, Serf and Nomad) implements SWIM with Lifeguard.
- **Raft and ZooKeeper** use simple heartbeats with timeouts, randomised in Raft's case.
- **Kubernetes** marks nodes unhealthy when their node Lease is not renewed in time, a heartbeat via a shared store.

In the interview

Q: Why not just use a 5-second heartbeat timeout?

A: One number cannot fit every network and load level. It is too slow in a quiet data centre and gives false alarms during congestion. Adaptive detectors like phi accrual set the bar from observed heartbeat timing.

Q: How does SWIM avoid $O(N^2)$ heartbeat traffic?

A: Each node probes just one random peer per period and spreads membership changes by piggybacking on those probes. Load per node stays constant and news reaches everyone in about log N rounds.

Q: Why does SWIM use indirect pings and a suspect state?

A: To avoid declaring a node dead because of one bad link or a short pause. Indirect pings test other paths, and the suspect state gives the node a chance to refute before it is removed.

TL;DR

You can only suspect failures, never prove them. Heartbeats with timeouts are simplest. Phi accrual turns lateness into a probability-based suspicion level that adapts to the network. SWIM probes one random peer per period, uses indirect probes and suspicion, and gossips results, giving constant per-node cost at large scale.

Sources

- Hayashibara, Défago, Yared, Katayama, "The phi accrual failure detector", SRDS 2004 (as implemented and documented in Akka): https://doc.akka.io/api/akka-core/2.9/akka/remote/PhiAccrualFailureDetector.html
- Apache Cassandra / DataStax docs, failure detection and `phi_convict_threshold`: https://docs.datastax.com/en/cassandra-oss/3.0/cassandra/architecture/archDataDistributeFailDetect.html
- Das, Gupta, Motivala, "SWIM: Scalable Weakly-consistent Infection-style Process Group Membership Protocol", DSN 2002: https://www.cs.cornell.edu/projects/Quicksilver/public_pdfs/SWIM.pdf
- HashiCorp, "Making Gossip More Robust with Lifeguard": https://www.hashicorp.com/blog/making-gossip-more-robust-with-lifeguard

Chain replication and CRAQ

What it is

Chain replication (Robbert van Renesse and Fred Schneider, OSDI 2004) arranges the replicas of a piece of data in a line called a **chain**. The first node is the **head**, the last is the **tail**. Writes enter at the head and flow down the chain; reads are answered by the tail. A separate, consensus-backed **configuration manager** (also called the master) decides who is in the chain.

CRAQ (Chain Replication with Apportioned Queries, Jeff Terrace and Michael Freedman, USENIX ATC 2009) keeps the same write path but lets **any** node in the chain answer reads, while still giving strongly consistent answers.

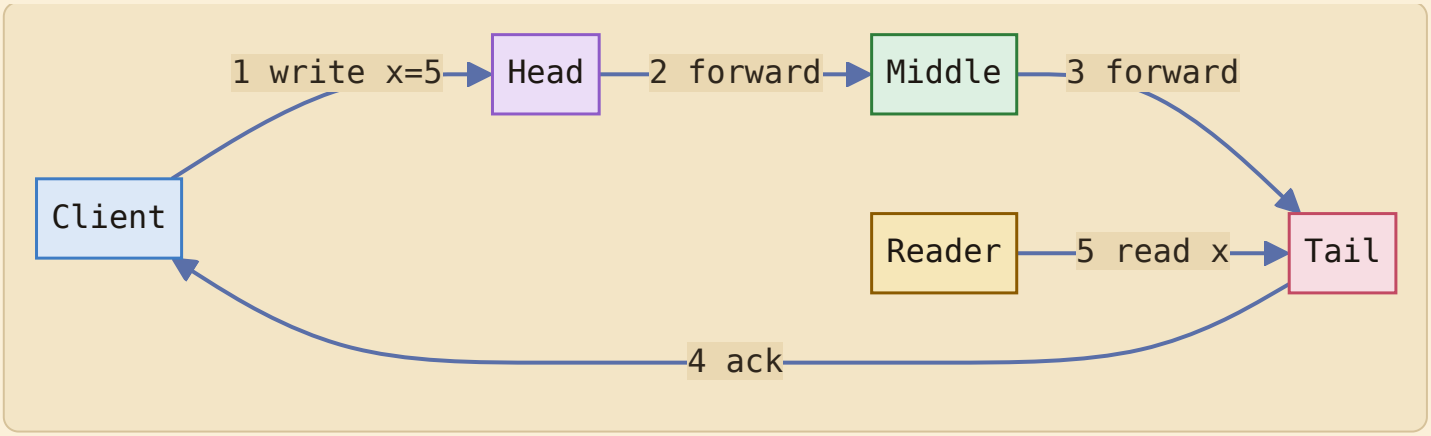
The problem it solves

Primary-backup replication makes one primary do everything: receive the write, send it to every backup, wait for acks, and serve reads. The primary's network card and CPU become the bottleneck. Chain replication spreads the work: each node sends each write to just one successor, and reads go to a different node (the tail) from writes (the head). It still offers **linearizability** (every read sees the latest completed write, as if there were one copy). CRAQ then solves chain replication's own weakness: the tail serving every read becomes a hot spot for read-heavy workloads.

How it works

Chain replication, step by step (chain H, M, T for head, middle, tail):

1. Client sends `put(x, 5)` to H.
2. H applies it (in its local log or store) and forwards it to M.
3. M applies it and forwards it to T.
4. T applies it. The write is now **committed**, because every replica has it. T replies to the client (or acks back up the chain so H can reply).
5. Reads `get(x)` go to T. Because T only has committed writes, every read returns committed data, and since there is one tail, reads and writes are totally ordered.



Failure handling (driven by the configuration manager):

- **Head fails**: its successor M becomes the new head. Writes that H had not yet forwarded are lost, but they were never acknowledged, so clients will retry.
- **Tail fails**: its predecessor M becomes the new tail. M already has every write T had, and possibly more, so nothing committed is lost.
- **Middle fails**: the manager links H directly to T. H re-sends to T every write it has forwarded but that is not yet acknowledged from downstream. Each node keeps a list of these **pending** writes until the tail's ack flows back.
- **Adding a node**: a new node copies state from the current tail, then is appended as the new tail.

Worked example with numbers. Each hop takes 0.5 ms inside a data centre. With 3 nodes, a write travels client to H, H to M, M to T, then T back to the client: 4 hops, roughly 2 ms. With primary-backup, the client sends to the primary, the primary sends to both backups in parallel, the acks come back and the primary replies: also 4 hops, about 2 ms, and it stays at 4 hops however many backups there are, while a 5-node chain needs 6. The difference is in the sending work: the primary sends 2 copies of every write while each chain node sends only 1. For 10,000 writes per second of 1 KB, the chain head sends 10 MB/s; a primary-backup leader with 4 backups would send 40 MB/s.

Complexity. Write latency grows with chain length: O(n) hops. Messages per write: n (one per link) plus acks. Storage: n full copies. A chain of n nodes tolerates n − 1 failures (the manager, not the chain, needs a majority).

CRAQ, step by step. Every node may store several **versions** of an object, each marked **clean** (known committed) or **dirty** (seen but not yet known committed).

1. A write travels head to tail as before. Each node stores the new version as **dirty**.
2. When the tail applies it, the version is committed. The tail sends an ack back up the chain.
3. As the ack passes, each node marks that version **clean** and deletes older versions.
4. A read can go to **any** node:
    - If the node's latest version is clean, it returns it immediately.
    - If it is dirty, the node asks the tail "what is the latest committed version number of x?" (a tiny **version query**, not a full read) and returns that version from its own store.

Worked example: x has version 7 (clean) everywhere. A write creates version 8. It has reached H and M (dirty) but not T. A reader at M sees a dirty latest version, asks T, learns "7", and returns version 7. One millisecond later T applies version 8, acks flow back, M marks 8 clean, and the next read at M returns 8 with no query. In a read-mostly workload most objects are clean most of the time, so read throughput scales with the number of nodes, not just one tail.

CRAQ can also relax to **eventual consistency** on request: return the newest local version, dirty or not, and skip the tail query.

Trade-offs

- **Higher write latency** than parallel fan-out: n sequential hops instead of 1 parallel round.
- **One slow node slows every write**, because every write passes through every node.
- **Needs an external configuration manager** (usually Paxos, Raft or ZooKeeper) to avoid split brain. Chain replication itself does not run consensus.

- **Strong consistency with simple logic**: no voting on each write, and the tail is a single point of truth for ordering.
- **CRAQ** fixes the read hot spot at the cost of multi-version storage and extra tail queries under write-heavy load.

Real-world examples

- **Hibari**, an open-source key-value store, uses chain replication with head, middle and tail "bricks".
- **FAWN-KV** (a research key-value store on low-power nodes) and CRAQ itself build on it.
- **HDFS** writes blocks through a **pipeline** of DataNodes, a chain-like pattern where each node forwards to the next.
- Course and industry write-ups cite chain-style replication in object stores and block storage; the pattern recurs wherever per-node bandwidth matters most.

In the interview

Q: Why are reads served by the tail in chain replication?

A: The tail is the only node guaranteed to hold just committed writes. Reading there gives linearizable reads without any coordination.

Q: How does CRAQ read from any node and stay consistent?

A: Nodes track whether each version is clean or dirty. Clean versions are returned directly. For dirty ones, the node asks the tail which version is committed and returns that one.

Q: When would you prefer quorum or Raft replication to a chain?

A: When write latency or tolerance of one slow replica matters more. A majority quorum commits after the fastest majority answers, while a chain waits for every node.

TL;DR

Chain replication pipes writes head to tail and serves reads from the tail, giving strong consistency and good throughput with simple logic, at the cost of latency proportional to chain length. CRAQ lets every node serve reads by tracking clean and dirty versions and asking the tail only when unsure.

Sources

- van Renesse, Schneider, "Chain Replication for Supporting High Throughput and Availability", OSDI 2004: https://www.usenix.org/conference/osdi-04/chain-replication-supporting-high-throughput-and-availability
- Terrace, Freedman, "Object Storage on CRAQ: High-Throughput Chain Replication for Read-Mostly Workloads", USENIX ATC 2009: https://www.usenix.org/conference/usenix-09/object-storage-craq-high-throughput-chain-replication-read-mostly-workloads
- Hibari documentation, architecture: https://hibari.readthedocs.io/en/latest/admin-guide/architecture.html
- Apache Hadoop HDFS architecture guide (replication pipelining): https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HdfsDesign.html

Hybrid logical clocks

What it is

A **hybrid logical clock (HLC)** gives every event a timestamp made of two parts: `l`, close to real **physical (wall-clock) time**, and `c`, a small **logical counter**. It was proposed by Sandeep Kulkarni, Murat Demirbas and colleagues in "Logical Physical Clocks" (OPODIS 2014).

Two older ideas it combines:

- **Physical timestamps** from each machine's clock. Easy to read and compare with human time, but machines disagree by milliseconds or more (**clock skew**), so a message can arrive "before" it was sent.
- **Lamport clocks**: plain counters bumped on every event and on every received message. They guarantee **causality** (if event e could have influenced f, e gets a smaller number), but the numbers have no

relation to real time.

The problem it solves

A distributed database wants timestamps that (1) never violate cause and effect, (2) fit in a normal 64-bit field, and (3) stay close enough to real time to answer "show me the data as of 10:00:00" or "snapshot at time t". Physical clocks fail (1). Lamport clocks fail (3). Vector clocks (see the version vectors section) satisfy (1) but grow with the number of nodes, failing (2). HLC satisfies all three.

How it works

Each node j keeps `l.j` (largest physical time it has heard of) and `c.j` (counter). `pt.j` is its own physical clock.

```
on local or send event:
    l_old = l
    l = max(l_old, pt)
    c = c + 1 if l == l_old else 0
    stamp event (and message) with (l, c)

on receive of message m with (l_m, c_m):
    l_old = l
    l = max(l_old, l_m, pt)
    if   l == l_old == l_m: c = max(c, c_m) + 1
    elif l == l_old:        c = c + 1
    elif l == l_m:          c = c_m + 1
    else:                   c = 0
    stamp event with (l, c)
```

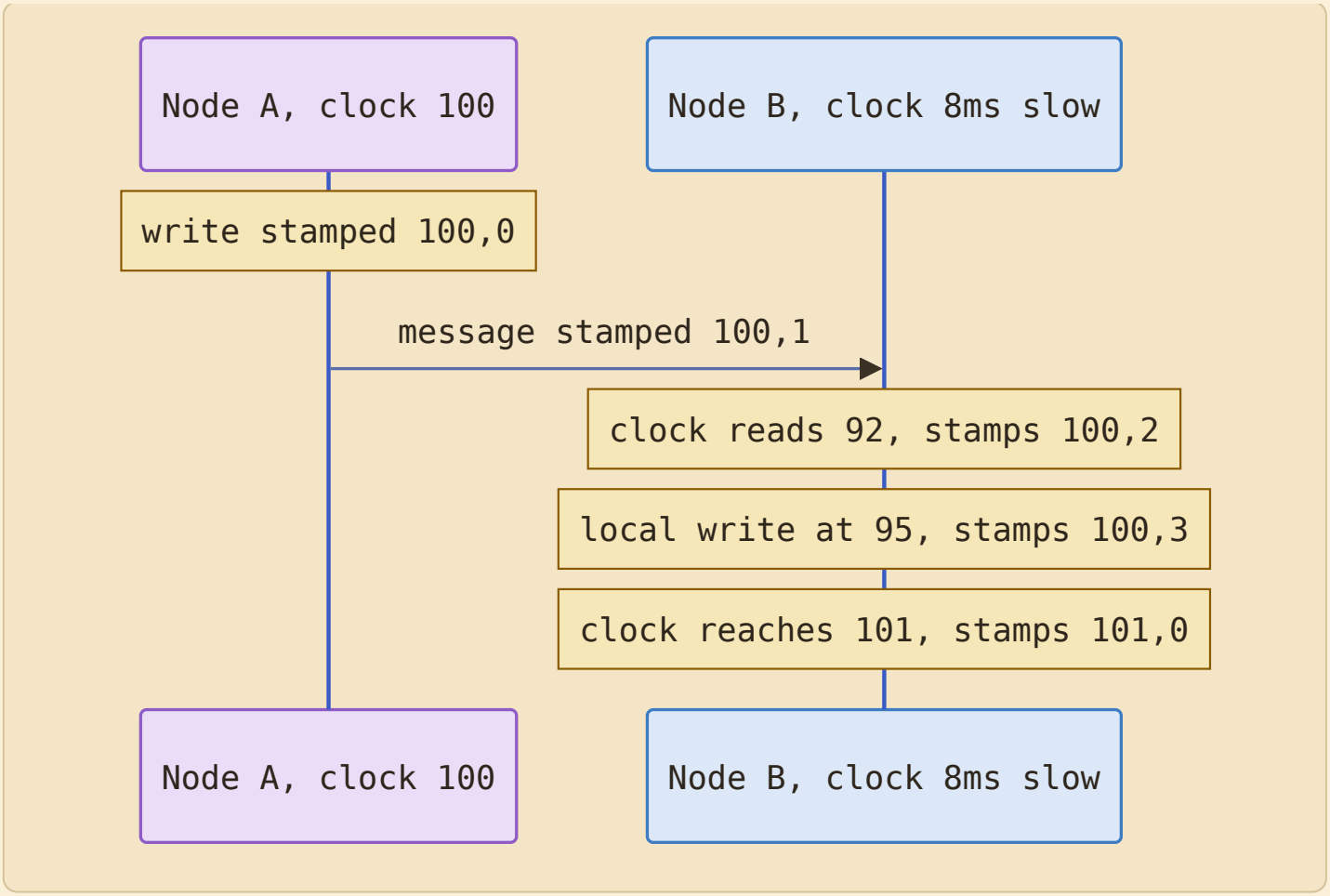
Compare timestamps by `l` first, then `c` (and node ID to break exact ties).

Worked example. Node A's clock reads 100 ms. Node B's clock is 8 ms slow. B's last stamp was (85, 0).

1. A writes a row at pt = 100: `l = max(0, 100) = 100`, `c = 0`. Stamp **(100, 0)**.
2. Still at pt = 100, A sends a message to B: `l` stays 100, so `c = 1`. Stamp **(100, 1)**.
3. B receives it when its clock reads 92: `l = max(85, 100, 92) = 100`, which equals the message's `l`, so `c = 1 + 1 = 2`. Stamp **(100, 2)**. With raw physical time, B would have stamped 92, *earlier* than the send at 100: causality broken. HLC keeps it in order.
4. B does a local write at pt = 95: `l = max(100, 95) = 100`, unchanged, so `c = 3`. Stamp **(100, 3)**.
5. B's clock reaches 101: `l = 101`, `c` resets to 0. Stamp **(101, 0)**. Once physical time passes the largest time heard, the counter goes back to zero.

Properties.

- If e happened before f, then HLC(e) < HLC(f). Same guarantee as a Lamport clock.
- `l` is never more than the clock skew ahead of a node's own physical time, so HLC stays within the skew bound of real time.
- `c` stays small (bounded by the number of nodes in theory, and tiny in practice). The paper packs it into 64 bits: 48 bits for `l` (the NTP-format time rounded) and 16 bits for `c`.
- O(1) time per event, O(1) space per timestamp, compared with O(N) for a vector clock.



What HLC does not do. It does not make clocks *accurate*. Two unrelated writes on different nodes 1 ms apart in real time can still be stamped in the "wrong" order if their clocks disagree by more than 1 ms. Systems that need "a transaction that starts after another commits sees it" (external consistency) add a rule on top, such as CockroachDB's **uncertainty interval**.

Trade-offs

- **Pros**: causally consistent, close to wall time, fixed 64-bit size, cheap. Great for MVCC (multi-version concurrency control) timestamps, snapshot reads and change-data-capture ordering.
- **Cons**: relies on clocks staying within a configured maximum skew. If a node's clock jumps far ahead, everyone's `l` follows it, so systems must reject or fence nodes with bad clocks.
- **Cannot detect concurrency**: unlike vector clocks, two HLC timestamps are always ordered, even for truly concurrent events, so HLC is not a conflict detector.

Real-world examples

- **CockroachDB** stamps transactions with HLC. Its `--max-offset` flag (default 500 ms) is the maximum allowed clock offset. A read treats any value with a timestamp within that offset above its own as **uncertain** and restarts at a higher timestamp. Nodes whose clocks drift too far shut themselves down.
- **MongoDB** uses a hybrid logical clock for **cluster time**, which powers causally consistent sessions across replica sets and shards.
- **YugabyteDB** uses HLC for its distributed transactions.

In the interview

Q: Why not just use NTP timestamps for ordering?

A: Clocks disagree by milliseconds or more and can jump backwards, so a reply can appear to happen before the request. HLC never goes backwards and always orders a received message after its send.

Q: HLC vs vector clock?

A: HLC is a fixed-size scalar that respects causality and stays near real time, but it cannot tell you two events were concurrent. Vector clocks detect concurrency but grow with the number of nodes.

Q: What does CockroachDB do about clock skew with HLC?

A: It sets a maximum offset. Reads that see values slightly newer than their timestamp, within that offset, cannot be sure of the order, so they restart at a later timestamp. Nodes whose clocks drift too far stop serving.

TL;DR

HLC = max physical time heard so far plus a small counter for ties. It respects cause and effect like a Lamport clock, stays within clock skew of real time, and fits in 64 bits. CockroachDB, MongoDB and YugabyteDB use it for transaction and snapshot timestamps.

Sources

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Google TrueTime and commit-wait

What it is

TrueTime is the clock service inside Google **Spanner**, a globally distributed SQL database (Corbett et al., OSDI 2012). Instead of returning a single time, `TT.now()` returns an **interval** `[earliest, latest]` that is guaranteed to contain the true absolute time. Half the interval's width is called **epsilon (ε)**, the uncertainty.

Commit-wait is the rule that uses it: after choosing a commit timestamp, the transaction waits until that timestamp is definitely in the past before making its writes visible.

The problem it solves

Spanner promises **external consistency** (the strongest form of linearizability for transactions): if transaction T1 commits before T2 starts, in real time, then T2's timestamp is larger, so T2 sees T1's writes. With ordinary clocks that can be off by milliseconds, a later transaction on a slow-clock server could receive a *smaller* timestamp than an earlier one on a fast-clock server, and a snapshot read could miss a write that, in the real world, had already finished. TrueTime makes clock error **explicit and bounded**, and commit-wait turns that bound into a guarantee.

How it works

Building the interval.

1. Each data centre has several **time masters**. Most have GPS receivers. The rest, called **Armageddon masters**, have atomic clocks. The two kinds fail in different ways.
2. A daemon on every machine polls several masters (every 30 seconds in the paper), rejects outliers with a variant of Marzullo's algorithm, and syncs the local clock.
3. Between polls, it assumes the local clock might drift at a worst-case rate (200 microseconds per second in the paper). After 30 s, that is up to 6 ms of possible drift, plus about 1 ms for network delay to the masters.
4. So ε follows a **sawtooth**: about 1 ms right after a poll, growing to about 7 ms just before the next. The paper reports ε is about 4 ms on average.

The API:

- `TT.now()` returns `[earliest, latest]`.

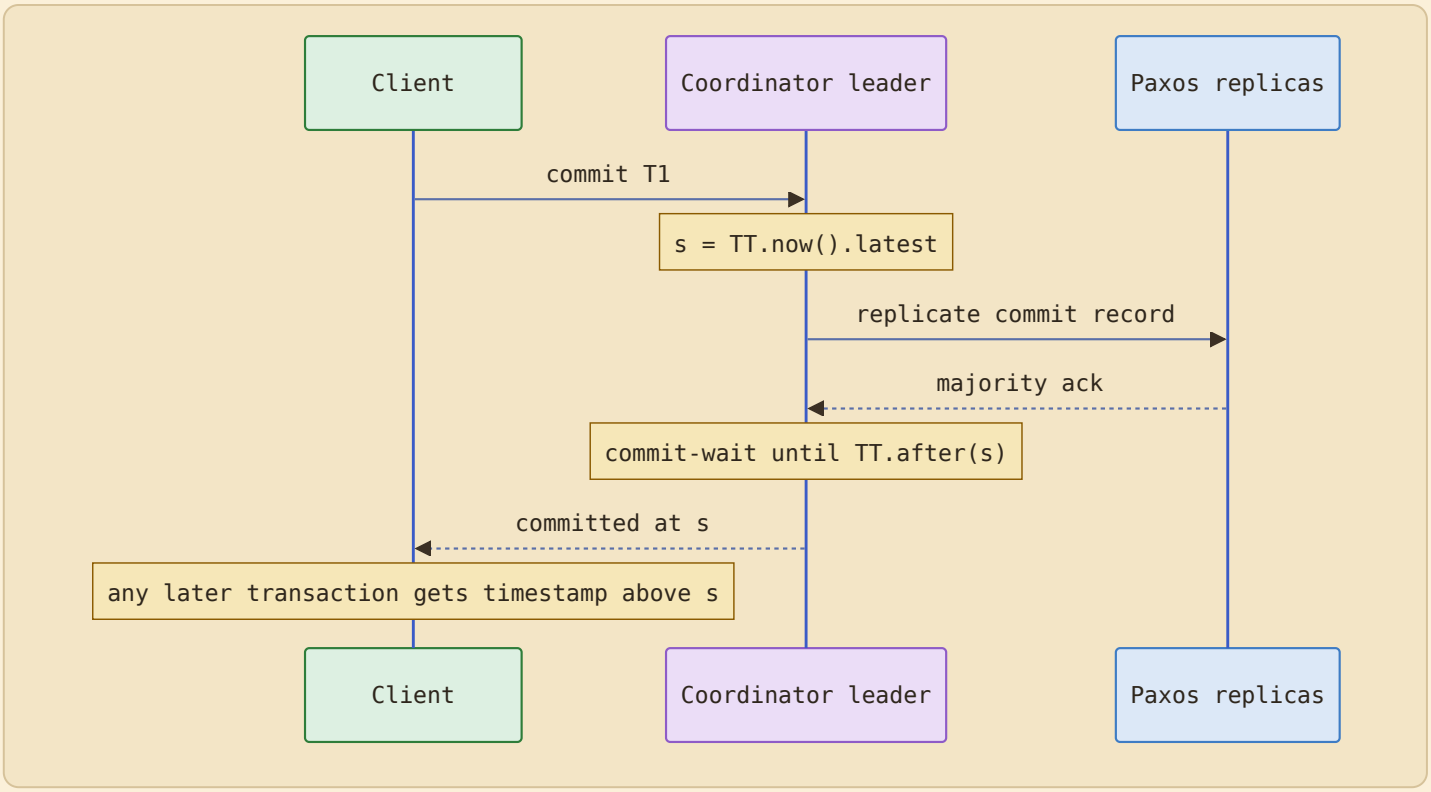
- `TT.after(t)` is true if t has definitely passed (`t < now.earliest`).
- `TT.before(t)` is true if t has definitely not arrived (`t > now.latest`).

Commit-wait, step by step.

1. A read-write transaction acquires its locks (Spanner uses two-phase locking).
2. The coordinator leader picks commit timestamp `s = TT.now().latest`. Because `latest` is at or after the true time, `s` is not in the past.
3. It replicates the commit record through Paxos (and runs two-phase commit if several shards are involved).
4. **Commit-wait:** it waits until `TT.after(s)` is true, so the true time is definitely later than `s`.
5. Only then does it release locks and tell the client "committed".

Worked example with ε = 4 ms.

- At true time 1000.000 s, `TT.now()` returns `[999.996, 1000.004]`. The coordinator picks s = 1000.004.
- It must wait until `earliest > 1000.004`, which happens at true time about 1000.008. That is about 2ε = 8 ms after it started.
- Paxos replication to other replicas runs in parallel during this wait (often longer than 8 ms across zones), so in practice much of the wait is hidden. The paper measured commit-wait at about 5 ms in a single-replica test.
- Any transaction T2 that starts after the client hears "committed" begins after true time 1000.008. Its timestamp is at least its own `TT.now().latest`, which is at least true time, so it is greater than 1000.004. T2 is guaranteed to be ordered after T1.



Why it pays off. Every committed write has a timestamp that matches real-time order, so a **read-only transaction** or **snapshot read** at timestamp t needs no locks: any replica that has caught up to t can answer it. Spanner leaders also use TrueTime to keep their **leader leases** (10 s by default) from overlapping.

Cost. Commit-wait adds roughly 2ε to write latency, O(1) work. Smaller ε means faster commits, which is why Google invests in GPS and atomic clock hardware.

Trade-offs

- **Write latency** includes the uncertainty. If time masters are unreachable, ε grows and so does every commit.

- **Specialised infrastructure**: GPS antennas and atomic clocks in every data centre, plus careful monitoring. Machines with clocks behaving worse than the assumed bound are evicted.
- **Lock-free consistent reads** anywhere in the world, at any past timestamp, are the big reward.
- **Alternatives without special hardware**: CockroachDB uses HLC with a larger maximum offset and restarts uncertain reads instead of waiting on every write. Cloud providers now expose clock error bounds too, for example AWS ClockBound returns an `[earliest, latest]` window much like TrueTime.

Real-world examples

- **Google Spanner** and Cloud Spanner (TrueTime plus commit-wait). Google's documentation notes that when uncertainty is smaller than the Paxos round trip, commit-wait does not add to commit latency.
- **CockroachDB** has a similar but opt-in "commit wait" for global tables, built on HLC and a configured clock offset instead of TrueTime.
- **AWS ClockBound** with Amazon Time Sync Service gives EC2 applications a bounded-uncertainty clock API.

In the interview

Q: What is commit-wait and why is it needed?

A: After picking a commit timestamp at the top of the uncertainty interval, the leader waits until that timestamp is certainly past before acknowledging. Then any transaction that starts afterwards must get a larger timestamp, giving external consistency.

Q: How long is the wait?

A: About twice the uncertainty, a few milliseconds in Spanner's environment (ε was 1 to 7 ms, about 4 ms on average, in the 2012 paper). It overlaps with Paxos replication, so the added latency is often small.

Q: Can I build this on AWS or with NTP?

A: Yes, if you have a trustworthy error bound. With NTP the bound is larger, so waits would be longer. That is why most open-source databases use HLC plus uncertainty restarts instead of waiting on every write.

TL;DR

TrueTime returns a time interval guaranteed to contain the true time, kept narrow (a few ms) by GPS and atomic clocks. Spanner stamps commits at the top of the interval and waits until that time has definitely passed. This makes timestamps follow real-time order, enabling globally consistent, lock-free snapshot reads.

Sources

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- AWS EC2 User Guide, "Compare timestamps for your Linux instances" (ClockBound): https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/compare-timestamps-with-clockbound.html
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Version vectors vs vector clocks (and dotted version vectors)

What it is

All three are lists of counters, one per participant, used to decide whether one thing **happened before** another or whether they are **concurrent** (neither knew about the other).

- A **vector clock** (Fidge and Mattern, 1988) tracks causality between **events** in processes. Every process keeps one counter per process and updates it on every local event, send and receive.
- A **version vector** (Parker and colleagues, 1983) tracks causality between **versions of data** on replicas. There is one counter per **replica** (the server that coordinates a write), and it changes only when data

is written or merged.

- A **dotted version vector (DVV)** (Preguiça, Baquero, Almeida and colleagues, 2010) is a version vector plus a **dot**: one `(replica, counter)` pair that names the exact write that created a value. It fixes false conflicts that plain version vectors create when many clients write through the same server.

Module 07 (*Vector Clocks*) explains how a leaderless store uses these labels to detect conflicting writes. Here we pin down the differences and the DVV improvement.

The problem it solves

In a leaderless or multi-leader database (Dynamo-style), two clients can update the same key on different replicas at the same time. The database must decide: is version B a newer edit of version A (keep only B), or were they made independently (keep both as **siblings** and let the application merge)? Timestamps cannot answer this (Module 00, *Clocks and ordering*). Counter vectors can.

How it works

Comparing any two vectors V and W (same for all three):

1. V ≤ W if every entry of V is less than or equal to the matching entry in W.
2. If V ≤ W and V ≠ W, then V happened before W, so W supersedes V.
3. If neither V ≤ W nor W ≤ V, they are **concurrent**: a conflict.

Example: `{A:2, B:1}` vs `{A:3, B:1}`: the first is older. `{A:2, B:1}` vs `{A:1, B:2}`: concurrent.

Vector clocks vs version vectors.

	Vector clock	Version vector
Orders	Events (messages, computations)	Versions of a data item
One entry per	Process	Replica that coordinates writes
Updated on	Every event, send and receive	Writes and merges only
Size	Grows with processes involved	Grows with replicas (usually 3 to 5)
Typical use	Debugging, causal message delivery	Conflict detection in replicated stores

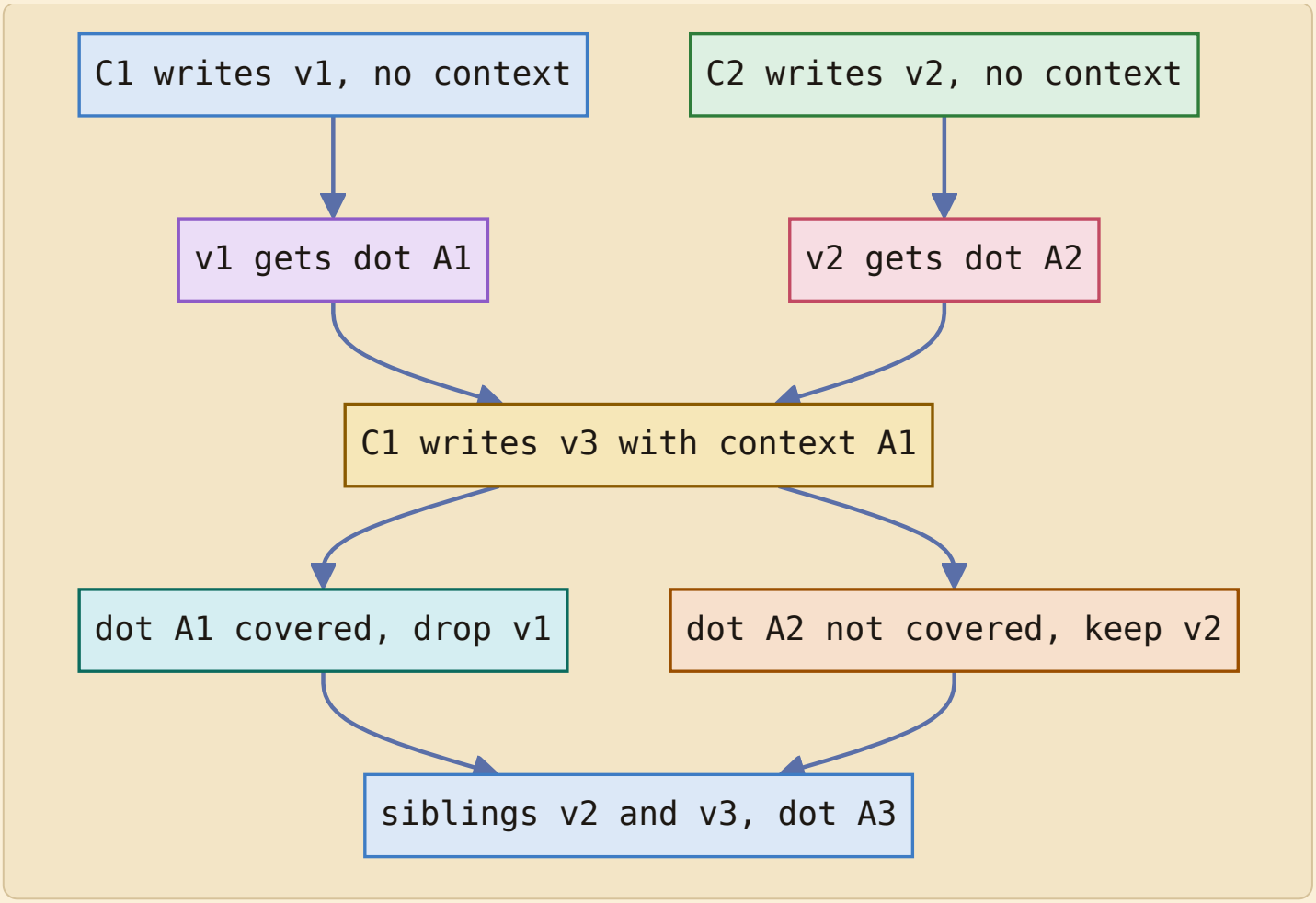
People often say "vector clock" for both, as the Dynamo paper does. The distinction matters because using **client IDs** as entries makes vectors grow with the number of clients (millions), while using **server IDs** keeps them small but loses information, which is the problem DVVs solve.

The sibling explosion problem, step by step. One replica, A, coordinates all writes to key `cart`. Version vectors use server IDs.

1. Client C1 writes `v1` with no context (it never read the key). A stores v1 with `{A:1}`.
2. Client C2, also without reading, writes `v2`. A cannot tell if v2 replaces v1, so it keeps both as siblings and bumps the vector to `{A:2}`. Both siblings now share one vector, `{A:2}`.
3. C1 now writes `v3`, sending the context it got back from step 1, `{A:1}`. C1 had seen only v1, so v3 should replace v1 but stay concurrent with v2.
4. A's vector `{A:2}` is newer than the context `{A:1}`, but A cannot tell *which* sibling the context covers, because they share one vector. To be safe it keeps v1, v2 and v3: three siblings. With retries and many clients, siblings pile up without bound. This is **sibling explosion**, and it comes from **false concurrency**.

The same steps with dotted version vectors.

1. v1 gets dot `(A,1)`. Store: `{v1@(A,1)}`, context returned `{A:1}`.
2. v2 (empty context) gets dot `(A,2)`. Store: `{v1@(A,1), v2@(A,2)}`.
3. v3 arrives with context `{A:1}`. The context **covers** dot (A,1), so v1 is discarded. It does not cover (A,2), so v2 stays. v3 gets dot `(A,3)`.
4. Store: `{v2@(A,2), v3@(A,3)}`. Exactly the two truly concurrent values. Correct.



Pseudo-code for a DVV-style write at replica r:

```
on write(value, context):           # context is a version vector
    siblings = [s for s in siblings if not covered(s.dot, context)]
    vv[r] = vv[r] + 1
    siblings.append(Sibling(value, dot=(r, vv[r])))
covered(dot, ctx): return dot.counter <= ctx.get(dot.replica, 0)
```

Complexity. Comparison is O(R) for R entries. A version vector costs O(R) space per key (R is the number of replicas, small). A DVV adds one dot per sibling. Vector clocks over processes cost O(P) for P processes, which can be large.

Trade-offs

- **Vector clocks** capture full causality between events, but grow with the number of processes, so they suit debugging and messaging more than storage.
- **Version vectors** keyed by replicas stay small but, without dots, create false conflicts when several clients write through one replica.
- **Client-keyed vectors** are precise but can grow without bound; Dynamo's paper truncated vectors once they reached a threshold (it gives 10 pairs), risking incorrect reconciliation.
- **DVVs** keep the small server-keyed size and remove false concurrency, at the cost of slightly more complex code.
- **All of them only detect conflicts.** The application or a **CRDT** (a conflict-free replicated data type, a structure that merges automatically) must resolve them. Many systems choose **last-write-wins** instead, which is simpler but silently drops concurrent writes.

Real-world examples

- **Amazon Dynamo** (2007) attached vector clocks of `(node, counter)` pairs to each object and returned siblings to the client for merging.

- **Riak** 2.0 added dotted version vectors as an option and recommends them to keep sibling counts bounded.
- **Project Voldemort** (LinkedIn) used vector clocks in a Dynamo-style design.
- **Apache Cassandra** deliberately avoids them and uses per-column last-write-wins timestamps.

In the interview

Q: What is the difference between a vector clock and a version vector?

A: A vector clock orders events between processes and ticks on every event and message. A version vector orders versions of data across replicas and ticks only on writes. Same comparison rule, different purpose and size.

Q: Why did Riak move to dotted version vectors?

A: With one counter per server, several clients writing through the same server made all their values share one vector, so the server couldn't tell which value a new write replaced. Siblings piled up. A dot names each value's exact write, so a new write's context can discard exactly the values it saw.

Q: Two versions have vectors `{A:3, B:1}` and `{A:2, B:2}`. What do you do?

A: Neither dominates, so they are concurrent. Keep both as siblings and merge them on read, with application logic, a CRDT, or last-write-wins if losing one is acceptable.

TL;DR

Vector clocks order events between processes; version vectors order data versions across replicas; both flag concurrency when neither vector dominates. Dotted version vectors tag each value with the exact write that created it, avoiding false conflicts and sibling explosion while keeping vectors small.

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- Riak engineering blog, "Vector Clocks Revisited Part 2: Dotted Version Vectors": https://riak.com/posts/technical/vector-clocks-revisited-part-2-dotted-version-vectors/

Consistent hashing variants: jump, rendezvous, Maglev and bounded loads

What it is

Module 0 introduced the **hash ring with virtual nodes**: servers and keys are hashed onto a circle, each key belongs to the next server clockwise, and each server owns many small arcs. That is one answer to a general question: *how do I map a key to one of N servers so that changing N moves as few keys as possible?* Any method that answers it is a **consistent hashing** scheme. This section covers four other well-known answers:

- **Jump consistent hash** — a few lines of arithmetic, no memory at all, for servers numbered 0 to N−1.
- **Rendezvous hashing**, also called **highest random weight (HRW)** — every server "bids" for the key and the highest bid wins.
- **Maglev hashing** — Google's load-balancer method, which precomputes a fixed-size lookup table so each lookup is a single array read.
- **Consistent hashing with bounded loads** — a ring (or similar) plus a hard cap so no server gets more than, say, 25% above the average.

Terms used below: a **key** is whatever you route on (user ID, URL, connection 5-tuple). A **bucket**, **backend** or **server** is the destination. **Disruption** or **movement** is the share of keys that change destination when the server set changes. The theoretical minimum, when going from N to N+1 servers, is 1/(N+1): the new server must take its fair share and nothing else should move.

The problem it solves

The ring works, but it has costs that matter at scale:

1. **Memory and lookup time.** With 1,000 servers and 200 virtual nodes each, the ring holds 200,000 points, and every lookup is a binary search over them.
2. **Imperfect balance.** Even with 100–200 virtual nodes, the biggest server typically carries several percent more than the smallest. More vnodes means more memory.
3. **Hot keys and uneven load.** The ring balances *keys*, not *load*. If one key is very popular, or keys are unevenly costly, a server can be overloaded no matter how evenly the arcs are spread.

Each variant attacks a different one of these.

How it works

1. Jump consistent hash (Lamping and Veach, Google, 2014).

The idea: imagine growing the cluster one bucket at a time, from 1 bucket to N. When bucket number b is added (going from b to b+1 buckets), each key should move to the new bucket with probability exactly 1/(b+1), and otherwise stay put. If you simulated that bucket by bucket it would cost O(N). The trick is a pseudo-random generator seeded by the key that lets you *jump* straight to the next bucket count at which this key would move, skipping all the ones where it stays. The expected number of jumps is about ln N, so for N = 1,000 it is about 7 steps.

```
function jump_hash(key, num_buckets):
    b = -1            # last bucket the key moved to
    j = 0             # next bucket count at which it might move
    while j < num_buckets:
        b = j
        key = key * 2862933555777941757 + 1          # 64-bit LCG step
        j = floor((b + 1) * 2^31 / ((key >> 33) + 1))
    return b
```

Worked example (computed with the published constant). Key 99999 maps to bucket 0 with 1 bucket, then bucket 1 for every cluster size from 2 to 7, and then jumps to bucket 7 when the 8th bucket appears, staying there for sizes 9 and 10. So this key moves exactly twice as the cluster grows from 1 to 10, and each time it moves *into the newly added bucket*, never sideways between old buckets. Running 100,000 keys from 10 to 11 buckets moved 9,042 of them (9.04%), almost exactly the ideal 1/11 = 9.09%.

Cost: O(ln N) time, O(1) memory, essentially perfect balance. The catch: buckets must be numbered 0..N−1 and you can only add or remove at the **end**. If bucket 3 of 10 dies, you cannot simply delete "3". The usual fix is to put jump hash on top of a layer that maps bucket numbers to replicas (bucket 3 is a replica group whose members can change), which is why the paper aims it at sharded storage rather than at caches where any node can vanish.

2. Rendezvous hashing / HRW (Thaler and Ravishankar, 1996–1998).

For each key, compute a score for every server, `score = hash(key, server)`, and send the key to the server with the highest score.

Worked example. Key `photo:7`, three servers, pretend hash outputs scaled to 0..1:

Server	hash("photo:7", server)
S1	0.62
S2	**0.91**
S3	0.17

The key goes to S2. If S2 dies, recompute among the survivors: S1 (0.62) wins. Keys whose top scorer was S1 or S3 are untouched, because their scores did not change. So exactly the dead server's keys move, and they spread evenly over all survivors (each key's runner-up is random). Adding S4 steals only the keys for which S4 scores highest, about 1/4 of them.

- **Replication for free:** the top 3 scores give 3 replicas, and the order is stable.
- **Weights:** a common weighted form is `score = -w / ln(h)` where h is the hash scaled to (0,1) and w is the server weight, so a server with weight 2 wins about twice as often.

- **Cost:** O(N) hash computations per lookup. For 10 servers that is nothing. For 10,000 it is too slow, and people use a tree of rendezvous choices (pick a group, then a server inside it).

3. Maglev hashing (Google, NSDI 2016).

Maglev is Google's software network load balancer, serving Google traffic since 2008. Each Maglev machine must send every packet of a TCP connection to the same backend, at line rate, so lookup must be a single array read. It builds a lookup table of prime size M (65,537 by default in the paper) where each slot holds a backend.

Steps:

1. For each backend i, hash its name twice: `offset = h1(name) mod M`, `skip = h2(name) mod (M-1) + 1`.
2. Its **preference list** is `permutation[i][j] = (offset + j * skip) mod M` for j = 0, 1, 2, ... Because M is prime, this visits every slot exactly once.
3. Fill the table in rounds: each backend in turn claims its next preferred slot that is still empty. Repeat until all M slots are full.
4. Lookup: `backend = table[hash(packet 5-tuple) mod M]`.

Worked example with M = 7 and three backends:

Backend	(offset, skip)	Preference list
B0	(3, 4)	3, 0, 4, 1, 5, 2, 6
B1	(0, 2)	0, 2, 4, 6, 1, 3, 5
B2	(3, 1)	3, 4, 5, 6, 0, 1, 2

Filling: B0 takes 3. B1 takes 0. B2 wants 3 (taken), takes 4. B0 wants 0, 4 (taken), takes 1. B1 takes 2. B2 takes 5. B0 wants 5, 2 (taken), takes 6. Table: `[B1, B0, B1, B0, B2, B2, B0]` — 3, 2, 2 slots, as even as 7 slots allow.

Now remove B1 and refill with only B0 and B2: `[B0, B0, B0, B0, B2, B2, B2]`. B1's two slots (0 and 2) moved, which is required. Slot 6 also moved from B0 to B2, which was not strictly required. That is Maglev's deliberate trade: **near-perfect balance and O(1) lookup in exchange for a little extra disruption**. Maglev hides the extra disruption with a per-machine connection-tracking table, so existing flows keep their backend. The paper measured that, with a 65,537-entry table, Karger-style ring hashing and rendezvous hashing would need 29.7% and 49.5% over-provisioning to absorb their imbalance, while Maglev was almost perfectly even. Envoy's documentation reports the flip side: when hosts are removed, Maglev moves about twice as many keys as its ring hash.

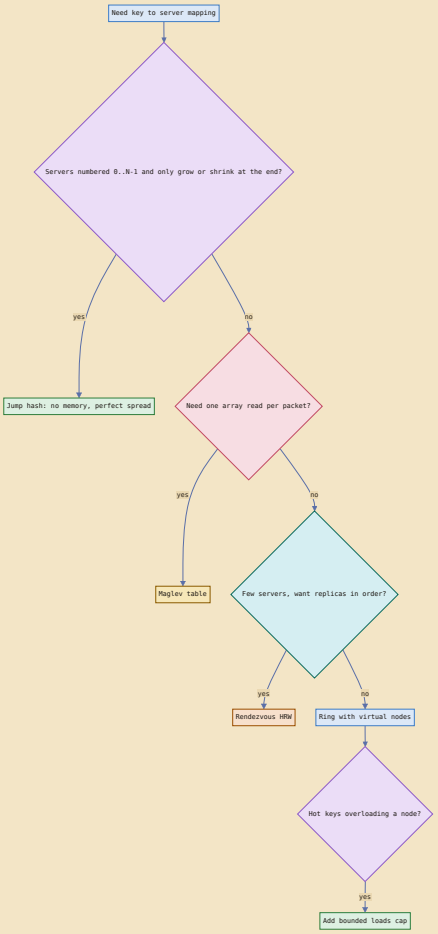
4. Consistent hashing with bounded loads (Mirrokni, Thorup, Zadimoghaddam, Google, 2016).

Pick a balance factor c = 1 + ε, for example 1.25. With m active keys (or requests) and n servers, no server may hold more than `ceil(c * m / n)`. Lookup:

1. Hash the key onto the ring as usual and find the first server clockwise.
2. If that server is below its cap, use it.
3. Otherwise keep walking clockwise to the next server with spare capacity.

Worked example: 12 active keys, 4 servers, ε = 0.25. Cap = ceil(1.25 × 12 / 4) = ceil(3.75) = 4. Suppose the plain ring sends 6 keys to A, 3 to B, 2 to C and 1 to D. A keeps its first 4, and the other 2 walk on to the next servers clockwise that are under 4. Final loads might be 4, 4, 3, 1 instead of 6, 3, 2, 1: the hottest server went from 2× the average (3) down to 1.33×.

The paper proves that the cap holds always, and that the number of keys moved per change stays small: about $O(1/\varepsilon^2)$ times the ideal for small ε. Vimeo implemented it in HAProxy for its video cache tier and reported cutting cache bandwidth by almost 8×; HAProxy exposes it as `hash-balance-factor` (a percentage such as 150 for 1.5× the average), and Google Cloud Pub/Sub used it internally.



Comparison against the ring with virtual nodes (N servers, V vnodes each, table size M):

Property	Ring + vnodes	Jump hash	Rendezvous (HRW)	Maglev	Bounded loads
Lookup time	O(log(N·V)) binary search	O(ln N)	O(N)	O(1) array read	Ring lookup plus a short walk
Memory	O(N·V) points	O(1)	O(N) server list	O(M), e.g. 65,537 slots	Ring plus live load counts
Keys moved on change	≈1/N	Exactly the minimum	Exactly the minimum	Minimum plus a little	More than the ring, bounded by ε
Balance	Good with many vnodes	Perfect	Good	Near perfect	Hard cap of (1+ε)× average
Remove an arbitrary server	Yes	No, only the last	Yes	Yes	Yes
Weights	More vnodes	Not natively	Weighted score	More table slots	Weighted caps
Typical home	Dynamo, Cassandra, memcached clients	Sharded storage, Guava library	Small caches, CDNs, replica placement	L4 load balancers, Envoy	Cache tiers with hot keys, HAProxy

Trade-offs

- **Jump** is the fastest and smallest but cannot drop a server from the middle, so it needs a stable numbering layer.
- **Rendezvous** is the simplest to reason about and gives ordered replicas for free, but lookup is linear in the number of servers.

- **Maglev** buys O(1) lookups and excellent balance with a fixed memory cost per service and slightly more churn. Rebuilding a bigger table also takes longer (the paper measured 1.8 ms at 65,537 entries versus 22.9 ms at 655,373).
- **Bounded loads** is the only one that looks at *actual load*, so it protects against hot keys, but it needs shared, current load numbers and gives up some stickiness: an overflowing key may land somewhere different from one moment to the next.

Real-world examples

- **Google Maglev** and **Envoy's Maglev load balancer** (default table size 65,537) for connection-level balancing.
- **Guava's** `Hashing.consistentHash` implements jump hash.
- **Rendezvous hashing** appears in caching proxies and in many object-placement schemes where replica order matters.
- **Vimeo with HAProxy** and **Google Cloud Pub/Sub** used consistent hashing with bounded loads.

In the interview

Q: You are building an L4 load balancer that must handle millions of packets per second. Ring hash or Maglev?

A: Maglev. Each packet becomes one array read into a precomputed table, and the table splits traffic almost perfectly evenly. The small extra key movement on backend changes is covered by a connection-tracking table that pins existing flows.

Q: Why not use jump hash for a memcached fleet?

A: Any cache node can die, and jump hash can only remove the highest-numbered bucket. Removing node 3 of 10 would renumber and remap most keys unless you add an indirection layer. A ring or rendezvous handles arbitrary removal directly.

Q: One celebrity key is overloading its cache node. Does adding virtual nodes help?

A: No. Virtual nodes balance how many keys each node owns, not how hot they are. Bounded loads caps each node at (1+ε) × average load and spills overflow clockwise. For a single extremely hot key you also replicate it or put a local cache in front.

Q: How does rendezvous hashing give you replicas?

A: Sort the servers by score for that key and take the top R. When one dies, the next in the list steps up, and every client computes the same order without coordination.

TL;DR

All consistent hashing schemes keep key movement near the 1/N minimum. Jump hash uses no memory but can only add or remove the last bucket. Rendezvous is simple, gives ordered replicas, and costs O(N) per lookup. Maglev answers each lookup with one array read and nearly perfect balance, at the price of slightly more churn. Bounded loads puts a hard cap on each server's load to absorb hot keys.

Sources

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HyperLogLog internals

What it is

HyperLogLog (HLL) is a **sketch**, a small fixed-size summary of a big stream, that estimates **cardinality**: the number of *distinct* items seen. It was published by Flajolet, Fusy, Gandouet and Meunier in 2007. With m small counters called **registers** its typical relative error is 1.04/√m. Redis uses m = 16,384 registers of 6 bits each, so about 12 KB answers "how many unique visitors?" for billions of visitors with about 0.81% error.

The problem it solves

Counting distinct items exactly means remembering every item you have seen, for example in a hash set. 100 million unique user IDs at 8 bytes each is 800 MB per counter, and you may want one counter per page per day. HLL replaces that with a few kilobytes, and two HLLs can be **merged** to get the distinct count of the union, which a plain count cannot do (you cannot add two distinct counts because users overlap).

How it works

The core intuition. Hash each item to a random-looking bit string. In a random bit string, a run of 1 leading zero happens half the time, 2 leading zeros a quarter of the time, k leading zeros about 1 in 2^k. If the longest run of leading zeros you have ever seen is 10, you have probably seen around 2^10 ≈ 1,000 distinct hashes. Duplicates do not matter, because the same item always gives the same hash. One counter is very noisy (one lucky hash ruins it), so HLL keeps many and averages them.

Steps (precision p, m = 2^p registers):

1. Hash the item to 64 bits.
2. Use the first p bits as the **register index** (which of the m registers this item belongs to).
3. In the remaining bits, find ρ ("rho"), the position of the first 1-bit, which is the number of leading zeros plus 1.
4. Set `register[index] = max(register[index], ρ)`. Registers only ever grow.
5. To estimate, combine the registers with a **harmonic mean**: `E = α_m · m² / Σ 2^(−register[j])`. α_m is a bias-correction constant, about 0.7213/(1 + 1.079/m) for m ≥ 128, and 0.673 for m = 16.
6. **Small-range correction:** if E ≤ 2.5·m and some registers are still 0, use **linear counting** instead: `m · ln(m / V)` where V is the number of zero registers. This is far more accurate when most registers are still empty.

Worked example 1 (update). p = 2, so 4 registers. An item hashes to `10 0001 1...`. The first two bits `10` select register 2. The rest starts with `0001`, three zeros then a 1, so ρ = 4. If register 2 held 2, it becomes max(2, 4) = 4. Another item hashing to `10 1...` has ρ = 1 and leaves register 2 at 4.

Worked example 2 (estimate). m = 16 registers with values `3,2,4,1,2,5,3,2,1,3,2,4,2,3,1,2`. Then Σ 2^(−R) = 3.656, and E = 0.673 × 16² / 3.656 ≈ 47. Since 47 > 2.5 × 16 = 40, the raw estimate stands: about 47 distinct items. With m = 16 the error is 1.04/4 = 26%, which is why real systems use thousands of registers.

Worked example 3 (small range). Registers `3,0,2,0,1,0,0,2,0,1,0,0,0,1,0,0`: the raw estimate is 14.2, below 40, and V = 10 registers are zero, so linear counting gives 16 × ln(16/10) ≈ 7.5, about 8 items.

Why a harmonic mean? Take four registers `2, 2, 2, 20`. The fourth saw one freak hash. The arithmetic mean of 2^R is (4 + 4 + 4 + 1,048,576)/4 ≈ 262,000, which is ruined. The harmonic mean is 4 / (0.25 + 0.25 + 0.25 + 2^(−20)) ≈ 5.3, which barely notices the outlier. The harmonic mean is dominated by the small values, which is what you want when the noise is occasional huge outliers.

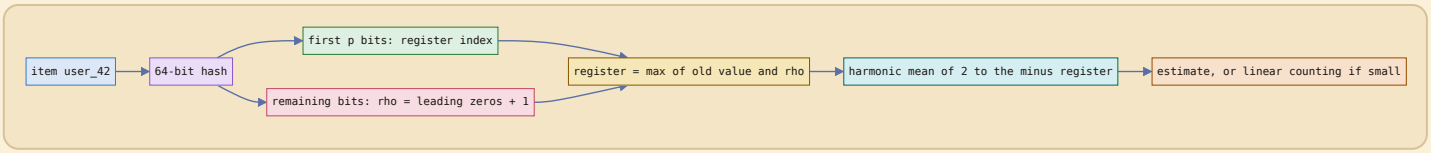
Error and size. Standard error ≈ 1.04/√m:

p	m	Error	Dense size at 6 bits/register
10	1,024	3.25%	768 bytes
14	16,384	0.81%	12 KB
16	65,536	0.41%	48 KB

Each 4× increase in memory halves the error. Six bits per register suffice because ρ never exceeds 64 for a 64-bit hash.

Sparse representation. A fresh 12 KB HLL that has seen 5 items is 99.9% zeros. **HyperLogLog++** (Heule, Nunkesser and Hall, Google, EDBT 2013) adds three improvements: a 64-bit hash so huge cardinalities need no large-range correction, an empirical bias-correction table for the awkward middle range, and a **sparse mode** that stores only the non-zero (index, ρ) pairs as a sorted, compressed list, using a higher precision (p' = 25) while sparse. When the list would grow larger than the dense array, it converts to dense. Redis does something similar with run-length opcodes: `XZERO:1000, VAL:2, ZERO:19, VAL:3 x2, XZERO:15362` encodes three non-zero registers in 7 bytes instead of 12 KB, and Redis switches to dense once the sparse form passes `hll-sparse-max-bytes` (3,000 bytes by default).

Merging. Two HLLs with the same p and the same hash function merge by taking the register-wise maximum: `merged[j] = max(a[j], b[j])`. The result is exactly the sketch you would have built from the union stream. So you can keep one HLL per hour and merge 24 of them for a daily count, or one per shard and merge at query time. Intersections are not directly supported: |A ∩ B| = |A| + |B| − |A ∪ B| works, but the errors of three estimates add up, so it is poor when the overlap is small.



Complexity: O(1) per add (one hash and one max), O(m) per estimate (Redis caches the last estimate and invalidates it only when a register changes), O(m) per merge, O(m) memory.

Trade-offs

- Error is relative, about 0.8% at 12 KB, and never zero. Fine for dashboards, not for billing.
- No deletions: registers only grow. To get "unique users in the last 7 days" you keep daily HLLs and merge the window.
- Cannot list the items, only count them.
- Merge requires identical p and hash function across all producers.
- Intersections and small overlaps are inaccurate. Theta or KMV sketches handle set operations better.

Real-world examples

- **Redis** `PFADD`, `PFCOUNT`, `PFMERGE`: up to 12 KB per key, 0.81% standard error.
- **Google** uses HLL++ internally and in BigQuery's approximate count functions.
- Analytics databases such as Druid, ClickHouse, Presto/Trino and Spark offer `approx_count_distinct`-style functions built on HLL variants.

In the interview

Q: How do you count daily unique visitors per page for 10 million pages?

A: One HLL per page per day. At 12 KB dense that is up to 120 GB, but most pages are small and stay in sparse form, so real usage is far lower. Weekly and monthly uniques come from merging daily sketches.

Q: Why can you merge HLLs but not add distinct counts?

A: Adding counts double-counts users who appear in both sets. An HLL register stores the maximum ρ seen, and max over the union is the max of the two maxes, so a register-wise max gives exactly the union's sketch.

Q: What does 1.04/√m mean in practice?

A: With 16,384 registers the estimate is typically within about ±0.81% of the truth (one standard error), and within about ±2.4% roughly 99% of the time (three standard errors).

TL;DR

HyperLogLog hashes items, uses some bits to pick a register and the run of leading zeros in the rest as evidence of how many distinct items there are, keeps the maximum per register, and combines registers with a bias-corrected harmonic mean. 12 KB gives about 0.8% error. Sketches merge by register-wise max, and a sparse encoding keeps small sketches tiny.

Sources

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Count-min sketch with conservative update

What it is

A **count-min sketch (CMS)** estimates *how many times* each item has appeared in a stream, using a small fixed grid of counters instead of one counter per item. It was introduced by Cormode and Muthukrishnan in 2005. Its estimates can be too high but never too low. **Conservative update** is a small change to the update rule, from Estan and Varghese's 2002 traffic-measurement work, that makes the over-estimates much smaller.

The problem it solves

You want per-key frequencies over a huge or unbounded key space: requests per IP for rate limiting, clicks per URL, queries per search term. An exact hash map needs memory proportional to the number of distinct keys, which might be hundreds of millions. A CMS uses a fixed size chosen from the accuracy you need, and it merges across machines by adding grids.

How it works

Structure: d rows by w columns of counters, and d independent hash functions, one per row.

Sizing: to guarantee that every estimate is at most the true count + ε·N (N = total of all counts) with probability at least 1 − δ, choose `w = ceil(e / ε)` and `d = ceil(ln(1/δ))`, where e ≈ 2.718. For ε = 0.001 and δ = 0.01: w = 2,719 and d = 5, so 13,595 counters, about 54 KB at 4 bytes each, however many distinct keys there are. (Some implementations use w = 2/ε. Redis's `CMS.INITBYPROB` with error 0.001 builds a 2,000-wide sketch.)

Standard update `add(x, c)`: for every row i, `table[i][h_i(x)] += c`. **Query** `estimate(x)`: `min over i of table[i][h_i(x)]`.

Why the minimum? Every counter x touches holds x's true count plus the counts of whatever else collided there. Collisions only add, so each counter is an over-estimate, and the smallest is the least polluted.

Worked example. w = 5, d = 3. The hash positions are:

Item	row 1	row 2	row 3
dog	3	4	0
cat	1	4	2
fox	1	0	3
owl (never seen)	1	4	0

Stream: dog ×5, then cat ×3, then fox ×2.

Standard update, final grid (columns 0..4):

```
row 1: [0, 5, 0, 5, 0]    cat 3 + fox 2 collide in column 1
row 2: [2, 0, 0, 0, 8]    dog 5 + cat 3 collide in column 4
row 3: [5, 0, 3, 2, 0]
```

- cat = min(5, 8, 3) = 3 (exact). dog = min(5, 8, 5) = 5 (exact). fox = min(5, 2, 2) = 2 (exact).
- owl, never seen = min(5, 8, 5) = **5**. That is a false over-estimate caused by collisions in every row.

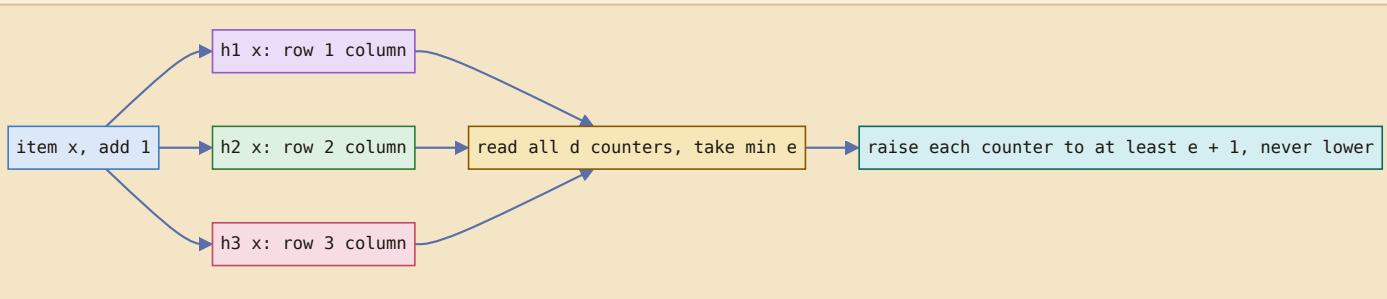
Conservative update. When adding c to x, first compute the current estimate `e = min of x's counters`. The true count is now at most e + c, so no counter needs to go *above* e + c. Raise each of x's counters to `max(counter, e + c)` and leave the rest alone.

```
function add_conservative(x, c):
    e = min(table[i][h_i(x)] for i in 1..d)
    for i in 1..d:
        table[i][h_i(x)] = max(table[i][h_i(x)], e + c)
```

Same stream with conservative update, one unit at a time:

- dog ×5: all its counters are 0, so they rise together to 5. Row 1 col 3 = 5, row 2 col 4 = 5, row 3 col 0 = 5.
- cat ×3: its counters are (0, 5, 0), estimate 0. After the first cat they become (1, 5, 1), because row 2's 5 is already ≥ 1. After three cats: (3, 5, 3). Row 2 col 4 **stays 5** instead of rising to 8.
- fox ×2: its counters (row 1 col 1 = 3, row 2 col 0 = 0, row 3 col 3 = 0) have estimate 0. After two foxes: (3, 2, 2). Row 1 col 1 **stays 3** instead of 5.

Now owl = min(3, 5, 5) = **3** instead of 5. Real items are still never under-estimated, and the noise from collisions shrinks. On real traffic Estan and Varghese reported that this kind of optimisation improved accuracy by about an order of magnitude.



Complexity: O(d) time per update and per query (d is usually 4–8), O(w·d) memory.

Trade-offs

- **Error is relative to the total stream size N**, not to the item. With ε = 0.001 and N = 10 million, any item can be over-estimated by up to 10,000. A CMS is excellent for heavy items and useless for rare ones. Redis's docs say to treat estimates below ε·N as noise.
- **Conservative update gives up mergeability and deletes.** Standard sketches merge by adding grids cell by cell, and support negative updates. A conservatively-updated sketch is no longer a plain sum, so adding two of them still gives an over-estimate but loses the formal error bound, and decrements are not allowed.
- **No item list.** A CMS cannot tell you *which* items are heavy. Pair it with a heap (next section).

- **Aging:** counts grow forever. Use time-bucketed sketches (one per minute) or periodically halve every counter.

Real-world examples

- **Redis** `CMS.INITBYPROB`, `CMS.INCRBY`, `CMS.QUERY`, `CMS.MERGE`.
- Network monitoring and DDoS detection on routers and switches, the original motivation for Estan and Varghese.
- The TinyLFU cache-admission policy, used by Caffeine in Java, uses a count-min-style frequency sketch with periodic halving to decide whether a new key deserves a cache slot.
- Stream processors (Flink, Spark) provide CMS helpers for approximate frequency queries.

In the interview

Q: Can a count-min sketch under-count?

A: Not with non-negative updates. Each counter holds the true count plus colliding counts, so the minimum is still at least the true count. It only over-counts.

Q: How do you size it for "error at most 0.1% of traffic with 99% confidence"?

A: ε = 0.001 gives w = ceil(e/0.001) = 2,719, and δ = 0.01 gives d = ceil(ln 100) = 5, so about 13,600 counters, roughly 54 KB with 32-bit counters.

Q: What does conservative update buy and cost?

A: It only raises the counters that are lower than the new lower-bound estimate, so collisions inflate counters much less, often making errors many times smaller. The cost is that you lose clean merging and deletions.

TL;DR

A count-min sketch is a d × w grid of counters. Adding an item increments one counter per row, and the estimate is the minimum over those counters. That estimate is never too low and is too high by at most ε·N with probability 1 − δ. Conservative update raises only the counters below min + c and sharply cuts over-counting, at the cost of mergeability and deletes.

Sources

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Heavy hitters: Misra-Gries, Space-Saving and streaming top-k

What it is

Heavy hitters are the items that make up a large share of a stream, for example every search term above 1% of all queries. **Top-k** asks for the k most frequent items. Two classic algorithms solve this with a fixed number of counters, keeping the *items themselves*, which a count-min sketch cannot do:

- **Misra-Gries** (Misra and Gries, 1982): k counters, and when they are full, decrement everyone.
- **Space-Saving** (Metwally, Agrawal and El Abbadi, 2005): k counters, and when they are full, evict the smallest and let the newcomer inherit its count.

The problem it solves

"What are the top 10 hashtags right now?" or "which 100 IPs send the most traffic?" An exact answer requires a counter per distinct item, which could be hundreds of millions. These algorithms keep only k entries and guarantee that every truly heavy item is among them, with bounded count error.

How it works

Stream for both examples (n = 10): `A B A C D A B A E A`. True counts are A = 5, B = 2, C = D = E = 1. Both use **k = 3 counters**.

Misra-Gries. For each item x:

1. If x has a counter, increment it.
2. Else if a counter slot is free, add x with count 1.
3. Else decrement **every** counter by 1 and drop those that reach 0. (x itself is not stored.)

Trace:

Arrival	Counters after
A, B, A, C	A:2, B:1, C:1
D (full)	decrement all: A:1 (B, C dropped)
A	A:2
B	A:2, B:1
A	A:3, B:1
E	A:3, B:1, E:1
A	A:4, B:1, E:1

Guarantee: each stored count is an **under-estimate** by at most n/(k+1) = 10/4 = 2.5. A is reported as 4 against a true 5. Any item with true frequency above n/(k+1) is guaranteed to be in the table. Each decrement step throws away k+1 occurrences (k counters plus the arriving item), so it can happen at most n/(k+1) times.

Space-Saving. For each item x:

1. If x has a counter, increment it.
2. Else if a slot is free, add x with count 1, error 0.
3. Else find the counter with the **minimum** count `min`, replace its item with x, set count = min + 1 and error = min.

Trace:

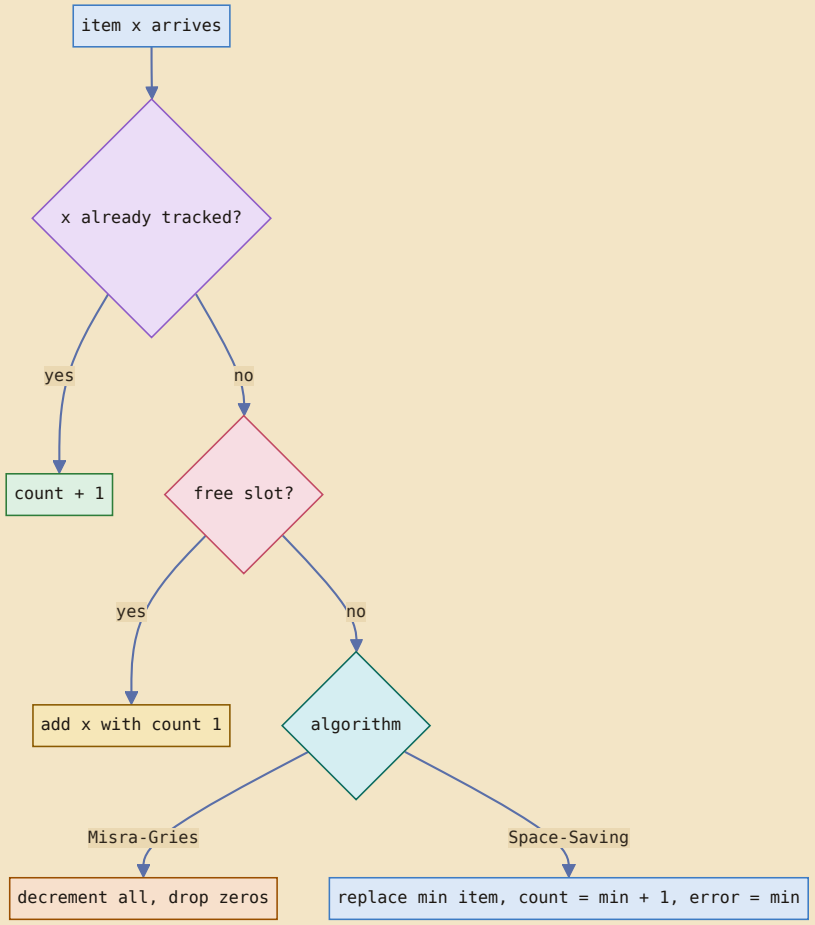
Arrival	Counters after (count, error)
A, B, A, C	A:2, B:1, C:1
D	evict B (min 1): A:2, C:1, D:2(err 1)
A	A:3, C:1, D:2
B	evict C (min 1): A:3, D:2, B:2(err 1)
A	A:4, D:2, B:2
E	evict D (min 2): A:4, B:2, E:3(err 2)
A	A:5, B:2, E:3

Counts here are **over-estimates**, and `count − error` is a guaranteed lower bound: E shows 3 with error 2, so its true count is between 1 and 3. A is exact at 5. The over-estimate is at most n/k. In practice Space-Saving is usually more accurate than Misra-Gries for the same k, especially on skewed (Zipf-like) data where a few items dominate. With a "stream summary" structure (buckets of equal counts in a linked list) each update is O(1). With a min-heap it is O(log k).

```
function space_saving(x):
    if x in counters: counters[x].count += 1
    elif len(counters) < k: counters[x] = (count=1, error=0)
    else:
        y = item with smallest count     # min-heap or stream-summary
        m = counters[y].count
        delete counters[y]
        counters[x] = (count=m+1, error=m)
```

Streaming top-k in practice: sketch plus heap. Another common pattern is a count-min sketch for frequencies plus a min-heap of size k for the current top items. On each arrival, update the CMS, read x's estimate, and if it beats the heap's minimum (or x is already in the heap) update the heap. Redis's `TOPK` commands use **HeavyKeeper** (Gong et al., USENIX ATC 2018), which pairs a sketch-like table with a heap and *decays* counters of colliding small items so large flows keep their place.

Distributed top-k. You cannot take the top 10 from each of 50 shards and merge them: an item that is 11th everywhere can be 1st globally. Options: (1) shards send Misra-Gries or Space-Saving summaries, which are designed to merge (add counts, then trim back to k), (2) shards send a larger top-k' (for example 10×k) with counts and the coordinator accepts a small error, or (3) partition by item (hash the hashtag to a shard) so each item is counted in exactly one place and local top-k lists merge exactly.



Complexity: O(k) memory for both. Misra-Gries is O(1) amortised per item with the right structure (O(k) naively). Space-Saving is O(1) with stream-summary or O(log k) with a heap.

Trade-offs

- **Misra-Gries** under-counts and is simple and mergeable, but its counts are less useful as numbers.
- **Space-Saving** over-counts but tracks an error per entry and is usually more accurate. Its ranking at the bottom of the list is noisy because evicted items re-enter with inflated counts.
- **CMS + heap** gives estimates for *any* item, not just tracked ones, but costs more memory and inherits CMS error relative to N.

- All of them count since the start. For "top right now" you need windows (one summary per minute, merged) or decay (next sections).

Real-world examples

- **Redis** `TOPK.ADD` / `TOPK.LIST` (HeavyKeeper).
- **Apache DataSketches** "frequent items" sketch, used in Druid and other analytics engines, is based on Misra-Gries-style reduction.
- Network gear and CDNs find "elephant flows" and abusive clients with these algorithms.
- Trending hashtags, top search queries, most-viewed products dashboards.

In the interview

Q: Design "top 100 search queries in the last hour" at 100,000 queries per second.

A: Hash each query to one of P aggregator partitions so a query is counted in one place. Each partition keeps a Space-Saving summary per minute (say 10,000 entries for top-100 accuracy). Every few seconds each partition merges its last 60 minute-summaries and publishes its top 100. A coordinator merges the P lists. That merge is exact because the partitions hold disjoint queries.

Q: Why not just take the top 10 from every server and merge?

A: Because a query that is just below the cut on every server can be the largest overall. Either partition by key, or send much larger lists or mergeable summaries.

Q: Misra-Gries says A = 4 but truth is 5. Is that a bug?

A: No. Misra-Gries always under-counts, by at most n/(k+1). Its promise is that every item above that threshold is in the table. If you need exact counts for the winners, do a second pass or keep exact counters for the candidates.

TL;DR

Misra-Gries and Space-Saving find frequent items with k counters. Misra-Gries decrements all counters when full and under-counts by at most n/(k+1). Space-Saving evicts the smallest counter, lets the newcomer inherit its count, records the error, and over-counts by at most n/k. For distributed top-k, partition by item or merge proper summaries, never just local top-k lists.

Sources

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- Redis `TOPK.ADD`: https://redis.io/commands/topk.add

Reservoir sampling: Algorithm R and the weighted variant

What it is

Reservoir sampling keeps a uniform random sample of exactly k items from a stream whose length you do not know in advance, in one pass and O(k) memory. The **reservoir** is the array of k slots. **Algorithm R** is the classic version, analysed in Vitter's 1985 paper "Random Sampling with a Reservoir". **Weighted reservoir sampling** (Efraimidis and Spirakis, 2006) lets heavier items be more likely to be picked.

The problem it solves

"Keep 1,000 representative log lines from today", "show 100 random requests from this endpoint for debugging", "sample 1% of traces but cap at 10,000 per minute". You cannot store the whole stream and

pick at the end, and you do not know how long it will be. Simple "keep each with probability p" sampling gives a random-size sample that can blow up on a busy day.

How it works

Algorithm R:

1. Put the first k items straight into the reservoir.
2. For item number i (counting from 1) with i > k, pick a random integer j uniformly in 1..i.
3. If j ≤ k, replace reservoir slot j with item i. Otherwise discard item i.

```
function reservoir(stream, k):
    R = []
    for i, x in enumerate(stream, start=1):
        if i <= k: R.append(x)
        else:
            j = random_int(1, i)
            if j <= k: R[j] = x
    return R
```

Worked example. k = 2, stream `a b c d e`.

- a, b fill the reservoir: `[a, b]`.
- c (i = 3): draw j from 1..3, say j = 1. 1 ≤ 2, so replace slot 1: `[c, b]`.
- d (i = 4): draw j = 4. 4 > 2, discard: `[c, b]`.
- e (i = 5): draw j = 2. Replace slot 2: `[c, e]`.

Why it is fair. Every item should end up in the sample with probability k/n = 2/5. Check item c: it enters with probability 2/3. It survives d unless d is kept (2/4) *and* d lands on c's slot (1/2), so it survives with 1 − 1/4 = 3/4. It survives e with 1 − (2/5)(1/2) = 4/5. Total: 2/3 × 3/4 × 4/5 = 2/5. The product telescopes like this for every position, so all items are equally likely.

Complexity: O(1) per item, O(k) memory. For very long streams, skip-ahead variants (Vitter's Algorithm Z, or Algorithm L) compute how many items to skip before the next replacement, so they draw far fewer random numbers.

Weighted variant (A-Res, Efraimidis and Spirakis). Each item has a weight w > 0.

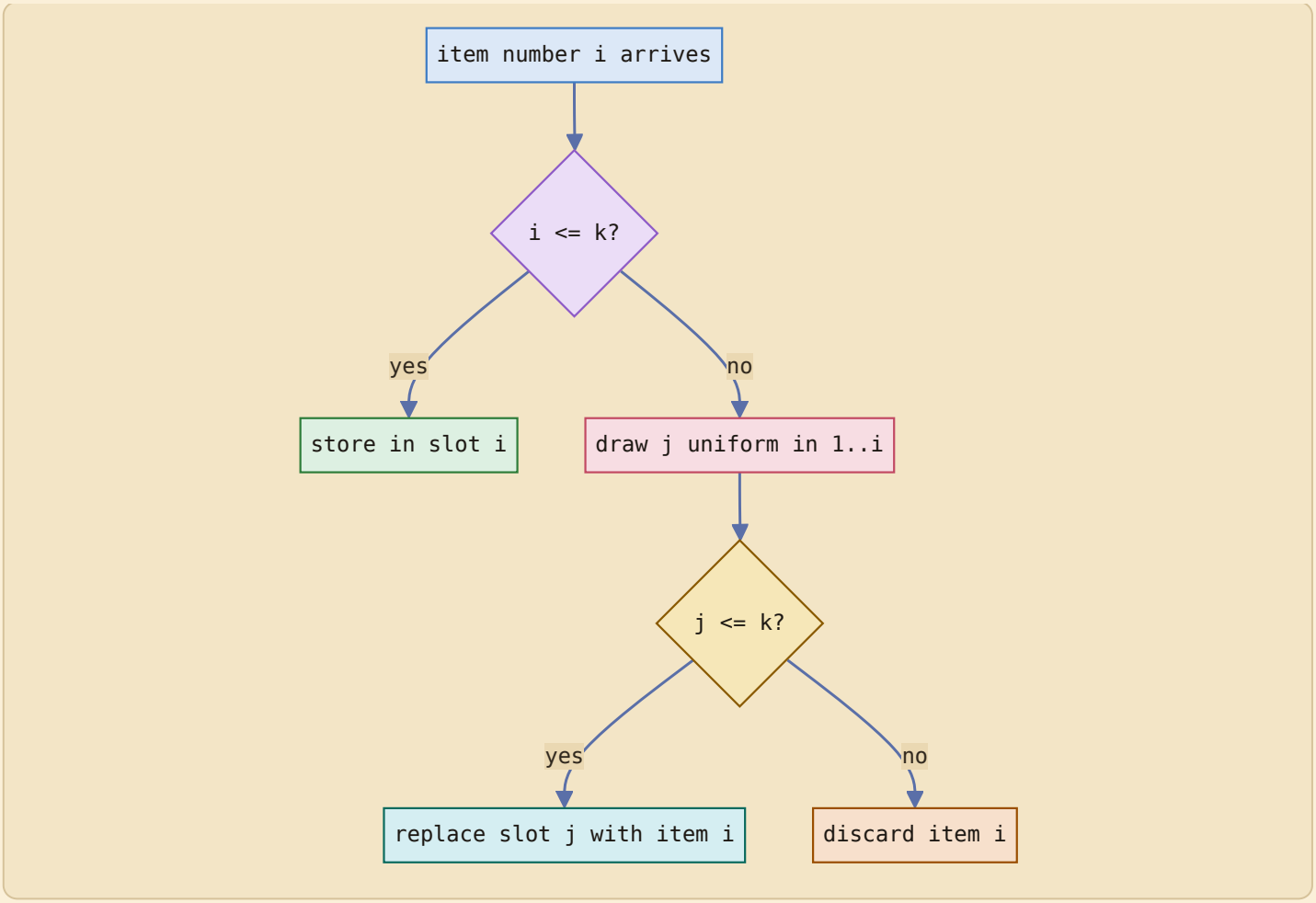
1. For each item draw u uniformly in (0, 1) and compute the key `u^(1/w)`.
2. Keep the k items with the **largest** keys, using a min-heap of size k.

Higher weights push the key toward 1, so heavy items tend to win, but luck still plays a part. Worked example, k = 2:

Item	weight w	u	key = u^(1/w)
A	1	0.50	0.500
B	4	0.30	0.740
C	2	0.80	0.894
D	1	0.90	0.900

The sample is {D, C}. B had the highest weight but drew a small u, which happens with the right probability. Use `log(u) / w` as the key (same order, numerically safer) when weights are large.

Distributed sampling. The "random key, keep the top k" form is **mergeable**: each shard keeps its top-k keys and the coordinator keeps the top k of the union. The result is exactly a sample of the whole stream. Uniform sampling is the special case where every weight is 1, so the key is just u.



Trade-offs

- Exactly k items regardless of volume, which is great for bounded memory and storage. The effective sampling rate drops as volume grows, so record n (items seen) to scale counts back up.
- Uniform samples miss rare but important events (errors, slow requests). Use weights, or a separate always-keep rule for errors.
- Algorithm R needs one random number per item. Skip-based versions are faster for huge streams.
- A sample is not a sketch: it cannot answer "how many distinct" or "how often did X occur" reliably for rare X.

Real-world examples

- **Dropwizard Metrics** (Java) uses reservoir-based histograms (uniform and exponentially decaying) to keep latency samples.
- Tail-based trace samplers and log samplers that cap traces per minute.
- Database `TABLESAMPLE` and approximate query engines use reservoir-style sampling to build statistics.

In the interview

Q: Pick 1 random line from a file of unknown length in one pass with O(1) memory.

A: Reservoir with k = 1: keep line 1, then replace the kept line with line i with probability 1/i. Every line ends up chosen with probability 1/n.

Q: How do you sample uniformly across 100 shards?

A: Give every item a random key u, keep the k items with the largest keys per shard, and take the top k of the union at the coordinator. For weighted sampling use u^(1/w) as the key.

Q: Why not just keep each request with probability 1%?

A: That sample's size grows with traffic and is unbounded during spikes. A reservoir guarantees exactly k items per window.

TL;DR

Algorithm R keeps the first k items, then keeps item i with probability k/i in a random slot, so every item ends up sampled with probability k/n in one pass and O(k) memory. Weighted sampling gives each item the key u^(1/w) and keeps the top k keys, which also merges across shards.

Sources

- Vitter, "Random Sampling with a Reservoir", ACM TOMS 11(1), 1985: https://doi.org/10.1145/3147.3165
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t-digest and percentile estimation

What it is

A **percentile** (or **quantile**) answers "what value are q% of observations below?". p99 latency is the value 99% of requests beat. A **t-digest** (Ted Dunning, refined with Otmar Ertl, 2019 paper) is a small summary of a distribution that answers any percentile query approximately, is very accurate near the extremes (p99, p99.9) and **merges** across machines without losing accuracy.

The problem it solves

Exact percentiles need every value sorted. With a billion requests a day across 500 servers that is impractical, so each server keeps a summary. The tempting shortcut, **averaging percentiles** from each server, is wrong. Example:

- Server A: 9,900 requests. 9,800 at 20 ms and 100 at 50 ms. Its p99 (the 9,801st value) is 50 ms.
- Server B: 100 requests, all at 1,000 ms. Its p99 is 1,000 ms.
- Average of the p99s: (50 + 1,000)/2 = **525 ms**.
- True p99 of all 10,000 requests: sort them, and the 9,900th value is 50 ms, so p99 = **50 ms**. (The 1,000 ms requests are exactly the slowest 1%, and they show up in p99.5.)

The average is 10× off. You cannot average or add percentiles. You need a summary of each distribution that can be merged, and *then* you query the merged summary.

How it works

Centroids. A t-digest is a sorted list of **centroids**, each a (mean, count) pair standing for a cluster of nearby values. The key rule is that **centroids near the middle of the distribution may be large, and centroids near the tails must be tiny**. The tails, where p99 lives, keep nearly every point. The middle, which you rarely ask about precisely, is summarised coarsely.

The size limit. A **scale function** k(q) sets how wide a centroid may be at quantile position q. A centroid covering quantiles q_left to q_right is allowed only if `k(q_right) − k(q_left) ≤ 1`. The function is steep near q = 0 and q = 1 and flat in the middle. A **compression** parameter δ (often 100) controls how many centroids exist overall. The original paper's simpler bound gives a feel for the numbers: a centroid at quantile q may hold about `4 · N · q · (1 − q) / δ` points. With N = 10,000 points and δ = 100:

Quantile q	Max points per centroid
0.5 (median)	4 × 10,000 × 0.25 / 100 = 100
0.1	4 × 10,000 × 0.09 / 100 = 36
0.01	4 × 10,000 × 0.0099 / 100 ≈ 4
0.001	≈ 0.4, so single points

So the median is known to within a bucket of ~100 points, while the extreme tail keeps individual values.

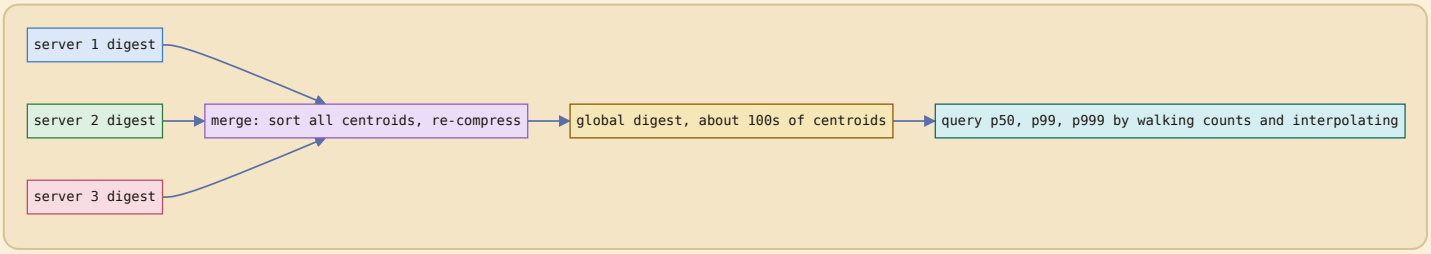
Adding points (merging variant):

1. Buffer incoming values. When the buffer fills, sort buffer and existing centroids together by mean.
2. Walk left to right, greedily merging each item into the current centroid if the size limit at that quantile still allows it. Otherwise start a new centroid.
3. The new centroid's mean is the weighted mean, and its count is the sum.

Querying p99:

1. Target rank = 0.99 × N.
2. Walk centroids accumulating counts until you pass the target rank.
3. **Interpolate** between the two neighbouring centroid means according to where the target falls. Exact minimum and maximum are kept for the ends.

Merging digests from many servers works the same way: concatenate all their centroids, sort, and run the same greedy merge. Because the size rule depends only on quantile position, the merged digest has the same accuracy guarantees, which is the property averaging percentiles lacks.



Complexity: amortised O(log δ) or better per insert with buffering, memory O(δ) centroids (Elasticsearch caps it at 20 × compression nodes, about 64 KB worst case at the default of 100), merge O(total centroids × log).

Alternatives worth naming:

- **HDR Histogram** (fixed log-scale buckets with a chosen number of significant digits): fast, mergeable by adding buckets, bounded relative error, needs a known value range. Very common for latency.
- **DDSketch** (Datadog): log-scale buckets with a guaranteed relative error, fully mergeable.
- **KLL** and other rank-error sketches (in Apache DataSketches): proven worst-case error bounds on rank.

Trade-offs

- t-digest is very accurate at the tails and small, but its error bounds are mostly empirical, and the result can depend a little on insertion order.
- Middle percentiles are less precise than tail percentiles, which is usually the right trade for latency.
- HDR or DDSketch are simpler and give hard relative-error guarantees, but need a range (HDR) or use more memory for wide ranges.
- Any sketch beats the wrong answer you get from averaging per-host percentiles.

Real-world examples

- **Elasticsearch / OpenSearch** `percentiles` aggregation uses t-digest by default (compression 100), with HDR histogram as an option.
- Many metrics libraries and stream processors offer t-digest. Prometheus **histograms** (fixed buckets, mergeable) are the bucket-based alternative, and Prometheus **summaries** compute quantiles per process, which cannot be aggregated across instances for exactly the reason shown above.
- Datadog uses DDSketch for distribution metrics.

In the interview

Q: Your dashboard averages p99 across 200 hosts. What is wrong?

A: Percentiles do not average. A host with few but slow requests drags the average far from the real p99, or a busy host's tail gets diluted. Ship mergeable sketches (t-digest, HDR, DDSketch) or bucketed histograms, merge them, then compute p99 once.

Q: Why is t-digest more accurate at p99.9 than at p50?

A: Its scale function forces centroids near the tails to hold very few points, often single values, while centroids near the median can hold many. Precision is spent where tail-latency questions are asked.

Q: Can you compute a weekly p99 from daily p99s?

A: No. Keep daily digests and merge seven of them, then query.

TL;DR

Never average percentiles. A t-digest summarises a distribution as sorted (mean, count) centroids that are tiny at the tails and larger in the middle, queries percentiles by walking counts and interpolating, and merges across hosts and time windows while keeping its accuracy. HDR Histogram and DDSketch are bucket-based alternatives with hard relative-error bounds.

Sources

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- t-digest reference implementation: https://github.com/tdunning/t-digest
- Elasticsearch percentiles aggregation: https://www.elastic.co/docs/reference/aggregations/search-aggregations-metrics-percentile-aggregation
- Prometheus histograms and summaries: https://prometheus.io/docs/practices/histograms/

Trending detection with decayed counts and z-scores

What it is

Trending detection finds items whose activity is *unusually high compared with their own normal*, not simply the most popular. "#WorldCup" with 50,000 mentions an hour every day is popular. A small band's name jumping from 20 to 2,000 mentions an hour is trending. The standard building blocks are **exponentially decayed counts**, which weight recent events more and let old ones fade smoothly, and **z-scores**, which measure how many standard deviations the current rate is above the usual rate.

The problem it solves

Raw counts always rank the same giants on top. Fixed windows ("count in the last hour") are expensive (you must expire old events) and jumpy (an event falls off a cliff exactly 60 minutes later). You want something that:

1. Reacts within minutes to a real surge.
2. Ignores items that are big but steady.
3. Uses O(1) state per item and O(1) work per event.
4. Does not fire on noise in tiny counts (2 mentions to 6 is "3×" but meaningless).

How it works

1. Exponentially decayed count. Keep for each item a value c and the time t_last it was updated. On a new event at time t:

```
c = c * 2^(-(t - t_last) / H) + 1     # H = half-life
t_last = t
```

After one half-life with no events, the count halves. Worked example with H = 10 minutes: 1,000 mentions arrive around 12:00, so c ≈ 1,000. By 12:10 with no new mentions c = 500, by 12:20 it is 250, by 12:30 it is 125. No window bookkeeping, no expiring of events, and the count reflects "recent activity". Keep two decayed counts with different half-lives, for example 5 minutes (short-term rate) and 24 hours (long-term baseline). A high short/long ratio means "much hotter than usual".

2. Baseline with an exponentially weighted moving average. Count events per item in fixed buckets (say per minute, giving x_t). Track an **exponentially weighted moving average (EWMA)** mean μ and variance σ²

with smoothing factor λ (NIST suggests 0.2–0.3 for control charts, and trending systems often use smaller values for longer memory):

```
diff  = x - mu
z     = diff / sqrt(var)            # score BEFORE updating the baseline
mu    = mu + lam * diff
var   = (1 - lam) * (var + lam * diff * diff)
```

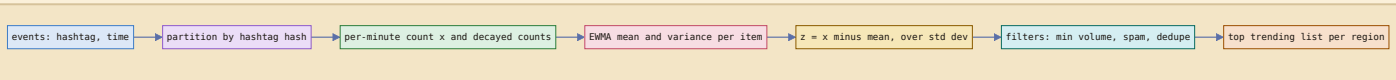
3. z-score and threshold. $z = (x - \mu)/\sigma$ says how unusual this minute is for *this item*. Flag as trending when z exceeds a threshold (for example 3) and the volume passes a minimum.

Worked example. $\lambda = 0.2$.

- A small topic has baseline $\mu = 100$ per minute, $\sigma^2 = 400$ ($\sigma = 20$). This minute $x = 180$.
  - $z = (180 - 100)/20 = \mathbf{4.0}$: trending.
  - Update: $\mu = 100 + 0.2 \times 80 = 116$. $\sigma^2 = 0.8 \times (400 + 0.2 \times 80^2) = 0.8 \times 1{,}680 = 1{,}344$, so $\sigma \approx 36.7$.
  - Next minute $x = 260$: $z = (260 - 116)/36.7 \approx \mathbf{3.9}$, still trending. The baseline is catching up, so a surge that persists for hours eventually stops being "trending", which is the intended behaviour.
- A giant topic has $\mu = 10{,}000$, $\sigma = 1{,}000$, and this minute $x = 10{,}500$. It has the higher raw count but $z = 0.5$: **not trending**.

4. Guard against small numbers. With $\mu = 2$ and $\sigma = 1$, a jump to 6 gives $z = 4$. Fixes: require a minimum count ($x \geq 50$), add a prior to the variance (`z = diff / sqrt(var + c)`), or score with a smoothed ratio `(x + β) / (μ + β)` with β around 10–50.

5. At scale. Per-item state for every hashtag is too much, so first find candidates with a heavy-hitter structure (Space-Saving or CMS + heap on the decayed counts), keep μ and σ^2 only for candidates and a long tail of recently-seen items, and partition by item hash across workers. Each worker emits its top trending scores every few seconds, and a coordinator merges them. Then add product filters: de-duplicate near-identical terms, remove spam and bot bursts, and localise by region.



Complexity: O(1) time and O(1) state per tracked item per event. Total memory depends on how many items you track, which is bounded with a heavy-hitter front end.

Trade-offs

- **Half-life / λ choice:** short half-lives react fast but flag noise. Long ones are stable but late. Many systems run two or three timescales together.
- **z-scores assume roughly normal noise.** Counts are bursty and follow daily cycles. Compare against the same hour last week, or model the seasonality, to avoid "every morning is a trend".
- **Small-count bias** needs explicit guards, or the trending list fills with tiny items.
- **Manipulation:** coordinated accounts can manufacture a spike. Count distinct users (HLL per item) rather than raw posts, and weight by account trust.
- Approximate structures (sketches) at the front cost a little accuracy for huge memory savings.

Real-world examples

- **Twitter / X trends** compare a topic's current volume with its own past volume rather than ranking raw popularity, and the company described rebuilding the trends back end on stream processing (Storm) to compute this in real time.
- **EWMA control charts** are a decades-old industrial quality-control method (NIST handbook) and the same math drives anomaly alerts in monitoring systems.
- News, e-commerce "trending now" and music charts use decayed scores so yesterday's hits fade smoothly.

In the interview

Q: Design trending hashtags for a social network.

A: Stream posts to workers partitioned by hashtag. Each worker keeps per-minute counts, a short-half-life decayed count and an EWMA mean and variance per tracked hashtag, with a Space-Saving front end to bound how many it tracks. Score by z-score with a minimum-volume floor and count distinct authors to resist spam. Every 30 seconds publish each worker's top N per region, and merge them at a coordinator into the cache that the app reads.

Q: Why not just rank by count in the last hour?

A: That shows the always-popular topics and nothing new. Trending means "unusual for this item", so normalise against the item's own baseline.

Q: Why exponential decay instead of a sliding window?

A: A decayed count is one number and one timestamp, updated in O(1), with no per-event storage or expiry. It also fades smoothly instead of dropping a burst exactly when it leaves the window.

TL;DR

Trending means "much higher than this item's own normal". Keep exponentially decayed counts (one number per item with a half-life) and an EWMA mean and variance per item, score each interval with $z = (x - \mu)/\sigma$, guard against small counts and spam, and bound the set of tracked items with a heavy-hitter sketch partitioned by item.

Sources

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- Finch, "Incremental calculation of weighted mean and variance" (2009): https://fanf2.user.srcf.net/hermes/doc/antiforgery/stats.pdf
- Coverage of Twitter's trends back-end rebuild (Storm): https://www.smartcompany.com.au/technology/twitter-overhauls-trending-topics-adds-more-contextual-information/
- Nikolov and Shah, trend detection research with Twitter data (MIT, 2012), summary: https://flowingdata.com/2013/06/19/twitter-trend-detection-algorithm/

LSM trees in depth

What it is

An **LSM tree** (log-structured merge tree) is a way of storing sorted key-value data on disk that turns every write into a fast, sequential append and does the sorting work later, in the background. It was described by Patrick O'Neil and colleagues in 1996 and is the storage engine behind LevelDB, RocksDB, Cassandra, ScyllaDB and HBase.

It has three moving parts:

- **Memtable:** an in-memory sorted structure (RocksDB's default is a skip list) that receives every new write.
- **SSTable** (sorted string table): an immutable file on disk holding key-value pairs sorted by key, plus a small index and a filter. "Immutable" means it is never edited after being written; changes go into newer files.
- **Compaction:** a background job that merges several SSTables into new ones, dropping overwritten values and deleted keys.

A **write-ahead log (WAL)** sits next to the memtable: an append-only file on disk that records each write first, so the memtable can be rebuilt after a crash.

The problem it solves

A B-tree updates data in place: changing one 100-byte row may mean reading and rewriting a whole 16 KB page at a random location on disk. Random writes are slow on spinning disks and wear out SSDs (solid-state drives). Workloads such as logging, metrics, messaging and time series write far more than they read. An LSM tree batches writes in memory and flushes them as large sequential files, so it can absorb much higher write rates on the same hardware.

How it works

Write path, step by step

1. Append the write (`put user:42 = "Ana"`) to the WAL and `fsync` it (force it to physical disk), or batch several writes per `fsync`.
2. Insert it into the memtable. Done: the client gets its acknowledgement. Cost: one sequential append plus an O(log n) in-memory insert.
3. When the memtable reaches its size limit (RocksDB default: 64 MB), it becomes **immutable** (read-only) and a fresh memtable takes new writes.
4. A background thread writes the immutable memtable out, in key order, as a new SSTable. This is a **flush**. The matching WAL segment can then be deleted.
5. A **delete** is just another write: a special marker called a **tombstone** that says "this key is gone as of now". The old value is removed only when compaction merges the tombstone with it.

Read path (point lookup), step by step

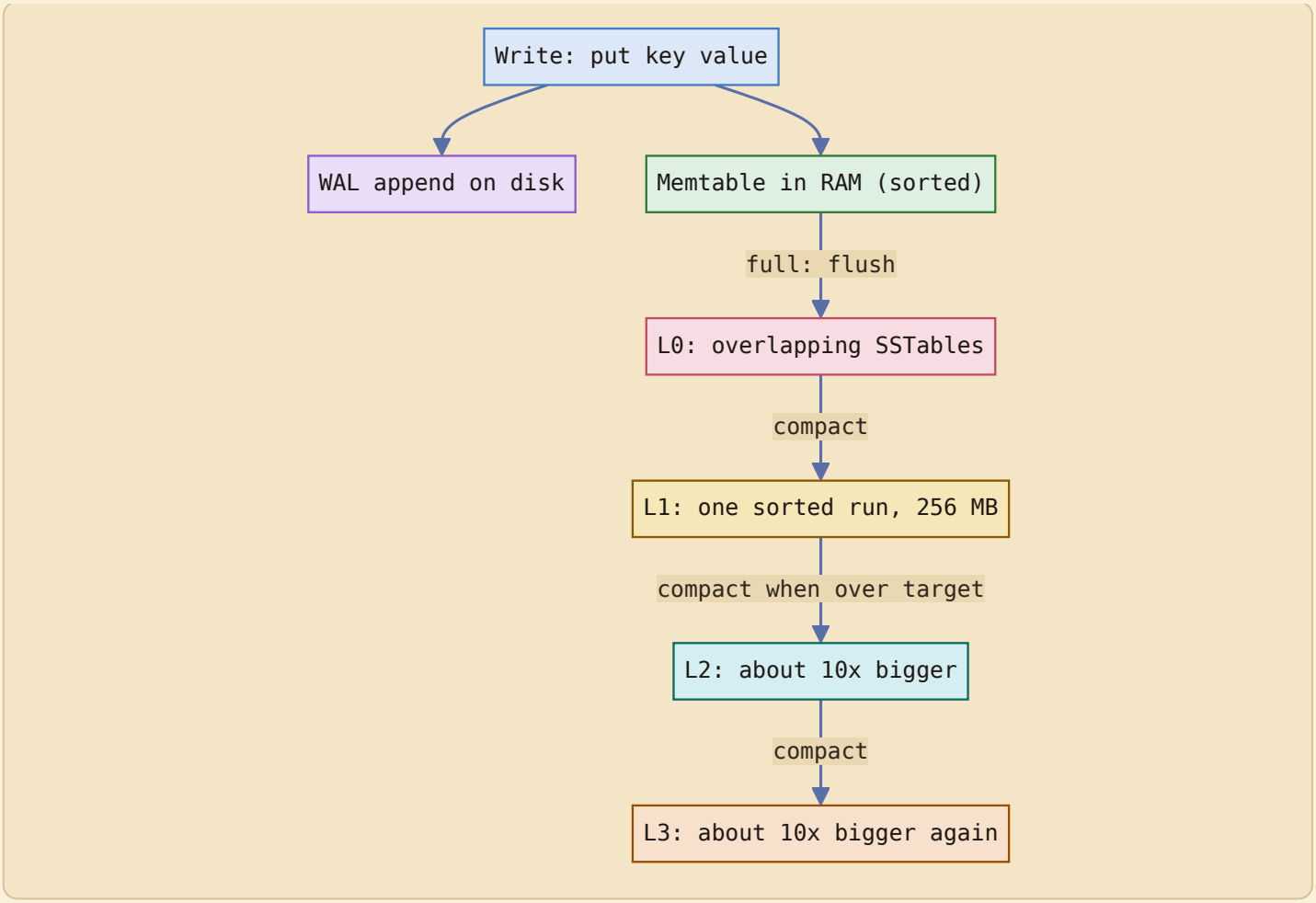
1. Check the active memtable, then any immutable memtables. Newest data wins.
2. Check SSTables from newest to oldest. For each SSTable, first ask its **Bloom filter**: a compact bit array that answers "definitely not here" or "maybe here". Skip the file on "definitely not".
3. On "maybe", use the file's index block to find the right 4 KB-ish data block, read it, and binary search inside.
4. Stop at the first file that contains the key (or a tombstone for it).

Bloom filter per SSTable. Each SSTable carries its own filter, built when the file is written. With about 10 bits per key the false-positive rate is about 1% (RocksDB measures about 0.95% for its cache-friendly variant). Worked example: a lookup for a missing key in a store with 8 SSTables would otherwise read 8 data blocks. With 1% filters it reads on average 8 x 0.01 = 0.08 blocks. That is why LSM point reads stay fast even with many files.

Compaction strategies. This is where LSM designs differ most.

- **Size-tiered compaction** (Cassandra's STCS, RocksDB's "universal"): group SSTables of similar size into tiers. When enough similar files pile up (Cassandra's default `min_threshold` is 4), merge them into one bigger file. Example: four 64 MB flushes merge into one ~256 MB file; four of those merge into one ~1 GB file.
- **Leveled compaction** (LevelDB, RocksDB default): disk is split into levels L0, L1, L2... L0 holds raw flushes whose key ranges overlap. From L1 down, each level is one sorted run: its files do not overlap, so a key lives in at most one file per level. Each level has a target size about 10x the previous one (the `max_bytes_for_level_multiplier`). When a level exceeds its target, pick one file, find the files in the next level whose key ranges overlap it, merge them, and write the result into the next level.

Worked example of leveled sizing with L1 = 256 MB and multiplier 10: L2 = 2.56 GB, L3 = 25.6 GB, L4 = 256 GB. The last level holds roughly 90% of the data, so the extra copies sitting in upper levels add only about 1/9 = 11% of overhead.



The three amplifications. These are the vocabulary for comparing storage engines.

- **Write amplification:** bytes written to disk divided by bytes the user wrote. Every compaction rewrites data. In leveled compaction, moving one file down can rewrite up to ~10 overlapping files in the next level, so the worst case is about the fanout (10) per level; across 4 to 5 levels that adds up to tens. Size-tiered rewrites each byte roughly once per tier, so it is much lower.
- **Read amplification:** number of disk reads per query. Leveled: at most one file per level plus the L0 files. Size-tiered: possibly many overlapping files per tier, so range scans especially suffer.
- **Space amplification:** bytes on disk divided by bytes of live data. Leveled: about 1.1. Size-tiered: old and new copies of a key can coexist across tiers, and a big merge needs room for inputs and outputs at the same time. RocksDB's universal compaction allows up to 200% extra by default, so 100 bytes of data may need up to 300 bytes of disk.

Complexity. Writes: O(log m) in-memory (m = memtable entries) plus amortised compaction I/O. Point reads: O(number of levels) filter checks, usually one disk read. Space: as above.

Trade-offs

- Excellent write throughput and good compression (sorted, immutable files compress well), in exchange for background compaction that steals disk bandwidth and CPU and can cause latency spikes when it falls behind.
- Leveled favours reads and space; size-tiered favours writes. Many systems offer both, plus hybrids (Cassandra's time-window compaction for time series).
- Deletes are not instant: tombstones linger until compaction reaches them, and a range scan over many tombstones is slow.
- Range scans must merge results from the memtable and every level, so they are costlier than in a B-tree.

Real-world examples

- **RocksDB** (Meta) underpins MyRocks, Kafka Streams state stores, CockroachDB's earlier storage, and many others. Default: skip-list memtable, leveled compaction, per-file Bloom filters.
- **Cassandra** defaults to size-tiered compaction and offers leveled and time-window strategies per table.
- **Bigtable/HBase** use the same memtable-plus-immutable-files idea.

In the interview

Q: Why are LSM trees faster for writes than B-trees?

A: Writes go to a sequential log and an in-memory table, and are flushed in big sorted batches. A B-tree must find and rewrite a page at a random disk location for each update.

Q: How does an LSM tree avoid checking every SSTable on a read?

A: Each SSTable has a Bloom filter. At 10 bits per key, about 99% of files that do not hold the key are skipped without any disk read. Leveled compaction also guarantees at most one candidate file per level below L0.

Q: Leveled or size-tiered for a write-heavy metrics store?

A: Size-tiered (or time-window): lower write amplification, and metrics are mostly appended and read in recent ranges. Leveled suits read-heavy or space-constrained workloads.

TL;DR

An LSM tree writes to a log and an in-memory sorted table, flushes it as immutable sorted files, and merges those files in the background. Bloom filters keep reads cheap. Leveled compaction gives low read and space amplification at a higher write cost; size-tiered does the opposite.

Sources

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- RocksDB wiki, Leveled Compaction: https://github.com/facebook/rocksdb/wiki/Leveled-Compaction
- RocksDB wiki, Compaction (amplification, universal compaction): https://github.com/facebook/rocksdb/wiki/Compaction
- RocksDB wiki, Bloom Filter: https://github.com/facebook/rocksdb/wiki/RocksDB-Bloom-Filter
- Apache Cassandra docs, Size Tiered Compaction Strategy: https://cassandra.apache.org/doc/latest/cassandra/managing/operating/compaction/stcs.html

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B+-trees

What it is

A **B+-tree** is a balanced search tree designed for storage that is read in fixed-size blocks called **pages** (InnoDB's default page is 16 KB; PostgreSQL's is 8 KB). Each node is one page and holds many keys, so the tree is very wide and very shallow. In a B+-tree, all actual records (or pointers to them) live in the **leaf** nodes at the bottom, and the leaves are linked left-to-right. The **internal** nodes above only hold separator keys that steer the search. "Balanced" means every leaf is at the same depth.

The problem it solves

Disks and SSDs read whole pages, and each page read is expensive compared with work in memory. A binary search tree with one key per node would need about 30 page reads to find one row among a billion. A B+-tree with hundreds of keys per node needs 3 or 4, and the top levels are almost always cached in RAM. Linked leaves also make range queries (`WHERE created_at BETWEEN ...`) and `ORDER BY` cheap: find the start, then walk sideways.

How it works

Fan-out arithmetic. Suppose an internal entry is an 8-byte key plus a 6-byte child page number, about 14 bytes. A 16 KB page then holds about 16,384 / 14 = roughly 1,170 children. With leaves holding about 15 rows of 1 KB each:

- Height 2: 1,170 leaves x 15 = about 17,500 rows.
- Height 3: 1,170 x 1,170 x 15 = about 20 million rows.
- Height 4: about 24 billion rows.

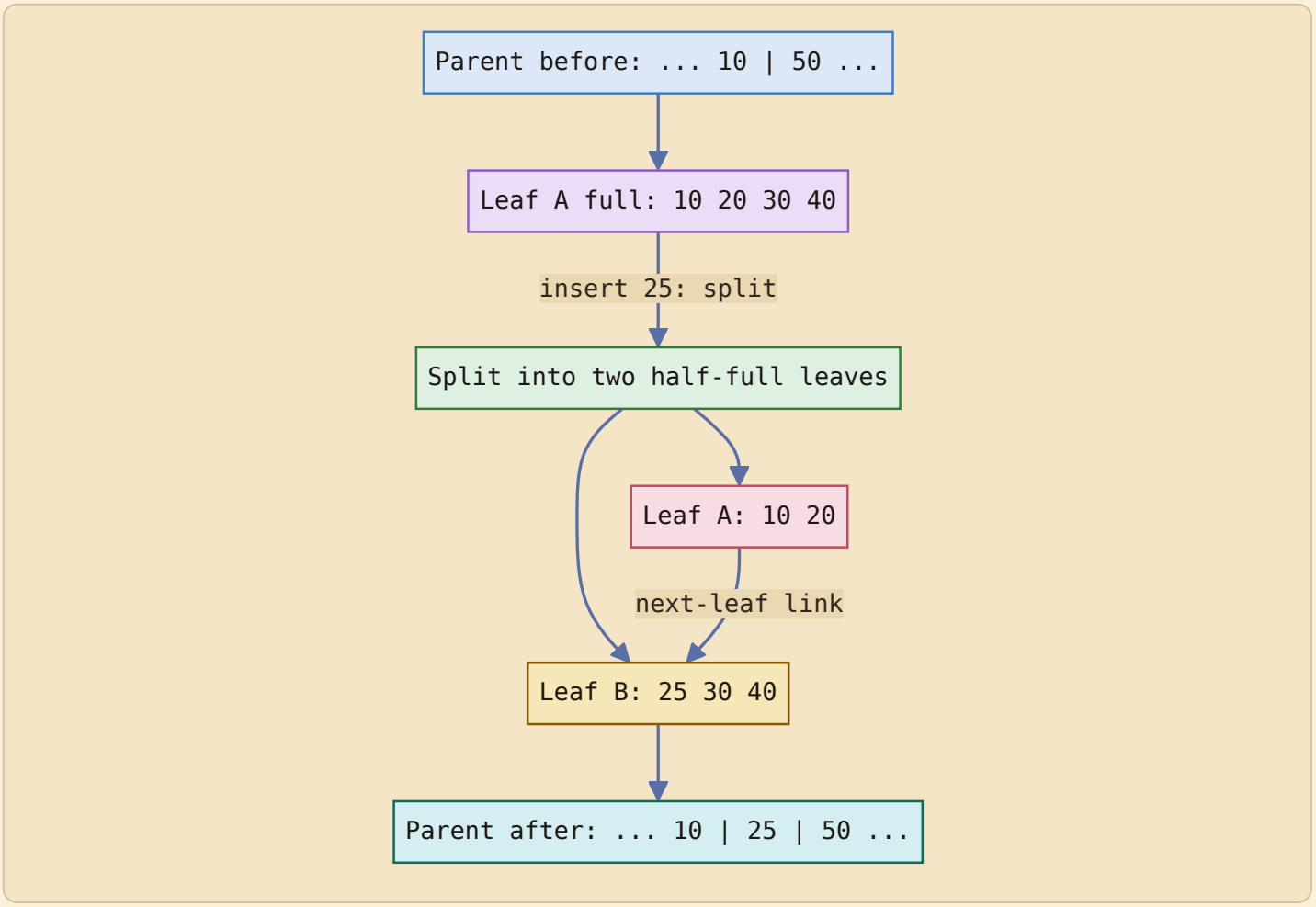
So any row is 3 to 4 page reads away, and usually only the leaf read misses the cache.

Search, step by step.

1. Start at the root page.
2. Binary search its keys to find the child whose range contains the target.
3. Repeat on that child until you reach a leaf.
4. Binary search the leaf. Cost: O(log_B n) page reads, where B is the fan-out, and O(log n) comparisons in total.

Insert with page split, step by step (tiny example, max 4 keys per leaf).

1. Search for the leaf where the key belongs. Leaf A holds [10, 20, 30, 40]. We insert 25.
2. If there is room, insert in sorted position and finish. Here the leaf is full.
3. **Split:** allocate a new leaf B. Keep the lower half in A ([10, 20]) and move the upper half to B ([25, 30, 40]).
4. Fix the sibling links: A points to B, B points to A's old right neighbour.
5. Insert a separator key (25, the first key of B) plus a pointer to B into the parent.
6. If the parent is also full, split it too and push a separator up. If the root splits, create a new root: this is the only way the tree grows taller, which is why all leaves stay at the same depth.



Delete is the mirror image: remove the key, and if a page drops below a threshold, borrow from or merge with a sibling. InnoDB tries to merge when a page falls below 50% full (its `MERGE_THRESHOLD` default).

Many systems are lazy here and leave half-empty pages.

Fill factor. The **fill factor** is how full pages are packed when an index is built or appended to. Leaving free space means future inserts into the middle of the range land in existing pages instead of triggering splits.

- PostgreSQL B-tree indexes default to a fill factor of 90: leaf pages are packed to 90% during a build and when appending new largest keys.
- InnoDB leaves 1/16 of each clustered-index page free. Sequential inserts produce pages about 15/16 full; random inserts produce pages between 1/2 and 15/16 full, because splits leave two half-full pages.

This is why a random UUID primary key in InnoDB causes more splits, more partly-empty pages and a bigger index than an auto-increment key, which always appends at the right edge.

Complexity. Search, insert and delete: O(log_B n) page I/Os. Range scan of k results: O(log_B n + k/B). Space: O(n), with typical page utilisation of roughly 50% to 100%.

Trade-offs

- Great reads and range scans; predictable latency (no background compaction).
- Every update rewrites a whole page in place, and the database must also write the change to its WAL first, so write amplification is high for small random updates.
- Random-key inserts cause splits and fragmentation; periodic rebuilds (`REINDEX`, `OPTIMIZE TABLE`) may be needed.
- Concurrency control inside the tree (latching pages during splits) is subtle; real engines use techniques such as latch crabbing or B-link trees.

Real-world examples

- **InnoDB** (MySQL): the table itself is a B+-tree keyed by the primary key (a **clustered index**: rows are stored in the leaves). Secondary indexes are separate B+-trees whose leaves hold the primary key.
- **PostgreSQL**: tables are unordered heaps; B-tree indexes point into the heap.
- **SQLite**, Oracle, SQL Server, and file systems such as NTFS, XFS and Btrfs all use B-tree variants.

In the interview

Q: Why do databases use B+-trees rather than binary trees or hash tables?

A: Pages are the unit of I/O, and a B+-tree packs hundreds of keys per page, so lookups take 3 to 4 page reads even for billions of rows. Unlike a hash table it keeps keys sorted, so range queries and ordering are cheap.

Q: Why is a random UUID a poor InnoDB primary key?

A: Inserts land on random leaves, causing page splits, half-full pages, more cache misses and a larger index. A monotonic key (auto-increment, or time-ordered UUIDv7) appends at the right edge.

Q: B+-tree or LSM tree?

A: B+-tree for read-heavy, range-heavy, latency-sensitive OLTP (online transaction processing). LSM for write-heavy workloads where some read and compaction cost is acceptable.

TL;DR

A B+-tree is a short, wide, balanced tree of pages with all data in linked leaves. Lookups take a handful of page reads, ranges are a sideways walk, and inserts split full pages and push separators upward. Fill factor leaves room to postpone splits.

Sources

- Bayer and McCreight, "Organization and Maintenance of Large Ordered Indexes", Acta Informatica, 1972, and Comer, "The Ubiquitous B-Tree", ACM Computing Surveys, 1979 (background).
- MySQL 8.0 Reference Manual, The Physical Structure of an InnoDB Index: https://dev.mysql.com/doc/refman/8.0/en/innodb-physical-structure.html

- PostgreSQL docs, CREATE INDEX (fillfactor): https://www.postgresql.org/docs/current/sql-createindex.html

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MVCC step by step

What it is

MVCC (multi-version concurrency control) is a technique where the database keeps several versions of a row at once, so that readers see a consistent **snapshot** (a frozen view of committed data as of some moment) while writers create new versions. The key promise: readers never block writers and writers never block readers. PostgreSQL, MySQL InnoDB, Oracle, SQL Server (with snapshot isolation), CockroachDB and many others use it.

The problem it solves

With plain locking, a long report query that reads a million rows would hold shared locks and stop every update to those rows until it finished, or updates would block the report. MVCC lets the report read the old versions while updates write new ones. It is the basis of **snapshot isolation**, the guarantee that every read in a transaction sees the same consistent snapshot.

How it works

Core ideas.

- Every transaction gets a **transaction ID** (XID), a number that increases over time.
- Every row version records which transaction created it and, if it has been replaced or deleted, which transaction did that.
- A snapshot records which transactions had committed when it was taken.
- A **visibility rule** decides, for each version, whether the snapshot should see it.
- Old versions that no snapshot can see any more are garbage-collected (PostgreSQL **vacuum**, InnoDB **purge**).

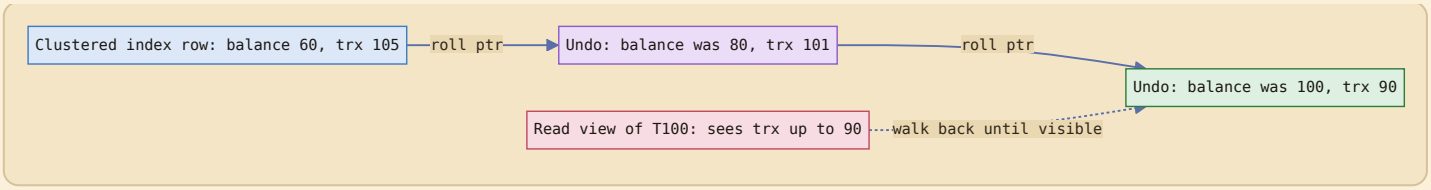
PostgreSQL: new versions in the table itself. Each row version (a **tuple**) has hidden header fields `xmin` (the XID that inserted it) and `xmax` (the XID that deleted or updated it, or empty). An `UPDATE` does not change the tuple in place: it sets `xmax` on the old tuple and inserts a new tuple with `xmin` = the updater's XID. The chain of versions lives in the table's heap.

Worked example. Row `account 7, balance 100` was inserted by XID 90, long committed.

1. T100 starts and takes a snapshot: "everything below 100 that committed is visible, T100 itself is in progress".
2. T101 runs `UPDATE ... SET balance = 80`. Old tuple: xmin 90, xmax 101. New tuple: xmin 101, xmax empty. T101 commits.
3. T100 reads account 7. Old tuple: created by 90 (committed, visible to T100), deleted by 101 (started after T100's snapshot, so the delete is invisible). Old tuple is visible: balance 100. New tuple: created by 101, invisible. T100 consistently sees 100.
4. T102 starts after T101 committed: it sees the new tuple, balance 80.

The visibility rule (simplified): a version is visible to snapshot S if its creator committed before S was taken (or is S's own transaction), and its deleter is either empty, aborted, or not yet committed as of S.

InnoDB: newest version in place, older ones rebuilt from undo. InnoDB updates the row in the clustered index in place. Each row carries a hidden 6-byte `DB_TRX_ID` (last transaction to modify it) and a 7-byte `DB_ROLL_PTR` (roll pointer) to an **undo log** record that describes how to rebuild the previous version. Following roll pointers backward gives the **version chain**. When a consistent read starts, InnoDB creates a **read view**: the list of transactions active at that moment, plus the bounds of IDs. To read a row, InnoDB checks `DB_TRX_ID` against the read view; if that change is not visible, it follows the roll pointer and applies undo records until it reaches a visible version. In REPEATABLE READ (InnoDB's default) the read view is created at the first consistent read in the transaction; in READ COMMITTED a fresh one is taken for each statement.



Garbage collection, step by step.

- **PostgreSQL VACUUM:** scan pages, find tuples whose `xmax` committed before the oldest snapshot still running, mark their space reusable, and remove index entries pointing to them. It also **freezes** very old tuples: XIDs are 32-bit and compared in a circle, so after about 2 billion transactions an old XID would appear to be "in the future". Freezing marks the tuple as visible to everyone forever. Autovacuum forces an anti-wraparound vacuum when a table's oldest unfrozen XID is 200 million transactions old (the default `autovacuum_freeze_max_age`).
- **HOT updates** (heap-only tuples): if an update changes no indexed column and the page has room, PostgreSQL puts the new version on the same page and skips new index entries. Lowering the table's fill factor makes this more likely.
- **InnoDB purge:** insert-undo records are dropped at commit. Update-undo records stay until no read view could need them, then a purge thread removes them and physically removes delete-marked rows.

Complexity. Reading a recent version: O(1) extra. Reading under a long-running snapshot: O(length of version chain). Space: proportional to the number of live old versions.

Trade-offs

- **PostgreSQL style** (append new versions): rollback is instant (the new tuple is just ignored), but tables and indexes bloat until vacuum runs, and every non-HOT update writes new index entries.
- **InnoDB style** (in place plus undo): the table stays compact and current reads are fast, but old snapshots must rebuild versions by walking undo, and a long transaction makes the undo log grow (the "history list length" problem).
- In both, one forgotten long-running transaction blocks cleanup for everyone.
- Snapshot isolation still allows **write skew** (see the next section), so it is not full serializability.

Real-world examples

- PostgreSQL: heap versions, `xmin` / `xmax`, VACUUM and autovacuum.
- MySQL InnoDB, Oracle: undo-based versions.
- CockroachDB and Spanner: versions keyed by timestamp in a key-value store, with garbage collection after a configurable time.

In the interview

Q: How do readers avoid blocking writers in PostgreSQL?

A: An update writes a new tuple and marks the old one with its XID. Readers use their snapshot to decide which tuple version is visible, so they never need to wait for the writer.

Q: Why can a long-running transaction hurt a database that is otherwise idle?

A: Its snapshot pins every old version created since it started. PostgreSQL cannot vacuum them and InnoDB cannot purge undo, so tables and undo logs grow and reads slow down.

Q: What is transaction ID wraparound?

A: PostgreSQL XIDs are 32-bit and compared modulo 2^32. Without freezing old tuples, after about 2 billion transactions old rows would look like they were created in the future and vanish. Vacuum freezing prevents it.

TL;DR

MVCC keeps multiple row versions tagged with transaction IDs. Each transaction reads through a snapshot, using visibility rules to pick the right version. PostgreSQL stores versions in the table and needs VACUUM; InnoDB updates in place, keeps old versions in undo logs and purges them later.

Sources

- PostgreSQL docs, Routine Vacuuming: https://www.postgresql.org/docs/current/routine-vacuuming.html
- PostgreSQL docs, Heap-Only Tuples: https://www.postgresql.org/docs/current/storage-hot.html
- MySQL 8.0 Reference Manual, InnoDB Multi-Versioning: https://dev.mysql.com/doc/refman/8.0/en/innodb-multi-versioning.html
- MySQL 8.0 Reference Manual, Consistent Nonlocking Reads: https://dev.mysql.com/doc/refman/8.0/en/innodb-consistent-read.html

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Two-phase locking and Serializable Snapshot Isolation

What it is

Both are ways to give **serializability**: the guarantee that concurrent transactions produce the same result as if they had run one at a time, in some order.

- **Two-phase locking (2PL)** is the classic, pessimistic approach: transactions take locks on data and wait for each other.
- **Serializable Snapshot Isolation (SSI)** is an optimistic approach built on MVCC snapshots: transactions run without blocking each other, and the database aborts one if it detects a pattern that could break serializability. It was introduced by Cahill, Röhm and Fekete (SIGMOD 2008) and has been PostgreSQL's SERIALIZABLE level since version 9.1.

The problem it solves

Snapshot isolation alone permits **write skew**: two transactions read overlapping data, each makes a decision based on what it read, and each writes to a *different* row, so no write-write conflict is detected.

Example: a hospital rule says at least one doctor must be on call. Alice and Bob are both on call. Each, at the same moment, checks "how many are on call?", sees 2, and takes themselves off call. Both commits succeed under snapshot isolation. Now nobody is on call. No serial order could produce this result.

How it works

2PL, step by step.

1. Before reading an item, take a **shared lock** (many readers can hold it). Before writing, take an **exclusive lock** (only one holder, no readers).
2. **Growing phase:** a transaction may acquire locks but not release any.
3. **Shrinking phase:** once it releases one lock, it may not acquire any more.
4. **Strict 2PL** (what databases actually do): hold all locks until commit or abort, which avoids other transactions reading uncommitted data.
5. To stop **phantoms** (new rows appearing that match a query's condition), lock ranges, not just rows: **predicate locks** or, in InnoDB, **next-key/gap locks** on index ranges.
6. **Deadlock** handling: T1 waits for T2 while T2 waits for T1. The database detects the cycle in its wait-for graph (or times out) and aborts one.

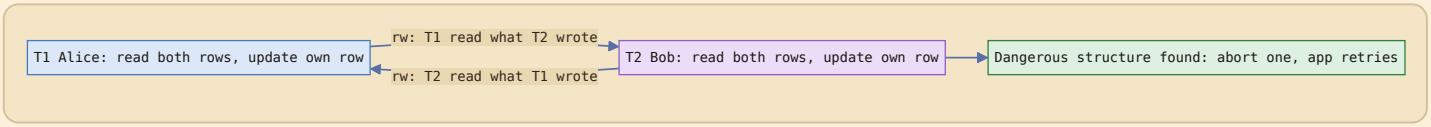
Applied to the doctors: Alice's transaction takes shared locks on the "on call" rows it reads. Bob's does too. Alice tries to update her row and needs an exclusive lock, but Bob holds a shared lock on it: wait. Bob tries to update his row: wait on Alice. Deadlock is detected, one is aborted, the other proceeds. Correct, but with blocking.

SSI, step by step.

1. Every transaction reads from an MVCC snapshot exactly as in snapshot isolation. No read blocks.
2. Reads also leave non-blocking markers called **SIREAD locks** (PostgreSQL shows them as `SIReadLock` in `pg_locks`) on the rows, pages or relations they read.
3. When a transaction writes something another concurrent transaction read, the database records a **rw-antidependency** (an edge "reader -> writer": the reader did not see the writer's change, so the reader must come first in any serial order).

4. Cahill's theorem: every snapshot-isolation anomaly contains a **dangerous structure**: two consecutive rw edges T1 -> T2 -> T3 among concurrent transactions (T1 and T3 may be the same). T2 is called the **pivot**.
5. When such a structure appears, abort one transaction with a serialization failure (SQLSTATE `40001`). The application retries it.

Doctors again: Alice reads Bob's row, Bob writes it (edge Alice -> Bob). Bob reads Alice's row, Alice writes it (edge Bob -> Alice). That is a cycle of two rw edges: one of them is aborted at commit. Neither ever waited.



False positives. SSI tracks a sufficient condition, not the exact cycle, and may coarsen tuple locks to page or table locks to save memory. So it sometimes aborts transactions that would have been safe. PostgreSQL's docs warn that sequential scans take relation-level predicate locks and raise the failure rate.

Complexity. 2PL: lock acquire O(1) per item, but waiting time depends on contention. SSI: O(1) bookkeeping per read and write, memory proportional to concurrent transactions' read sets, and retries cost a whole transaction.

Trade-offs

- **2PL:** simple guarantee, no retries for anomalies, but readers block writers, throughput collapses under contention, and deadlocks must be handled.
- **SSI:** read-only and low-conflict workloads run at nearly snapshot-isolation speed, but applications *must* retry on `40001`, aborts waste work under high contention, and some aborts are false positives.
- Neither removes the need for good transaction design: keep transactions short and touch rows in a consistent order.

Real-world examples

- **PostgreSQL** SERIALIZABLE = SSI (since 9.1); REPEATABLE READ = plain snapshot isolation.
- **MySQL InnoDB** SERIALIZABLE = lock-based: plain `SELECT`s become locking reads, with next-key locks to prevent phantoms.
- **SQL Server** offers both a lock-based SERIALIZABLE and a separate SNAPSHOT level.

In the interview

Q: What is write skew and which isolation level prevents it?

A: Two transactions read the same data, then each update a different row based on what they read, breaking an invariant. Snapshot isolation allows it; serializable (2PL or SSI) prevents it. A workaround under weaker levels is `SELECT ... FOR UPDATE` on the rows the decision depends on.

Q: Why is SSI called optimistic?

A: It lets transactions run without blocking and only checks for dangerous dependency patterns, aborting at the end if needed. 2PL is pessimistic: it blocks up front in case of conflict.

Q: What must the application do differently under SSI?

A: Treat SQLSTATE 40001 as normal and retry the whole transaction, discarding anything computed in the failed attempt.

TL;DR

2PL achieves serializability by locking: grow locks, hold them until commit, block and break deadlocks. SSI runs on MVCC snapshots without blocking, tracks read-write dependencies, and aborts a transaction when two consecutive rw-edges form a dangerous structure. 2PL pays in waiting; SSI pays in retries.

Sources

- Cahill, Röhm, Fekete, "Serializable Isolation for Snapshot Databases", SIGMOD 2008: https://courses.cs.washington.edu/courses/csep544/09sp/resources/fekete-sigmod2008.pdf
- PostgreSQL docs, Transaction Isolation: https://www.postgresql.org/docs/current/transaction-iso.html
- PostgreSQL wiki, SSI: https://wiki.postgresql.org/wiki/SSI

Skip lists

What it is

A **skip list** is a sorted linked list with extra "express lanes" stacked on top. The bottom level links every element in order. Each element is also randomly promoted to higher levels, so level 1 links about every 4th element (with promotion probability 1/4), level 2 every 16th, and so on. Searching starts in the top lane and drops down when it would overshoot. William Pugh published it in 1990 as a simpler alternative to balanced trees.

The problem it solves

A plain sorted linked list costs O(n) to search. Balanced trees (red-black, AVL) give O(log n) but need rotations to stay balanced, which are fiddly to code and hard to make concurrent. A skip list gets expected O(log n) search, insert and delete using only coin flips and pointer updates, and its bottom level is already an ordered list, so range scans are trivial.

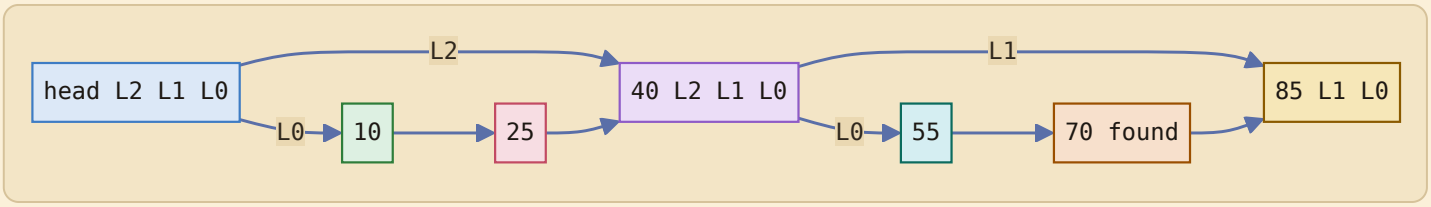
How it works

Structure. Each node has a key (and value) plus an array of forward pointers, one per level it belongs to. A head node has pointers at every level. A node's height is chosen at insert time: start at 1, and keep adding a level with probability p (Redis uses p = 1/4) up to a maximum (Redis: 32).

Search for 70, step by step. Keys: 10, 25, 40, 55, 70, 85. Suppose 40 and 85 reached level 1, and 40 reached level 2.

1. Start at the head, top level (2). Next is 40, which is less than 70: move to 40.
2. At 40, level 2: next is nothing. Drop to level 1.
3. Level 1: next is 85, greater than 70. Drop to level 0.
4. Level 0: next is 55, less than 70: move. Next is 70: found.

We touched 40, 55 and 70 instead of walking from 10.



Insert, step by step.

1. Search as above, but remember at each level the last node visited before dropping down (the `update` array).
2. Pick a random height h for the new node.
3. For each level 0 to h-1, splice the node in after `update[level]`.

```
insert(key):
  x = head
  for lvl from top down to 0:
    while x.next[lvl] and x.next[lvl].key < key:
      x = x.next[lvl]
    update[lvl] = x
  h = 1
  while random() < p and h < MAX: h += 1
  node = new Node(key, h)
  for lvl in 0..h-1:
    node.next[lvl] = update[lvl].next[lvl]
    update[lvl].next[lvl] = node
```

Delete uses the same `update` array and unsplices the node at each of its levels.

Complexity. Expected search, insert and delete: O(log n), with expected number of levels log base 1/p of n. With p = 1/4 and one million items, that is log4(10^6), about 10 levels. Space: each node has on average 1/(1 - p) pointers, so 1.33 pointers per node at p = 1/4 (versus 2 at p = 1/2, and 2 for a binary tree). The worst case is O(n), but it is astronomically unlikely and does not depend on input order, only on the random number generator.

Rank queries. Store on each forward pointer a **span**: how many bottom-level nodes it jumps over. Summing spans along the search path gives the element's position (rank) in O(log n). This is exactly what Redis does.

Trade-offs

- Simple code, no rotations, easy to reason about; ordered iteration for free.
- Expected, not guaranteed, O(log n). Fine in practice.
- Pointer-chasing across random memory locations is less cache-friendly than a B-tree with packed arrays; for on-disk data B-trees win.
- Lock-free concurrent versions are comparatively easy (Java's `ConcurrentSkipListMap`), because inserts only touch a few local pointers.

Real-world examples

- **Redis sorted sets** (`ZADD`, `ZRANGE`, `ZRANK`, leaderboards). Once a sorted set grows beyond 128 entries or any member exceeds 64 bytes (the `zset-max-listpack-entries` and `zset-max-listpack-value` defaults), Redis stores it as a skip list plus a hash table. The hash table maps member to score for O(1) `ZSCORE`; the skip list orders by score for ranges and ranks. Each node also has a `backward` pointer for reverse iteration. Small sets use a compact flat encoding (a listpack) instead.
- **Why Redis chose a skip list** over a balanced tree, per its author: comparable memory use (tunable through p), range iteration that is at least as cache-friendly as trees, and much simpler code, which made adding O(log n) `ZRANK` via spans a small change.
- **LevelDB and RocksDB memtables** default to skip lists.
- Java's `ConcurrentSkipListMap` and `ConcurrentSkipListSet`.

In the interview

Q: How would you build a real-time leaderboard with "top 10" and "what is my rank"?

A: A Redis sorted set: `ZADD` to update scores, `ZREVRANGE 0 9` for the top 10, `ZREVRANK` for a player's rank, all O(log n) thanks to the skip list with spans. Shard by game or region if one set gets too large.

Q: Why not use a balanced binary tree?

A: Same asymptotic cost, but a skip list is simpler to implement and to extend (rank via spans), needs no rebalancing, and naturally supports ordered range scans along the bottom level.

Q: What is the expected height of a skip list?

A: About log base 1/p of n. For p = 1/4 and a million entries, around 10 levels.

TL;DR

A skip list is a sorted linked list with randomly built express lanes, giving expected O(log n) search, insert and delete without rebalancing. Redis sorted sets pair one with a hash table: the hash for score lookup, the skip list (with spans) for ranges and ranks.

Sources

- Pugh, "Skip Lists: A Probabilistic Alternative to Balanced Trees", CACM 33(6), 1990: https://homepage.cs.uiowa.edu/~ghosh/skip.pdf
- Redis source, `server.h` (ZSKIPLIST_MAXLEVEL, ZSKIPLIST_P, zskiplistNode): https://github.com/redis/redis/blob/unstable/src/server.h
- Redis default configuration, `redis.conf` (zset-max-listpack-*): https://github.com/redis/redis/blob/unstable/redis.conf
- Redis docs, Sorted sets: https://redis.io/docs/latest/develop/data-types/sorted-sets/
- antirez on why Redis uses skip lists (Hacker News): https://news.ycombinator.com/item?id=1171423

Cache eviction beyond LRU

What it is

An **eviction policy** decides which item to throw out when a cache is full. **LRU** (least recently used) evicts the item untouched for the longest. This section covers three better-behaved families:

- **CLOCK:** a cheap approximation of LRU.
- **ARC** (Adaptive Replacement Cache): balances recency and frequency and tunes itself.
- **TinyLFU / W-TinyLFU:** an admission filter based on approximate frequency counts, used by the Java library Caffeine.

Two terms: **recency** is how recently an item was used; **frequency** is how often. **Hit ratio** is the fraction of requests served from the cache.

The problem it solves

LRU has two weaknesses:

1. **Cost:** every hit must move the item to the front of a linked list, which needs a lock and touches shared memory on every read.
2. **Scan pollution:** one big sequential scan (a backup, a report) reads many items once each and pushes all the genuinely popular items out. LRU only knows recency, not frequency.

How it works

CLOCK, step by step. Arrange cache slots in a circle, each with a **reference bit**, and keep a pointer (the "hand").

1. On a hit, set the item's bit to 1. Nothing moves, so hits are nearly free.
2. To evict, look at the slot under the hand. If its bit is 1, clear it to 0 and advance (a "second chance"). If its bit is 0, evict that item.

Example: slots A(1) B(0) C(1) D(1), hand at A. Evict: A has 1, clear and move. B has 0: evict B. A cleared bit means "not used since the hand last passed". PostgreSQL's shared buffer cache uses a "clock sweep" variant with small usage counters instead of a single bit.

ARC, step by step (Megiddo and Modha, IBM, FAST 2003). For a cache of c pages, ARC keeps four lists:

- **T1:** pages seen once recently (recency).
- **T2:** pages seen at least twice recently (frequency).
- **B1, B2: ghost lists** holding only the keys (no data) of pages recently evicted from T1 and T2.
- A target size **p** for T1, adjusted on the fly.

1. Hit in T1 or T2: move the page to the front of T2.

2. Miss, but key found in ghost list B1: we evicted a recency page too early, so increase p (give T1 more room), then load the page into T2.
3. Miss found in B2: we evicted a frequency page too early, so decrease p.
4. Complete miss: put the page in T1.
5. When evicting, take from T1 if it is larger than p, else from T2, and record the key in the matching ghost list.

A one-off scan only ever touches T1, so it cannot flush T2. In the paper's SPC1-like benchmark with a 4 GB cache, LRU achieved a 9.19% hit ratio and ARC 20%. ARC was patented by IBM; PostgreSQL briefly used it in 8.0 and replaced it (2Q in 8.0.2, then clock sweep in 8.1) partly to avoid the patent.

TinyLFU and W-TinyLFU, step by step (Einziger, Friedman and Manes).

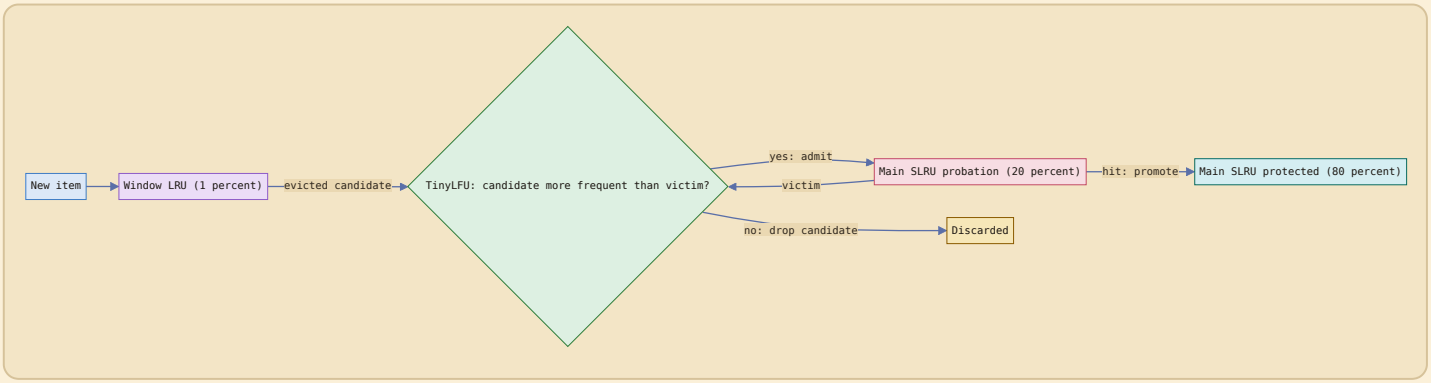
TinyLFU is an **admission policy**: when a new item arrives and the cache is full, it decides whether the newcomer deserves to replace the eviction victim at all.

1. Every access increments the item's count in a **count-min sketch**: a small 2D array of counters indexed by several hash functions. The estimate is the minimum of the item's counters, so it can overcount slightly but never undercounts. Caffeine uses 4-bit counters.
2. **Aging (reset):** after W accesses (Caffeine uses W = 10 x cache size), halve every counter, so old popularity fades.
3. A **doorkeeper** (a small Bloom filter) absorbs first-time items so one-hit wonders do not consume sketch counters.
4. **Admission:** compare the estimated frequency of the candidate with that of the victim. Admit the candidate only if it is more frequent.

Plain TinyLFU rejects new items that are about to become popular but have no history yet (sudden bursts). **W-TinyLFU** adds a **window**:

1. New items enter a small LRU window (1% of capacity in the paper).
2. Items evicted from the window become candidates for the **main cache** (99%), which is a **segmented LRU** (SLRU): 20% probation, 80% protected. Items hit while in probation are promoted to protected.
3. When main is full, the TinyLFU filter compares the window's candidate with probation's victim, and the more frequent one stays.
4. Caffeine goes further and resizes the window with **hill climbing**: it nudges the window bigger or smaller and keeps the change if the hit ratio improves.

Worked example: cache of 100 items: window 1, probation 20, protected 79. A backup job reads 10,000 cold keys. Each enters the window and is immediately a candidate; its estimated frequency is about 1, while probation's victim has, say, 6. Every cold key is rejected, and the hot set survives.



Complexity. CLOCK: O(1) hit, amortised O(1) eviction, 1 bit per entry. ARC: O(1) per request, plus ghost metadata for up to c extra keys. W-TinyLFU: O(1) per request plus a sketch of a few bytes per entry (Caffeine's docs cite about 8 bytes per entry for the 4-bit sketch).

Trade-offs

- CLOCK: very cheap and concurrent-friendly, but only approximates LRU and is still not scan-resistant on its own.
- ARC: self-tuning and scan-resistant, but uses ghost lists and is more complex; patent history limited adoption.

- W-TinyLFU: near-optimal hit ratios across many traces with tiny metadata, but frequency estimates are approximate and the design has more knobs (window size, sample size).
- Any fancy policy only pays off if the workload has skew; for uniform random access, everything performs about the same.

Real-world examples

- **PostgreSQL** shared buffers: clock sweep with usage counts.
- **ARC**: ZFS uses an ARC-derived cache (also called the ARC).
- **Caffeine** (W-TinyLFU) is used by Cassandra, Kafka, HBase, Solr, Neo4j, Druid and frameworks such as Spring Cache and Quarkus.

In the interview

Q: Why is LRU vulnerable to scans and how do you fix it?

A: LRU treats one access the same as many, so a big one-time scan evicts popular items. Fix with a policy that considers frequency: ARC's separate T1/T2 lists, SLRU segments, or a TinyLFU admission filter that refuses cold items.

Q: What does TinyLFU use instead of exact counters?

A: A count-min sketch with small counters that are halved periodically. It estimates frequency for far more keys than the cache holds, in a few bytes per entry.

Q: Why does CLOCK scale better than LRU under concurrency?

A: A hit just sets a bit; there is no list reordering, so readers do not contend on a shared lock.

TL;DR

CLOCK makes LRU cheap with a reference bit and a sweeping hand. ARC keeps recency and frequency lists plus ghosts and adapts the split. W-TinyLFU puts a small LRU window in front of an SLRU main cache and admits items only if a count-min sketch says they are more popular than the victim, which beats LRU on skewed and scan-heavy workloads.

Sources

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- PostgreSQL 8.1 release notes (clock sweep): https://www.postgresql.org/docs/release/8.1.0/

Hierarchical timing wheels

What it is

A **timing wheel** is a data structure for managing a very large number of timers (timeouts, retries, scheduled tasks) cheaply. Picture a clock face divided into slots. Each slot holds a list of timers due in that time interval, and a pointer advances one slot per **tick** (a fixed time unit, such as 1 ms). A **hierarchical** timing wheel stacks several wheels with coarser ticks, like the seconds, minutes and hours hands of a clock. George Varghese and Tony Lauck described the family in 1987.

The problem it solves

Servers track enormous numbers of timers: every open connection has an idle timeout, every request has a deadline, every retry has a back-off. Most are cancelled before they fire (the response arrived in time). The usual alternative, a **priority queue** (a heap ordered by deadline), costs O(log n) to add and to cancel. A timing wheel makes add and cancel O(1) and fires timers in batches.

How it works

A single (simple) wheel. n slots, tick u. A timer due in d ticks (d < n) goes in slot `(current + d) mod n`. Each tick: advance the pointer and fire everything in the new slot. Cancel: remove the timer from its slot's doubly linked list in O(1). Limit: it only covers n x u of time.

Hashed wheel. For longer delays, put the timer in slot `(current + d) mod n` anyway and store how many full rotations it must wait. Each pass decrements the count and fires timers at zero. Simple, but long timers are visited on every lap.

Hierarchical wheels, step by step. Example modelled on Kafka: tick 1 ms, 20 slots per wheel.

- Level 0: 20 slots of 1 ms, covering 20 ms.
- Level 1: 20 slots of 20 ms, covering 400 ms.
- Level 2: 20 slots of 400 ms, covering 8 s. (Kafka creates higher "overflow" wheels on demand.)

Add a timer due 350 ms from now (time 0):

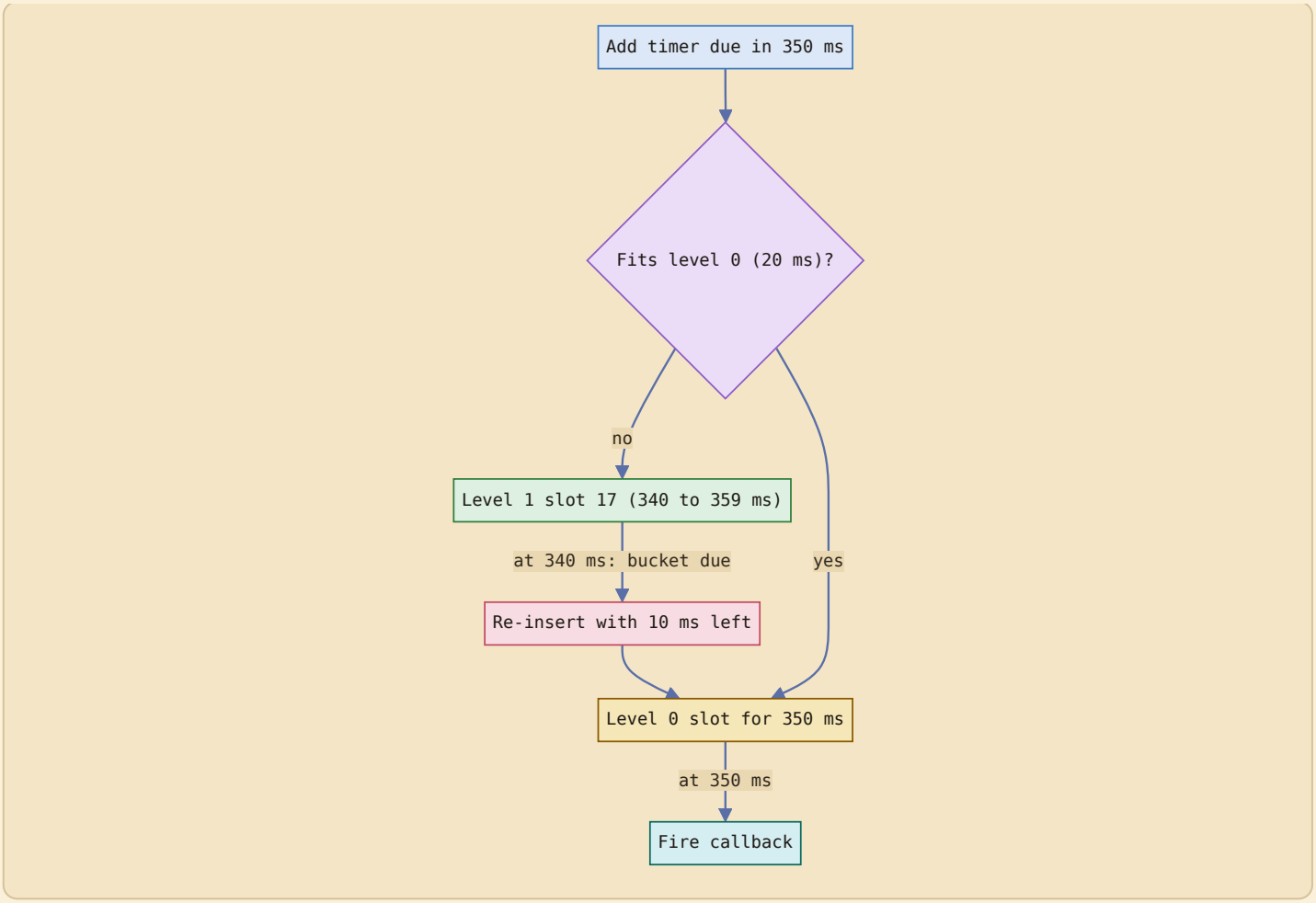
1. Too far for level 0 (20 ms). Fits level 1 (400 ms). Slot = 350 / 20 = 17 (rounded down), the bucket that covers 340 to 359 ms.
2. At t = 340 ms that level-1 bucket comes due. Its timers are not fired yet; they are **re-inserted** with their remaining time. Our timer has 10 ms left, so it lands in level 0 slot 10 (relative).
3. At t = 350 ms level 0 reaches that slot and the timer fires.

Each timer moves down at most once per level, so add costs O(m) where m is the number of levels (tiny), cancel is O(1), and per-tick work is proportional only to timers that actually come due.

Avoiding wasted ticks. Ticking every millisecond when nothing is due wastes CPU. Kafka keeps only the *buckets* (not individual timers) in a Java `DelayQueue` ordered by bucket expiry, so the timer thread sleeps until the next non-empty bucket is due.

```
add(timer):
  if timer.deadline < currentTime + tick: run(timer) and return
  if timer.deadline < currentTime + tick * slots:
    bucket = buckets[(timer.deadline / tick) mod slots]
    bucket.append(timer)
    if bucket expiry changed: delayQueue.offer(bucket)
  else:
    overflowWheel().add(timer)   # coarser wheel, created lazily

advanceTo(t):  # called when delayQueue hands back a due bucket
  currentTime = t rounded down to tick
  for each timer in bucket: add(timer)   # re-insert lower or run
```



Complexity. Add: O(m) for m levels (O(1) for a single wheel). Cancel: O(1). Per tick: O(1) plus the timers that expire or cascade. Space: O(number of slots + timers).

Trade-offs

- Precision is limited to the tick (and coarser on upper levels if they fire in place). Good for timeouts, wrong for high-precision scheduling.
- Re-inserting (cascading) timers from upper wheels costs work in bursts; the Linux kernel's 2015 rework chose to let timers on coarse levels fire in place, slightly late, to avoid cascading.
- A heap is simpler and exact; it wins when timers are few or must fire precisely.
- Memory for the slot arrays is fixed even when idle.

Real-world examples

- **Apache Kafka** "purgatory" (requests waiting for acks or data) moved to hierarchical timing wheels. In Confluent's benchmark with a high timeout, the old design saturated at about 25,000 requests per second while the new one reached about 105,000.
- **Netty** `HashedWheelTimer`: default 100 ms tick, 512-slot wheel, for I/O timeouts that need not be exact.
- **Linux kernel** timer wheel for low-resolution timeouts (network, I/O).
- Job schedulers and delay queues built on Redis or in-process often use the same idea.

In the interview

Q: You need ten million connection idle timeouts, most of which get reset. What data structure?

A: A hashed or hierarchical timing wheel: O(1) add and cancel, batch firing per tick. A heap costs O(log n) for every reset and cancel.

Q: What do you give up compared with a priority queue?

A: Precision: timers fire at tick granularity. Plus some cascade work as timers move from coarse wheels to fine ones.

Q: How do you avoid the CPU cost of ticking every millisecond?

A: Only wake when a non-empty bucket is due, for example by keeping buckets in a delay queue, as Kafka does.

TL;DR

A timing wheel is a circular array of time slots with a moving pointer; timers drop into the slot for their deadline. Stacking wheels with coarser ticks covers long delays with O(1) cancel and near-O(1) add, which is why Kafka, Netty and the Linux kernel use them for huge numbers of mostly-cancelled timeouts.

Sources

- Varghese and Lauck, "Hashed and Hierarchical Timing Wheels", SOSP 1987: http://www.cs.columbia.edu/~nahum/w6998/papers/sosp87-timing-wheels.pdf
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- Netty API, HashedWheelTimer: https://netty.io/4.1/api/io/netty/util/HashedWheelTimer.html
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Distributed snapshots: Chandy-Lamport and Flink checkpoint barriers

What it is

A **distributed snapshot** is a record of the state of every process in a distributed system, plus the messages in flight between them, that together form a **consistent cut**: a moment that *could* have happened, where no message is recorded as received without also being recorded as sent. The **Chandy-Lamport algorithm** (K. Mani Chandy and Leslie Lamport, 1985) takes such a snapshot without stopping the system. Apache Flink's **checkpoint barriers** are a variant tailored to streaming dataflows.

The problem it solves

There is no global clock, so you cannot say "everyone record your state at 12:00:00.000". If each process just records whenever it likes, the result can be inconsistent. Example: two bank branches, P and Q, each hold $100. P sends $10 to Q. If P records after sending ($90) and Q records before receiving ($100), the snapshot shows $190 and $10 has vanished. A consistent snapshot is needed for checkpointing (restart after a crash), detecting deadlock or termination, and debugging.

How it works

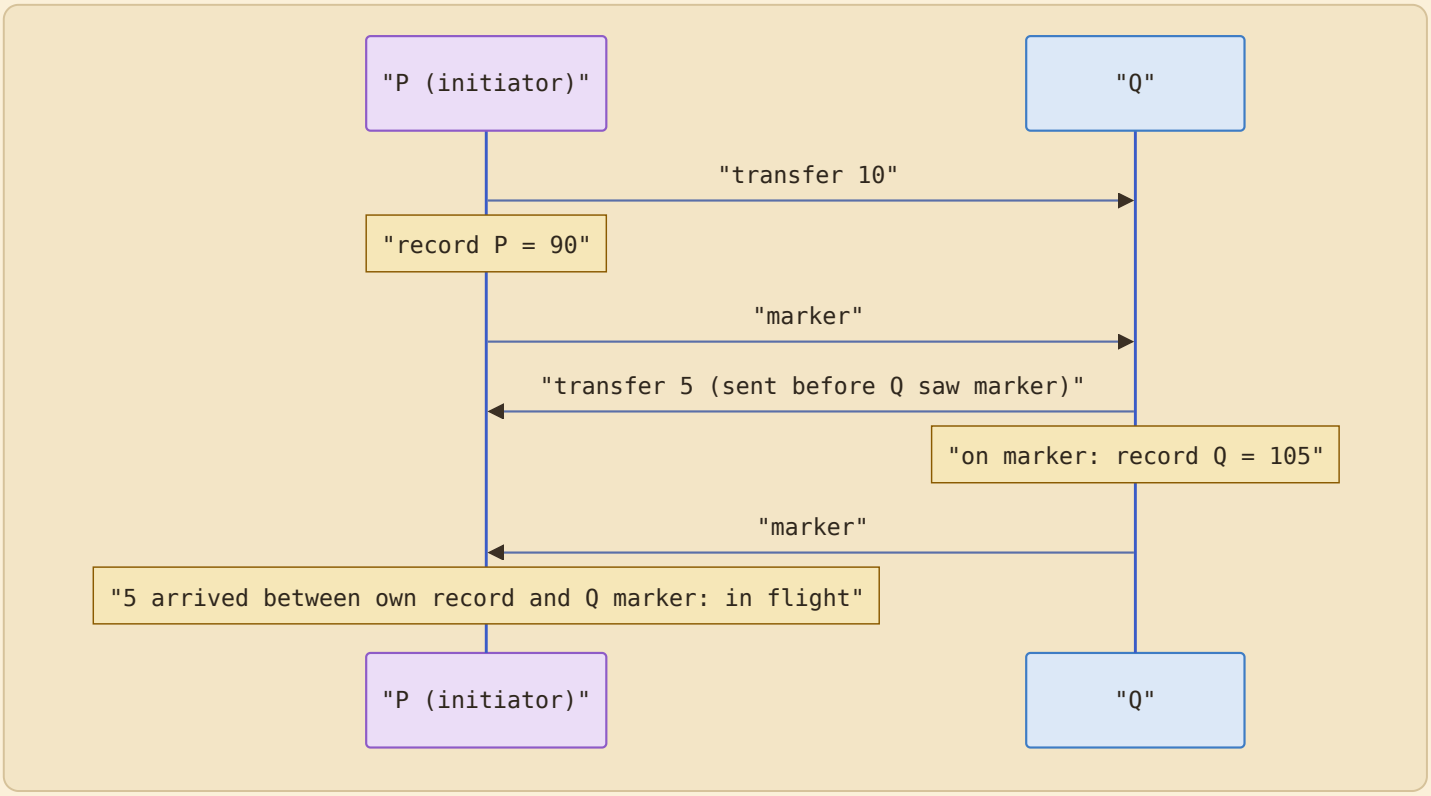
Assumptions. Channels between processes are reliable and FIFO (first in, first out: messages arrive in the order sent). A special **marker** message is used to separate "before the snapshot" from "after".

Chandy-Lamport, step by step.

1. The **initiator** records its own state, then sends a marker on every outgoing channel before sending anything else.
2. When a process receives a marker on channel C for the first time:
    - record its own state,
    - record channel C's state as empty,
    - send a marker on every outgoing channel,
    - start recording every message that arrives on each *other* incoming channel.
3. When a process that has already recorded receives a marker on channel C: stop recording C. The messages recorded on C between the process's own snapshot and this marker are C's in-flight state.
4. The snapshot is complete when every process has received a marker on every incoming channel.

Worked example with the bank. P is initiator.

1. P sends $10 to Q (P now $90), then records "P = 90" and sends a marker to Q. Q, meanwhile, sends $5 to P (Q now $95).
2. Q receives the $10 (Q = $105), then the marker (FIFO guarantees the $10 arrived first). Q records "Q = 105", records channel P->Q as empty, sends a marker to P.
3. P receives Q's $5 *after* its own snapshot and before Q's marker, so P records it as in flight on channel Q->P.
4. P receives Q's marker: done. Snapshot: P = 90, Q = 105, in flight Q->P = 5. Total 200. Consistent.



Flink checkpoint barriers, step by step. A Flink job is a graph of sources, operators and sinks running in parallel. Flink's mechanism is inspired by Chandy-Lamport and described in Carbone et al., "Lightweight Asynchronous Snapshots for Distributed Dataflows" (2015).

1. The **checkpoint coordinator** (in the JobManager) asks every source to start checkpoint n.
2. Each source records its position (for example, Kafka partition offsets) and injects **barrier n** into its output stream. Barriers flow in line with records and never overtake them.
3. An operator with one input, on seeing barrier n, snapshots its state (asynchronously to durable storage such as S3) and forwards the barrier.
4. An operator with several inputs performs **alignment**: once barrier n arrives on one input, it buffers further records from that input until barrier n has arrived on all inputs. Then it snapshots and forwards the barrier. This keeps the cut consistent, giving exactly-once state.
5. Each sink acknowledges when it has seen barrier n on all inputs. When all acknowledgements arrive, checkpoint n is complete.
6. On failure, Flink restores every operator's state from the latest complete checkpoint and rewinds sources to the recorded offsets.

Why no in-flight channel state? Because the data graph is acyclic and sources are replayable: aligned barriers guarantee no record is "between" operators at the cut, so channel state is empty. (Cycles in the graph do need recorded channel state, as in the original algorithm.)

Unaligned checkpoints (Flink 1.11, 2020): under backpressure, alignment can stall for a long time. Instead, the operator forwards the barrier as soon as the first copy arrives and stores the overtaken in-flight buffers as part of the checkpoint, which is closer to classic Chandy-Lamport. Checkpoints finish fast regardless of throughput, at the cost of more I/O.

At-least-once mode simply skips alignment: records after the barrier may be processed before the snapshot, so they are replayed and may appear twice after recovery.

Complexity. Messages: one marker per channel, O(E) for E channels. Storage: total process state plus in-flight messages (zero in aligned Flink). Time: roughly the time for markers to cross the graph's longest path, plus the time to persist state.

Trade-offs

- No global pause, but snapshots cost bandwidth and storage; frequent checkpoints shorten replay after failure but add overhead.
- Aligned barriers: no channel state, but alignment adds latency and stalls under backpressure.
- Unaligned: fast checkpoints under backpressure, but larger checkpoints and more I/O.
- End-to-end exactly-once also needs a replayable source and a transactional or idempotent sink (for example Flink's two-phase-commit Kafka sink); the snapshot alone covers only internal state.

Real-world examples

- **Apache Flink**: barriers, aligned and unaligned checkpoints, savepoints (always aligned).
- Distributed debuggers and deadlock or termination detectors, the original motivation in the 1985 paper.
- Similar marker-based ideas appear in other stream processors and in consistent backups of sharded systems.

In the interview

Q: Why can't each node just snapshot its own state at 12:00?

A: Clocks drift and messages are in flight, so you may record a message as received but not sent, or lose one. You need a consistent cut, which markers (or barriers) provide.

Q: How does Flink achieve exactly-once state?

A: Sources inject barriers and record offsets, operators align barriers across inputs and snapshot state, and on failure everything rolls back to the last complete checkpoint and sources replay from the stored offsets.

Q: What problem do unaligned checkpoints solve?

A: Under backpressure, aligned barriers queue behind slow data and checkpoints take very long or time out. Unaligned checkpoints let the barrier overtake buffered data and save that data as part of the snapshot.

TL;DR

Chandy-Lamport takes a consistent global snapshot without stopping anything: record local state, flood markers, and record messages that arrive on a channel before its marker. Flink adapts it with barriers injected at sources and aligned at multi-input operators, plus replayable sources, to give exactly-once state; unaligned checkpoints trade larger snapshots for speed under backpressure.

Sources

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- Carbone et al., "Lightweight Asynchronous Snapshots for Distributed Dataflows" (arXiv 1506.08603): https://arxiv.org/abs/1506.08603
- Apache Flink docs, Stateful Stream Processing: https://nightlies.apache.org/flink/flink-docs-stable/docs/concepts/stateful-stream-processing/
- Apache Flink 1.11.0 release announcement (unaligned checkpoints): https://flink.apache.org/2020/07/06/apache-flink-1.11.0-release-announcement/

Inverted index

What it is

An **inverted index** is the data structure behind almost every search box. A normal ("forward") index maps a document to the words it contains. An inverted index flips that around: it maps each word to the list of documents that contain it.

- A **term** is one searchable unit, usually a lower-cased word after cleanup.
- The **dictionary** (or term dictionary) is the sorted list of all distinct terms.
- A **posting** is one entry saying "this term appears in document N" (optionally with how often and at which positions).
- A **posting list** is all the postings for one term, sorted by document ID.

> **Analogy:** the index at the back of a textbook. You look up "photosynthesis" and get a list of page numbers. You never read the whole book to find it.

The problem it solves

Scanning every document for every query is hopeless at scale. With 1 billion documents of 10 KB each (10 TB), even a machine reading 1 GB/s would need almost 3 hours per query. An inverted index lets a query touch only the posting lists for its own terms, which is usually a tiny fraction of the data, and turns "find documents with apple AND pie" into "intersect two sorted lists of integers".

How it works

Step 1: Analyse the text. Each document goes through an **analyser**: split into tokens (tokenise), lower-case, optionally drop very common "stop words" (the, a, of) and reduce words to a root form (**stemming**: "apples" becomes "apple").

Step 2: Build postings. Take three tiny documents:

Doc ID	Text
1	Red apple pie
2	Green apple
3	Red car

Emit (term, docID) pairs, sort them by term then docID, and group:

Term	Posting list
apple	1, 2
car	3
green	2
pie	1
red	1, 3

The query "red AND apple" intersects [1, 3] with [1, 2] by walking both lists with two pointers: result [1]. Because lists are sorted, intersection costs O(length of list A + length of list B), and with **skip pointers** (stored jumps such as "the 128th entry is doc 9,412") it can leap over long runs instead of stepping through them.

Step 3: Write immutable segments. Real engines such as Lucene (used by Elasticsearch, OpenSearch and Solr) do not update one giant index in place. New documents collect in a memory buffer; when it fills or a refresh timer fires, the buffer is written to disk as a small, complete, **immutable segment** (a mini inverted index that never changes once written). A search queries every segment and merges the results.

A delete does not rewrite a segment; it just marks the document as dead in a small "live documents" bitmap, and an update is a delete plus a new add.

Step 4: Merge segments. Thousands of tiny segments would make every query slow (one lookup per segment) and keep dead documents on disk forever. A background **merge policy** combines small segments into bigger ones, dropping deleted documents as it copies. Lucene's default `TieredMergePolicy` groups segments of similar size into tiers; Solr documents its defaults as about 10 segments per tier, merging 10 at once, with a cap of about 5 GB per merged segment.

Worked numbers: suppose each flush produces a 10 MB segment.

1. After 10 flushes there are ten 10 MB segments, so they merge into one 100 MB segment.
2. After ten 100 MB segments exist, they merge into one 1 GB segment.
3. Growth stops near the 5 GB cap.

Each document is therefore rewritten about log10(5,000 MB / 10 MB) = log10(500), roughly 2.7, so about 3 times over its life. That repeated rewriting is called **write amplification**, and it is the price paid for append-only segments.

Step 5: Compress posting lists. A popular term can appear in hundreds of millions of documents, so lists must be small. Two tricks combine:

- **Delta (gap) encoding:** store the difference from the previous doc ID instead of the ID itself. Sorted IDs give small, positive gaps.
- **Varint (variable-length integer, also called VByte):** store each number in as few bytes as possible. Each byte carries 7 bits of the number; the top bit says "more bytes follow" (1) or "this is the last byte" (0). Numbers below 128 take 1 byte, below 16,384 take 2 bytes.

Worked example: posting list [3, 7, 150, 152, 1000].

1. Gaps: 3, 4, 143, 2, 848.
2. 3, 4 and 2 are below 128, so 1 byte each.
3. 143 = 1 x 128 + 15. Low 7 bits = 15, so the first byte is 15 + 128 (continuation bit) = 143 = 0x8F, the second byte is 1 = 0x01. Two bytes.
4. 848 = 6 x 128 + 80. First byte 80 + 128 = 208 = 0xD0, second byte 6 = 0x06. Two bytes.
5. Total: 1 + 1 + 2 + 1 + 2 = 7 bytes, against 5 x 4 = 20 bytes as plain 32-bit integers. That is almost 3 times smaller, and real lists with dense terms compress far better.

```
encode_varint(n):
    out = []
    while n >= 128:
        out.append((n % 128) + 128)   # low 7 bits, continuation bit set
        n = n // 128
    out.append(n)                     # last byte, top bit clear
    return out
```

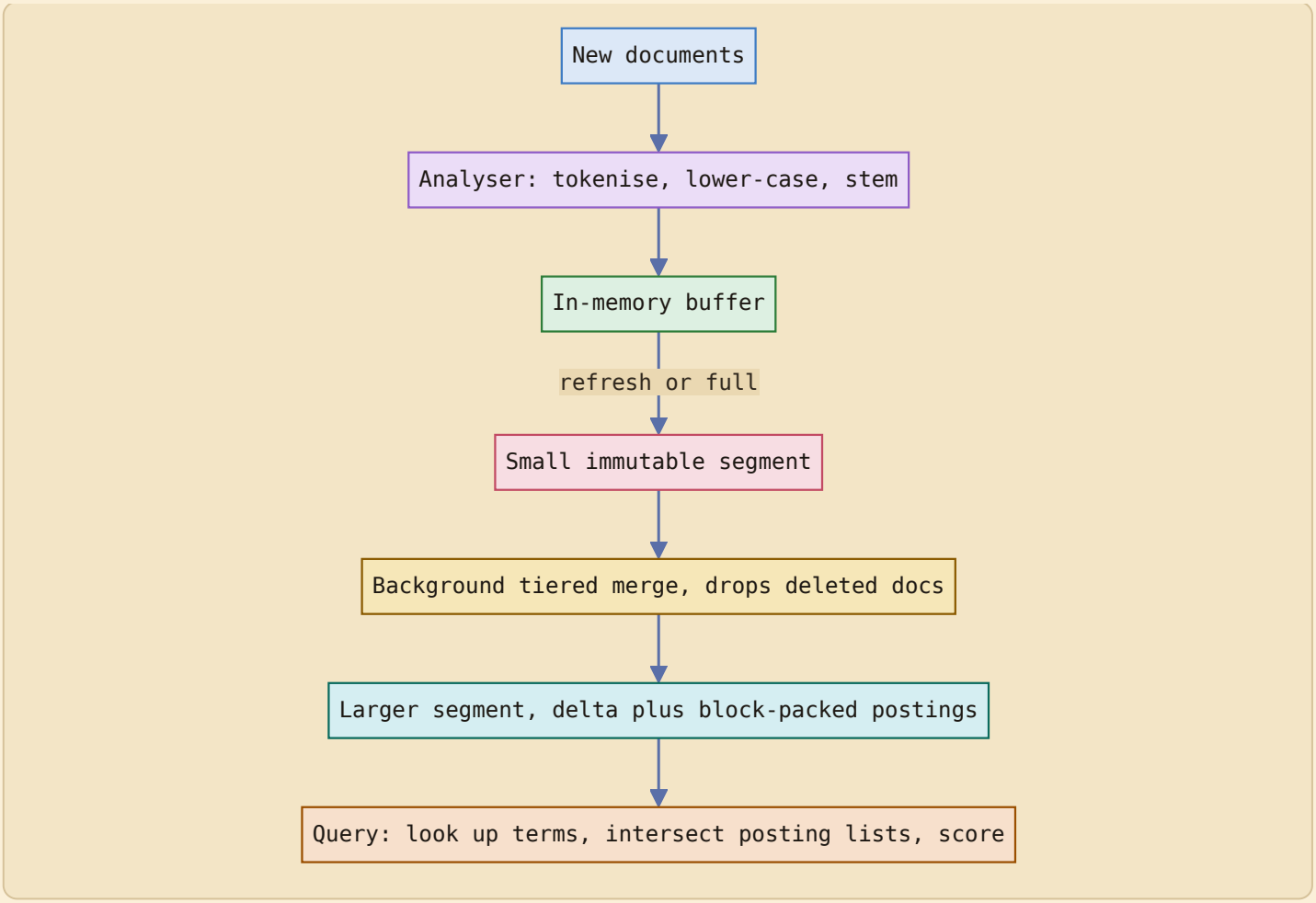
Modern Lucene goes further: it packs gaps in blocks of 128 integers using the smallest bit width that fits the largest gap in the block, and uses varints only for the leftover tail. Block packing decodes much faster on modern CPUs (central processing units) than byte-at-a-time varints.

Step 6: Partition across machines. When the index outgrows one machine there are two ways to split it:

- **Document-partitioned (local index):** each shard holds a complete inverted index for a subset of documents. A query is sent to every shard (**scatter-gather**), each returns its local top 10, and a coordinator merges them.
- **Term-partitioned (global index):** each shard owns whole posting lists for a subset of terms ("apple" lives on shard 7). A query only visits the shards owning its terms.

Worked comparison: 1 billion documents over 100 shards.

- Document-partitioned: a two-word query fans out to all 100 shards, each scans posting lists for its 10 million documents, returns 10 hits, and the coordinator merges 1,000 candidates into a top 10. Load is even, but every query touches every shard, and the slowest shard sets the response time.
- Term-partitioned: the same query touches 2 shards, but if "apple" appears in 50 million documents, that shard must ship (or intersect against) a 50-million-entry list. Popular terms create **hot spots** (overloaded shards), and indexing one document must update many shards.



Complexity: building is dominated by sorting (term, doc) pairs, O(P log P) for P postings. A query costs roughly the total length of the posting lists it reads, less with skip pointers. Space is roughly proportional to P after compression, often a fraction of the raw text size.

Trade-offs

- **Freshness vs throughput:** refreshing every second gives near-real-time search but creates many small segments and more merge work. Refreshing every 30 seconds is cheaper but staler.
- **Compression vs CPU:** tighter encodings save disk and memory but cost decode time on every query. Block packing is the usual sweet spot.
- **Document vs term partitioning:** document partitioning is simpler, balances load and keeps indexing local, at the cost of fan-out to every shard. Term partitioning reduces fan-out but suffers hot terms and expensive multi-term queries. Nearly all production systems choose document partitioning.
- **Merging vs I/O (input/output):** aggressive merging keeps queries fast but burns disk bandwidth and can starve indexing.

Real-world examples

- **Lucene, Elasticsearch, OpenSearch, Solr:** immutable segments, tiered merging, delta plus block-packed postings, document-partitioned shards.
- **Google (2003 paper by Barroso, Dean and Holzle):** the index was split into shards of documents so that one query could run in parallel across many machines, with each shard replicated for throughput.
- **Database full-text indexes**, such as PostgreSQL's GIN (generalised inverted index), use the same term-to-rows idea.

In the interview

Q: Why are Lucene segments immutable?

A: Immutability means no locks for readers, files can be cached aggressively by the operating system, and compression can be tuned once at write time. The cost is that deletes become tombstones and the

system needs background merges to reclaim space.

Q: How do you split a 10 TB index?

A: By document, into shards sized so each fits comfortably on a node (tens of GB), and replicate each shard for read throughput and failover. Queries scatter to all shards and a coordinator merges the top results. I would mention term partitioning only to explain why hot terms make it a poor default.

Q: Why delta-encode before varint?

A: Varint saves space only when numbers are small. Raw doc IDs grow into the millions, but gaps between sorted IDs are small, especially for common terms, so most gaps fit in one byte.

TL;DR

An inverted index maps each term to a sorted list of document IDs. Engines write it as immutable segments, merge them in the background, compress postings with delta encoding plus varint or block packing, and shard by document so every query scatters to all shards and gathers a merged top-k.

Sources

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Ranking with TF-IDF and BM25

What it is

Once the inverted index finds the documents that match, **ranking** decides which order to show them in. Two classic scoring formulas dominate keyword search:

- **TF-IDF** (term frequency times inverse document frequency) scores a document higher when the query word appears often in it (TF) and when the word is rare across the whole collection (IDF).
- **BM25** ("Best Matching 25", from the Okapi research system) is a refined TF-IDF that stops rewarding repetition after a while and adjusts for document length. It is the default scoring function in Lucene, Elasticsearch and OpenSearch.

Terms used below: **N** is the number of documents in the collection. **df** (document frequency) is how many documents contain a term. **tf** (term frequency) is how many times the term appears in one document. **dl** is a document's length in words, and **avgdl** is the average length.

The problem it solves

A query like "apple pie" may match millions of pages. Users look at the first few results, so the order is the product. Simple matching ("does it contain the words?") gives no order at all. TF-IDF and BM25 are cheap, need no training data, and give a sensible order out of the box. They also serve as the first stage in modern systems, where a machine-learned model re-ranks the top few hundred BM25 results.

How it works

The intuition, in three ideas:

1. A document that mentions "apple" five times is probably more about apples than one that mentions it once (term frequency).
2. A match on a rare word ("apple") tells you more than a match on a common word ("the") (inverse document frequency).

3. A long document mentions everything more often just by being long, so its counts should be discounted (length normalisation, BM25 only).

Worked collection (N = 4), query "apple pie":

Doc	Text	Length dl
D1	apple pie recipe apple	4
D2	apple phone review	3
D3	cherry pie	2
D4	best pie shop in town today	6

df(apple) = 2 (D1, D2), df(pie) = 3 (D1, D3, D4), avgdl = (4 + 3 + 2 + 6) / 4 = 3.75.

Step 1: TF-IDF. Use idf = log10(N / df) and score = sum over query terms of tf x idf.

- idf(apple) = log10(4/2) = 0.301
- idf(pie) = log10(4/3) = 0.125
- D1: 2 x 0.301 + 1 x 0.125 = **0.727**
- D2: 1 x 0.301 = **0.301**
- D3: 1 x 0.125 = **0.125**
- D4: 1 x 0.125 = **0.125**

Two problems show up. D3 and D4 tie, although D3 is a two-word document squarely about pie and D4 is a six-word document that mentions it in passing. And if D1 said "apple" 50 times, its score would grow 25-fold, which rewards spam.

Step 2: BM25. For each query term in a document:

```
idf(t)  = ln(1 + (N - df + 0.5) / (df + 0.5))          # Lucene's variant, never negative
norm(d) = 1 - b + b * dl / avgdl                        # 1.0 for an average-length doc
score(t, d) = idf(t) * tf * (k1 + 1) / (tf + k1 * norm(d))
score(q, d) = sum of score(t, d) over query terms t
```

The two knobs: **k1** controls how fast repeated terms saturate (default 1.2), and **b** controls how strongly length matters (default 0.75, where 0 means ignore length and 1 means full normalisation).

IDFs:

- idf(apple) = ln(1 + 2.5/2.5) = ln 2 = 0.693
- idf(pie) = ln(1 + 1.5/3.5) = ln 1.4286 = 0.357

Length factors (k1 x norm):

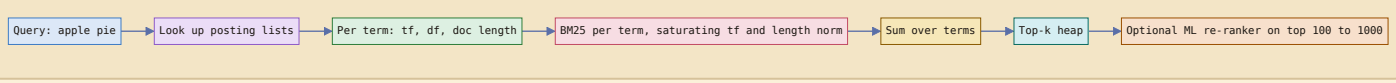
- D1: norm = 0.25 + 0.75 x 4/3.75 = 1.05, so k1 x norm = 1.26
- D2: norm = 0.25 + 0.75 x 0.8 = 0.85, so 1.02
- D3: norm = 0.25 + 0.75 x 0.533 = 0.65, so 0.78
- D4: norm = 0.25 + 0.75 x 1.6 = 1.45, so 1.74

Scores (k1 + 1 = 2.2):

- D1: apple = 0.693 x (2 x 2.2)/(2 + 1.26) = 0.693 x 1.350 = 0.936. pie = 0.357 x 2.2/(1 + 1.26) = 0.357 x 0.973 = 0.347. Total **1.28**.
- D2: apple = 0.693 x 2.2/(1 + 1.02) = 0.693 x 1.089 = **0.755**.
- D3: pie = 0.357 x 2.2/(1 + 0.78) = 0.357 x 1.236 = **0.441**.
- D4: pie = 0.357 x 2.2/(1 + 1.74) = 0.357 x 0.803 = **0.286**.

Ranking: D1 > D2 > D3 > D4. The tie is broken in favour of the short, focused document.

Step 3: See saturation. For D1's length factor, the tf part tf x 2.2 / (tf + 1.26) gives 0.97 at tf = 1, 1.35 at tf = 2, 1.95 at tf = 10, and can never exceed k1 + 1 = 2.2 however many times the word is repeated. Keyword stuffing stops paying.



Complexity: scoring is O(total postings read) with a heap of size k, O(log k) per insert. Engines avoid scoring every match with tricks such as WAND and block-max WAND, which skip documents whose best possible score cannot enter the current top k.

Trade-offs

- **No semantics:** BM25 only matches exact (analysed) words. "Car" will not match "automobile" unless you add synonyms or combine it with vector search ("hybrid search").
- **Tuning:** k1 and b are rarely tuned, but short fields (titles) and long fields (bodies) often want different b values. Elastic's guidance is that defaults are a reasonable start and any change must be measured on real queries.
- **Per-shard statistics:** each shard computes df and avgdl from its own documents, so the same document can score slightly differently on different shards. Elasticsearch offers a "DFS" mode that gathers global statistics first, at the cost of an extra round trip.
- **Cheap and explainable:** no training data needed, and a score can be broken down term by term, which is valuable for debugging.

Real-world examples

- Lucene switched its default similarity from classic TF-IDF to BM25 in version 6 (2016), and Elasticsearch and OpenSearch inherit it.
- Retrieval-augmented generation (RAG) systems commonly run BM25 alongside an embedding search and fuse the two result lists, because BM25 handles rare identifiers, product codes and names well.
- Web search engines historically used BM25-style features as just one input among hundreds to a learned ranking model.

In the interview

Q: Why does BM25 beat raw TF-IDF?

A: Two reasons: term frequency saturates (the tenth mention adds much less than the second), and length normalisation stops long documents winning just because they contain more words. Both are controlled by k1 and b.

Q: Where does BM25 sit in a modern search stack?

A: As the cheap first stage. It retrieves the top few hundred candidates from the inverted index in milliseconds, then a learned re-ranker (gradient-boosted trees or a cross-encoder neural model) re-orders that short list using many more features.

Q: What happens if you set b = 0?

A: Length is ignored, so long documents gain an advantage. That can be right for fields of roughly fixed length, such as titles or tags.

TL;DR

TF-IDF scores a document by how often it contains the query terms, weighted by how rare those terms are. BM25 adds saturation (k1, default 1.2) and length normalisation (b, default 0.75), which makes it robust enough to be the default first-stage ranker in Lucene-based search.

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Near-duplicate detection with SimHash, MinHash and LSH

What it is

Near-duplicate detection finds items that are almost, but not exactly, the same: two copies of a news article with different adverts, a product listing reposted with one changed word, the same web page under two URLs. Exact duplicates are easy (compare a cryptographic hash such as SHA-256). Near-duplicates need hashes where similar inputs give similar outputs. Three tools do this:

- **SimHash** turns a document into a short bit string (often 64 bits) so that similar documents differ in only a few bit positions. Closeness is measured by **Hamming distance**, the number of positions where two bit strings differ.
- **MinHash** turns a document into a short list of numbers (a **signature**) so that the fraction of matching numbers estimates the **Jaccard similarity**: the size of the overlap of two sets divided by the size of their union.
- **LSH (locality-sensitive hashing)** is a way to bucket those signatures so that similar items land in the same bucket and you only compare items that share one, instead of comparing every pair.

The problem it solves

Comparing every pair is quadratic. With 1 billion pages there are about 5 x 10^17 pairs, which no cluster can check. Crawlers need to decide "have I seen this page already?" in milliseconds per page, and data pipelines (for example, de-duplicating training data for language models) need to cluster billions of documents. Both need a compact fingerprint plus an index that finds close fingerprints without scanning everything.

How it works

Step 1: Turn documents into features. A common choice is **shingles**: overlapping sequences of k words or characters. "the cat sat on" with 2-word shingles gives {the cat, cat sat, sat on}. Similar documents share most shingles.

Step 2a: SimHash (Charikar 2002, used at Google by Manku, Jain and Das Sarma). Toy version with 8-bit hashes and weighted features:

Feature	Weight	8-bit hash
apple	2	10110010
pie	1	01110100
recipe	1	10011001

1. Keep 8 counters, one per bit position, all starting at 0.
2. For each feature and each bit: add its weight if the bit is 1, subtract it if the bit is 0.
3. Sums per bit: +2, -2, +2, +4, -2, -2, 0, -2.
4. Output 1 where the sum is positive, else 0: **10110000**.

Now document B swaps "recipe" for "easy" (hash 00011101, weight 1). Its sums are 0, -2, +2, +4, -2, 0, 0, -2, giving **00110000**. Hamming distance from A: 1 bit. An unrelated document would differ in about half the bits (4 of 8, or 32 of 64).

Finding close SimHashes fast. Manku and colleagues reported that for about 8 billion pages, 64-bit SimHashes with a threshold of k = 3 differing bits worked well. The lookup trick is the pigeonhole principle: split the 64 bits into 4 blocks of 16. If two fingerprints differ in at most 3 bits, at least one of the 4 blocks must match exactly. So keep 4 tables, each keyed by one block, look up the new fingerprint's 4 blocks exactly, and only check full Hamming distance on the few hits.

Step 2b: MinHash (Broder 1997, built for AltaVista). Treat each document as a set of shingles. A = {a, b, c, d}, B = {b, c, d, e}. True Jaccard = |{b, c, d}| / |{a, b, c, d, e}| = 3/5 = **0.6**.

Pick several hash functions. For each, a document's MinHash value is the smallest hash among its shingles. The key fact: the probability that two sets have the same minimum equals their Jaccard similarity.

Hash	a	b	c	d	e	min of A	min of B	Match?
h1	3	7	1	9	5	1 (c)	1 (c)	yes
h2	2	8	6	4	1	2 (a)	1 (e)	no
h3	9	2	5	7	3	2 (b)	2 (b)	yes
h4	4	6	8	3	9	3 (d)	3 (d)	yes

Estimate = 3 matches / 4 hashes = 0.75, against the true 0.6. With 4 hashes the estimate is noisy. The standard error is about sqrt(J(1 - J)/n), so 100 hashes give sqrt(0.6 x 0.4 / 100), about 0.05. A 100-number signature (400 bytes at 4 bytes each) summarises a document of any size.

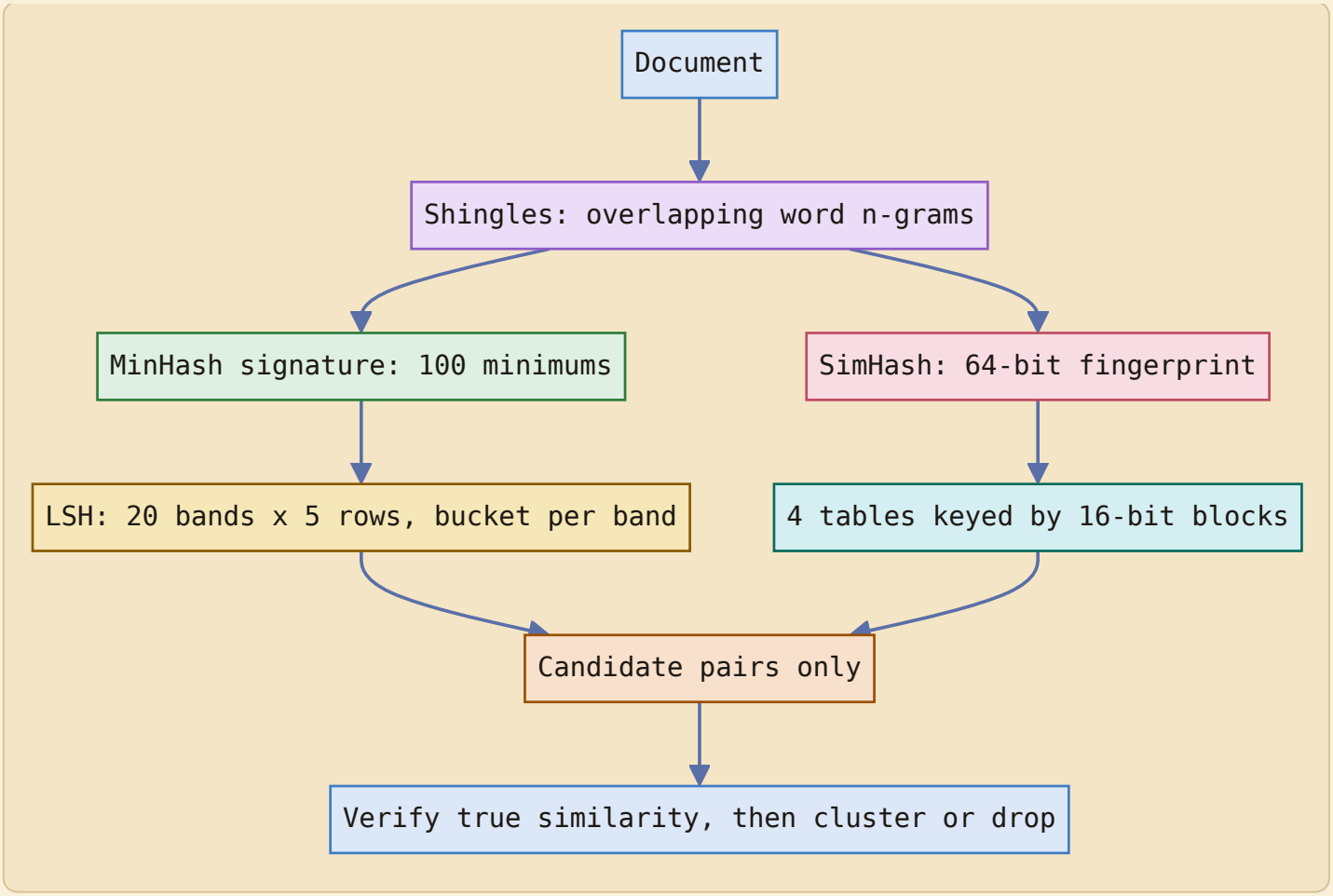
Step 3: LSH banding. Signatures still need pairwise comparison. Split each 100-number signature into **b = 20 bands of r = 5 rows**. Hash each band to a bucket. Two documents become a **candidate pair** if any one band matches exactly. For two documents with Jaccard s:

- P(one band matches) = s^r
- P(candidate) = 1 - (1 - s^r)^b

Similarity s	s^5	P(candidate) with b = 20
0.3	0.0024	0.047
0.5	0.031	0.47
0.8	0.328	0.9996

This is an S-shaped curve with its steep part near (1/b)^(1/r) = (1/20)^(1/5), about 0.55. Pairs above 0.8 are almost always caught; pairs below 0.3 are rarely compared. More rows per band raises the threshold (fewer false positives), more bands lowers it (fewer misses).

```
for doc in docs:
    sig = [min(h_i(s) for s in shingles(doc)) for i in 1..100]
    for band in 0..19:
        key = hash(band, sig[band*5 : band*5+5])
        for other in buckets[key]: candidates.add((doc, other))
        buckets[key].append(doc)
verify each candidate pair with exact Jaccard (or signature agreement)
```



Complexity: MinHash costs O(n x number of shingles) per document for n hashes, and O(n) space. LSH insert and lookup are O(b) bucket operations per document, so the whole job is close to linear instead of quadratic. SimHash costs O(features x 64) per document and 8 bytes of storage.

Trade-offs

- **SimHash vs MinHash:** SimHash is tiny (8 bytes) and good for "almost identical" pages with weighted features. MinHash estimates Jaccard directly, behaves predictably at moderate similarities (0.5 to 0.8) and is easier to tune with banding.
- **Banding parameters:** the b and r choice trades false positives (wasted verification work) against false negatives (missed duplicates). Choose them from the target threshold.
- **Shingle size:** small shingles make unrelated documents look similar; large shingles make one changed word break many shingles. Word 5-grams or character 5- to 9-grams are common starting points.
- **Memory:** LSH tables for billions of documents are large; systems often run it as a batch job (Spark, MapReduce) rather than an online index.

Real-world examples

- **AltaVista:** Broder's shingling and MinHash sketches clustered the whole crawl to remove near-duplicate pages.
- **Google crawling:** the Manku paper used 64-bit SimHash with k = 3 over roughly 8 billion pages.
- **Large-language-model data pipelines** de-duplicate web text with MinHash plus LSH before training, since repeated text wastes compute and encourages memorisation.
- **Plagiarism checkers and news aggregators** group syndicated copies of the same story.

In the interview

Q: How would a web crawler avoid storing the same page under many URLs?

A: Compute a 64-bit SimHash of each fetched page's text. Store fingerprints in a few tables keyed by fixed bit blocks; for a new page, look up each block, then check full Hamming distance on the hits. If any is within 3 bits, treat it as a duplicate and skip or canonicalise it.

Q: Why not just compare MinHash signatures pairwise?

A: Pairwise is still quadratic in the number of documents. LSH banding puts likely-similar documents in a shared bucket so you only compare within buckets, which makes the whole job near-linear.

Q: How do you pick b and r?

A: Fix the signature length n = b x r, then pick values so the threshold (1/b)^(1/r) sits just below the similarity you care about, and check the false-positive and false-negative rates with the 1 - (1 - s^r)^b formula.

TL;DR

Hash documents so that similar ones collide. SimHash gives a 64-bit fingerprint compared by Hamming distance; MinHash gives a signature whose agreement rate estimates Jaccard similarity. LSH banding then finds candidate pairs without an all-pairs comparison, turning a quadratic problem into a near-linear one.

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Vector search with HNSW, IVF and product quantization

What it is

Vector search finds the items whose **embeddings** are closest to a query's embedding. An embedding is a list of numbers (a **vector**, for example 768 numbers long) produced by a machine-learning model so that similar meaning gives nearby vectors. "Closest" is measured by a **distance** or **similarity**: Euclidean (L2) distance, inner product, or cosine similarity.

Exact search compares the query with every vector. **Approximate nearest neighbour (ANN)** search accepts missing a few true neighbours in exchange for huge speed-ups. Quality is measured by **recall@k**: of the true k nearest neighbours, what fraction did the search return? Recall@10 = 0.95 means 9.5 of the true top 10 on average.

Three building blocks cover most systems:

- **HNSW** (Hierarchical Navigable Small World): a layered graph you walk greedily towards the query.
- **IVF** (inverted file index): cluster the vectors and only search the few clusters nearest the query.
- **PQ** (product quantization): compress each vector to a few bytes so far more fit in memory.

The problem it solves

Brute force does not scale. 100 million vectors x 768 dimensions x 4 bytes = **307 GB** of raw floats, and each query needs 100 million x 768, about 77 billion multiply-adds. Semantic search, retrieval-augmented generation, recommendations and image search need answers in tens of milliseconds, often over billions of vectors. ANN indexes trade a little recall for orders of magnitude less latency and memory.

How it works

Step 1: Brute force baseline. Compute the distance to every vector and keep the top k in a heap. O(N x d) per query. Use it for small sets (under roughly 100,000 vectors) and as the ground truth when measuring recall.

Step 2: IVF, search only nearby clusters.

1. **Train:** run k-means (an algorithm that finds nlist cluster centres) on a sample. With N = 1,000,000, a common starting point is nlist close to sqrt(N), so 1,000 centres.
2. **Assign:** put each vector into the list of its nearest centre, like an inverted index where the "terms" are clusters. Each list holds about 1,000 vectors on average.
3. **Query:** compare the query with the 1,000 centres, pick the closest **nprobe** of them (say 10), and scan only those lists: 10 x 1,000 = 10,000 vectors plus 1,000 centres, about **1% of the brute-force work**.
4. **Tune:** raising nprobe increases recall and latency roughly linearly. A true neighbour sitting just across a cluster boundary is missed if its cluster is not probed, which is why nprobe = 1 gives poor recall.

Step 3: PQ, shrink the vectors. Split each vector into m sub-vectors, run k-means separately in each sub-space with 256 centres, and store only the centre number (1 byte) per sub-vector.

Tiny example with 4 dimensions, m = 2 sub-vectors and 4 centres each (2 bits):

- Vector x = [0.1, 0.9, 0.4, 0.3] splits into [0.1, 0.9] and [0.4, 0.3].
- Codebook 1: c0 = [0,0], c1 = [0,1], c2 = [1,0], c3 = [1,1]. Nearest to [0.1, 0.9] is c1 (squared distance 0.01 + 0.01 = 0.02).
- Codebook 2: c0 = [0.5,0.5], c1 = [0,0], c2 = [1,0], c3 = [0.5,0]. Nearest to [0.4, 0.3] is c0 (0.01 + 0.04 = 0.05, versus 0.10 for c3).
- Stored code: (1, 0). Four floats (16 bytes) became 4 bits.

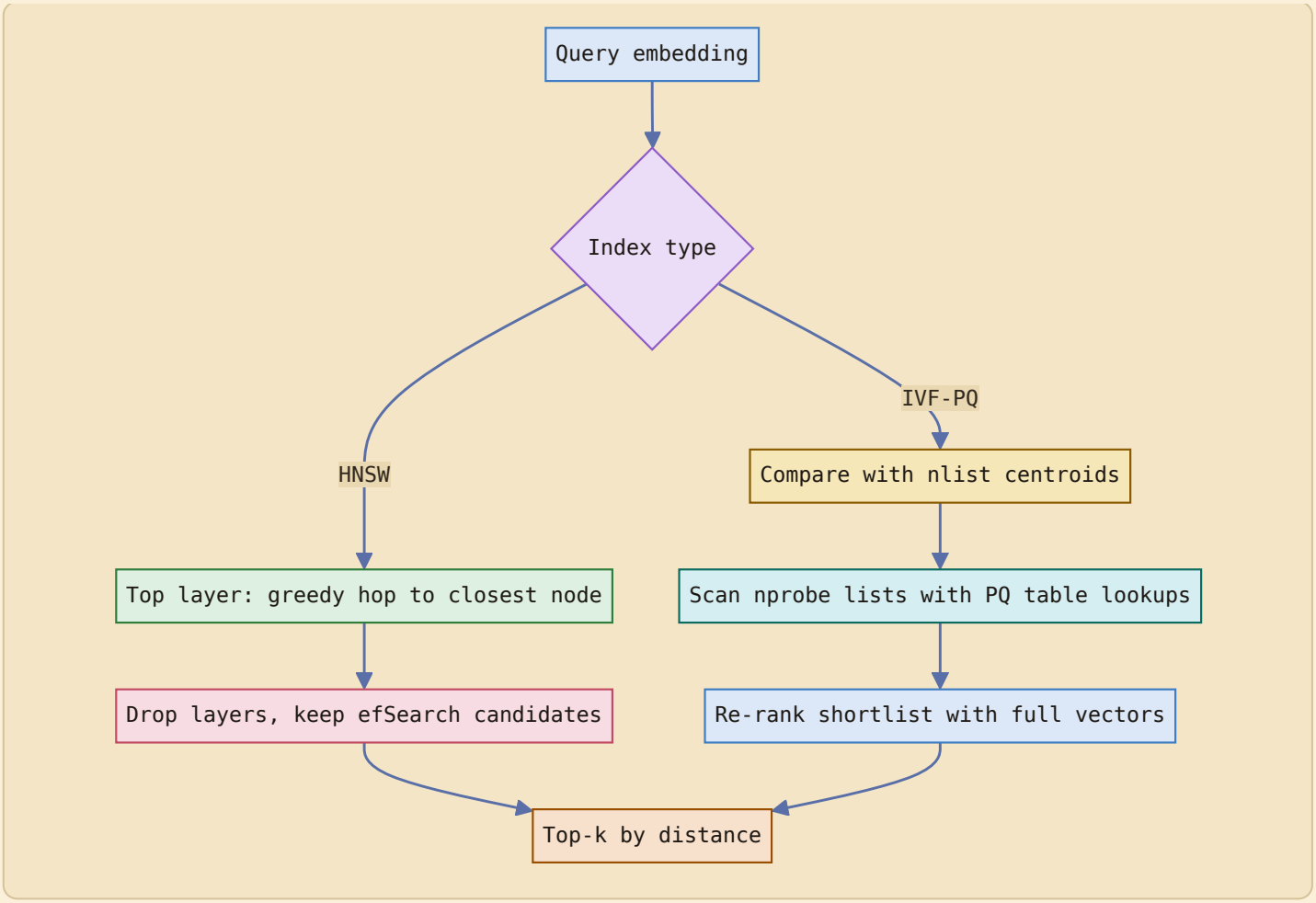
Asymmetric distance computation (ADC): at query time, compute the distance from each query sub-vector to every centre in its codebook once (an m x 256 table). The distance to any stored vector is then m table lookups added together, no floating-point vector maths.

Real numbers: 768 dimensions, m = 96 sub-vectors of 8 dimensions, 8 bits each, gives **96 bytes per vector** instead of 3,072, a 32x reduction. 100 million vectors shrink from 307 GB to about **9.6 GB**, which fits in one server's memory. Combining the two gives **IVF-PQ**: cluster with IVF, store residuals with PQ, and optionally **re-rank** the top few hundred with the full vectors kept on SSD.

Step 4: HNSW, walk a graph.

1. Each vector becomes a node. When inserted, it is linked to about **M** of its nearest existing nodes (M is commonly 16; the bottom layer allows 2M links).
2. Each node is also given a random top layer, with exponentially fewer nodes on higher layers, like an express-lane skip list.
3. **Search:** start at the single entry point on the top layer, greedily move to whichever neighbour is closest to the query until no neighbour is closer, then drop a layer and repeat. On the bottom layer, keep a candidate list of size **efSearch** (for example 64) and explore until it stops improving, then return the best k.
4. **Build quality** is set by **efConstruction** (candidate list size while inserting); higher means better graphs and slower indexing.

Search cost grows roughly with log N, and HNSW typically gives the best recall for a given latency when everything fits in RAM. Memory: with M = 16, the bottom layer stores 32 neighbour IDs x 4 bytes = 128 bytes per vector on top of the raw vector. For 100 million 768-dimension vectors that is about 13 GB of links plus 307 GB of vectors.



Complexity summary (N vectors, d dimensions):

Method	Query time	Memory per vector	Notes
Brute force	O(N d)	4d bytes	100% recall
IVF (flat lists)	O(nlist d + nprobe (N/nlist) d)	4d bytes	recall set by nprobe
IVF-PQ	same scan, but m lookups per vector	m bytes plus IDs	lowest memory, lower recall
HNSW	roughly O(log N) hops x M x d	4d + about 8M bytes	best recall/latency in RAM, slow build

Trade-offs

- **Recall vs latency:** every method has a dial (nprobe for IVF, efSearch for HNSW). Plot recall@10 against p99 latency and choose a point, such as 0.95 recall at 10 ms.
- **Memory vs recall:** PQ cuts memory 10 to 60 times but loses precision. Re-ranking a shortlist with full vectors recovers much of it.
- **Updates and deletes:** HNSW inserts are cheap-ish but deletes are awkward (tombstones, periodic rebuild). IVF needs retraining when the data distribution drifts.
- **Filtering:** "nearest products in stock under 20 dollars" interacts badly with ANN; filtering after search can return too few results, and filtering during graph traversal can disconnect the graph.
- **Build time:** HNSW builds are CPU-heavy; IVF-PQ needs a training pass but builds quickly.

Real-world examples

- **FAISS** (Meta's library) implements flat, IVF, PQ, IVF-PQ and HNSW indexes, plus GPU versions.
- **pgvector, Elasticsearch, OpenSearch, Redis, Qdrant, Weaviate and Milvus** all offer HNSW indexes; several also offer IVF or quantised variants.

- **Retrieval-augmented generation** stores document-chunk embeddings and retrieves the top 5 to 50 chunks per question.
- **Recommendation candidate generation** retrieves items whose embeddings are near the user's embedding (see Feed ranking below).

In the interview

Q: You have 1 billion 768-dimension embeddings. What index?

A: Raw floats would be about 3 TB, so not HNSW on full vectors in RAM. I would use IVF-PQ: tens of thousands of clusters, 64 to 96 byte PQ codes (64 to 96 GB total, sharded across a few machines), probe a few dozen lists, then re-rank the top few hundred candidates with full vectors from SSD.

Q: What do efSearch and nprobe control?

A: Both are query-time recall vs latency knobs. efSearch is how many candidates HNSW keeps while exploring the bottom layer; nprobe is how many IVF clusters are scanned. Neither needs a rebuild to change.

Q: How do you measure whether the ANN index is good enough?

A: Compute exact top-k with brute force for a sample of real queries, then measure recall@k and latency percentiles of the ANN index on the same queries, and also watch end-to-end quality metrics, because 0.9 recall may be fine if the downstream re-ranker is strong.

TL;DR

Vector search finds nearest embeddings approximately. HNSW walks a layered graph and gives the best recall for latency when vectors fit in memory. IVF searches only the closest clusters. PQ compresses vectors to tens of bytes using per-subspace codebooks. Real systems combine them and tune nprobe or efSearch for recall vs latency vs memory.

Sources

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Shortest paths for maps: Dijkstra, A* and contraction hierarchies

What it is

A road map is a **graph**: **nodes** are intersections and **edges** are road segments, each with a **weight** (length in metres or, more usefully, travel time in seconds). The **shortest path** problem asks for the route between two nodes with the smallest total weight. Three algorithms build on each other:

- **Dijkstra's algorithm** grows outward from the start in order of distance, settling one node at a time. It is exact for non-negative weights.
- **A*** (A-star) is Dijkstra with a **heuristic**, an estimate of the remaining distance to the goal, that steers the search towards the target.
- **Contraction hierarchies (CH)** precompute **shortcut** edges once, so that each query explores only a few hundred nodes even on a continental map.

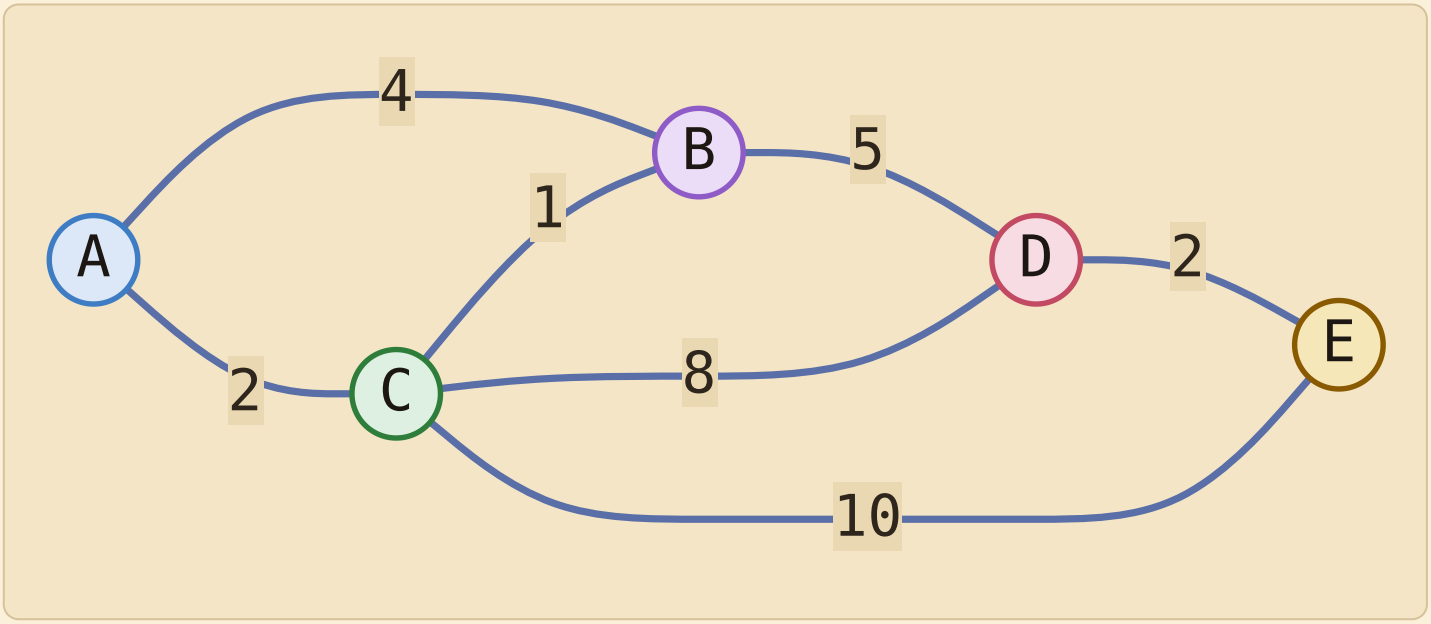
The problem it solves

A continental road network has tens of millions of nodes. Plain Dijkstra from London to Rome settles millions of them, which takes seconds; a navigation service handling thousands of requests per second,

plus distance-matrix requests (ETAs from 100 drivers to 100 riders = 10,000 routes), needs answers in milliseconds. Road networks have structure (motorways matter more than cul-de-sacs) that clever preprocessing can exploit.

How it works

The worked graph (undirected, weights in minutes):



Goal: shortest path from A to E.

Step 1: Dijkstra. Keep a tentative distance for each node (infinity at first) and a **priority queue** (a min-heap that always returns the smallest distance). Repeatedly take the closest unsettled node and **relax** its edges: if going through it gives a neighbour a shorter distance, update that neighbour.

```
dist[s] = 0, all others infinity, push (0, s)
while queue not empty:
    (d, u) = pop_min(queue)
    if u settled: continue
    mark u settled, if u == target: stop
    for (v, w) in edges(u):
        if d + w < dist[v]:
            dist[v] = d + w, prev[v] = u, push (dist[v], v)
```

Pop	Settled distance	Updates
A	0	B = 4, C = 2
C	2	B = 3 (via C beats 4), D = 10, E = 12
B	3	D = 8 (3 + 5 beats 10)
D	8	E = 10 (8 + 2 beats 12)
E	10	done

Answer: **A to C to B to D to E, 10 minutes**, found by following `prev` back from E. Note that the direct A-B road (4) lost to the detour through C (3). Complexity with a binary heap: O((V + E) log V) time, O(V) space.

Step 2: A*. Pop nodes by f = g + h, where g is the distance so far and h is the heuristic estimate to the goal. On maps h is usually straight-line distance divided by the maximum road speed.

- A heuristic is **admissible** if it never overestimates the true remaining distance. Admissible means A* still returns the optimal path.
- It is **consistent** (monotone) if for every edge u to v, h(u) <= w(u, v) + h(v). Consistency means each node is settled once, like Dijkstra.

Heuristic values: h(A) = 7, h(B) = 6, h(C) = 7, h(D) = 2, h(E) = 0. True remaining distances are A 10, B 7, C 8, D 2, so every h is at or below the truth (admissible), and checking each edge confirms consistency (for example C to B: 7 <= 1 + 6).

Pop (f)	g	Pushes (g, f)
A (7)	0	B (4, 10), C (2, 9)
C (9)	2	B (3, 9), D (10, 12), E (12, 12)
B (9)	3	D (8, 10)
D (10)	8	E (10, 10)
E (10)	10	done

Same answer. On this tiny graph the saving is nil, but on a real map Dijkstra explores a disc around the start while A* explores an ellipse stretched towards the goal. A weak heuristic (h = 0) turns A* back into Dijkstra; an overestimating one is faster but can return a worse route.

Step 3: Contraction hierarchies. Two phases.

Preprocessing (once, minutes to hours for a continent):

1. Order nodes by **importance** (heuristics such as how many shortcuts removing it would add; motorway junctions end up important, dead ends unimportant).
2. **Contract** nodes from least to most important. Removing node v, for every pair of its remaining neighbours u and w: if the shortest u-to-w path goes through v, add a shortcut edge u-w with that weight. If a **witness path** (an equally short path avoiding v) exists, skip it.

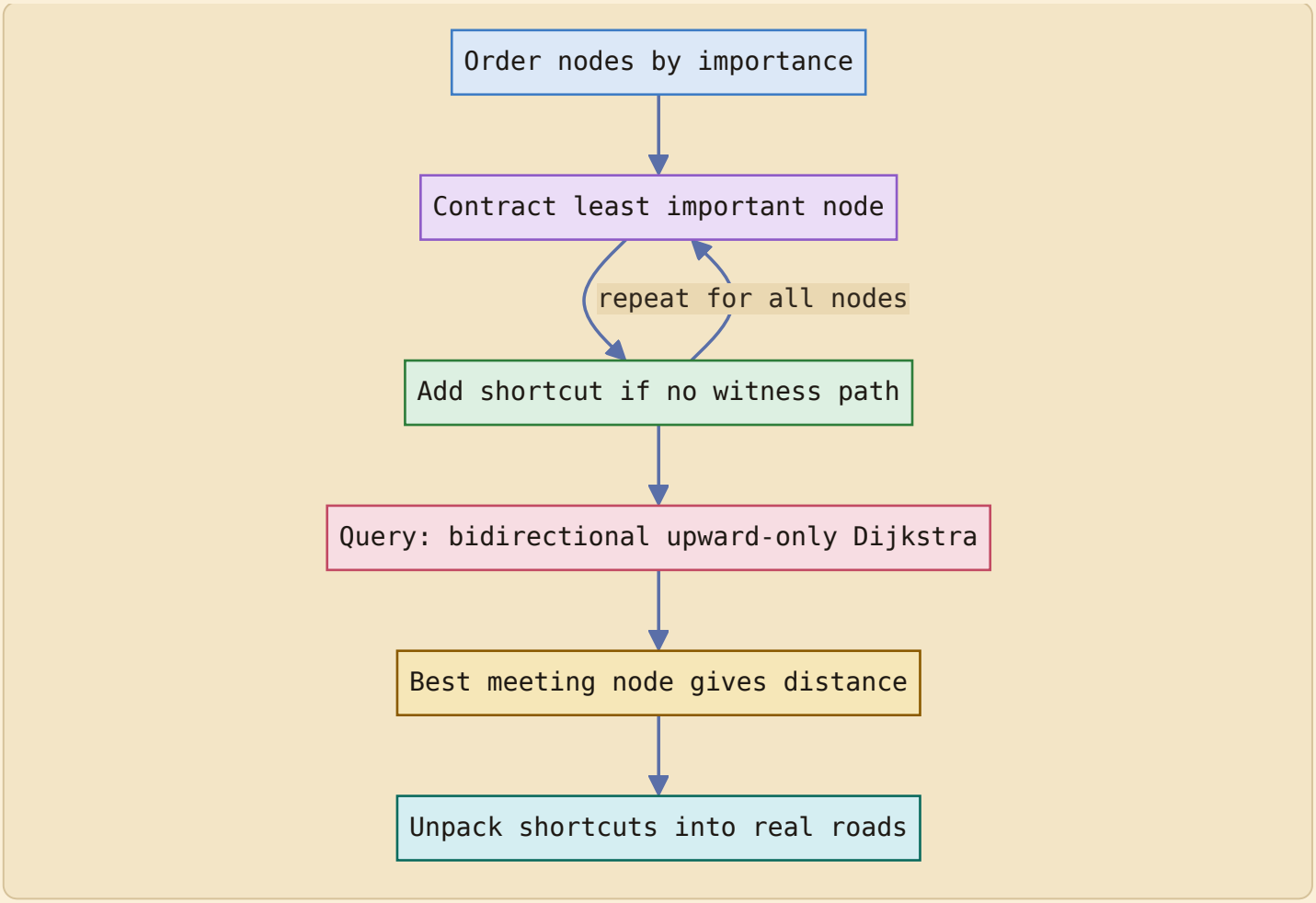
On our graph, order C (rank 1), A (2), B (3), D (4), E (5):

- Contract C (neighbours A, B, D, E). A-B via C = 3, shorter than the direct 4 and no witness, so **add shortcut A-B = 3 (via C)**. A-D via C = 10, but witness A-B-D = 9, skip. A-E via C = 12, witness A-B-D-E = 11, skip. B-D via C = 9 > direct 5, skip. B-E via C = 11, witness B-D-E = 7, skip. D-E via C = 18, skip.
- Contract A: only neighbour left is B, no pairs, no shortcuts. Same for B (only D left) and D (only E).

Query: run a bidirectional Dijkstra where the forward search from the start and the backward search from the target both only follow edges **upward** to higher-ranked nodes.

- Forward from A: up-edges are A-B (shortcut, 3). From B: B-D, total 8. From D: D-E, total 10.
- Backward from E: E is the top node, nothing higher.
- Best meeting node: E with 10 + 0 = **10**. Unpack the shortcut A-B into A-C-B: route A-C-B-D-E.

Every shortest path in the original graph goes "up then down" in rank, which is why the restricted search is still exact. On continental road networks the CH authors report queries that visit only a few hundred nodes and take well under a millisecond, against millions of settled nodes for plain Dijkstra.



Trade-offs

- **Dijkstra:** no preprocessing, handles any change instantly, but slow on big graphs.
- **A*:** easy speed-up when good geometry is available; benefit shrinks when weights are travel times on fast motorways (straight-line over top speed is a weak estimate).
- **CH:** fastest queries and excellent for distance matrices, but the hierarchy must be rebuilt when weights change. Live traffic therefore pushes systems towards **customisable** techniques such as multi-level Dijkstra (MLD) or customisable CH, which separate the slow metric-independent step from a fast reweighting step.
- **Memory:** shortcuts roughly double the edge count, which is acceptable in practice.

Real-world examples

- **OSRM** (Open Source Routing Machine, used with OpenStreetMap data) ships both a CH pipeline and an MLD pipeline. Its wiki recommends MLD for most uses because it supports frequent traffic updates, and CH where raw query speed for very large distance matrices matters most.
- **GraphHopper** offers CH, A* and a landmark-based A* variant.
- Commercial map providers combine hierarchical preprocessing with live traffic layers, and ride-hailing ETA systems run millions of shortest-path queries per minute for dispatch.

In the interview

Q: Why not just run Dijkstra for every route request?

A: On a continental graph it settles millions of nodes per query, which means seconds of CPU per route. Preprocessing such as contraction hierarchies lets each query touch only a few hundred nodes, so queries drop below a millisecond.

Q: What makes an A* heuristic admissible, and why does it matter?

A: It never overestimates the true remaining cost. That guarantees the first time the goal is popped its distance is optimal. Straight-line distance divided by the maximum speed is the standard admissible choice for travel-time maps.

Q: How do you handle live traffic with contraction hierarchies?

A: Full re-contraction is too slow to run every minute. Use a customisable variant: precompute the node order or partition once, then recompute only the edge weights when traffic changes (OSRM's MLD, or customisable CH), and fall back to A* on small regions for very fresh incidents.

TL;DR

Dijkstra settles nodes in distance order and is exact but slow on huge graphs. A* adds an admissible heuristic to aim the search at the goal. Contraction hierarchies precompute shortcuts by contracting unimportant nodes, so a bidirectional upward-only search answers continental queries in under a millisecond, at the cost of rebuilding (or customising) when weights change.

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Feed ranking

What it is

Feed ranking decides which posts, out of everything a user could see, appear in their home feed and in what order. A **chronological feed** simply shows newest first. A **ranked feed** scores every eligible post by how likely the user is to find it valuable and shows the best first.

Key terms:

- **Candidate generation (retrieval):** quickly picking a few hundred to a few thousand plausible posts out of millions.
- **Ranking (scoring):** running a heavier model on those candidates to predict engagement such as P(like), P(comment), P(share).
- **Re-ranking:** adjusting the scored list for goals the model does not capture: diversity, freshness, policy, ads.
- **In-network** posts come from accounts the user follows; **out-of-network** posts come from accounts they do not.

The problem it solves

A user following 500 accounts may have thousands of unseen posts, and a platform with out-of-network recommendations has hundreds of millions. A user opens the app and scrolls maybe 20 to 50 items. Chronological order buries the post from a close friend under bursts from prolific accounts. Ranking every post with a big neural model is impossible within a 200 ms budget, so the work is split into stages, each cheaper per item and handling more items than the next.

How it works

Step 1: EdgeRank, the classic intuition. Facebook's early News Feed (around 2010) was publicly described as summing, over the interactions ("edges") attached to a story, three factors:

- **Affinity** (u): how close the viewer is to the author, from past interactions. It is one-directional.

- **Weight** (w): how valuable the edge type is. A comment counts more than a like; photos counted more than text.
- **Time decay** (d): older edges count less.

Score = sum over edges of u x w x d. Worked example, with decay d = 1 / (1 + hours old):

- Post P from a close friend (u = 0.9), one comment (w = 4), 1 hour old (d = 0.5): 0.9 x 4 x 0.5 = **1.8**.
- Post Q from an acquaintance (u = 0.2), three likes (w = 1 each), 1 hour old: 3 x (0.2 x 1 x 0.5) = **0.3**.

P ranks first, even though Q has more raw engagement. The hand-tuned formula was later replaced by machine-learned models, but the three ideas (relationship, content value, recency) remain as features.

Step 2: Candidate generation. Several **sources** each return a few hundred items, cheaply:

1. Recent posts from followed accounts (read from a precomputed timeline cache or a fan-out store).
2. Posts liked by people similar to you (collaborative filtering).
3. **Embedding retrieval:** a **two-tower model** learns one embedding for the user and one for each post so that their dot product predicts engagement. Post embeddings go into an ANN index (see Vector search) and the user embedding retrieves the nearest few hundred.
4. Trending or topic-based posts.

Twitter's open-sourced 2023 pipeline, for example, gathered around 1,500 candidates per request from a pool of hundreds of millions, roughly half in-network and half out-of-network.

Step 3: Light ranking (optional). A small model (logistic regression or a tiny network) trims 1,500 candidates to about 500 using cheap features.

Step 4: Heavy ranking. A large model predicts several probabilities for each candidate using hundreds of **features**: author-viewer affinity, post age, media type, the viewer's past behaviour, the post's early engagement. The predictions are combined with business-chosen weights into one value score.

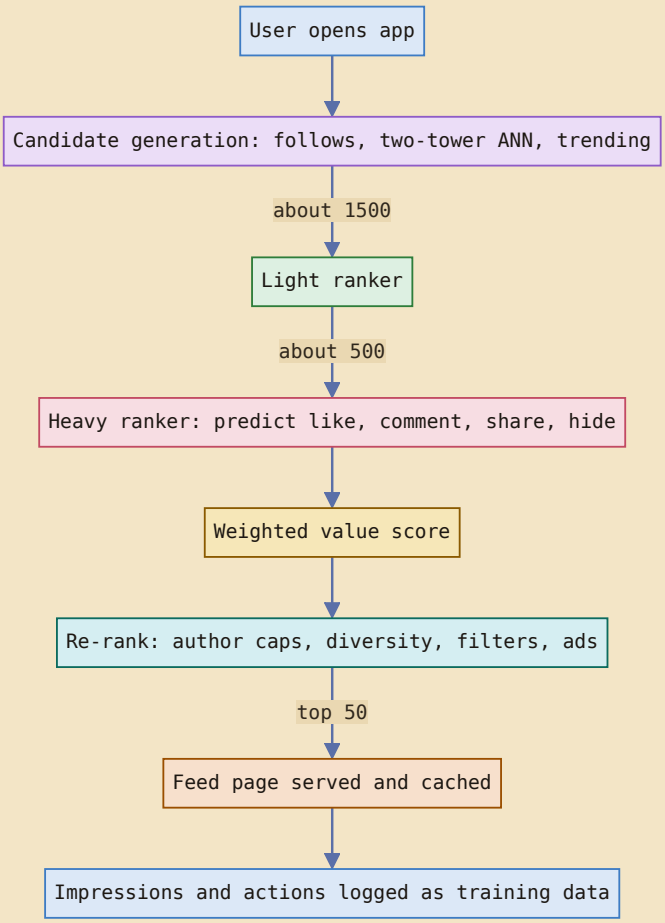
Illustrative weights: score = 1 x P(like) + 5 x P(comment) + 10 x P(share) - 50 x P(hide).

Post	P(like)	P(comment)	P(share)	P(hide)	Score
Friend's photo	0.30	0.05	0.01	0.001	0.30 + 0.25 + 0.10 - 0.05 = **0.60**
Viral meme	0.40	0.02	0.03	0.005	0.40 + 0.10 + 0.30 - 0.25 = **0.55**
Brand ad-like post	0.10	0.01	0.00	0.002	0.10 + 0.05 + 0 - 0.10 = **0.05**

The meme has the highest like probability, but the friend's photo wins because comments are weighted highly and the meme is more likely to be hidden. Changing these weights changes what the platform optimises for, which is a product decision, not just an engineering one.

Step 5: Re-ranking for diversity and rules. A pure score sort can show five posts from one prolific author in a row. Common fixes:

- **Author caps:** for example, no more than 2 consecutive posts from one author, or at most N per page. Example: sorted list A1 (0.9), A2 (0.85), A3 (0.8), B1 (0.7), C1 (0.6). With "no more than 2 consecutive from the same author" the feed becomes A1, A2, B1, A3, C1.
- **Maximal marginal relevance (MMR):** choose the next item maximising lambda x score - (1 - lambda) x similarity to items already chosen.
- **Freshness boosts, filters** (blocked users, already seen, policy violations) and **ad insertion** at fixed slots.




```
def build_feed(user):
    cands = union(follow_source(user, 500), ann_source(user_emb(user), 800), trending(200))
    cands = dedupe_and_filter(cands, user)                # blocked, seen, policy
    cands = top_n(cands, light_model.score, 500)
    for c in cands: c.p = heavy_model.predict(user, c)     # several probabilities
    for c in cands: c.value = dot(WEIGHTS, c.p)
    return rerank_with_diversity(sort_desc(cands, "value"))[:50]
```

Complexity and budget: each stage handles fewer items with a more expensive model. A typical budget might be 20 to 40 ms for retrieval, 50 to 100 ms for heavy ranking on GPU or CPU batches, and a few ms for re-ranking, all inside a 200 to 300 ms request.

Trade-offs

- **Engagement vs well-being:** optimising clicks can promote outrage or clickbait. Negative signals (hide, report) and survey-based "was this worth your time?" labels counteract it.
- **Freshness vs quality:** new posts have no engagement history (cold start); ranking must explore new content or it never gets a chance.
- **Precompute vs on demand:** ranking at request time uses the freshest signals but costs compute on every open; precomputing feeds is cheaper but stale.
- **Feedback loops:** the model is trained on what it chose to show, so it can reinforce its own biases. Random exploration and position-bias correction help.
- **Explainability:** a weighted sum of predicted actions is easier to reason about than a single opaque score.

Real-world examples

- **YouTube** (Covington, Adams and Sargin, 2016) described the canonical two-stage design: a deep candidate generation network narrowing millions of videos to hundreds, followed by a deep ranking network.

- **Twitter/X** open-sourced its "For You" pipeline in 2023: candidate sources, a light ranker, a heavy neural ranker, then heuristics and filters.
- **Facebook News Feed** evolved from the EdgeRank formula to multi-task neural models predicting many actions combined into a value score.
- **Instagram, TikTok and LinkedIn** all publicly describe multi-stage retrieve-then-rank systems.

In the interview

Q: Why split feed ranking into stages?

A: Cost. The heavy model may take a millisecond or more per item; scoring millions of posts per request is impossible. Cheap retrieval narrows to about a thousand, a light model to a few hundred, and only those get the expensive model, keeping the whole request within a couple of hundred milliseconds.

Q: How would you stop one popular creator dominating the feed?

A: Add a re-ranking pass after scoring: per-author caps in a window, or MMR that penalises similarity to already-chosen items, plus a diversity metric tracked in A/B tests.

Q: What would you log, and why?

A: Every impression with its position, the candidate source, model scores, and the subsequent actions (or lack of them). That is the training data for the next model and lets you correct for position bias and measure each source's contribution.

TL;DR

Modern feeds are a funnel: candidate generation pulls about a thousand posts from follows, embedding retrieval and trends; a light ranker trims; a heavy multi-task model predicts actions combined into a weighted value score; re-ranking adds diversity, filters and ads. EdgeRank's affinity x weight x time decay is the intuition that still lives on as features.

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